RCA
Solid
State

RCA
Solid
State

Industrial · Industrial · Industrial · Industrial · Industrial · Ind
Bipolar · BiMOS · I2L · Gold CHIP · MOS · Bipolar · BiMOS · I2L · Gold C
Consumer · Consumer · Consumer · Consumer · Consumer · Consumer · C

Linear Integrated Circuits

SSD-240A

RCA
Linear Integrated Circuits

This DATABOOK contains complete technical information on the full line of RCA standard commercial linear integrated circuits and MOS field-effect transistors for both industrial and consumer applications. An Index to Devices provides a complete listing of types, together with an indication of package options available for each of them.

The pages immediately following the Index to Devices include photographs of the packages used for RCA linear integrated circuits and MOS/FET's, a product-classification chart, recommended operating and handling considerations, a list of special terms and symbols used in the characterization of RCA linear integrated circuits and MOS/FET's, and a cross-reference directory that indicates RCA types recommended as direct replacements for other manufacturers' types.

Three separate data sections provide definitive ratings and electrical characteristics for (1) Linear Integrated Circuits for Industrial Applications, (2) Linear Integrated Circuits for Consumer Applications, and (3) MOS Field-Effect Transistors (MOS/FET's). Data pages for individual devices are included as nearly as possible in alpha-numerical sequence of type numbers. Because some devices are grouped together to show similarity of function or data, individual type numbers may be out of sequence. If you don't find the data on a specific type where you expect it to be, check the Index to Devices.

The DATABOOK also includes dimensional outlines for all currently available packages and selected RCA Application Notes on RCA Linear Integrated Circuits and MOS/FET's.

Table of Contents

	Page
Index to Devices	3
Packages	5
Product Classification Chart	7
Operating and Handling Considerations	8
Terms and Symbols	10
Cross-Reference Directory	12
Linear Integrated Circuits for Industrial Applications—Technical Data	17
Linear Integrated Circuits for Consumer Applications—Technical Data	291
MOS Field-Effect Transistors—Technical Data	433
Dimensional Outlines	475
Application Notes	483
RCA Sales Offices, Manufacturers' Representatives, and Authorized Distributors	559

RCA Solid State

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The device data shown for some types are indicated as preliminary. **Preliminary data** are intended for guidance purposes in evaluating devices for equipment design. Such data are shown for types currently being designed for inclusion in our standard line of commercially available products. For current information on the status of preliminary programs, please contact your local RCA sales office.

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Printed in USA/11-78

Type Number	Package Suffix						Data Bulletin File No.	Page
CA101	T	S	—	—	—	—	786	18
CA101A	T	S	G	—	—	—	786	18
CA107	T	S	G	—	—	—	785	22
CA111	T	S	G	—	—	—	797	25
CA124	E	G	—	—	—	—	796	29
CA139	E	G	—	—	—	—	795	32
CA139A	E	G	—	—	—	—	795	32
CA158	T	S	G	—	—	—	1019	35
CA158A	T	S	G	—	—	—	1019	35
CA201	T	S	—	—	—	—	786	18
CA201A	T	S	G	—	—	—	786	18
CA207	T	S	G	—	—	—	785	22
CA211	T	S	G	—	—	—	797	25
CA224	E	G	—	—	—	—	796	29
CA239	E	G	—	—	—	—	795	32
CA239A	E	G	—	—	—	—	795	32
CA258	T	S	G	—	—	—	1019	35
CA258A	T	S	G	—	—	—	1019	35
CA270	W	—	—	—	—	—	879E	292
CA301A	T	S	E*	G	H	—	786	18
CA307	T	S	E*	G	H	—	785	22
CA311	T	S	E*	G	—	—	797	25
CA324	E	G	H	GH	—	—	796	29
CA339	E	G	H	GH	—	—	795	32
CA339A	E	G	—	—	—	—	795	32
CA358	T	S	G	H	GH	—	1019	35
CA358A	T	S	G	—	—	—	1019	35
CA555	T	S	E*	G	—	—	834	42
CA555C	T	S	E*	G	H	—	834	42
CA723	T	E	—	—	—	—	788	46
CA723C	T	E	H	—	—	—	788	46
CA741	T	S	E*	G	L	—	531	50
CA741C	T	S	E*	G	H	GH	531	50
CA747	T	E	G	—	—	—	531	50
CA747C	T	E	G	H	GH	—	531	50
CA748	T	S	E*	G	—	—	531	50
CA748C	T	S	E*	G	H	GH	531	50
CA758	E	—	—	—	—	—	760	295
CA810	Q	QM	—	—	—	—	1154	298
CA810A	Q	QM	—	—	—	—	1154	298
CA920	E	—	—	—	—	—	1132	301
CA1190	Q	—	—	—	—	—	1155	303
CA1310	E	—	—	—	—	—	761	306
CA1352	E	—	—	—	—	—	961	309
CA1391	E*	—	—	—	—	—	981	310
CA1394	E*	—	—	—	—	—	981	310
CA1398	E	—	—	—	—	—	686	312
CA1458	T	S	E*	G	H	GH	531	50
CA1541	D	H	—	—	—	—	536	54
CA1558	T	S	E*	G	—	—	531	50
CA2002	⊕	M	—	—	—	—	1156	314
CA2004	⊕	M	—	—	—	—	1105	317
CA2111A	E	Q	—	—	—	—	612	57
CA2904	G	—	—	—	—	—	1019	35
CA3000	■	H	—	—	—	—	121	59
CA3001	■	H	—	—	—	—	122	61
CA3002	■	H	—	—	—	—	123	64
CA3004	■	H	—	—	—	—	124	66
CA3005	■	H	—	—	—	—	125	69
CA3006	■	—	—	—	—	—	125	69

Type Number	Package Suffix						Data Bulletin File No.	Page
CA3007	■	—	—	—	—	—	126	72
CA3008	▲	—	—	—	—	—	316	74
CA3008A	▲	—	—	—	—	—	310	78
CA3010	■	—	—	—	—	—	316	74
CA3010A	■	—	—	—	—	—	310	78
CA3011	■	—	—	—	—	—	128	82
CA3012	■	H	—	—	—	—	128	82
CA3013	■	—	—	—	—	—	129	84
CA3014	■	—	—	—	—	—	129	84
CA3015	■	L	H	—	—	—	316	74
CA3015A	■	—	—	—	—	—	310	78
CA3016	▲	—	—	—	—	—	316	74
CA3016A	▲	—	—	—	—	—	310	78
CA3018	■	L	H	—	—	—	338	87
CA3018A	■	—	—	—	—	—	338	87
CA3019	■	H	—	—	—	—	236	90
CA3020	■	H	—	—	—	—	339	92
CA3020A	■	—	—	—	—	—	339	92
CA3021	■	—	—	—	—	—	243	96
CA3022	■	—	—	—	—	—	243	96
CA3023	■	H	—	—	—	—	243	96
CA3026	■	H	—	—	—	—	338	99
CA3028A	■	S	L	—	H	—	382	103
CA3028B	■	S	—	—	—	—	382	103
CA3029	●	—	—	—	—	—	316	74
CA3029A	●	—	—	—	—	—	310	78
CA3030	●	—	—	—	—	—	316	74
CA3030A	●	—	—	—	—	—	310	78
CA3035	■	VI	H	—	—	—	274	319
CA3036	■	—	—	—	—	—	275	108
CA3037	†	—	—	—	—	—	316	74
CA3037A	†	—	—	—	—	—	310	78
CA3038	†	—	—	—	—	—	316	74
CA3038A	†	—	—	—	—	—	310	78
CA3039	■	L	H	—	—	—	343	109
CA3040	■	—	—	—	—	—	363	111
CA3041	◆	—	—	—	—	—	318	320
CA3042	◆	—	—	—	—	—	319	323
CA3043	■	H	—	—	—	—	331	326
CA3044	■	VI	—	—	—	—	340	328
CA3045	†	F	L	H	—	—	341	114
CA3046	†	—	—	—	—	—	341	114
CA3048	●	—	—	—	—	—	377	117
CA3049	T	L	H	—	—	—	611	120
CA3050	†	—	—	—	—	—	361	124
CA3051	●	—	—	—	—	—	361	124
CA3052	●	—	—	—	—	—	387	331
CA3053	■	S	—	—	—	—	382	103
CA3054	●	L	H	—	—	—	388	99
CA3058	†	—	—	—	—	—	490	127
CA3059	●	H	—	—	—	—	490	127
CA3060	D	E	H	—	—	—	537	132
CA3060A	D	—	—	—	—	—	537	132
CA3060B	D	—	—	—	—	—	537	132
CA3064	E	—	—	—	—	—	396	335
CA3065	◆	—	—	—	—	—	412	337
CA3066	◆	—	—	—	—	—	466	340
CA3067	◆	—	—	—	—	—	466	340
CA3068	◆	—	—	—	—	—	467	343
CA3070	●	—	—	—	—	—	468	345

Index to Devices — Linear IC's

Type Number	Package Suffix						Data Bulletin File No.	Page
CA3071	•	—	—	—	—	—	468	345
CA3072	•	—	—	—	—	—	468	345
CA3075	♦	H	—	—	—	—	429	138
CA3076	■	H	—	—	—	—	430	140
CA3078	T	S	—	H	—	—	535	142
CA3078A	T	S	—	—	—	—	535	142
CA3079	•	—	—	—	—	—	490	127
CA3080	■	E*	S	H	—	—	475	146
CA3080A	■	E*	S	—	—	—	475	146
CA3081	•	F	H	—	—	—	480	152
CA3082	•	F	H	—	—	—	480	152
CA3083	•	F	L	H	—	—	481	154
CA3084	•	L	H	—	—	—	482	156
CA3085	■	E*	S	L	H	—	491	159
CA3085A	■	S	E*	—	—	—	491	159
CA3085B	■	S	—	—	—	—	491	159
CA3086	•	F	—	—	—	—	483	162
CA3088	E	—	—	—	—	—	560	352
CA3089	E	—	—	—	—	—	561	354
CA3090A	Q	—	—	—	—	—	684	357
CA3091	D	H	—	—	—	—	534	164
CA3093	E	H	—	—	—	—	533	170
CA3094	T	S	E*	H	—	—	598	173
CA3094A	T	S	E*	—	—	—	598	173
CA3094B	T	S	—	—	—	—	598	173
CA3095	E	H	—	—	—	—	591	181
CA3096	E	H	—	—	—	—	595	186
CA3096A	E	—	—	—	—	—	595	186
CA3096C	E	—	—	—	—	—	595	186
CA3097	E	H	—	—	—	—	633	193
CA3098	T	S	E*	H	—	—	896	199
CA3099	E	H	—	—	—	—	620	203
CA3100	T	S	F	H	—	—	625	206
CA3102	E	H	—	—	—	—	611	120
CA3118	T	H	—	—	—	—	532	210
CA3118A	T	—	—	—	—	—	532	210
CA3120	E	—	—	—	—	—	907	359
CA3121	E	—	—	—	—	—	688	363
CA3123	E	—	—	—	—	—	631	367
CA3125	E	—	—	—	—	—	685	369
CA3126	Q	—	—	—	—	—	860	370
CA3127	E	—	—	—	—	—	662	215
CA3128	Q	—	—	—	—	—	1161	372
CA3130	T	S	E*	H	—	—	817	218
CA3130A	T	S	E*	—	—	—	817	218
CA3130B	T	S	—	—	—	—	817	218
CA3131	EM	—	—	—	—	—	1157	373
CA3132	EM	—	—	—	—	—	1157	373
CA3134	G	GM	GQM	—	—	—	1097	374
CA3135	G	—	—	—	—	—	1021	378

Type Number	Package Suffix						Data Bulletin File No.	Page
CA3136	E	—	—	—	—	—	1158	383
CA3137	E	—	—	—	—	—	970	386
CA3138	G	GH	—	—	—	—	1131	228
CA3138A	G	—	—	—	—	—	1131	228
CA3139	E	Q	—	—	—	—	905	390
CA3140	T	S	E*	H	—	—	957	230
CA3140A	T	S	E*	—	—	—	957	230
CA3140B	T	S	—	—	—	—	957	230
CA3141	E	—	—	—	—	—	906	242
CA3142	E	—	—	—	—	—	907	359
CA3143	E	—	—	—	—	—	1138	393
CA3144	G	—	—	—	—	—	1137	395
CA3146	E	H	—	—	—	—	532	210
CA3146A	E	—	—	—	—	—	532	210
CA3151	G	—	—	—	—	—	1160	399
CA3153	G	—	—	—	—	—	1142	402
CA3159	G	—	—	—	—	—	1136	408
CA3160	T	S	E*	H	—	—	976	244
CA3160A	T	S	E*	—	—	—	976	244
CA3160B	T	S	—	—	—	—	976	244
CA3161	E	—	—	—	—	—	1079	256
CA3162	E	—	—	—	—	—	1080	258
CA3163	G	—	—	—	—	—	1092	410
CA3164	E	—	—	—	—	—	1139	262
CA3166	E	—	—	—	—	—	1110	412
CA3168	E	—	—	—	—	—	1140	415
CA3170	E	—	—	—	—	—	1129	417
CA3172	G	—	—	—	—	—	1130	420
CA3183	E	H	—	—	—	—	532	210
CA3183A	E	—	—	—	—	—	532	210
CA3189	E	—	—	—	—	—	1046	422
CA3221	G	—	—	—	—	—	1057	427
CA3240	E*	E1§	—	—	—	—	1050	265
CA3240A	E*	E1§	—	—	—	—	1050	265
CA3290	T	S	E*	E1§	—	—	1049	274
CA3290A	T	S	E*	E1§	—	—	1049	274
CA3290B	T	S	—	—	—	—	1049	274
CA3401	E	G	H	GH	—	—	630	280
CA3600	E	—	—	—	—	—	619	282
CA3724	G	GH	—	—	—	—	884	287
CA3725	G	GH	—	—	—	—	884	287
CA6078A	T	S	H	—	—	—	592	289
CA6741	T	S	—	—	—	—	592	289

■ No designated suffix letter for this type in TO-5-style package
▲ No designated suffix letter for this type in ceramic flat package
• No designated suffix letter for this type in dual-in-line plastic package
† No designated suffix letter for this type in dual-in-line ceramic package
♦ No designated suffix letter for this type in quad-in-line plastic package
* In 8-lead dual-in-line Mini-DIP package
§ In 14-lead dual-in-line plastic package
⊕ No designated suffix letter for this type in TO-220-style package with vertical-mount lead form

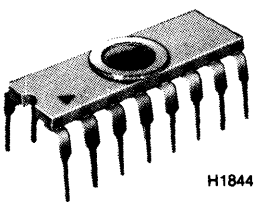
Index to Devices — MOS/FET's

Type Number	Package	Data Bulletin File No.	Page
3N128	TO-72	309	434
3N138	TO-72	283	436
3N139	TO-72	284	437
3N140	TO-72	285	438
3N141	TO-72	285	438
3N142	TO-72	286	441
3N143	TO-72	309	434
3N152	TO-72	314	442
3N153	TO-72	320	443
3N154	TO-72	335	444
3N159	TO-72	336	459
3N187	TO-72	326	446
3N200	TO-72	436	450
3N204	TO-72	959	453
3N205	TO-72	959	453
3N206	TO-72	959	453
3N211	TO-72	875	458

Type Number	Package	Data Bulletin File No.	Page
3N212	TO-72	875	458
3N213	TO-72	875	458
40467A	TO-72	324	462
40468A	TO-72	323	463
40559A	TO-72	323	463
40600	TO-72	333	464
40601	TO-72	333	464
40602	TO-72	333	464
40603	TO-72	334	465
40604	TO-72	334	465
40673	TO-72	381	466
40819	TO-72	463	467
40820	TO-72	464	468
40821	TO-72	464	468
40822	TO-72	465	470
40823	TO-72	465	470
40841	TO-72	489	471

Packages

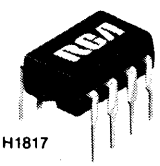
D Suffix
Dual-In-Line Welded-Seal Ceramic Package



H1844

14 and 16-lead versions

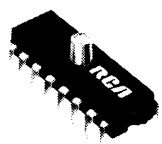
E Suffix
Dual-In-Line Plastic Package



H1817

8, 14, and 16-lead versions

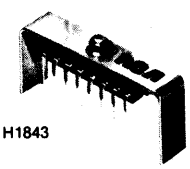
E Suffix
Power Stud Plastic Dual-In-Line Package



H1828

CA3134E only

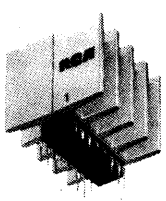
EM Suffix
Modified 16-lead Dual-In-Line Plastic Package



H1843

CA3134EM only

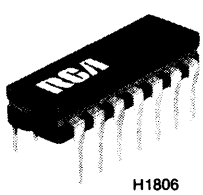
EM Suffix
Modified 16-lead Dual-In-Line Plastic Package



H1827

CA3131EM, CA3132EM only

F Suffix
Dual-In-Line Frit-Seal Ceramic Package

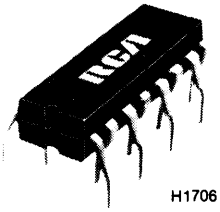


H1806

14 and 16-lead versions

Packages

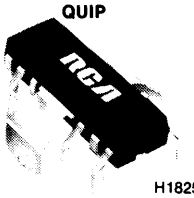
Q Suffix
Quad-In-Line Plastic Package (QUIP)



H1706

14 and 16-lead versions

Q Suffix
Modified 16-lead QUIP



H1825

CA810Q, CA810AQ, and CA1190Q only (A flat wing-tab version, QM suffix is also available for the CA810, CA810A).

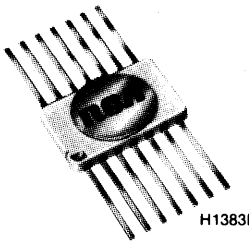
VERSA-V TO-220 Style Plastic Package with Vertical-Mount Lead Form



H1887

CA2002, CA2004 only (Versions with Horizontal-Mount Lead Form—CA2002M and CA2004M are also available)

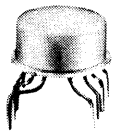
K Suffix
Ceramic Flat Package



H1383R1

14-lead version

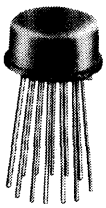
S Suffix
TO-5 Style Package with Dual-In-Line Formed Leads (DIL-CAN)



H1787

8-lead version

T Suffix
TO-5 Style Package with Straight Leads



H1463

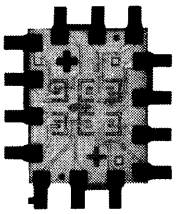
8, 10, and 12-lead versions

V1 Suffix
TO-5 Style Package with Radial Formed Leads

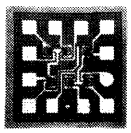


H1561

L Suffix
Beam-Lead



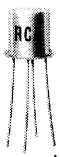
H Suffix
Chip



92CS-22137

8, 10, and 12-lead versions

JEDEC TO-72



H1299

MOS/FET's only

Notes:

1. Some types may have an additional "M" suffix following the package designation suffix, i.e., CA3131EM. The additional "M" suffix simply indicates that the device is a mechanical variant of the basic package type.

2. RCA Linear integrated circuits are provided in chip form to allow customer design of special and complex circuits to suit individual needs. Linear chips are electrically identical to and offer the features of their counterparts, sealed in ceramic, TO-5, and plastic packages.

Product Classification Chart

Industrial Circuits

OPERATIONAL AMPLIFIERS

General Purpose

Single Unit	Dual Unit
CA101	CA158
CA107	CA258
CA201	CA358
CA207	CA747
CA301	CA1458
CA307	CA1558
CA741	CA2904
CA748	Quad Unit
CA6741•	CA124
	CA224
	CA324
	CA3401

General Purpose Wideband

Single Unit
CA3008
CA3010
CA3015
CA3016
CA3029
CA3030
CA3037
CA3038
CA3100*
CA3130*
CA3140*
CA3160*
Dual Unit
CA3240*

Variable

High Current
CA3094
Micropower
CA3060
CA3078
CA3080
CA6078•

DIFFERENTIAL AMPLIFIERS

CA3000
CA3001
CA3004
CA3005
CA3006
CA3007
CA3026
CA3028
CA3049
CA3050
CA3051
CA3053
CA3054
CA3102

ARRAYS

Amplifier/Diode

Amplifier
CA3026
CA3035
CA3048
CA3049
CA3052
CA3054
CA3060
CA3102
Diode
CA3019
CA3039
CA3141

Transistor

CA3018	CA3095
CA3036	CA3096
CA3045	CA3097
CA3046	CA3118
CA3050	CA3127
CA3051	CA3138
CA3081	CA3146
CA3082	CA3183
CA3083	CA3600▲
CA3084	CA3724
CA3086	CA3725
CA3093	

VOLTAGE REGULATORS

CA723
CA3085

ZERO-VOLTAGE SWITCHES

CA3058
CA3059
CA3079

VOLTAGE COMPARATORS

Single Unit
CA111
CA211
CA311
CA3098+
CA3099+
Dual Unit
CA3290*
Quad Unit
CA139
CA239
CA339

SPECIAL-FUNCTION CIRCUITS

A/D Converter
CA3162
BCD-to-7-Segment Decoder/Driver
CA3161
Memory Sense Amplifier
CA1541
Four-Quadrant Multiplier
CA3091
Timer
CA555
Programmable Schmitt Trigger
CA3098

MOS/FET's

Single Gate	Dual Gate
3N128	3N140
3N138	3N141
3N139	3N159
3N142	Dual Gate
3N143	Protected
3N152	3N187
3N153	3N200
3N154	40819

Consumer Circuits

BROADBAND (VIDEO) AMPLIFIERS

CA3002
CA1352
CA3020
CA3021
CA3022
CA3023
CA3040

MULTIPLEX DECODERS

CA758
C1310
CA3090A

AM/FM COMMUNICATIONS CIRCUITS

CA2111A
CA3011
CA3012
CA3013
CA3014
CA3043
CA3075
CA3076
CA3088
CA3089
CA3123
CA3163
CA3189

AUDIO CIRCUITS

Preamplifiers
CA3036
CA3052
Drivers
CA3094
Power Amplifiers
CA810
CA2002
CA2004
CA3131
CA3132

FM IF CIRCUITS

Subsystems
CA2111A
CA3013
CA3014
CA3043
CA3075
CA3089
CA3189
Gain Blocks
CA3011
CA3012
CA3076

TV RECEIVER CIRCUITS

Tuning	Chroma Systems
CA3163	CA1398
CA3166	CA3066
CA3168	CA3067
AFT	CA3070
CA3044	CA3071
CA3064	CA3072
CA3139	CA3121
Sound IF	CA3125
CA1190	CA3126
CA2111A	CA3128
CA3041	CA3151
CA3042	CA3170
CA3065	Luminance
CA3134	Processors
PIX IF	CA3135
CA270	CA3143
CA1352	CA3144
CA3068	Horizontal
CA3136	Systems
Remote Control	CA1391
CA3035	CA1394
"Jungle" Circuits	CA920A
CA3120	CA3159
CA3142	CA3172

MOS/FET's

Single Gate	Dual Gate
40467A	Protected
40468A	3N204
40559A	3N205
Dual Gate	3N206
40600	3N211
40601	3N212
40602	3N213
40603	40673
40604	40820
	40821
	40822
	40823
	40841

• Low-noise versions of CA741 and CA3078 * BiMOS types ▲CMOS type + Programmable

Operating and Handling Considerations

Solid state devices are being designed into an increasing variety of electronic equipment because of their high standards of reliability and performance. However, it is essential that equipment designers be mindful of good engineering practices in the use of these devices to achieve the desired performance.

This Note summarizes important operating recommendations and precautions which should be followed in the interest of maintaining the high standards of performance of linear integrated circuits and MOS field-effect transistors.

The ratings included in RCA data bulletins are based on the Absolute Maximum Rating System, which is defined by the following Industry Standard (JEDEC) statement:

Absolute-Maximum Ratings are limiting values of operating and environmental conditions applicable to any electron device of a specified type as defined by its published data, and should not be exceeded under the worst probable conditions.

The device manufacturer chooses these values to provide acceptable serviceability of the device, taking no responsibility for equipment variations, environmental variations, and the effects of changes in operating conditions due to variations in device characteristics.

The equipment manufacturer should design so that initially and throughout life no absolute-maximum value for the intended service is exceeded with any device under the worst probable operating conditions with respect to supply-voltage variation, equipment component variation, equipment control adjustment, load variation, signal variation, environmental conditions, and variations in device characteristics.

It is recommended that equipment manufacturers consult RCA whenever device applications involve unusual electrical, mechanical or environmental operating conditions.

GENERAL CONSIDERATIONS

The design flexibility provided by integrated circuits and MOS/FET's makes possible their use in a broad range of applications and under many different operating conditions. When incorporating these devices in equipment, designers should anticipate the rare possibility of device failure and make certain that no safety hazard would result from such an occurrence.

The small size of these devices provides obvious advantages to the designers of electronic equipment. However, it should be recognized that these compact devices usually provide only relatively small insulation area between adjacent leads and the metal envelope. When these devices are used in moist or contaminated atmospheres, therefore, supplemental protection must be provided to prevent the development of electrical conductive paths across the relatively small insulating surfaces.

Devices should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent damage to the devices.

TESTING PRECAUTIONS

In common with many electronic components, solid-state devices should be operated and tested in circuits which have reasonable values of current limiting resistance, or other forms of effective current overload protection. Failure to observe these precautions can cause excessive internal heating of the device resulting in destruction and/or possible shattering of the enclosure.

MOUNTING

Integrated circuits are normally supplied with lead-tin plated leads to facilitate soldering into circuit boards. In those relatively few applications requiring welding of the device leads, rather than soldering, the devices may be obtained with gold or nickel plated Kovar leads.* It should be recognized that this type of plating will not provide complete protection against lead corrosion in the presence of high humidity and mechanical stress. The aluminum-foil-lined cardboard "sandwich pack" employed for static protection of the flat-pack also provides some additional protection against lead corrosion, and it is recommended that the devices be stored in this package until used.

When integrated circuits are welded onto printed circuit boards or equipment, the presence of moisture between the closely spaced terminals can result in conductive paths that may impair device performance in high-impedance applications. It is therefore recommended that conformal coatings or potting be provided as an added measure of protection against moisture penetration.

In any method of mounting integrated circuits which involves bending or forming of the device leads, it is extremely important that the lead be supported and clamped between the bend and the package seal, and that bending be done with care to avoid damage to lead plating. In no case should the radius of the bend be less than the diameter of the lead, or in the case of rectangular leads, such as those used in RCA 14-lead and 16-lead flat-packages, less than the lead thickness. It is also extremely important that the ends of the bent leads be straight to assure proper insertion through the holes in the printed-circuit board.

MOS FIELD-EFFECT TRANSISTORS

Insulated-Gate Metal Oxide-Semiconductor Field-Effect Transistors (MOS FETs), like bipolar high-frequency transistors, are susceptible to gate insulation damage by the electrostatic discharge of energy through the devices. Electrostatic discharges can occur in an MOS FET if a type with an unprotected gate is picked up and the static charge, built in the handler's body capacitance, is discharged through the device. With proper handling and applications procedures, however, MOS transistors are currently being extensively used in production by numerous equipment manufacturers in military, industrial, and consumer applica-

* MIL-38510A, paragraph 3.5.6.1(a), lead material.

tions, with virtually no problems of damage due to electrostatic discharge.

In some MOS FETs, diodes are electrically connected between each insulated gate and the transistor's source. These diodes offer protection against static discharge and in-circuit transients without the need for external shorting mechanisms. MOS FETs which do not include gate-protection diodes can be handled safely if the following basic precautions are taken:

1. Prior to assembly into a circuit, all leads should be kept shorted together either by the use of metal shorting springs attached to the device by the vendor, or by the insertion into conductive material such as "ECCOSORB* LD26" or equivalent.
   (NOTE: Polystyrene *insulating* "SNOW" is not sufficiently conductive and should not be used.)
2. When devices are removed by hand from their carriers, the hand being used should be grounded by any suitable means, for example, with a metallic wristband.
3. Tips of soldering irons should be grounded.
4. Devices should never be inserted into or removed from circuits with power on.

*Trade Mark: Emerson and Cumming, Inc.

SOLID STATE CHIPS

Solid state chips, unlike packaged devices, are non-hermetic devices, normally fragile and small in physical size, and therefore, require special handling considerations as follows:

1. Chips must be stored under proper conditions to insure that they are not subjected to a moist and/or contaminated atmosphere that could alter their electrical, physical, or mechanical characteristics. After the shipping container is opened, the chip must be stored under the following conditions:
   A. Storage temperature, 40°C max.
   B. Relative humidity, 50% max.
   C. Clean, dust-free environment.
2. The user must exercise proper care when handling chips to prevent even the slightest physical damage to the chip.
3. During mounting and lead bonding of chips the user must use proper assembly techniques to obtain proper electrical, thermal, and mechanical performance.
4. After the chip has been mounted and bonded, any necessary procedure must be followed by the user to insure that these non-hermetic chips are not subjected to moist or contaminated atmosphere which might cause the development of electrical conductive paths across the relatively small insulating surfaces. In addition, proper consideration must be given to the protection of these devices from other harmful environments which could conceivably adversely affect their proper performance.

Terms and Symbols

A	closed-loop voltage gain
A_{AF}	audio amplifier gain
A_{DIFF}	differential voltage gain
ACC	automatic chroma control
AFC	automatic frequency control
AFT	automatic fine tuning
AGC	automatic gain control
AMR	am rejection
A_{OL}	open-loop voltage gain
A_V	amplifier voltage gain
b_{fs}	small-signal, common-source, forward transfer susceptance (imaginary part of corresponding admittance; see y_{fs})
b_{is}	small-signal, common-source, input susceptance (imaginary part of corresponding admittance; see y_{is})
b_{os}	small-signal, common-source, output susceptance (imaginary part of corresponding admittance, see y_{os})
b_{rs}	small-signal, common-source, reverse transfer susceptance (imaginary part of corresponding admittance, see y_{rs})
BW	bandwidth (unity gain)
BW_{OL}	open-loop bandwidth
C_{BI}	base-to-substrate capacitance
C_{CB}	collector-to-base capacitance
C_{EB}	emitter-to-base capacitance
C_{EXT}	external capacitance
C_{FB}	feedback capacitance
C_I	input capacitance
C_{ios}	small-signal output capacitance
C_{is}	small-signal input capacitance
C_{iss}	small-signal, common-source short-circuit input capacitance
C_{I-O}	input-to-output capacitance; data in/out capacitance
CMMR	common-mode rejection ratio
C_O	output capacitance
C_{os}	feedthrough capacitance
C_{oss}	small-signal, common-source short-circuit output capacitance
C_{QP}	charge-pump capacitance
C_{rss}	small-signal, common-source short-circuit, reverse transfer capacitance
e_i	input sensitivity
E_N	I/F noise voltage
e_N	low-frequency noise voltage; equivalent short-circuit input noise voltage ($\mu V/\sqrt{Hz}$)
$e_{N(total)}$	wideband noise voltage referenced to input
e_{O1}/e_{O2}	channel separation
E_{ON}	broadband output noise voltage
f_{CL}	clock input frequency
f_{max}	maximum operating frequency
f_p	charge-pump input-pulse frequency
f_t	unity-gain crossover frequency; gain-bandwidth product
f_ϕ	input-pulse frequency
G_p	power gain
G_m	forward transconductance (large-signal)
h_{FE}	static forward-current transfer ratio (beta)
h_{fe}	small-signal forward-current transfer ratio
I^+	dc supply current
I^-	dc supply current
I_A	amplifier supply current
I_{ABC}	amplifier bias current
I_{AGC}	AGC source current
I_B	base current
I_C	collector current
I_{CBO}	collector cutoff current
I_{CEO}	collector cutoff current
$I_{CE(OFF)}$	output leakage current
I_D	drain current
$I_{D(ON)}$	dc on-state drain current
I_{DARK}	dark current
I_{DF}	diode forward current
I_{DDO}	supply current for drain supply voltage (V_{DD})
I_{DS}	zero-gate (bias) drain current (dual-gate types)
I_{DSS}	zero-gate (bias) drain current (single-gate types)
I_F	forward current
I_G	channel (input) gate lead current
I_{GR}	channel (input) gate reverse current
I_{GS}	gate terminal current (single-gate types)
I_{G1S}	gate-No.1 terminal current dual-gate types
I_{G2S}	gate-No. 2 terminal current dual-gate types
I_{GSSF}	gate-to-source forward leakage current, all other terminals shorted to source (dual-gate types).
I_{G1SSF}	gate-No.1 source forward leakage current, all other terminals shorted to source (dual-gate types).
I_{G2SSF}	gate-No. 2-to-source forward leakage current, all other terminals shorted to source (dual-gate types).
I_{GSSR}	gate-to-source reverse leakage current, all other terminals shorted to source (single-gate types).
I_{G1SSR}	gate-No. 1-to-source reverse leakage current, all other terminals shorted to source (dual-gate types).
I_{G2SSR}	gate-No.2-to-source reverse leakage current, all other terminals shorted to source (dual-gate types).
I_{GT}	gate trigger current; gate terminal current
I_I	input current
I_{IB}	input bias current
I_{IBC}	internal bias current
I_{IO}	input offset current
αI_{IO}	average temperature coefficient of input offset current
$\Delta I_{IO}/\Delta T$	temperature coefficient of input offset current (drift)
I_{LIM}	short-circuit limiting current
I_{MTR}	current-mirror transfer ratio
I_N	I/F noise current
i_N	equivalent open-circuit noise current (pA/$\sqrt{Hz}$)
I_O	output current
$I_{O(DIFF)}$	differential output current (sink)
I_{OO}	output offset current
I_{OOL}	output leakage current, low
I_{OM}	peak output current
$\lvert I_{OM}\rvert$	magnitude of peak output current
I_{OM}^+	maximum output current (source)
I_{OM}^-	maximum output current (sink)
I_p	photo current
I_{p-p}	peak-to-peak output current
I_Q	total quiescent current
I_{QPL}	charge-pump input current
I_R	dc reverse (leakage) current
I_{REFO}	supply current for reference supply voltage
I_{SSO}	strobe load current voltage (V_{SS})
I_{SXO}	supply current for supply voltage
I_{TH}	threshold current
I_{TOTAL}	total supply current
k_N	normalized factor ($k_N = k/k_r$)
MAG	maximum available power gain
MUG	maximum useable power gain (unneutralized)
NF	noise factor
P_O	power output
P_D	device dissipation
P_{SRR}	power supply rejection ratio
$r_{ds(off)}$	small-signal drain-to-source off-state resistance
$r_{ds(on)}$	static drain-to-source on-state resistance
R_{GS}	gate leakage-current resistance
R_O	output resistance
R_o	low-frequency output resistance
r_o	small-signal output resistance
r_{oss}	small-signal, short-circuit, common-source output resistance
R_I	differential input resistance
r_i	small-signal input resistance
r_{iss}	small-signal, short-circuit, common-source input resistance
R_i	low-frequency input resistance
R_{ON}	ON resistance; the ON-state resistance of an analog switch at specified input and load conditions.
ΔR_{ON}	ΔON resistance; the difference in ON-state resistance between any 2 analog switches at specified input and load conditions.
S/N	signal-to-noise ratio
SR	slew rate
T_A	ambient temperature
t_d	delay time
t_{DR}	differential recovery time
t_f	fall time
$t_{f\phi}$	input-pulse rise time
THD	total harmonic distortion
t_{off}	turn-off time
t_{on}	turn-on time
t_r	rise time
$t_{R\phi}$	input-pulse rise time
t_{rr}	reverse recovery time
t_S	setup time
t_{STG}	storage time
t_W	pulse width
V^+	DC positive supply voltage
V^-	DC negative supply voltage
V_{ABC}	amplifier bias voltage
V_{BB}	substrate voltage
V_{BE}	base-to-emitter voltage

$V_{BE(sat)}$	base-to-emitter saturation voltage
$V_{(BR)CBO}$	collector-to-base breakdown voltage
$V_{(BR)CES}$	collector-to-emitter breakdown voltage
$V_{(BR)DI}$	dc breakdown voltage between diode and substrate
$V_{(BR)R}$	dc reverse breakdown voltage
$V_{(BR)EBO}$	emitter-to-base breakdown voltage
$V_{(BR)GSSF}$	dc gate-to-source forward breakdown voltage, all other terminals shorted to source (single-gate types)
$V_{(BR)G1SSF}$	dc gate-No.1-to-source forward breakdown voltage, all other terminals shorted to source (dual-gate types)
$V_{(BR)G2SSF}$	dc gate No.2-to-source forward breakdown voltage, all other terminals shorted to source (dual-gate types)
$V_{(BR)GSSR}$	dc gate-to-source reverse breakdown voltage, all other terminals shorted to source (single-gate types)
$V_{(BR)G2SSR}$	dc gate-No.2-to-source reverse breakdown voltage, all other terminals shorted to source (dual-gate types)
V_{CBO}	collector-to-base voltage
V_{CC}	drain supply voltage used as a second positive supply voltage. It is $\leqslant V_{DD}$ and referenced to V_{SS}
VCO	voltage controlled oscillator
V_{CEO}	collector-to-emitter voltage
$V_{CEO(sus)}$	collector-to-emitter sustaining voltage
V_{CIO}	collector-to-substrate voltage
V_{CP}	charge pump voltage
V_{DD}	drain supply voltage (the most positive supply voltage; always referenced to ground)
V_{DG}	drain-to-gate voltage (single-gate types)
V_{DG1}	drain-to-gate-No.1 voltage (dual-gate types)
V_{DG2}	drain-to-gate-No.2 voltage (single-gate types)
V_{DIO}	diode-to-substrate voltage
V_{DR}	diode reverse voltage
V_{DS}	drain-to-source voltage
V_{EE}	source voltage (the most negative supply voltage in a 3-supply voltage system)
V_F	dc forward voltage
$\Delta V_F/\Delta T$	temperature coefficient of forward voltage drop
V_{GH}	channel gate input voltage, high level
V_{GL}	channel gate input voltage, low level
V_{GS}	gate-to-source voltage
$V_{GS(TH)}$	gate-to-source threshold voltage
$V_{GS(Off)}$	gate-to-source cutoff voltage (single-gate types)
V_{G1S}	gate-No.1-to-source voltage (dual-gate type)
$V_{G1S(Off)}$	gate-No.1-to-source cutoff voltage (dual-gate types)
V_{G2S}	gate-No.2-to-source voltage (dual-gate types)
$V_{G2S(off)}$	gate-No.2-to-source cutoff voltage (dual-gate types)
V_I	input voltage
$V_{I(Lim)}$	input limiting voltage
V_{ICR}	common-mode input voltage range
V_{IL}	input-voltage, low level
V_{IH}	input-voltage, high level
V_{IO}	input offset voltage
$\lvert V_{IO}\rvert$	magnitude of input offset voltage
$\Delta V_{IO}/\Delta T$	temperature coefficient of magnitude of input offset voltage
$\Delta V_{IO}/\Delta T$	temperature coefficient of input offset voltage drift
$\Delta V_{IO}/\Delta V^+$	positive input-offset-voltage sensitivity
$\Delta V_{IO}/\Delta V^-$	negative input-offset-voltage sensitivity
αV_{IO}	average temperature coefficient of input-offset voltage
$V_{i(Lim)}$	input limiting voltage (knee)
V_{knee}	protective diode knee voltage (protected gate types)
V_N	output noise voltage
V_O	output voltage
$\Delta V_O/\Delta V^-$	dc supply voltage sensitivity
$\Delta V_O/\Delta V^+$	dc supply voltage sensitivity
$V_{O(rms)}$	open-loop output voltage swing
ΔV_O	output voltage temperature coefficient
$V_{Op\text{-}p}$	output voltage swing
$V_{O(af)}$	recovered af voltage
V_{OL}	output voltage, low level; the voltage level at an output when the input logic conditions have been set to establish logic LOW output.
V_{OO}	output offset voltage
V_{OH}	output voltage, high level; the voltage level at an output when the input logic conditions have been set to establish a logic HIGH output.
V_{OM}^+	maximum output voltage
V_{OM}^-	maximum output voltage
V_{QP}	charge pump voltage
V_{QPL}	charge pump input voltage, low level
V_{QPH}	charge-pump input voltage, high level
V_{REF}	reference voltage
V_{REG}	regulated supply voltage
V_{RR}	supply voltage rejection ratio
V_{TH}	input threshold voltage
V_Z	zener voltage
Y_{fs}	magnitude of small-signal, common-source, short-circuit forward transfer admittance (transadmittance)
Y_{is}	small-signal, common-source, short-circuit, input-admittance (conductance, real part of admittance; susceptance, imaginary part of admittance)
Y_{os}	small-signal, common-source, short-circuit, output admittance
$\lvert Y_{rs}\rvert$	magnitude of small-signal, common-source, short-circuit, reverse transadmittance
$<Y_{rs}$	phase angle of small-signal, common-source, short-circuit, reverse transadmittance
$(-)_{rs}$	angle of reverse transadmittance, common-source circuit
Z_1	input impedance
Z_O	output impedance
Z_Z	zener impedance
ϕ	phase angle
ϕ	phase margin
η	efficiency
ϕL	open-loop phase lag

Cross-Reference Directory for Linear Integrated Circuits

Industry Type	RCA Replacement Type
AD101AH	CA101AT
AD101AN	CA101AG,CA101AE
AD201AH	CA201AT
AD201AN	CA201AG,CA201AE
AD301AH	CA301AT
AD301AN	CA301AG,CA301AE
AD741H	CA741T
AD741N	CA741G,CA741E
AD741CH	CA741CT
AD741CN	CA741CG,CA741CE
AD2020	CA3162E
AMLM101AD	CA101AG
AMLM101AH	CA101AT
AMLM101H	CA101T
AMLM107D	CA107G
AMLM107H	CA107T
AMLM111D	CA111G
AMLM111H	CA111T
AMLM201AD	CA201AG
AMLM201AH	CA201AT
AMLM201D	CA201G
AMLM201H	CA201T
AMLM207D	CA207G
AMLM207H	CA207T
AMLM211D	CA2111G
AMLM211H	CA2111T
AMLM301AD	CA301AG
AMLM301AH	CA301AT
AMLM301H	CA301T
AMLM307D	CA307G
AMLM307H	CA307T
AMLM311D	CA311G
AMLM311H	CA311T
AM723HC	CA723CT
AM723HM	CA723T
AM741DC	CA741CG
AM741DM	CA741G
AM741HC	CA741CT
AM741HM	CA741T
AM747DC	CA747CG
AM747DM	CA747G
AM747HC	CA747CT
AM747HM	CA747T
AM748DC	CA748CG
AM748HC	CA748CT
AM748DM	CA748G
AM1458H	CA1458T
AM1558H	CA1558T
DH3724CN	CA3724G
DH3725CN	CA3725F
FPQ3724	CA3724G
FPQ3725	CA3725G
HA1-2111-2	CA111G
HA1-2211-4	CA211G
HA1-2311-5	CA311G
HA1-2630	CA3020
HA1-2650	CA1558G, CA1558E
HA1-2655	CA1458G, CA1458E
HA1-2720	CA6078
HA2-2111-2	CA111T
HA2-2111-4	CA211T
HA2-2311-5	CA311T
HA2-2520	CA3100T
HA2-2650	CA1558T
HA2-2655	CA1458T

Industry Type	RCA Replacement Type
HA2-2720	CA3078E
ITT1352N	CA1352E
ITT3064C	CA3064T
ITT3064N	CA3064E
ITT3065N	CA3065E
L4001M9	2N5756
LM100	CA3085E
LM101AD	CA101AG
LM101ADE	CA101AG
LM101AF	CA101AG
LM101AH	CA101AT
LM101AJ	CA101AG
LM101AJG	CA101AG
LM101AL	CA101AT
LM101AN	CA101AG,CA101AE
LM101AP	CA101AG, CA101AE
LM101AT	CA101AT
LM101D	CA101G
LM101F	CA101G
LM101H	CA101T
LM101J	CA101G
LM101N	CA101G, CA101E
LM101T	CA101T
LM107D	CA107G
LM107DE	CA107G
LM107F	CA107G
LM107H	CA107T
LM111JG	CA111G
LM107N	CA107G, CA107E
LM111P	CA111G, CA111E
LM107T	CA107T
LM111D	CA111G
LM111H	CA111T
LM111L	CA111T
LM111N	CA111G, CA111E
LM111T	CA111T
LM111V	CA111G,CA111E
LM124D	CA124G
LM124F	CA124G
LM124J	CA124G
LM124N	CA124G, CA124E
LM139AD	CA139AG
LM139AF	CA139AG
LM139AJ	CA139AG
LM139AN	CA139AG, CA139AE
LM139A	CA139G,CA139E
LM139D	CA139G
LM139F	CA139G
LM139J	CA139G
LM139N	CA139G, CA139E
LM158AH	CA158AT
LM158AN	CA158AG, CA158AE
LM158AT	CA158AT
LM158JG	CA158G
LM158L	CA158T
LM158N	CA158G, CA158E
LM158P	CA158G, CA158E
LM158T	CA158T
LM201AD	CA201AG
LM201AF	CA201G
LM201AH	CA201AT
LM201AJ	CA201AG
LM201AJG	CA201AG
LM201AL	CA201AT
LM201AN	CA201AG, CA201AE

Industry Type	RCA Replacement Type
LM201AP	CA201AG,CA201AE
LM201AT	CA201AT
LM201AV	CA201AG, CA201AE
LM201D	CA201G
LM201F	CA201G
LM201H	CA201T
LM201J	CA201G
LM201N	CA201G, CA201E
LM201N-14	CA201G,CA201E
LM201T	CA201T
LM207D	CA207G
LM207F	CA207G
LM207H	CA207T
LM207J	CA207G
LM207N	CA207G, CA207E
LM207T	CA207T
LM211D	CA211G
LM211H	CA211T
LM211N	CA211G, CA211E
LM211T	CA211T
LM224A	CA224G,CA224E
LM224D	CA224G
LM224F	CA224G
LM224N	CA224G,CA224E
LM224T	CA224G
LM239AD	CA239AG
LM239AF	CA239AG
LM239AJ	CA239AG
LM239AN	CA239AG, CA239AE
LM239A	CA239G, CA239E
LM239D	CA239G
LM239F	CA239G
LM239J	CA239G
LM239N	LM239G, CA239E
LM258AH	CA258AT
LM258AN	CA258AG, CA258AE
LM258AT	CA258AT
LM258AJ	CA258AG
LM258JG	CA258G
LM258H	CA258T
LM258L	CA258T
LM258N	CA258G, CA258E
LM258P	CA258G, CA258E
LM258T	CA258T
LM301AD	CA301AG
LM301AH	CA301AT
LM301AF	CA301AG
LM301AJ	CA301AG
LM301AJG	CA301AG
LM301AL	CA301AT
LM301AN	CA301AG, CA301AE
LM301AP	CA301AG, CA301AE
LM301AT	CA301AT
LM301AV	CA301AG, CA301AE
LM301T	CA301T
LM307DE	CA307G
LM307D	CA307G
LM307F	CA307G
LM307H	CA307T
LM307N	CA307G, CA307E
LM307T	CA307T
LM311D	CA311G
LM311F	CA311G
LM311H	CA311T
LM311JG	CA311G

Cross-Reference Directory for Linear Integrated Circuits

Industry Type	RCA Replacement Type
LM311L	CA311T
LM311N	CA311G, CA311E
LM311N-14	CA311G,CA311E
LM311P	CA311G, CA311E
LM311T	CA311T
LM318H	CA3130T
LM324AD	CA324AG
LM324AN	CA324AG, CA324AF
LM324D	CA324G
LM324F	CA324G
LM324J	CA324G
LM324N	CA324G, CA324E
LM339AD	CA339AG
LM339AF	CA339AG
LM339AJ	CA339AG
LM339AN	CA339AG, CA339AE
LM339A	CA339G, CA339E
LM339D	CA339G
LM339F	CA339G
LM339J	CA339G
LM339N	CA339G, CA339E
LM358AH	CA358AT
LM358AN	CA358AG, CA358AE
LM358AT	CA358AT
LM358JG	CA358G
LM358H	CA358
LM358L	CA358T
LM358N	CA358G, CA358E
LM358P	CA358G, CA358E
LM358T	CA358T
LM393N	CA3290E
LM555CH	CA555CT
LM555CN	CA555CG, CA555CE
LM555H	CA555T
LM555N	CA555G, CA555E
LM723CD	CA723CE
LM723CH	CA723CT
LM723CN	CA723CE
LM723D	CA723E
LM723H	CA723T
LM723N	CA723E
LM741CH	CA741CT
LM741CJ	CA741CG
LM741CN	CA741CG, CA741CE
LM741H	CA741T
LM741N	CA741G, CA741E
LM746N	CA3072
LM747CD	CA747CG
LM747CH	CA747CT
LM747CJ	CA747CG
LM747CN	CA747CG, CA747CE
LM747D	CA747G
LM747H	CA747T
LM747J	CA747G
LM748CH	CA748CT
LM748CJ	CA748CG
LM748CN	CA748CG, CA748CE
LM748H	CA748T
LM748J	CA748G
LM1310N	CA1310E
LM1391N	CA1391E
LM1394N	CA1394E
LM1458H	CA1458T
LM1458J	CA1458G
LM1458N	CA1458G, CA1458E
LM1558H	CA1558T
LM1558J	CA1558G
LM1558N	CA1558G, CA1558E
LM1800N	CA758E
LM1820N	CA3123E
LM1845N	CA3120E
LM2111N	CA2111AE
LM2901N	CA339G
LM2904N	CA2904G
LM2904P	CA2904G
LM3011H	CA3011
LM3018H	CA3018
LM3018AH	CA3018A
LM3019H	CA3019
LM3026H	CA3026
LM3028AH	CA3028A
LM3028B	CA3028B
LM3039H	CA3039
LM3045D	CA3045
LM3046N	CA3046
LM3053H	CA3053
LM3054N	CA3054
LM3064H	CA3064T
LM3064N	CA3064E
LM3065N	CA3065
LM3066N	CA3066
LM3067N	CA3067
LM3070N	CA3070
LM3071N	CA3071
LM3075N	CA3075
LM3086N	CA3086
LM3089N	CA3089E, CA3189E
LM3126N	CA3126E
LM3146AN	CA3146AE
LM3401N	CA3401G, CA3401E
MC1310P	CA1310E
MC1352P	CA1352E
MC1357P	CA2111AE
MC1357PQ	CA2111AQ
MC1358P	CA3065
MC1364G	CA3064T
MC1364P	CA3064E
MC1370P	CA3070
MC1371P	CA3071
MC1375P	CA3075
MC1389P	CA3089E, CA3189E
MC1391P	CA1391E
MC1394P	CA1394E
MC1398P	CA1398E
MC1455G	CA555CT
MC1455P1	CA555CG, CA555CE
MC1455U	CA555CG
MC1458JG	CA1458G
MC1458G	CA1458T
MC1458L	CA1458T
MC1458P	CA1458G, CA1458E
MC1458P1	CA1458G, CA1458E
MC1458T	CA1458T
MC1541L	CA1541D
MC1555G	CA555T
MC1555P1	CA555CG, CA555CE
MC1555U	CA555G
MC1558JG	CA1558T
MC1558G	CA1558T
MC1558L	CA1558T
MC1558P	CA1558G, CA1558E
MC1558P1	CA1558G, CA1558E
MC1558T	CA1558T
MC1558U	CA1558G
MC1723CG	CA723CT
MC1723CP	CA723CE
MC1723G	CA723T
MC1741CG	CA741CT
MC1741CL	CA741CG
MC1741CP1	CA741CG, CA741CE
MC1741CP2	CA741CG,CA741CE
MC1741G	CA741T
MC1741L	CA741G
MC1741U	CA741G
MC1747CG	CA747CT
MC1747CL	CA747CG
MC1747G	CA747T
MC1747L	CA747G
MC1748CG	CA748CT
MC1748CP1	CA748CG, CA748CE
MC1748CU	CA748CG
MC1748G	CA748T
MC1748U	CA748G
MC3346P	CA3046
MC3386P	CA3086
MC3401L	CA3401G
MC3401P	CA3401E
MLM101AG	CA101AT
MLM101AU	CA101AG
MLM107G	CA101T
MLM107U	CA101G
MLM111G	CA111T
MLM111U	CA111G
MLM124L	CA124G
MLM139AL	CA139AG
MLM139L	CA139G
MLM158G	CA158T
MLM158P1	CA158G, CA158E
MLM158U	CA158G
MLM201AG	CA201AT
MLM201AP1	CA201AG, CA201AE
MLM201AU	CA201AG
MLM207G	CA207T
MLM207U	CA207G
MLM211G	CA211T
MLM211U	CA211G
MLM224L	CA224G
MLM224P	CA224G, CA224E
MLM239AL	CA239AG
MLM239AP	CA239AG, CA239AE
MLM239L	CA239G
MLM239P	CA239G, CA239E
MLM258G	CA258T
MLM258U	CA258G
MLM301AD	CA301AG
MLM301AG	CA301AT
MLM301AP1	CA301AG, CA301AE
MLM301AU	CA301AG
MLM307G	CA307T
MLM307P1	CA307G, CA307E
MLM307U	CA307G
MLM311G	CA311T
MLM311P1	CA311G, CA311E
MLM311U	CA311G
MLM324L	CA324G, CA324E

Cross-Reference Directory for Linear Integrated Circuits

Industry Type	RCA Replacement Type
MLM324P1	CA324G
MLM339AL	CA339AG
MLM339AP	CA339AG, CA339AE
MLM339L	CA339G
MLM339P	CA339G, CA339E
MLM358G	CA358T
MLM358P1	CA358G, CA358E
MLM358U	CA358G
MPQ3724	CA3724G
MPQ3725	CA3725G
NE555JG	CA555CG
NE555P	CA555CG,CA555CE
NE555L	CA555T
NE555T	CA555CT
NE555V	CA555CG, CA555CE
PM741J	CA741T
PM741CJ	CA741CT
PM741Y	CA741G
PM741CY	CA741CG
PM747K	CA747T
PM747CK	CA747CT
PM747Y	CA747G
PM747CY	CA747CG
Q2T3725	CA3725G
RC555DE	CA555CG
RC555NB	CA555CG, CA555CE
RC555T	CA555CT
RC723DB	CA723CE
RC723T	CA723CT
RC1458DE	CA1458G
RC1458NB	CA1458G, CA1458E
RC1458T	CA1458T
RC3401DB	CA3401G, CA3401E
RC741DB	CA741CG,CA741CE
RC741DC	CA741CG
RC741DE	CA741CG
RC741NB	CA741CG, CA741CE
RC741T	CA741T
RC747DC	CA747CG
RC747DB	CA747CG, CA747CE
RC747T	CA747T
RM555DE	CA555G
RM555T	CA555T
RM723T	CA723T
RM741DC	CA741G
RM741DE	CA741G
RM741T	CA741T
RM747DC	CA747G
RM747T	CA747T
RM1558DE	CA1558G
RM1558T	CA1558T
SE555JG	CA555G
SE555L	CA555T
SE555N	CA555G, CA555E
SE555P	CA555G, CA555E
SE555T	CA555T
SE9300	RCA120
SE9301	RCA121
SE9302	RCA122
SE9303	2N6384
SE9304	2N6385
SFC2101A	CA101AT
SFC2107M	CA107T
SFC211M	CA111T
SFC2201A	CA201AT
SFC2207	CA207T
SFC2211	CA211T
SFC2301A	CA301AT
SFC2301ADC	CA301AE
SFC2307	CA307T
SFC2311	CA311T
SFC2741C	CA741CT
SFC2741M	CA741T
SFC2748DC	CA748CE
SFC2748C	CA748CT
SG101AD	CA101AG
SG101D	CA101G
SG107D	CA107G
SG111D	CA111G
SG111M	CA111G, CA111E
SG111T	CA111T
SG201AD	CA201AG
SG201AM	CA201AG
SG201N	CA201G, CA201E
SG201M	CA201G, CA201E
SG207D	CA207G
SG207N	CA207G, CA207E
SG207T	CA207T
SG211D	CA211G
SG211M	CA211G, CA211E
SG211T	CA211T
SG301AM	CA301AG, CA301AE
SG301AT	CA301AT
SG301N	CA301G, CA301E
SG301T	CA301T
SG307D	CA307G
SG307N	CA307G, CA307E
SG307T	CA307T
SG311D	CA311G
SG311M	CA311G, CA311E
SG311T	CA311T
SG723CN	CA723CE
SG723CT	CA723CT
SG723T	CA723T
SG741CD	CA741CG
SG741CM	CA741CG
SG741CN	CA741CG,CA741CE
SG741CT	CA741CT
SG741D	CA741G
SG741T	CA741T
SG747CD	CA747CG
SG747CN	CA747CG, CA747CE
SG747CT	CA747CT
SG747D	CA747G
SG747T	CA747T
SG748CM	CA748CG
SG748CN	CA748CG, CA748CE
SG748CT	CA748CT
SG748T	CA748T
SG1458M	CA1458G, CA1458E
SG1458T	CA1458T
SG1558T	CA1558T
SG3018T	CA3018
SG3018AT	CA3018A
SG3058J	CA3058D
SG3059J	CA3059D
SG3079J	CA3079D
SG3081N	CA3081E
SG3081J	CA3081F
SG3082N	CA3082E
SG3082J	CA3082F
SG3083J	CA3083F
SG3401N	CA3401G, CA3401E
SN52101AJ	CA101AG
SN52101AL	CA101AT
SN52101AN	CA101AG
SN52101AP	CA101AG, CA101AE
SN52107L	CA107T
SN52107P	CA107G, CA107E
SN52111L	CA111T
SN5211P	CA111G, CA111E
SN52555L	CA555T
SN52555P	CA555G, CA555E
SN52558L	CA1558T
SN52558P	CA1558G, CA1558E
SN52723N	CA723E
SN52723L	CA723T
SN52741J	CA741CG
SN52741L	CA741CT
SN52741N	CA741CG,CA741CE
SN52741P	CA741CG, CA741CE
SN52747L	CA747T
SN52747N	CA747G, CA747E
SN52748J	CA748G
SN52748L	CA748T
SN52748N	CA748G,CA748E
SN52748P	CA748G, CA748E
SN72301AJ	CA301AG
SN72301AL	CA301AT
SN72301AN	CA301AG,CA301AE
SN72301AP	CA301AG
SN72307L	CA307T
SN72307N	CA307G,CA307E
SN72307P	CA307G, CA307E
SN72311L	CA311T
SN72311P	CA311G, CA311E
SN72555L	CA555CT
SN72555P	CA555CG, CA555CE
SN72558L	CA1458T
SN72558P	CA1458G, CA1458E
SN72723N	CA723CE
SN72723L	CA723CT
SN72741J	CA741CG
SN72741L	CA741CT
SN72741N	CA741CG,CA741CE
SN72741P	CA741CG, CA741CE
SN72747J	CA747CG
SN72747L	CA747CT
SN72747N	CA747CG, CA747CE
SN72748J	CA748CG
SN72748L	CA748CT
SN72748N	CA748CG,CA748CE
SN72748P	CA748CG, CA748CE
SN76115N	CA1310E
SN76116N	CA758E
SN76242N	CA3070
SN76243AN	CA3071
SN76264N	CA3072
SN76266N	CA3066
SN76267N	CA3067
SN76298N	CA1398E
SN76564N	CA3064
SN76565N	CA3064E
SN76635N	CA3123E
SN76650N	CA1352E

Cross-Reference Directory for Linear Integrated Circuits

Industry Type	RCA Replacement Type
SN76666N	CA3065
SN76675N	CA3075
SN76676P	CA3076
SN76689N	CA3089E, CA3189E
SP3724	CA3724G
SP3725	CA3725G
SSS101AJ	CA101AT
SSS101AP	CA101AG, CA101AE
SSS107J	CA107T
SSS107P	CA107G, CA107E
SSS201AJ	CA201AT
SSS201AP	CA201AG, CA201AE
SSS207J	CA207T
SSS301AJ	CA301AT
SSS301AP	CA301AG, CA301AE
SSS741CJ	CA741CT
SSS1458J	CA1458T
SSS1558J	CA1558T
TBA810S	CA810Q
TBA810AS	CA810QM
TDA2002V	CA2002
TDA2002H	CA2002M
TBB0747	CA747CT
TBB0748	CA748CT
TBB0748B	CA748CE
TBB1458B	CA1458E
TBC0747	CA747T
TCA270	CA270
TDA3081N	CA3081
TDA3082N	CA3082
TDA3083N	CA3083
TDB0723	CA723CT
TDB0723A	CA723CE
TDC0723	CA723T
U5B7741312	CA741T
U5B7741393	CA741CT
U5B7748312	CA748T
U5B7748393	CA748CT
U5R7723312	CA723T
U5R7723393	CA723CT
U6A7723393	CA723CG, CA723CE
U9T7758393	CA1458G
U9T7741393	CA741CG, CA741CE
ULN2111A	CA2111AE
ULN2111N	CA2111AQ
ULN2114A	CA3072
ULN2124A	CA3070
ULN2125A	CA3120E
ULN2127A	CA3071
ULN2129A	CA3075
ULN2137A	CA3123E
ULN2165A	CA3065
ULN2210A	CA1310E
ULN2212B	CA3012
ULN2262A	CA3126Q
ULN2264A	CA3064
ULN2266A	CA3066
ULN2267A	CA3067
ULN2269A	CA3121E
ULN2289A	CA3089E, CA3189E
ULN2298A	CA1398E
ULX2244A	CA758E
µA101AH	CA101AT
µA101AD	CA101AG
µA101D	CA101G
µA101H	CA101T
µA107H	CA107T
µA111H	CA111T
µA111R	CA111G
µA201AD	CA201AG
µA201AH	CA201AT
µA201D	CA201G
µA201H	CA201AT
µA207H	CA207T
µA301AD	CA301AG
µA301AH	CA301AT
µA307H	CA307T
µA307T	CA307G, CA307E
µA301AT	CA301AG, CA301AE
µA311H	CA311T
µA311R	CA311G
µA311T	CA311G, CA311E
µA555HC	CA555CT
µA555HM	CA555T
µA555TC	CA555CG, CA555CE
µA720PC	CA3123E
µA723CA	CA723CE
µA723CK	CA723CT
µA723CL	CA723CT
µA723CN	CA723CE
µA723DM	CA723E
µA723HC	CA723CT
µA723HM	CA723T
µA723K	CA723T
µA723MN	CA723E
µA723ML	CA723T
µA723PC	CA723CE
µA741CJG	CA741CG
µA741CJ	CA741CG
µA741CN	CA741CG,CA741CE
µA741CL	CA741T
µA741CP	CA741CG, CA741CE
µA741CT	CA741CT
µA741DC	CA741G
µA741DM	CA741G
µA741HC	CA741CT
µA741HM	CA741T
µA741MJG	CA741G
µA741MJ	CA741G
µA741ML	CA741T
µA741MN	CA741G,CA741E
µA741MP	CA741G, CA741E
µA741PC	CA741G, CA741E
µA748PC	CA3072
µA747CA	CA747CE
µA747CJ	CA747CG
µA747CK	CA747CT
µA747CL	CA747CT
µA747CN	CA747CG, CA747CE
µA747DC	CA747CG
µA747DM	CA747G
µA747HC	CA747CT
µA747HM	CA747T
µA747MJ	CA747G
µA747ML	CA747T
µA747MN	CA747G, CA747E
µA747PC	CA747G, CA747E
µA747A	CA747E
µA747K	CA747T
µA748CJG	CA748G
µA748CJ	CA748CG
µA748CL	CA748T
µA748CN	CA748G,CA748E
µA478CP	CA748G,CA748E
µA748CT	CA748CT
µA748DC	CA748CG
µA748DM	CA748G, CA748E
µA748HC	CA748CT
µA748HM	CA748T
µA748MJG	CA748G
µA748MJ	CA748G
µA748ML	CA748T
µA748MN	CA748G,CA748E
µA748MP	CA748G, CA748E
µA748T	CA748T
µA748TC	CA748CG, CA748CE
µA758PC	CA758E
µA780PC	CA3070
µA781PC	CA3071
µA787PC	CA3126Q
µA1391T	CA1391E
µA1394T	CA1394E
µA1458HC	CA1458T
µA1458R1	CA1458G
µA1458HC	CA1458G, CA1458E
µA1558HM	CA1558T
µA3018HM	CA3018
µA3018AHM	CA3018A
µA3019HM	CA3019
µA3026HM	CA3026
µA3036HM	CA3036
µA3039HM	CA3039
µA3045DM	CA3045
µA3046DC	CA3046
µA3064HC	CA3064T
µA3064PC	CA3064E
µA3065PC	CA3065
µA3066PC	CA3066
µA3075PC	CA3075
µA3086DC	CA3086F
µA3089E	CA3089E, CA3189E
µA3401P	CA3401G, CA3401E
µPC151A	CA741CT
µPC151C	CA741CG, CA741CE
µPC157A	CA301AT
µPC157C	CA301AG, CA301AE
µPC251A	CA747CT
µPC251C	CA1458G, CA1458E
µPC301AC	CA301AG, CA301AE
µPC311C	CA311G, CA311E
µPC324C	CA324G, CA324E
µPC339C	CA339G, CA339E
µPC741C	CA741CG, CA741CE
µPC1458C	CA1458G, CA1458E

Linear Integrated Circuits for Industrial Applications

Technical Data

CA101, CA201, CA301 Types

Operational Amplifiers

For Commercial, Industrial, and Military Applications

RCA-CA101, CA101A, CA201, CA201A, and CA301A are general-purpose, high-gain operational amplifiers for use in military, industrial, and commercial applications.

These types, which are externally phase compensated, permit a choice of operation for optimum high-frequency performance at a selected gain; unity-gain compensation can be obtained with a single 30-pF capacitor. Types CA101A and CA201A have all the desirable features and characteristics of the CA101 and CA201, respectively, plus superior input-offset characteristics, and improved noise performance.

All types are available in hermetic gold-CHIP dual-in-line plastic packages (G suffix), 8-lead TO-5 style packages with standard leads (T suffix), and with dual-in-line formed leads ("DIL-CAN", S suffix). The CA301A is also available in the 8-lead dual-in-line plastic package ("MINI-DIP", E suffix), and in chip form (H suffix).

"G" Suffix Types—Hermetic Gold-CHIP Dual-In-Line Plastic Package
"E" Suffix Types—Standard Dual-In-Line Plastic Package
"T" and "S" Suffix Types—TO-5 Style Package

Features:

- **Short-circuit protection and latch-free operation**
- **Unity-gain phase compensation with a single 30-pF capacitor**
- **Replacement for industry types 101, 101A, 201, 201A, 301A**

Applications:

- **Long-interval integrator**
- **Timers**
- **Sample and hold circuits**
- **Summing amplifiers**
- **Multivibrators**
- **Comparators**
- **Instrumentation**
- **AC/DC converters**
- **Inverting amplifiers**
- **Sine- & square-wave generators**
- **Capacitance multipliers & simulated inductors**

Maximum Ratings, *Absolute-Maximum Values at $T_A = 25^oC$*

DC SUPPLY VOLTAGE (between V^+ and V^- terminals):	
CA101, CA101A, CA201, CA201A	44 V
CA301A	36 V
DC INPUT VOLTAGE	±15 V
(For supply voltage less than ± 15 V, the Input Voltage rating is equal to the DC Supply Voltage)	
DIFFERENTIAL INPUT VOLTAGE	±30 V
OUTPUT SHORT-CIRCUIT DURATION	Indefinite*
DEVICE DISSIPATION:	
Up to $T_A = 75^oC$	500 mW
Above $T_A = 75^oC$	derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating –	
CA101, CA101A	−55 to +125 °C
CA201A	−25 to +85 °C
CA201, CA301A	0 to +70 °C
Storage (All types)	−65 to +150 °C
LEAD TEMPERATURE (During Soldering):	
At a distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.	+265 °C

* At $T_A \leq 70^oC$ and $T_C \leq 125^oC$ (CA101);
$T_A \leq 75^oC$ and $T_C \leq 125^oC$ (CA101A, CA201A);
$T_A \leq 55^oC$ and $T_C \leq 70^oC$ (CA201, CA301A).

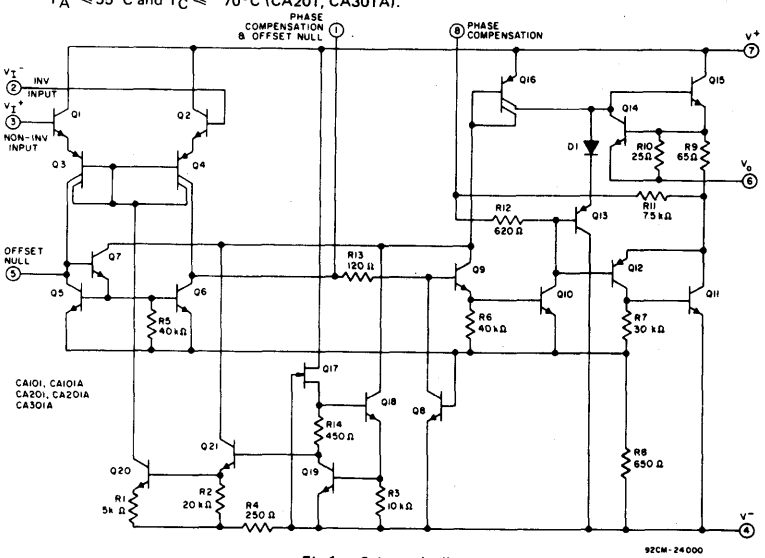
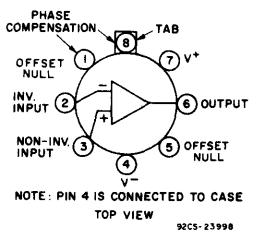
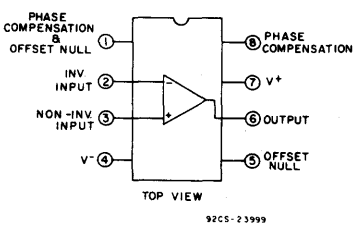


Fig.1 – Schematic diagram.



a – TO-5 style package for all types

T-Suffix
S-Suffix



b – Plastic package for CA301A

G-Suffix
E-Suffix

Fig.2 – Functional diagrams.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	TEST CONDITIONS▲ Supply Voltage ($V^{\pm}$) = 5 to 15 V		CA101 Min.	CA101 Typ.	CA101 Max.	CA201 Min.	CA201 Typ.	CA201 Max.	UNITS	CA101A CA201A Min.	CA101A CA201A Typ.	CA101A CA201A Max.	CA301A Min.	CA301A Typ.	CA301A Max.	UNITS
Input Offset Voltage V_{IO}	T_A=25°C	$R_S \leq 10k\Omega$	–	1	5	–	2	7.5	mV	–	–	–	–	–	–	mV
		$R_S \leq 50k\Omega$	–	–	–	–	–	–		–	0.7	2	–	2	7.5	
		$R_S \leq 10k\Omega$	–	–	6	–	–	10		–	–	–	–	–	–	
		$R_S \leq 50k\Omega$	–	–	–	–	–	–		–	–	3	–	–	10	
Average Temperature Coefficient of Input Offset Voltage αV_{IO}		$R_S \leq 10k\Omega$	–	6	–	–	10	–	µV/°C	–	–	–	–	–	–	µV/°C
		$R_S \leq 50\Omega$	–	3	–	–	6	–		–	–	–	–	–	–	
			–	–	–	–	–	–		–	3	15	–	6	30	
Average Temperature Coefficient of Input Offset Current αI_{IO}	−55°C to +25°C		–	–	–	–	–	–	nA/°C	–	0.02	0.2	–	–	–	nA/°C
	0°C to +25°C		–	–	–	–	–	–		–	–	–	–	0.02	0.6	
	+25°C to +70°C		–	–	–	–	–	–		–	–	–	–	0.01	0.3	
	+25°C to +125°C		–	–	–	–	–	–		–	0.01	0.1	–	–	–	
Input Offset Current I_{IO}	T_A=0°C		–	–	–	–	150	750	nA	–	–	–	–	–	–	nA
	T_A=25°C		–	40	200	–	100	500		–	1.5	10	–	3	50	
	T_A=70°C		–	–	–	–	50	400		–	–	–	–	–	–	
	T_A=125°C		–	10	200	–	–	–		–	–	–	–	–	–	
			–	–	–	–	–	–		–	–	20	–	–	70	
	T_A=−55°C		–	100	500	–	–	–		–	–	–	–	–	–	
Input Bias Current I_{IB}	T_A=−55°C		–	0.28	1.5	–	–	–	µA	–	–	–	–	–	–	µA
	T_A=0°C		–	–	–	–	0.32	2		–	–	–	–	–	–	
	T_A=25°C		–	0.12	0.5	–	0.25	1.5		–	0.03	0.075	–	0.07	0.25	
			–	–	–	–	–	–		–	–	0.1	–	–	0.3	
Supply Current $I^{\pm}$	T_A=25°C	$V^{\pm}$=15V	–	–	–	–	–	–	mA	–	–	–	–	1.8	3	mA
		$V^{\pm}$=20V	–	1.8	3	–	1.8	3		–	1.8	3	–	–	–	
	T_A=125°C	$V^{\pm}$=20V	–	1.2	2.5	–	–	–		–	1.2	2.5	–	–	–	
Open-Loop Differential Voltage Gain A_{OL}	T_A=25°C V_O=±10V	$V^{\pm}$=15V $R_L \geq 2k\Omega$	50	160	–	20	150	–	V/mV	50	160	–	25	160	–	V/mV
	V_O=±10V	$V^{\pm}$=15V $R_L \geq 2k\Omega$	25	–	–	15	–	–		25	–	–	15	–	–	
Input Resistance R_I	T_A=25°C		0.3	0.8	–	0.1	0.4	–	MΩ	1.5	4	–	0.5	2	–	MΩ
Output Voltage Swing V_{OPP}	$V^{\pm}$=15V	R_L=10kΩ	±12	±14	–	±12	±14	–	V	±12	±14	–	±12	±14	–	V
	$V^{\pm}$=15V	R_L=2kΩ	±10	±13	–	±10	±13	–		±10	±13	–	±10	±13	–	
Common-Mode Input-Voltage Range V_{ICR}	$V^{\pm}$=15V		±12	–	–	±12	–	–	V	–	–	–	±12	–	–	V
	$V^{\pm}$=20V		–	–	–	–	–	–		±15	–	–	–	–	–	
Common-Mode Rejection Ratio CMRR		$R_S \leq 10k\Omega$	70	90	–	65	90	–	dB	–	–	–	–	–	–	dB
		$R_S \leq 50k\Omega$	–	–	–	–	–	–		80	96	–	70	90	–	
Supply-Voltage Rejection Ratio PSRR		$R_S \leq 10k\Omega$	70	90	–	70	90	–	dB	–	–	–	–	–	–	dB
		$R_S \leq 50k\Omega$	–	–	–	–	–	–		80	96	–	70	90	–	

▲ Characteristics applicable over operating temperature range (T_A) as shown below, unless otherwise specified: CA101, CA101A: −55 to +125°C; CA201A: −25 to +85°C; CA201, CA301A: 0 to 70°C

	CA101	CA201	CA101A	CA201A	CA301A	
Max. V_{IO} (T_A = 25°C)	5	7.5	2	2	7.5	mV
Max. I_{IO} (T_A = 25°C)	200	500	10	10	50	nA
Min. A_{OL} (T_A = 25°C)	50	20	50	50	25	V/mV
T_A Range (Operating)	−55 to +125	0 to +70	−55 to +125	−25 to +85	0 to +70	°C
Slew Rate (Summing ampl.)	–	–	10	10	10	V/µs

TYPICAL STATIC CHARACTERISTICS

Type CA101

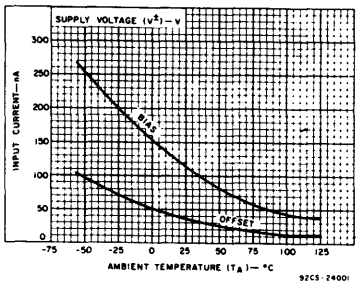


Fig.3 – Input current (I_{IO}, I_{IB}) vs. temperature.

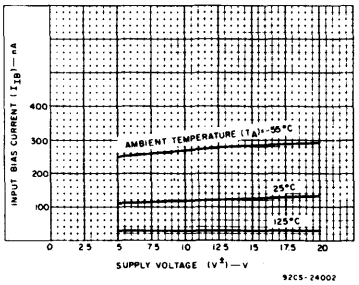


Fig.4 – Input bias current vs. supply voltage.

Types CA101, CA101A, and CA201A

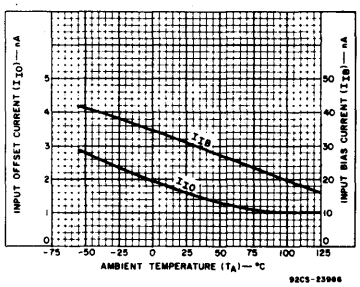


Fig. 5 – Input current (I_{IO}, I_{IB}) vs. temperature (CA101A and CA201A only).

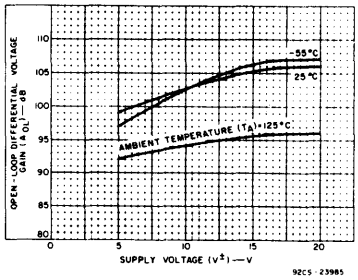


Fig.6 – Voltage gain vs. supply voltage.

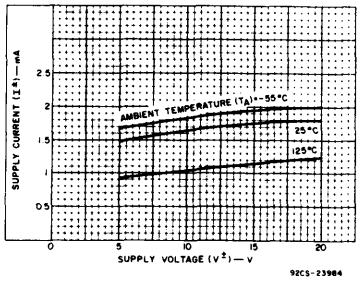


Fig.7 – Supply characteristics.

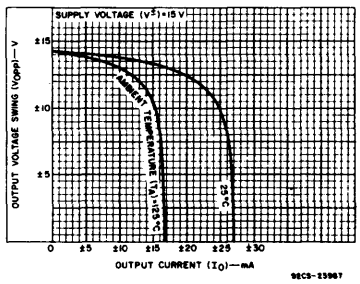


Fig.8 – Output characteristics.

Type CA301A

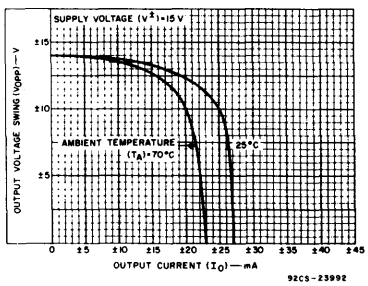


Fig. 9 – Output characteristics.

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPES CA101A AND CA201A

Single-Pole Compensation

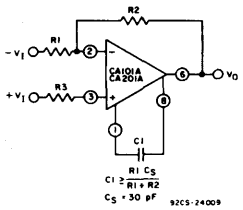


Fig. 10 – Test circuit employing single-pole compensation.

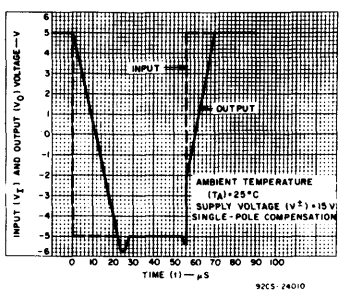


Fig. 11 – Voltage follower (V_I, V_O) pulse response.

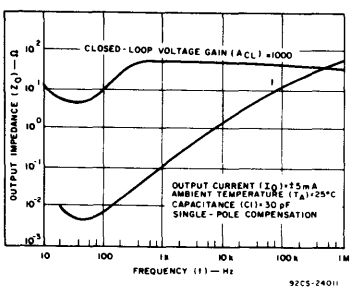


Fig. 12 – Closed-loop output impedance vs. frequency.

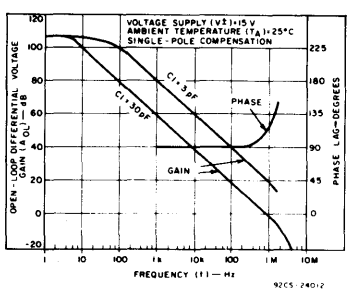


Fig. 13 – Voltage gain and phase lag vs. frequency.

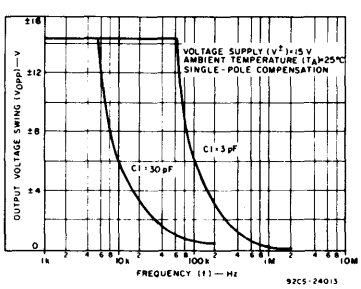


Fig. 14 – Output voltage swing vs. frequency.

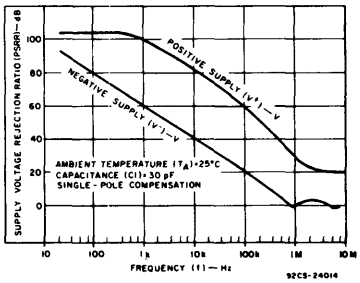


Fig. 15 — Supply voltage rejection ratio vs. frequency.

Two-Pole Compensation

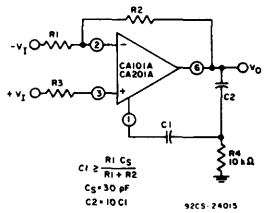


Fig. 16 — Test circuit employing two-pole compensation.

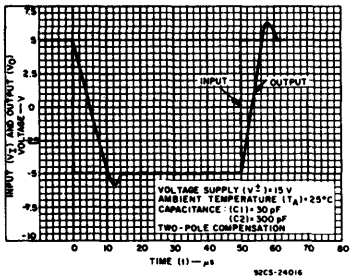


Fig. 17 — Voltage follower pulse response.

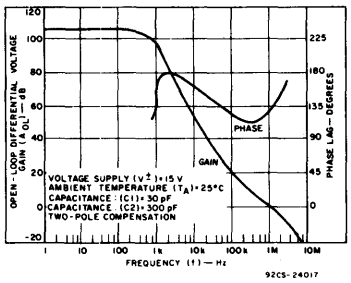


Fig. 18 — Voltage gain and phase lag vs. frequency.

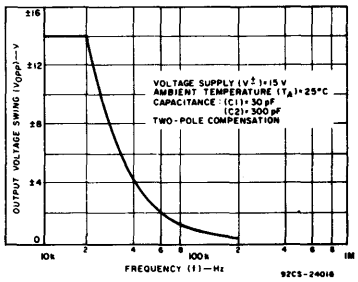


Fig. 19 — Output voltage swing vs. frequency.

Feed-Forward Compensation

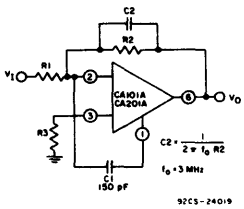


Fig. 20 — Test circuit employing feedforward compensation.

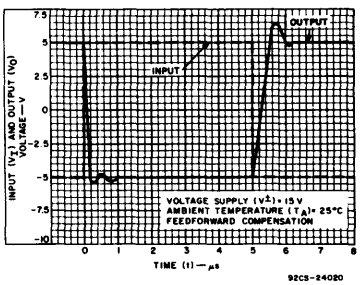


Fig. 21 — Inverter pulse response.

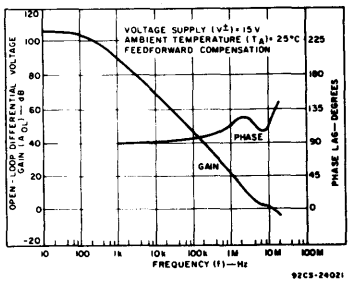


Fig. 22 — Voltage gain and phase lag vs. frequency.

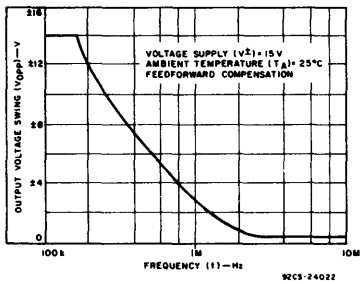


Fig. 23 — Output voltage swing vs. frequency.

CA101A AND CA201A

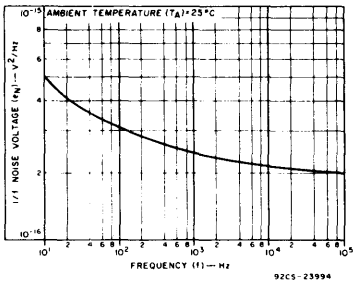


Fig. 24 — 1/f noise voltage vs. frequency.

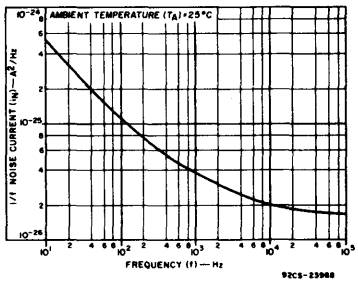


Fig. 25 — 1/f noise current vs. frequency.

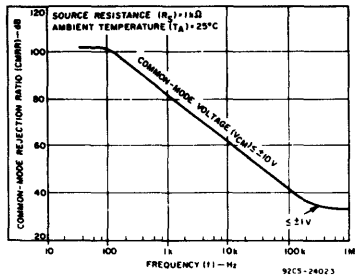


Fig. 26 — Common-mode rejection ratio vs. frequency.

CA107, CA207, CA307 Types
Operational Amplifiers

For Military, Industrial, and Commercial Applications

RCA-CA107, CA207, CA307 are general-purpose operational amplifiers intended for use in military, industrial, and commercial applications. A 30-pF on-chip capacitor provides internal frequency compensation. Low input current over temperature range (100 nA max.) for the CA107 and CA207 make these types especially well suited for applications such as long interval timers and sample-and-hold circuits.

All types are available in hermetic gold-CHIP dual-in-line plastic packages (G suffix), 8-lead TO-5 style packages with standard leads (T suffix), and with dual-in-line formed leads ("DIL-CAN", S suffix). The CA307 is also available in the 8-lead dual-in-line plastic package ("MINI-DIP", E suffix), and in chip form (H suffix).

The CA107, CA207, and CA307 are direct replacements for industry types 107, 207, and 307 in packages with similar terminal arrangements.

"G" Suffix Types—Hermetic Gold-CHIP in Dual-In-Line Plastic Package
"E" Suffix Types—Standard Dual-In-Line Plastic Package
"T" and "S" Suffix Types—TO-5 Style Package

Applications:

- **Long-interval integrators**
- **Timers**
- **Sample-and-hold circuits**
- **Summing amplifiers**
- **Multivibrators**

Type \ Feature	Max. V_{IO} (mV)	Max. I_{IO} (nA)	Max. I_{IB} (nA)	Temp. Range (T_A) °C	Package (Suffix)
CA107	3	20	100	−55 to +125	G, S, T
CA207	3	20	100	−25 to +85*	G, S, T
CA307	10	70	300	0 to +70▲	G, E, S, T

*Types CA207G, S, and T can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temp. range of −25 to +85°C.

▲Types CA307G, E, S, and T can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temp. range of 0 to 70°C.

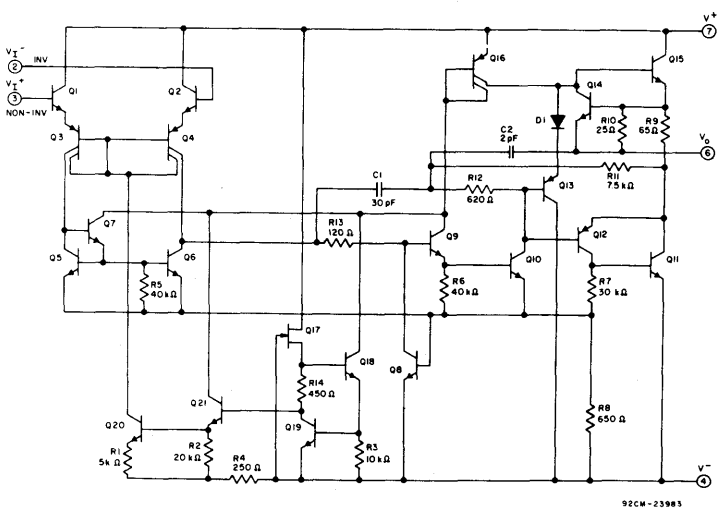
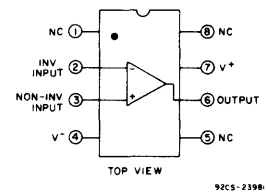
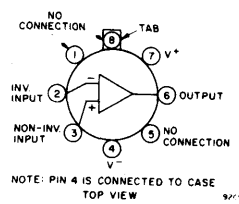


Fig. 1 — Schematic diagram of CA107, CA207, and CA307.



Functional diagram for plastic package.



Functional diagram for TO-5 style packages.

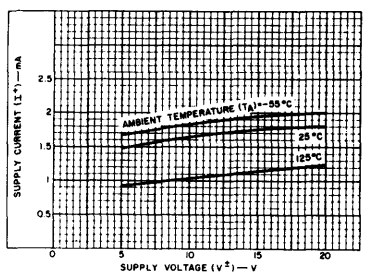


Fig. 2 — Supply current vs. supply voltage.

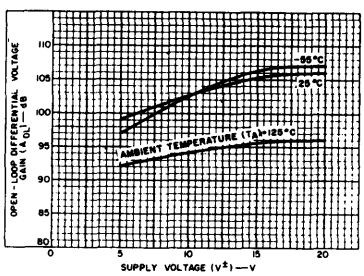


Fig. 3 — Open-loop differential voltage gain vs. supply voltage.

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

DC SUPPLY VOLTAGE (Between V⁺ and V⁻ Terminals):	
CA107, CA207	44 V
CA307	36 V
DC INPUT VOLTAGE	±15 V
(For supply voltages less than ±15 V, the absolute maximum input voltage is equal to the supply voltage)	
DIFFERENTIAL INPUT VOLTAGE	±30 V
OUTPUT SHORT-CIRCUIT DURATION*	Indefinite
DEVICE DISSIPATION UP TO T_A = 70°C	500 mW
Above T_A = 70°C Derate linearly at	6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating – CA107	−55°C to +125°C
CA207	−25°C to +85°C▲
CA307	0°C to +70°C†
Storage – All Types	−65°C to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

*For type CA307 continuous short circuit is allowed for Case Temperature to +70°C and ambient temperature to +55°C.

▲Types CA207G, S, and T can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of −25 to +85°C.

†Types CA307G, E, S, and T can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of 0 to 70°C.

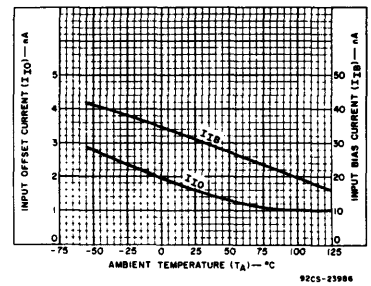


Fig. 4 – Input offset and input bias currents vs. ambient temperature.

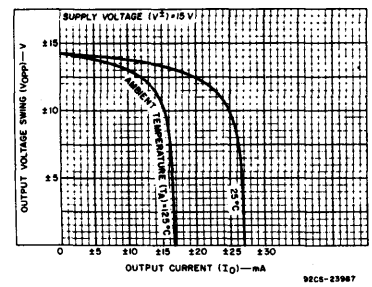


Fig. 5 – Output voltage swing vs. output current.

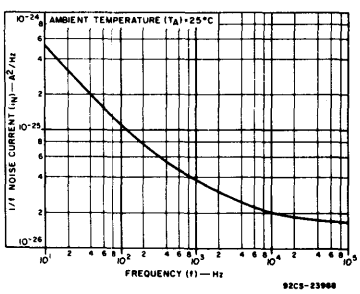


Fig. 6 – 1/f noise current vs. frequency.

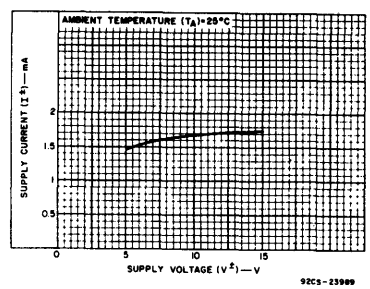


Fig. 7 – Supply current vs. supply voltage.

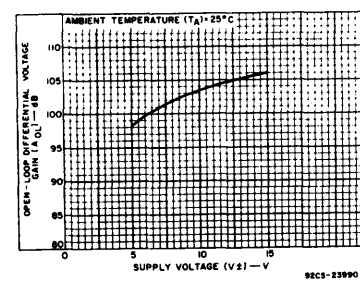


Fig. 8 – Open-loop differential voltage gain vs. supply voltage.

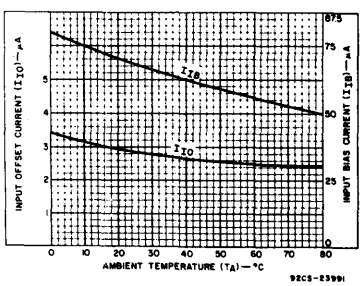


Fig. 9 – Input offset and input bias current vs. ambient temperature.

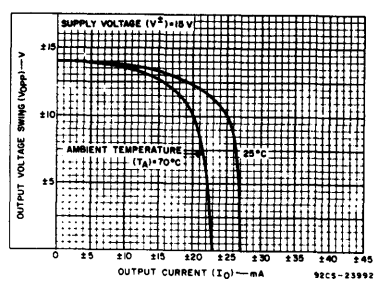


Fig. 10 – Output voltage swing vs. output current.

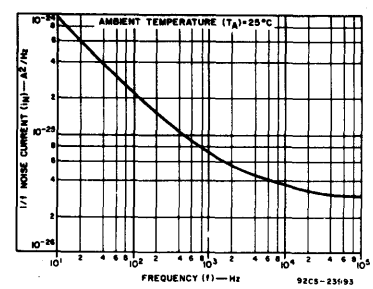


Fig. 11 – 1/f noise current vs. frequency.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS[A]	LIMITS						UNITS
	Supply Voltage (V±) = 5 V to 15 V (CA307) 5 V to 20 V (CA107,CA207)	CA107 CA207			CA307			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage, V_{IO}	T_A = 25°C, $R_S \leq 50\,k\Omega$	–	0.7	2	–	2	7.5	mV
	$R_S \leq 50\,k\Omega$	–	–	3	–	–	10	
Average Temperature Coefficient of Input Offset Voltage, αV_{IO}		–	3	15	–	6	30	μV/°C
Input Offset Current, I_{IO}		–	–	20	–	–	70	nA
	T_A = 25°C	–	1.5	10	–	3	50	
Average Temperature Coefficient of Input Offset Current, αI_{IO}	See Note 1	–	0.01	0.1	–	0.01	0.3	nA/°C
	See Note 2	–	0.02	0.2	–	0.02	0.6	
Input Bias Current, I_{IB}		–	–	100	–	–	300	nA
	T_A = 25°C	–	30	75	–	70	250	
Supply Current, $I^{\pm}$	T_A = +125°C, $V^{\pm}$ = 20 V	–	1.2	2.5	–	–	–	mA
	T_A = 25°C, $V^{\pm}$ = 20 V, (CA307 $V^{\pm}$ = 15 V)	–	1.8	3	–	1.8	3	
Open-Loop Differential Voltage Gain, A_{OL}	$V^{\pm}$ = 15 V, V_O = ±10 V, $R_L \geq 2\,k\Omega$	25	–	–	15	–	–	V/mV
	$V^{\pm}$ = 15 V, V_O = ±10 V $R_L \geq 2\,k\Omega$, T_A = 25°C	50	160	–	25	160	–	
Input Resistance, R_I	T_A = 25°C	1.5	4	–	0.5	2	–	MΩ
Output Voltage Swing, V_{OPP}	$V^{\pm}$ = 15 V, R_L=10 kΩ	±12	±14	–	±12	±14	–	V
	$V^{\pm}$ = 15 V, R_L = 2 kΩ	±10	±13	–	±10	±13	–	
Input Voltage Range, V_{ICR}	$V^{\pm}$ = 20 V, (CA307 $V^{\pm}$ = 15 V)	±15	–	–	±12	–	–	V
Common-Mode Rejection Ratio, CMRR	$R_S \leq 50\,k\Omega$	80	96	–	70	90	–	dB
Supply-Voltage Rejection Ratio, PSRR	$R_S \leq 50\,k\Omega$	80	96	–	70	96	–	dB

Note 1: For CA107, +25, to +125°C; For CA207, +25 to +85°C: For CA307, +25 to 70°C.

Note 2: For CA107, −55 to +25°C; For CA207, −25 to +25°C: For CA307, 0 to +25°C.

[A] Characteristics applicable over operating temperature range as shown below unless otherwise specified.

CA107 – T_A = −55 to +125°C

CA207 – T_A = −25 to +85°C

CA307 – T_A = 0 to 70°C

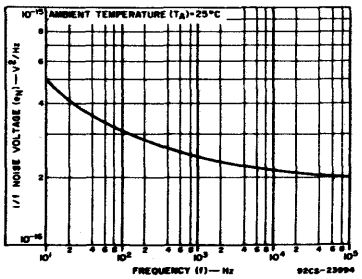


Fig. 12 – 1/f noise voltage vs. frequency.

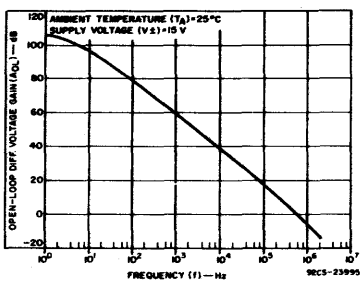


Fig. 13 – Open-loop differential voltage gain vs. frequency.

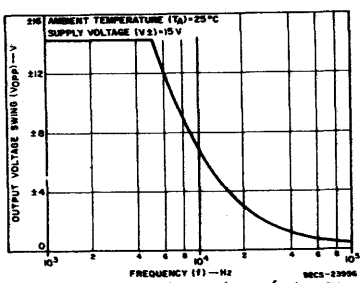


Fig. 14 – Output voltage swing vs. frequency.

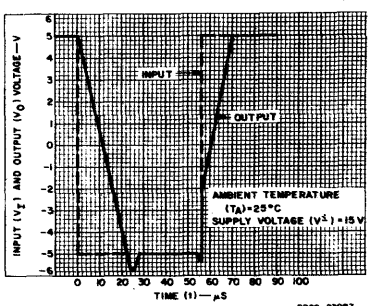


Fig. 15 – Voltage follower pulse response.

Voltage Comparators

For Commercial and Industrial Applications

- "G" Suffix Types—Hermetic Gold-CHIP in Dual-In-Line Plastic Package
- "E" Suffix Types—Standard Dual-In-Line Plastic Package
- "T" and "S" Suffix Types—TO-5 Style Package

Applications

- Multivibrators
- Positive and negative peak detectors
- Crystal oscillators
- Zero-crossing detectors
- Solenoid, relay, and lamp drivers

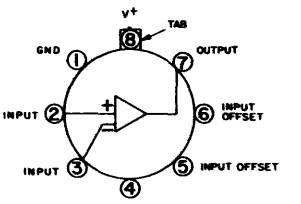
Features

- Single- or dual-supply operation
- Power consumption — 135 mW at ±15 V
- Strobe capability
- Low input-offset current:
  CA111, CA211 — 4 nA(typ.)
  CA311 — 6 nA(typ.)
- Differential input-voltage range — ±30 V
- Directly interchangeable with National Semiconductor LM111, LM211, and LM311 Series types

The RCA-CA111, CA211, and CA311 are monolithic voltage comparators that operate from dual supplies up to ±15 V, or from single supplies down to 5 V. This single-supply capability makes the outputs of these devices compatible with RTL, DTL, TTL, and MOS circuits. In addition, they can drive lamps or relays, and switch voltages up to 50 V (CA311, 40 V) at currents as high as 50 mA.

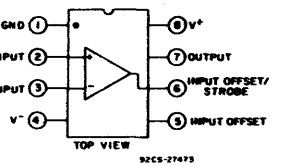
The inputs and the outputs of the CA111, CA211, and CA311 can be isolated from system ground, allowing the output to drive loads referred to ground, V^+, or V^-.

All types are available in hermetic gold-CHIP dual-in-line plastic packages (G suffix), 8-lead TO-5 style packages with standard leads (T suffix), and with dual-in-line formed leads ("DIL-CAN", S suffix). The CA311 is also available in the 8-lead dual-in-line plastic package ("MINI-DIP",E suffix), and in chip form (H suffix).



NOTE: PIN 4 IS CONNECTED TO CASE 92CS-24379

Functional diagram for TO-5 style package.



Functional diagram for plastic package.

Type \ Feature	Max. V_{IO} (mV)	Max. I_{IO} (nA)	Max. I_{IB} (nA)	Temp. Range (T_A) °C	Package (Suffix)
CA111	3	10	100	−55 to +125	G,S,T
CA211	3	10	100	−25 to +85▲	G,S,T
CA311	7.5	50	250	0 to +70†	G,E,S,T

MAXIMUM RATINGS, *Absolute Maximum Values at T_A = 25°C*

DC SUPPLY VOLTAGE (between V^+ and V^- terminals) . . . . . 36 V
DC INPUT VOLTAGE* . . . . . ±15 V
DIFFERENTIAL INPUT VOLTAGE . . . . . ±30 V
OUTPUT TO NEGATIVE SUPPLY VOLTAGE (V_{7-4}):
  CA111, CA211 . . . . . 50 V
  CA311 . . . . . 40 V
GROUND TO NEGATIVE SUPPLY VOLTAGE (V_{1-4}) . . . . . 30 V
OUTPUT SHORT-CIRCUIT DURATION . . . . . 10 s
DEVICE DISSIPATION:
  Up to T_A = 25°C . . . . . 500 mW
  Above T_A = 25°C . . . . . derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:
  Operating:
    CA111 . . . . . −55 to +125°C
    CA211 . . . . . −25 to +85°C▲
    CA311 . . . . . 0 to +70°C†
  Storage, all types . . . . . −65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):
  At distance 1/16 ±1/32 in. (1.59 ±0.79 mm)
  from case for 10 seconds max. . . . . +265°C

*This rating applies for ±15 V supplies. The positive input-voltage limit is 30 V above the negative supply. The negative input-voltage limit is equal to the negative supply voltage or 30 V below the positive supply. The negative input-voltage limit is equal to the negative supply voltage or 30 V below the positive supply, whichever is less.

▲ Types CA211G,S, and T can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of −25 to +85°C.

† Types CA311G,E,S and T can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of 0 to 70°C.

TYPICAL CHARACTERISTICS – ALL TYPES

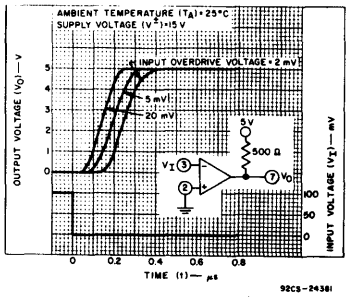


Fig. 1 – Response time for various input overdrive voltages–positive input.

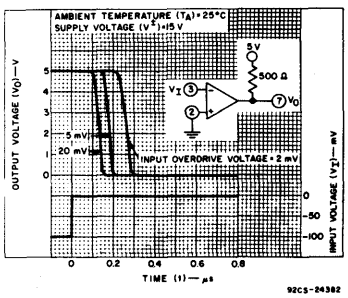


Fig. 2 – Response time for various input overdrive voltages–negative input.

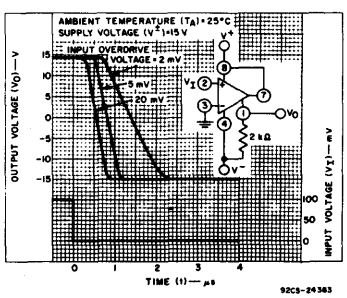


Fig. 4 – Response time for various input overdrive voltages–positive input.

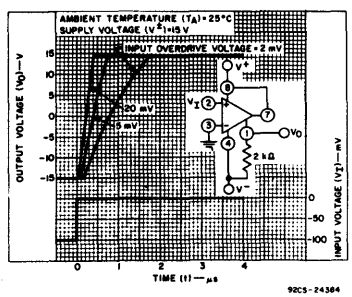


Fig. 5 – Response time for various input overdrive voltages–negative input.

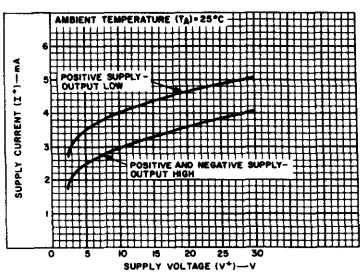


Fig. 7 – Supply current vs. supply voltage.

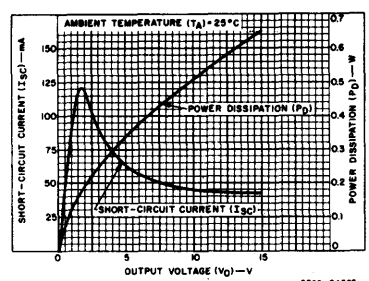


Fig. 6 – Output limiting characteristics.

TYPICAL CHARACTERISTCS –CA111, CA211

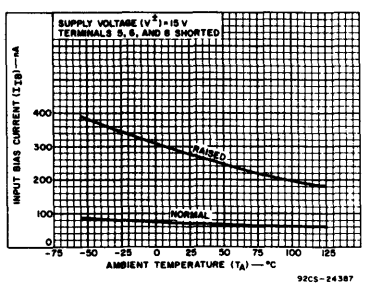


Fig. 8 – Input bias current vs. ambient temperature.

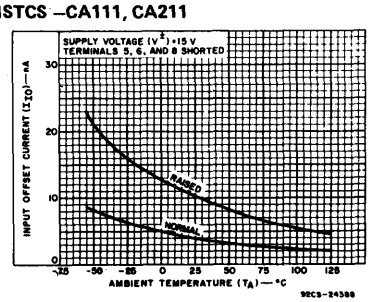


Fig. 9 – Input offset current vs. ambient temperature.

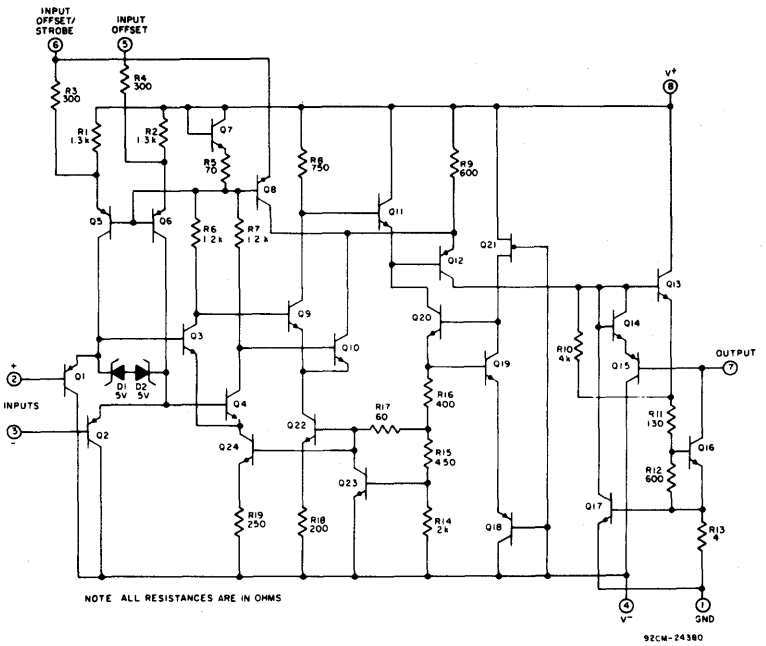


Fig. 3 – Schematic diagram for CA111, CA211, and CA311.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	TEST CONDITIONS: SUPPLY VOLTAGE ($V^{\pm}$) = 15 V UNLESS OTHERWISE SPECIFIED		LIMITS: CA111 CA211 TYP.	CA111 CA211 MAX.	CA311 TYP.	CA311 MAX.	UNITS
Input Offset Voltage, V_{IO}	$R_s \leq 5\ k\Omega$, Note 2	T_A=25°C	0.7	3	2	7.5	mV
		Note 1	–	4	–	10	
Saturation Voltage	V_I = −5 mV, I_O = 50 mA (For CA311, $V_I \leq$ −10 mV)	T_A=25°C	0.75	1.5	–	–	V
	$V^+ \geq$ 4.5 V, V^-=0, $V_I \leq$ −6 mV, $I_{SINK} \leq$ 8 mA (For CA311, $V_I \leq$ −10 mV)	Note 1	0.23	0.4	–	–	
Input Voltage Range, V_{IPP}		Note 1	±14	–	±14	–	V
Input Offset Current, I_{IO}	Note 2	T_A = 25°C	4	10	6	50	nA
		Note 1	–	20	–	70	
Input Bias Current, I_{IB}	Note 2	T_A=25°C	60	100	100	250	nA
		Note 1	–	150	–	300	
Positive Supply Current, I^+		T_A = 25°C	5.1	6	5.1	7.5	mA
Negative Supply Current, I^-		T_A=25°C	4.1	5	4.1	5	mA
Output Leakage Current	$V_I \geq$ 5 mV, V_O = 35 V (For CA311, $V_I \geq$ −10 mV)	T_A=25°C	0.2	10	–	–	nA
		Note 1	0.1	0.5	–	–	μA
Strobe On Current		T_A=25°C	3	–	3	–	mA
Voltage Gain, A		T_A=25°C	200	–	200	–	V/mV
Response Time	100 mV Input Step with 5 mV overdrive voltage	T_A=25°C	200	–	200	–	ns

Note 1: Ambient temperature (T_A) over applicable operating temperature range as shown below.

CA111	CA211	CA311
−55 to +125°C	−25 to +85°C	0 to +70°C

Note 2: The input offset characteristics given are the values required to drive the output to within 1 V of either supply with a 1-mA load. These characteristics define an error band which takes into account the worst-case effects of voltage gain and input impedance. The input offset voltage, input offset current, and input bias current specifications apply for any supply voltage from a 5 V single supply up to a ±15 V dual supply.

TYPICAL CHARACTERISTICS – CA111, CA211 (CONT'D)

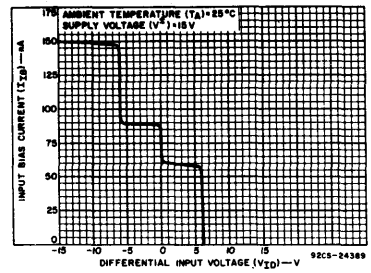


Fig. 10 – Input characteristcs.

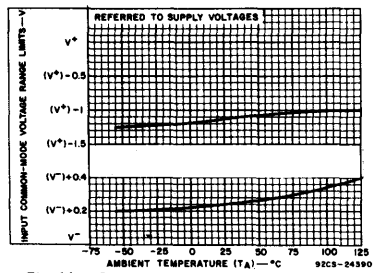


Fig. 11 – Common-mode voltage range limits vs. ambient temperature.

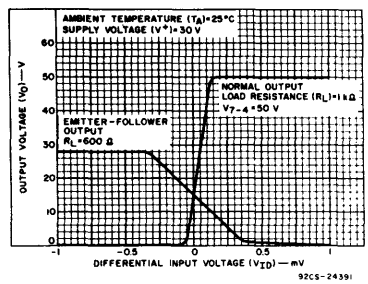


Fig. 12 – Transfer function.

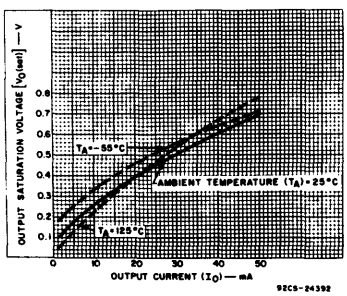


Fig. 13 – Output saturation voltage vs. output current.

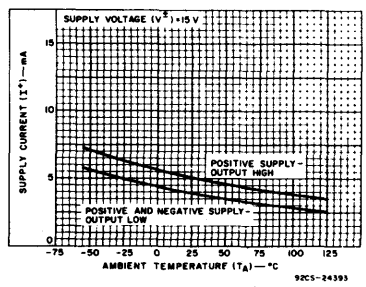


Fig. 14 – Supply current vs. ambient temperature.

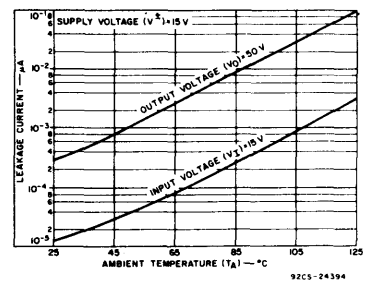


Fig. 15 – Input and output leakage current vs. ambient temperature.

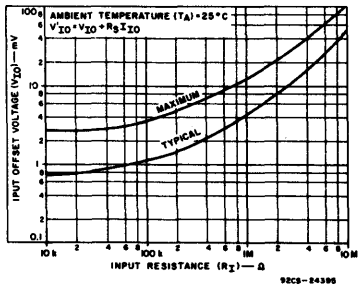


Fig. 16 – Offset error.

TYPICAL CHARACTERISTICS – CA311

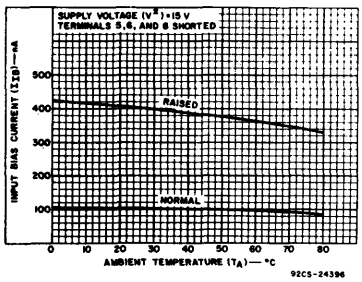


Fig. 17 – Input bias current vs. ambient temperature.

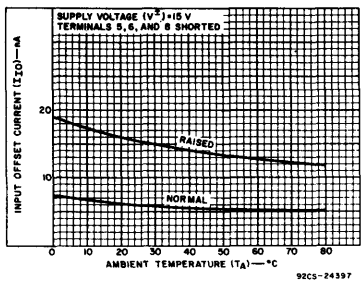


Fig. 18 – Input offset current vs. ambient temperature.

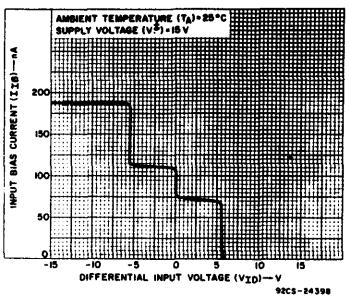


Fig. 19 – Input characteristics.

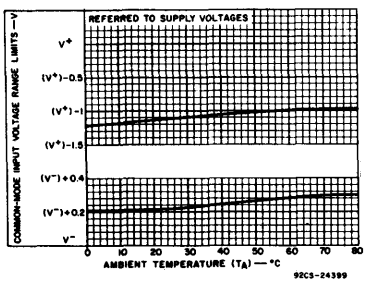


Fig. 20. – Common-mode voltage range limits vs. ambient temperature.

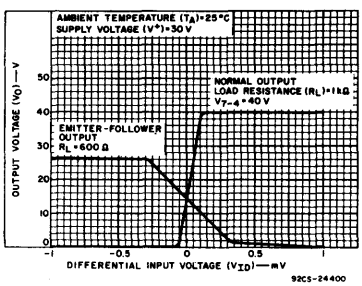


Fig. 21 – Transfer function.

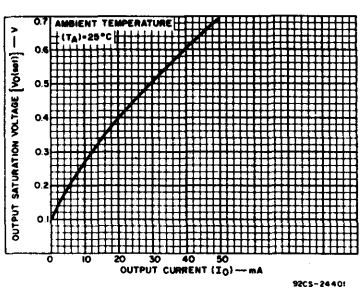


Fig. 22 – Output saturation voltage vs. output current.

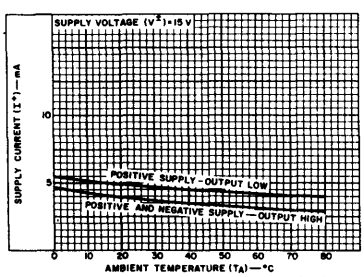


Fig. 23 – Supply current vs. ambient temperature.

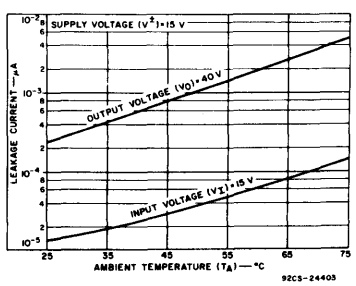


Fig. 24 – Input and output leakage current vs. ambient temperature.

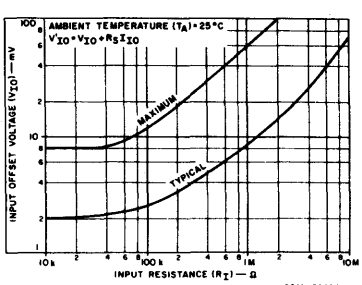


Fig. 25 – Offset error.

Quad Operational Amplifiers

"E" Suffix Types: Standard Dual-In-Line Plastic Package
"G" Suffix Types: Hermetic Gold-Chip Dual-In-Line Plastic Package

For Commercial, Industrial, and Military Applications

The RCA-CA124, -CA224, and -CA324 consist of four independent, high-gain operational amplifiers on a single monolithic substrate. An on-chip capacitor in each of the amplifiers provides frequency compensation for unity gain. These devices are designed specifically to operate from either single or dual supplies, and the differential voltage range is equal to the power-supply voltage. Low power drain and an input common-mode voltage range of from 0 V to $V^+ - 1.5$ V (single-supply operation) make the CA124, CA224, and CA324 suitable for battery operation.

The CA124, CA224, and CA324 are supplied in a 14-lead dual-in-line plastic package (E suffix), or in a hermetic gold-chip 14-lead dual-in-line plastic package (G suffix) to provide true hermetic performance. The CA324 is also available in chip form (H suffix), and as a hermetic gold-chip (HG suffix).

Features:

- **Operation from single or dual supplies**
- **Unity-gain bandwidth . . . . . 1 MHz (typ.)**
- **DC voltage gain . . . . . 100 dB (typ.)**
- **Input bias current . . . . . 45 nA (typ.)**
- **Input offset voltage . . . . . 2 mV (typ.)**
- **Input offset current . . . . . 5 nA (typ.) for CA224, CA324**
  3 nA (typ.) for CA124
- **Replacement for industry types 124, 224, 324**

Applications

- **Summing amplifiers**
- **Multivibrators**
- **Oscillators**
- **Transducer amplifiers**
- **DC gain blocks**

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

SUPPLY VOLTAGE	32 V or ±16 V
DIFFERENTIAL INPUT VOLTAGE	±32 V
INPUT VOLTAGE	−0.3 V to +32 V
INPUT CURRENT ($V_I < -0.3$ V)†	50 mA
OUTPUT SHORT CIRCUIT TO GROUND ($V^+ \leq 15$ V)*	Continuous
DEVICE DISSIPATION:	
Up to $T_A = 55^oC$	750 mW
Above $T_A = 55^oC$	derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

*The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device.

†This input current will only exist when the voltage at any of the input leads is driven negative. This current is due to the collector-base junction of the input p-n-p transistors becoming forward biased and thereby acting as input diode clamps. In addition to this diode action, there is also lateral n-p-n parasitic transistor action on the IC chip. This transistor action can cause the output voltages of the amplifiers to go to the V^+ voltage level (or to ground for a large overdrive) for the time duration that an input is driven negative. This transistor action is not destructive and normal output states will re-establish when the input voltage, which was negative, again returns to a value greater than −0.3 V dc.

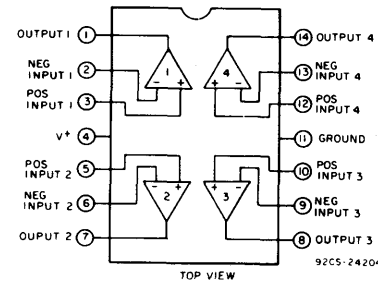


Fig. 1 – Functional diagram.

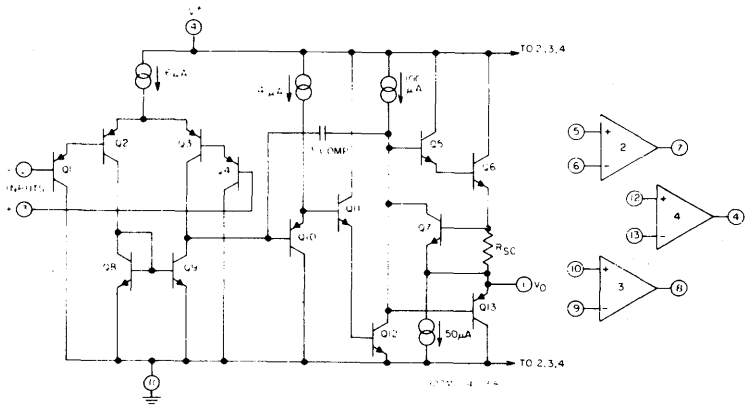


Fig. 2–Schematic diagram–one of four operational amplifiers.

ELECTRICAL CHARACTERISTICS (Values apply for each operational amplifier)

CHARACTERISTIC	TEST CONDITIONS	CA124 LIMITS			CA224, CA324 LIMITS			UNITS
	Supply Voltage (V^+) = 5 V Unless Otherwise Specified	Min.	Typ.	Max.	Min.	Typ.	Max.	
	T_A = 25°C							
Input Offset Voltage, V_{IO}	Note 3	–	2	5	–	2	7	mV
Output Voltage Swing, V_{OPP}	R_L = 2 kΩ	0	–	V^+–1.5	0	–	V^+–1.5	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, V^+=30 V	0	–	V^+–1.5	0	–	V^+–1.5	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	3	30	–	5	50	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	45	150	–	45	250	nA
Output Current (Source), I_O	V_I^+=+1 V, V_I^-=0 V, V^+=15 V	20	40	–	20	40	–	mA
Output Current (Sink), I_O	V_I^+=0 V, V_I^-=1 V, V^+=15 V	10	20	–	10	20	–	mA
	V_I^+=0 V, V_I^-=1 V, V_O=200 mV	12	50	–	12	50	–	μA
Large-Signal Voltage Gain, A	$R_L \geq 2$ kΩ, V^+=15 V (For large V_O swing)	94	100	–	88	100	–	dB
Common-Mode Rejection Ratio, CMRR	DC	70	85	–	65	70	–	dB
Power Supply Rejection Ratio, PSRR	DC	65	100	–	65	100	–	dB
Amplifier-to-Amplifier Coupling	f=1 to 20 kHz (Input referred)	–	–120	–	–	–120	–	dB
		T_A = –55 to +125°C			T_A = –40 to +85°C (CA224), T_A = 0 to 70°C (CA324)			
Input Offset Voltage, V_{IO}	Note 3	–	–	7	–	–	9	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	R_S = 0	–	7	–	–	7	–	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	–	100	–	–	150	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	–	–	10	–	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	–	300	–	–	500	nA
Supply Current, I^+	R_L = ∞ On All Ampl.	–	0.8	2	–	0.8	2	mA
Input Common-Mode Voltage Range, V_{ICR}	V^+ = 30 V	0	–	V^+–2	0	–	V^+–2	V
Large-Signal Voltage Gain, A	$R_L \geq 2$ kΩ, V^+=15 V (For large V_O swing)	88	–	–	83	–	–	dB
Output Voltage Swing: High-Level, V_{OH}	R_L=2 kΩ, V^+=30 V	26	–	–	26	–	–	V
	R_L=10 kΩ	27	28	–	27	28	–	
Low-Level, V_{OL}	R_L=10 kΩ	–	5	20	–	5	20	mV
Output Current: Source, I_O	V_I^+=1 V_{DC}, V_I^-=0, V^+=15 V	10	20	–	10	20	–	mA
Sink, I_O	V_I^-=1 V_{DC}, V_I^+=0, V^+=15 V	5	8	–	5	8	–	mA
Differential Input Voltage	Note 2	–	–	V^+	–	–	V^+	V

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is V^+ – 1.5 V, but either or both inputs can go to +32 V without damage.

NOTE 3: V_O = 1.4 V_{DC}, R_s = 0 Ω with V^+ from 5 V to 30 V; and over the full input common-mode voltage range (0 V to V^+ – 1.5 V).

TYPICAL CHARACTERISTICS CURVES

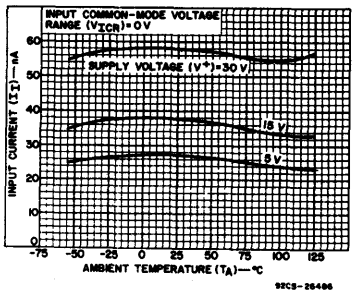


Fig. 3—Input current vs. ambient temperature.

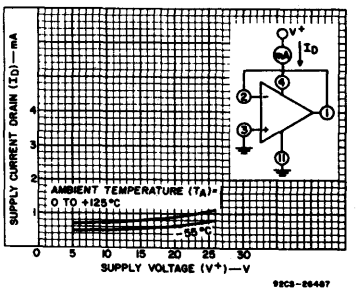


Fig. 4—Supply current drain vs. supply voltage.

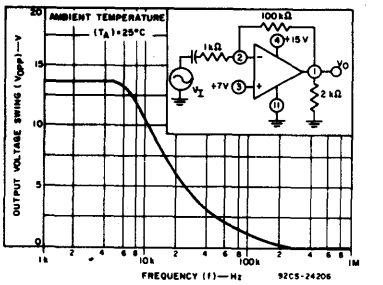


Fig. 5—Large-signal frequency response.

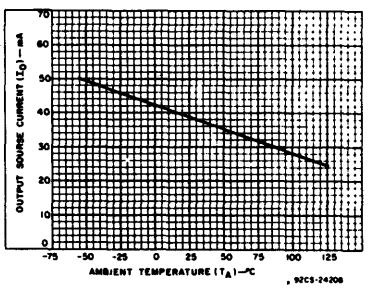


Fig. 6—Output current vs. ambient temperature.

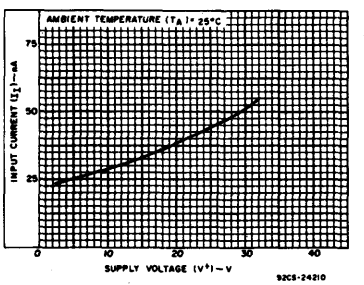


Fig. 7—Input current vs. supply voltage.

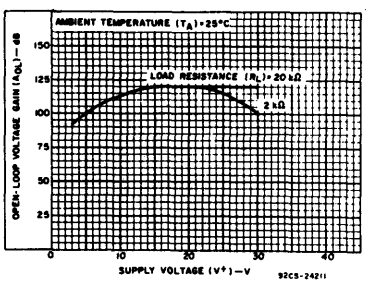


Fig. 8—Voltage gain vs. supply voltage.

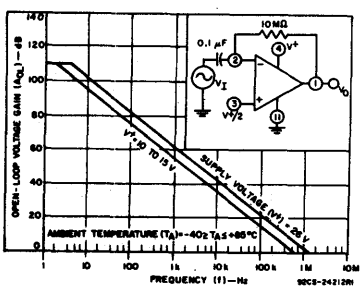


Fig. 9—Open-loop frequency response.

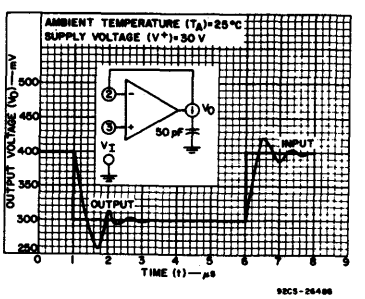


Fig. 10—Voltage follower pulse response (small signal).

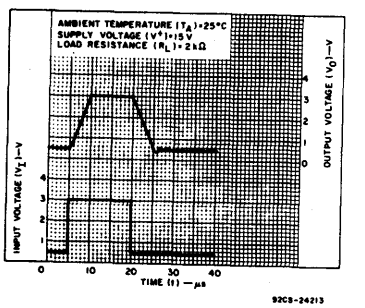


Fig. 11—Voltage follower pulse response.

CA139, CA239, CA339 Types

Quad Voltage Comparators

For Industrial, Commercial, and Military Applications

"E" Suffix Types: Standard Dual-In-Line Plastic Package
"G" Suffix Types: Hermetic Gold-Chip Dual-In-Line Plastic Package

The RCA-CA139, -CA239, -CA339, -CA139A, -CA239A, and -CA339A types consist of four independent single- or dual-supply voltage comparators on a single monolithic substrate. The common-mode input voltage range includes ground even when operated from a single supply, and the low power supply current drain makes these comparators suitable for battery operation. These types were designed to directly interface with TTL and and CMOS.

Types CA139A, CA239A, and CA339A have all the features and characteristics of their prototype counter parts CA139, CA239, and CA339 plus an even lower input-offset-voltage characteristic. These devices are supplied in a 14-lead dual-in-line plastic package (E suffix), or in a 14-lead dual-in-line plastic package with a hermetic chip (G suffix), to provide true hermetic performance. The CA339 is also available in chip form (H suffix), and as a hermetic chip (HG suffix).

Features:

- **Operation from single or dual supplies**
- **Common-mode input-voltage range to ground**
- **Output voltage compatible with TTL, DTL, ECL, MOS, and CMOS**
- **Differential input-voltage range equal to the supply voltage**
- **Maximum input-offset voltage (V_{IO}):**
  CA139A, CA239A, CA339A – 2 mV
  CA139, CA239, CA339 – 5 mV
- **Replacement for industry types 139, 239 339, 139A, 239A, and 339A**

Applications:

- **Square-wave generators**
- **Time-delay generators**
- **Pulse generators**
- **Multivibrators**
- **High-voltage digital logic gates**
- **A/D converters**
- **MOS clock timers**

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C:*

DC SUPPLY VOLTAGE	36 V or ±18 V
DC DIFFERENTIAL INPUT VOLTAGE	±36 V
INPUT VOLTAGE	–0.3 V to +36 V
INPUT CURRENT (V_I < –0.3 V)*	50 mA
OUTPUT SHORT CIRCUIT TO GROUND▲	
(Single Supply)	Continuous
DEVICE DISSIPATION:	
Up to T_A = 55°C	750 mW
Above T_A = 55°C	derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	–55 to +125°C
Storage	–65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

* Inputs must not go more negative than –0.3 V.

▲ Short circuits from the output to V^+ can cause excessive heating and eventual destruction. The maximum output current independent of V^+ is approximately 20 mA.

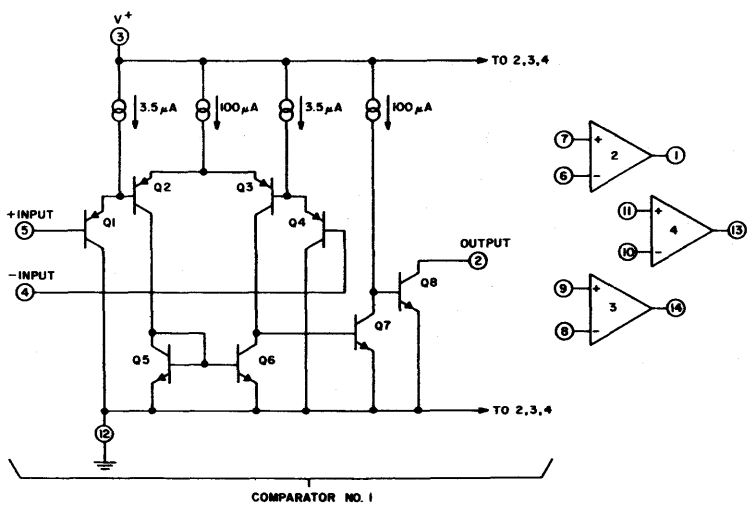


Fig. 1 –Schematic diagram.

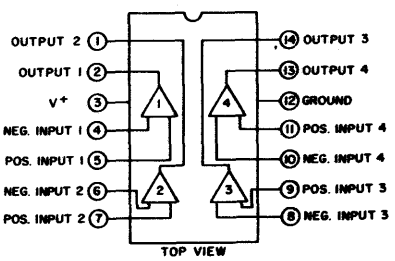


Fig. 2 –Functional diagram.

TYPICAL CHARACTERISTICS

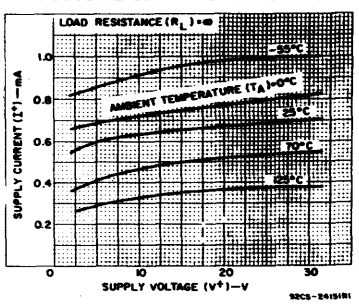


Fig. 3–Supply current vs. supply voltage.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS $V^+ = 5$ V, Unless otherwise indicated		CA139 Min.	CA139 Typ.	CA139 Max.	CA139A Min.	CA139A Typ.	CA139A Max.	UNITS
Input Offset Voltage (V_{IO}) At Output Switch Point $V \cong 1.4$ V	V_{REF} = 1.4 V, R_S = 0	25°C	–	2	5	–	1	2	mV
		Note 1	–	–	9	–	–	4	
Differential Input Voltage (V_{ID})	Keep all inputs ⩾0 V for V^- (If used), Notes 1, 2		–	–	36	–	–	36	V
Saturation Voltage (V_{sat})	V_I^- = 1 V, V_I^+ = 0 V, I_{SINK} ⩽ 4 mA	25°C	–	250	500	–	250	500	mV
		Note 1	–	–	700	–	–	700	
Common-Mode Input Voltage Range (V_{ICR})	Note 3	25°C	0	–	$V^+-1.5$	0	–	$V^+-1.5$	V
		Note 1	0	–	V^+-2	0	–	V^+-2	
Input Offset Current (I_{IO})	$I_I^+ - I_I^-$	25°C	–	3	25	–	3	25	nA
		Note 1	–	–	100	–	–	100	
Input Bias Current (I_{IB})	I_I^+ or I_I^- with Output in Linear Range	25°C	–	25	100	–	25	100	nA
		Note 1	–	–	300	–	–	300	
Supply Current (I^+)	$R_L = \infty$ on all comparators, T_A = 25°C		–	0.8	2	–	0.8	2	mA
Output Leakage Current	$V_I^+ \geqslant 1$ V, V_I^- = 0, V_O = 5 V	25°C	–	0.1	–	–	0.1	–	nA
	$V_I^+ \geqslant 1$ V, V_I^- = 0, V_O = 30 V	Note 1	–	–	1	–	–	1	μA
Output Sink Current	$V_I^- \geqslant 1$ V, V_I^+ = 0, $V_O \leqslant +1.5$ V, T_A = 25°C		6	16	–	6	16	–	mA
Voltage Gain (A_{OL})	$R_L \geqslant 15$ kΩ V^+=15 V, T_A = 25°C		–	200	–	50	200	–	V/mV
Large Signal Response Time	V_I = TTL Logic Swing, V_{REF} = +1.4 V, V_{RL} = 50 V, R_L = 5.1 kΩ, T_A = 25°C		–	300	–	–	300	–	ns
Response Time See Figs. 5 & 6	V_{RL} = 5 V, R_L = 5.1 kΩ, T_A = 25°C		–	1.3	–	–	1.3	–	μs

Note 1: Ambient Temperature (T_A) applicable over operating temperature range as shown below.

CA139, CA139A (−55 to +125°C) | CA239, CA239A (−25 to +85°C) | CA339, CA339A (0 to +70°C)

Note 2: The comparator will provide a proper output state even if the positive swing of the inputs exceeds the power supply voltage level, if the other input remains within the common-mode voltage range. The low input voltage state must not be less than −0.3 V (or 0.3 V below the magnitude of the negative power supply, if used).

Note 3: The upper end of the common-mode voltage range is (V^+) − 1.5 V, but either or both inputs can go to +30 V without damage.

TYPICAL CHARACTERISTICS (Cont'd)

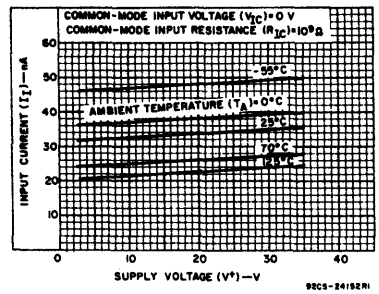


Fig. 4—Input current vs. supply voltage.

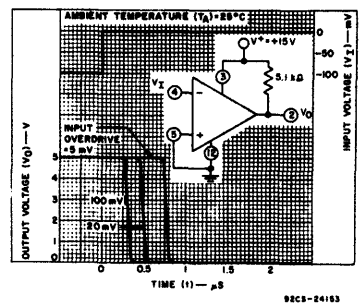


Fig. 5—Response time for various input overdrives—negative transition.

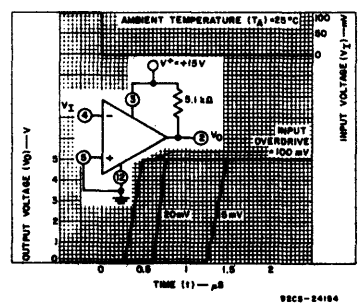


Fig. 6—Response time for various input overdrives—positive transition.

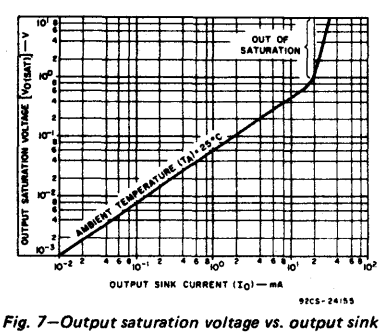


Fig. 7—Output saturation voltage vs. output sink current.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS $V^+ = 5$ V Unless otherwise indicated		LIMITS CA239, CA339 Min.	Typ.	Max.	LIMITS CA239A, CA339A Min.	Typ.	Max.	UNITS
Input Offset Voltage (V_{IO}) At Output Switch Point V ≅ 1.4 V	V_{REF} = 1.4 V, R_S = 0	25°C	–	2	5	–	1	2	mV
		Note 1	–	–	9	–	–	4	
Differential Input Voltage (V_{ID})	Keep all inputs ≥0 V for V^- (If used), Notes 1, 2		–	–	36	–	–	36	V
Saturation Voltage (V_{sat})	V_I^- = 1 V, V_I^+ = 0 V, I_{SINK} ≤ 4 mA	25°C	–	250	500	–	250	500	mV
		Note 1	–	–	700	–	–	700	
Common-Mode Input Voltage Range (V_{ICR})	Note 3	25°C	0	–	V^+–1.5	0	–	V^+–1.5	V
		Note 1	0	–	V^+–2	0	–	V^+–2	
Input Offset Current (I_{IO})	$I_I^+ - I_I^-$	25°C	–	5	50	–	5	50	nA
		Note 1	–	–	150	–	–	150	
Input Bias Current (I_{IB})	I_I^+ or I_I^- with Output in Linear Range	25°C	–	25	250	–	25	250	nA
		Note 1	–	–	400	–	–	400	
Supply Current (I^+)	R_L = ∞ on all comparators, T_A = 25°C		–	0.8	2	–	0.8	2	mA
Output Leakage Current	V_I^+ ≥1 V, V_I^- = 0, V_O = 5 V	25°C	–	0.1	–	–	0.1	–	nA
	V_I^+ ≥1 V, V_I^- = 0, V_O = 30 V	Note 1	–	–	1	–	–	1	μA
Output Sink Current	V_I^- ≥1 V, V_I^+ = 0, V_O ≤+1.5 V, T_A = 25°C		6	16	–	6	16	–	mA
Voltage Gain (A_{OL})	R_L ≥15 kΩ, V^+=15 V, T_A = 25°C		–	200	–	50	200	–	V/mV
Large Signal Response Time	V_I = TTL Logic Swing, V_{REF} = +1.4 V, V_{RL} = 50 V, R_L = 5.1 kΩ, T_A = 25°C		–	300	–	–	300	–	ns
Response Time See Figs. 5 & 6	V_{RL} = 5 V, R_L = 5.1 kΩ, T_A = 25°C		–	1.3	–	–	1.3	–	μs

Note 1: Ambient Temperature (T_A) applicable over operating temperature range as shown below.
CA139 / CA139A (–55 to +125°C) CA239 / CA239A (–25 to +85°C) CA339 / CA339A (0 to +70°C)

Note 2: The comparator will provide a proper output state even if the positive swing of the inputs exceeds the power supply voltage level, if the other input remains within the common-mode voltage range. The low input voltage state must not be less than –0.3 V (or 0.3 V below the magnitude of the negative power supply, if used).

Note 3: The upper end of the common-mode voltage range is (V^+) – 1.5 V, but either or both inputs can go to +30 V without damage.

CA158, CA158A, CA258, CA258A, CA358, CA358A, CA2904 Types

Dual Operational Amplifiers

For Commercial, Industrial, and Military Applications

The RCA-CA158, –CA158A, –CA258, –CA258A, –CA358, –CA358A, and CA-2904 types consist of two independent, high gain, internally frequency compensated operational amplifiers which are designed specifically to operate from a single power supply over a wide range of voltages. They may also be operated from split power supplies. The supply current is basically independent of the supply voltage over the recommended voltage range.

These devices are particularly useful in interface circuits with digital systems and can be operated from the single common 5 Vdc power supply. They are also intended for transducer amplifiers, dc gain blocks and many other conventional op amp circuits which can benefit from the single power supply capability.

The CA158, CA158A, CA258, CA258A, CA358 and CA358A types are supplied in hermetic gold-CHIP 8-lead dual-in-line plastic packages (G suffix), 8-lead TO-5 style packages with standard leads (T suffix), and with dual-in-line formed leads (DIL-CAN, S suffix). The CA2904 is supplied only in the gold-CHIP plastic package (G suffix).

The CA158, CA158A, CA258, CA258A, CA358, CA358A, and CA2904 types are an equivalent to or a replacement for the industry types 158, 158A, 258, 258A, 358, 358A, and 2904.

Features:

- **Internal frequency compensation for unity gain**
- **High dc voltage gain – 100 dB typ.**
- **Wide bandwidth at unity gain – 1 MHz typ.**
- **Wide power supply range:**
  - **Single supply . . . . . 3 to 30 V**
  - **Dual supplies . . . . . ± 1.5 to ± 15 V**
- **Low supply current – 1.5 mA typ.**
- **Low input bias current**
- **Low input offset voltage and current**
- **Input common-mode voltage range includes ground**
- **Differential input voltage range equal to V^+ range**
- **Large output voltage swing – 0 to V^+ –1.5 V**

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

SUPPLY VOLTAGE, V^+:	
CA2904	26 V or ± 13 V
Other Types	32 V or ± 16 V
DIFFERENTIAL INPUT VOLTAGE:	
CA2904	± 26 V
Other Types	± 32 V
INPUT VOLTAGE	–0.3 V to V^+ V
INPUT CURRENT ($V_I < -0.3$ V) +	50 mA
OUTPUT SHORT CIRCUIT TO GROUND ($V^+ \leq 15$ V)*	Continuous
DEVICE DISSIPATION:	
Up to $T_A = 55^oC$	630 mW
Above $T_A = 55^oC$	derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	–55 to + 125°C
Storage	–65 to + 150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+ 300 °C

+ This input current will only exist when the voltage at any of the input leads is driven negative. This current is due to the collector-base junction of the input p-n-p transistors becoming forward biased and thereby acting as input diode clamps. In addition to this diode action, there is also lateral n-p-n parasitic transistor action on the IC chip. This transistor action can cause the output voltages of the amplifiers to go to the V^+ voltage level (or to ground for a large overdrive) for the time duration that an input is driven negative. This transistor action is not destructive and normal output states will re-establish when the input voltage, which was negative, again returns to a value greater than –0.3 V dc.

* The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

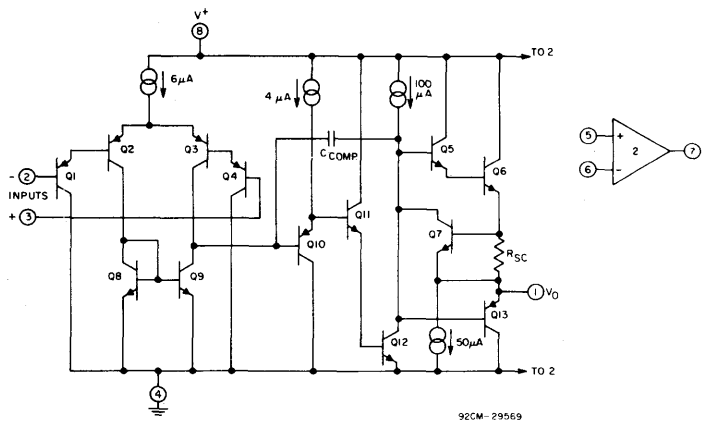


Fig. 1 – Schematic diagram – one of two operational amplifiers.

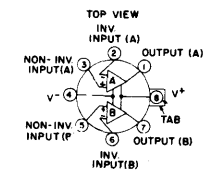


Fig. 2 – Functional diagram for CA158, CA258, and CA358 S- and T-suffix types.

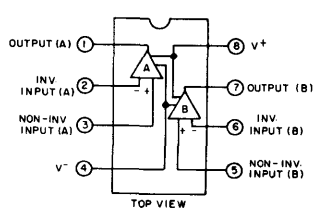


Fig. 3 – Functional diagram for CA158, CA258, CA358, and CA2904 G-suffix types.

ELECTRICAL CHARACTERISTICS (Values Apply For Each Operational Amplifier)

CHARACTERISTIC	TEST CONDITIONS Supply Voltage (V^+) = 5 V Unless Otherwise Specified	LIMITS CA158A (G, T, S) Min.	Typ.	Max.	UNITS
	T_A = 25°C				
Input Offset Voltage, V_{IO}	Note 3	–	1	2	mV
Output Voltage Swing, V_{OPP}	R_L = 2 kΩ	0	–	V^+ –1.5	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, V^+ = 30 V	0	–	V^+ –1.5	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	2	10	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	20	50	nA
Output Current (Source), I_O	V_I^+ = +1 V, V_I^- = 0 V, V^+ = 15 V	20	40	–	mA
Output Current (Sink), I_O	V_I^+ = 0 V, V_I^- = 1 V, V^+=15 V	10	20	–	mA
	V_I^+ = 0 V, V_I^- = 1 V, V_O = 200 mV	12	50	–	μA
Short Circuit Output Current	R_L = 0 (to Ground) Note 4	–	40	60	mA
Large Signal Voltage Gain, A_{OL}	$R_L \geq$ 2 kΩ, V^+ = 15 V (For large V_O swing)	50	100	–	V/mV
Common-Mode Rejection Ratio, CMRR	DC	70	85	–	dB
Power Supply Rejection Ratio, PSRR	DC	65	100	–	dB
Amplifier-to-Amplifier Coupling	f = 1 to 20 kHz (Input referred)	–	–120	–	dB
	T_A = –55 to +125°C				
Input Offset Voltage, V_{IO}	Note 3	–	–	4	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	R_S = 0	–	7	15	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	–	30	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	200	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	40	100	nA
Input Common-Mode Voltage Range, V_{ICR}	V^+ = 30 V, Note 2	0	–	V^+–2	V
Supply Current, I^+	R_L = ∞ On All Ampl.	–	0.7	1.2	mA
	R_L = ∞, V^+ = 30 V	–	1.5	3	

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is V^+ – 1.5 V, but either or both inputs can go the + 32 V without damage.

NOTE 3: V_O = 1.4 V_{DC}, R_S = 0 Ω with V^+ from 5 V to 30 V, and over the full input common-mode voltage range (0 V to V^+ – 1.5 V).

NOTE 4: The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

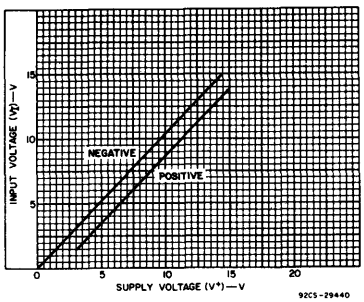


Fig.4 – Input voltage range as a function of supply voltage.

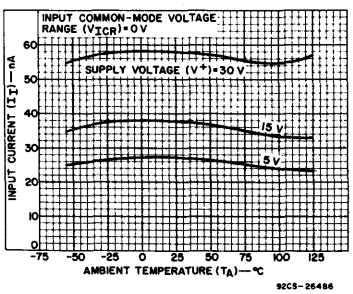


Fig.5 – Input current as a function of ambient temperature.

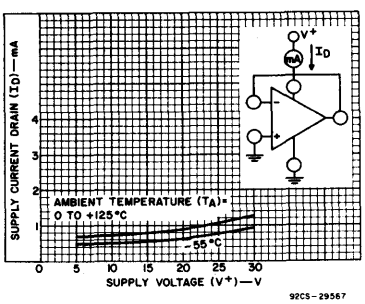


Fig.6 – Supply current drain as a function of supply voltage.

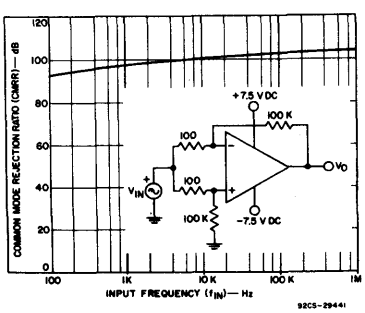


Fig.7 – Common mode rejection ratio as a function of input frequency.

ELECTRICAL CHARACTERISTICS (Values Apply for Each Operational Amplifier)

CHARACTERISTIC	TEST CONDITIONS Supply Voltage (V^+) = 5 V Unless Otherwise Specified	LIMITS CA258A (G, T, S) Min.	Typ.	Max.	UNITS
	T_A = 25°C				
Input Offset Voltage, V_{IO}	Note 3	–	1	3	mV
Output Voltage Swing, V_{OPP}	R_L = 2 kΩ	0	–	V^+ –1.5	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, V^+ = 30 V	0	–	V^+ –1.5	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	2	15	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	40	80	nA
Output Current (Source), I_O	V_I^+ = +1 V, V_I^- = 0 V, V^+ = 15 V	20	40	–	mA
Output Current (Sink), I_O	V_I^+ = 0 V, V_I^- = 1 V, V^+=15 V	10	20	–	mA
	V_I^+ = 0 V, V_I^- = 1 V, V_O = 200 mV	12	50	–	μA
Short Circuit Output Current	R_L = 0 (to Ground) Note 4	–	40	60	mA
Large Signal Voltage Gain, A_{OL}	$R_L \geq$ 2 kΩ, V^+ = 15 V (For large V_O swing)	50	100	–	V/mV
Common-Mode Rejection Ratio, CMRR	DC	70	85	–	dB
Power Supply Rejection Ratio, PSRR	DC	65	100	–	dB
Amplifier-to-Amplifier Coupling	f = 1 to 20 kHz (Input referred)	–	–120	–	dB
	T_A = –25 to +85°C				
Input Offset Voltage, V_{IO}	Note 3	–	–	4	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	R_S = 0	–	7	15	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	–	30	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	200	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	40	100	nA
Input Common-Mode Voltage Range, V_{ICR}	V^+ = 30 V, Note 2	0	–	V^+–2	V
Supply Current, I^+	R_L = ∞ On All Ampl.	–	0.7	1.2	mA
	R_L = ∞, V^+ = 30 V	–	1.5	3	

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is V^+ – 1.5 V, but either or both inputs can go the + 32 V without damage.

NOTE 3: V_O = 1.4 V_{DC}, R_S = 0 Ω with V^+ from 5 V to 30 V, and over the full input common-mode voltage range (0 V to V^+ – 1.5 V).

NOTE 4: The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

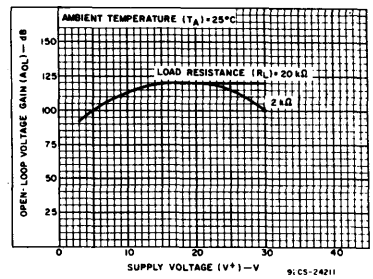


Fig.8 – Voltage gain as a function of supply voltage.

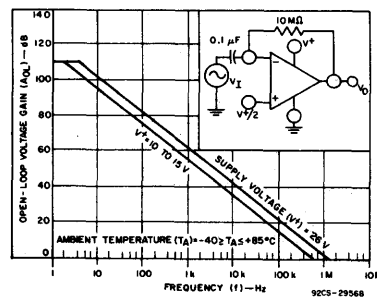


Fig.9 – Open-loop frequency response.

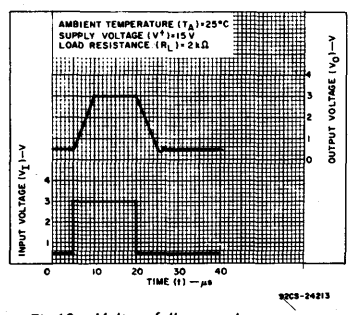


Fig.10 – Voltage follower pulse response.

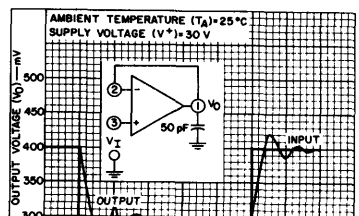


Fig.11 – Voltage follower pulse response (small signal).

ELECTRICAL CHARACTERISTICS (Values Apply for Each Operational Amplifier)

CHARACTERISTIC	TEST CONDITIONS Supply Voltage (V^+) = 5 V Unless Otherwise Specified	LIMITS CA358A (G, T, S) Min.	Typ.	Max.	UNITS
T_A = 25°C					
Input Offset Voltage, V_{IO}	Note 3	–	2	3	mV
Output Voltage Swing, V_{OPP}	R_L = 2 kΩ	0	–	V^+ –1.5	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, V^+ = 30 V	0	–	V^+ –1.5	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	5	30	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	45	100	nA
Output Current (Source), I_O	V_I^+ = +1 V, V_I^- = 0 V, V^+ = 15 V	20	40	–	mA
Output Current (Sink), I_O	V_I^+ = 0 V, V_I^- = 1 V, V^+=15 V	10	20	–	mA
	V_I^+ = 0 V, V_I^- = 1 V, V_O = 200 mV	12	50	–	μA
Short Circuit Output Current	R_L = 0 (to Ground) Note 4	–	40	60	mA
Large Signal Voltage Gain, A_{OL}	$R_L \geq$ 2 kΩ, V^+ = 15 V (For large V_O swing)	25	100	–	V/mV
Common-Mode Rejection Ratio, CMRR	DC	65	85	–	dB
Power Supply Rejection Ratio, PSRR	DC	65	100	–	dB
Amplifier-to-Amplifier Coupling	f = 1 to 20 kHz (Input referred)	–	–120	–	dB
T_A = 0 to +70°C					
Input Offset Voltage, V_{IO}	Note 3	–	–	5	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	R_s = 0	–	7	20	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	–	75	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	300	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	40	200	nA
Input Common-Mode Voltage Range, V_{ICR}	V^+ = 30 V, Note 2	0	–	V^+–2	V
Supply Current, I^+	R_L = ∞ On All Ampl.	–	0.7	1.2	mA
	R_L = ∞, V^+ = 30 V	–	1.5	3	

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is V^+ – 1.5 V, but either or both inputs can go the + 32 V without damage.

NOTE 3: V_O = 1.4 V_{DC}, R_s = 0 Ω with V^+ from 5 V to 30 V, and over the full input common-mode voltage range (0 V to V^+ – 1.5 V).

NOTE 4: The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

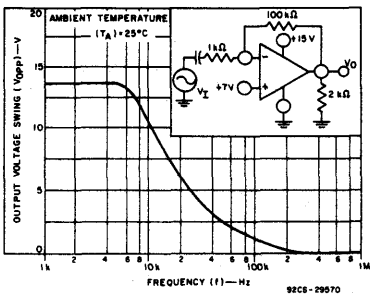


Fig. 12 – Large-signal frequency response.

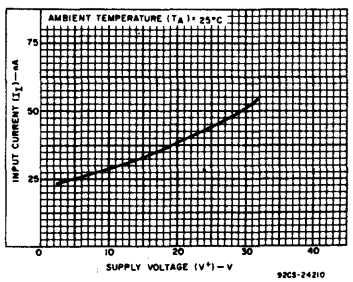


Fig. 13 – Input current as a function of supply voltage.

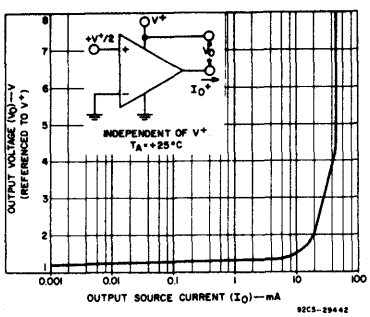


Fig. 14 – Output source current characteristics.

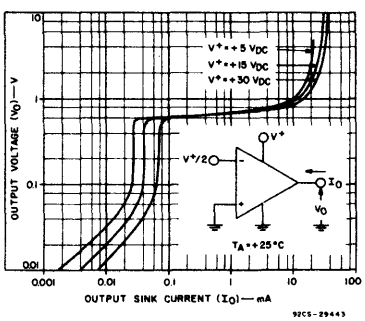


Fig. 15 – Output sink current characteristics.

ELECTRICAL CHARACTERISTICS (Values Apply for Each Operational Amplifier)

CHARACTERISTIC	TEST CONDITIONS Supply Voltage (V^+) = 5 V Unless Otherwise Specified	LIMITS CA158 (G, T, S) CA258 (G, T, S) Min.	Typ.	Max.	UNITS
	T_A = 25°C				
Input Offset Voltage, V_{IO}	Note 3	–	2	5	mV
Output Voltage Swing, V_{OPP}	R_L = 2 kΩ	0	–	V^+ –1.5	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, V^+ = 30 V	0	–	V^+ –1.5	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	3	30	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	45	150	nA
Output Current (Source), I_O	V_I^+ = +1 V, V_I^- = 0 V, V^+ = 15 V	20	40	–	mA
Output Current (Sink), I_O	V_I^+ = 0 V, V_I^- = 1 V, V^+=15 V	10	20	–	mA
	V_I^+ = 0 V, V_I^- = 1 V, V_O = 200 mV	12	50	–	μA
Short Circuit Output Current	R_L = 0 (to Ground) Note 4	–	40	60	mA
Large Signal Voltage Gain, A_{OL}	$R_L \geq$ 2 kΩ, V^+ = 15 V (For large V_O swing)	50	100	–	V/mV
Common-Mode Rejection Ratio, CMRR	DC	70	85	–	dB
Power Supply Rejection Ratio, PSRR	DC	65	100	–	dB
Amplifier-to-Amplifier Coupling	f = 1 to 20 kHz (Input referred)	–	–120	–	dB
	T_A = –55 to + 125°C (CA158); T_A = –25 to +85°C (CA258)				
Input Offset Voltage, V_{IO}	Note 3	–	–	7	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	R_S = 0	–	7	–	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	–	100	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	–	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	40	300	nA
Input Common-Mode Voltage Range, V_{ICR}	V^+ = 30 V, Note 2	0	–	V^+–2	V
Supply Current, I^+	R_L = ∞ On All Ampl.	–	0.7	1.2	mA
	R_L = ∞, V^+ = 30 V	–	1.5	3	

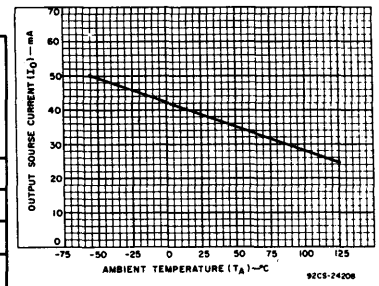


Fig.16 – Output current as a function of ambient temperature.

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is V^+ – 1.5 V, but either or both inputs can go the + 32 V without damage.

NOTE 3: V_O = 1.4 V_{DC}, R_s = 0 Ω with V^+ from 5 V to 30 V, and over the full input common-mode voltage range (0 V to V^+ – 1.5 V).

NOTE 4: The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

ELECTRICAL CHARACTERISTICS (Values Apply for Each Operational Amplifier)

CHARACTERISTIC	TEST CONDITIONS	LIMITS CA358 (G, T, S)			UNITS
	Supply Voltage (V^+) = 5 V Unless Otherwise Specified	Min.	Typ.	Max.	
	T_A = 25°C				
Input Offset Voltage, V_{IO}	Note 3	–	2	7	mV
Output Voltage Swing, V_{OPP}	R_L = 2 kΩ	0	–	V^+ –1.5	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, V^+ = 30 V	0	–	V^+ –1.5	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	5	50	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	45	250	nA
Output Current (Source), I_O	V_I^+ = +1 V, V_I^- = 0 V, V^+ = 15 V	20	40	–	mA
Output Current (Sink), I_O	V_I^+ = 0 V, V_I^- = 1 V, V^+=15 V	10	20	–	mA
	V_I^+ = 0 V, V_I^- = 1 V, V_O = 200 mV	12	50	–	μA
Short Circuit Output Current	R_L = 0 (to Ground) Note 4	–	40	60	mA
Large Signal Voltage Gain, A_{OL}	$R_L \geq$ 2 kΩ, V^+ = 15 V (For large V_O swing)	25	100	–	V/mV
Common-Mode Rejection Ratio, CMRR	DC	65	70	–	dB
Power Supply Rejection Ratio, PSRR	DC	65	100	–	dB
Amplifier-to-Amplifier Coupling	f = 1 to 20 kHz (Input referred)	–	–120	–	dB
	T_A = 0 to +70°C				
Input Offset Voltage, V_{IO}	Note 3	–	–	9	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	R_S = 0	–	7	–	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	–	150	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	–	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	40	500	nA
Input Common-Mode Voltage Range, V_{ICR}	V^+ = 30 V, Note 2	0	–	V^+–2	V
Supply Current, I^+	R_L = ∞ On All Ampl.	–	0.7	1.2	mA
	R_L = ∞, V^+ = 30 V	–	1.5	3	

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is $V^+ - 1.5$ V, but either or both inputs can go the + 32 V without damage.

NOTE 3: V_O = 1.4 V_{DC}, R_s = 0 Ω with V^+ from 5 V to 30 V, and over the full input common-mode voltage range (0 V to $V^+ - 1.5$ V).

NOTE 4: The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of thedevice. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

ELECTRICAL CHARACTERISTICS (Values Apply for Each Operational Amplifier)

CHARACTERISTIC	TEST CONDITIONS	LIMITS CA2904G			UNITS
	Supply Voltage (V^+) = 5 V Unless Otherwise Specified	Min.	Typ.	Max.	
	T_A = 25°C				
Input Offset Voltage, V_{IO}	Note 3	–	2	7	mV
Output Voltage Swing, V_{OPP}	$R_L \geq 10\ k\Omega$	0	–	$V^+ -1.5$	V
Input Common-Mode Voltage Range, V_{ICR}	Note 2, $V^+ = 30$ V	0	–	$V^+ -1.5$	V
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	5	50	nA
Input Bias Current, I_{IB}	I_I^+ or I_I^-, Note 1	–	45	250	nA
Output Current (Source), I_O	$V_I^+ = +1$ V, $V_I^- = 0$ V, $V^+ = 15$ V	20	40	–	mA
Output Current (Sink), I_O	$V_I^+ = 0$ V, $V_I^- = 1$ V, $V^+ = 15$ V	10	20	–	mA
Short Circuit Output Current	$R_L = 0$ (to Ground) Note 4	–	40	60	mA
Large Signal Voltage Gain, A_{OL}	$R_L \geq 2\ k\Omega$, $V^+ = 15$ V (For large V_O swing)	–	100	–	V/mV
Common-Mode Rejection Ratio, CMRR	DC	50	70	–	dB
Power Supply Rejection Ratio, PSRR	DC	50	100	–	dB
Amplifier-to-Amplifier Coupling	f = 1 to 20 kHz (Input referred)	–	–120	–	dB
	T_A = –40 to + 85°C				
Input Offset Voltage, V_{IO}	Note 3	–	–	10	mV
Temperature Coefficient of Input Offset Voltage, αV_{IO}	$R_s = 0$	–	7	–	μV/°C
Input Offset Current, I_{IO}	$I_I^+ - I_I^-$	–	45	200	nA
Temperature Coefficient of Input Offset Current, αI_{IO}		–	10	–	pA/°C
Input Bias Current, I_{IB}	I_I^+ or I_I^-	–	40	500	nA
Input Common-Mode Voltage Range, V_{ICR}	$V^+ = 30$ V, Note 2	0	–	$V^+ -2$	V
Supply Current, I^+	$R_L = \infty$ On All Ampl.	–	0.7	1.2	mA
	$R_L = \infty$, $V^+ = 30$ V	–	1.5	3	

NOTE 1: Due to the p-n-p input stage the direction of the input current is out of the IC. No loading change exists on the input lines because this current is essentially constant, independent of the state of the output.

NOTE 2: The input signal voltages and the input common-mode voltage should not be allowed to go negative by more than 0.3 V. The positive limit of the common-mode voltage range is $V^+ - 1.5$ V, but either or both inputs can go the + 32 V without damage.

NOTE 3: $V_O = 1.4\ V_{DC}$, $R_s = 0\ \Omega$ with V^+ from 5 V to 30 V, and over the full input common-mode voltage range (0 V to $V^+ - 1.5$ V).

NOTE 4: The maximum output current is approximately 40 mA independent of the magnitude of V^+. Continuous short circuits at $V^+ > 15$ V can cause excessive power dissipation and eventual destruction. Short circuits from the output to V^+ can cause overheating and eventual destruction of the device. Destructive dissipation can result from simultaneous short circuits on both amplifiers.

CA555, CA555C Types

Timers

For Timing Delays & Oscillator Applications in Commercial, Industrial, and Military Equipment

The RCA-CA555 and CA555C are highly stable timers for use in precision timing and oscillator applications. As timers, these monolithic integrated circuits are capable of producing accurate time delays for periods ranging from microseconds through hours. These devices are also useful for astable oscillator operation and can maintain an accurately controlled free-running frequency and duty cycle with only two external resistors and one capacitor.

The circuits of the CA555 and CA555C may be triggered by the falling edge of the waveform signal, and the output of these circuits can source or sink up to a 200-milliampere current or drive TTL circuits.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE	18	V
DEVICE DISSIPATION:		
Up to T_A = 55°C	600	mW
Above T_A = 55°C Derate linearly	5	mW/°C
AMBIENT TEMPERATURE RANGE (All Types):		
Operating		
CA555	−55 to +125	°C
CA555C	0 to 70	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (During Soldering):		
At distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.	+265	°C

The CA555 and CA555C are supplied in hermetic IC Gold-CHIP 8-lead dual-in-line plastic packages (G Suffix), standard 8-lead TO-5 style packages (T suffix), 8-lead TO-5 style packages with dual-in-line formed leads (DIL-CAN, S suffix), 8-lead dual-in-line plastic packages (MINI-DIP, E suffix), and in chip form (H suffix). These types are direct replacements for industry types in packages with similar terminal arrangements e.g.SE555 and NE555, MC1555 and MC1455, respectively. The CA555 type circuits are intended for applications requiring premium electrical performance. The CA555C type circuits are intended for applications requiring less stringent electrical characteristics.

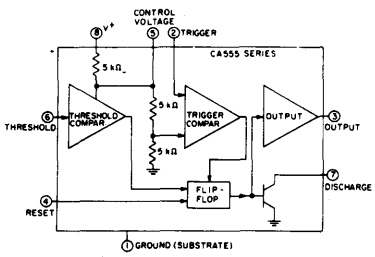


Fig. 1 — Functional diagram of the CA555 series.

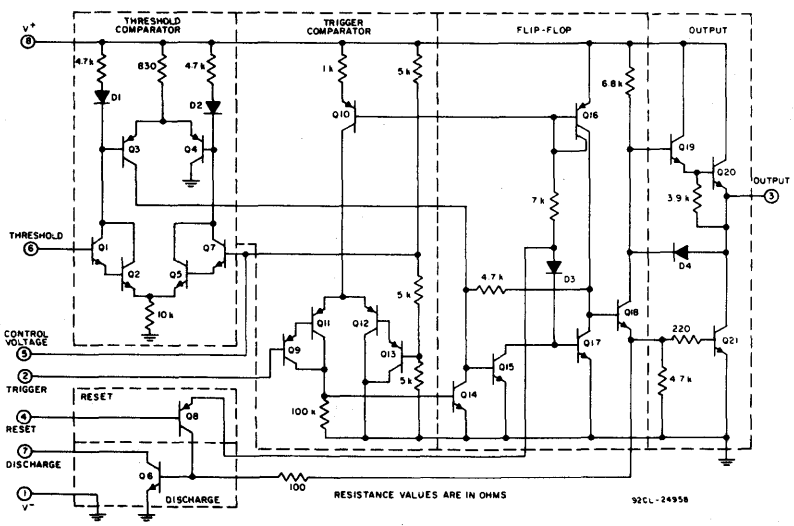


Fig. 2 — Schematic diagram of the CA555 and CA555C.

CA555G, CA555CG:

Hermetic Gold-CHIP 8-Lead Dual-In-Line Plastic Package (MINI-DIP)

CA555T, CA555CT:

Standard 8-Lead TO-5 Style Package

CA555S, CA555CS:

Standard 8-Lead TO-5 Style Package With Formed Leads (DIL-CAN)

CA555E, CA555CE:

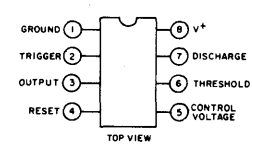
8-Lead Dual-In-Line Plastic Package (MINI-DIP)

Features:

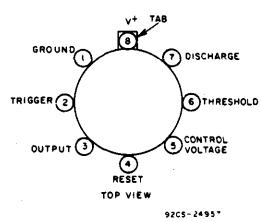
- **Accurate timing from microseconds through hours**
- **Astable and monostable operation**
- **Adjustable duty cycle**
- **Output capable of sourcing or sinking up to 200 mA**
- **Output capable of driving TTL devices**
- **Normally ON and OFF outputs**
- **High-temperature stability —0.005%/°C**
- **Directly interchangeable with SE555, NE555, MC1555, and MC1455**

Applications:

- **Precision timing**
- **Sequential timing**
- **Time-delay generation**
- **Pulse generation**
- **Pulse-width and position modulation**
- **Pulse detector**



a. MINI-DIP plastic package TO-5 style package with formed leads



b. TO-5 style package

Fig. 3 — Terminal assignment diagrams.

ELECTRICAL CHARACTERISTICS, *At T_A = 25°C, V^+ = 5 to 15 V unless otherwise specified*

CHARACTERISTIC	TEST CONDITIONS	LIMITS						UNITS
		CA555			CA555C			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
DC Supply Voltage, V^+		4.5	–	18	4.5	–	16	V
DC Supply Current (Low State)*, I^+	V^+ = 5 V, R_L = ∞	–	3	5	–	3	6	mA
	V^+ = 15 V, R_L = ∞	–	10	12	–	10	15	mA
Threshold Voltage, V_{TH}		–	(2/3)V^+	–	–	(2/3)V^+	–	V
Trigger Voltage	V^+ = 5 V	1.45	1.67	1.9	–	1.67	–	V
	V^+ = 15 V	4.8	5	5.2	–	5	–	
Trigger Current		–	0.5	–	–	0.5	–	μA
Threshold Current▲, I_{TH}		–	0.1	0.25	–	0.1	0.25	μA
Reset Voltage		0.4	0.7	1.0	0.4	0.7	1.0	V
Reset Current		–	0.1	–	–	0.1	–	mA
Control Voltage Level	V^+ = 5 V	2.9	3.33	3.8	2.6	3.33	4	V
	V^+ = 15 V	9.6	10	10.4	9	10	11	V
Output Voltage Drop: Low State, V_{OL}	V^+ = 5 V, I_{SINK} = 5 mA	–	–	–	–	0.25	0.35	V
	I_{SINK} = 8 mA	–	0.1	0.25	–	–	–	
	V^+ = 15 V, I_{SINK} = 10 mA	–	0.1	0.15	–	0.1	0.25	
	I_{SINK} = 50 mA	–	0.4	0.5	–	0.4	0.75	V
	I_{SINK} = 100 mA	–	2.0	2.2	–	2.0	2.5	
	I_{SINK} = 200 mA	–	2.5	–	–	2.5	–	
High State, V_{OH}	V^+ = 5 V, I_{SOURCE} = 100 mA	3.0	3.3	–	2.75	3.3	–	V
	V^+ = 15 V, I_{SOURCE} = 100 mA	13.0	13.3	–	12.75	13.3	–	
	I_{SOURCE} = 200 mA	–	12.5	–	–	12.5	–	
Timing Error (Monostable): Initial Accuracy	R_1, R_2 = 1 to 100 kΩ, C = 0.1 μF, Tested at V^+ = 5 V, V^+ = 15 V	–	0.5	2	–	1	–	%
Frequency Drift with Temperature		–	30	100	–	50	–	p/m/°C
Drift with Supply Voltage		–	0.05	0.2	–	0.1	–	%/V
Output Rise Time, t_r		–	100	–	–	100	–	ns
Output Fall Time, t_f		–	100	–	–	100	–	ns

* When the output is in a high state, the dc supply current is typically 1 mA less than the low-state value.

▲ The threshold current will determine the sum of the values of R_1 and R_2 to be used in Fig. 16 (astable operation): the maximum total $R_1 + R_2$ = 20 MΩ.

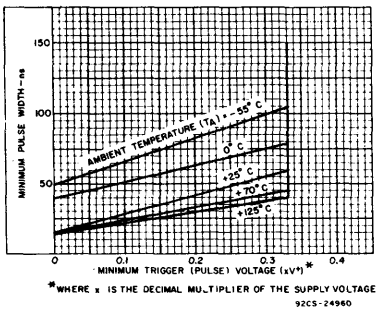


Fig. 4 – Minimum pulse width vs. minimum trigger voltage.

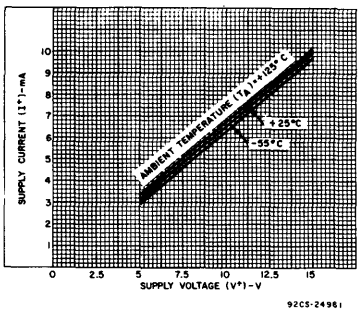


Fig. 5 – Supply current vs. supply voltage.

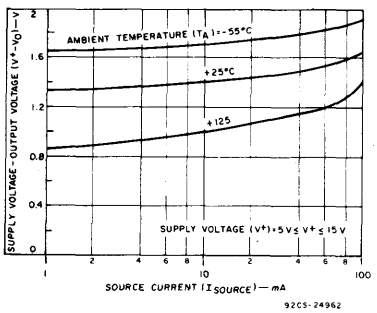


Fig. 6 – Output voltage drop (high state) vs. source current.

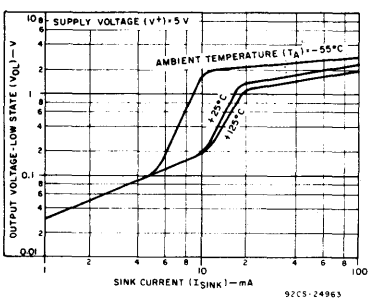


Fig.7 – Output voltage-low state vs. sink current at V^+ = 5 V.

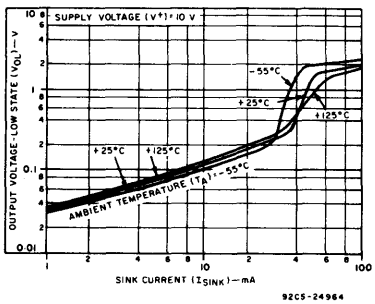


Fig.8 – Output voltage-low state vs. sink current at V^+ = 10 V.

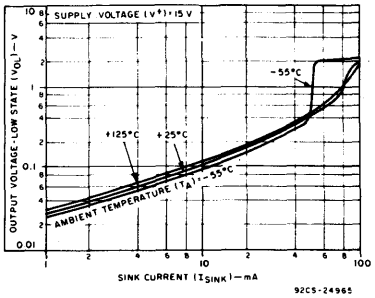


Fig.9 – Output voltage-low state vs. sink current at V^+ = 15 V.

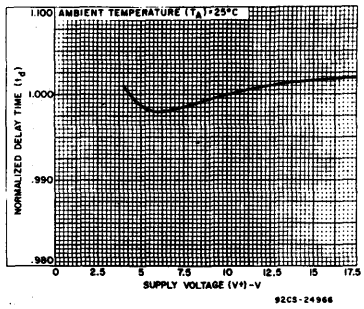


Fig.10 – Delay time vs. supply voltage.

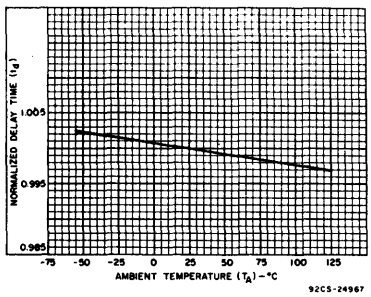


Fig.11 – Delay time vs. temperature.

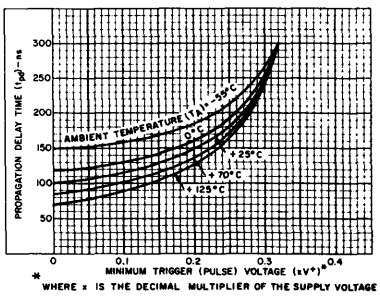


Fig.12 – Propagation delay time vs. trigger voltage.

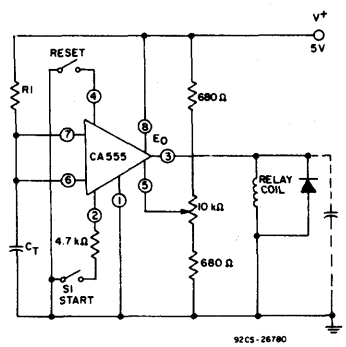


Fig.13 – Reset timer (monostable operation).

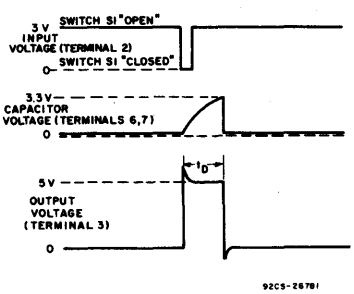


Fig.14 – Typical waveforms for reset timer.

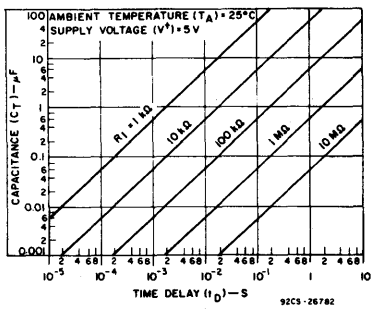


Fig.15 – Time delay vs. resistance and capacitance.

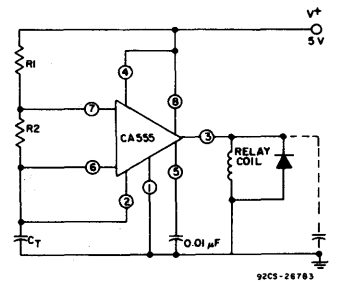


Fig.16 – Repeat cycle timer (astable operation).

TYPICAL APPLICATIONS

Reset Timer (Monostable Operation)

Fig.13 shows the CA555 connected as a reset timer. In this mode of operation capacitor C_T is initially held discharged by a transistor on the integrated circuit. Upon closing the "start" switch, or applying a negative trigger pulse to terminal 2, the integral timer flip-flop is "set" and releases the short circuit across C_T which drives the output voltage "high" (relay energized). The action allows the voltage across the capacitor to increase exponentially with the time constant $t = R_1C_T$. When the voltage across the capacitor equals 2/3 V^+, the comparator resets the flip-flop which in turn discharges the capacitor rapidly and drives the output to its low state.

Since the charge rate and threshold level of the comparator are both directly proportional to V^+, the timing interval is relatively independent of supply voltage variations. Typically, the timing varies only 0.05% for a 1 volt change in V^+.

Applying a negative pulse simultaneously to the reset terminal (4) and the trigger terminal (2) during the timing cycle discharges C_T and causes the timing cycle to restart. Momentarily closing only the reset switch during the timing interval discharges C_T, but the timing cycle does not restart.

Fig.14 shows the typical waveforms generated during this mode of operation, and Fig.15 gives the family of time delay curves with variations in R_1 and C_T.

Repeat Cycle Timer (Astable Operation)

Fig.16 shows the CA555 connected as a repeat cycle timer. In this mode of operation, the total period is a function of both R_1 and R_2:

$$T = 0.693(R_1 + 2R_2)C_T = t_1 + t_2$$

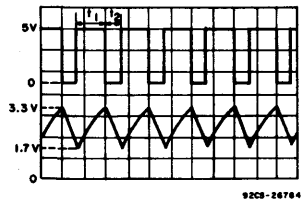
where $t_1 = 0.693(R_1 + R_2)\,C_T$

and $t_2 = 0.693(R_2)C_T$

The duty cycle is:

$$\frac{t_2}{t_1 + t_2} = \frac{R_2}{R_1 + 2R_2}$$

Typical waveforms generated during this mode of operation are shown in Fig. 17. Fig. 18 gives the family of curves of free running frequency with variations in the value of $(R_1 + 2R_2)$ and C_T.



Top Trace: Output voltage (2V/div. and 0.5 ms/div.)
Bottom Trace: Capacitor voltage (1 V/ div. and 0.5 ms/div.)

Fig. 17 – Typical waveforms for repeat cycle timer.

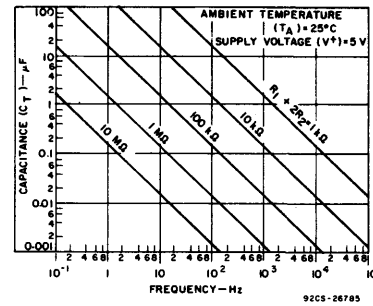


Fig. 18 – Free running frequency of repeat cycle timer with variation in capacitance and resistance.

CA723 Types

Voltage Regulators

For Regulated Output Voltages Adjustable from 2 V to 37 V at Output Currents up to 150 mA Without External Pass Transistors

RCA-CA723 and CA723C are silicon monolithic integrated circuits designed for service as voltage regulators at output voltages ranging from 2 to 37 volts at currents up to 150 milliamperes.

Each type includes a temperature-compensated reference amplifier, an error amplifier, a power series pass transistor, and a current-limiting circuit. They also provide independently accessible inputs for adjustable current limiting and remote shutdown and, in addition, feature low standby current drain, low temperature drift, and high ripple rejection.

The CA723 and CA723C may be used with positive and negative power supplies in a wide variety of series, shunt, switching, and floating regulator applications. They can provide regulation at load currents greater than 150 milliamperes and in excess of 10 amperes with the use of suitable n-p-n or p-n-p external pass transistors.

The CA723 and CA723C are supplied in the 10-lead TO-5-style ceramic package (T suffix), and the 14-lead dual-in-line plastic package (E suffix), and are direct replacements for industry types 723, 723C, μA723, and μA723C in packages with similar terminal arrangements. They are also available in chip form ("H" suffix).

All types are rated for operation over the full military-temperature range of −55°C to +125°C.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE (Between V^+ and V^- Terminals)	40	V
PULSE VOLTAGE FOR 50-ms PULSE WIDTH (Between V^+ and V^- Terminals)	50	V
DIFFERENTIAL INPUT–OUTPUT VOLTAGE	40	V
DIFFERENTIAL INPUT VOLTAGE: Between Inverting and Non-Inverting Inputs	±5	V
Between Non-Inverting Input and V^-	8	V
CURRENT FROM ZENER DIODE TERMINAL (V_Z)	25	mA
CURRENT FROM VOLTAGE REFERENCE TERMINAL (V_{REF})	15	mA

DEVICE DISSIPATION:		
Up to T_A = 25°C –		
CA723T, CA723CT	800	mW
CA723E, CA723CE	1000	mW
Above T_A = 25°C –		
CA723T, CA723CT Derate linearly	6.3	mW/°C
CA723E, CA723CE Derate linearly	8.3	mW/°C
AMBIENT TEMPERATURE RANGE (All Types):		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (During Soldering): At a distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.	+265	°C

Features:

- **Up to 150 mA output current**
- **Positive and negative voltage regulation**
- **Regulation in excess of 10A with suitable pass transistors**
- **Input and output short-circuit protection**
- **Load and line regulation: 0.03%**
- **Direct replacement for 723 and 723C industry types**
- **Adjustable output voltage: 2 to 37 V**

Applications:

- **Series and shunt voltage regulator**
- **Floating regulator**
- **Switching voltage regulator**
- **High-current voltage regulator**
- **Temperature controller**

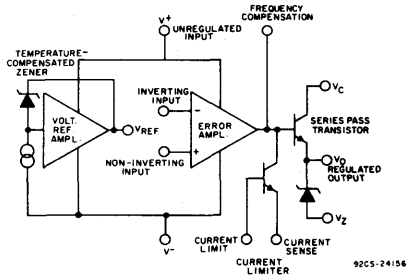


Fig. 1 – Functional diagram of the CA723 and CA723C.

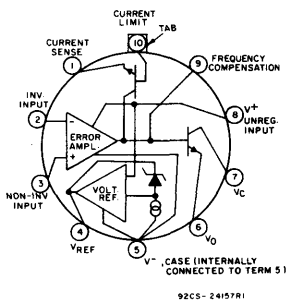


Fig. 2 – Terminal arrangement of the CA723T and CA723CT in the TO-5 style package.

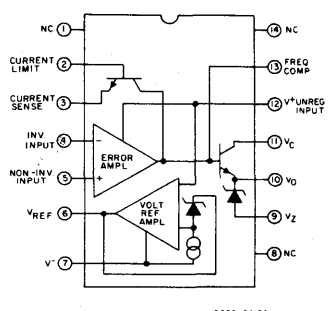


Fig. 3 – Terminal arrangement of the CA723E and CA723CE in the dual-in-line plastic package.

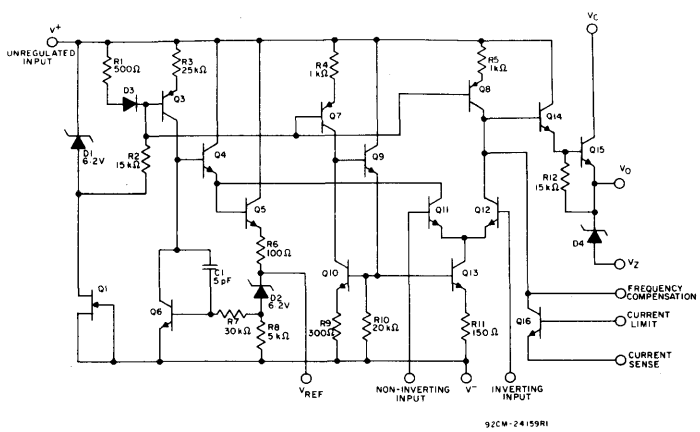


Fig. 4 – Equivalent schematic diagram of the CA723 and CA723C.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, $V^+ = V_C = V_I$ = 12 V, V^- = 0, V_O = 5 V, I_L = 1 mA, C_1 = 100 pF, C_{REF} = 0, R_{SCP} = 0, unless otherwise specified. Divider impedance R_1R_2 R_1+R_2 at non-inverting input, Term. 5, = 10 kΩ (see Fig. 23).

CHARACTERISTIC	TEST CONDITIONS	LIMITS						UNITS
		CA723			CA723C			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
Quiescent Regulator Current, I_Q	I_L = 0, V_I = 30 V	–	2.3	3.5	–	2.3	4	mA
Input Voltage Range, V_I		9.5	–	40	9.5	–	40	V
Output Voltage Range, V_O		2	–	37	2	–	37	V
Differential Input-Output Voltage, V_I-V_O		3	–	38	3	–	38	V
Reference Voltage, V_{REF}		6.95	7.15	7.35	6.8	7.15	7.5	V
Line Regulation (See Note 1)	V_I = 12 to 40 V	–	0.02	0.2	–	0.1	0.5	%V_O
	V_I = 12 to 15 V	–	0.01	0.1	–	0.01	0.1	
	V_I = 12 to 15 V, T_A = −55 to +125°C	–	–	0.3	–	–	–	
	V_I = 12 to 15 V, T_A = 0 to 70°C	–	–	–	–	–	0.3	
Load Regulation (See Note 1)	I_L = 1 to 50 mA	–	0.03	0.15	–	0.03	0.2	%V_O
	I_L = 1 to 50 mA, T_A = −55 to +125°C	–	–	0.6	–	–	–	
	I_L = 1 to 50 mA, T_A = 0 to 70°C	–	–	–	–	–	0.6	
Output-Voltage Temp. Coefficient, $\wedge V_O$	T_A = −55 to +125°C	–	0.002	0.015	–	–	–	%/°C
	T_A = 0 to 70°C	–	–	–	–	0.003	0.015	
Ripple Rejection (See Note 2)	f = 50 Hz to 10 kHz	–	74	–	–	74	–	dB
	f = 50 Hz to 10 kHz, C_{REF} = 5 μF	–	86	–	–	86	–	
Short-Circuit Limiting Current, I_{LIM}	R_{SCP} = 10 Ω, V_O = 0	–	65	–	–	65	–	mA
Equivalent Noise RMS Output Voltage, V_N (See Note 2)	BW = 100 Hz to 10 kHz, C_{REF} = 0	–	20	–	–	20	–	μV
	BW = 100 Hz 10 kHz, C_{REF} = 5 μF	–	2.5	–	–	2.5	–	

Note 1: Line and load regulation specifications are given for condition of a constant chip temperature. For high-dissipation conditions, temperature drifts must be separately taken into account.

Note 2: For C_{REF}, see Fig. 23.

TYPICAL CHARACTERISTICS CURVES FOR TYPE CA723

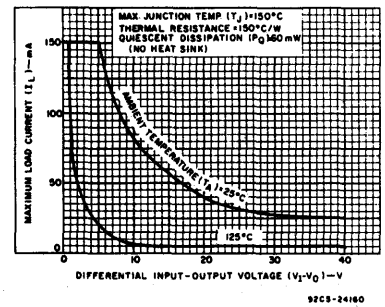


Fig. 5 – Max. load current vs differential input-output voltage.

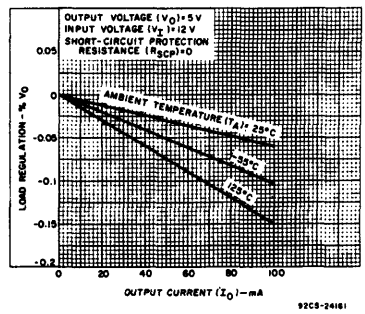


Fig. 6 – Load regulation without current limiting.

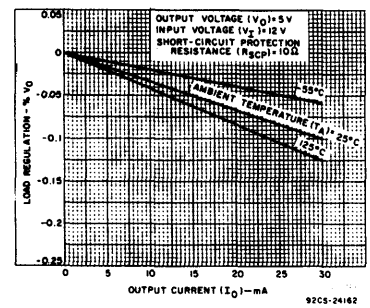


Fig. 7 – Load regulation with current limiting.

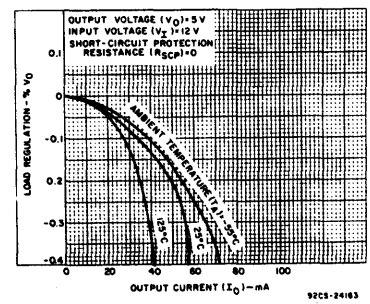


Fig. 8 – Load regulation with current limiting.

TYPICAL CHARACTERISTICS CURVES FOR TYPE CA723 (Cont'd)

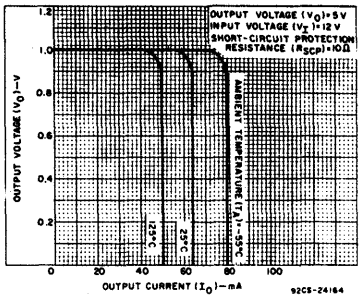


Fig. 9 – Current limiting characteristics.

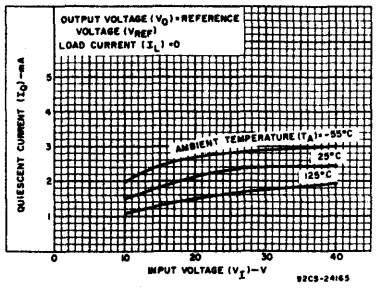


Fig. 10 – Quiescent current vs. input voltage.

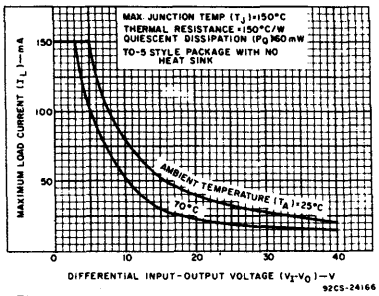


Fig. 11 – Max. load current vs differential input-output voltage CA723CT.

TYPICAL CHARACTERISTICS CURVES FOR TYPE CA723C

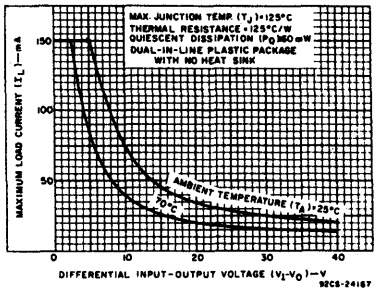


Fig. 12 – Max. load current vs differential input-output voltage for CA723CE.

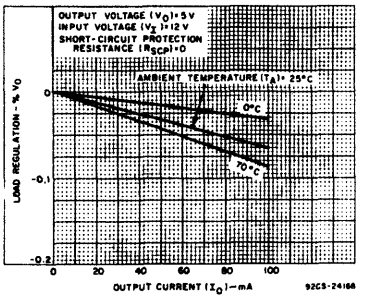


Fig. 13 – Load regulation without current limiting.

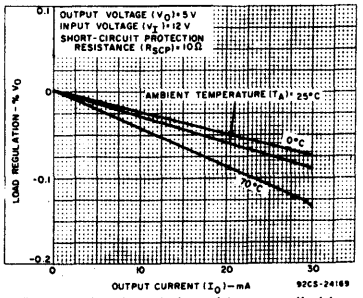


Fig. 14 – Load regulation with current limiting.

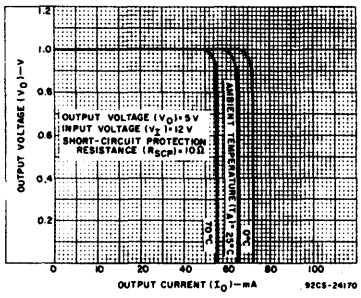


Fig. 15 – Current limiting characteristics.

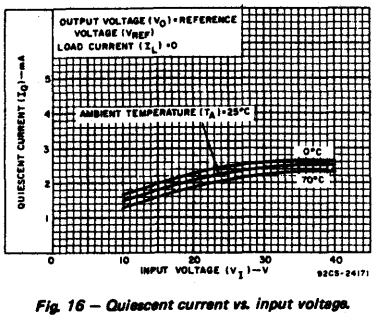


Fig. 16 – Quiescent current vs. input voltage.

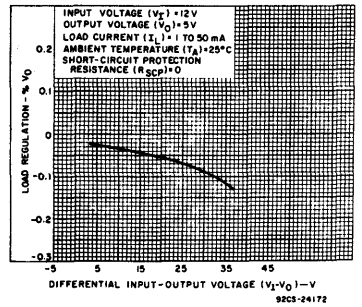


Fig. 17 – Load regulation vs. differential input-output voltage.

TYPICAL CHARACTERISTICS CURVES FOR TYPES CA723 AND CA723C

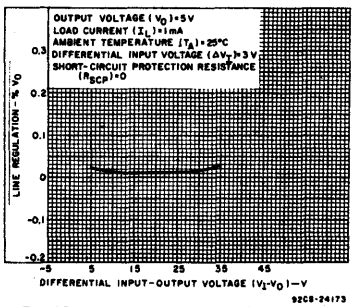


Fig. 18 – Line regulation vs. differential input-output voltage.

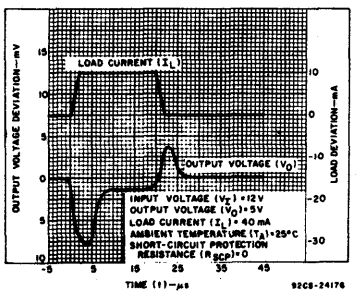


Fig. 19 – Line transient response.

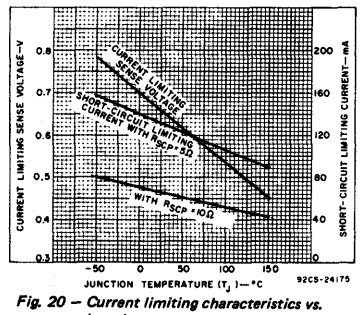


Fig. 20 – Current limiting characteristics vs. junction temperature.

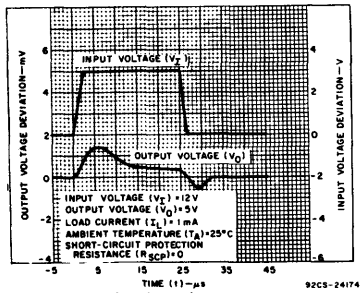


Fig. 21 – Load transient response.

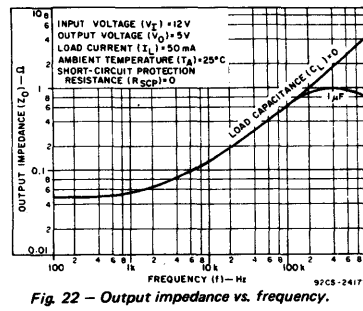
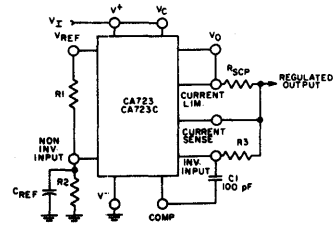


Fig. 22 – Output impedance vs. frequency.

TYPICAL APPLICATION CIRCUITS

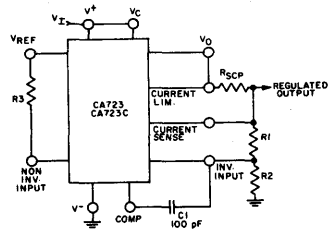


CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	5 V
LINE REGULATION (ΔVI = 3 V)	0.5 mV
LOAD REGULATION (ΔIL = 50 mA)	1.5 mV

Note: R3 = R1 R2 / R1+R2 for minimum temperature drift

Fig. 23 – Low-voltage regulator circuit (V_O = 2 to 7 volts).



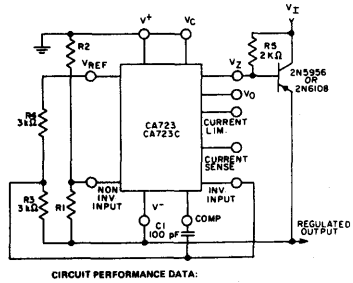
CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	15 V
LINE REGULATION (ΔVI = 3 V)	1.5 mV
LOAD REGULATION (ΔIL = 50 mA)	4.5 mV

Note: R3 = R1 R2 / R1+R2 for minimum temperature drift

R3 may be eliminated for minimum component count.

Fig. 24 – High-voltage regulator circuit (V_O = 7 to 37 volts).

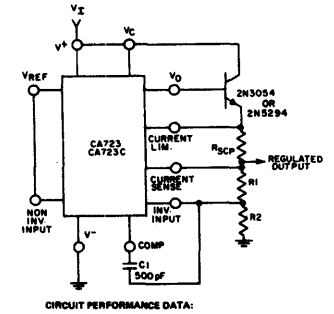


CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	−15 V
LINE REGULATION (ΔVI = 3 V)	1 mV
LOAD REGULATION (ΔIL = 100 mA)	2 mV

Note: For applications employing the TO-5 style package and where VZ is required, an external 6.2-volt zener diode should be connected in series with VO (Terminal 6).

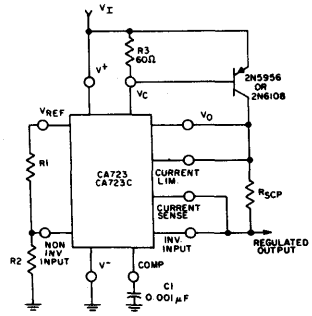
Fig. 25 – Negative-voltage regulator circuit.



CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	15 V
LINE REGULATION (ΔVI = 3 V)	1.5 mV
LOAD REGULATION (ΔIL = 1 A)	15 mV

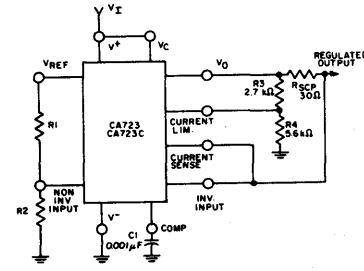
Fig. 26 – Positive-voltage-regulator circuit (with external n-p-n pass transistor).



CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	5 V
LINE REGULATION (ΔVI = 3 V)	0.5 mV
LOAD REGULATION (ΔIL = 1 A)	5 mV

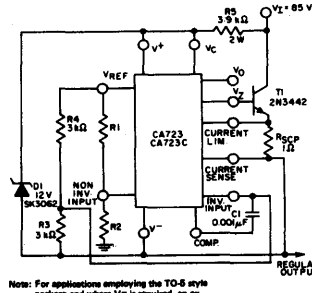
Fig. 27 – Positive voltage-regulator circuit (with external p-n-p pass transistor).



CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	5 V
LINE REGULATION (ΔVI = 3 V)	0.5 mV
LOAD REGULATION (ΔIL = 10 mA)	1 mV
SHORT-CIRCUIT CURRENT	20 mA

Fig. 28 – Foldback current-limiting circuit.



Note: For applications employing the TO-5 style package and where VZ is required, an external 6.2-volt zener diode should be connected in series with VO (Terminal 6).

CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	50 V
LINE REGULATION (ΔVI = 20 V)	15 mV
LOAD REGULATION (ΔIL = 50 mA)	20 mV

Fig. 29 – Positive-floating regulator circuit.

CA741, CA747, CA748, CA1458, CA1558 Types

Operational Amplifiers

High-Gain Single and Dual Operational Amplifiers
For Military, Industrial and Commercial Applications

"G" Suffix Types—Hermetic Gold-CHIP in Dual-In-Line Plastic Package
"E" Suffix Types—Standard Dual-In-Line Plastic Package
"T" and "S" Suffix Types—TO-5 Style Package

The RCA-CA1458, CA1558 (dual types); CA741C, CA741 (single-types); CA747C, CA747 (dual types); and CA748C, CA748 (single types) are general-purpose, high-gain operational amplifiers for use in military, industrial, and commercial applications.

These monolithic silicon integrated-circuit devices provide output short-circuit protection and latch-free operation. These types also feature wide common-mode and differential-mode signal ranges and have low-offset voltage nulling capability when used with an appropriately valued potentiometer. A 5-megohm potentiometer is used for offset nulling types CA748C, CA748 (See Fig. 10); a 10-kilohm potentiometer is used for offset nulling types CA741C, CA741, CA747CE, CA747CG, CA747E, CA747G (See Fig. 9); and types CA1458, CA1558, CA747CT, have no specific terminals for offset nulling. Each type consists of a differential-input amplifier that effectively drives a gain and level-shifting stage having a complementary emitter-follower output.

This operational amplifier line also offers the circuit designer the option of operation with internal or external phase compensation.

Types CA748C and CA748, which are externally phase compensated (terminals 1 and 8) permit a choice of operation for improved bandwidth and slew-rate capabilities. Unity gain with external phase compensation can be obtained with a single 30-pF capacitor. All the other types are internally phase-compensated.

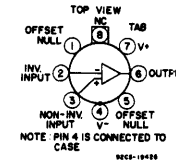
RCA's manufacturing process makes it possible to produce IC operational amplifiers with low-burst ("popcorn") noise characteristics. Type CA6741, a low-noise version of the CA741, gives limit specifications for burst noise in the data bulletin, File No. 530. Contact your RCA Sales Representative for information pertinent to other operational amplifier types that meet low-burst noise specifications.

Features:

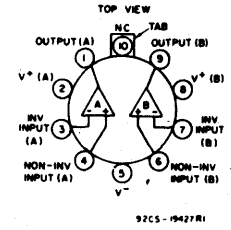
- Input bias current (all types): 500 nA max.
- Input offset current (all types): 200 nA max.

Applications:

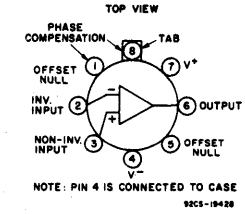
- Comparator
- DC amplifier
- Integrator or differentiator
- Multivibrator
- Narrow-band or band-pass filter
- Summing amplifier



1a.—CA741CS, CA741CT, CA741S, & CA741T with internal phase compensation.



1b.—CA747CT and CA747T with internal phase compensation.



1c.—CA748CS, CA748CT, CA748S, and CA748T with external phase compensation.

Fig. 1 — Functional diagrams.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C:*

DC Supply Voltage (between V^+ and V^- terminals):	
CA741C, CA747C▲, CA748C, CA1458▲	36 V
CA741, CA747▲, CA748, CA1558▲	44 V
Differential Input Voltage	±30 V
DC Input Voltage*	±15 V
Output Short-Circuit Duration	Indefinite
Device Dissipation:	
Up to 70°C (CA741C, CA748C)	500 mW
Up to 75°C (CA741, CA748)	500 mW
Up to 30°C (CA747)	800 mW
Up to 25°C (CA747C)	800 mW
Up to 30°C (CA1558)	680 mW
Up to 25°C (CA1458)	680 mW
For Temperatures Indicated Above	Derate linearly 6.67 mW/°C
Voltage between Offset Null and V^- (CA741C, CA741, CA747CE, CA747CG)	±0.5 V
Ambient Temperature Range:	
Operating – CA741, CA747E, CA748, CA1558	−55 to +125 °C
CA741C, CA747C, CA748C, CA1458	0 to +70 °C†
Storage	−65 to +150 °C
Lead Temperature (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	265 °C

* If Supply Voltage is less than ± 15 volts, the Absolute Maximum Input Voltage is equal to the Supply Voltage.

▲ Voltage values apply for each of the dual operational amplifiers.

† All types in any package style can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of 0 to +70°C.

RCA Type No.	No. of Ampl.	Phase Comp.	Offset Voltage Null	Min. A_{OL}	Max. V_{IO} (mV)	Operating-Temperature Range (°C)
CA1458	dual	int.	no	20k	6	0 to +70▲
CA1558	dual	int.	no	50k	5	−55 to +125
CA741C	single	int.	yes	20k	6	0 to +70▲
CA741	single	int.	yes	50k	5	−55 to +125
CA747C	dual	int.	yes*	20k	6	0 to +70▲
CA747	dual	int.	yes*	50k	5	−55 to +125
CA748C	single	ext.	yes	20k	6	0 to +70▲
CA748	single	ext.	yes	50k	5	−55 to +125

*In the 14-lead dual-in-line plastic package only.

▲All types in any package style can be operated over the temperature range of −55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of 0 to +70°C.

ORDERING INFORMATION

When ordering any of these types, it is important that the appropriate suffix letter for the package required be affixed to the type number. For example: If a CA1458 in a straight-lead TO-5 style package is desired, order CA1458T.

Type No.	PACKAGE TYPE AND SUFFIX LETTER										FIG. No.
	TO-5 STYLE			PLASTIC		Gold-CHIP PLASTIC		CHIP	Gold-CHIP	BEAM-LEAD	
	8L	10L	DIL-CAN	8L	14L	8L	14L				
CA1458	T		S	E		G		H	GH		1d, 1h
CA1558	T		S	E		G					1d, 1h
CA741C	T		S	E		G		H	GH		1a, 1e
CA741	T		S	E		G				L	1a, 1e
CA747C		T			E		G	H	GH		1b, 1f
CA747		T			E		G				1b, 1f
CA748C	T		S	E		G		H	GH		1c, 1g
CA748	T		S	E		G					1c, 1g

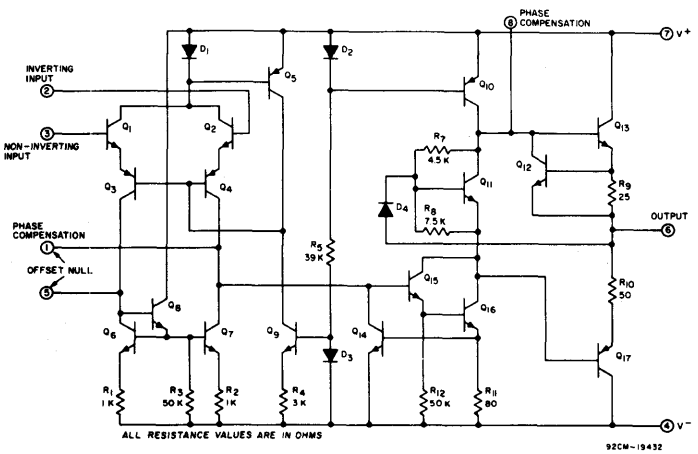
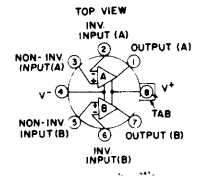
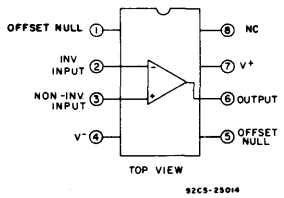


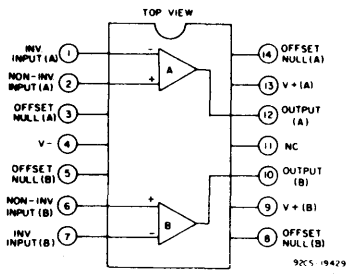
Fig.2—Schematic diagram of operational amplifier with external phase compensation for CA748C and CA748.



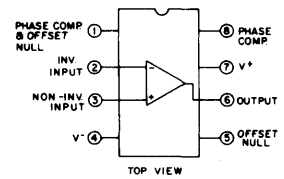
1d.—CA1458S,CA1458T,CA1558S, and CA1558T and internal phase compensation.



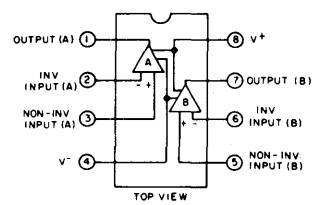
1e.—CA741CE,CA741CG,CA741E, and CA741G with internal phase compensation.



1f.—CA747CE,CA747CG,CA747E, and CA747G with internal phase compensation.



1g.—CA748CE,CA748CG,CA748E, and CA748G with external phase compensation.



1h.—CA1458E,CA1458G,CA1558E, and CA1558G with internal phase compensation.

Fig. 1 — Functional Diagrams (Cont'd)

CA741, CA747, CA748, CA1458, CA1558 Types

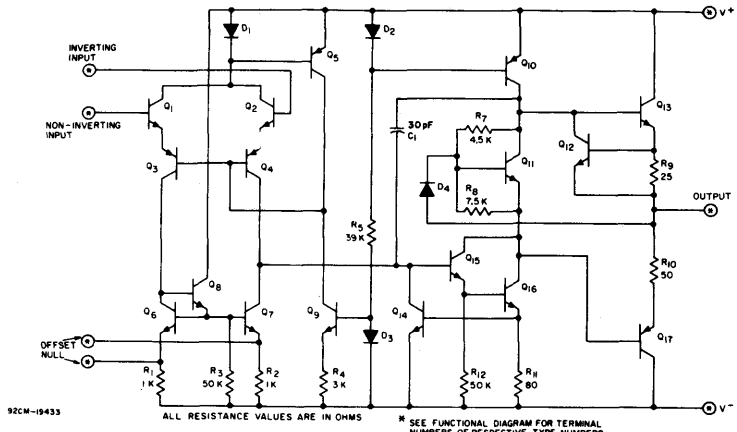


Fig.3—Schematic diagram of operational amplifiers with internal phase compensation for CA741C, CA741, and for each amplifier of the CA747C, CA747, CA1458, and CA1558.

ELECTRICAL CHARACTERISTICS

For Equipment Design

CHARACTERISTIC	TEST CONDITIONS Supply Voltage, $V^+ = 15$ V, $V^- = -15$ V	Ambient Temperature, T_A	LIMITS CA741 CA747* CA748 CA1558* Min.	Typ.	Max.	UNITS
Input Offset Voltage, V_{IO}	$R_S = \leq 10\,k\Omega$	25 °C	–	1	5	mV
		−55 to +125 °C	–	1	6	
Input Offset Current, I_{IO}		25 °C	–	20	200	nA
		−55 °C	–	85	500	
		+125 °C	–	7	200	
Input Bias Current, I_{IB}		25 °C	–	80	500	nA
		−55 °C	–	300	1500	
		+125 °C	–	30	500	
Input Resistance, R_I			0.3	2	–	MΩ
Open-Loop Differential Voltage Gain, A_{OL}	$R_L \geq 2\,k\Omega$, $V_O = \pm 10$ V	25 °C	50,000	200,000	–	
		−55 to +125 °C	25,000	–	–	
Common-Mode Input Voltage Range, V_{ICR}		−55 to +125 °C	±12	±13	–	V
Common-Mode Rejection Ratio, CMRR	$R_S \leq 10\,k\Omega$	−55 to +125 °C	70	90	–	dB
Supply Voltage Rejection Ratio, PSRR	$R_S \leq 10\,k\Omega$	−55 to +125 °C	–	30	150	μV/V
Output Voltage Swing, V_{OPP}	$R_L \geq 10\,k\Omega$	−55 to +125 °C	±12	±14	–	V
	$R_L \geq 2\,k\Omega$	−55 to +125 °C	±10	±13	–	
Supply Current, $I^{\pm}$		25 °C	–	1.7	2.8	mA
		−55 °C	–	2	3.3	
		+125 °C	–	1.5	2.5	
Device Dissipation, P_D		25 °C	–	50	85	mW
		−55 °C	–	60	100	
		+125 °C	–	45	75	

* Values apply for each section of the dual amplifiers.

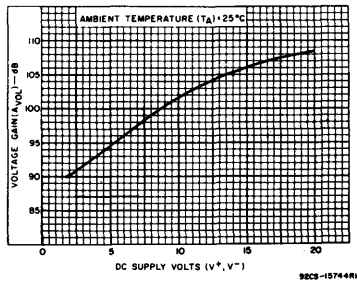


Fig.4—Open-loop voltage gain vs. supply voltage for all types except CA748 and CA748C.

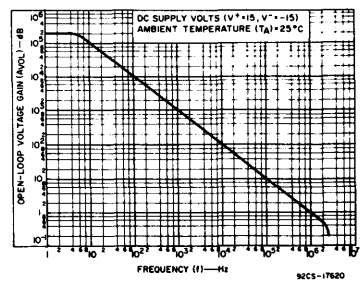


Fig.5—Open-loop voltage gain vs. frequency for all types except CA748 and CA748C.

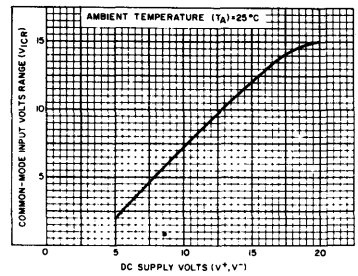


Fig.6—Common-mode input voltage range vs. supply voltage for all types.

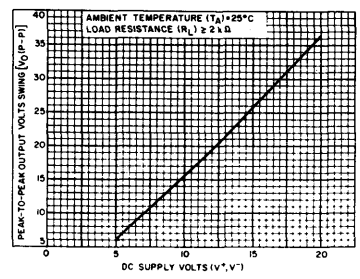


Fig.7—Peak-to-peak output voltage vs. supply voltage for all types except CA748 and CA748C.

ELECTRICAL CHARACTERISTICS
For Equipment Design

CHARACTERISTIC	TEST CONDITIONS Supply Voltage, $V^+ = 15$ V, $V^- = -15$ V	Ambient Temperature, T_A	LIMITS CA741C CA747C* CA748C CA1458* Min.	Typ.	Max.	UNITS
Input Offset Voltage, V_{IO}	$R_S \leq 10$ kΩ	25 °C	–	2	6	mV
		0 to 70 °C	–	–	7.5	
Input Offset Current, I_{IO}		25 °C	–	20	200	nA
		0 to 70 °C	–	–	300	
Input Bias Current, I_{IB}		25 °C	–	80	500	nA
		0 to 70 °C	–	–	800	
Input Resistance, R_I			0.3	2	–	MΩ
Open-Loop Differential Voltage Gain, A_{OL}	$R_L \geq 2$ kΩ $V_O = \pm 10$ V	25 °C	20,000	200,000	–	
		0 to 70 °C	15,000	–	–	
Common-Mode Input Voltage Range, V_{ICR}		25 °C	±12	±13	–	V
Common-Mode Rejection Ratio, CMRR	$R_S \leq 10$ kΩ	25 °C	70	90	–	dB
Supply-Voltage Rejection Ratio, PSRR	$R_S \leq 10$ kΩ	25 °C	–	30	150	μV/V
Output Voltage Swing, V_{OPP}	$R_L \geq 10$ kΩ	25 °C	±12	±14	–	V
	$R_L \geq 2$ kΩ	25 °C	±10	±13	–	
		0 to 70 °C	±10	±13	–	
Supply Current, $I^{\pm}$		25 °C	–	1.7	2.8	mA
Device Dissipation, P_D		25 °C	–	50	85	mW

* Values apply for each section of the dual amplifiers.

ELECTRICAL CHARACTERISTICS
Typical Values Intended Only for Design Guidance

CHARACTERISTIC	TEST CONDITIONS $V\pm = \pm 15$ V	TYP. VALUES ALL TYPES	UNITS
Input Capacitance, C_I		1.4	pF
Offset Voltage Adjustment Range		±15	mV
Output Resistance, R_O		75	Ω
Output Short-Circuit Current		25	mA
Transient Response: Rise Time, t_r	Unity gain $V_I = 20$ mV $R_L = 2$ kΩ $C_L \leq 100$ pF	0.3	μs
Overshoot		5	%
Slew Rate, SR: Closed-loop	$R_L \geq 2$ kΩ	0.5	V/μs
Open-loop▲		40	

▲ Open-loop slew rate applies only for types CA748C and CA748.

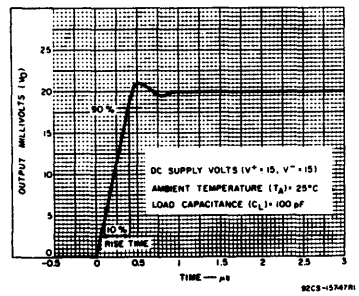


Fig.8—Output voltage vs. transient response time for CA741C and CA741.

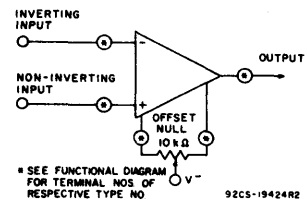


Fig.9—Voltage-offset null circuit for CA741C, CA741, CA747CE, CA747CG, CA747E, and CA747G.

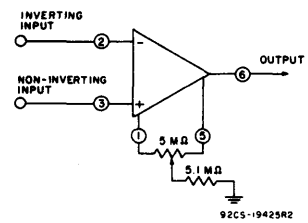


Fig.10—Voltage-offset null circuit for CA748C and CA748.

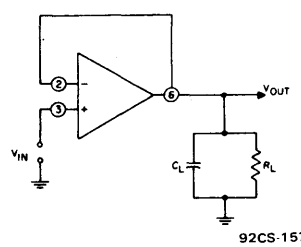


Fig.11—Transient response test circuit for all types.

CA1541D

Dual-Input Memory Sense Amplifier

RCA-CA1541D , a monolithic silicon integrated circuit, is a dual-input memory sense amplifier intended for core memory applications.

The sense amplifier, consisting of two differential input amplifiers, a common second stage amplifier, and an output logic gate (See Fig. 1), converts low-level core-memory "1" pulses to saturated logic-level output pulses. Either one of the input amplifiers may be gated ON with a saturated logic signal so that an incoming "1" pulse of positive or negative polarity can be detected from either of two sense lines.

The CA1541D features an external switching threshold adjustment, plus its gate and strobe inputs are compatible with saturated logic levels. The sense amplifier is suitable for operation with core memories having cycle times equal to or greater than 0.4 μs and is unilaterally interchangeable with industry types 1541L and 1441.

The CA1541D is supplied in 14-lead dual-in-line ceramic package and is rated for operation over the full military temperature range of −55°C to +125°C.

Features

- Complete dual input core memory sense amplifier
- Two available outputs: —Saturated logic output
  —Linear output (positive output for either polarity input)
- Nominal threshold voltage: 17 mV
- Adjustable threshold: 10 to 35 mV
- Low threshold uncertainty range: ±3 mV
- Fast overload recovery time: —Differential-Mode: 15 ns typ.
  —Common-Mode: 30 ns typ.
- Independent channel gate and strobe terminals compatible with saturated logic levels
- Suitable for core memories having cycle times ≥ 0.4 μs
- Input offset voltage: 6 mV max.

MAXIMUM RATINGS, Absolute Maximum Values, at T_A = 25°C

Except for Differential Input Voltage, all voltages are measured with respect to ground (Term. 8).

DC Supply Voltage:	
V^+ (Term. 2)	+10 V
V^- (Term. 7)	−10 V
Differential Input Voltage	±5 V
Common-Mode Input Voltage	±5 V
"A" or "B"-Gate Input Voltage*	V^- to V^+
Strobe Terminal Voltage	V^- to +6V
Output Terminal Load Current	±25 mA
Device Dissipation:	
Up to T_A = 75°C	750 mW
Above T_A = 75°C	Derate Linearly 8 mW/°C
Ambient Temperature Range:	
Operating	−55 to +125 °C
Storage	−65 to +150 °C
Lead Temperature (during soldering):	
At distance not less than 1/32 inch (0.79 mm) from case for 10 seconds max.	+265°C

*Note: The "A" or "B"-Gate Input Voltage is also referred to, as the Channel-Gate Input Voltage.

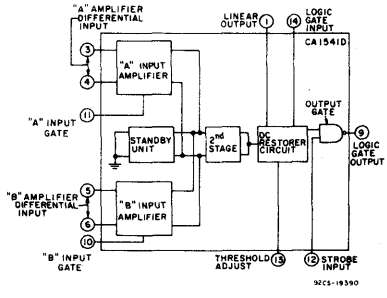


Fig. 1 — Functional block diagram of the CA1541D.

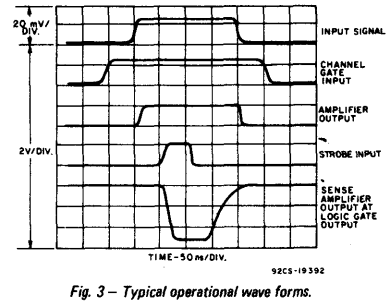


Fig. 3 — Typical operational wave forms.

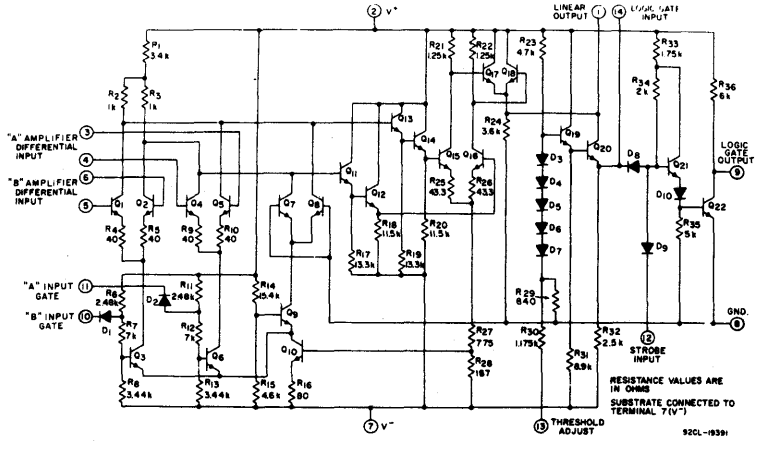


Fig. 2 — Schematic diagram of the CA1541D.

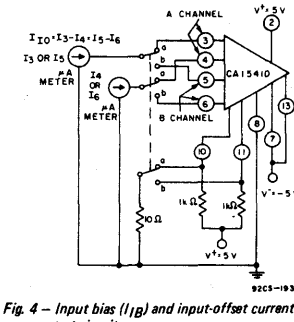


Fig. 4 — Input bias (I_{IB}) and input-offset current (I_{IO}) test circuit.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS: $V^+ = 5V$, $V^- = -5V$; V_{TH} ADJ. = $-5V \pm 1\%$, (Term. 13); $C_{EXT} = 0.01\ \mu F$; $T_A = 25^oC$ (unless indicated otherwise)		LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Static (DC) Characteristics							
Power Dissipation	P_D			–	140	180	mW
Input Offset Current	I_{IO}			–	1	2	μA
Input Bias Current:							
$T_A = 25^oC$	I_{IB}		$V_5 = V_6 = V_3 = V_4 = 0$	–	5	25	μA
$T_A = -55^oC$				–	–	50	
Output Voltage:							
High	V_{OH}	$I_{OM} = 200\ \mu A$		3	–	–	V
Low –							
$T_A = 25^oC$	V_{OL}	$V_{14} = 5\ V$, $I_9 = 10\ mA$		–	–	350	mV
$T_A = 125^oC$				–	–	400	
Strobe Load Current	I_S	$V_{12} = 0$		–	–	1.5	mA
Strobe Reverse Current:							
$T_A = 25^oC$	I_{SR}	$V_{12} = 5V$		–	–	2	μA
$T_A = 125^oC$				–	–	25	
Input Gate Load Current	I_G	$V_{10} = V_{11}$ 0		–	–	2.5	mA
Input Gate Reverse Current:							
$T_A = 25^oC$	I_{GR}	$V_{10} = V_{11} = 5V$		–	–	2	μA
$T_A = 125^oC$				–	–	25	
Switching Characteristics							
Input Threshold Voltage:							
$T_A = 25^oC$	V_{TH}			14	17	20	mV
$T_A = -55$ to 125^oC				12	17	22	
Input Offset Voltage	V_{IO}			–	1	6	mV
Input Gate Voltage:							
High	V_{GH}	$V_3 = V_5 = 25$ mV,		–	1.6	–	V
Low	V_{GL}	$V_4 = V_6 = 0$		–	0.7	–	
Common-Mode Range:							
Input Gate High	V_{CM}			–	±1.5	–	V
Input Gate Low				–	±1.5	–	
Differential-Mode Range:							
Input Gate High	V_{DH}			–	±600	–	mV
Input Gate Low	V_{DL}			–	±1.5	–	V
Propagation Delay:							
Input to Amplifier Output	t_{IA}	$V_3 = 25$ mV (pulsed),		–	10	15	ns
Input to Output	t_{IO}	$V_{12} = 2V$		–	20	30	
Strobe to Output	t_{SO}	$V_3 = V_4 = V_5 = V_6 = 0$, $V_{12} = 2V$ (pulsed)		–	15	20	
Gate Input to Amplifier Output	t_{GA}	$V_{11} = 2V$ (pulsed)		–	10	15	
Gate Input to Amplifier Input	t_{GI}	$V_3 = 25$ mV		–	30	35	
Common-Mode Recovery Time:							
Input Gate High	t_{CMR}	$V_3 = V_5 = 1.5\ V$		–	15	30	ns
Input Gate Low				–	15	30	
Differential-Mode Recovery Time:							
Input Gate High	t_{DR}	$V_3 = V_5 = 400$ mV		–	30	–	ns
Input Gate Low				–	0	–	

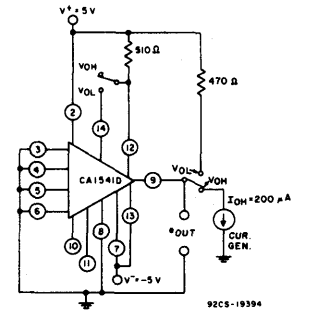
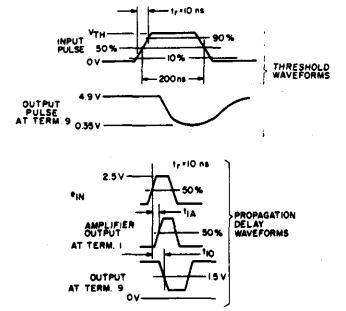


Fig. 5 – Test circuit for measurement of low (V_{OL}) and high (V_{OH}) output voltage levels.



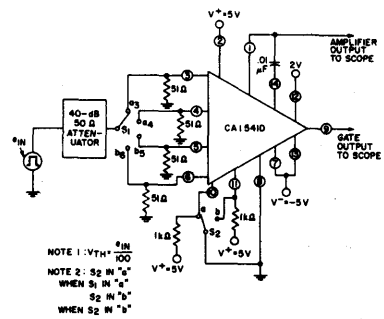


Fig. 6 – Threshold propagation delay, gate and input-offset test circuit with associated pulse wave forms.

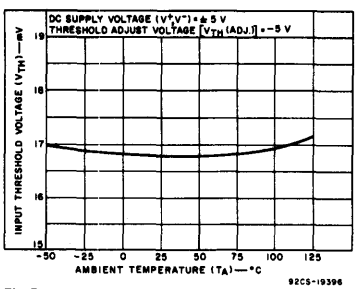


Fig. 7a – Input V_{TH} vs. T_A.

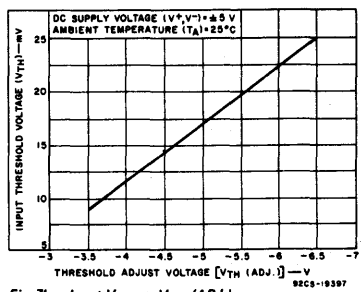


Fig. 7b – Input V_{TH} vs. V_{TH} (ADJ.).

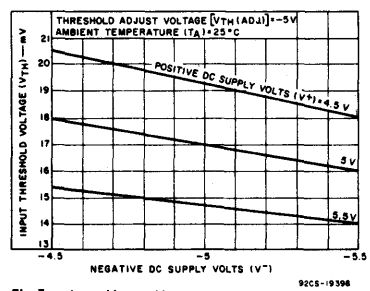


Fig. 7c – Input V_{TH} vs. V^-.

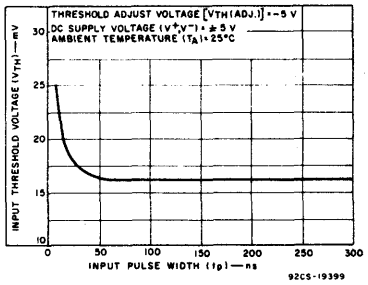
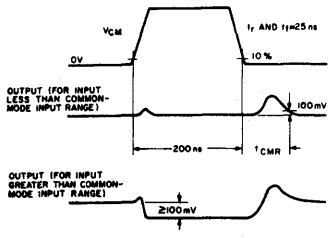


Fig. 7d – Input V_{TH} vs. input pulse width.



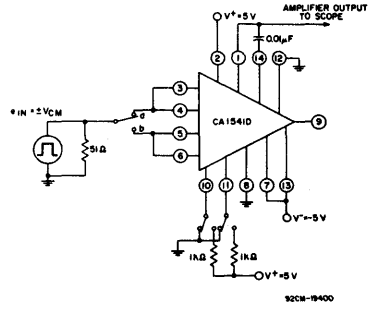


Fig. 8 – Common-mode input range test circuit with associated pulse wave forms.

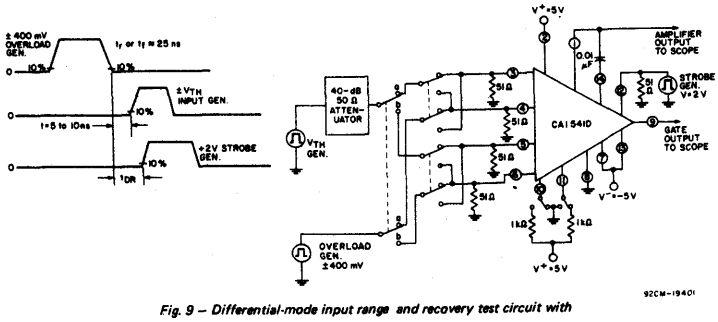


Fig. 9 – Differential-mode input range and recovery test circuit with associated pulse wave forms.

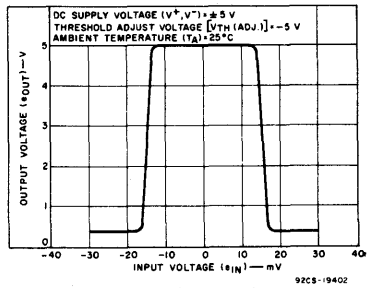
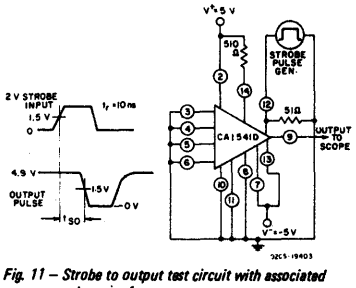
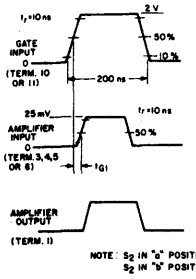


Fig. 10 – Input-output transfer characteristics.





NOTE: S_2 IN "a" POSITION WHEN S_1 IN EITHER "a" POSITION
S_2 IN "b" POSITION WHEN S_1 IN EITHER "b" POSITION

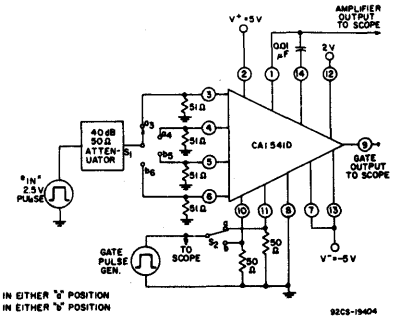


Fig. 11 – Strobe to output test circuit with associated pulse wave-forms.

Fig. 12 – Gate input to amplifier input (t_{GI}) test circuit with associated pulse wave forms.

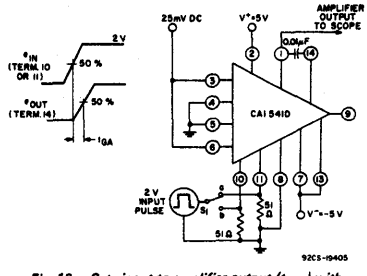


Fig. 13 – Gate input to amplifier output (t_{GA}) with associated pulse wave forms.

FM IF Amplifier-Limiter and Quadrature Detector

For FM IF and TV Sound IF Applications

The CA2111A, on a single monolithic chip, provides a multi-stage wideband amplifier-limiter, a quadrature detector, and an emitter-follower output stage. This device is designed for use in FM receivers and in the sound IF sections of TV receivers. In addition, an output terminal is provided which allows the use of the amplifier-limiter as a straight 60-dB wideband amplifier.

The amplifier-limiter features the excellent limiting characteristics of 3 cascaded differential amplifiers.

The quadrature detector requires only one coil in the associated outboard circuit and therefore, tuning is a simple procedure.

A unique feature of the CA2111A is its exceptionally low AFC voltage drift over the full operating-temperature range.

This device can be supplied in either dual-in-line or quad-in-line 14-lead plastic packages (CA2111AE and CA2111AQ, respectively).

Features:

- **Direct replacement for ULN2111A and MC1357**
- **Good sensitivity: Input limiting voltage (knee) (400 μV typ. at 10.7 MHz; 250 μV typ. at 4.5 MHz and 5.5 MHz)**
- **Excellent AM rejection (45 dB typ. at 10.7 MHz)**
- **Provision for output from 3-stage IF amplifier section**
- **Low harmonic distortion**
- **Quadrature detection permits simplified single-coil tuning**
- **Extremely low AFC voltage drift over full operating-temperature range**
- **Minimum number of external parts required**

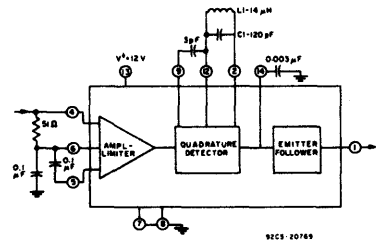


Fig. 1—Block diagram of CA2111A and associated outboard components.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A=25°C*

DC Supply Voltage [between terminals 13 (V^+) and 7 (V^-)]	16	V
Device Dissipation:		
Up to T_A = 60°C	600	mW
Above T_A = 60°C	derate linearly 6.7	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10s max.	+265	°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
DC Voltage:						
At Terminal 1	V_1	V^+ = 12V	–	5.4	–	
		= 8V	–	3.7	–	
At Terminals 4, 5, 6, 10	$V_{4,5,6,10}$	V^+ = 8V	–	1.35	–	V
At Terminals 2, 12	$V_{2,12}$		–	3.5	–	
DC Current (into Terminal 13)						
At V^+ = 8V			–	14	–	mA
At V^+ = 12V	I_{13}		–	16	–	
Amplifier Input Resistance	R_4	f_o = 10.7 MHz	–	7	–	kΩ
Amplifier Input Capacitance	C_4		–	11	–	pF
Detector Input Resistance	R_{12}		–	70	–	kΩ
Detector Input Capacitance	C_{12}		–	2.7	–	pF
Amplifier Output Resistance	R_{10}		–	60	–	Ω
Detector Output Resistance	R_1		–	200	–	Ω
De-Emphasis Resistance	R_{14}		–	8.8	–	kΩ

DYNAMIC ELECTRICAL CHARACTERISTICS at T_A = 25°C
FM Modulation Frequency = 400 Hz, Source Resistance = 50Ω

CHARACTERISTIC	SYMBOL	TEST CONDITIONS: f_o = 10.7 MHz, Δf = ± 75 KHz, V^+ = 12V		TEST CONDITIONS: f_o = 10.7 MHz, Δf = ± 75 KHz, V^+ = 8V		TEST CONDITIONS: f_o = 4.5 MHz, Δf = ± 25 KHz, V^+ = 12V		TEST CONDITIONS: f_o = 5.5 MHz, Δf = ± 50 KHz, V^+ = 12V		UNITS	TEST CIRCUIT OR CHARACTERISTIC CURVES FIG. NO.
		LIMITS TYP.	LIMITS MAX.	LIMITS TYP.	LIMITS MAX.	LIMITS TYP.	LIMITS MAX.	LIMITS TYP.	LIMITS MAX.		
AMPL-LIMITER											
Input Limiting Threshold Voltage	V_i(lim) (4)	400	600	400	600	250	400	250	400	V (RMS)	7, 6, 8, 9
AM Rejection‡ *	AMR(1)	45	–	37	–	36	–	40	–	dB	2, 7, 5, 6
Ampl. Voltage Gain▲	A_V(10)	55	–	55	–	60	–	60	–	dB	7
DETECTOR											
Recovered Audio‡ Output Voltage	V_o(AF) (1)	0.48	–	0.3	–	0.72	–	1.2	–	V (RMS)	6, 7, 8, 9
Total Harmonic‡ Distortion	THD(1)	1	–	1	–	1.5	–	3	–	%	7

‡V_i = 10 mV (RMS) ▲$\Delta V_i \leq 50\ \mu V$ (rms) *100% FM, 30% AM

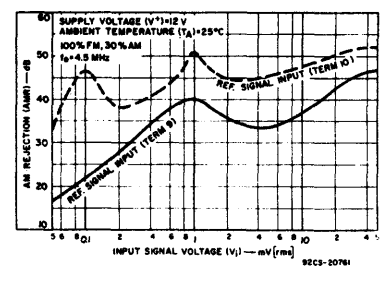


Fig. 2—AM rejection vs input voltage (4.5 MHz).

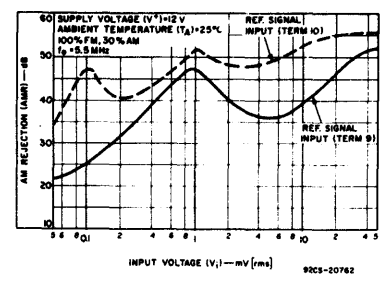


Fig. 3—AM rejection vs input voltage (5.5 MHz).

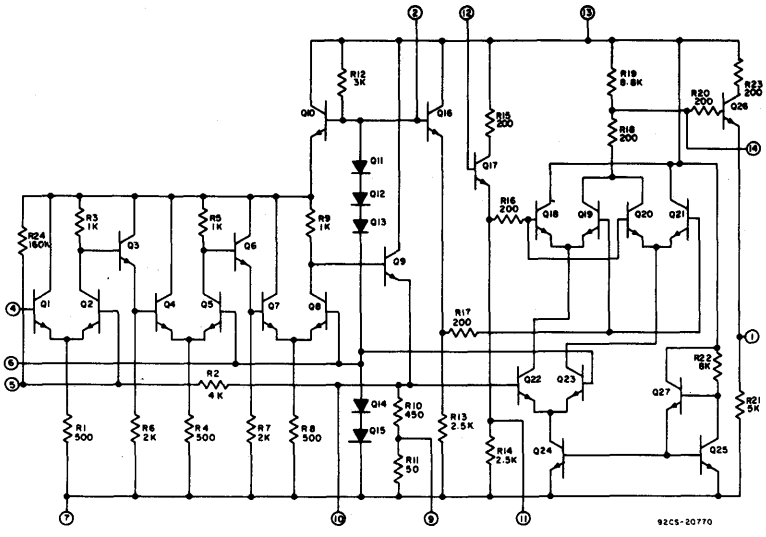


Fig. 4 –Circuit schematic–CA2111A

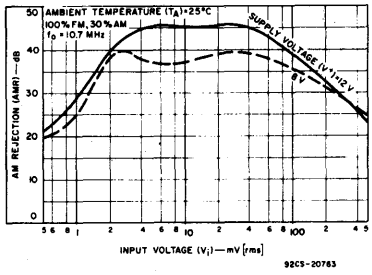


Fig. 5 –AM rejection vs input voltage (10.7 MHz).

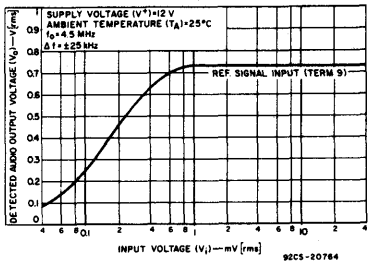
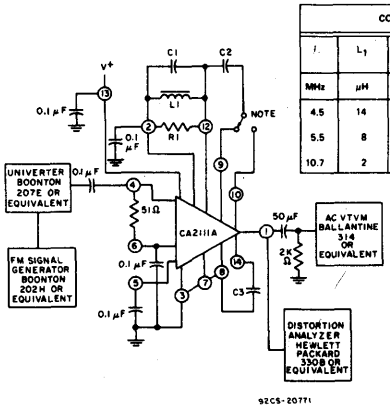


Fig. 6 –Detected audio output vs input voltage (4.5 MHz).



COMPONENT VALUES							DETECTOR TRANSFER CHARACTERISTICS	
f	L_1	C_1	R_1	Q	C_2	C_3	UPPER PEAK	LOWER PEAK
MHz	μH	pF	KΩ	–	pF	μF	MHz	MHz
4.5	14	120	20	30	3	0.003	4.58	4.42
5.5	8	100	20	30	3	0.003	5.63	5.37
10.7	2	120	3.9	20	4.7	0.01	10.9	10.5

NOTE:
Input to the quadrature coil can be from either terminal 9 or terminal 10. Terminal 9 is normally used because it lessens the possibility of overloads during tuning. The use of terminal 10 increases the limiting sensitivity significantly and has been used successfully in these tests.

Fig. 7 –Test circuit.

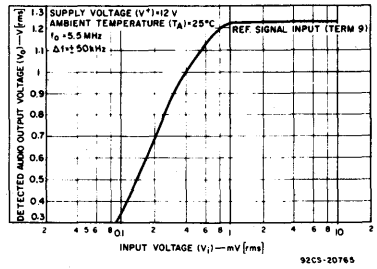


Fig. 8–Detected audio output vs input voltage (5.5 MHz).

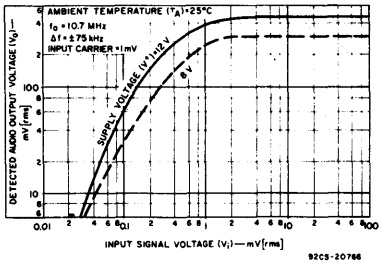


Fig. 9–Detected audio output voltage vs input voltage (10.7 MHz).

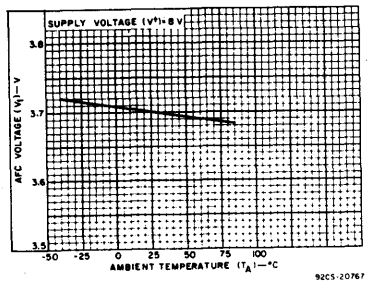


Fig. 10–AFC voltage vs ambient temp.

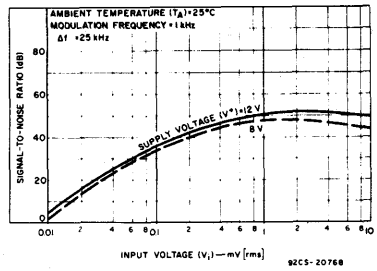


Fig. 11–Signal-to-noise ratio vs input voltage.

DC Amplifier

- Designed for use in Communication, Telemetry, Instrumentation, and Data-Processing Equipment
- Balanced differential-amplifier configuration with controlled constant-current source to provide outstanding versatility
- Built-in temperature stability for operation from -55°C to +125°C
- Companion Application Note, ICAN 5030 "Applications of RCA CA3000 Integrated Circuit DC Amplifier" covers characteristics of different *operating modes, frequency considerations, 10 MHz narrow band* tuned amplifier design, crystal oscillator design, and many other application aids
- 10-Lead hermetic TO-5 style package

HIGHLIGHTS

- Input Impedance . . . . . 195 kΩ typ.
- Voltage Gain. . . . . 30 dB typ.
- Common-Mode Rejection Ratio . . . 98 dB typ.
- Input Offset Voltage. . . . . 1.4 mV typ.
- Push-Pull Input and Output
- Frequency Capability
  DC to 30 MHz (with external C and R)
- Wide AGC Range. . . . . 90 dB typ.

APPLICATIONS

- Schmitt Trigger
- RC-Coupled Feedback Amplifier
- Mixer
- Comparator
- Modulator
- Crystal Oscillator
- Sense Amplifier

ABSOLUTE-MAXIMUM VOLTAGE LIMITS

at T_{FA} = 25°C

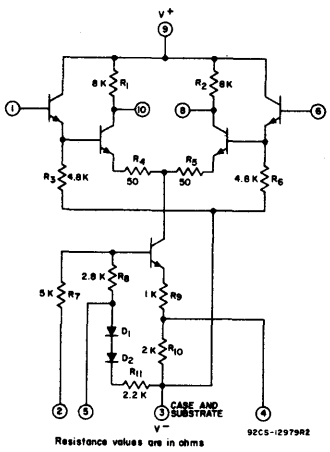
OPERATING-TEMPERATURE RANGE . . . −55°C to +125°C
STORAGE-TEMPERATURE RANGE . . . −65°C to +150°C
LEAD-TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

MAXIMUM POWER SUPPLY VOLTAGE — 16 or ±8 V

MAXIMUM SINGLE-ENDED INPUT-SIGNAL VOLTAGE . . ±4 V
MAXIMUM COMMON-MODE INPUT-SIGNAL VOLTAGE . . ±2 V
MAXIMUM DEVICE DISSIPATION:
From −55°C to 85°C. . . . . 450 mW
Above 85°C . . . . . Derate 5 mW/°C

ELECTRICAL CHARACTERISTICS, at T_{FA} = 25°C, V_{CC} = +6V, V_{EE} = -6V, unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.4 & No.5 Not Connected Unless Specified		TEST CIRCUITS	LIMITS TYPE CA3000				TYPICAL CHARACTERISTICS CURVES
				Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS									
Input Offset Voltage	V_{IO}				-	1.4	5	mV	2
Input Offset Current	I_{IO}				-	1.2	10	μA	2
Input Bias Current	I_{IB}				-	23	36	μA	3
Quiescent Operating Voltage	V_8 or V_{IO}	TERMINALS 4	5						
		NC	NC		-	2.6	-	V	4
		NC	VEE		-	4.2	-	V	4
		VEE	NC		-	-1.5	-	V	4
		VEE	VEE		-	0.6	-	V	4
Device Dissipation	P_D	NC	NC		-	30	-	mW	NONE
DYNAMIC CHARACTERISTICS									
Differential Voltage Gain Single-Ended Input	A_{DIFF}	Single-Ended Output f = 1 kHz		6	28	32	-	dB	5
		Double-Ended Output f = 1 kHz		6	-	38	-	dB	5
Bandwidth at -3 dB Point	BW	V_i = 10 mV, R_s = 1 kΩ			-	650	-	kHz	7
Maximum Output Voltage Swing	$V_{OUT(P-P)}$	f = 1 kHz		6	-	6.4	-	V(P-P)	NONE
Common-Mode Rejection Ratio	CMRR	f = 1 kHz		9	70	98	-	dB	8
Single-Ended Input Impedance	Z_{IN}	f = 1 kHz		11	70K	195K	-	Ω	10
Single-Ended Output Impedance	Z_{OUT}	f = 1 kHz		13	5.5K	8K	10.5K	Ω	12
Total Harmonic Distortion	THD	R_S=1kΩ f = 1 kHz V_O=42V_{p-p}			-	0.2	5	%	14
AGC Range (Maximum Voltage Gain to Complete Cutoff)	AGC	f = 1 kHz		15	80	90	-	dB	NONE



Resistance values are in ohms

Fig.1 SCHEMATIC DIAGRAM

STATIC CHARACTERISTICS FOR TYPE CA3000

INPUT OFFSET VOLTAGE AND CURRENT vs TEMPERATURE

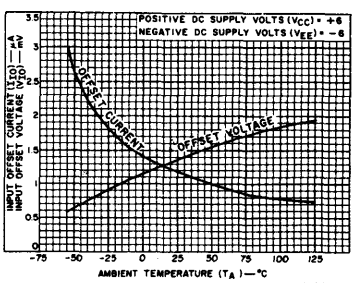


Fig.2

INPUT BIAS CURRENT vs TEMPERATURE

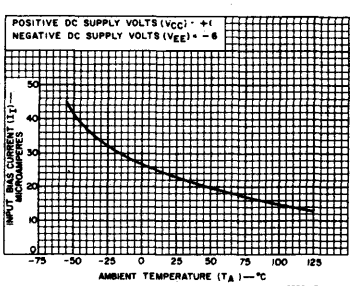


Fig.3

QUIESCENT OPERATING VOLTAGE vs TEMPERATURE

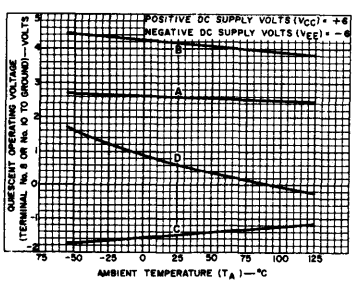


Fig.4

DYNAMIC CHARACTERISTICS AND TEST CIRCUIT FOR TYPE CA3000

DIFFERENTIAL VOLTAGE GAIN vs TEMPERATURE

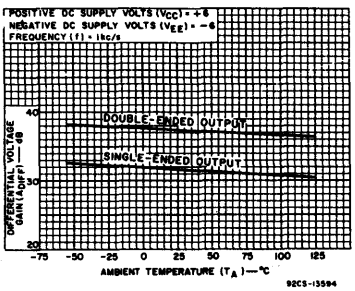


Fig.5

DIFFERENTIAL VOLTAGE GAIN AND MAXIMUM OUTPUT VOLTAGE SWING TEST CIRCUIT

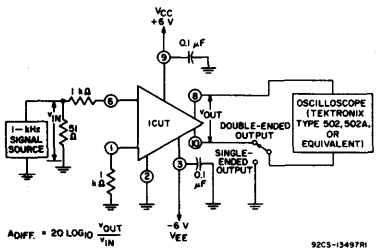


Fig.6

BANDWIDTH AT -3 dB POINT vs TEMPERATURE

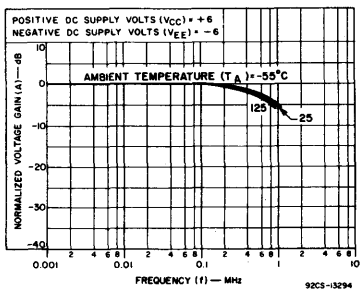


Fig.7

COMMON-MODE REJECTION RATIO vs TEMPERATURE

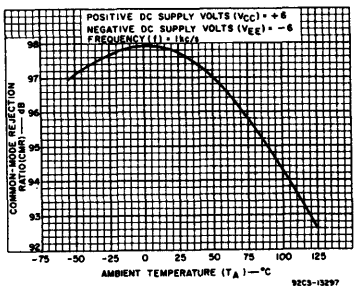


Fig.8

COMMON-MODE REJECTION RATIO TEST CIRCUIT

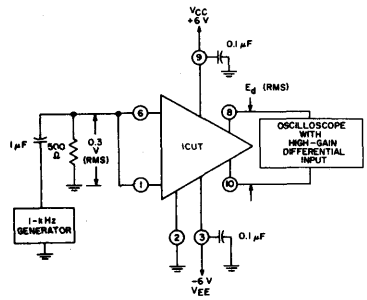


Fig.9

SINGLE-ENDED INPUT IMPEDANCE vs TEMPERATURE

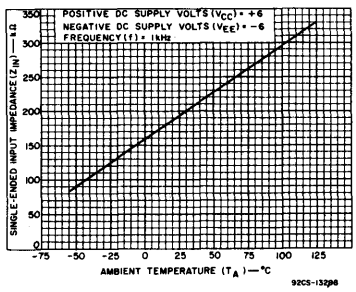


Fig.10

SINGLE-ENDED INPUT IMPEDANCE TEST CIRCUIT

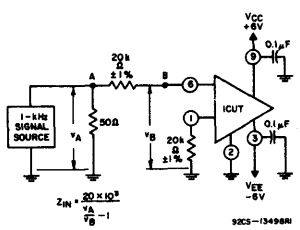


Fig.11

SINGLE-ENDED OUTPUT IMPEDANCE vs TEMPERATURE

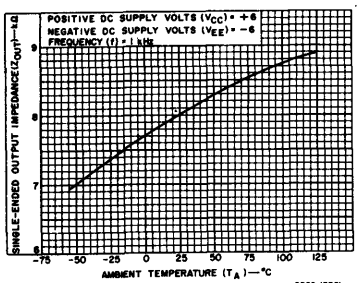


Fig.12

SINGLE-ENDED OUTPUT IMPEDANCE TEST CIRCUIT

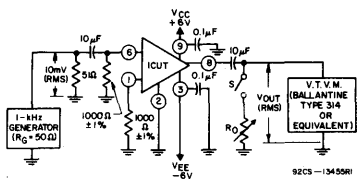


Fig.13

TOTAL HARMONIC DISTORTION vs TEMPERATURE

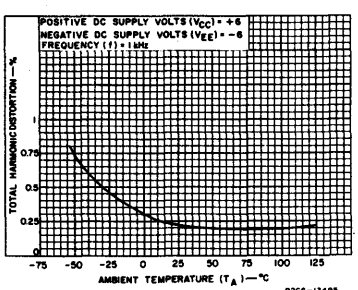


Fig.14

AGC RANGE TEST CIRCUIT

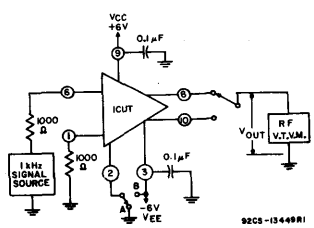


Fig.15

Video and Wideband Amplifier

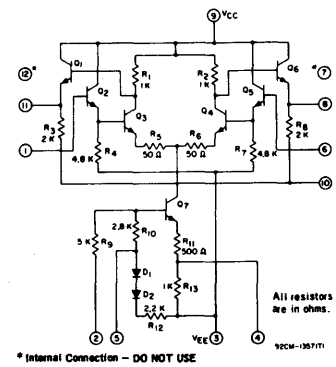
- Designed for use in Video Systems and Communication Equipment
- Balanced differential amplifier configuration with controlled constant-current source provides outstanding versatility
- Built-in temperature stability for operation from -55°C to +125°C
- Emitter follower input & output
- Companion Application Note ICAN5038 "Application of the RCA-CA3001 Integrated-Circuit Video Amplifier", covers different operating modes, gain control, distortion, swing capability, 3 stage amplifier design, and a Schmitt trigger study.
- 12-Lead Hermetic TO-5 Style Package

HIGHLIGHTS

- Push-Pull Input & Output
- AGC Range ..... 60 dB typ.
- Bandwidth ..... 29 MHz
- Input Resistance ..... 150 kΩ typ.
- Output Resistance ..... 45 Ω typ.
- Voltage Gain ..... 19 dB typ.
- Input Offset Voltage .... 1.5 mV typ.

APPLICATIONS

- Schmitt Trigger
- Mixer
- Modulator
- DC, IF, & Video Amplifier



* Internal Connection – DO NOT USE

Fig.1 - Schematic Diagram.

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS at T_A = 25°C

Indicated voltage or current limits for each terminal can be applied under the specified conditions for other terminals.
All Voltages are with respect to ground (common terminal of Positive and Negative DC Supplies).

TERMINAL	VOLTAGE OR CURRENT LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-2.5	+2.5	2, 6 3, 10 9	0 -6 +6
2	-8.5	0	1, 6 3, 10 9	0 -8.5 +6
3	-10	0	1, 2, 6 9 10	0 +6 -6
4	-8.5	0	1, 2, 6 9 10	0 +6 -6
5	-6	0	1, 2, 6 3, 10 9	0 -6 +6
6	-2.5	+2.5	1, 2 3, 10 9	0 -6 +6
7	INTERNAL CONNECTION DO NOT USE			

TERMINAL	VOLTAGE OR CURRENT LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
8	25 mA		1, 2, 6, 10 3 9 200-Ω RESISTOR CONNECTED BETWEEN TERMINALS No.8 & No.10	0 -6 +6
9	0	+10	1, 2, 6, 10 3	0 -6
10	-10	0	1, 2, 6 3 9	0 -6 +6
11	25 mA		1, 2, 6, 10 3 9 200-Ω RESISTOR CONNECTED BETWEEN TERMINALS No.10&No.11	0 -6 +6
12	INTERNAL CONNECTION DO NOT USE			
CASE	INTERNALLY CONNECTED TO TERMINAL No.3 (SUBSTRATE) DO NOT GROUND			

OPERATING TEMPERATURE RANGE	-55°C to +125°C
STORAGE TEMPERATURE RANGE	-65°C to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C
MAXIMUM SINGLE-ENDED INPUT-SIGNAL VOLTAGE	± 4 V
MAXIMUM COMMON-MODE INPUT-SIGNAL VOLTAGE	± 2.5 V
MAXIMUM DEVICE DISSIPATION:	
-55 to 85°C	450 mW
Above 85°C	Derate linearly 5 mW/°C

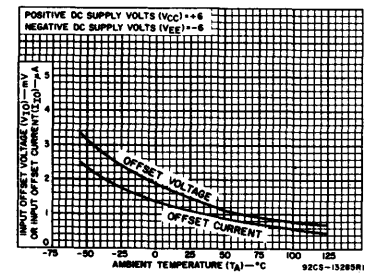


Fig.2 - Input offset voltage and current vs. temperature.

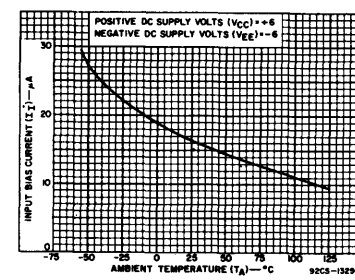


Fig.3 - Input bias current vs. temperature.

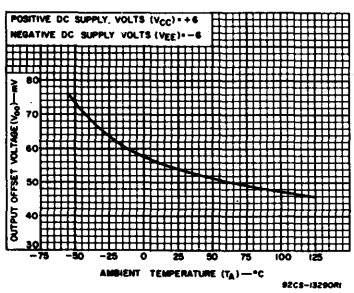


Fig. 4 - Output offset voltage vs. temperature.

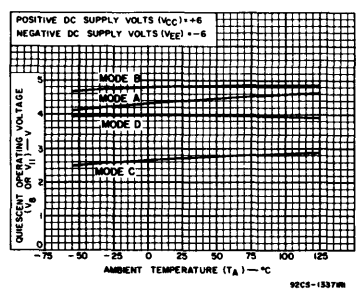


Fig. 5 - Quiescent operating voltage vs. temperature.

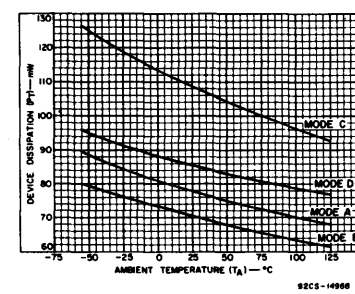


Fig. 6 - Device dissipation vs. temperature.

CA3001

ELECTRICAL CHARACTERISTICS, AT T_A = 25°C, V_{CC} = +6 V, V_{EE} = −6 V

CHARACTERISTICS (See Page 2 for Definitions of Terms)	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.4 and No.5 Not Connected Unless Specified			TEST CIRCUITS	LIMITS TYPE CA3001				TYPICAL CHARACTERISTICS CURVES
					Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS:										
Input Offset Voltage	V_{IO}				12	-	1.5	-	mV	2
Input Offset Current	I_{IO}				13	-	1	10	μA	2
Input Bias Current	I_I				13	-	16	36	μA	3
Output Offset Voltage	V_{OO}	R_S = 1 kΩ				-	54	300	mV	4
Quiescent Operating Voltage	V_8 OR V_{11}	TERMINALS MODE	4	5						
		A	NC	NC		3.8	4.4	5	V	5
		B	NC	V_{EE}		-	4.8	-	V	5
		C	V_{EE}	NC		-	2.7	-	V	5
		D	V_{EE}	V_{EE}		-	4	-	V	5
Device Dissipation	P_D	A	NC	NC		60	78	120	mW	6
		B	NC	V_{EE}		-	71	-	mW	6
		C	V_{EE}	NC		-	110	-	mW	6
		D	V_{EE}	V_{EE}		-	86	-	mW	6
DYNAMIC CHARACTERISTICS:										
Differential Voltage Gain (Single-ended input and output)	A_{DIFF}	f 1.75 MHz f 20 MHz				16 10	19 14	- -	dB dB	7, 8
Bandwidth at -3 dB Point	BW	R_S = 50Ω				16	29	-	MHz	NONE
Maximum Output Voltage Swing	$V_{OUT(P-P)}$	R_S = 50Ω f = 1.75 MHz				-	5	-	V_{P-P}	NONE
Noise Figure	NF	f 1.75 MHz, R_S 1 KΩ			11	-	5	8	dB	
		f 11.7 MHz, R_S 1 KΩ			11	-	7.7	-	dB	9
Common-Mode Rejection Ratio	CMRR	f 1 KHz				70	88	-	dB	10
Input Impedance Components:										
Parallel Input Resistance	R_{IN}	f 1.75 MHz				50	140	-	KΩ	11
Parallel Input Capacitance	C_{IN}	f 1.75 MHz				-	3.4	7	pF	11
Output Resistance	R_{OUT}	f 1.75 MHz				-	45	70	Ω	NONE
AGC Range (Maximum voltage gain to complete cutoff)	AGC	f 1.75 MHz				55	60	-	dB	NONE

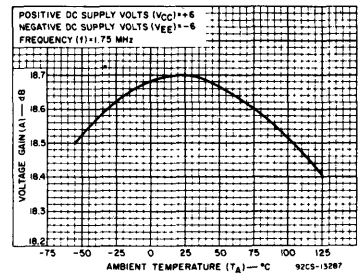


Fig. 7 - Differential voltage gain vs. temperature.

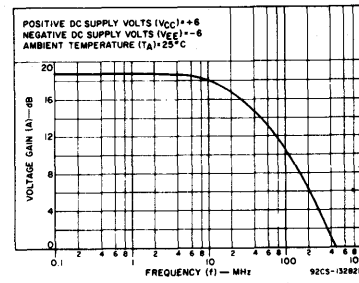


Fig. 8 - Differential voltage gain vs. frequency.

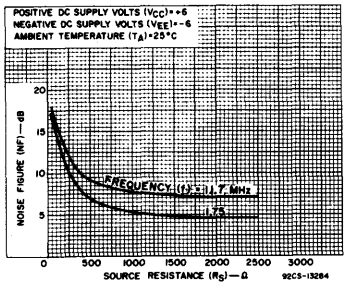


Fig. 9 - Noise figure vs. source resistance and frequency.

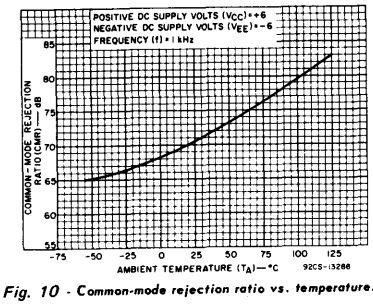


Fig. 10 - Common-mode rejection ratio vs. temperature.

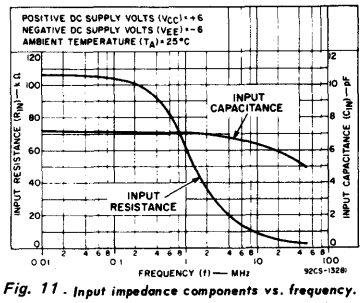
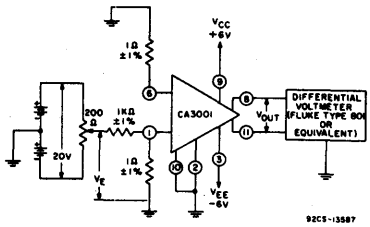


Fig. 11 - Input impedance components vs. frequency.



1. Adjust V_E for $V_{OUT(DC)} = 0 \pm 0.1$ V. 2. Measure V_E and record input offset voltage (V_{IO}) in mV as $V_{IO} = \frac{V_E}{1000}$

Fig. 12 - Input offset voltage test circuit.

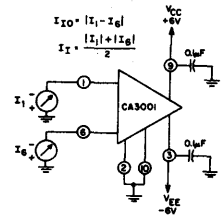
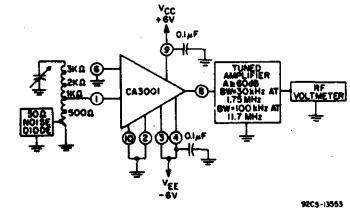
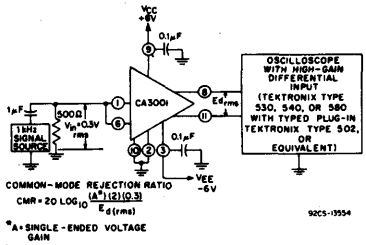


Fig. 13 - Input offset current and input bias current test circuit.



* Separate tuned input circuits are used for 1.75 MHz and 11.7 MHz. Source-resistance matching taps adjusted with circuit tuned to resonance and with 50-ohm resistor connected to simulate noise diode.

Fig. 14 . Noise figure test circuit.



*A = SINGLE-ENDED VOLTAGE GAIN

Fig. 15 - Common-mode rejection ratio test circuit.

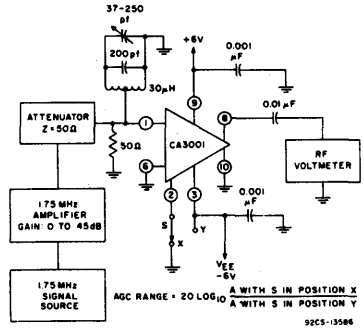


Fig. 16 - AGC range test circuit.

CA3002

IF Amplifier

- Designed for use in Communication Equipment
- Balanced differential amplifier configuration with controlled constant-current source provides outstanding versatility
- 10-Lead hermetic TO-5 style package
- Built-in temperature stability for operation from -55°C to +125°C
- Companion Application Note ICAN-5036 "Application of the RCA-3002 Integrated-Circuit IF Amplifier" covers different operating modes, cross modulation, gain control, 4-stage amplifier design, and an envelope and product detector analysis.

HIGHLIGHTS

- Input Resistance . . . . . 100 kΩ typ.
- Output Resistance . . . . . 70 Ω typ.
- Voltage Gain . . 24 dB typ. @ 1.75 MHz
- Push-Pull Input, Single-Ended Output
- -3 dB Bandwidth . . . . . 11 MHz typ.
- AGC Range . . . . . 80 dB typ.
- Useful Frequency Range DC to . . 15 MHz

APPLICATIONS

- Product Detector
- AM Detector
- IF & Video Amplifier
- Schmitt Trigger

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS, at T_A = 25°C

COMMON-MODE INPUT SIGNAL VOLTAGE	±2 V
MAXIMUM POWER SUPPLY VOLTAGE	16 V or ±8 V
OPERATING-TEMPERATURE RANGE	-55°C to +125°C
STORAGE-TEMPERATURE RANGE	-65°C to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C
MAXIMUM INPUT-SIGNAL VOLTAGE	± 4 V
MAXIMUM DEVICE DISSIPATION:	
-55 to 85°C	450 mW
Above 85°C	Derate linearly 5 mW/°C

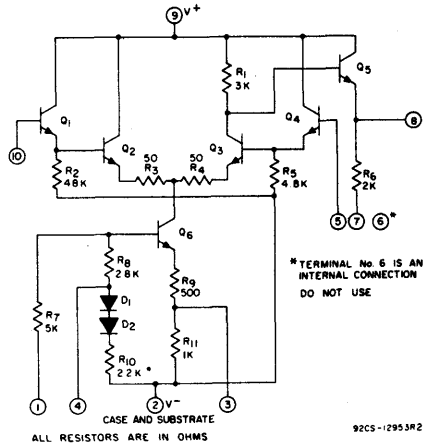


Fig. 1 — Schematic diagram.

STATIC CHARACTERISTICS AND TEST CIRCUITS

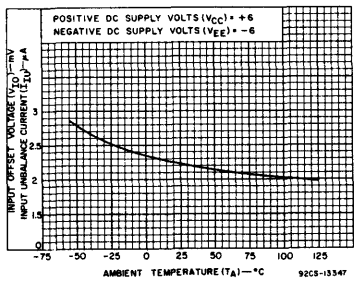


Fig.2 - Input unbalance voltage & current vs temperature.

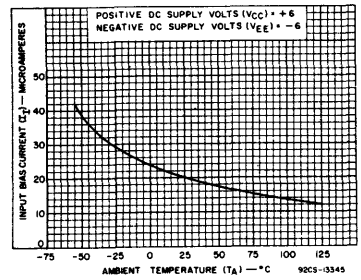


Fig.3 - Input bias current vs temperature.

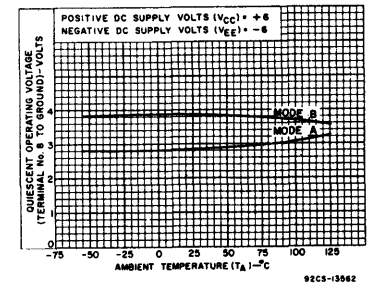


Fig.4 - Quiescent operating voltage vs temperature.

STATIC CHARACTERISTICS AND TEST CIRCUITS

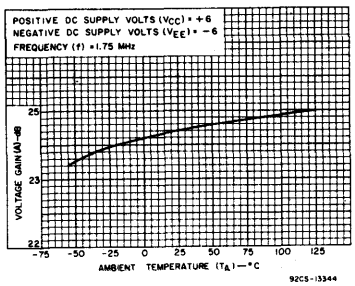


Fig. 5a - Differential voltage gain vs temperature.

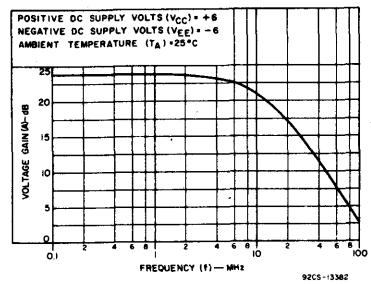


Fig. 5b - Differential voltage gain vs frequency.

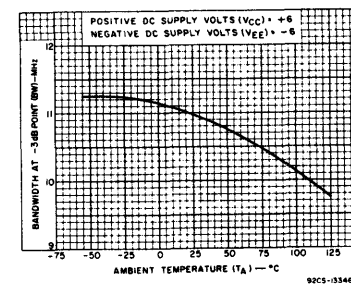


Fig. 6 - Bandwidth at -3 dB point vs temperature.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C, V_{CC} = +6 V, V_{EE} = -6 V

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS TERMINALS No.3 & No.4 NOT CONNECTED UNLESS OTHERWISE NOTED	TEST CIRCUITS	LIMITS CA3002				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS:								
Input Offset Voltage	V_{IO}		4	-	2.2	-	mV	2
Input Unbalance Current	I_{IU}			-	2.2	10	μA	2
Input Bias Current	I_I			-	20	35	μA	3
Quiescent Operating Voltage		MODE A: TERMINAL 2 = V_{EE}, TERMINAL 4 = NC		-	2.8	-	V	4
		MODE B: TERMINAL 2 = V_{EE}, TERMINAL 4 = V_{EE}		-	3.9	-	V	4
Device Dissipation	P_T			-	55	-	mW	None
DYNAMIC CHARACTERISTICS:								
Differential Voltage Gain (Single-Ended Input and Output)	A_{DIFF}	V_{IN} = 10 mV, f = 1.75 MHz, R_S = 50Ω		19	24	-	dB	5 & 5
Bandwidth at -3 dB Point	BW	R_S = 50Ω, V_{IN} = 10 mV		-	11	-	MHz	6
Maximum Output Voltage Swing	$V_{OUT(P-P)}$	-		-	5.5	-	V_{P-P}	None
Noise Figure	NF	f = 1.75 MHz, R_S = 1 kΩ	8	-	4	8	dB	7
Input Impedance Components: Parallel Input Resistance	R_{IN}	f = 1.75 MHz	None	-	100k	-	Ω	None
Parallel Input Capacitance	C_{IN}	f = 1.75 MHz	None	-	4	-	pF	None
Output Resistance	R_{OUT}	f = 1.75 MHz	14	-	70	-	Ω	9a & 9b
AGC Range (Maximum Voltage Gain to Complete Cutoff	AGC	f = 1.75 MHz	13	60	80	-	dB	12

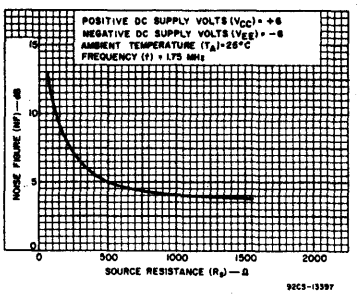
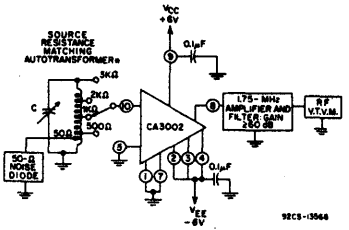


Fig. 7 - Noise figure vs source resistance.



*Taps are adjusted to provide indicated equivalent values of R_S with tank tuned to resonance at 1.75 MHz, and a 50-Ω resistor connected to simulate the noise diode.

Fig. 8 - Noise figure.

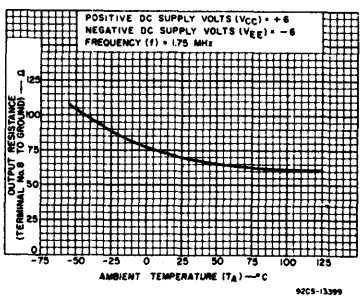


Fig. 9a - Output resistance vs temperature.

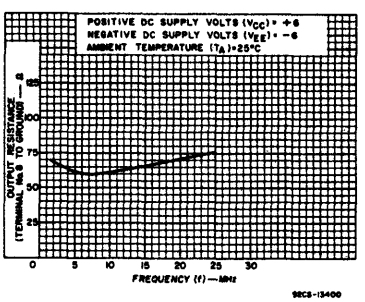


Fig. 9b - Output resistance vs frequency.

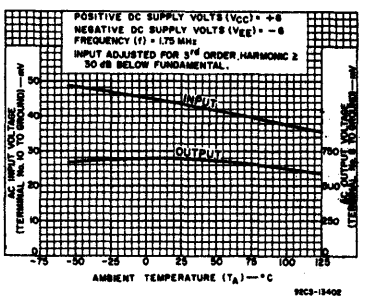
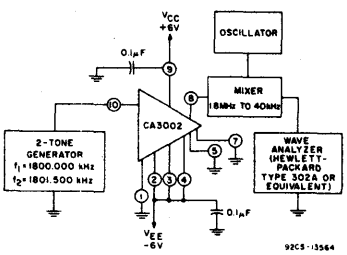


Fig. 10 - Input level for — 30 dB intermodulation vs. temperature



1) Increase both input-signal tones until the $2f_2$-f_1 and $2f_1$-f_2 output-signal voltages are 30 dB below the f_1 and f_2 output-signal voltages.
2) Measure rms values of the input and output signal voltages.
3) The measured input signal voltage is that value when the 3rd-harmonic intermodulation products are 30 dB below the fundamental outputs.

Fig. 11 - Intermodulation Test Circuit.

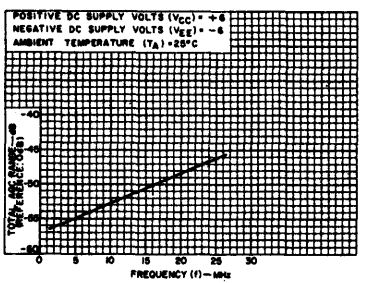
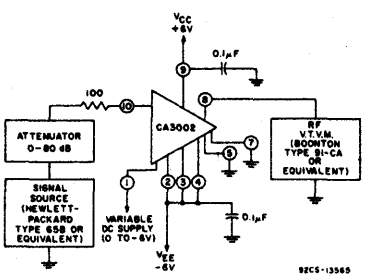


Fig. 12 - AGC range vs frequency.



1) Set attenuator at 80 dB attenuation.
2) Set variable dc supply voltage at 0 V.
3) Increase signal input voltage until RF V.T.V.M. indicates 5 mV output.
4) Set variable dc supply voltage at -6 V.
5) Adjust attenuator until RF V.T.V.M. again indicates 5 mV output.
6) Change in attenuator setting in dB is total AGC Range.

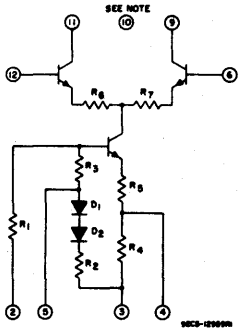
Fig. 13 - AGC range.

CA3004

RF Amplifier

- Designed for use in Communications Equipment
- Balanced Differential-Amplifier Configuration with Controlled Constant-Current Source Provides Unexcelled Versatility
- Built-in Temperature Stability for Operation from -55° C to +125° C
- Similar to RCA CA3005 and CA3006, plus Emitter-Degeneration Resistors to Provide More Linear Transfer Characteristic and Increased Input-Signal Handling Capability
- Companion Application Note ICAN 5022 "Application of RCA CA3004, CA3005, and CA3006 Integrated Circuit RF Amplifiers", covers characteristics of different operating modes, noise performance, cross-modulation, mixer, AGC, limiter, detector, and amplifier design considerations.
- 12-Lead Hermetic TO-5 Style Package
- Push-Pull Input and Output
- Wide and Narrow-Band Amplifier
- AGC
- Detector
- Operation from DC to 100 Mc/s
- Mixer
- Limiter
- Modulator
- RF, IF, and Video Frequency Capability

SCHEMATIC DIAGRAM FOR CA3004



NOTE: Connect Terminal No. 10 to most positive dc supply voltage used for circuit.

Fig. 1

ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_{FA} = 25°C

Voltage limits shown for each terminal can be applied under the indicated circuit conditions for other terminals. All voltages are with respect to GROUND (common terminal of Positive and Negative DC Supplies)

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	NO CONNECTION			
2	-9.5	0	6	0
			12	0
			3	-9.5
			9	+6
			10	+6
			11	+6
3	-12	0	2	0
			6	0
			9	+6
			10	+6
			11	+6
			12	0
4	-12	0	2	0
			6	0
			9	+6
			10	+6
			11	+6
			12	0
5	-6	0	2,6,12	0
			3	-6
			9	+6
			10	+6
			11	+6
6	-3.5	+3.5	2	0
			3	-6
			9	+6
			10	+6
			11	+6
			12	0

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
7	NO CONNECTION			
8	NO CONNECTION			
9	0	+12	2	0
			3	-6
			6	0
			10	+6
			11	+6
			12	0
10	0	+12	2	0
			3	-6
			6	0
			9	+6
			11	+6
			12	0
11	0	+12	2	0
			3	-6
			6	0
			10	+6
			11	+6
			12	0
12	-3.5	+3.5	2	0
			3	-6
			6	0
			9	+6
			10	+6
			11	+6
CASE	INTERNALLY CONNECTED TO TERMINAL NO.3 (SUBSTRATE) DO NOT GROUND			

OPERATING-TEMPERATURE RANGE . . . . . -55°C to +125°C

STORAGE-TEMPERATURE RANGE . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. . . . . +265°C

MAXIMUM SINGLE-ENDED INPUT-SIGNAL VOLTAGE . . . . . ±3.5 V

MAXIMUM COMMON-MODE INPUT-SIGNAL VOLTAGE . . . . . -2.5 V, +3.5 V

MAXIMUM DEVICE DISSIPATION . . . . . 300 mW

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPE CA3004 (Figs. 2 to 8)

INPUT OFFSET VOLTAGE AND CURRENT VS TEMPERATURE

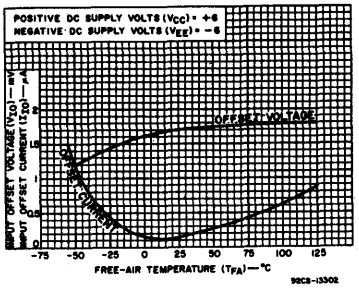


Fig. 2

INPUT BIAS CURRENT VS TEMPERATURE

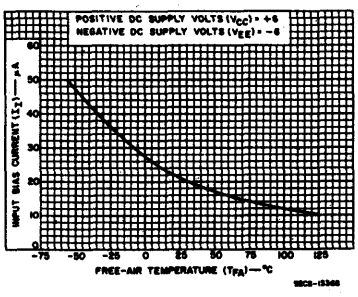


Fig. 3

ELECTRICAL CHARACTERISTICS, at T_{FA} = 25° C, V_{CC} = +6V, V_{EE} = -6 V unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.4 and No.5 Open Unless Otherwise Specified		TEST CIRCUIT	LIMITS TYPE CA3004				TYPICAL CHARACTERISTICS CURVES
				Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS									
Input Offset Voltage	V_{IO}			Fig.4	–	1.7	5	mV	Fig.2
Input Offset Current	I_{IO}			Fig.5	–	0.125	5	μA	Fig.2
Input Bias Current	I_I			Fig.5	–	21	40	μA	Fig.3
Quiescent Operating Current	I_9 or I_{11}	TERMINALS 4	5						
		NC	NC	Fig.8	–	1	–	mA	Fig.6
		V_{EE}	NC	Fig.8	–	2.7	–	mA	Fig.6
		NC	V_{EE}	Fig.8	–	0.45	–	mA	Fig.6
		V_{EE}	V_{EE}	Fig.8	–	1.25	–	mA	Fig.6
Quiescent Operating Current Ratio	I_9/I_{11}			Fig.8	–	1.1	–	–	Fig.7
Device Dissipation	P_T			Fig.8	–	26	–	mW	NONE
DYNAMIC CHARACTERISTICS									
Power Gain	G_P	f = 100 Mc/s		Fig.11	10	12	–	dB	Fig.9
Noise Figure	NF	f = 100 Mc/s		Fig.11	–	6.3	9	dB	Fig.10
Common Mode Rejection Ratio	CMR	f = 1 Kc/s		Fig.13	–	98	–	dB	Fig.12
AGC Range (Max. Voltage Gain to Complete Cutoff)	AGC	f = 1.75 Mc/s		Fig.14	-60	–	–	dB	NONE

DEFINITIONS OF TERMS

Input Offset Voltage

The difference in the dc voltages which must be applied to the input terminals to obtain equal quiescent operating voltages (zero output offset voltage) at the output terminals.

Input Offset Current

The difference in the currents at the two input terminals when the quiescent operating voltages at the two output terminals are equal.

Input Bias Current

The average value (one-half the sum) of the currents at the two input terminals when the quiescent operating voltages at the two output terminals are equal.

Quiescent Operating Current

The average (dc) value of the current in either output terminal.

Quiescent Operating Current Ratio

The ratio of the Quiescent operating currents in the two output terminals.

Device Dissipation

The total power drain of the device with no signal applied and no external load current.

Power Gain

The ratio of the signal power developed at the output of the device to the signal power applied to the input, expressed in dB.

Noise Figure

The ratio of the total noise power of the device and a resistive signal source to the noise power of the signal source alone, the signal source representing a generator of zero impedance in series with the source resistance.

Common-Mode Rejection Ratio

The ratio of the full differential voltage gain to the common-mode voltage gain.

Common-Mode Voltage Gain

The ratio of the signal voltages developed between the two output terminals to the signal voltage applied to the two input terminals connected in parallel for ac.

Differential Voltage Gain

The ratio of the change in output voltage at either output terminal with respect to ground, to a change in input voltage at either input terminal with respect to ground, with the other input terminal at ac ground.

AGC Range

The total change in voltage gain (from maximum gain to complete cutoff) which may be achieved by application of the specified range of dc voltage to the AGC input terminal of the device

INPUT OFFSET VOLTAGE TEST CIRCUIT

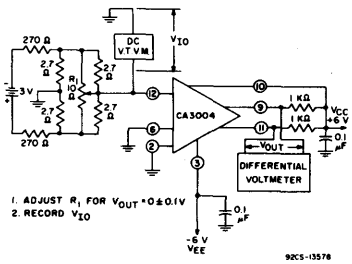


Fig.4

INPUT OFFSET CURRENT AND BIAS CURRENT TEST CIRCUIT

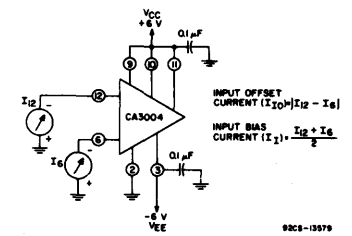


Fig.5

QUIESCENT OPERATING CURRENT VS TEMPERATURE

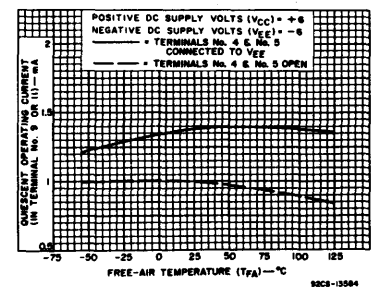


Fig.6

QUIESCENT OPERATING CURRENT RATIO VS TEMPERATURE

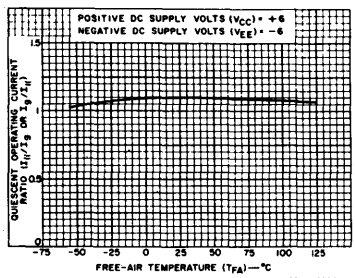
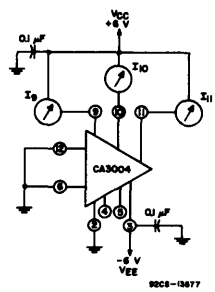


Fig.7

TEST CIRCUIT FOR TYPE CA3004

QUIESCENT OPERATING CURRENT, QUIESCENT OPERATING CURRENT RATIO, AND DEVICE DISSIPATION TEST CIRCUIT



$P_T = V_{CC}(I_9 + I_{10} + I_{11}) + V_{EE} I_3$

Fig.8

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPE CA3004 (Figs. 9 to 14)

POWER GAIN VS FREQUENCY

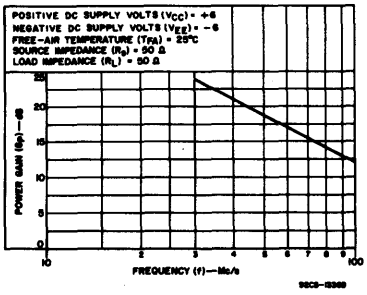


Fig. 9

NOISE FIGURE VS FREQUENCY

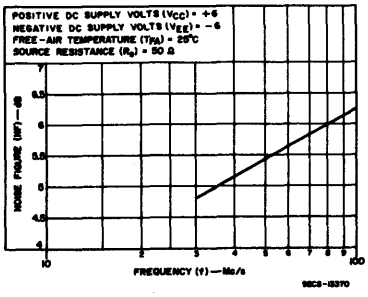


Fig. 10

100 Mc/s POWER GAIN AND NOISE FIGURE TEST CIRCUIT

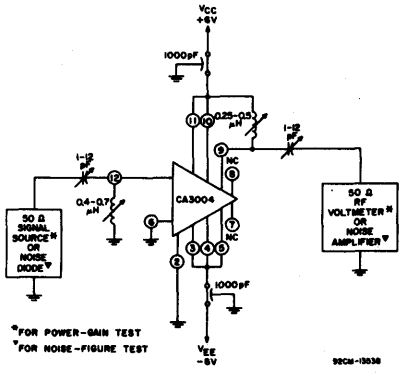


Fig. 11

COMMON-MODE REJECTION RATIO VS TEMPERATURE

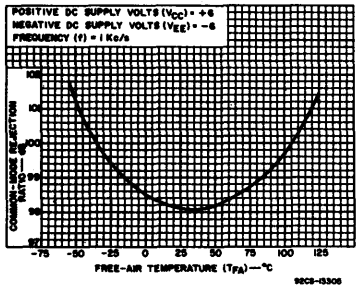


Fig. 12

COMMON-MODE REJECTION RATIO TEST CIRCUIT

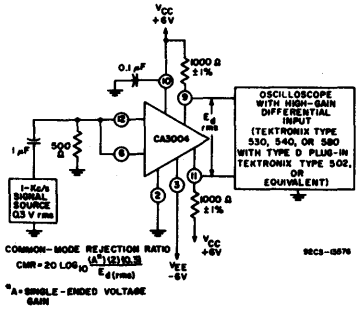


Fig. 13

AGC RANGE TEST CIRCUIT

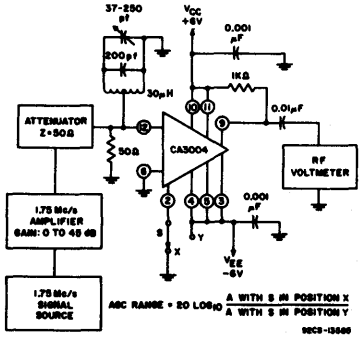
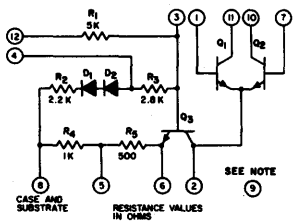


Fig. 14

RF Amplifiers

- Designed for use in Communications Equipment
- Balanced Differential Amplifier Configuration with Controlled Constant-Current Source to Provide Unexcelled Versatility
- Built-in Temperature Stability for Operation from -55° C to +125° C
- Companion Application Note, ICAN 5022 "Application of RCA CA3004, CA3005, and CA3006 Integrated Circuit RF Amplifiers", covers characteristics of different operating modes, noise performance, cross-modulation, mixer, AGC limiter, detector, and amplifier design considerations.
- 12-Lead Hermetic TO-5 Style Package.
- Push-Pull Input and Output
- Wide and Narrow Band Amplifier
- AGC
- Detector
- RF, IF, and Video Frequency Capability
- Operation from DC to 100 MHz
- Mixer
- Limiter
- Modulator
- Cascode Amplifier

SCHEMATIC DIAGRAM FOR CA3005 AND CA3006



NOTE: Connect Terminal No.9 to most positive dc supply voltage used for circuit.

Fig.1

ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_{FA} = 25°C

Voltage limits shown for each terminal can be applied under the indicated voltage conditions for other terminals.
All voltages are with respect to GROUND (common terminal of Positive and Negative DC Supplies)

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-3.5	+3.5	7	0
			8	-6
			9	+6
			10	+6
			11	+6
			12	0
2	TEST POINT: DO NOT APPLY VOLTAGE FROM EXTERNAL SOURCE			
3	-9.5	0	1	0
			7	0
			8	-9.5
			9	+6
			10	+6
			11	+6
			12	0
4	-6	0	1	0
			7	0
			8	-6
			9	+6
			10	+6
			11	+6
			12	0
5	-12	0	1	0
			7	0
			9	+6
			10	+6
			11	+6
			12	0
6	-6	0	1	0
			7	0
			9	+6
			10	+6
			11	+6
			12	-6
7	-3.5	+3.5	1	0
			8	-6
			9	+6
			10	+6
			11	+6
			12	0

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
8	-12	0	1	0
			7	0
			9	+6
			10	+6
			11	+6
			12	0
9	0	+12	1	0
			7	0
			8	-6
			10	+6
			11	+6
			12	0
10	0	+12	1	0
			7	0
			8	-6
			9	+6
			11	+6
			12	0
11	0	+12	1	0
			7	0
			8	-6
			9	+6
			10	+6
			12	0
12	-9.5	0	8	-9.5
			9	+6
			10	+6
			11	+6
CASE	Internally connected to Terminal No.8 (substrate) DO NOT GROUND			

OPERATING-TEMPERATURE RANGE . . . . . -55°C to +125°C
STORAGE-TEMPERATURE RANGE . . . . . -65°C to +150°C
LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C
MAXIMUM SINGLE-ENDED INPUT-
SIGNAL VOLTAGE . . . . . ±3.5 V
MAXIMUM COMMON-MODE INPUT-
SIGNAL VOLTAGE . . . . . -2.5 V, +3.5 V
MAXIMUM DEVICE DISSIPATION . . . . . 300 mW

INPUT OFFSET VOLTAGE AND CURRENT

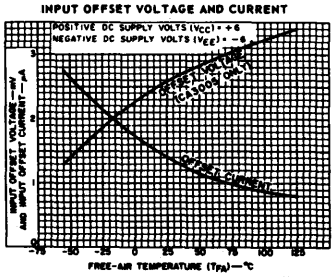


Fig.2

INPUT OFFSET VOLTAGE TEST CIRCUIT

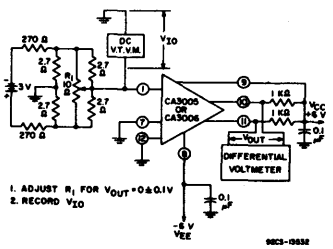


Fig.3

INPUT BIAS CURRENT

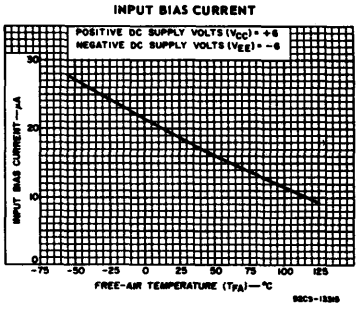


Fig.4

ELECTRICAL CHARACTERISTICS, at T_A = 25° C, V_{CC} = +6V, V_{EE} = -6V

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.3,4,5, and 6 Not Connected Except Where Noted	TEST CIRCUITS	LIMITS TYPE CA3005			LIMITS TYPE CA3006				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
STATIC CHARACTERISTICS											
Input Offset Voltage	V_{IO}		Fig.3	–	2.6	5	–	0.8	1	mV	Fig.2
Input Offset Current	I_{IO}		Fig.4	–	1.4	–	–	1.4	–	μA	Fig.2
Input Bias Current	I_{IB}		Fig.4	–	19	40	–	19	40	μA	Fig.5
Quiescent Operating Current	I_{10} or I_{11}	TERMINALS 4 / 5: NC / NC	Fig. 10	–	1	–	–	1	–	mA	Fig.6
		NC / -VEE	Fig. 10	–	2.7	–	–	2.7	–	mA	NONE
		-VEE / NC	Fig. 10	–	0.45	–	–	0.45	–	mA	NONE
		-VEE / -VEE	Fig. 10	–	1.25	–	–	1.25	–	mA	Fig.6
Quiescent Operating Current Ratio	$\frac{I_{10}}{I_{11}}$		Fig. 10	–	1.05	–	–	1.05	–	–	Fig.7
Device Dissipation	P_T		Fig. 10	–	26	–	–	26	–	mW	NONE
DYNAMIC CHARACTERISTICS											
Power Gain	G_p	f = 100 MHz, Cascode Configuration	Fig. 11	16	20	–	16	20	–	dB	Fig.9
		f = 100 MHz, Differential-Ampl. Configuration	Fig.12	14	16	–	14	16	–	dB	Fig.11
Noise Figure	NF	f = 100 MHz, Cascode Configuration	Fig. 11	–	7.8	9	–	7.8	9	dB	Fig.13
		f = 100 MHz, Differential Ampl. Configuration	Fig.12	–	7.8	9	–	7.8	9	dB	Fig.14
Common-Mode Rejection Ratio	CMR	f = 1 kHz		–	101	–	–	101	–	dB	Fig.15
AGC Range (Max. Voltage Gain to Complete Cutoff)	AGC	f = 1.75 MHz		-60	–	–	-60	–	–	dB	NONE

POWER-GAIN (CASCODE CONFIGURATION)

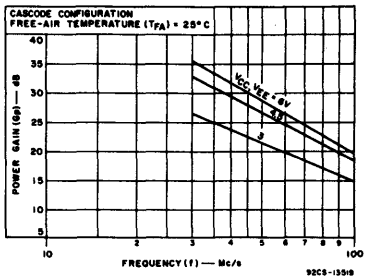


Fig. 7

POWER-GAIN (DIFFERENTIAL-AMPLIFIER CONFIGURATION)

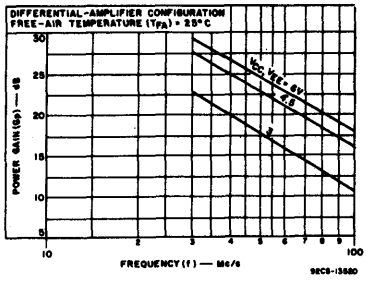
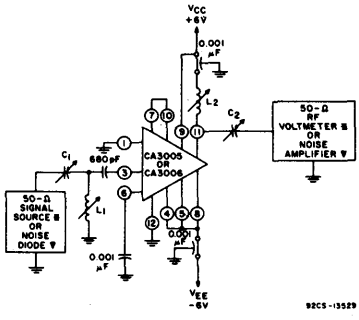


Fig. 8

NOISE FIGURE AND POWER GAIN TEST CIRCUIT (CASCODE CONFIGURATION)



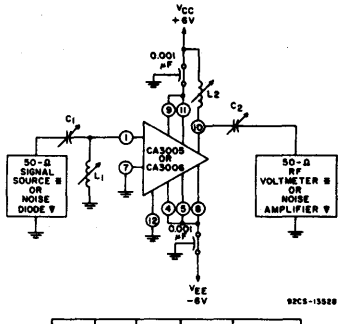
f Mc/s	C_1 pF	C_2 pF	L_1 μH	L_2 μH
30	14-150	5-40	0.3-0.6	0.8-1.4
100	5-40	5-40	0.07-0.12	0.15-0.3

* FOR POWER-GAIN TEST

▼ FOR NOISE-FIGURE TEST

Fig. 10

NOISE FIGURE AND POWER-GAIN TEST CIRCUIT (DIFFERENTIAL AMPLIFIER CONFIGURATION)



f Mc/s	C_1 pF	C_2 pF	L_1 μH	L_2 μH
30	5-40	1.5-20	1.2-2	1.2-2
100	1-12	1-12	0.4-0.7	0.25-0.5

* FOR POWER-GAIN TEST

▼ FOR NOISE-FIGURE TEST

Fig. 11

QUIESCENT OPERATING CURRENT

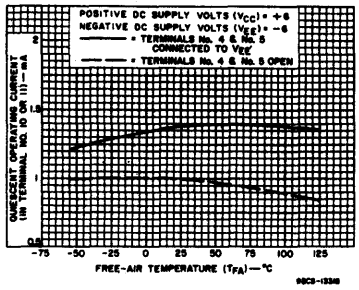


Fig. 5

QUIESCENT OPERATING CURRENT RATIO

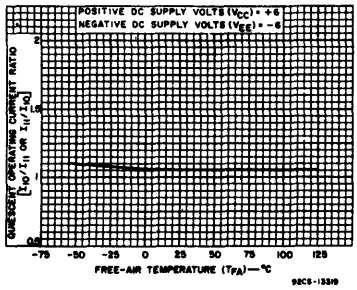


Fig. 6

100-Mc/s NOISE FIGURE VS. V_{EE} (CASCODE CONFIGURATION)

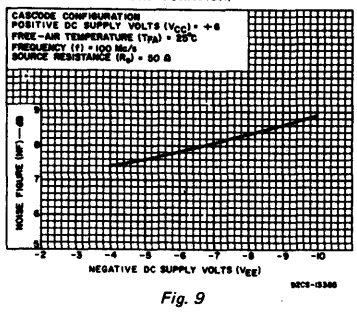


Fig. 9

100 Mc/s NOISE FIGURE VS. V_{EE} (DIFFERENTIAL AMPLIFIER CONFIGURATION)

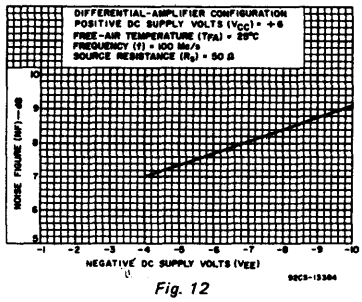
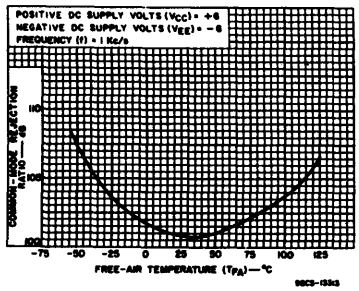


Fig. 12

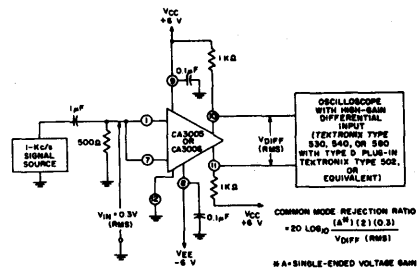
COMMON-MODE-REJECTION RATIO



92CS-13343

Fig. 13

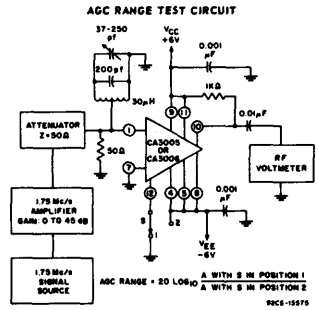
COMMON-MODE-REJECTION RATIO TEST CIRCUIT



92CM-13134

Fig. 14

AGC RANGE TEST CIRCUIT



92CS-15575

Fig. 15

CA3007

AF Amplifier

- Designed for use in Sound Systems and Communication Equipment
- Balanced differential-amplifier configuration with controlled constant-current source provides for both audio amplification and phase inversion
- Built-in temperature stability for operation from -55°C to +125°C
- Eliminates need for audio driver transformer
- Companion Application Note, ICAN 5037 "Application of the RCA-CA3007 Integrated Circuit Audio Driver" covers design of a dual supply audio driver in a direct-coupled audio amplifier, and a single supply audio driver in a capacitor-coupled audio amplifier
- **Supplied in the hermetic 12-lead TO-5 style package**

HIGHLIGHTS

- Input Impedance. . . . 4 kΩ typ.
- Output Impedance . . . 60 Ω typ.
- Power Gain . . . . . 22 dB typ.
- Push-Pull Input & Output
- Direct Coupling to Class B Audio Output Stage

APPLICATIONS

- Audio Amplifier
- Audio Driver

OPERATING-TEMPERATURE RANGE	-55°C to +125°C
STORAGE-TEMPERATURE RANGE	-65°C to +150°C
LEAD TEMPERATURE (During Soldering) At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C
MAXIMUM SINGLE-ENDED INPUT-SIGNAL VOLTAGE	±2.5 V
MAXIMUM COMMON-MODE INPUT-SIGNAL VOLTAGE	±2.5 V
MAXIMUM DEVICE DISSIPATION	300 mW

ELECTRICAL CHARACTERISTICS, at T_{FA} = 25°C, V_{CC} = +6 V, V_{EE} = -6 V,

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Pin 4 Not Connected Unless Otherwise Noted	TEST CIRCUITS	LIMITS TYPE CA3007				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS								
Input Unbalance Voltage	V_{IU}		3	–	0.57	5	mV	2
Input Unbalance Current	I_{IU}		3	–	0.57	5	μA	2
Input Bias Current	I_I		3	–	11	34	μA	4
Quiescent Operating Voltage	V_8 or V_{10}		3	–	0.87	–	V	5
Device Dissipation	P_T		3	–	30	–	mW	NONE
DYNAMIC CHARACTERISTICS								
Power Gain	G_P	f = 1 Kc/s	6	20	22	–	dB	NONE
Total Harmonic Distortion	THD	f = 1 Kc/s	6	–	0.28	–	%	NONE
Input Impedance	Z_{IN}	f = 1 Kc/s	7	–	4K	–	Ω	NONE
Common-Mode Rejection Ratio	CMR	f = 1 Kc/s	9(A) 9(B)	–	77	–	dB	8

SCHEMATIC DIAGRAM

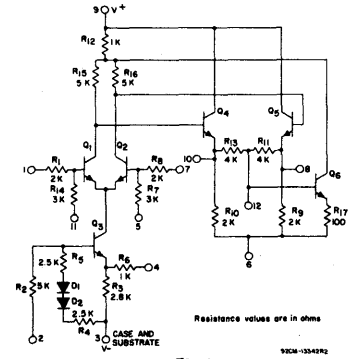


Fig.1

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUIT FOR CA3007

INPUT UNBALANCE VOLTAGE AND CURRENT vs TEMPERATURE

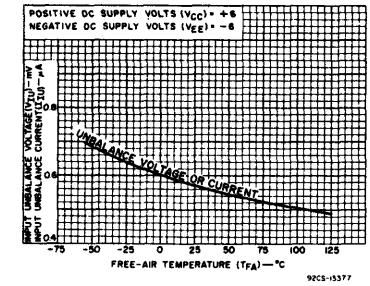
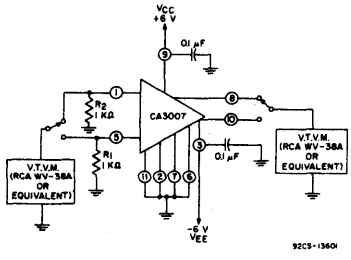


Fig.2

INPUT UNBALANCE VOLTAGE & CURRENT, INPUT BIAS CURRENT, QUIESCENT OPERATING VOLTAGE, AND DEVICE DISSIPATION TEST CIRCUIT



R_1 and R_2 matched to ±1%.

$P_T = V_{CC}I_9 + V_{EE}I_3$

I_9 = Direct Current into Terminal No.9

I_3 = Direct Current out of Terminal No.3

Fig.3

INPUT BIAS CURRENT vs TEMPERATURE

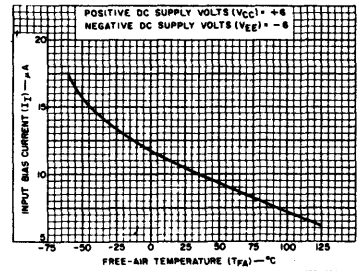


Fig.4

QUIESCENT OPERATING VOLTAGE vs TEMPERATURE

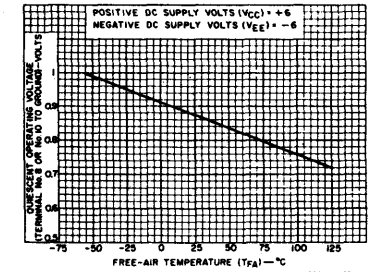


Fig.5

ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_A = 25° C

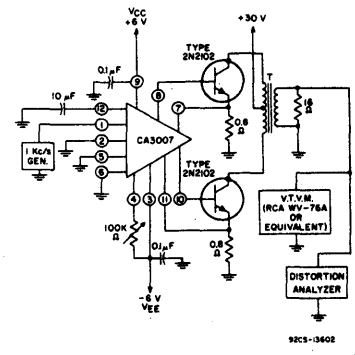
Indicated voltage limits for each terminal can be applied under the specified operating conditions for other terminals. All voltages are with respect to ground ($-V_{CC}$, $+V_{EE}$, or common terminal of Positive and Negative DC supplies).

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-2.5	+2.5	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
2	-8	0	3	-8
			6	0
			7	0
			9	+6
			11	0
3	-10	0	6	0
			7	0
			9	+6
			11	0
4	-8.5	0	6	0
			7	0
			9	+6
			11	0
5	-2.5	+2.5	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
6	-3	0	2	0
			3	-6
			7	0
			9	+6
			11	0
7	-2.5	+2.5	1	0
			2	0
			3	-6
			5	0
			6	0
			9	+6

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
8	-2	0	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
9	0	+10	2	0
			3	-6
			6	0
			7	0
			11	0
10	-2	0	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
11	-2.5	+2.5	1	0
			2	0
			3	-6
			6	0
			7	0
			9	+6
12	-2	0	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
CASE	INTERNALLY CONNECTED TO TERMINAL No.3 (SUBSTRATE) DO NOT GROUND			

TYPICAL DYNAMIC CHARACTERISTIC AND TEST CIRCUITS FOR CA3007

POWER GAIN AND TOTAL HARMONIC DISTORTION TEST CIRCUIT



T (Output Transformer):
Primary Impedance = 2000 Ω C.T.
Secondary Impedance = 16 Ω
Efficiency = 45% approx.
(STANCOR TYPE TA-10 OR EQUIVALENT)

Fig.6

INPUT IMPEDANCE TEST CIRCUIT

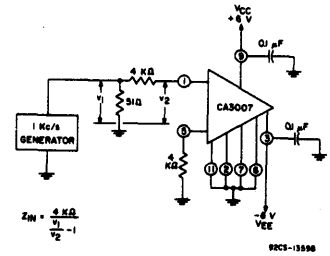


Fig.7

COMMON-MODE REJECTION RATIO vs TEMPERATURE

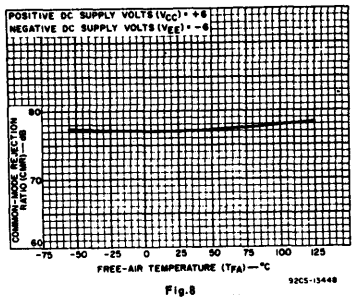
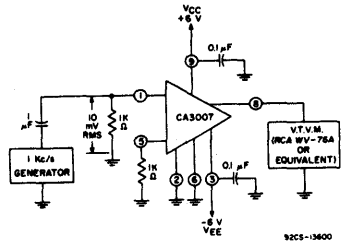
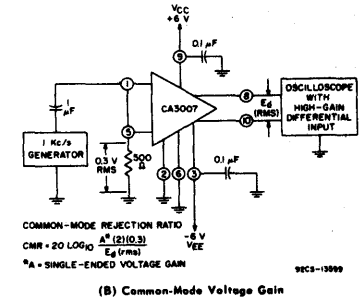


Fig.8

COMMON-MODE REJECTION-RATIO TEST CIRCUITS



(A) Single-Ended Differential Voltage Gain



(B) Common-Mode Voltage Gain

Fig.9

CA3008, CA3010, CA3015, CA3016, CA3029,CA3030, CA3037, CA3038

Operational Amplifiers

6-VOLT TYPES	12-VOLT TYPES	PACKAGE
CA3008	CA3016	14-Lead Flat Pack
CA3010	CA3015	12-Lead TO-5 Style
CA3029	CA3015L	Beam-Lead Device
CA3037	CA3030	14-Lead Plastic Dual In-Line (TO-116)
	CA3038	14-Lead Ceramic Dual In-Line (TO-116)

HIGHLIGHTS

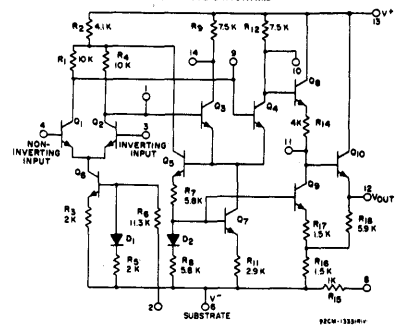
	6 V Types	12 V Types	
• Open-Loop Voltage Gain	60	70	dB typ.
• Common-Mode Rejection Ratio	94	103	dB typ.
• Output Impedance	200	92	Ω typ.
• Input Offset Voltage	1	1	mV typ.
• Static Power Drain at ± 12 V	-	175	mW typ.
± 6 V	30	30	mW typ.
± 3 V	7	7	mW typ.

- All types are electrically identical within their voltage groups
- The CA3105 is available in a sealed-junction Beam-Lead version (CA3015L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications."
- Designed for use in Telemetry, Data-Processing, Instrumentation, and Communication Equipment
- Built-in temperature stability from -55°C to +125°C for flatpack, TO-5 style, and ceramic dual in-line packages; 0°C to +70°C for plastic dual in-line package
- Companion Application Notes ICAN-5290, "Integrated Circuit Operational Amplifiers"; ICAN-5213, "Application of the RCA-CA3015, CA3016 Integrated Circuit Operational Amplifiers"; and ICAN-5015, "Application of the RCA-CA3008, CA3010 Integrated Circuit Operational Amplifiers".

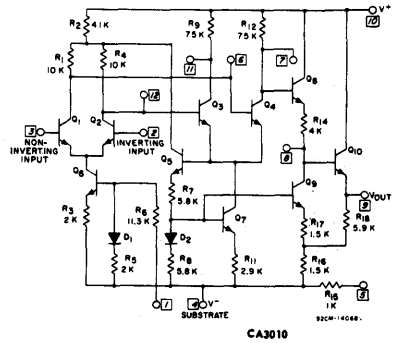
APPLICATIONS

- Narrow-Band and Band-pass Amplifier
- Operational Functions
- Feedback Amplifier
- DC and Video Amplifier
- Multivibrator
- Oscillator
- Comparator
- Servo Driver
- Scaling Adder
- Balanced Modulator-Driver

SCHEMATIC DIAGRAMS



CA3008 CA3016 CA3029 CA3030 CA3037 CA3038



CA3010 CA3015

Fig.1

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS, T_A = 25°C

Voltage or current limits shown for each terminal can be applied under the indicated voltage or other circuit conditions for other terminals

All voltages are with respect to ground (common terminal of Positive and Negative DC Supplies)

Terminal		Voltage or Current Limits		Circuit Conditions		
CA3010	CA3008 CA3029 CA3037	Negative	Positive	Terminal		Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
				CA3010	CA3008 CA3029 CA3037	
1	2	-8 V	0 V	4 10	6 13	-8 +6
2	3	-4 V	+1 V	1 3 4 10	2 4 6 13	0 0 -6 +6
3	4	-4 V	+1 V	1 2 4 10	2 3 6 13	0 0 -6 +6
	5	NO CONNECTION				
4	6	-10 V	0 V	1 10	2 13	0 +6
	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
7	10	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
9	12	30 mA		4 10 200 Ω Between Terminals 6 & 12 (CA3008, CA3029, CA3037) 4 & 9 (CA3010)	6 13	-6 +6
10	13	0 V	+10 V	1 4	2 6	0 -6
11	14	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
CASE		Internally connected to Terminal No.4, CA3010 (Substrate) DO NOT GROUND				

Terminal		Voltage or Current Limits		Circuit Conditions		
CA3015	CA3016 CA3030 CA3038	Negative	Positive	Terminal		Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
				CA3015	CA3016 CA3030 CA3038	
1	2	-16 V	0 V	4 10	6 13	-16 +12
2	3	-8 V	+1 V	1 3 4 10	2 4 6 13	0 0 -12 +12
3	4	-8 V	+1 V	1 2 4 10	2 3 6 13	0 0 -12 +12
	5	NO CONNECTION				
4	6	-20 V	0 V	1 10	2 13	0 +12
	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
7	10	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
9	12	30 mA		4 10 400 Ω Between Terminals 6 & 12 (CA3016, CA3030, CA3038) 4 & 9 (CA3015)	6 13	-12 +12
10	13	0 V	+20 V	1 4	2 6	0 -12
11	14	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
CASE		Internally connected to Terminal No.4, CA3015 (Substrate) DO NOT GROUND				

	CA3008 CA3016 CA3037	CA3010 CA3015 CA3038	CA3029 CA3030
OPERATING TEMPERATURE RANGE	-55°C to +125°C	-40°C to +85°C	
STORAGE TEMPERATURE RANGE	-65°C to +150°C	-65°C to +150°C	

	CA3016 CA3030 CA3038	CA3015	CA3008 CA3029	CA3010 CA3037
MAXIMUM SIGNAL VOLTAGE		-8 V to +1 V	-4 V to +1 V	
MAXIMUM DEVICE DISSIPATION		600 mW	300 mW	

ELECTRICAL CHARACTERISTICS at T_A = 25°C

Characteristics	Symbols	Special Test Conditions Terminal No.8 (CA3008, CA3016, CA3029, CA3030, CA3037, CA3038) Terminal No.5 (CA3010, CA3015) Not Connected Unless Otherwise Specified	Test Circuit Fig.	CA3008 CA3010 CA3029 CA3037 Min.	Typ.	Max.	CA3016 CA3015 CA3030 CA3038 Min.	Typ.	Max.	Units	Typical Characteristic Curves Fig.
STATIC CHARACTERISTICS:											
Input Offset Voltage	V_{IO}	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	4	- -	1.08 -	5 -	- -	- 1.37	- 5	mV	2
Input Offset Current	I_{IO}	= +6V = -6V = +12V = -12V	5	- -	0.54 -	5 -	- -	- 1.07	- 5	μA	2
Input Bias Current	I_{IB}	= +6V = -6V = +12V = -12V	5	- -	5.3 -	12 -	- -	- 9.6	- 24	μA	3
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V_{CC}$	= +6V = -6V = +12V = -12V	4	- -	0.10 -	1 -	- -	- 0.096	- 0.5	mV/V	none
Negative	$\Delta V_{IO}/\Delta V_{EE}$	= +6V = -6V = +12V = -12V		- -	0.26 -	1 -	- -	- 0.156	- 0.5		
Device Dissipation	P_D	= +6 V = -6 V = +12V = -12V	4	- -	30 -	- -	- -	- 175	- -	mW	none
		[5] shorted to [9] V_{CC} = +6V, V_{EE} = -6V 8 shorted to 12 V_{CC} = +12V, V_{EE} = -12V		- -	102 -	- -	- -	- 500	- -		
DYNAMIC CHARACTERISTICS: All tests at f = 1 kHz except BW_{OL}											
Open-Loop Differential Voltage Gain	A_{OL}	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	8	57 -	60 -	- -	- 66	- 70	- -	dB	6 & 7
Open-Loop Bandwidth at -3 dB Point	BW_{OL}	= +6V = -6V = +12V = -12V	8	200 -	300 -	- -	- 200	- 320	- -	kHz	6 & 7
Common-Mode Rejection Ratio	CMRR	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	11	70 -	94 -	- -	- 80	- 103	- -	dB	12
Maximum Output-Voltage Swing	$V_{O(P-P)}$	= +6V = -6V = +12V = -12V	8	4 -	6.75 -	- -	- 12	- 14	- -	V_{P-P}	9 & 10
Input Impedance	Z_{IN}	= +6V = -6V = +12V = -12V	14	10 -	14 -	- -	- 5	- 7.8	- -	kΩ	13
Output Impedance	Z_{OUT}	= +6V = -6V = +12V = -12V	15	- -	200 -	- -	- -	- 92	- -	Ω	15
Common-Mode Input-Voltage Range	V_{ICR}	= +6V = -6V = +12V = -12V	11	0.5 to -4 -	- -	- -	- 0.65 to -8	- -	- -	V	none

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS

INPUT OFFSET VOLTAGE AND CURRENT

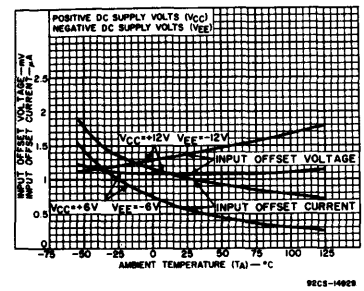


Fig.2

INPUT BIAS CURRENT

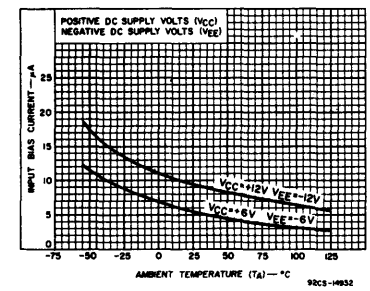


Fig.3

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038;
Italic Numbers in Square Boxes are for CA3010, CA3015

INPUT OFFSET VOLTAGE, INPUT OFFSET VOLTAGE SENSITIVITY, AND DEVICE DISSIPATION TEST CIRCUIT

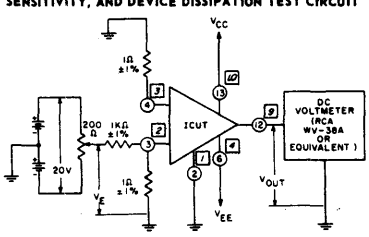


Fig.4

Procedure:

Input Offset Voltage

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Measure V_E and record Input Offset Voltage in millivolts as $V_E/1000$.

Input Offset Voltage Sensitivity

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Increase $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
3. Decrease $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
4. Divide the difference between V_{OUT} measured in steps 2 and 3 by the change in V_{CC} in steps 2 and 3.

$$\frac{V_{OUT}}{V_{CC}} = \frac{V_{OUT}\ (\text{Step 2}) - V_{OUT}\ (\text{Step 3})}{2\ \text{volts}}$$

5. Refer the reading to the input by dividing by Open Loop Voltage Gain (A_{OL}).

$$V_{IO}/V_{CC} = \frac{V_{OUT}/V_{CC}}{A_{OL}}$$

6. Repeat procedures 1 through 5 for the Negative Supply (V_{EE}).
7. Device Dissipation

$P_T = V_{CC}I_C + V_{EE}I_E$
I_C = Direct Current into Terminal 13 or 10
I_E = Direct Current out of Terminal 6 or 4

INPUT OFFSET CURRENT AND INPUT BIAS CURRENT TEST CIRCUIT

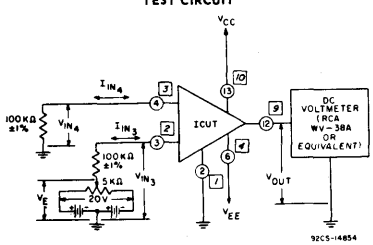


Fig.5

Procedure:

Input Bias Current and Input Offset Current

1. Adjust V_E for $|V_{OUT}| < 0.1$ V DC.
2. Measure and record V_E and V_{IN_4}.
3. Calculate the Input Bias Current using the following equation:

$$I_{I4} = \frac{V_{IN_4}}{100\ k\Omega}$$

4. Calculate the Input Offset Current using the following equation:

$$I_{IO} = V_E/100\ k\Omega$$

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038;
Italic Numbers in Square Boxes are for CA3010, CA3015

OPEN-LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3008, CA3010, CA3015, CA3016, CA3037, CA3038

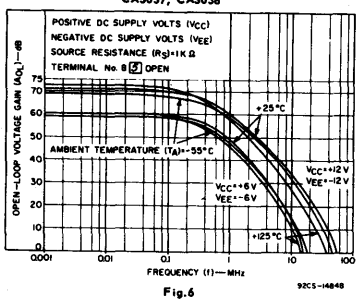


Fig.6

OPEN-LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3029 AND CA3030

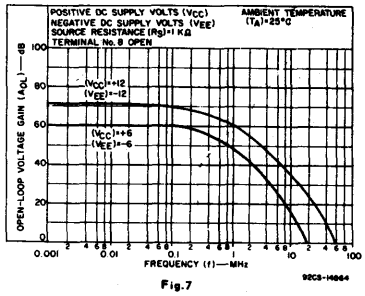
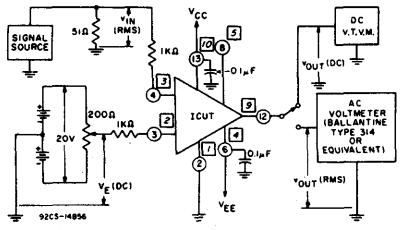


Fig.7

OPEN-LOOP DIFFERENTIAL VOLTAGE GAIN, MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE, AND OPEN-LOOP BANDWIDTH AT −3 dB POINT TEST CIRCUIT



Procedure:

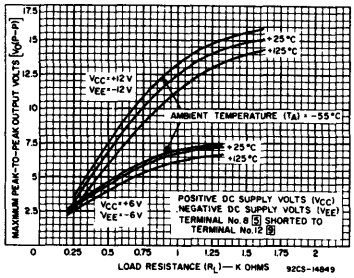
1. Adjust V_E for V_{OUT} = ±0.1 V DC.
2. Measure Open-Loop Differential Voltage Gain (A_{OL}) at f = 1 kHz.

$$A_{OL} = 20 \log_{10} \frac{V_{OUT}}{V_{IN}}$$

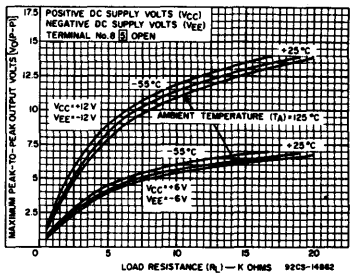
3. Measure Maximum Peak-to-Peak Output Voltage at f = 1 kHz.
4. Measure Open-Loop Bandwidth at -3 dB Point.
   Reference Level = A_{OL} at 1 kHz.

Fig.8

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3008, CA3010, CA3015, CA3016, CA3037, CA3038



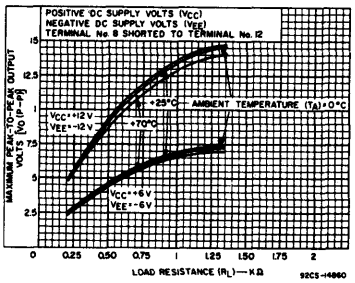
(a)



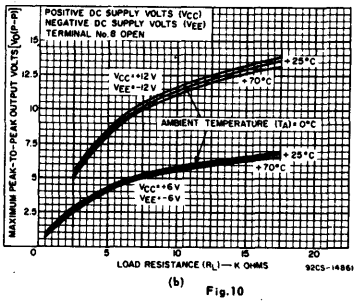
(b)

Fig.9

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3029 AND CA3030



(a)

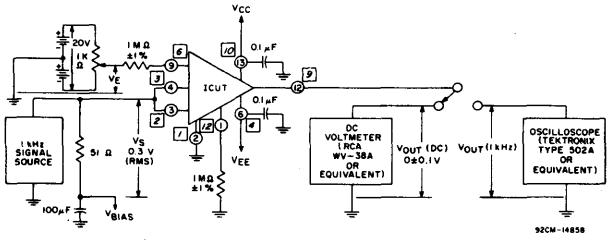


(b)

Fig.10

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038;
Italic Numbers in Square Boxes are for CA3010, CA3015

COMMON-MODE REJECTION RATIO AND COMMON-MODE INPUT-VOLTAGE-RANGE TEST CIRCUIT



Procedures:

Common-Mode Rejection Ratio:

1. Set V_{BIAS} = 0. Adjust V_E for $V_{OUT(DC)}$ = 0 ± 0.1 V.
2. Apply 1-kHz sinusodial input signal and adjust for V_S = 0.3 V (RMS).
3. Measure and record the RMS value of V_{OUT}. An oscilloscope is used for this measurement so that the output signal may be visually separated from noise output.
4. Calculate Common-Mode Voltage Gain:

$$A_{CM} = V_{OUT}/V_S$$

$$A_{CM} \text{ in dB} = -20 \log_{10} V_S/V_{OUT}$$

5. Calculate Common-Mode Rejection Ratio:

$$CMR \text{ in dB} = A_{DIFF} \text{ in dB} - A_{CM} \text{ in dB}.$$

Common-Mode Input-Voltage Range:

1. Calculate and record CMR for various positive and negative values of V_{BIAS} within the maximum limits shown on Page 2. The Common-Mode Input-Voltage Range limits are those values of V_{BIAS} at which CMR is 6 dB less than that calculated in Step 5 of the procedure given above.

Fig.11

COMMON-MODE REJECTION RATIO vs. FREQUENCY

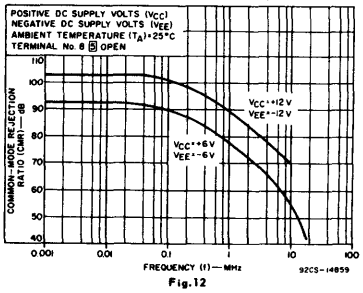


Fig.12

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038;
Italic Numbers in Square Boxes are for CA3010, CA3015

SINGLE-ENDED INPUT IMPEDANCE vs. TEMPERATURE

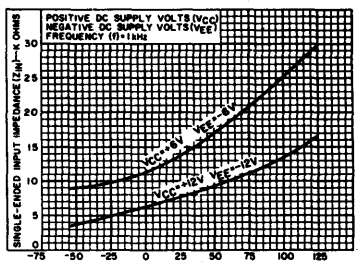


Fig.13

SINGLE-ENDED INPUT IMPEDANCE TEST CIRCUIT

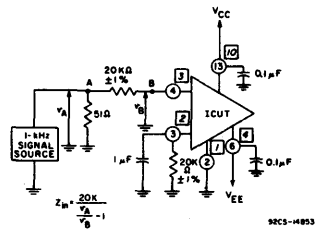


Fig.14

OUTPUT IMPEDANCE vs. TEMPERATURE

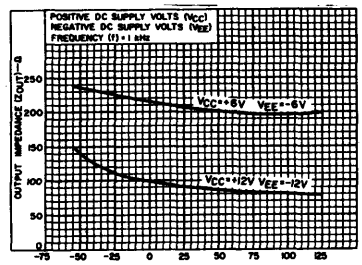


Fig.15

OUTPUT IMPEDANCE TEST CIRCUIT

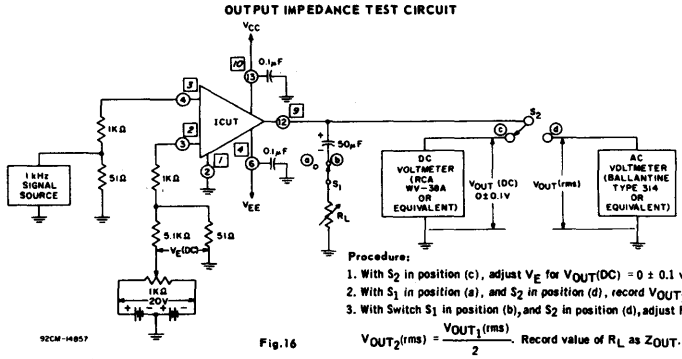


Fig.16

Procedure:

1. With S_2 in position (c), adjust V_E for $V_{OUT(DC)} = 0 \pm 0.1$ volt.
2. With S_1 in position (a), and S_2 in position (d), record $V_{OUT_1(rms)}$.
3. With Switch S_1 in position (b), and S_2 in position (d), adjust R_L until $V_{OUT_2(rms)} = \frac{V_{OUT_1(rms)}}{2}$. Record value of R_L as Z_{OUT}.

CA3008A, CA3010A, CA3015A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A

Operational Amplifiers

6-VOLT TYPES	12-VOLT TYPES	PACKAGE
CA3008A	CA3016A	14-Lead Flat Pack
CA3010A	CA3015A	12-Lead TO-5 Style
CA3029A	CA3030A	14-Lead Plastic Dual In-Line (TO-116)
CA3037A	CA3038A	14-Lead Ceramic Dual In-Line (TO-116)

- These new types have all the desirable features and characteristics of their prototypes plus lower noise figures and improved input characteristics for offset voltage, offset current, bias current, and impedance.
- All types are electrically identical within their voltage groups
- Designed for use in Telemetry, Data-Processing, Instrumentation, and Communication Equipment
- Built-in temperature stability from -55°C to +125°C for Flatpack, TO-5 style, and ceramic dual in-line packages; 0°C to +70°C for plastic dual in-line package
- Companion Application Notes ICAN-5290, "Integrated Circuit Operational Amplifiers"; ICAN-5213, "Application of the RCA-CA3015, CA3016 Integrated Circuit Operational Amplifiers"; and ICAN-5015, "Application of the RCA-CA3008, CA3010 Integrated Circuit Operational Amplifiers" cover Bode characteristics, phase compensation, frequency shaping, and amplifier design.

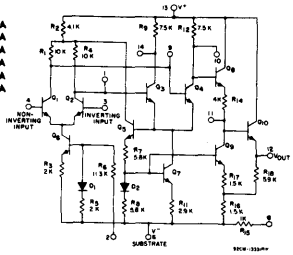
HIGHLIGHTS

	6 V Types	12 V Types	
• Open-Loop Voltage Gain	60	70	dB typ.
• Common-Mode Rejection Ratio	94	103	dB typ.
• Input Impedance	20	10	k typ.
• Input Offset Voltage	0.9	1	mV typ.
• Input Offset Current	0.3	0.5	A typ.
• Input Bias Current	2.5	4.7	A typ.
• Static Power Drain at 12 V	-	175	mW typ.
at 6 V	30	30	mW typ.
at 3 V	7	7	mW typ.

APPLICATIONS

- Narrow-Band and Band-pass Amplifier
- Operational Functions
- Feedback Amplifier
- DC and Video Amplifier
- Multivibrator
- Oscillator
- Comparator
- Servo Driver
- Scaling Adder
- Balanced Modulator-Driver

CA3008A
CA3016A
CA3029A
CA3030A
CA3037A
CA3038A



CA3010A
CA3015A

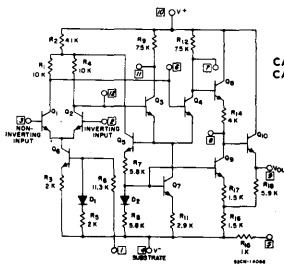


Fig.1

SCHEMATIC DIAGRAMS

ELECTRICAL CHARACTERISTICS at T_A = 25°C

Characteristics	Symbols	Special Test Conditions Terminal No.8 (CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A), Terminal No.5 (CA3010A, CA3015A) Not Connected Unless Otherwise Specified	Test Circuit Fig.	CA3008A CA3010A CA3029A CA3037A Min.	Typ.	Max.	CA3016A CA3015A CA3030A CA3038A Min.	Typ.	Max.	Units	Typical Characteristic Curves Fig.
STATIC CHARACTERISTICS:											
Input Offset Voltage	V_{IO}	V_{CC} = +6V, V_{EE} = -6V = +12V, = -12V	4	- -	0.9 -	2 -	- -	- 1	- 2	mV	2
Input Offset Current	I_{IO}	= +6V, = -6V = +12V, = -12V	5	- -	0.3 -	1.5 -	- -	- 0.5	- 1.6	μA	2
Input Bias Current	I_{IB}	= +6V, = -6V = +12V, = -12V	5	- -	2.5 -	4 -	- -	- 4.7	- 6	μA	3
Input Offset Voltage Sensitivity: Positive	$\triangle V_{IO}/\triangle V_{CC}$	= +6V, = -6V = +12V, = -12V	4	- -	0.10 -	1 -	- -	- 0.096	- 0.5	mV/V	none
Negative	$\triangle V_{IO}/\triangle V_{EE}$	= +6V, = -6V = +12V, = -12V		- -	0.26 -	1 -	- -	- 0.156	- 0.5		
Device Dissipation	P_D	= +6 V, = -6 V = +12V, = -12V	4	- -	40 -	- -	- -	- 175	- -	mW	none
		5 shorted to 9, V_{CC} = +6V, V_{EE} = -6V		-	102	-	-	-	-		
		8 shorted to 12, V_{CC} = +12V, V_{EE} = -12V		-	-	-	-	500	-		
DYNAMIC CHARACTERISTICS: All tests at f = 1 kHz except BW_{OL}											
Open-Loop Differential Voltage Gain	A_{OL}	V_{CC} = +6V, V_{EE} = -6V = +12V, = -12V	8	57 -	60 -	- -	- 66	- 70	- -	dB	6 & 7
Open-Loop Bandwidth at -3 dB Point	BW_{OL}	= +6V, = -6V = +12V, = -12V	8	200 -	300 -	- -	- 200	- 320	- -	kHz	6 & 7
Slew Rate	SR	V_{CC} = +6V V_{EE} = -6V, R_S = 1 kΩ = +12V, = -12V	none	- -	3 -	- -	- -	- 7	- -	V/μs	none
Common-Mode Rejection Ratio	CMR	V_{CC} = +6V, V_{EE} = -6V = +12V, = -12V	11	70 -	94 -	- -	- 80	- 103	- -	dB	12
Maximum Output-Voltage Swing	$V_{O(P-P)}$	= +6V, = -6V = +12V, = -12V	8	4 -	6.75 -	- -	- 12	- 14	- -	V_{P-P}	9 & 10
Input Impedance	Z_{IN}	= +6V, = -6V = +12V, = -12V	14	15 -	20 -	- -	- 7.5	- 10	- -	kΩ	13
Output Impedance	Z_{OUT}	= +6V, = -6V = +12V, = -12V	15	- -	160 -	- -	- -	- 85	- -	Ω	16
Common-Mode Input-Voltage Range	V_{ICR}	= +6V, = -6V	11	+0.5 -4	- -	-	-	-	-	V	none
		= +12V, = -12V		-	-	-	+0.65 -8	-	-		
Noise Figure	NF	V_{CC} = +3V, V_{EE} = -3V, R_S = 1 kΩ	18	-	6.3	9	-	6.3	9	dB	17
		= +6V, = -6V		-	8.3	12	-	8.3	12		
		= +9V, = -9V		-	-	-	-	10	14		
		= +12V, = -12V		-	-	-	-	11	16		

LEAD TEMPERATURE (During Soldering): **ALL TYPES**

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

CA3008A, CA3010A, CA3015A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS, T_A = 25°C

Voltage or current limits shown for each terminal can be applied under the indicated voltage or other circuit conditions for other terminals

All voltages are with respect to ground (common terminal of Positive and Negative DC Supplies)

Terminal		Voltage or Current Limits		Circuit Conditions	
CA3010A	CA3008A, CA3029A, CA3037A	Negative	Positive	Terminal	Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
				CA3010A	CA3008A, CA3029A, CA3037A
1	2	-8 V	0 V	4 10	6 13	-8 +6
2	3	-4 V	+1 V	1 3 4 10	2 4 6 13	0 0 -6 +6
3	4	-4 V	+1 V	1 2 4 10	2 3 6 13	0 0 -6 +6
-	5	NO CONNECTION				
4	6	-10 V	0 V	1 10	2 13	0 +6
-	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
7	10	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
9	12	30 mA		4 10 200 Ω Between Terminals 6 & 12 (CA3008A, CA3029A, CA3037A) 4 & 9 (CA3010A)	6 13	-6 +6
10	13	0 V	+10 V	1 4	2 6	0 -6
11	14	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
CASE		Internally connected to Terminal No.4, CA3010A (Substrate) DO NOT GROUND			

Terminal		Voltage or Current Limits		Circuit Conditions	
CA3015A	CA3016A, CA3030A, CA3038A	Negative	Positive	Terminal	Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
				CA3015A	CA3016A, CA3030A, CA3038A
1	2	-16 V	0 V	4 10	6 13	-16 +12
2	3	-8 V	+1 V	1 3 4 10	2 4 6 13	0 0 -12 +12
3	4	-8 V	+1 V	1 2 4 10	2 3 6 13	0 0 -12 +12
-	5	NO CONNECTION				
4	6	-20 V	0 V	1 10	2 13	0 +12
-	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
7	10	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL			
9	12	30 mA		4 10 400 Ω Between Terminals 6 & 12 (CA3016A, CA3030A, CA3038A) 4 & 9 (CA3015A)	6 13	-12 +12
10	13	0 V	+20 V	1 4	2 6	0 -12
11	14	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
CASE		Internally connected to Terminal No.4, CA3015A (Substrate) DO NOT GROUND			

	CA3008A, CA3010A, CA3016A, CA3015A, CA3037A, CA3038A	CA3029A, CA3030A
OPERATING TEMPERATURE RANGE	-55°C to +125°C	-40°C to +85°C
STORAGE TEMPERATURE RANGE	-65°C to +200°C	-65°C to +150°C

	CA3016A, CA3030A	CA3015A, CA3038A	CA3008A, CA3029A	CA3010A, CA3037A
MAXIMUM SIGNAL VOLTAGE	-8 V to +1 V		-4 V to +1 V	
MAXIMUM DEVICE DISSIPATION	600 mW		300 mW	

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A;
Italic Numbers are for CA3010A, CA3015A

INPUT OFFSET VOLTAGE AND CURRENT

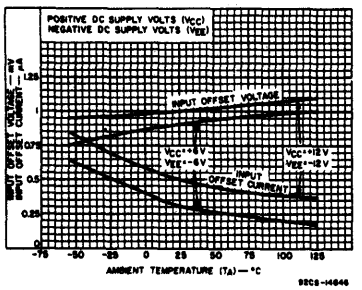


Fig.2

INPUT BIAS CURRENT

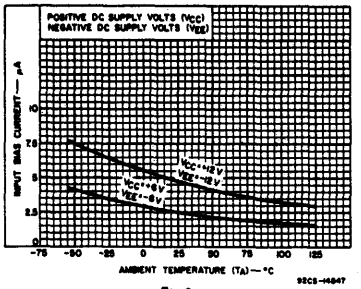


Fig.3

INPUT OFFSET VOLTAGE, INPUT OFFSET VOLTAGE SENSITIVITY, AND DEVICE DISSIPATION TEST CIRCUIT

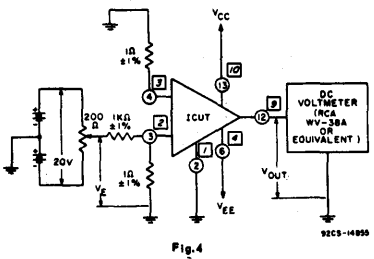


Fig.4

Procedure:

Input Offset Voltage

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Measure V_E and record Input Offset Voltage in millivolts as $V_E/1000$.

Input Offset Voltage Sensitivity

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Increase $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
3. Decrease $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
4. Divide the difference between V_{OUT} measured in steps 2 and 3 by the change in V_{CC} in steps 2 and 3.

$$\frac{V_{OUT}}{V_{CC}} = \frac{V_{OUT}\text{ (Step 2)} - V_{OUT}\text{ (Step 3)}}{2\text{ volts}}$$

5. Refer the reading to the input by dividing by Open Loop Voltage Gain (A_{OL}).

$$V_{IO}/V_{CC} = \frac{V_{OUT}/V_{CC}}{A_{OL}}$$

6. Repeat procedures 1 through 5 for the Negative Supply (V_{EE}).
7. Device Dissipation

$P_T = V_{CC}I_C + V_{EE}I_E$

I_C = Direct Current into Terminal 13 or 10

I_E = Direct Current out of Terminal 6 or 4

INPUT OFFSET CURRENT AND INPUT BIAS CURRENT TEST CIRCUIT

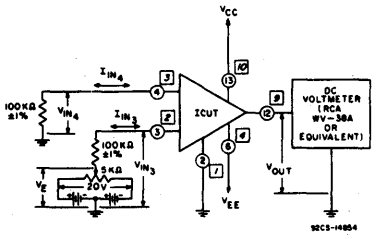


Fig.5

Procedure:

Input Bias Current and Input Offset Current

1. Adjust V_E for $|V_{OUT}| < 0.1$ V DC.
2. Measure and record V_E and V_{IN_4}
3. Calculate the Input Bias Current using the following equation:

$$I_{I4} = \frac{V_{IN_4}}{100\text{ k}\Omega}$$

4. Calculate the Input Offset Current using the following equation:

$$I_{IO} = V_E/100\text{ k}\Omega$$

OPEN LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3008A, CA3010A, CA3015A, CA3016A, CA3037A, CA3038A

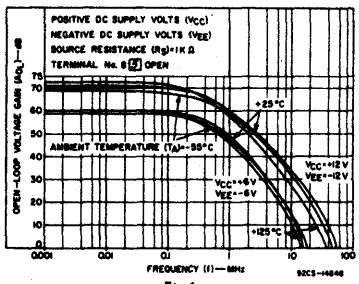


Fig.6

OPEN LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3029A AND CA3030A.

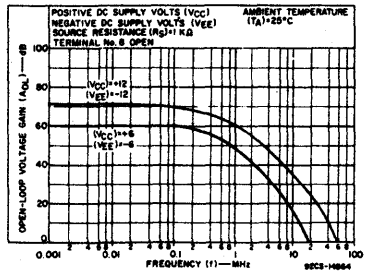
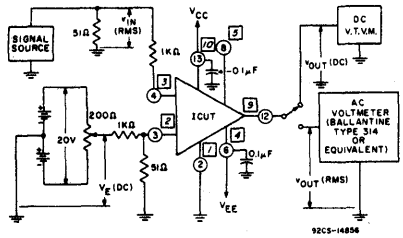


Fig.7

OPEN-LOOP DIFFERENTIAL VOLTAGE GAIN, MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE, AND OPEN-LOOP BANDWIDTH AT −3 POINT TEST CIRCUIT



Procedure:

1. Adjust V_E for V_{OUT} = ±0.1 V DC.
2. Measure Open-Loop Differential Voltage Gain (A_{OL}) at f = 1 kHz

$$A_{OL} = 20 \log_{10} \frac{V_{OUT}}{V_{IN}}$$

3. Measure Maximum Peak-to-Peak Output Voltage at f = 1 kHz
4. Measure Open-Loop Bandwidth at -3 dB Point
   Reference Level = A_{OL} at 1 kHz

Fig.8

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3008A, CA3010A, CA3015A, CA3016A, CA3037A, CA3038A

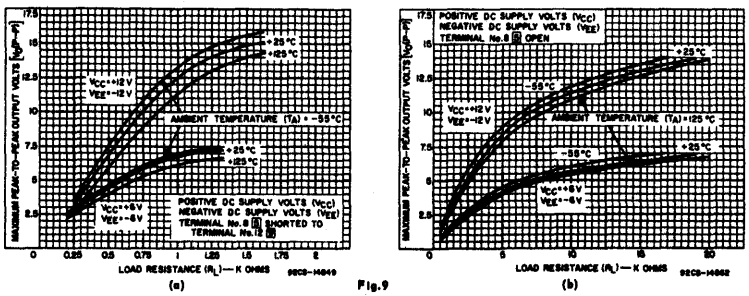


Fig.9

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3029A AND CA3030A

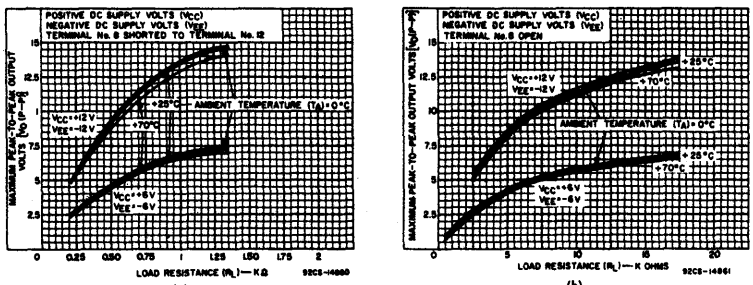


Fig.10

COMMON-MODE REJECTION RATIO AND COMMON-MODE INPUT-VOLTAGE-RANGE TEST CIRCUIT

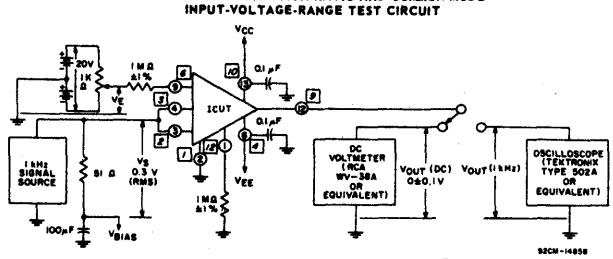


Fig.11

Procedures:

Common-Mode Rejection Ratio:

1. Set V_{BIAS} = 0. Adjust V_E for V_{OUT}(DC) = 0 ± 0.1 V.
2. Apply 1-kHz sinusoidal input signal and adjust for V_S = 0.3 V (RMS).
3. Measure and record the RMS value of V_{OUT}. An oscilloscope is used for this measurement so that the output signal may be visually separated from noise output.
4. Calculate Common-Mode Voltage Gain:

$$A_{CM} = V_{OUT}/V_S$$

$$A_{CM} \text{ in dB} = -20 \log_{10} V_S/V_{OUT}$$

5. Calculate Common-Mode Rejection Ratio:

$$CMR \text{ in dB} = A_{DIFF} \text{ in dB} - A_{CM} \text{ in dB}.$$

Common-Mode Input-Voltage Range:

1. Calculate and record CMR for various positive and negative values of V_{BIAS} within the maximum limits shown on Page 2. The Common-Mode Input-Voltage Range limits are those values of V_{BIAS} at which CMR is 6 dB less than that calculated in Step 5 of the procedure given above.

COMMON-MODE REJECTION RATIO vs. FREQUENCY

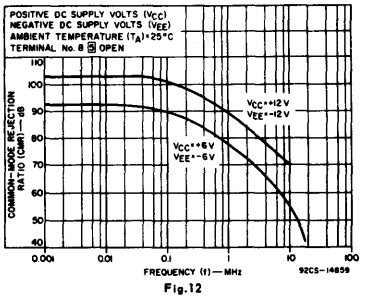


Fig.12

SINGLE-ENDED INPUT IMPEDANCE vs. TEMPERATURE

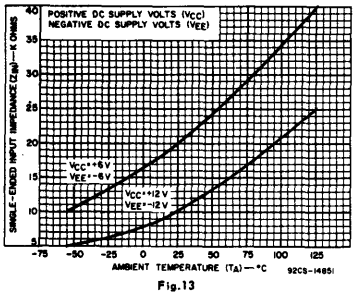


Fig.13

SINGLE-ENDED INPUT IMPEDANCE TEST CIRCUIT

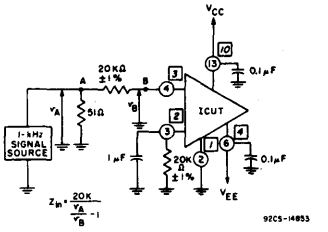


Fig.14

OUTPUT IMPEDANCE TEST CIRCUIT

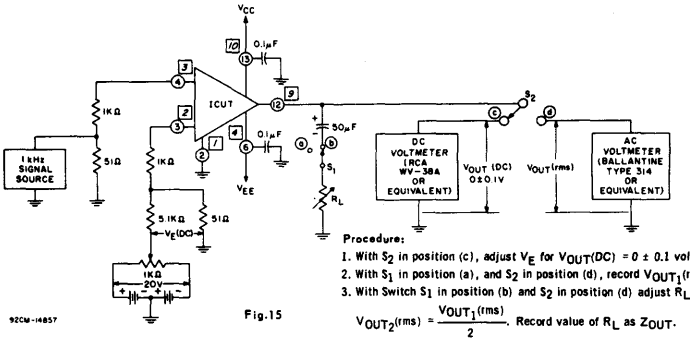
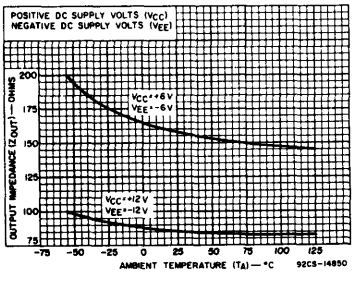


Fig.15

Procedure:

1. With S_2 in position (c), adjust V_E for $V_{OUT}(DC) = 0 \pm 0.1$ volt.
2. With S_1 in position (a), and S_2 in position (d), record $V_{OUT_1}(rms)$.
3. With Switch S_1 in position (b) and S_2 in position (d) adjust R_L until $V_{OUT_2}(rms) = \frac{V_{OUT_1}(rms)}{2}$. Record value of R_L as Z_{OUT}.



OUTPUT IMPEDANCE vs. TEMPERATURE

Fig.16

NOISE FIGURE vs. FREQUENCY

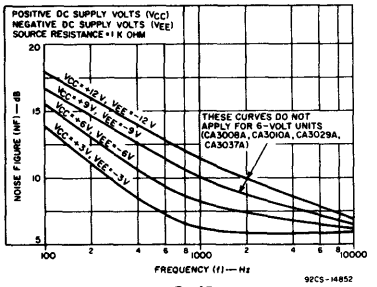


Fig.17

CA3011, CA3012

Wideband Amplifiers

FEATURES & APPLICATIONS

- exceptionally high amplifier gain: power gain at 4.5 MHz – 75 dB typ.
- excellent limiting characteristics – input limiting voltage (knee) = 600 μV typ. at 10.7 MHz
- wide frequency capability – 100 kHz to > 20 MHz
- supplied in the hermetic 10-lead TO-5 style package

ABSOLUTE-MAXIMUM VOLTAGE LIMITS AT T_A = 25° C

Indicated voltage limits for each terminal can be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 8).

NOTE: TERMINALS 6, 7, AND 9 OF RCA-CA3011 AND CA3012 ARE USED FOR INTERNAL CONNECTIONS. DO NOT APPLY VOLTAGES OR MAKE EXTERNAL CONNECTIONS TO THESE TERMINALS.

CA3011

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS						
			1	2	3	4	5	8	10
1	-3	+3	–	Same as 1	Do Not Apply External Voltage	+2.5 to +7.5	+7.5	Ground	+7.5
2	-3	+3	Same as 2	–		+2.5 to +7.5	+7.5	Ground	+7.5
3	-3	+3	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Ground	+7.5
4	+2.5	+7.5	-3 to +3	Same as 1		–	+7.5	Ground	+7.5
5	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	–	Ground	+7.5
8	-3	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Ground	+7.5
10	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Ground	–
CASE	INTERNALLY CONNECTED TO TERMINAL NO.8 (GROUND TERMINAL)								

CA3012

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS						
			1	2	3	4	5	8	10
1	-3	+3	–	Same as 1	Do Not Apply External Voltage	+2.5 to +10	+10	Ground	+10
2	-3	+3	Same as 2	–		+2.5 to +10	+10	Ground	+10
3	-3	+3	-3 to +3	Same as 1		+2.5 to +10	+10	Ground	+10
4	+2.5	+10	-3 to +3	Same as 1		–	+10	Ground	+10
5	0	+13	-3 to +3	Same as 1		+2.5 to +10	–	Ground	+10
8	-3	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Ground	+10
10	0	+13	-3 to +3	Same as 1		+2.5 to +10	+10	Ground	–
CASE	INTERNALLY CONNECTED TO TERMINAL NO.8 (GROUND TERMINAL)								

OPERATING-TEMPERATURE RANGE ..... -55 to +125° C
STORAGE-TEMPERATURE RANGE ..... -65 to +150° C
LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C
MAXIMUM INPUT-SIGNAL VOLTAGE:
Between Terminals 1 and 2 ..... ±3 V
MAXIMUM DEVICE DISSIPATION ..... 300 mW
RECOMMENDED MINIMUM DC SUPPLY VOLTAGE (V_{CC}) .. 5.5 V

Example of Use of LIMITS TABLE:

For RCA-3012, a maximum voltage of ±3 volts may be applied to Terminal 1 under the following conditions:

Terminal 2 is at the same dc potential as Terminal 1
Terminal 3: do not apply external voltage
Terminal 4 is at any dc potential between +2.5 and +10 volts
Terminal 5 is at a dc potential of +10 volts
Terminals 6, 7, and 9 are at 0 dc potential (NOT USED)
Terminal 8 is at dc ground potential
Terminal 10 is at a dc potential of +10 volts

BLOCK DIAGRAM OF TYPICAL FM RECEIVER USING RCA-CA3011 OR CA3012 INTEGRATED CIRCUIT WIDE-BAND AMPLIFIER

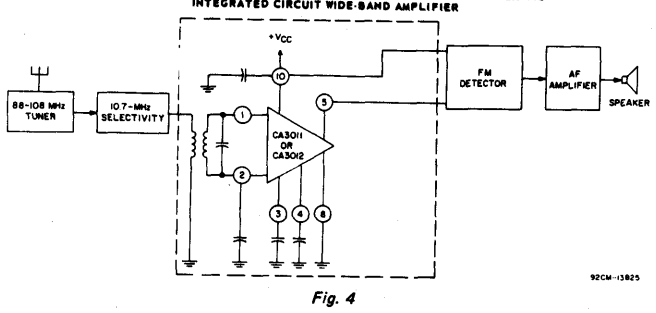


Fig. 4

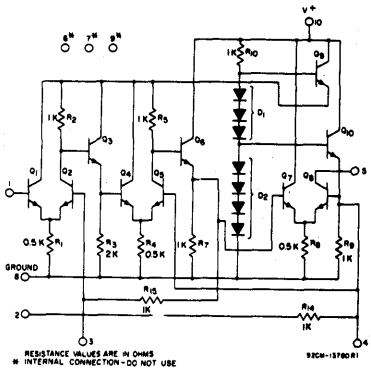


Fig. 1 – Schematic diagram for CA3011 and CA3012.

INPUT-IMPEDANCE COMPONENTS VS FREQUENCY

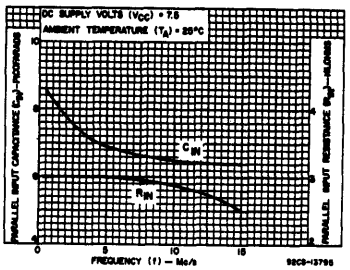


Fig. 2

OUTPUT-IMPEDANCE COMPONENTS VS FREQUENCY

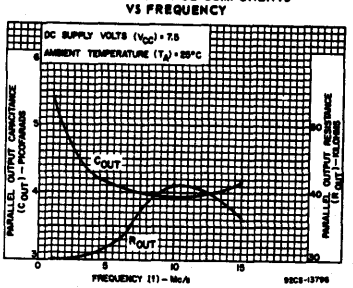


Fig. 3

VOLTAGE GAIN AND INPUT LIMITING VOLTAGE VS FREQUENCY

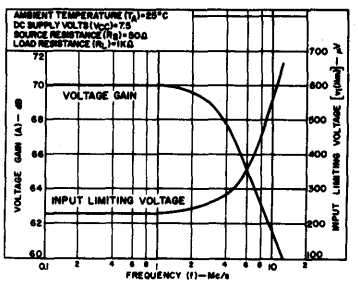


Fig. 5

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS				LIMITS						UNITS	TYPICAL CHARACTERISTICS CURVES
		SETUP & PROCEDURE	FREQUENCY f	DC SUPPLY VOLTAGE V_{CC}	AMBIENT TEMPERATURE T_A	RCA CA3011			RCA CA3012				
		Fig.	Mc/s	Volts	°C	Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
Total Device Dissipation*	P_T	6	-	6	-55	-	80	-	66	80	135	mW	
					+25	60	90	133	66	90	121	mW	
					+125	-	70	-	65	70	121	mW	
				7.5	-55	-	130	-	97	130	190	mW	
					+25	95	120	187	97	120	167	mW	
					+125	-	100	-	95	100	167	mW	
				10	-55	-	-	-	150	210	275	mW	
					+25	-	-	-	150	190	255	mW	
					+125	-	-	-	150	160	255	mW	
Voltage Gain**	A	9	1	6	-55	-	55	-	50	55	-	dB	
					+25	60	66	-	60	66	-	dB	
					+125	-	61	-	50	61	-	dB	
				7.5	-55	-	59	-	55	59	-	dB	
					+25	65	70	-	65	70	-	dB	
					+125	-	65	-	55	65	-	dB	
				10	-55	-	-	-	55	61	-	dB	
					+25	-	-	-	65	71	-	dB	
					+125	-	-	-	55	66	-	dB	
		9	4.5	7.5	+25	60	67	-	60	67	-	dB	5
			10.7	7.5	+25	55	61	-	55	61	-	dB	
Input-Impedance Components: Parallel Input Resistance	R_{IN}	7	4.5	7.5	+25	-	3	-	-	3	-	kΩ	2
Parallel Input Capacitance	C_{IN}	7	4.5	7.5	+25	-	7	-	-	7	-	pF	2
Output Impedance Components: Parallel Output Resistance	R_{OUT}	8	4.5	7.5	+25	-	31.5	-	-	31.5	-	kΩ	3
Parallel Output Capacitance	C_{OUT}	8	4.5	7.5	+25	-	4.2	-	-	4.2	-	pF	3
Noise Figure	NF	10	4.5	7.5	+25	-	8.7	-	-	8.7	-	dB	
Input Limiting Voltage (Knee)	$V_{i(lim)}$	9	4.5	7.5	+25	-	300	450	-	300	400	μV	

* The total current drain may be determined by dividing P_T by V_{CC}.

** Recommended minimum dc supply voltage (V_{CC}) is 5.5 V. Nominal load current flowing into terminal 5 is 1.5 mA at 7.5 V.

DISSIPATION TEST SETUP

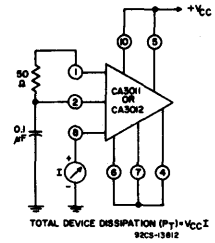


Fig. 6

INPUT-IMPEDANCE COMPONENTS TEST SETUP

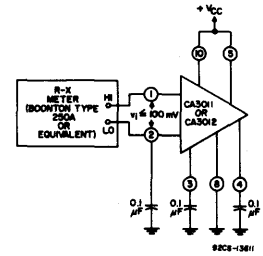


Fig. 7

OUTPUT-IMPEDANCE COMPONENTS TEST SETUP

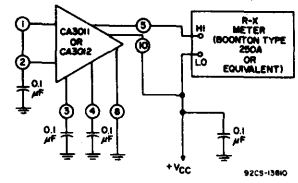
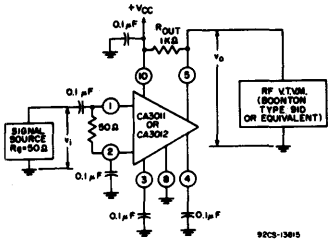


Fig. 8

VOLTAGE-GAIN TEST SETUP



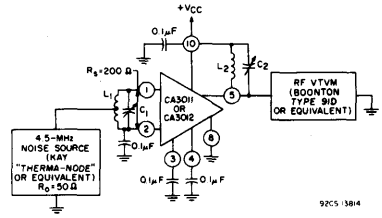
PROCEDURES

A - Voltage Gain:
1) Set input frequency at desired value, v_i = 100 μV rms.
2) Record v_o.
3) Calculate Voltage Gain A from $A = 20 \log_{10} v_o/v_i$
4) Repeat Steps 1, 2, and 3 for each frequency and/or for temperature desired.

B - Input Limiting Voltage (Knee):
1) Repeat Steps A1 and A2, using v_i = 100 mV
2) Decrease v_i to the level at which v_o is 3 dB below its value for v_i = 100 mV.
3) Record v_i as Input Limiting Voltage (Knee).

Fig. 9

NOISE FIGURE TEST SETUP



L_1 = 82 μH, center-tapped
L_2 = 2.36 μH
C_1,C_2 = Arco Type 423 padder, or equivalent

Fig. 10

CA3013, CA3014
Wideband Amplifier-Discriminators

SCHEMATIC DIAGRAM FOR CA3013 AND CA3014

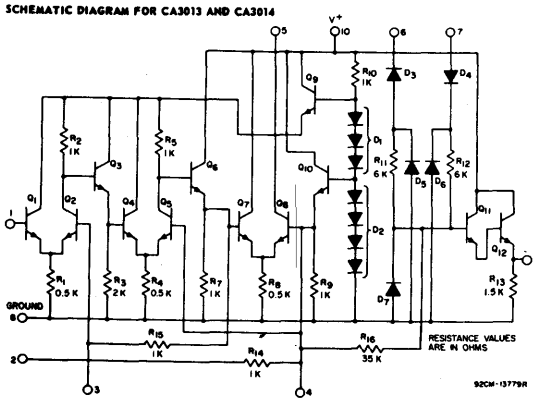


Fig. 1

BLOCK DIAGRAM OF TYPICAL TELEVISION RECEIVER USING RCA INTEGRATED-CIRCUIT SOUND-IF AMPLIFIER AND DETECTOR SECTION

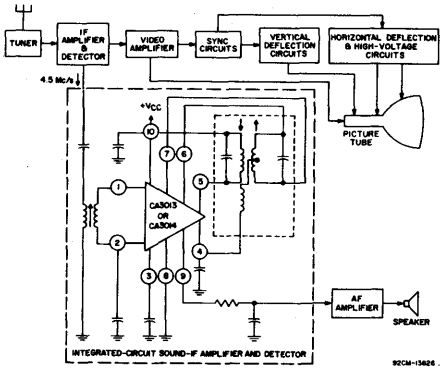


Fig. 2

FEATURES & APPLICATIONS:

- **exceptionally high gain:**
  power gain at 4.5 MHz — 75 dB typ.
- **excellent limiting characteristics — input limiting voltage (knee) = 300 μV typ. at 4.5 MHz**
- **excellent AM rejection: > 50 dB at 4.5 MHz**
- **high audio-voltage recovery — 220 mV typ. at 4.5 MHz 25 kHz deviation**
- **wide frequency capability — 100 kHz to > 20 MHz**
- **comprehensive circuit functions:**
  if amplifier, AM and noise limiter, FM detector, audio preamplifier
- **supplied in the hermetic 10-lead TO-5 style package**

TYPICAL CHARACTERISTICS AND TEST SETUPS

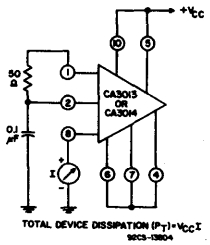


Fig.3

ABSOLUTE-MAXIMUM VOLTAGE LIMITS AT T_A = 25° C

Indicated voltage limits for each terminal can be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 8).

CA3013

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS									
			1	2	3	4	5	6	7	8	9	10
1	-3	+3	-	Same as 1	Do Not Apply External Voltage	+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
2	-3	+3	Same as 2	-		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
3	-3	+3	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
4	+2.5	+7.5	-3 to +3	Same as 1		-	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
5	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	-	Same as 4	Same as 4	Ground	AF Output	+7.5
6	+2.5	+7.5	-3 to +3	Same as 1		Same as 6	+7.5	-	Same as 4	Ground	AF Output	+7.5
7	+2.5	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	-	Ground	AF Output	+7.5
8	-3	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
9	0	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	-	+7.5
10	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	-
CASE	INTERNALLY CONNECTED TO TERMINAL No.8 (GROUND TERMINAL)											

CA3014

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS									
			1	2	3	4	5	6	7	8	9	10
1	-3	+3	-	Same as 1	Do Not Apply External Voltage	+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
2	-3	+3	Same as 2	-		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
3	-3	+3	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
4	+2.5	+10	-3 to +3	Same as 1		-	+10	Same as 4	Same as 4	Ground	AF Output	+10
5	0	+13	-3 to +3	Same as 1		+2.5 to +10	-	Same as 4	Same as 4	Ground	AF Output	+10
6	+2.5	+10	-3 to +3	Same as 1		Same as 6	+10	-	Same as 4	Ground	AF Output	+10
7	+2.5	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	-	Ground	AF Output	+10
8	-3	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
9	0	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	-	+10
10	0	+13	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	-
CASE	INTERNALLY CONNECTED TO TERMINAL No.8 (GROUND TERMINAL)											

OPERATING-TEMPERATURE RANGE	-55 to +125°C
STORAGE-TEMPERATURE RANGE	-65 to +150°C
LEAD TEMPERATURE (During Soldering): At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max	+265°C
MAXIMUM INPUT-SIGNAL VOLTAGE: Between Terminals 1 and 2	± 3 V
MAXIMUM DEVICE DISSIPATION	300 mW
RECOMMENDED MINIMUM DC SUPPLY VOLTAGE (V_{CC})	5.5 V

Example of use of LIMITS TABLE:

For RCA-CA3013, a maximum voltage of ±3 volts may be applied to Terminal 1 under the following conditions:

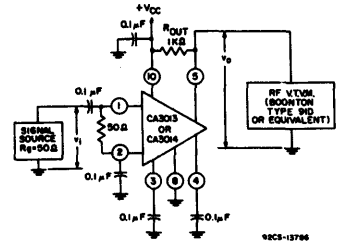
Terminal 2 is at the same dc potential as Terminal 1
Terminal 3: do not apply external voltage
Terminal 4 is at any dc potential between +2.5 and +7.5 volts
Terminal 5 is at a dc potential of +7.5 volts
Terminals 6 and 7 are at the same dc potential as Terminal 4
Terminal 8 is at dc ground potential
Terminal 9 is used as the af output terminal
Terminal 10 is at a dc potential of +7.5 volts

ELECTRICAL CHARACTERISTICS (See Page 8 for Definitions of Terms)	SYMBOLS	TEST CONDITIONS				LIMITS						UNITS	TYPICAL CHARACTERISTICS CURVES
		SETUP & PROCEDURE	FREQUENCY f	DC SUPPLY VOLTAGE V_{CC}	AMBIENT TEMPERATURE T_A	RCA CA3013			RCA CA3014				
		Fig.	Mc/s	volts	°C	Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
Total Device Dissipation*	P_T	3	–	6	-55	–	80	–	73	80	120	mW	
					+25	60	90	133	73	90	110	mW	
					+125	–	70	–	60	70	110	mW	
		3	–	7.5	-55	–	130	–	106	130	170	mW	
					+25	87	120	187	106	120	150	mW	
					+125	–	100	–	90	100	150	mW	
		3	–	10	-55	–	–	–	165	210	250	mW	
					+25	–	–	–	165	190	230	mW	
					+125	–	–	–	150	160	230	mW	
Voltage Gain**	A	4	1	6	-55	–	55	–	50	55	–	dB	
					+25	60	66	–	60	66	–	dB	
					+125	–	61	–	50	61	–	dB	
		4	1	7.5	-55	–	59	–	55	59	–	dB	
					+25	65	70	–	65	70	–	dB	
					+125	–	65	–	55	65	–	dB	
		4	1	10	-55	–	–	–	55	61	–	dB	
					+25	–	–	–	65	71	–	dB	
					+125	–	–	–	55	66	–	dB	
		4	4.5	7.5	+25	60	67	–	60	67	–	dB	5
			10.7	7.5	+25	55	60	–	55	60	–	dB	
Input-Impedance Components:													
Parallel Input Resistance	R_{IN}	6	4.5	7.5	+25	–	3	–	–	3	–	kΩ	7
Parallel Input Capacitance	C_{IN}	6	4.5	7.5	+25	–	7	–	–	7	–	pF	7
Output-Impedance Components:													
Parallel Output Resistance	R_{OUT}	8	4.5	7.5	+25	–	31.5	–	–	31.5	–	kΩ	9
Parallel Output Capacitance	C_{OUT}	8	4.5	7.5	+25	–	4.2	–	–	4.2	–	pF	9
Noise Figure	NF	10	4.5	7.5	+25	–	8.7	–	–	8.7	–	dB	11
Input Limiting Voltage (Knee)	$v_i(lim)$	14	4.5	7.5	+25	–	300	450	–	300	400	μV	13
Recovered AF Voltage	$v_o(af)$	14	4.5	6	+25	–	155	–	–	155	–	mV	13
				7.5	+25	128	188	–	135	188	–	mV	
				10	+25	–	–	–	–	220	–	mV	
Amplitude-Modulation Rejection	AMR	15	4.5	7.5	+25	–	50	–	–	50	–	dB	–
Discriminator Output Resistance	$R_O(disc)$	–	4.5	7.5	+25	–	60	–	–	60	–	Ω	–
Total Harmonic Distortion	THD	14	4.5	7.5	+25	–	1.8	–	–	1.8	–	%	12

* Total current drain may be determined by dividing P_T by V_{CC}.

** Recommended minimum dc supply voltage (V_{CC}) is 5.5 V. Nominal load current flowing into terminal 5 is 1.5 mA at 7.5 V.

VOLTAGE-GAIN TEST SETUP



PROCEDURE:

1) Set input frequency at desired value, v_i = 100 μV rms.
2) Record v_o.
3) Calculate Voltage Gain A from A = 20 $\log_{10} v_o/v_i$.
4) Repeat Steps 1, 2, and 3 for each frequency and/or temperature desired.

Fig. 4

VOLTAGE GAIN vs. FREQUENCY

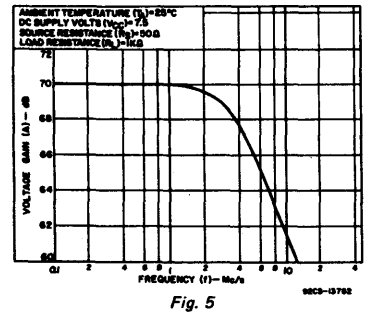


Fig. 5

INPUT-IMPEDANCE COMPONENTS TEST SETUP

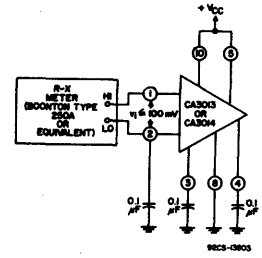


Fig. 6

INPUT-IMPEDANCE COMPONENTS vs. FREQUENCY

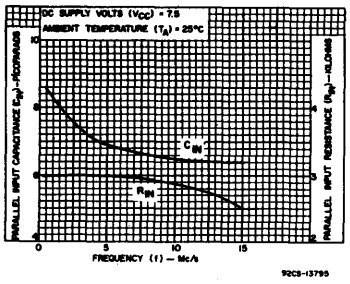


Fig. 7

OUTPUT-IMPEDANCE COMPONENTS TEST SETUP

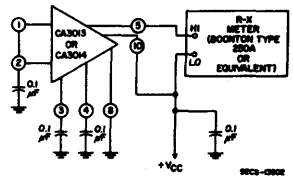


Fig. 8

OUTPUT-IMPEDANCE COMPONENTS vs. FREQUENCY

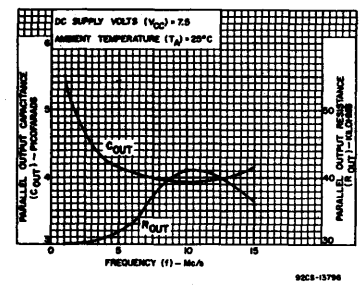
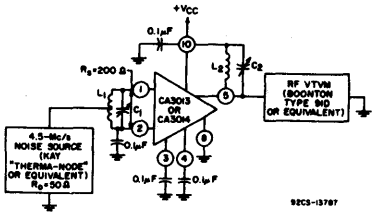


Fig. 9

NOISE FIGURE TEST SETUP



L_1 = 82 μH, center-tapped

L_2 = 2.36 μH

C_1, C_2 = Arco Type 423 padder, or equivalent

Fig. 10

NOISE FIGURE vs. DC SUPPLY VOLTAGE

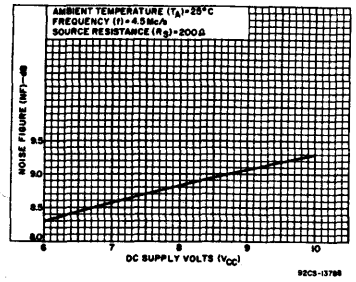


Fig. 11

TOTAL HARMONIC DISTORTION vs. DC SUPPLY VOLTAGE

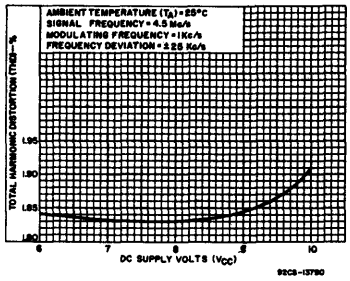
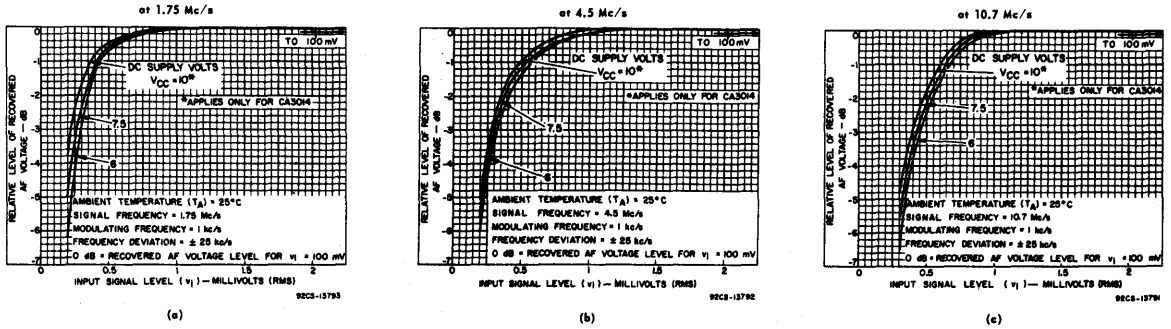


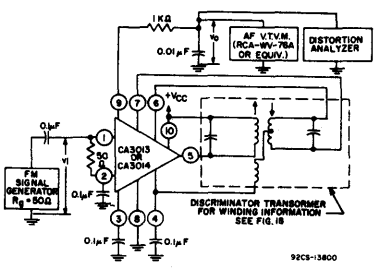
Fig. 12

INPUT LIMITING VOLTAGE (KNEE) AND RECOVERED AF VOLTAGE



(a) (b) (c)

Fig. 13



PROCEDURE:

A - Recovered-AF Voltage Output:

1) Set input frequency = 4.5 Mc/s, v_i = 100 mV rms, modulating frequency = 1 kc/s, frequency deviation = ±25 kc/s.

2) Record v_o as Recovered-AF Voltage Output.

B - Input Limiting Voltage (Knee):

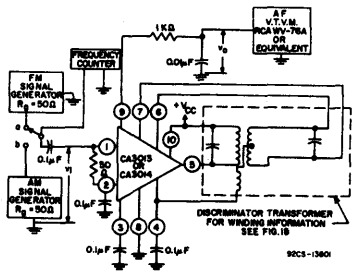
1) Repeat Steps A1 and A2, using v_i = 100 mV rms.

2) Decrease v_i to the level at which v_o is 3 dB below its value for v_i = 100 mV.

3) Record v_i as Input Limiting Voltage (Knee).

INPUT LIMITING VOLTAGE, RECOVERED AF VOLTAGE, AND TOTAL HARMONIC DISTORTION TEST SETUP

Fig. 14



PROCEDURE:

1) With Switch S in position "a", set input frequency = 4.5 Mc/s, v_i = 10 mV rms, modulating frequency = 1 kc/s, frequency deviation = ±25 kc/s.

2) Record v_o.

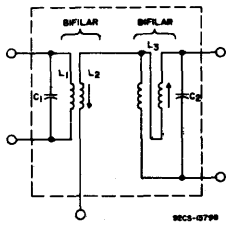
3) Place Switch S in position "b", and set input frequency = 4.5 Mc/s, v_i = 10 mV rms, modulating frequency = 1 kc/s, % modulation = 50.

4) Measure v_o, and record value in dB below value in Step 2 as AM Rejection.

AM-REJECTION TEST SETUP

Fig. 15

DISCRIMINATOR TRANSFORMER SCHEMATIC



(a)

CONSTRUCTION DETAILS OF DISCRIMINATOR TRANSFORMERS SHOWN IN FIGS. 2, 14 AND 15

Coil-Form Outside Diameter = 7/32 inch

Slugs: Radio Industries, Inc. Type "E" Material, or equivalent

Wire Type: "GRIPEZE"*, or equivalent

Operating Frequency Mc/s	Wire Size (AWG #)	Turns L_1▲	Turns L_2▲	Turns L_3	C_1 pF	C_2 pF
1.75	40	44	20	44 total (22 bifilar wound)	820	820
4.5	36	18	7	22 total (11 bifilar wound)	560	330
10.7	36	18	18	18 total (9 bifilar wound)	100	100

* Registered Trade Mark, Phelps-Dodge Copper Products.

▲ wound bifilar.

NOTE: The mutual coupling between L_1 and L_3 is adjusted for the desired degree of linearity.

(b)

Fig. 16

General-Purpose Transistor Arrays

TWO ISOLATED TRANSISTORS AND A DARLINGTON-CONNECTED TRANSISTOR PAIR

For Low-Power Applications at Frequencies from DC Through the VHF Range

The CA3018 and CA3018A consist of four general purpose silicon n-p-n transistors on a common monolithic substrate.

Two of the four transistors are connected in the Darlington configuration. The substrate is connected to a separate terminal for maximum flexibility.

The transistors of the CA3018 and the CA3018A are well suited to a wide variety of applications in low-power systems in the DC through VHF range. They may be used as discrete transistors in conventional circuits but in addition they provide the advantages of close electrical and thermal matching inherent in integrated circuit construction.

The CA3018A is similar to the CA3018 but features tighter control of current gain, leakage, and offset parameters making it suitable for more critical applications requiring premium performance.

APPLICATIONS

- General use in signal processing systems in DC through VHF range
- Custom designed differential amplifiers
- Temperature compensated amplifiers
- See RCA Application Note, ICAN-5296 "Application of the RCA CA3018 Integrated-Circuit Transistor Array" for suggested Applications.

FEATURES

- Matched monolithic general purpose transistors
- H_{FE} matched ± 10%
- V_{BE} matched ± 2 mV CA3018A (± 5mV CA3018)
- Operation from DC to 120 MHz
- Wide operating current range
- CA3018A performance characteristics controlled from 10 μA to 10mA
- Low noise figure - - 3.2 dB typical at 1KHz
- Full military temperature range capability (-55 to + 125°C)
- The CA3018 is available in a sealed-junction Beam Lead version (CA3018L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".
- Supplied in the hermetic 12-lead TO-5 style package.

Maximum Ratings, Absolute-Maximum Values, at TA=25°C

	CA3018	CA3018A	
Power Dissipation, P:			
Any one transistor	300	300	mW
Total package	450	450	mW
Derate at 5 mW/°C for $T_A > 85°C$			
Temperature Range:			
Operating	-55 to + 125	-55 to + 125	°C
Storage	-65 to + 150	-65 to + 150	°C
LEAD TEMPERATURE (During Soldering)			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265		°C

The following ratings apply for each transistor in the device:

	CA3018	CA3018A	
Collector-to-Emitter Voltage, V_{CEO} *	15	15	V
Collector-to-Base Voltage, V_{CBO}	20	30	V
Collector-to-Substrate Voltage, V_{CIO}*	20	40	V
Emitter-to-Base Voltage, V_{EBO}	5	5	V
Collector Current, I_C	50	50	mA

*The collector of each transistor of the CA3018 and CA3018A is isolated from the substrate by an integral diode. *The substrate (terminal 10) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.*

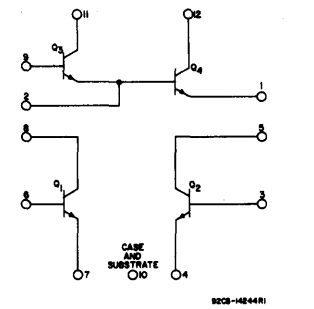


Fig. 1 - Schematic Diagram for CA3018 and CA3018A

STATIC CHARACTERISTICS

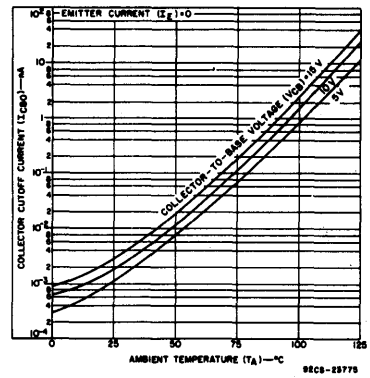


Fig.2 - Typical Collector-To-Base Cutoff Current vs Ambient Temperature for Each Transistor.

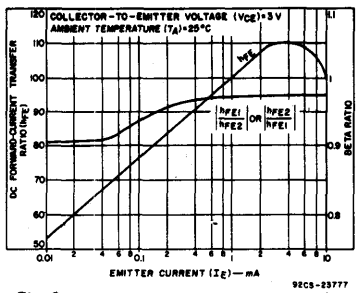


Fig. 3 - Typical Static Forward Current-Transfer Ratio and Beta Ratio for Transistors Q_1 and Q_2 vs Emitter Current.

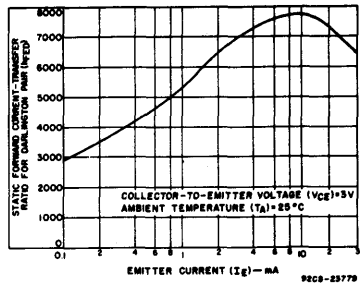


Fig. 4 - Typical Static Forward Current - Transfer Ratio for Darlington-connected Transistors Q_3 and Q_4 vs Emitter Current.

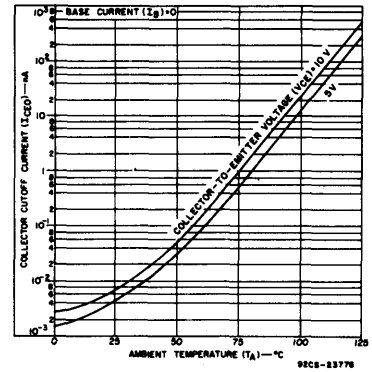


Fig. 7 - Typical Collector-To-Emmitter Cutoff Current vs Ambient Temperature for Each Transistor.

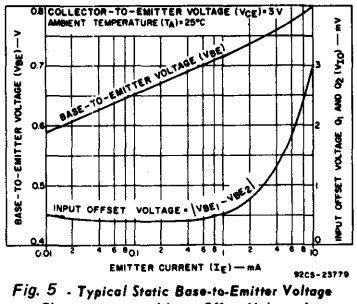


Fig. 5 - Typical Static Base-to-Emitter Voltage Characteristic and Input Offset Voltage for Q_1 and Q_2 vs Emitter Current.

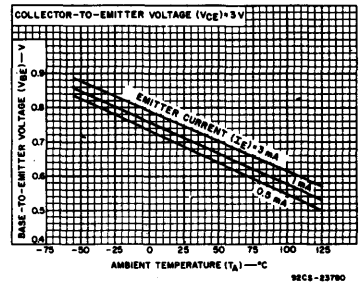


Fig. 6 - Typical Base-To-Emitter Voltage Characteristic for Each Transistor vs Ambient Temperature

CA3018, CA3018A

Characteristics apply for each transistor in the CA3018 and CA3018A as specified.

ELECTRICAL CHARACTERISTICS at T_A = 25°C	SYMBOLS	SPECIAL TEST CONDITIONS	CA3018 LIMITS			CA3018A LIMITS			Units	CHARACTERISTICS CURVES
			Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
STATIC CHARACTERISTICS										
Collector-Cutoff Current	I_{CBO}	V_{CB}=10V, I_E=0	–	0.002	100	–	0.002	40	nA	2
Collector-Cutoff Current	I_{CEO}	V_{CE}=10V, I_B=0	–	See Curve	5	–	See Curve	0.5	μA	7
Collector-Cutoff Current Darlington Pair	I_{CEOD}	V_{CE}=10V, I_B=0	–	–	–	–	–	5	μA	–
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C=1mA, I_B=0	15	24	–	15	24	–	V	–
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C=10μA, I_E=0	20	60	–	30	60	–	V	–
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E=10μA, I_C=0	5	7	–	5	7	–	V	–
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C=10μA, I_{CI}=0	20	60	–	40	60	–	V	–
Collector-to-Emitter Saturation Voltage	V_{CES}	I_B=1mA, I_C=10mA	–	0.23	–	–	0.23	0.5	V	–
Static Forward Current Transfer Ratio	h_{FE}	V_{CE}=3V, I_C=10mA I_C= 1mA I_C=10μA	– 30 –	100 100 54	– 200 –	50 60 30	100 100 54	– 200 –	– – –	3
Magnitude of Static-Beta Ratio (Isolated Transistors Q_1 and Q_2)		V_{CE}=3V, I_{C1}=I_{C2}=1mA	0.9	0.97	–	0.9	0.97	–	–	3
Static Forward Current Transfer Ratio Darlington Pair (Q_3 & Q_4)	h_{FED}	V_{CE}=3V I_C= 1mA I_C=100μA	1500 –	5400 –	– –	2000 1000	5400 2800	– –	–	4
Base-to-Emitter Voltage	V_{BE}	V_{CE}=3V I_E=1mA I_E=10mA	– –	0.715 0.800	– –	0.600 –	0.715 0.800	0.800 0.900	V	5
Input Offset Voltage	$\lvert V_{BE_1} - V_{BE_2} \rvert$	V_{CE}=3V, I_E=1mA	–	0.48	5	–	0.48	2	mV	5,8
Temperature Coefficient: Base-to-Emitter Voltage Q_1,Q_2	$\left\lvert \frac{\Delta V_{BE}}{\Delta T} \right\rvert$	V_{CE}=3V, I_E=1mA	–	1.9	–	–	1.9	–	mV/°C	6
Base (Q_3)-to-Emitter (Q_4) Voltage-Darlington Pair	V_{BED} (V_{3-1})	V_{CE}=3V I_E=10mA I_E= 1mA	– –	1.46 1.32	– –	– 1.10	1.46 1.32	1.60 1.50	V	9
Temperature Coefficient: Base-to-Emitter Voltage Darlington Pair-Q_3,Q_4	$\left\lvert \frac{\Delta V_{BED}}{\Delta T} \right\rvert$	V_{CE}=3V, I_E=1mA	–	4.4	–	–	4.4	–	mV/°C	10
Temperature Coefficient: Magnitude of Input-Offset Voltage	$\left\lvert \frac{V_{BE_1} - V_{BE_2}}{\Delta T} \right\rvert$	V_{CC}=+6V, V_{EE}=-6V, I_{C1}=I_{C2}=1mA	–	10	–	–	10	–	μV/°C	–

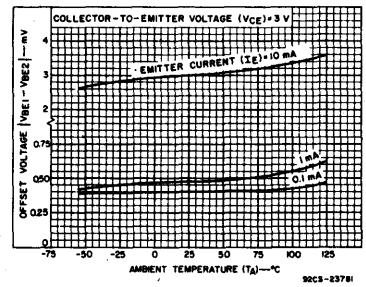


Fig.8 - Typical Offset Voltage Characteristic vs Ambient Temperature

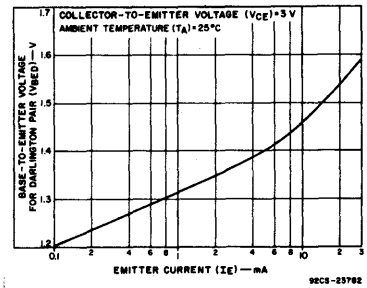


Fig.9 - Typical Static Input Voltage Characteristic for Darlington Pair (Q_3 and Q_4) vs Emitter Current

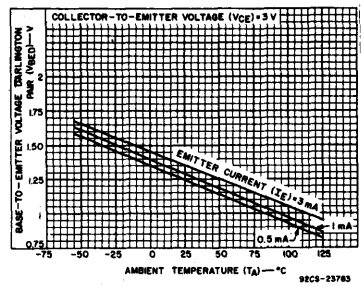


Fig.10 - Typical Static Input Voltage Characteristic for Darlington Pair (Q_3 and Q_4) vs Ambient Temperature.

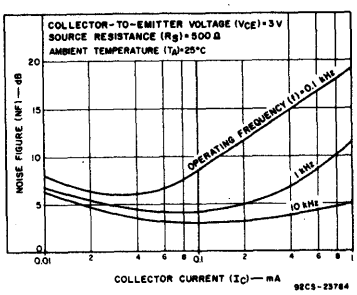


Fig.11(a) - Noise Figure vs Collector Current, R_S = 500 Ω.

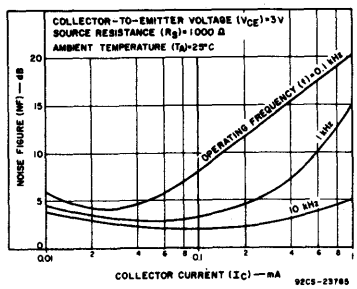


Fig.11(b) - Noise Figure vs Collector Current, R_S = 1 KΩ.

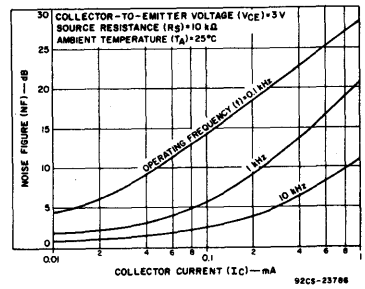


Fig.11(c) - Noise Figure vs Collector Current, R_S = 10 KΩ.

ELECTRICAL CHARACTERISTICS, (CONT'D)

DYNAMIC CHARACTERISTICS			CA3018			CA3018A				
Low Frequency Noise Figure	NF	f=1 KHz, V_{CE}=3V, I_C=100μA Source resistance=1 KΩ	–	3.25	–	–	3.25	–	dB	11(b)
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:										
Forward Current-Transfer Ratio	h_{fe}	f=1kHz, V_{CE}=3V, I_C=1mA	–	110	–	–	110	–	–	12
Short-Circuit Input Impedance	h_{ie}		–	3.5	–	–	3.5	–	KΩ	12
Open-Circuit Output Impedance	h_{oe}		–	15.6	–	–	15.6	–	μmho	12
Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		–	1.8×10^{-4}	–	–	1.8×10^{-4}	–	–	12
Admittance Characteristics:										
Forward Transfer Admittance	Y_{fe}	f=1MHz, V_{CE}=3V, I_C=1mA	–	31-j1.5	–	–	31-j1.5	–	mmho	13
Input Admittance	Y_{ie}		–	0.3+j0.04	–	–	0.3+j0.04	–	mmho	14
Output Admittance	Y_{oe}		–	0.001+j0.03	–	–	0.001+j0.03	–	mmho	15
Reverse Transfer Admittance	Y_{re}		See Curve			See Curve			mmho	16
Gain-Bandwidth Product	f_T	V_{CE}=3V, I_C=3mA	300	500	–	300	500	–	MHz	17
Emitter-to-Base Capacitance	C_{EB}	V_{EB}=3V, I_E=0	–	0.6	–	–	0.6	–	pF	–
Collector-to-Base Capacitance	C_{CB}	V_{CB}=3V, I_C=0	–	0.58	–	–	0.58	–	pF	–
Collector-to-Substrate Capacitance	C_{CI}	V_{CI}=3V, I_C=0	–	2.8	–	–	2.8	–	pF	–

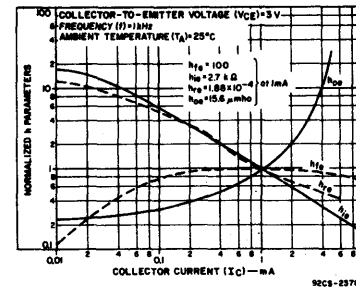


Fig.12 - Forward Current-Transfer Ratio (h_{fe}), Short-Circuit Input Impedance (h_{ie}), Open-Circuit Output Impedance (h_{oe}), and Open-Circuit Reverse Voltage-Transfer Ratio (h_{re}) vs Collector Current

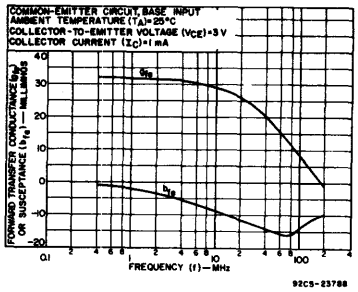


Fig.13 - Forward Transfer Admittance (Y_{fe})

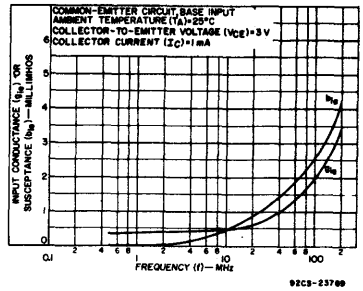


Fig.14 - Input Admittance (Y_{ie})

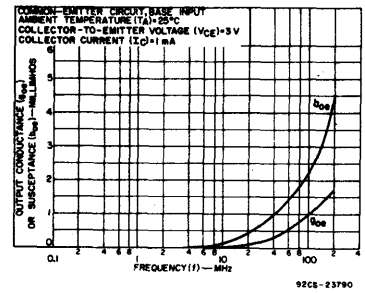


Fig.15 - Output Admittance (Y_{oe})

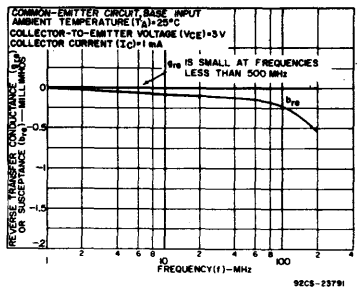


Fig.16 - Reverse Transfer Admittance (Y_{re})

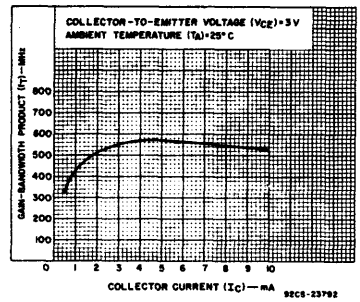


Fig.17 - Typical Gain-Bandwidth Product (f_T) vs Collector Current

CA3019

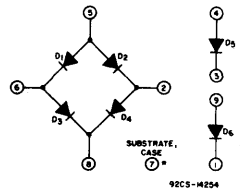
Ultra-Fast Low-Capacitance Matched Diodes

For Applications in Communications and Switching Systems

The RCA-CA3019 consists of six ultra-fast, low capacitance diodes on a common monolithic substrate. Integrated circuit construction assures excellent static and dynamic matching of the diodes, making the array extremely useful for a wide variety of applications in communication and switching systems.

Four of the diodes are internally connected as a "quad" and two are independently accessible. The substrate is internally connected to the 10-lead TO-5-style case.

For applications such as balanced modulators or ring modulators where capacitive balance is important, the substrate should be returned to a DC potential which is significantly more negative (with respect to the active diodes) than the peak signal applied.



*Connect to most negative circuit potential.

Fig. 1 – Schematic Diagram.

Features:

- Excellent Diode Match
- Low Leakage Current
- Low Pedestal Voltage when Gating
- Companion Application Note, ICAN-5299: "Application of the RCA-CA3019 Integrated-Circuit Diode Array"

Applications:

- Modulator
- Mixer
- Balanced Modulator
- Analog Switch
- Diode Gate for Chopper-Modulator Applications

Absolute-Maximum Ratings:

DISSIPATION:
- Any one diode unit . . . 20 max. mW
- Total for device . . . . 120 max. mW

TEMPERATURE RANGE:
- Storage . . . . . –65 to +200 °C
- Operating . . . . . –55 to +125 °C

DC Forward Current, I_F . . 25 mA

Peak Recurrent Forward Current, I_f . . . . . 100 mA

Peak Forward Surge Current, I_f (surge) . . . 100 mA

VOLTAGE: See Table

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

Characteristics Apply for Each Diode Unit, Unless Otherwise Specified

CHARACTERISTICS	SPECIAL TEST CONDITIONS	LIMITS TYPE CA3019			Units
		Min.	Typ.	Max.	
DC Forward Voltage Drop	DC Forward Current (I_F) = 1 mA	–	0.73	0.78	V
DC Reverse Breakdown Voltage	DC Reverse Current (I_R) = –10 μA	4	6	–	V
DC Reverse Breakdown Voltage Between any Diode Unit and Substrate	DC Reverse Current (I_R)= –10 μA	25	80	–	V
DC Reverse (Leakage) Current	DC Reverse Voltage (V_R) =–4 V	–	0.0055	10	μA
DC Reverse (Leakage) Current Between any Diode Unit and Substrate	DC Reverse Voltage (V_R) = –4 V	–	0.010	10	μA
Magnitude of Diode Offset Voltage (Difference in DC Forward Voltage Drops of any Two Diode Units)	DC Forward Current (I_F)= 1 mA	–	1	5	mV
Single Diode Capacitance	Frequency (f) = 1 MHz DC Reverse Voltage (V_R) = –2V	–	1.8	–	pF
Diode Quad-to-Substrate Capacitance	Frequency (f) = 1 MHz DC Reverse Voltage (V_R) between Terminal 2,5,6, or 8 of Diode Quad and Terminal 7 (Substrate) = –2 V				
	Terminal 2 or 6 to Terminal 7	–	4.4	–	pF
	Terminal 5 or 8 to Terminal 7	–	2.7	–	pF
Series Gate Switching Pedestal Voltage		–	10	–	mV

Absolute-Maximum Voltage Limits:

TERM.	VOLTAGE LIMITS		CONDITIONS	
	NEG.	POS.	TERM.	VOLT.
1	–3	+12	7	–6
2	–3	+12	7	–6
3	–3	+12	7	–6
4	–3	+12	7	–6
5	–3	+12	7	–6
6	–3	+12	7	–6
7	–18	0	1,2, 3,6, 8	0
8	–3	+12	7	–6
9	–3	+12	7	–6
10	NO CONNECTION			
CASE	INTERNALLY CONNECTED TO TERMINAL 7 DO NOT GROUND			

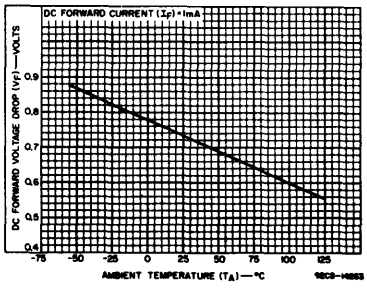


Fig. 2 – DC forward voltage drop (any diode) as a function of temperature.

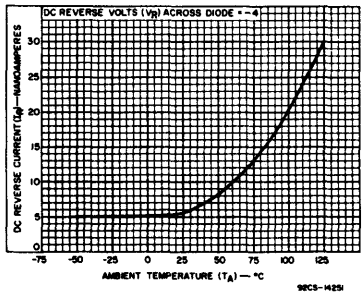


Fig. 3 – Reverse (leakage) current (any diode) as a function of temperature.

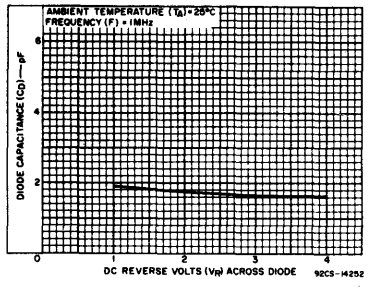


Fig. 4 – Diode capacitance (any diode) as a function of reverse voltage.

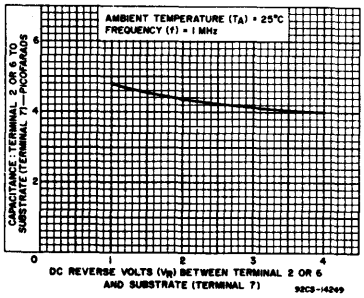


Fig. 5 – Diode quad-to-substrate capacitance as a function of reverse voltage.

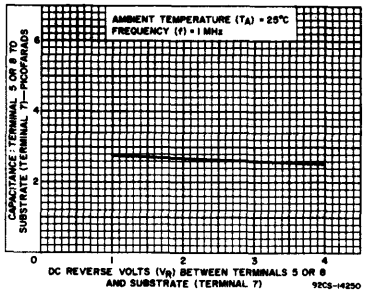


Fig. 6 – Diode quad-to-substrate capacitance as a function of reverse voltage.

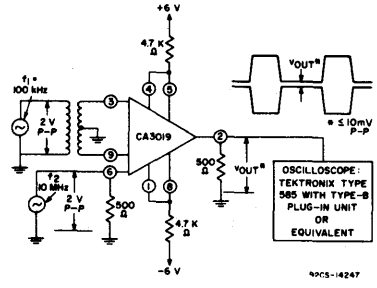


Fig. 7 – Series gate switching test setup.

CA3020, CA3020A
Multi-Purpose Wideband Power Amplifiers

The RCA-CA3020 and CA3020A are Integrated-Circuit, Multistage, Multipurpose, Wide-Band Power Amplifiers on a single monolithic silicon chip. They employ a highly versatile and stable direct-coupled circuit configuration featuring wide frequency range, high voltage and power gain, and high power output. These features plus inherent stability over a wide temperature range make the CA3020 and CA3020A extremely useful for a wide variety of applications in military, industrial, and commercial equipment.

For Military, Industrial, and Commercial Equipment at Frequencies up to 8 MHz

The CA3020 and CA3020A are particularly suited for service as Class B power amplifiers. The CA3020A can provide a maximum power output of 1 watt from a 12-volt DC supply with a typical power gain of 75 dB. The CA3020 provides 0.5 watt power output from a 9-volt supply with the same power gain.

These types are supplied in hermetically sealed, TO-5 style 12-lead packages.

SCHEMATIC DIAGRAM FOR CA3020 AND CA3020A

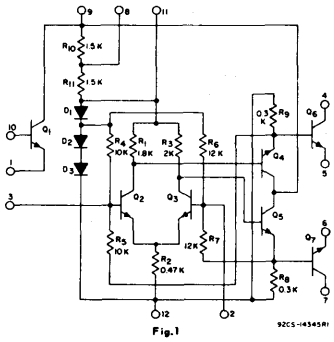


Fig.1

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturers in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ± 30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

ABSOLUTE-MAXIMUM RATINGS:

	WITHOUT HEAT SINK	WITH HEAT SINK
DISSIPATION:		
At T_A = 25°C	1 W	
Above T_A = 25°C	derate linearly 6.7 mW/°C	
At T_C = 25°C		2 W
At T_C = 25°C to T_C = 55°C		2 W
Above T_C = 55°C		derate linearly 16.7 mW/°C

TEMPERATURE RANGE:

Operating . . . . . -55°C to +125°C
Storage . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 1 with respect to terminal 12 is 0 to +10 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12
1		*	*	*	*	*	*	*	▲ 0 -10/-12	+3 Note 1	*	+10 0
2			*	*	*	*	*	*	*	*	*	+2 -2
3				*	*	*	*	*	*	*	*	+2 -2
4					▲ +18/+25 0	*	*	*	*	*	*	▲ +18/+25 0
5						*	*	*	*	*	*	+3 Note 2
6							▲ 0 -18/-25	*	*	*	*	+3 Note 2
7								*	*	*	*	▲ +18/+25 0
8									Note 3	*	*	Note 3 0
9										+10 0	Note 1 0	+10/+12 0
10											*	+10 0
11												*
12												REF. SUBSTRATE

Note 1: This voltage is established by the maximum current rating.

Note 2: The emitters of Q_6 and Q_7 may be returned to a negative voltage supply through emitter resistors. Current into terminal No.9 should not be exceeded and the total device dissipation should not be exceeded.

Note 3: Terminal No.8 may be connected to terminals Nos.9, 11, or 12.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

▲ Higher value is for CA3020A.

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	-	20
2	-	-
3	-	-
4	300	-
5	-	300
6	-	300
7	300	-
8	-	-
9	20	-
10	1	-
11	20	-
12	-	-

FEATURES

- High power output - class B amplifier --
  CA3020 .... 0.5 watt typ. at V_{CC} = +9V
  CA3020A ... 1.0 watt typ. at V_{CC} = +12V
- Wide frequency range --
  Up to 8 MHZ with resistive loads
- High power gain ..... 75db typ.
- Single power supply for class B operation with transformer --
  CA3020 ..... 3 to 9V
  CA3020A ..... 3 to 12V
- Built-in temperature-tracking voltage regulator provides stable operation over -55°C to +125°C temperature range

APPLICATIONS

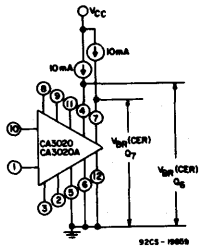
- AF power amplifiers for portable and fixed sound and communications systems
- Servo-control amplifiers
- Wide-band linear mixers
- Video power amplifiers
- Transmission-line driver amplifiers (balanced and unbalanced)
- Fan-in and fan-out amplifiers for computer logic circuits
- Lamp-control amplifiers
- Motor-control amplifiers
- Power multivibrator
- Power switches
- Companion Application Note, ICAN 5766 "Application of CA3020 and CA3020A Integrated Circuit Multipurpose Wide-Band Power Amplifiers."

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

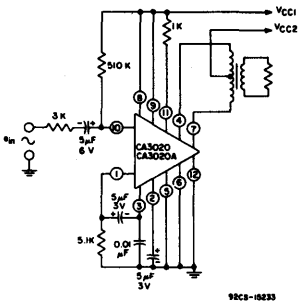
CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			LIMITS CA3020			LIMITS CA3020A			UNITS
		CIRCUIT AND PROCEDURE	DC SUPPLY VOLTAGE								
		FIG.	V_{CC1}	V_{CC2}	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.	
Collector-to-Emitter Breakdown Voltage, Q_6 & Q_7 at 10 mA	$V_{(BR)CER}$	2a	-	-	18	-	-	25	-	-	V
Collector-to-Emitter Breakdown Voltage, Q_1 at 0.1 mA	$V_{(BR)CEO}$	-	-	-	10	-	-	10	-	-	V
Idle Currents, Q_6 & Q_7	I_6 IDLE I_7 IDLE	4	9.0	2.0	-	5.5	-	-	5.5	-	mA
Peak Output Currents, Q_6 & Q_7	I_6PK I_7PK	4	9.0	2.0	140	-	-	180	-	-	mA
Cutoff Currents, Q_6 & Q_7	I_6 CUTOFF I_7 CUTOFF	4	9.0	2.0	-	-	1.0	-	-	1.0	mA
Differetial Amplifier Current Drain	I_{CC1}	4	9.0	9.0	6.3	9.4	12.5	6.3	9.4	12.5	mA
Total Current Drain	I_{CC1} + I_{CC2}	4	9.0	9.0	8.0	21.5	35.0	14.0	21.5	30.0	mA
Differential Amplifier Input Terminal Voltages	V_2 V_3	4	9.0	2.0	-	1.11	-	-	1.11	-	V
Regulator Terminal Voltage	V_{11}	4	9.0	2.0	-	2.35	-	-	2.35	-	V
Q, Cutoff (Leakage) Currents: Collector-to-Emitter	I_{CEO}	-	10.0	-	-	-	100	-	-	100	μA
Emitter-to-Base	I_{EBO}		3.0	-	-	-	0.1	-	-	0.1	
Collector-to-Base	I_{CBO}		3.0	-	-	-	0.1	-	-	0.1	
Forward Current Transfer Ratio, Q_1 at 3 mA	h_{FE1}	-	6.0	-	30	75	-	30	75	-	
Bandwidth at -3 dB Point	BW		6.0	6.0	-	8	-	-	8	-	MHz
Maximum Power Output	$P_{O(MAX)}$	6	6.0	6.0	200	300[a]	-	200	300[a]	-	mW
			9.0	9.0	400	550[a]	-	400	550[a]	-	
			9.0	12.0	-	-	-	800	1000[b]	-	
Sensitivity for P_{OUT} = 400 mW	e_{IN}	6	9.0	9.0	-	35[a]	55	-	-	-	mV
Sensitivity for P_{OUT} = 800 mW	e_{IN}	6	9.0	12.0	-	-	-	-	50[b]	100	mV
Input Resistance---- Terminal 3 to Ground	R_{IN3}	9	6.0	6.0	-	1000	-	-	1000	-	Ω
Junction-to-Case Thermal Resistance	θ_{J-C}	-	-	-	-	-	60	-	-	60	°C/W

a R_{CC} = 130 Ω

b R_{CC} = 200 Ω



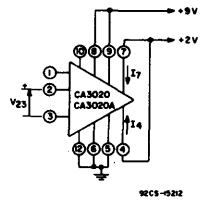
a. Collector-to-emitter breakdown voltage (Q_6 & Q_7) circuit



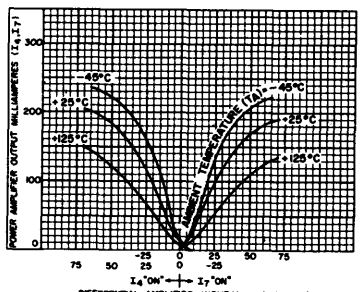
b. Typical audio amplifier circuit utilizing the CA3020 or CA3020A as an audio preamplifier and class B power amplifier

Fig.2

TYPICAL TRANSFER CHARACTERISTICS



a. Test Setup



b. Characteristics with R_{10} shorted out

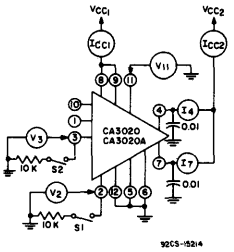
Fig.3

TYPICAL PERFORMANCE DATA

An External Radiator is Recommended for High Ambient Temperature Operation

CHARACTERISTICS	SYMBOLS	CA3020	CA3020A	UNITS
Power Supply Voltage	V_{CC_1}	9.0	9.0	V
	V_{CC_2}	9.0	12.0	
Zero Signal Current Diff. Ampl.	I_{CC_1}	15	15	mA
Zero Signal Current Output Ampl.	I_{CC_2}	24	24	
Maximum Signal Current Diff. Ampl.	I_{CC_1}	16	16.6	mA
Maximum Signal Current Output Ampl.	I_{CC_2}	125	140	
Maximum Power Output at THD = 10%	Po	550	1000	mW
Sensitivity	e_{IN}	35	45	mV
Power Gain	G_P	75	75	dB
Input Resistance	R_{IN}	55	55	kΩ
Efficiency	η	45	55	%
Signal-to-Noise Ratio	S/N	70	66	dB
THD at 150 mW level		3.1	3.3	%
Test Signal Frequency from 600Ω Generator		1000	1000	Hz
Equivalent Collector-to-Collector Load Resistance	R_{CC}	130	200	Ω

STATIC CURRENT AND VOLTAGE TEST CIRCUIT

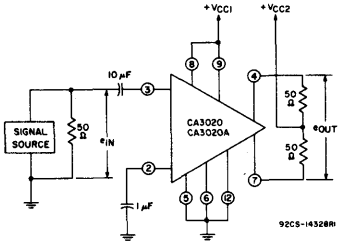


CURRENTS OR VOLTAGES	S1	S2
$I_{4\text{-IDLE}}$	open	open
$I_{7\text{-IDLE}}$	open	open
$I_{4\text{-PEAK}}$	open	close
$I_{7\text{-PEAK}}$	close	open
$I_{4\text{-CUTOFF}}$	close	open
$I_{7\text{-CUTOFF}}$	open	close

CURRENTS OR VOLTAGES	S1	S2
I_{CC1}	open	open
I_{CC2}	open	open
V_2	open	open
V_3	open	open
V_{11}	open	open

Fig.4

MEASUREMENT OF BANDWIDTH AT -3 dB POINTS

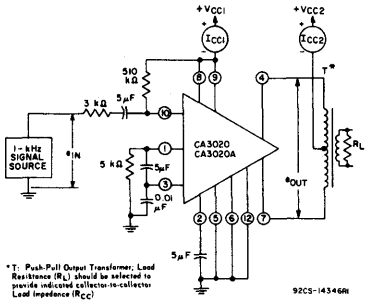


PROCEDURES:

1. Apply desired value of V_{CC_1} and V_{CC_2}
2. Apply 1 kHz input signal and adjust for e_{IN} = 5 mV (rms)
3. Record the resulting value of e_{OUT} in dB (reference value)
4. Vary input-signal frequency, keeping e_{IN} constant at 5 mV, and record frequencies above and below 1 kHz at which e_{OUT} decreases 3 dB below reference value.
5. Record bandwidth as frequency range between -3 dB points.

Fig.5

MEASUREMENTS OF ZERO-SIGNAL DC CURRENT DRAIN, MAXIMUM-SIGNAL DC CURRENT DRAIN, MAXIMUM POWER OUTPUT, CIRCUIT EFFICIENCY, SENSITIVITY, AND TRANSDUCER POWER GAIN



PROCEDURES:

Zero-Signal DC Current Drain

1. Apply desired Value of V_{CC_1} and V_{CC_2} and reduce e_{IN} to 0V
2. Record resulting values of I_{CC_1} and I_{CC_2} in mA as Zero-Signal DC Current Drain.

Fig.10

Maximum-Signal DC Current Drain, Maximum Power Output, Circuit Efficiency, Sensitivity, and Transducer Power Gain

1. Apply desired value of V_{CC_1} and V_{CC_2} and adjust e_{IN} to the value at which the Total Harmonic Distortion in the output of the amplifier = 10%
2. Record resulting value of I_{CC_1} and I_{CC_2} in mA as Maximum-Signal DC Current Drain
3. Determine resulting amplifier power output in watts and record as Maximum Power Output (P_{OUT})
4. Calculate Circuit Efficiency (η) in % as follows:

$$\eta = 100\,\frac{P_{OUT}}{V_{CC_1}I_{CC_1}+V_{CC_2}I_{CC_2}}$$

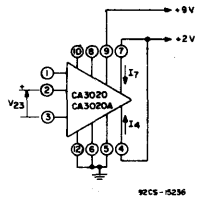
where P_{OUT} is in watts, V_{CC_1} and V_{CC_2} are in volts, and I_{CC_1} and I_{CC_2} are in amperes.

5. Record value of e_{IN} in mV (rms) required in Step 1 as Sensitivity (e_{IN})
6. Calculate Transducer Power Gain (G_p) in dB as follows:

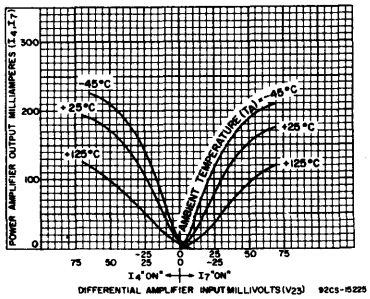
$$G_p = 10\log_{10}\frac{P_{OUT}}{P_{IN}}$$

where P_{IN} (in mW) $= \dfrac{e_{IN}^2}{3000 + R_{IN(10)}}$

Fig.6



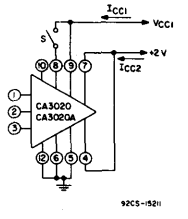
a. Test Setup



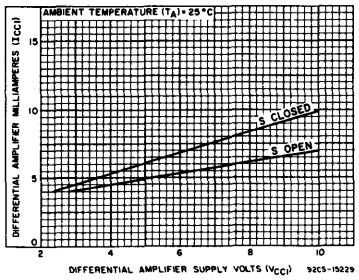
b. Characteristics with R_{10} in circuit

Fig.7

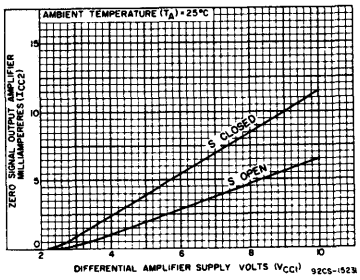
ZERO SIGNAL AMPLIFIER CURRENT vs DIFFERENTIAL AMPLIFIER SUPPLY VOLTAGE



a. Test Setup



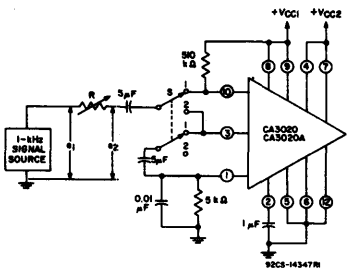
b. Differential Amplifier Characteristics



c. Output Amplifier Characteristics

Fig.8

MEASUREMENT OF INPUT RESISTANCE



PROCEDURES:

Input Resistance Terminal 10 to Ground ($R_{IN_{10}}$)

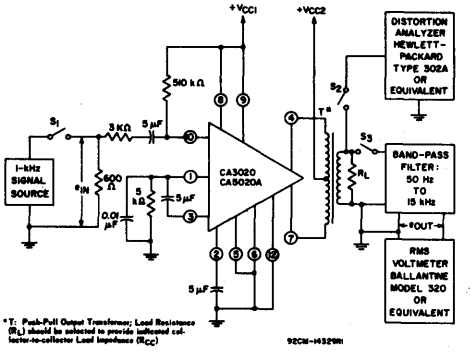
1. Apply desired value of V_{CC_1} and V_{CC_2} and set S in Position 1
2. Adjust 1-kHz input for desired signal level of measurement
3. Adjust R for $e_2 = e_1/2$
4. Record resulting value of R as $R_{IN_{10}}$

Input Resistance Terminal 3 to Ground (R_{IN_3})

1. Apply desired value of V_{CC_1} and V_{CC_2} set S in Position 2
2. Adjust 1-kHz input for desired signal level of measurement
3. Adjust R for $e_2 = e_1/2$
4. Record resulting value of R as R_{IN_3}

Fig.9

MEASUREMENT OF SIGNAL-TO-NOISE RATIO AND TOTAL HARMONIC DISTORTION



PROCEDURES:

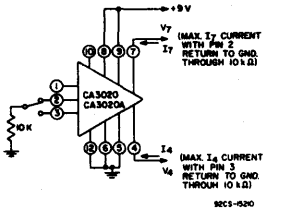
Signal-to-Noise Ratio

1. Close S_1 and S_3; open S_2
2. Apply desired values of V_{CC_1} and V_{CC_2}
3. Adjust e_{IN} for an amplifier output of 150mW and record resulting value of E_{OUT} in dB as e_{OUT_1} (reference value)
4. Open S_1 and record resulting value of e_{OUT} in dB as e_{OUT_2}
5. Signal-to-Noise Ratio (S/N) = $20\log_{10} \frac{e_{OUT_1}}{e_{OUT_2}}$.

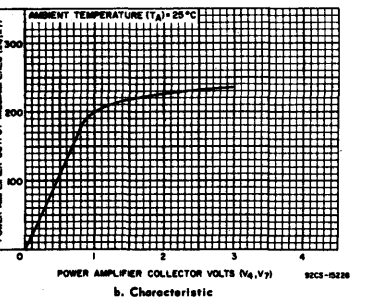
Total Harmonic Distortion

1. Close S_1 and S_2; open S_3
2. Apply desired values of V_{CC_1} and V_{CC_2}
3. Adjust e_{IN} for desired level amplifier output power
4. Record Total Harmonic Distortion (THD) in %

Fig.10



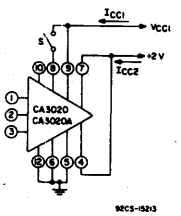
a. Test Setup



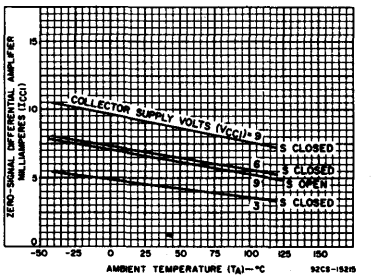
b. Characteristic

Fig.11

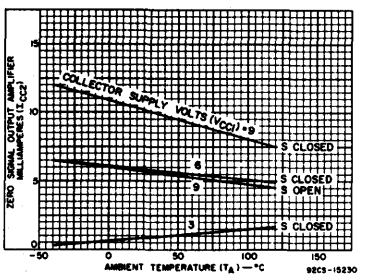
ZERO SIGNAL AMPLIFIER CURRENT vs AMBIENT TEMPERATURE



a. Test Setup



b. Differential Amplifier Characteristics



c. Output Amplifier Characteristics

Fig.12

CA3021, CA2022, CA2023
Low-Power Video and Wideband Amplifiers

RCA-CA3021, CA3022, and CA3023 are low-power integrated-circuit wideband amplifiers with a wide range of applications in industrial, military, and commercial communications equipment. Each consists of a multistage amplifier circuit and unconnected diodes on a single chip, hermetically sealed in a 12-lead TO-5 style package. The diodes may be connected to provide limiting in FM applications.

The CA3021, CA3022, and CA3023 have the same maximum ratings, and differ principally in dissipation (dc power requirements) and bandwidth capability. All three devices are designed for operation over the temperature range from -55° C to +125° C.

APPLICATIONS

- Gain-Controlled Linear Amplifiers
- AM/FM IF Amplifiers • Video Amplifiers • Limiters

SCHEMATIC DIAGRAM FOR CA3021, CA3022, AND CA3023

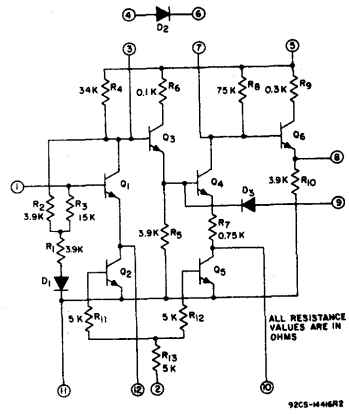


Fig. 1

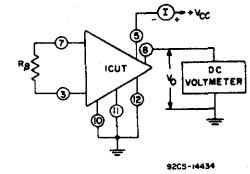
ABSOLUTE-MAXIMUM RATINGS:

OPERATING-TEMPERATURE RANGE	-55°C to +125°C	
STORAGE-TEMPERATURE RANGE	-65°C to +150°C	
LEAD TEMPERATURE (During Soldering): At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C	
DEVICE DISSIPATION, P_T	120 max.	mW
INPUT-SIGNAL VOLTAGE	-3, +3 max.	V
DC VOLTAGES AND CURRENTS	See Table Below	

HIGHLIGHTS

- **Low DC Power Drain:** P_D { CA3021 = 4 mW typ. / CA3022 = 12.5 mW typ. / CA3023 = 35 mW typ. } at V_{CC} = 6 V
- **Excellent frequency response:** -3 dB BW { CA3021 = 2.4 MHz typ. / CA3022 = 7.5 MHz typ. / CA3023 = 16 MHz typ.
- **High Voltage Gain:** A { CA3021 = 56 dB typ. at 0.5 MHz / CA3022 = 57 dB typ. at 2.5 MHz / CA3023 = 53 dB typ. at 5 MHz
- **Wide AGC Range:** 33 dB typ.
- **Only one power supply (4.5 to 12 V) required**
- **Hermetically Sealed 12-Lead TO-5-style package**
- **Operation from -55° C to +125° C**

TEST SETUP FOR MEASUREMENT OF DEVICE DISSIPATION AND QUIESCENT OUTPUT VOLTAGE



$P_T = V_{CC}$ (I)

Fig. 2

	VOLTAGE OR CURRENT LIMITS		CIRCUIT CONDITIONS	
TERMINAL	NEGATIVE	POSITIVE	TERMINAL	CONDITIONS
1	-3V	+3V	1	Connected to Voltage Source through 100Ω Resistor
			5	+12V
			10, 11, 12	Ground
2	-3V	+12V	5	+12V
			10, 11, 12	Ground
3	0V	+12V	5	+12V
			10, 11, 12	Ground
4	-12V	+12V	6, 11	Ground
	10 max. mA			
5	0V	+18V	10, 11, 12	Ground
6	-12V	+12V	5, 11	Ground
	10 max. mA			

	VOLTAGE OR CURRENT LIMITS		CIRCUIT CONDITIONS	
TERMINAL	NEGATIVE	POSITIVE	TERMINAL	CONDITIONS
7	0V	+12V	5	+12V
			10, 11, 12	Ground
8	20 max. mA		5	+12V
			10, 11, 12	Ground
9	-0.5V	+3V	5	+12V
			10, 11, 12	Ground
10	0V	+4V	2,5	+12V
			11	Ground
11	-6V	+12V	2	Ground
			5	+12V
12	0V	+4V	2,5	+12V
			11	Ground

DEVICE DISSIPATION VS DC SUPPLY VOLTAGE FOR CA3021

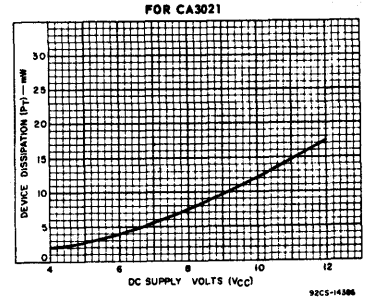


Fig. 3(a)

DEVICE DISSIPATION VS DC SUPPLY VOLTAGE FOR CA3022

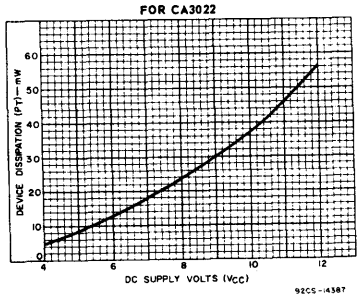


Fig. 3(b)

DEVICE DISSIPATION VS DC SUPPLY VOLTAGE FOR CA3023

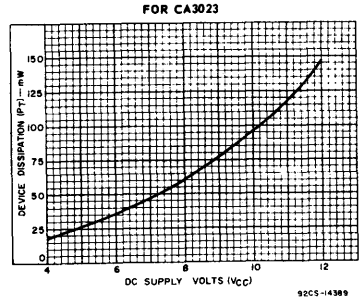


Fig. 3(c)

DEVICE DISSIPATION VS TEMPERATURE FOR CA3021, CA3022, AND CA3023

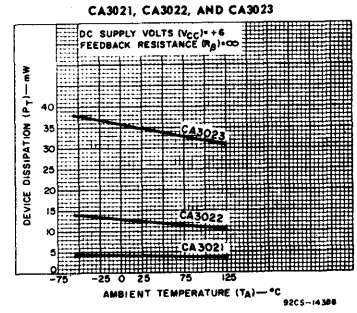
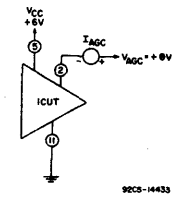


Fig. 3(d)

ELECTRICAL CHARACTERISTICS, at T_A = 25° C, V_{CC} = +6V, unless otherwise specified

CHARACTERISTIC		SYMBOL	TEST CONDITIONS			LIMITS									UNITS	TYPICAL CHARACTERISTIC CURVE
			TEST SETUP AND PROCEDURE	FEEDBACK RESISTANCE (R_B) BETWEEN TERMINALS 3 AND 7	FREQUENCY f	CA3021 (TA5219)			CA3022 (TA5236)			CA3023 (TA5218)				
			Fig.	Ω	MHz	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	Units	Fig.
Device Dissipation		P_T	2	∞	-	1	4	8	-	-	-	-	-	-	mW	3a,d
				∞	-	-	-	-	5	12.5	24	-	-	-	mW	3b,d
				∞	-	-	-	-	-	-	-	24	35	48	mW	3c,d
Quiescent Output Voltage		V_O	2	39k	-	-	2.2	-	-	-	-	-	-	-	V	-
				10k	-	-	-	-	-	1.9	-	-	-	-	V	
				4.7k	-	-	-	-	-	-	-	-	1.3	-	V	
AGC Source Current		I_{AGC}	4	V_{AGC} =+6V		-	0.8	-	-	0.8	-	-	0.8	-	mA	-
Voltage Gain		A	5	560k	0.5	50	56	-	-	-	-	-	-	-	dB	6a
				39k	0.8	40	46	-	-	-	-	-	-	-	dB	6a,d
				39k	2.5	-	-	-	50	57	-	-	-	-	dB	6b
				10k	3	-	-	-	40	44	-	-	-	-	dB	6b,d
				18k	5	-	-	-	-	-	-	50	53	-	dB	6c
				4.7k	10	-	-	-	-	-	-	40	44	-	dB	6c,d
Bandwidth at -3 dB Point		BW	5	39k	-	0.8	2.4	-	-	-	-	-	-	-	MHz	6a
				10k	-	-	-	-	3	7.5	-	-	-	-	MHz	6b
				4.7k	-	-	-	-	-	-	-	10	16	-	MHz	6c
Input-Impedance Components	Input Resistance	R_{IN}	7	39k	1	-	4000	-	-	-	-	-	-	-	Ω	-
				10k	5	-	-	-	-	1300	-	-	-	-	Ω	
				4.7k	10	-	-	-	-	-	-	-	300	-	Ω	
	Input Capacitance	C_{IN}	7	39k	1	-	11	-	-	-	-	-	-	-	pF	-
				10k	5	-	-	-	-	18	-	-	-	-	pF	
				4.7k	10	-	-	-	-	-	-	-	13	-	pF	
Output Resistance		R_{OUT}	8	39k	1	-	300	-	-	-	-	-	-	-	Ω	-
				10k	5	-	-	-	-	120	-	-	-	-	Ω	
				4.7k	10	-	-	-	-	-	-	-	100	-	Ω	
Noise Figure		NF	9	39k	1	-	4.2	8.5	-	-	-	-	-	-	dB	-
				10k	1	-	-	-	-	4.4	8.5	-	-	-	dB	
				4.7k	1	-	-	-	-	-	-	-	6.5	8.5	dB	
AGC Range		AGC	10	-	1	-	33	-	-	-	-	-	-	-	dB	-
				-	5	-	-	-	-	33	-	-	-	-	dB	
				-	10	-	-	-	-	-	-	-	33	-	dB	
Maximum Output Voltage (RMS Value)		V_{out}	5	39k	1	-	0.6	-	-	-	-	-	-	-	V(rms)	-
				10k	5	-	-	-	-	0.7	-	-	-	-	V(rms)	
				4.7k	10	-	-	-	-	-	-	-	0.5	-	V(rms)	

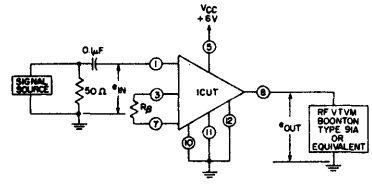
TEST SETUP FOR MEASUREMENT OF AGC SOURCE CURRENT



I_{AGC} IS THE CURRENT FLOWING INTO TERMINAL 2.

Fig. 4

TEST SETUP FOR MEASUREMENTS OF VOLTAGE-GAIN, -3dB BANDWIDTH, AND MAXIMUM OUTPUT VOLTAGE



PROCEDURES

Voltage Gain:

(a) Set e_{in} = 0.5 mV at frequency specified, read e_{out} Voltage Gain

(A) = 20 $\text{Log}_{10} \frac{e_{out}}{e_{in}}$

Bandwidth:

(a) Set e_{out} to a convenient reference voltage at f = 100 kHz and record corresponding value of e_{in}.

(b) Increase the frequency, keeping e_{in} constant until e_{out} drops 3-dB. Record Bandwidth.

Fig. 5

VOLTAGE GAIN VS FREQUENCY FOR CA3021

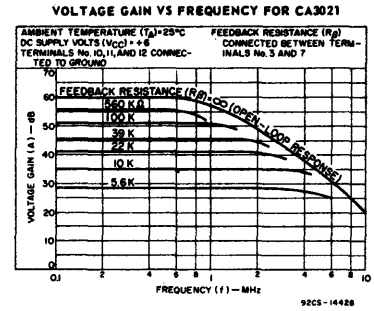


Fig. 6(a)

VOLTAGE GAIN VS FREQUENCY FOR CA3022

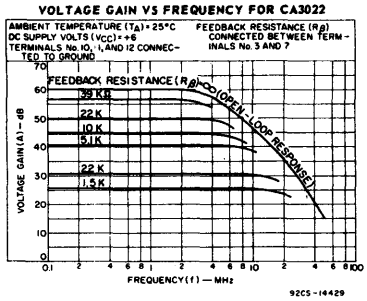


Fig. 6(b)

VOLTAGE GAIN VS FREQUENCY FOR CA3023

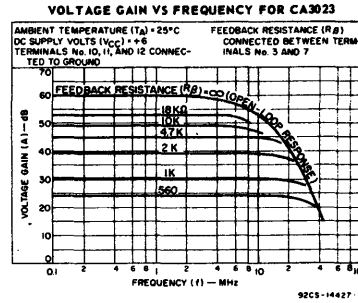


Fig. 6(c)

VOLTAGE GAIN VS TEMPERATURE FOR CA3021, CA3022, AND CA3023

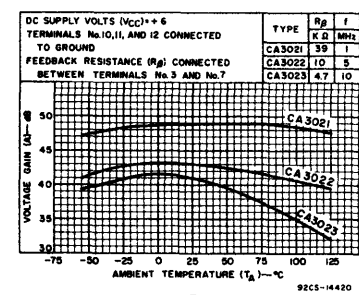
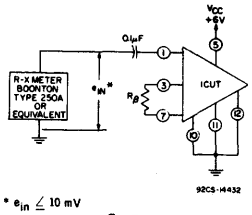


Fig. 6(d)

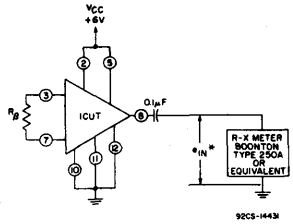
TEST SETUP FOR MEASUREMENT OF INPUT-IMPEDANCE COMPONENTS



* $e_{in} \leq 10$ mV

Fig.7

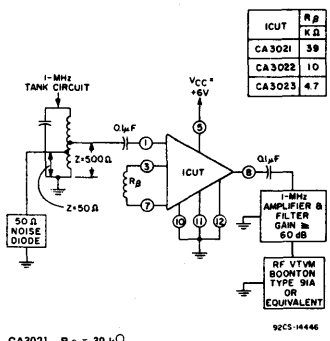
TEST SETUP FOR MEASUREMENT OF OUTPUT RESISTANCE



* $e_{in} \leq 10$ mV

Fig.8

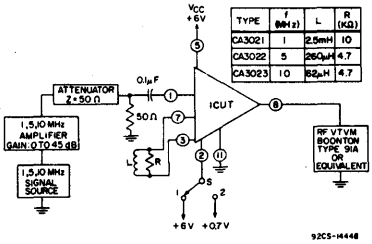
TEST SETUP FOR MEASUREMENT OF NOISE FIGURE



CA3021 - R_β = 39 kΩ
CA3022 - R_β = 10 kΩ
CA3023 - R_β = 4.7 kΩ

Fig.9

TEST SETUP FOR MEASUREMENT OF AGC RANGE



$$\text{AGC RANGE} = 20 \log_{10} \frac{\text{A WITH S IN POSITION 1}}{\text{A WITH S IN POSITION 2}}$$

(A = VOLTAGE GAIN)

	f MHz
CA3021	1
CA3022	5
CA3023	10

Fig.10

Dual Independent Differential Amplifiers

For Low-Power Applications at Frequencies from DC to 120 MHz

The CA3026 and CA3054 each consists of two independent differential amplifiers with associated constant-current transistors on a common monolithic substrate. The six n-p-n transistors which comprise the amplifiers are general purpose devices which exhibit low 1/f noise and a value of f_T in excess of 300 MHz. These features make the CA3026 and CA3054 useful from dc to 120 MHz. Bias and load resistors have been omitted to provide maximum application flexibility.

The monolithic construction of the CA3026 and CA3054 provides close electrical and thermal matching of the amplifiers. This feature makes these devices particularly useful in dual channel applications where matched performance of the two channels is required.

APPLICATIONS

- Dual sense amplifiers
- Dual Schmitt triggers
- Multifunction combinations -- RF Mixer, Oscillator; Converter-IF
- IF amplifiers (differential and or cascode)
- Product detectors
- Doubly balanced modulators and demodulators
- Balanced quadrature detectors
- Cascade limiters
- Synchronous detectors
- Pairs of balanced mixers
- Synthesizer mixers
- Balanced (push-pull) cascode amplifiers

FEATURES

- Two differential amplifiers on a common substrate
- Independently accessible inputs and outputs
- Maximum input offset voltage -- ±5 mV
- Full military temperature range capability -- -55°C to +125°C
- Limited temperature range -- 0°C to 85°C for CA3054

- **The CA3054 is available in a sealed-junction Beam-Lead version (CA3054L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".**
- **CA3026—Hermetic 12-lead TO-5 package**
- **CA3054—14-lead dual-in-line plastic package**

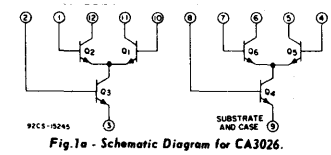


Fig.1a - Schematic Diagram for CA3026.

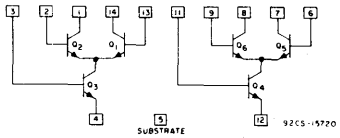


Fig.1b - Schematic Diagram for CA3054.

CAUTION: Substrate MUST be maintained negative with respect to all collector terminals of this device. See Maximum Voltage Ratings chart.

MAXIMUM RATINGS, ABSOLUTE-MAXIMUM VALUES, AT T_A = 25°C

	CA3026	CA3054	
Power Dissipation, P:			
Any one transistor	300	300	mW
Total package	600	750	mW
For T_A > 55°C	Derate at 5	6.67	mW/°C
Temperature Range:			
Operating	-55 to + 125		°C
Storage	-65 to + 150		°C
Lead Temperature (During Soldering):			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265		°C

The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

* The collector of each transistor of the CA3026 and CA3054 is isolated from the substrate by an integral diode. *The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide for normal transistor action. The substrate should be maintained at signal (AC) ground by means of a suitable grounding capacitor, to avoid undesired coupling between transistors.*

Maximum Voltage Ratings

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 1† and horizontal terminal 3† is +15 to -5 volts.

† For CA3026; corresponding terminals for CA3054 are vertical terminal 2 and horizontal terminal 4.

CA3054 TERMINAL No. →		13	14	1	2	3	4	6	7	8	9	11	12	5
↓	CA3026 TERMINAL No. →	10	11	12	1	2	3	4	5	6	7	8	Note 1 9	Note 1 9
13	10		0 -20	*	+5 -5	*	+15 -5	*	*	*	*	*	*	*
14	11			*	*	*	+20 0	*	*	*	*	*	*	+20 0
1	12				+20 0	*	+20 0	*	*	*	*	*	*	+20 0
2	1					*	+15 -5	*	*	*	*	*	*	*
3	2						+1 -5	*	*	*	*	*	*	*
4	3							*	*	*	*	*	*	*
6	4								0 -20	*	+5 -5	*	+15 -5	*
7	5									*	*	*	*	+20 0
8	6										+20 0	*	*	+20 0
9	7											*	+15 -5	*
11	8												+1 -5	*
12	9													*
5	9													Ref Substrate

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

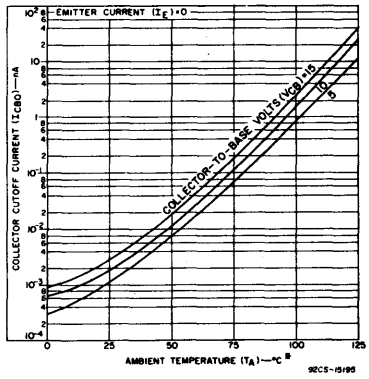
Note 1: In the CA3026 terminal No.9 is connected to the emitter of Q_4, the reference substrate, and the case; therefore, the case should not be grounded. Two terminal 9 columns (CA3026) appear in the voltage rating chart because it is a composite chart for both the CA3026 and the CA3054. Wherever an asterisk is shown in one column 9 and a rating is shown in the other column 9, the asterisk should be ignored.

Maximum Current Ratings

CA3054 TERMINAL No.●	CA3026 TERMINAL No.	I_{IN} mA	I_{OUT} mA
13	10	5	0.1
14	11	50	0.1
1	12	50	0.1
2	1	5	0.1
3	2	5	0.1
4	3	0.1	-50
6	4	5	0.1
7	5	50	0.1
8	6	50	0.1
9	7	5	0.1
11	8	5	0.1
12	9	0.1	50

● Terminal No.10 of CA3054 is not used

TYPICAL STATIC CHARACTERISTICS



* For CA3054: use data from 0°C to 85°C only

Fig.2 - Collector-to-base cutoff current vs ambient temperature for each transistor.

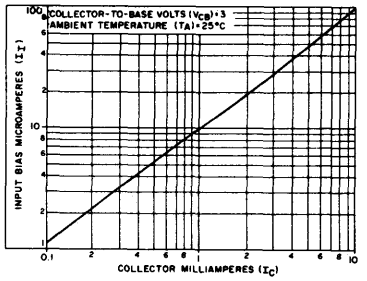


Fig.3 - Input bias current characteristic vs collector current for each transistor.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIR-CUIT	CA3026 CA3054 LIMITS			UNITS	TYPICAL CHARAC-TERISTICS CURVES
			FIG.	MIN.	TYP.	MAX.		FIG.
STATIC CHARACTERISTICS								
For Each Differential Amplifier								
Input Offset Voltage	V_{IO}	V_{CB} = 3 V	-	-	0.45	5	mV	6
Input Offset Current	I_{IO}		-	-	0.3	2	μA	7
Input Bias Current	I_I		-	-	10	24	μA	3
Quiescent Operating Current Ratio	$\frac{I_{C(Q1)}}{I_{C(Q2)}}$ or $\frac{I_{C(Q5)}}{I_{C(Q6)}}$	$I_{E(Q3)} = I_{E(Q4)}$ = 2 mA	-	-	0.98 to 1.02	-	-	3
Temperature Coefficient Magnitude of Input-Offset Voltage	$\frac{\lvert\Delta V_{IO}\rvert}{\Delta T}$		-	-	1.1	-	μV/°C	5
For Each Transistor								
DC Forward Base-to-Emitter Voltage	V_{BE}	V_{CB} = 3 V, I_C = 50 μA	-	-	0.630	0.700	V	6
		V_{CB} = 3 V, I_C = 1 mA	-	-	0.715	0.800		
		V_{CB} = 3 V, I_C = 3 mA	-	-	0.750	0.850		
		V_{CB} = 3 V, I_C = 10 mA	-	-	0.800	0.900		
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CB} = 3 V, I_C = 1 mA	-	-	-1.9	-	μV/°C	4
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	-	-	0.002	100	nA	2
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	-	15	24	-	V	-
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	-	20	60	-	V	-
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	-	20	60	-	V	-
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	-	5	7	-	V	-
DYNAMIC CHARACTERISTICS								
Common-Mode Rejection Ratio For Each Amplifier	CMR	V_{CC} = 12 V, V_{EE} = -6 V, V_x = -3.3 V, f = 1 kHz	8a	-	100	-	dB	8b
AGC Range, One Stage	AGC		9a	-	75	-	dB	9b
Voltage Gain, Single Stage Double-Ended Output	A		9a	-	32	-	dB	9b
AGC Range, Two Stage	AGC		10a	-	105	-	dB	10b
Voltage Gain, Two Stage Double-Ended Output	A		10a	-	60	-	dB	10b
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics: (For Single Transistor)								
Forward Current-Transfer Ratio	h_{fe}	f = 1 kHz, V_{CE} = 3 V, I_C = 1 mA	-	-	110	-	-	11
Short-Circuit Input Impedance	h_{ie}		-	-	3.5	-	kΩ	11
Open-Circuit Output Impedance	h_{oe}		-	-	15.6	-	μmho	11
Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		-	-	1.8×10^{-4}	-	-	11

DYNAMIC CHARACTERISTICS CONT'D

1/f Noise Figure (For Single Transistor)	NF	f = 1 kHz, V_{CE} = 3 V	-	-	3.25	-	dB	-
Gain-Bandwidth Product (For Single Transistor)	f_T	V_{CE} = 3 V, I_C = 3 mA	-	-	550	-	MHz	12
Admittance Characteristics; Differential Circuit Configuration: (For Each Amplifier)								
Forward Transfer Admittance	y_{21}	V_{CB} = 3 V, Each Collector $I_C \approx$ 1.25 mA, f = 1 MHz	-	-	-20+j0	-	mmho	13a
Input Admittance	y_{11}		-	-	0.22+j0.1	-	mmho	13b
Output Admittance	y_{22}		-	-	0.01+j0	-	mmho	13c
Reverse Transfer Admittance	y_{12}		-	-	-0.003+j0	-	mmho	13d
Admittance Characteristics; Cascode Circuit Configuration: (For Each Amplifier)								
Forward Transfer Admittance	y_{21}	V_{CB} = 3 V, Total Stage $I_C \approx$ 2.5 mA, f = 1 MHz	-	-	68-j0	-	mmho	14a
Input Admittance	y_{11}		-	-	0.55+j0	-	mmho	14b
Output Admittance	y_{22}		-	-	0+j0.02	-	mmho	14c
Reverse Transfer Admittance	y_{12}		-	-	0.004-j0.005	-	μmho	14d
Noise Figure	NF	f = 100 MHz	-	-	8	-	dB	-

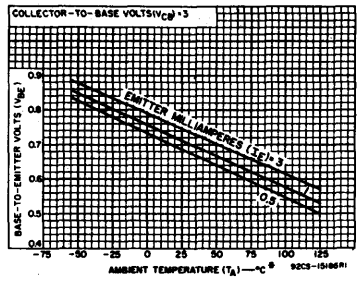


Fig.4 - Base-to-emitter voltage characteristic for each transistor vs ambient temperature.

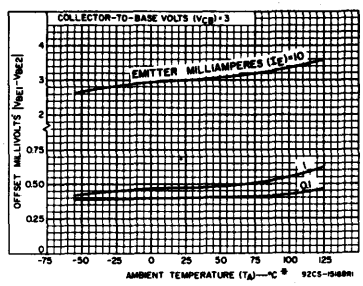


Fig.5 - Offset voltage characteristic vs ambient temperature for differential pairs.

* For CA3054: use data from 0°C to 85°C only

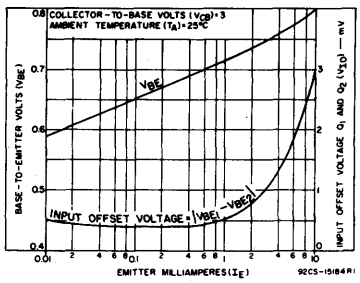


Fig.6 - Static base-to-emitter voltage characteristic and input offset voltage for differential pairs vs emitter current.

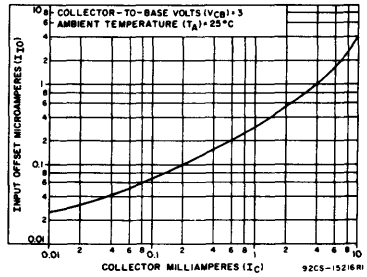
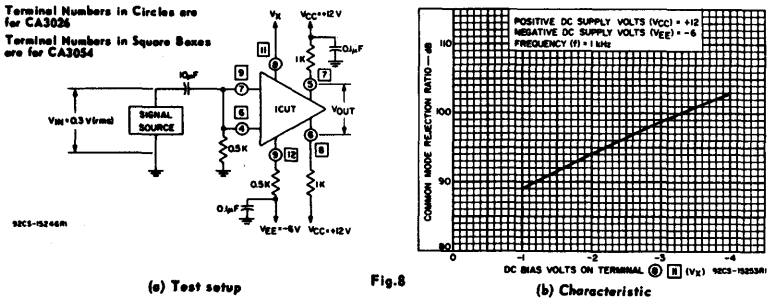


Fig.7 - Input offset current for matched differential pairs vs collector current.

TYPICAL DYNAMIC CHARACTERISTICS

COMMON MODE REJECTION RATIO

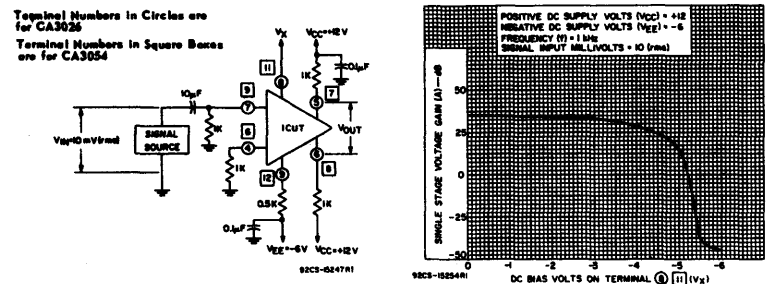


(a) Test setup

Fig.8

(b) Characteristic

SINGLE-STAGE VOLTAGE GAIN

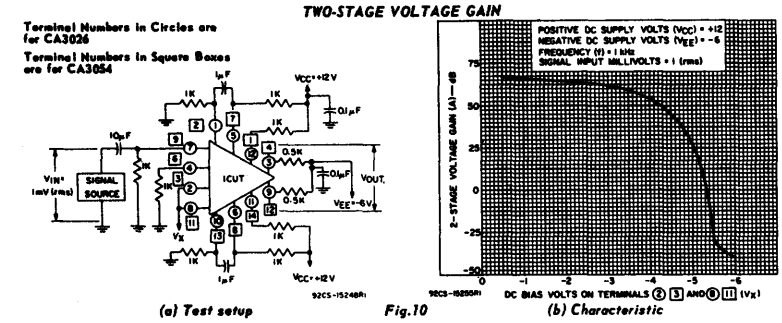


(a) Test setup

Fig.9

(b) Characteristic

TWO-STAGE VOLTAGE GAIN



(a) Test setup

Fig.10

(b) Characteristic

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

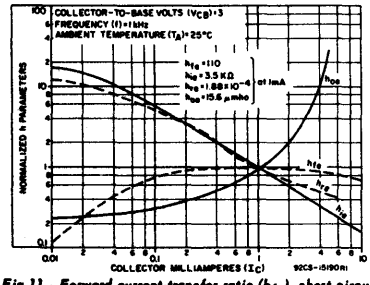


Fig.11 - Forward current-transfer ratio (h_{fe}), short-circuit input impedance (h_{ie}), open-circuit output impedance (h_{oe}), and open-circuit reverse voltage-transfer ratio (h_{re}) vs collector current for each transistor.

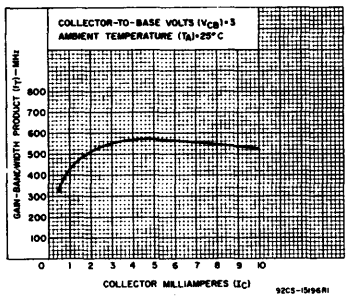


Fig.12 - Gain-bandwidth product (fT) vs collector current.

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH DIFFERENTIAL AMPLIFIER

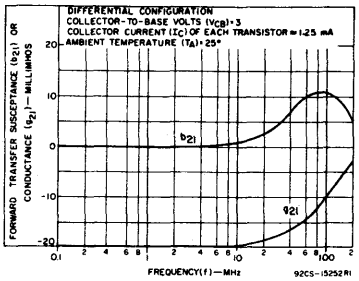


Fig.13(a) - Forward transfer admittance (Y_{21}) vs frequency.

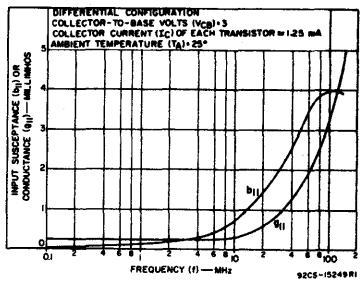


Fig.13(b) - Input admittance (Y_{11}).

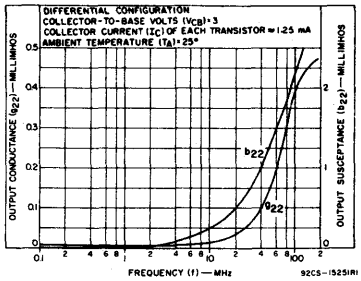


Fig.13(c) - Output admittance (Y_{22}) vs frequency.

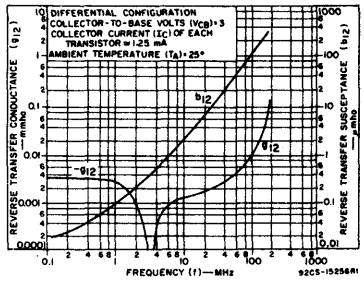


Fig.13(d) - Reverse transfer admittance (Y_{12}) vs frequency.

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH CASCODE AMPLIFIER

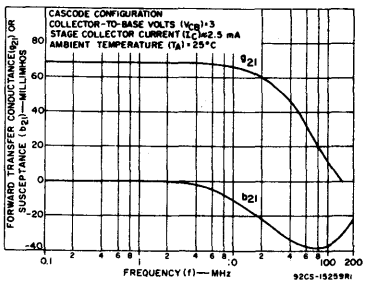


Fig.14(a) - Forward transfer admittance (Y_{21}) vs frequency.

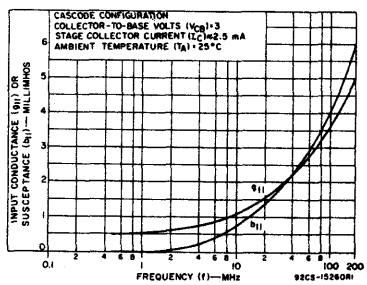


Fig.14(b) - Input admittance (Y_{11}) vs frequency.

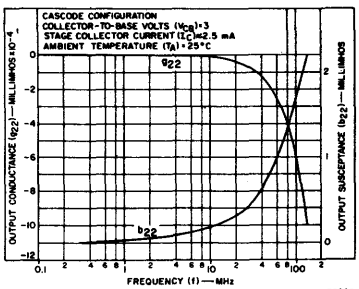


Fig.14(c) - Output admittance (Y_{22}) vs frequency.

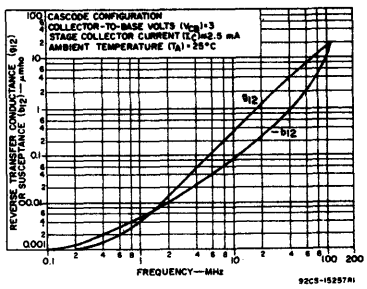


Fig.14(d) - Reverse transfer admittance (Y_{12}) vs frequency.

DIFFERENTIAL/CASCODE AMPLIFIERS

For Communications and Industrial Equipment at Frequencies from DC to 120 MHz

The CA3028A and CA3028B are differential/cascode amplifiers designed for use in communications and industrial equipment operating at frequencies from dc to 120 MHz.

The CA3028B is like the CA3028A but is capable of premium performance particularly in critical dc and differential amplifier applications requiring tight controls for input offset voltage, input offset current, and input bias current.

The CA3053 is similar to the CA3028A and CA3028B but is recommended for IF amplifier applications.

ABSOLUTE MAXIMUM RATINGS AT T_A = 25°C

DISSIPATION:
- At T_A up to 55°C (CA3028AF, CA3028BF, CA3053F) ..... 750 mW
- At T_A > 55°C (CA3028AF, CA3028BF, CA3053F) ..... Derate linearly 6.67 mW/°C
- At T_A up to 85°C (CA3028A, CA3028B, CA3053) ..... 450 mW
- At T_A > 85°C (CA3028A, CA3028B, CA3053) Derate linearly 5 mW/°C

AMBIENT-TEMPERATURE RANGE:
- Operating ..... −55°C to +125°C
- Storage ..... −65°C to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16 ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max. ..... +265°C

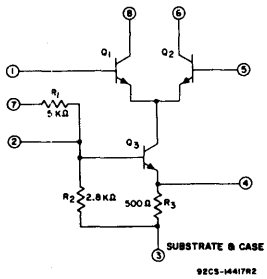


Fig.1 - Schematic diagram for CA3028A, CA3028B and CA3053.

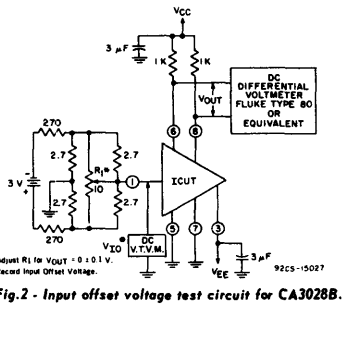


Fig.2 - Input offset voltage test circuit for CA3028B.

APPLICATIONS

- RF and IF Amplifiers (Differential or Cascode)
- DC, Audio, and Sense Amplifiers
- Converter in the Commercial FM Band
- Oscillator • Mixer • Limiter
- Companion Application Note, ICAN 5337 "Application of the RCA CA3028 Integrated Circuit Amplifier in the HF and VHF Ranges." This note covers characteristics of different operating modes, noise performance, mixer, limiter, and amplifier design considerations.

FEATURES

- Controlled for Input Offset Voltage, Input Offset Current, and Input Bias Current (CA3028B)
- Balanced Differential Amplifier Configuration with Controlled Constant-Current Source to Provide Unexcelled Versatility
- Single- and Dual-Ended Operation
- Operation from DC to 120 MHz
- Balanced-AGC Capability
- Wide Operating-Current Range

The CA3028A, CA3028B, and CA3053 are available in the packages shown below. When ordering these devices, it is important to add the appropriate suffix letter to the device.

Package 8-Lead TO-5	Suffix Letter	CA3028A	CA3028B	CA3053
TO-5	T	√	√	√
With Dual-In-Line Formed Leads (DIL-CAN)	S	√	√	√
Beam-Lead	L	√		
Chip	H	√		

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

TERMINAL No.	1	2	3	4	5	6	7	8
1		0 to -15▲	0 to -15▲	0 to -15▲	+5 to -5	*	*	+20⊕ to 0
2			+5 to -11	+5 to -1	+15♦ to 0	*	+15♦ to 0	*
3‡				+10 to 0	+15♦ to 0	+30⊕ to 0	+15♦ to 0	+30⊕ to 0
4					+15♦ to 0	*	*	*
5						+20⊕ to 0	*	*
6							*	*
7								*
8								

This chart gives the range of voltages which can be applied to the terminals listed horizontally with respect to the terminals listed vertically. For example, the voltage range of the horizontal terminal 4 with respect to terminal 2 is -1 to +5 volts.

‡ Terminal #3 is connected to the substrate and case.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe, if the specified voltage limits between all other terminals are not exceeded.

▲ Limit is -12V for CA3053

⊕ Limit is +15V for CA3053

♦ Limit is +12V for CA3053

● Limit is +24V for CA3028A and +18V for CA3053

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	0.6	0.1
2	4	0.1
3	0.1	23
4	20	0.1
5	0.6	0.1
6	20	0.1
7	4	0.1
8	20	0.1

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CIRCUIT Fig.	SPECIAL TEST CONDITIONS		LIMITS TYPE CA3028A			LIMITS TYPE CA3028B			LIMITS TYPE CA3053			UNITS	TYPICAL CHARACTERISTICS CURVES Fig.
					Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.		
STATIC CHARACTERISTICS			$+V_{CC}$	$-V_{EE}$											
Input Offset Voltage	V_{IO}	2	6V 12V	6V 12V	- -	- -	- -	- -	0.98 0.89	5 5	- -	- -	- -	mV	4
Input Offset Current	I_{IO}	3a	6V 12V	6V 12V	- -	- -	- -	- -	0.56 1.06	5 6	- -	- -	- -	μA	4
Input Bias Current	I_I	3a	6V 12V	6V 12V	- -	16.6 36	70 106	- -	16.6 36	40 80	- -	- -	- -	μA	5a
		3b	9V 12V	-	- -	- -	- -	- -	- -	- -	- -	29 36	85 125		5b
Quiescent Operating Current	I_6 or I_8	3a	6V 12V	6V 12V	0.8 2	1.25 3.3	2 5	1 2.5	1.25 3.3	1.5 4	- -	- -	- -	mA	6a 7
		3b	9V 12V	-	- -	- -	- -	- -	- -	- -	1.2 2.0	2.2 3.3	3.5 5.0		6b
AGC Bias Current (Into Constant-Current Source Terminal No.7)	I_7	8a	12V 12V	V_{AGC} = +9 V_{AGC} = +12	- -	1.28 1.65	- -	- -	1.28 1.65	- -	- -	- -	- -	mA	8b
		-	9V 12V	-	- -	- -	- -	- -	- -	- -	- -	1.15 1.55	- -		-
Input Current (Terminal No.7)	I_7	-	6V 12V	6V 12V	0.5 1	0.85 1.65	1 2.1	0.5 1	0.85 1.65	1 2.1	- -	- -	- -	mA	-
Device Dissipation	P_T	3a	6V 12V	6V 12V	24 120	36 175	54 260	24 120	36 175	42 220	- -	- -	- -	mW	9
		3b	9V 12V	-	- -	- -	- -	- -	- -	- -	- -	50 100	80 150		-

ELECTRICAL CHARACTERISTICS at T_A = 25°C (cont'd)

CHARACTERISTIC	SYMBOL	TEST CIRCUIT Fig.	SPECIAL TEST CONDITIONS		LIMITS TYPE CA3028A Min.	Typ.	Max.	LIMITS TYPE CA3028B Min.	Typ.	Max.	UNITS	TYPICAL CHARACTERISTICS CURVE Fig.
DYNAMIC CHARACTERISTICS												
Power Gain	G_P	10a	f = 100 MHz	Cascode	16	20	-	16	20	-	dB	10b
		11a,d	V_{CC} = +9V	Diff.-Ampl.	14	17	-	14	17	-		11b,e
		10a	f = 10.7 MHz	Cascode	35	39	-	35	39	-	dB	10b
		11a	V_{CC} = +9V	Diff.-Ampl.	28	32	-	28	32	-		11b
Noise Figure	NF	10a	f = 100 MHz	Cascode	-	7.2	9	-	7.2	9	dB	10c
		11a,d	V_{CC} = +9V	Diff.-Ampl.	-	6.7	9	-	6.7	9		11c,e
Input Admittance	Y_{11}	-	f = 10.7 MHz V_{CC} = +9V	Cascode				-	0.6 + j 1.6	-	mmho	12
		-		Diff.-Ampl.				-	0.5 + j 0.5	-		13
Reverse Transfer Admittance	Y_{12}	-		Cascode				-	0.0003 - j0	-	mmho	14
		-		Diff.-Ampl.				-	0.01 - j0.0002	-		15
Forward Transfer Admittance	Y_{21}	-		Cascode				-	99 - j18	-	mmho	16
		-		Diff.-Ampl.				-	-37 + j0.5	-		17
Output Admittance	Y_{22}	-		Cascode				-	0. + j0.08	-	mmho	18
		-		Diff.-Ampl.				-	0.04 + j0.23	-		19
Power Output (Untuned)	P_o	20a	f = 10.7 MHz	Diff.-Ampl. 50 Ω Input, Output	-	5.7	-	-	5.7	-	mW	20b
AGC Range (Max. Power Gain to Full Cutoff)	AGC	21a	V_{CC} = +9V	Diff.-Ampl.	-	62	-	-	62	-	dB	21b
Voltage Gain at f = 10.7 MHz	A	22a	f = 10.7 MHz	Cascode	-	40	-	-	40	-	dB	22b
		22c	V_{CC} = +0V R_L = 1 kΩ	Diff.-Ampl.	-	30	-	-	30	-		22d
Voltage Gain Differential at f = 1 kHz		23	V_{CC} = +6V, V_{EE} = -6V, R_L = 2 kΩ		-	-	-	35	38	42	dB	-
			V_{CC} = +12V, V_{EE} = -12V R_L = 1.6 kΩ		-	-	-	40	42.5	45		
Max. Peak-to-Peak Output Voltage at f = 1 kHz	V_o(P-P)	23	V_{CC} = +6V, V_{EE} = -6V, R_L = 2 kΩ		-	-	-	7	11.5	-	V_{P-P}	-
			V_{CC} = +12V, V_{EE} = -12V R_L = 1.6 kΩ		-	-	-	15	23	-		
Bandwidth at -3 dB point	BW	23	V_{CC} = +6V, V_{EE} = -6V, R_L = 2 kΩ		-	-	-	-	7.3	-	MHz	-
			V_{CC} = +12V, V_{EE} = -12V R_L = 1.6 kΩ		-	-	-	-	8	-		
Common-Mode Input-Voltage Range	V_{CMR}	24	V_{CC} = +6V, V_{EE} = -6V V_{CC} = +12V, V_{EE} = -12V		-	-	-	±2.5 ±5	(-3.2 - 4.5) (-7 - 9)	4 7	V	-
Common-Mode Rejection Ratio	CMR	24	V_{CC} = +6V, V_{EE} = -6V V_{CC} = +12V, V_{EE} = -12V		-	-	-	60 60	110 90	-	dB	-
Input Impedance at f = 1 kHz	Z_{IN}		V_{CC} = +6V, V_{EE} = -6V V_{CC} = +12V, V_{EE} = -12V		-	-	-	-	5.5 3	-	kΩ	-
Peak-to-Peak Output Current	I_{P-P}		V_{CC} = +9V	f = 10.7 MHz	2	4	7	2.5	4	6	mA	-
			V_{CC} = +12V	e_{in} = 400 mV Diff.-Ampl.	3.5	6	10	4.5	6	8		

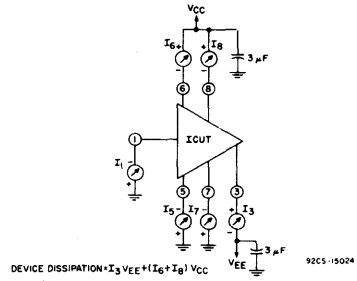


Fig.3a - Input offset current, input bias current, device dissipation, and quiescent operating current test circuit for CA3028A and CA3028B.

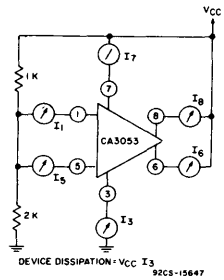


Fig.3b - Input bias current, device dissipation, and quiescent operating current test circuit for CA3053.

ELECTRICAL CHARACTERISTICS at T_A = 25°C (cont'd)

CHARACTERISTIC	SYMBOL	TEST CIRCUIT Fig.	SPECIAL TEST CONDITIONS		LIMITS TYPE CA3053 Min.	Typ.	Max.	UNITS	TYPICAL CHARACTERISTICS CURVE Fig.
DYNAMIC CHARACTERISTICS									
Power Gain	G_P	10a	f = 10.7 MHz	Cascode	35	39	-	dB	
		11a	V_{CC} = +9V	Diff.-Ampl.	28	32	-		
Input Admittance	Y_{11}	-	f = 10.7 MHz V_{CC} = +9V	Cascode	-	0.6 + j 1.6	-	mmho	12
		-		Diff.-Ampl.	-	0.5 + j 0.5	-		13
Reverse Transfer Admittance	Y_{12}	-		Cascode	-	0.0003 - j0	-	mmho	14
		-		Diff.-Ampl.	-	0.01 - j0.0002	-		15
Forward Transfer Admittance	Y_{21}	-		Cascode	-	99 - j18	-	mmho	16
		-		Diff.-Ampl.	-	-37 + j0.5	-		17
Output Admittance	Y_{22}	-		Cascode	-	0. + j0.08	-	mmho	18
		-		Diff.-Ampl.	-	0.04 + j0.23	-		19
Voltage Gain at f = 10.7 MHz	A	22a	f = 10.7 MHz	Cascode	-	40	-	dB	22b
		22c	V_{CC} = +0V R_L = 1 kΩ	Diff.-Ampl.	-	30	-		22d
Peak-to-Peak Output Current	P-P		V_{CC} = +9V	f = 10.7 MHz	2	4	7	mA	
			V_{CC} = +12V	e_{in} = 400 mV Diff.-Ampl.	3.5	6	10		

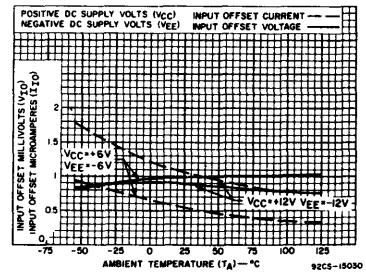


Fig.4 - Input offset voltage and input offset current for CA3028B.

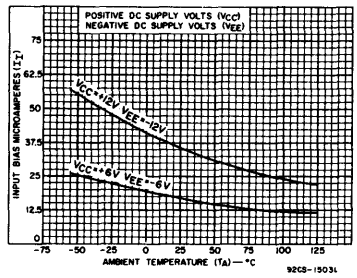


Fig.5a - Input bias current vs. ambient temperature for CA3028A and CA3028B.

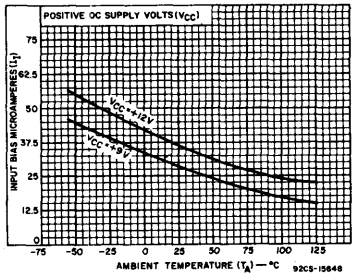


Fig.5b - Input bias current vs. ambient temperature for CA3053.

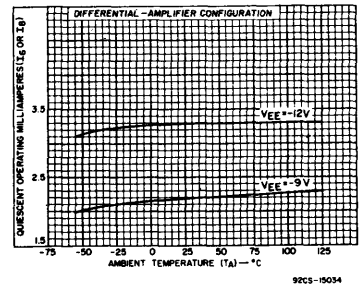


Fig.6a - Quiescent operating current vs. ambient temperature for CA3028A and CA3028B.

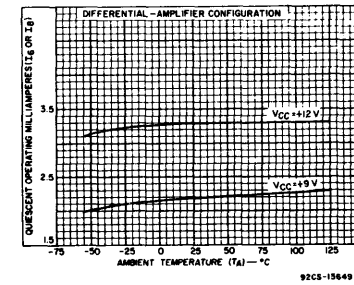


Fig.6b - Quiescent operating current vs. ambient temperature for CA3053.

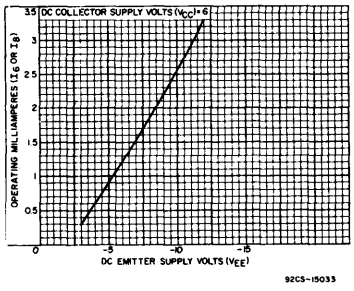


Fig.7 - Operating current vs. V_{EE} voltage for CA3028A and CA3028B.

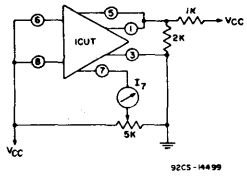


Fig.8a - AGC bias current test circuit (differential-amplifier configuration) for CA3028A and CA3028B.

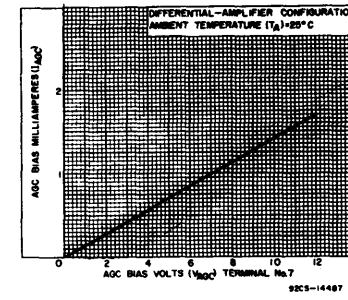


Fig.8b - AGC bias current vs. bias volts (terminal No.7) for CA3028A and CA3028B.

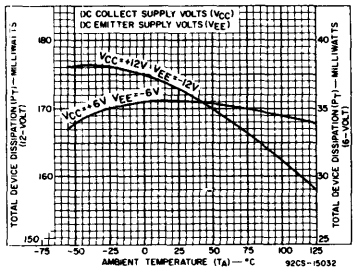
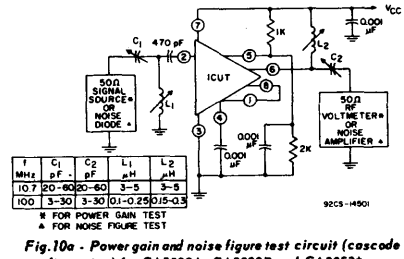


Fig.9 - Device dissipation vs. temperature for CA3028A and CA3028B.



f MHz	C1 pF	C2 pF	L1 µH	L2 µH
10.7	20-60	20-60	3-5	3-5
100	3-30	3-30	0.1-0.25	0.15-0.3

* FOR POWER GAIN TEST
▲ FOR NOISE FIGURE TEST

Fig.10a - Power gain and noise figure test circuit (cascode configuration) for CA3028A, CA3028B and CA3053.*

* 10.7 MHz Power Gain Test Only.

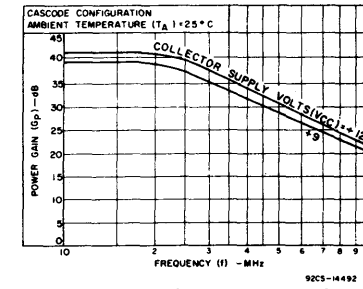


Fig.10b - Power gain vs. frequency (cascode configuration) for CA3028A and CA3028B.

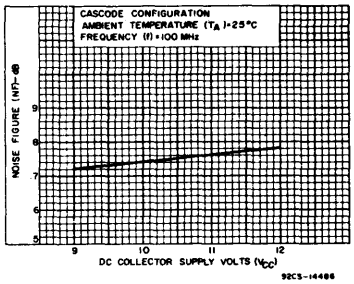
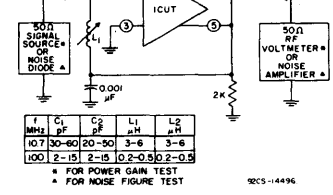


Fig.10c - 100 MHz noise figure vs. collector supply volts (cascode configuration) for CA3028A and CA3028B.

TYPICAL NOISE FIGURE AND POWER GAIN TEST CIRCUITS AND CHARACTERISTICS



f MHz	C1 pF	C2 pF	L1 µH	L2 µH
10.7	30-60	20-50	3-6	3-6
100	2-15	2-15	0.2-0.5	0.2-0.5

* FOR POWER GAIN TEST
▲ FOR NOISE FIGURE TEST

Fig.11a - Power gain and noise figure test circuit (differential-amplifier configuration and terminal No.7 connected to V_{CC}) for CA3028A, CA3028B and CA3053.*

* 10.7 MHz Power Gain Test Only.

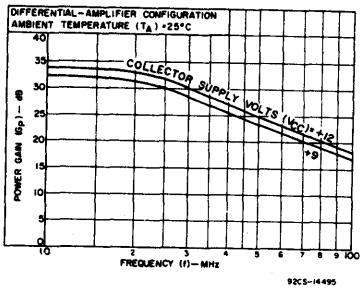


Fig.11b - Power gain vs. frequency (differential-amplifier configuration) for CA3028A and CA3028B.

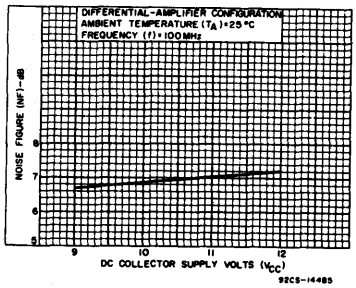


Fig.11c - 100 MHz noise figure vs. collector supply voltage (differential-amplifier configuration) for CA3028A and CA3028B.

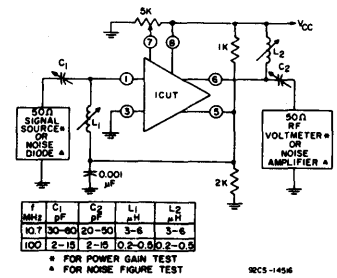


Fig.11d - Power gain and noise figure test circuit (differential-amplifier configuration for CA3028A and CA3028B.

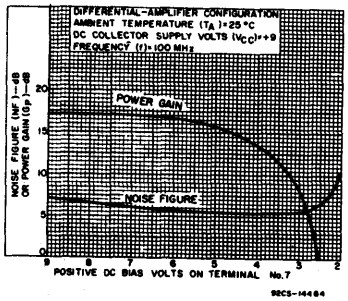


Fig.11e - 100 MHz noise figure and power gain vs. base-to-emitter bias (terminal No.7) for CA3028A and CA3028B.

TYPICAL ADMITTANCE PARAMETERS

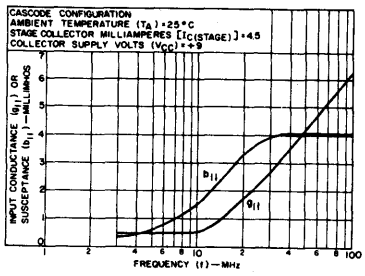


Fig.12 - Input admittance (Y_{11}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

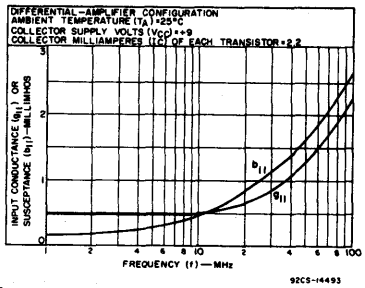


Fig.13 - Input admittance (Y_{11}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

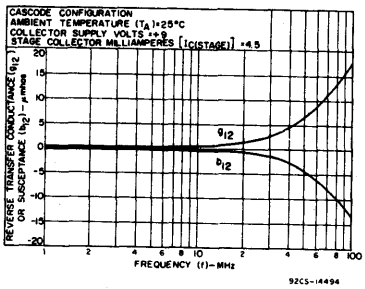


Fig.14 - Reverse transadmittance (Y_{12}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

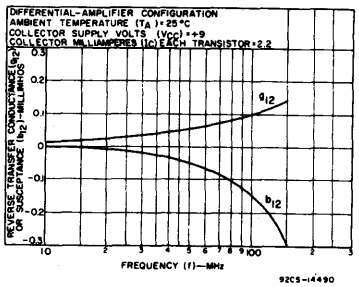


Fig.15 - Reverse transadmittance (Y_{12}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

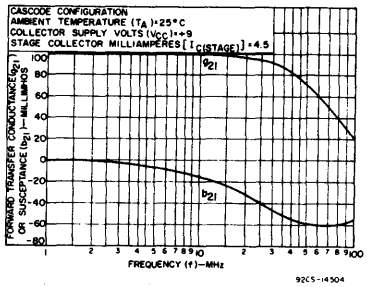


Fig.16 - Forward transadmittance (Y_{21}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

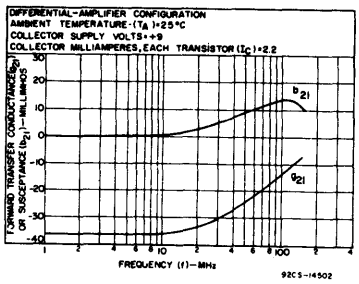


Fig.17 - Forward transadmittance (Y_{21}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

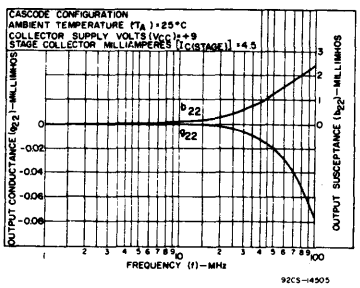


Fig.18 - Output admittance (Y_{22}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

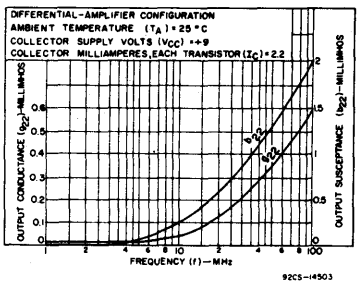


Fig.19 - Output admittance (Y_{22}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

TYPICAL TEST CIRCUITS AND CHARACTERISTICS

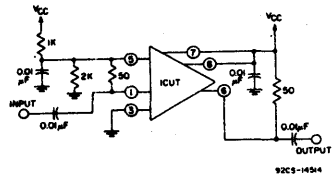


Fig.20a - Output power test circuit for CA3028A and CA3028B.

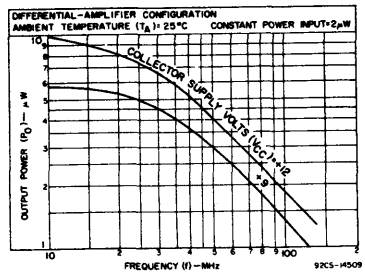
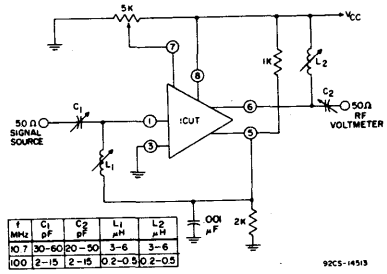


Fig.20b - Output power vs. frequency — 50 Ω input and 50 Ω output (differential-amplifier configuration) for CA3028A and CA3028B.



f MHz	C1 pF	C2 pF	L1 μH	L2 μH
10.7	30-60	20-50	3-6	3-6
100	2-15	2-15	0.2-0.5	0.2-0.5

Fig.21a - AGC range test circuit (differential amplifier) for CA3028A and CA3028B.

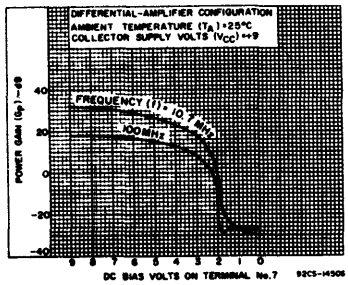


Fig.21b - AGC characteristics for CA3028A and CA3028B.

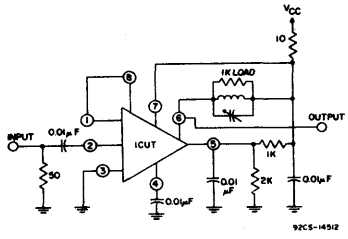


Fig.22a - Transfer characteristic (voltage gain) test circuit (10.7 MHz) cascode configuration for CA3028A, CA3028B and CA3053.

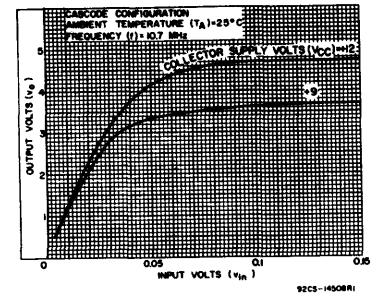


Fig.22b - Transfer characteristics (cascode configuration) for CA3028A, CA3028B and CA3053.

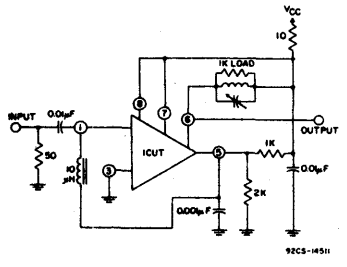


Fig.22c - Transfer characteristic (voltage gain) test circuit (10.7 MHz) differential-amplifier configuration for CA3028A, CA3028B and CA3053.

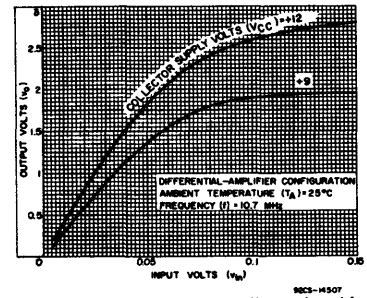
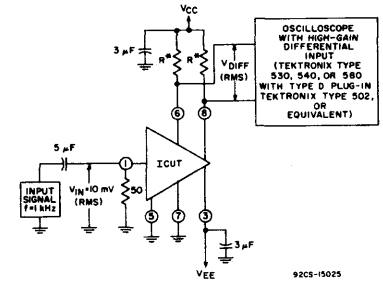
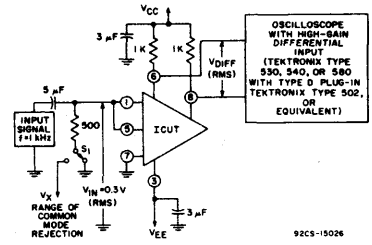


Fig.22d - Transfer characteristics (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.



* For R = 1.6 kΩ — (V_{CC} = 12V, V_{EE} = -12V)
For R = 2 kΩ — (V_{CC} = 6V, V_{EE} = -6V)

Fig.23 - Differential voltage gain, maximum peak-to-peak output voltage, and bandwidth test circuit for CA3028B.



For CMR test: S_1 to ground
For input common-mode voltage range test: S_1 to V_X

Common mode rejection ratio = $20 \log_{10} \frac{(A^*)\,(2)\,(0.3)}{V_{DIFF\,(RMS)}}$

* A = Single-ended voltage gain.

Fig.24 - Common-mode rejection ratio and common-mode input-voltage range test circuit for CA3028B.

CA3036
DUAL DARLINGTON ARRAY

- Two independent low-noise wide-band amplifier channels
- Particularly useful for preamplifier and low-level amplifier applications in single-channel and stereo systems
- Wide application in low-noise industrial instrumentation amplifiers
- Hermetically sealed, all-welded 10-lead TO-5-style metal package

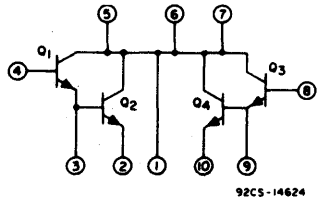


Fig.1 – Schematic Diagram for CA3036.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS		SYMBOLS	TEST CONDITIONS	LIMITS TYPE CA3036 Min.	Typ.	Max.	UNITS
For Each Transistor (Q_1, Q_2, Q_3, Q_4)	Collector-Cutoff Current	I_{CBO}	V_{CB} = 5V, I_E = 0	–	–	0.5	μA
	Collector-Cutoff Current	I_{CEO}	V_{CE} = 10 V, I_B = 0	–	–	5	μA
	Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	15	20	–	V
	Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	30	44	–	V
	Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	5	6	–	V
For Either Input Transistor (Q_1 or Q_3)	Static Forward Current-Transfer Ratio	h_{FE}	I_{C1} or I_{C3} = 1 mA	30	82	–	–
For Either Darlington Pair (Q_1, Q_2 or Q_3, Q_4)	Emitter-to-Base Breadkown Voltage	$V_{(BR)EBO(D)}$	I_{E2} or I_{E4} = 10 μA	10	12.6	–	V
	Static Forward Current-Transfer Ratio	$h_{FE(D)}$	$I_{C1} + I_{C2}$ or $I_{C3} + I_{C4}$ = 1 mA	1000	4540	–	–
For Each Input Transistor (Q_1 or Q_3)	Short-Circuit Forward Current-Transfer Ratio	h_{fe}	f = 1 kHz, I_{C1} or I_{C3} = 1 mA	–	82	–	–
	Short-Circuit Input Impedance	h_{ie}		–	2.6K	–	Ω
	Open-Circuit Output Admittance	h_{oe}		–	7	–	μmho
	Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		–	9.8 x 10-5	–	–
For Either Darlington Pair (Q_1, Q_2 or Q_3, Q_4)	Short-Circuit Forward Current-Transfer Ratio	$h_{fe(D)}$	f = 1 kHz, $I_{C1} + I_{C2}$ or $I_{C3} + I_{C4}$ = 1 mA	–	1300	–	–
	Short-Circuit Input Impedance	$h_{ie(D)}$		–	82K	–	Ω
	Open-Circuit Output Admittance	$h_{oe(D)}$		–	108	–	μmho
	Open-Circuit Reverse Voltage-Transfer Ratio	$h_{re(D)}$		–	2.7 x 10-3	–	–
	Voltage Gain	$A_{(D)}$		–	26	–	dB
	Power Gain	$G_{p(D)}$		–	47	–	dB
	Noise Voltage See Fig.3 for Test Circuit	E_N	f = 100 Hz	–	0.2	3	$\frac{\mu V(rms)}{\sqrt{f(Hz)}}$
			f = 1 kHz	–	0.05	0.3	
			f = 10 kHz	–	0.012	0.1	
For Either Input Transistor (Q_1 or Q_3)	Forward Transfer Admittance	y_{fe}	f = 50 MHz, I_{C1} or I_{C3} = 2 mA	–	0.68 + j 7.9	–	mmho
	Input Admittance (Output Short-Circuited)	y_{ie}		–	4.14 + j 5.95	–	mmho
	Output Admittance (Input Short-Circuited)	y_{oe}		–	1.94 + j 2.64	–	mmho
	Reverse Transfer Admittance (Input Short-Circuited)	y_{re}		–	Negligible	–	mmho
For either Darlington Pair (Q_1, Q_2 or Q_3, Q_4)	Input Admittance (Output Short-Circuited)	$y_{ie(D)}$	f = 50 MHz, $I_{C1} + I_{C2}$ or $I_{C3} + I_{C4}$ = 2 mA	–	1.71 + j 2.8	–	mmho
	Output Admittance (Input Short-Circuited)	$y_{oe(D)}$		–	3.96 + j 2.6	–	mmho
	Gain-Bandwidth Product	$f_{T(D)}$		150	200	–	MHz

HIGHLIGHTS

- Matched transistors with emitter-follower outputs
- Low-noise performance
- 200-MHz gain-bandwidth product
- Operation from -55°C to +125°C

APPLICATIONS

- Stereo phonograph preamplifiers
- Low-level stereo and single channel amplifier stages
- Low-noise, emitter-follower differential amplifiers
- Operational amplifier drivers

MAXIMUM RATINGS, Absolute-Maximum Values:

POWER DISSIPATION, P:

Any one transistor	300 max. mW
Total for array	600 max. mW

TEMPERATURE RANGE:

Operating	–55 to +125 °C
Storage	–65 to +150 °C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265 °C

The following ratings apply for each transistor in the array:

Collector-to-Emitter Voltage, V_{CEO}	15 max. V
Collector-to-Base Voltage, V_{CBO}	30 max. V
Emitter-to-Base Voltage, V_{EBO}	5 max. V
Collector Current, I_C	50 max. mA

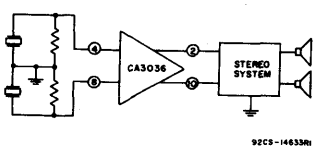


Fig.2 - Block Diagram of Stereo System using CA3036 as Phono Preamplifier.

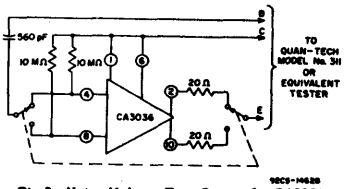


Fig.3 - Noise Voltage Test Circuit for CA3036.

Diode Array

Six Matched Diodes on a Common Substrate

ULTRA-FAST LOW-CAPACITANCE MATCHED DIODES

For Applications in Communications and Switching Systems

The RCA-CA3039 consists of six ultra-fast, low capacitance diodes on a common monolithic substrate. Integrated circuit construction assures excellent static and dynamic matching of the diodes, making the array extremely useful for a wide variety of applications in communication and switching systems.

Five of the diodes are independently accessible, the sixth shares a common terminal with the substrate.

For applications such as balanced modulators or ring modulators where capacitive balance is important, the substrate should be returned to a DC potential which is significantly more negative (with respect to the active diodes) than the peak signal applied.

APPLICATIONS

- Balanced modulators or demodulators
- Ring modulators
- High speed diode gates
- Analog switches

FEATURES

- Excellent reverse recovery time – 1 ns typ.
- Matched monolithic construction – V_F matched within 5 mV
- Low diode capacitance – C_D = 0.65 pF typical at V_R = –2 V
- The CA3039 is available in a sealed-junction Beam-Lead version (CA3039L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".

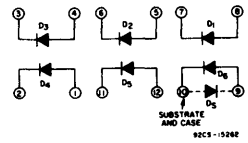


Fig. 1 - Schematic Diagram for CA3039

ABSOLUTE MAXIMUM RATINGS AT T_A = 25°C

DISSIPATION:	
Any one diode unit	100 mW
Total for device	600 mW
For T_A > 55°C	derate linearly 5.7 mW/°C
TEMPERATURE RANGE:	
Operating	–55 to +125°C
Storage	–65 to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+ 265°C
PEAK INVERSE VOLTAGE, PIV for: D_1–D_5	5 V
D_6	0.5 V
PEAK DIODE-TO-SUBSTRATE VOLTAGE, V_{DI} for D_1–D_5 (term. 1,4,5,8 or 12 to term. 10)	+ 20, –1 V
DC FORWARD CURRENT, I_F	25 mA
PEAK RECURRENT FORWARD CURRENT, I_f	100 mA
PEAK FORWARD SURGE CURRENT, $I_{f(surge)}$	100 mA

ELECTRICAL CHARACTERISTICS, at T_A = 25° C

Characteristics apply for each diode unit, unless otherwise specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	CHARACTERISTIC CURVES FIG.
DC Forward Voltage Drop	V_F	I_F = 50 μA	–	0.65	0.69	V	2
		1 mA	–	0.73	0.78	V	
		3 mA	–	0.76	0.80	V	
		10 mA	–	0.81	0.90	V	
DC Reverse Breakdown Voltage	$V_{(BR)R}$	I_R = –10 μA	5	7	–	V	–
DC Reverse Breakdown Voltage Between any Diode Unit and Substrate	$V_{(BR)R}$	I_R = –10 μA	20	–	–	V	–
DC Reverse (Leakage) Current	I_R	V_R = –4 V	–	0.016	100	nA	3
DC Reverse (Leakage) Current Between any Diode Unit and Substrate	I_R	V_R = –10 V	–	0.022	100	nA	4
Magnitude of Diode Offset Voltage (Difference in DC Forward Voltage Drops of any Two Diode Units)	$\lvert V_{F_1} - V_{F_2} \rvert$	I_F = 1 mA	–	0.5	5	mV	2
Temperature Coefficient of $\lvert V_{F_1} - V_{F_2} \rvert$	$\frac{\Delta \lvert V_{F_1} - V_{F_2} \rvert}{\Delta T}$	I_F = 1 mA	–	1	–	μV/°C	5
Temperature Coefficient of Forward Drop	$\frac{\Delta V_F}{\Delta T}$	I_F = 1 mA	–	–1.9	–	mV/°C	6
DC Forward Voltage Drop for Anode-to-Substrate Diode (D_S)	V_F	I_F = 1 mA	–	0.65	–	V	–
Reverse Recovery Time	t_{rr}	I_F = 10 mA, I_R = 10 mA	–	1	–	ns	–
Diode Resistance	R_D	f = 1 kHz, I_F = 1 mA	25	30	45	Ω	7
Diode Capacitance	C_D	V_R = –2 V, I_F = 0	–	0.65	–	pF	8
Diode-to-Substrate Capacitance	C_{DI}	V_{DI} = +4 V, I_F = 0	–	3.2	–	pF	9

TYPICAL CHARACTERISTICS

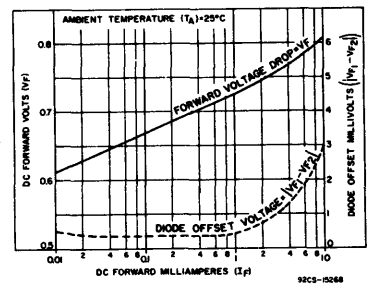


Fig. 2 - DC forward voltage drop (any diode) and diode offset voltage vs DC forward current

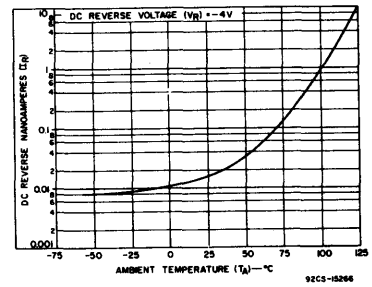


Fig. 3 - DC reverse (leakage) current (diodes 1,2,3,4,5) vs temperature

This device is supplied in the hermetic 12-lead TO-5 style package.

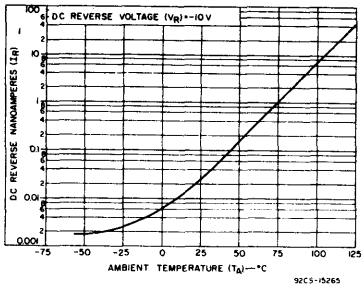


Fig. 4 - DC reverse (leakage) current between diodes (1,2,3,4,5) and substrate vs temperature

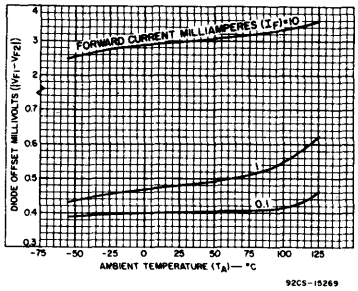


Fig. 5 - Diode offset voltage (any diode) vs temperature

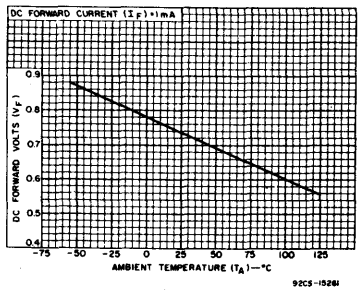


Fig. 6 - DC forward voltage drop (any diode) vs temperature

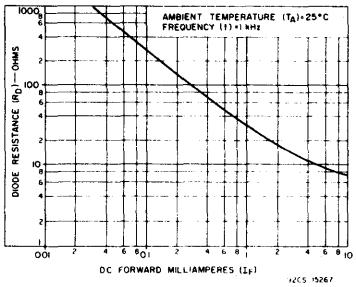


Fig. 7 - Diode resistance (any diode) vs DC forward current

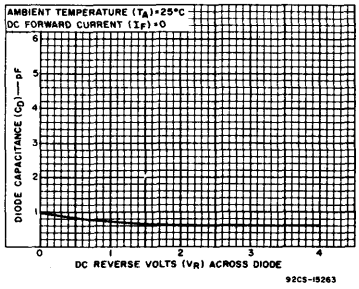


Fig. 8 - Diode capacitance (diodes 1,2,3,4,5) vs reverse voltage

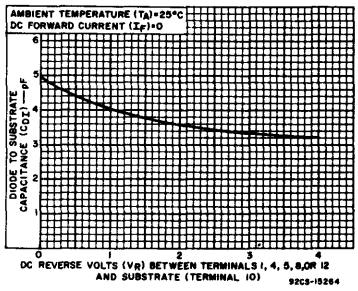


Fig. 9 - Diode-to-substrate capacitance vs reverse voltage

Video and Wideband Amplifier

For Industrial and Commercial Equipment at Frequencies up to 200 MHz

The RCA CA3040 is a monolithic silicon integrated circuit designed to meet the requirements of a wide variety of applications requiring high gain and wide bandwidth. The cascode-connected differential amplifier achieves a double-ended gain of 37 dB with a typical 3 dB bandwidth of 55 MHz. Emitter-Follower input and output stages provide the desirable high input impedance and low output impedance for coupling to other circuits.

The CA3040 includes two biasing options, allowing the user to optimize his design over the entire military temperature range of -55 to +125°C. **Bias Mode A** yields a substantially constant voltage at the output terminals for applications using DC coupling to succeeding stages or requiring maximum dynamic range over the temperature range. DC output voltage varies less than 0.1 volt (typically) over the entire temperature range while gain varies ±2 dB. **Bias Mode B** provides extremely stable gain over the temperature range. Gain variation is 0 dB (typically) in this Bias Mode. DC variation is ±0.8 volt.

Provisions are also made for stabilizing the operating point for either single or split power supplies.

FEATURES

- High Differential Push-Pull Voltage Gain ..... 37 dB typ.
  Single-Ended Voltage Gain ..... 31 dB typ.
- Wide (3dB) Bandwidth ..... 55 MHz typ.
- Balanced Input and Output
- High Input Resistance ..... 150 kΩ typ.
- Low Output Resistance ..... 125 Ω typ.
- Bias Options for Temperature Compensation:
  Bias Mode A: "Constant" Voltage
  Bias Mode B: "Constant" Gain
- **Supplied in the hermetic 12-lead TO-5 style package**

APPLICATIONS

- Video Amplifier
- Modulator
- Mixer
- Schmitt Trigger
- IF Amplifier
- DC Amplifier
- Sense Amplifier

ABSOLUTE-MAXIMUM RATINGS

DISSIPATION * ..... 450 mW
Derating factor for T_A > 85°C ..... 5 mW/°C
TEMPERATURE RANGE:
Operating ..... -55°C to +125°C
Storage ..... -65°C to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. ..... +265°C

* Limitation imposed by the thermal resistance of package.

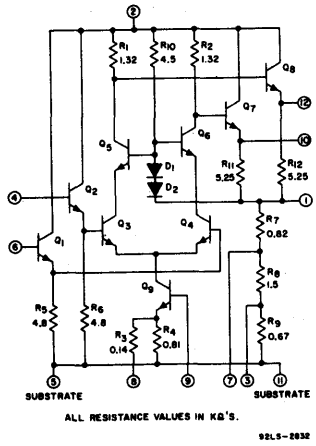


Fig.1 - Schematic Diagram for CA3040

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 2 with respect to terminal 11 is 0 to +14 volts.

TERMINAL No.	1	2	3	4	5▲	6	7	8	9	10	11▲	12
1		0 -14	*	*	+14 0	*	+10 -10	*	*	*	+14 0	*
2			*	+14 0	+14 0	+14 0	*	*	*	+14 0	+14 0	+14 0
3				*	+5 -3	*	*	*	*	*	+5 -3	*
4					*	+3 -3	*	*	*	*	*	*
5▲					▲	*	+10 -3	*	+3 -7	*	0 Note 1	*
6							*	*	*	*	*	*
7								*	*	*	+10 -3	*
8									+3 -3	*	*	*
9										*	+7 -3	*
10											*	*
11▲											▲	*
12												

▲ Reference Substrate

Note 1: External connection required for proper operation.

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	5	5
2	-	-
3	5	5
4	1	0.1
5	-	-
6	1	0.1
7	5	5
8	5	5
9	1	0.1
10	-	10
11	-	-
12	-	10

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

STATIC CHARACTERISTICS TEST CIRCUITS

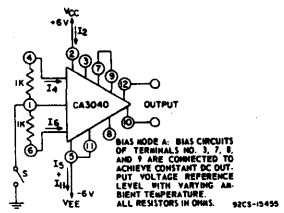


Fig.2(a) - Bias Mode A

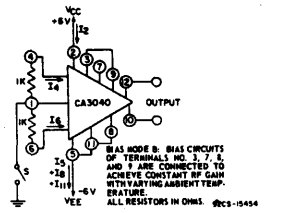


Fig.2(b) - Bias Mode B

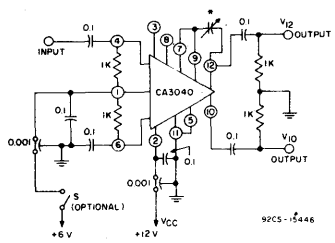
CA3040

ELECTRICAL CHARACTERISTICS AT T_A = 25°C Unless Otherwise Specified

Characteristics	Symbols	Test Circuits Fig.	Special Test Conditions	Limits Min.	Limits Typ.	Limits Max.	Units
STATIC CHARACTERISTICS	V_{CC} = +6V, V_{EE} = -6V						
Output Voltage	V_{10} or V_{12}	2(a) 2(b)	Bias Mode Switch A or B: Closed	1.4	2.7	3.7	V
Base Bias Voltage	V_9	2(a)	Bias Mode A Switch Closed	-	-1.7	-	V
		2(b)	Bias Mode B Switch Closed	-	-1.7	-	V
Input Bias Reference Voltage	V_1	2(a) 2(b)	Bias Mode Switch A or B: Open	-1	-	+1	V
Input Bias Current	I_4, I_6	2(a) 2(b)	Bias Mode Switch A or B: Closed	-	15	45	μA
Input Unbalance Current	$\lvert I_6 - I_4 \rvert$	2(a) 2(b)	Bias Mode Switch A or B: Closed	-	-	6	μA
Power Supply Current Drain	I_2 or $I_5 + I_{11}$	2(a)	Mode A Switch open or closed	4.7	8.5	15.5	mA
	I_2 or $I_5 + I_8 + I_{11}$	2(b)	Mode B Switch open or closed				
DYNAMIC CHARACTERISTICS	V_{CC} = +12V, V_{EE} = 0, Split Voltage Supply (Optional) = +6V						
Differential Voltage Gain							
Single-Ended Input Differential Output	$A_{DIFF(DE)}$	3(a)	f = 1 MHz, R_S = 50 Ω	34	37	-	dB
Single-Ended Input and Output	$A_{DIFF(SE)}$	3(a)	f = 1 MHz, R_S = 50 Ω	28	31	-	dB
-3dB Bandwidth	BW	3(a)	R_S = 50 Ω	40	55	-	MHz
Differential Voltage Gain Balance	$A_{DIFF(SE)10} - A_{DIFF(SE)12}$	3(a)	f = 1 MHz	-1	0	+1	dB
Output Voltage Swing	V_8 or V_{10} RMS	3(a)	f = 1 MHz, R_S = 50 Ω	-	0.5	-	V_{RMS}
Noise Figure	NF	3(a)	(Note 1) f = 30 MHz, R_S = 400 Ω	-	7.5	9	dB
Parallel Input Resistance	R_I	3(a)	f = 1 MHz	-	150	-	kΩ
Parallel Input Capacitance	C_I	3(a)		-	2.2	-	pF
Output Resistance	R_O	3(a)		-	125	-	Ω
TEMPERATURE DEPENDENT CHARACTERISTICS Temperature coefficients for ambient temperature: -55°C ≤ T_A ≤ +125°C							
Output Voltage	$\frac{\Delta V_{10} \text{ or } \Delta V_{12}}{^oC}$	3(a)	Bias Mode A	-	0	-	mV/°C
		3(b)	Bias Mode B	-	6.4	-	mV/°C
Power Supply Current Drain	$\Delta I_2/^oC$	3(a)	Bias Mode A	-	5	-	μA/°C
Differential Voltage Gain	$A_{DIFF}/^oC$	3(a)	Bias Mode A	-	0.0166	-	dB/°C
		3(b)	Bias Mode B	-	0	-	

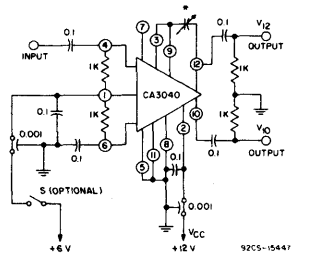
Note 1: Replace 1-kΩ resistors between Term. 1 and 4 and Term. 1 and 6 with suitable chokes so that reactance at 30 MHz exceeds 5kΩ

DYNAMIC CHARACTERISTICS TEST CIRCUITS



*VARIABLE CAPACITANCE (0.5-1.0 μF) ADJUSTMENT FOR EQUAL 3dB BANDWIDTH AT AMPLIFIER OUTPUTS, TERMINALS 10 AND 12.
ALL RESISTORS IN OHMS.
ALL CAPACITORS IN MICROFARADS (UNLESS OTHERWISE INDICATED).
BIAS MODE A IS AS DEFINED IN FIG 2(a)

Fig.3(a) - Bias Mode A



*SEE FIG 3(a)
BIAS MODE B IS AS DEFINED IN FIG 2(b)
ALL RESISTORS IN OHMS
ALL CAPACITORS IN MICROFARADS (UNLESS OTHERWISE INDICATED).

Fig.3(b) - Bias Mode B

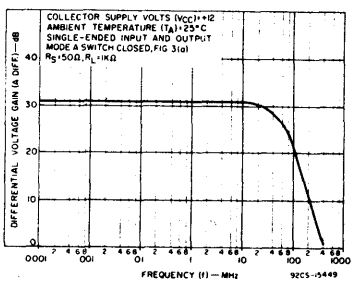


Fig.4 - Differential Voltage Gain vs Frequency

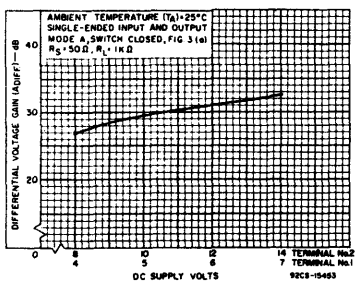


Fig.5 - Differential Voltage Gain vs DC Supply Voltages

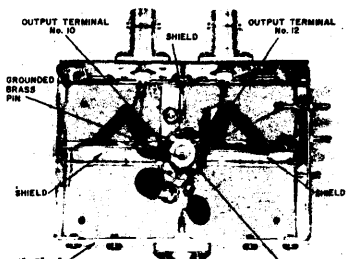


Fig.6 - Test Circuit Layout

OPERATING CONSIDERATIONS

General

The CA3040 is designed to provide flexibility in the selection of power supply configurations and to provide the circuit designer the choice between two modes of temperature-compensated performance. Mode A, which provides constant DC output voltage, is recommended for most applications. The control of the operating point provided by this mode maintains the dynamic range of the device while gain variation over most of the range is less than ±1 dB. Mode B provides constant gain for applications where this consideration is critical, but will exhibit a reduction of dynamic range at the temperature extremes.

Power Supply Considerations

Figures 2 and 3 illustrate the use of the CA3040 with balanced dual supplies and single power supplies, respectively. Both figures demonstrate that the inputs may be directly referenced to the center point of the supply (ground in Fig.2) by closing the included switch. This is the natural connection in Fig.2. This connection is optional, however, and need not be made. Use of this connection in Fig.3 implies the presence of another DC supply or a "stiff" bleeder. If such a source is present its use is suggested in order to maintain maximum common mode range. Dynamic performance and dynamic range of the output circuit are unaffected by the choice of biasing scheme used so that in most cases direct connection of Terminal No.1 to the center point of the supply is not required. Where direct connection is not used, Terminals No.4 and No.6 must be biased from Terminal No.1 for proper operation.

High-Frequency Considerations

Stable high-frequency operation requires that proper high-frequency construction techniques be followed. The photograph of Fig.6 illustrates the precautions taken in the construction of the test circuit of Fig.3.

Extreme caution is required because of the extended gain bandwidth capability of the device. Oscillations have been observed in the 400-to-800 MHz range when precautions were not taken. In addition to normal considerations of shielding, parts layout, and isolation, the following specific suggestions are made:

1. Use sockets only when necessary. Sockets, when used, must provide shielding within the pin circle. The socket shown in the chassis of Fig.6 is a Barnes MG-1201, or equivalent, modified by drilling a 1/8" hole in the center and inserting a grounded brass pin.
2. Do not bypass Terminal No.9 in normal operation. Fig.3 shows the use of neutralization between Terminal No.9 and one output to balance the amplifier at high frequencies. Experience shows that stable operation, while possible, is difficult to achieve if Terminal No.9 is bypassed to ground.
3. In DC testing, 1 kΩ, 1/4 W carbon resistors should be soldered directly to the socket Terminals No.4 and No.6 to suppress parasitic oscillations. All current carrying connections are made at the other end of the resistors. Direct sensing of Terminal No.4 or No.6 voltage should not be attempted.

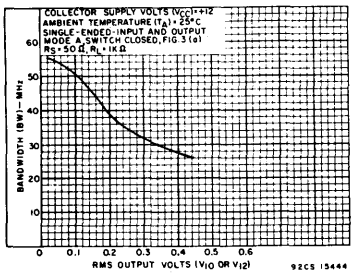


Fig.7 - 3dB Bandwidth vs Single-Ended Output Voltage

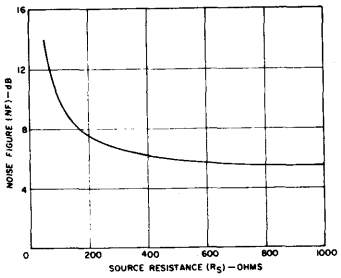


Fig.8 - Noise Figure (NF) vs Source Impedance

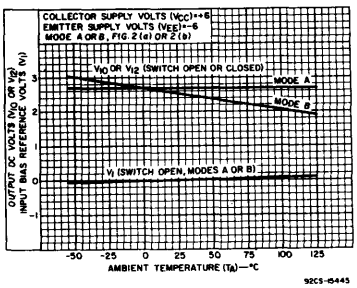


Fig.9 - Output Volts or Input Bias Reference Volts vs Ambient Temperature

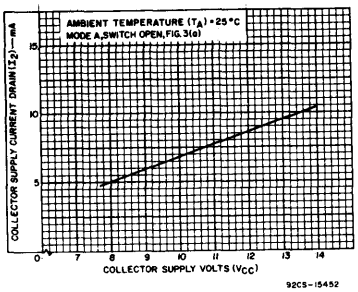


Fig.10 - Collector Supply Current Drain (I_2) vs Collector Supply Voltage (V_{CC})

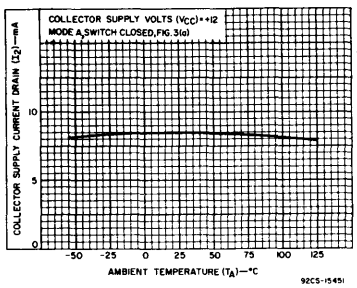


Fig.11 - Collector Supply Current Drain (I_2) vs Ambient Temperature

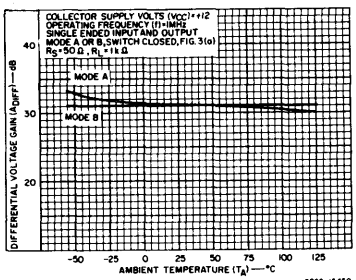


Fig.12 - Single-Ended Differential Voltage Gain vs Ambient Temperature

CA3045, CA3046 Types

General-Purpose Transistor Arrays For Low-Power Applications at Frequencies from DC through the VHF Range

THREE ISOLATED TRANSISTORS AND ONE DIFFERENTIALLY-CONNECTED TRANSISTOR PAIR

The CA3045 and CA3046 each consist of five general-purpose silicon n-p-n transistors on a common monolithic substrate. Two of the transistors are internally connected to form a differentially-connected pair.

The transistors of the CA3045 and CA3046 are well suited to a wide variety of applications in low power systems in the DC through VHF range. They may be used as discrete transistors in conventional circuits. However, in addition, they provide the very significant inherent integrated circuit advantages of close electrical and thermal matching.

The CA3045 is supplied in a 14-lead dual-in-line hermetic (welded-seal) ceramic package and the CA3045F in a 14-lead dual-in-line hermetic (frit-seal) ceramic package.

The CA3046 is electrically identical to the CA3045 but is supplied in a dual-in-line plastic package for applications requiring only a limited temperature range.

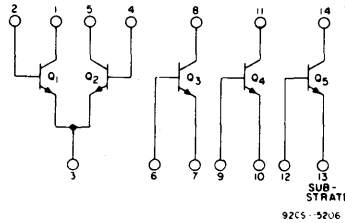


Fig. 1 - Schematic diagram.

ABSOLUTE MAXIMUM RATINGS AT T_A = 25°C

	CA3045		CA3045F, CA3046		
	Each Transistor	Total Package	Each Transistor	Total Package	
Power Dissipation:					
T_A up to 55°C	–	–	300	750	mW
T_A > 55°C	–	–	Derate at 6.67		mW/°C
T_A up to 75°C	300	750	–	–	mW
T_A > 75°C	Derate at 8		–	–	mW/°C
Collector-to-Emitter Voltage, V_{CEO}	15	–	15	–	V
Collector-to-Base Voltage, V_{CBO}	20	–	20	–	V
Collector-to-Substrate Voltage, V_{CIO}*	20	–	20	–	V
Emitter-to-Base Voltage, V_{EBO}	5	–	5	–	V
Temperature Range:					
Operating	–55 to +125		–55 to +125		°C
Storage	–65 to +150		–65 to +150		°C
Lead Temperature (During Soldering): At distance 1/16 ±1/32" (1.59 ±0.79 mm) from case for 10 seconds max:	+265		+265		°C

* The collector of each transistor of the CA3045 and CA3046 is isolated from the substrate by an integral diode. *The substrate (terminal 13) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.*

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

Characteristics apply for each transistor in the CA3045 and CA3046 as specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS Type CA3045 Type CA3046 MIN.	TYP.	MAX.	UNITS
STATIC CHARACTERISTICS						
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	20	60	-	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	15	24	-	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	20	60	-	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	5	7	-	V
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	-	0.002	40	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10 V, I_B = 0	-	See curve	0.5	μA
Static Forward Current-Transfer Ratio (Static Beta)	h_{FE}	V_{CE} = 3 V, I_C = 10 mA	-	100	-	-
		V_{CE} = 3 V, I_C = 1 mA	40	100	-	-
		V_{CE} = 3 V, I_C = 10 μA	-	54	-	-
Input Offset Current for Matched Pair Q_1 and Q_2: $\lvert I_{IO_1} - I_{IO_2} \rvert$		V_{CE} = 3 V, I_C = 1 mA	-	0.3	2	μA
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3 V, I_E = 1 mA	-	0.715	-	V
		V_{CE} = 3 V, I_E = 10 mA	-	0.800	-	
Magnitude of Input Offset Voltage for Differential Pair $\lvert V_{BE_1} - V_{BE_2} \rvert$		V_{CE} = 3 V, I_C = 1 mA	-	0.45	5	mV
Magnitude of Input Offset Voltage for Isolated Transistors $\lvert V_{BE_3} - V_{BE_4} \rvert$, $\lvert V_{BE_4} - V_{BE_5} \rvert$, $\lvert V_{BE_5} - V_{BE_3} \rvert$		V_{CE} = 3 V, I_C = 1 mA	-	0.45	5	mV
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 3 V, I_C = 1 mA	-	-1.9	-	mV/°C
Collector-to-Emitter Saturation Voltage	V_{CES}	I_B = 1 mA, I_C = 10 mA	-	0.23	-	V
Temperature Coefficient: Magnitude of Input-Offset Voltage	$\frac{\lvert \Delta V_{IO} \rvert}{\Delta T}$	V_{CE} 3 V, I_C 1 mA	-	1.1	-	μV/°C

FEATURES

- Two matched pairs of transistors
  V_{BE} matched ±5 mV
  Input offset current 2 μA max. at I_C = 1 mA
- 5 general purpose monolithic transistors
- Operation from DC to 120 MHz
- Wide operating current range
- Low noise figure - - 3.2 dB typ. at 1 kHz
- Full military temperature range for CA3045
  -55 to +125°C
- The CA3045 is available in a sealed-junction Beam-Lead version (CA3045L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".

APPLICATIONS

- General use in all types of signal processing systems operating anywhere in the frequency range from DC to VHF
- Custom designed differential amplifiers
- Temperature compensated amplifiers
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Integrated-Circuit Transistor Array" for suggested applications.

STATIC CHARACTERISTICS

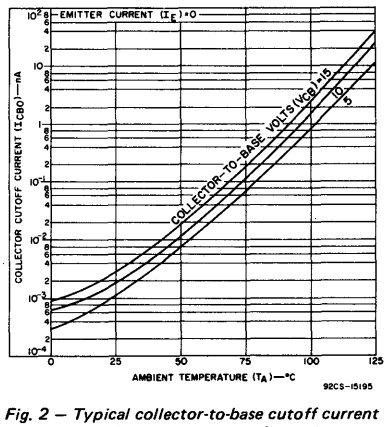


Fig. 2 – Typical collector-to-base cutoff current vs ambient temperature for each transistor.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

Characteristics apply for each transistor in the CA3045 and CA3046 as specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS Type CA3045 Type CA3046 MIN.	TYP.	MAX.	UNITS
DYNAMIC CHARACTERISTICS						
Low-Frequency Noise Figure	NF	**f = 1 kHz, V_{CE} = 3V, I_C = 100 μA Source Resistance = 1 kΩ**	-	3.25	-	dB
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:						
Forward Current-Transfer Ratio	h_{fe}	f = 1 kHz, V_{CE} = 3 V, I_C = 1 mA	-	110	-	-
Short-Circuit Input Impedance	h_{ie}		-	3.5	-	kΩ
Open-Circuit Output Impedance	h_{oe}		-	15.6	-	μmho
Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		-	1.8×10^{-4}	-	-
Admittance Characteristics:						
Forward Transfer Admittance	Y_{fe}	f = 1 MHz, V_{CE} = 3 V, I_C = 1 mA	-	31-j1.5	-	-
Input Admittance	Y_{ie}		-	0.3+j0.04	-	-
Output Admittance	Y_{oe}		-	0.001+j0.03	-	-
Reverse Transfer Admittance	Y_{re}		-	See curve	-	-
Gain-Bandwidth Product	f_T	V_{CE} = 3 V, I_C = 3 mA	300	550	-	-
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 3 V, I_E = 0	-	0.6	-	pF
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 3 V, I_C = 0	-	0.58	-	pF
Collector-to-Substrate Capacitance	C_{CI}	V_{CS} = 3 V, I_C = 0	-	2.8	-	pF

STATIC CHARACTERISTICS

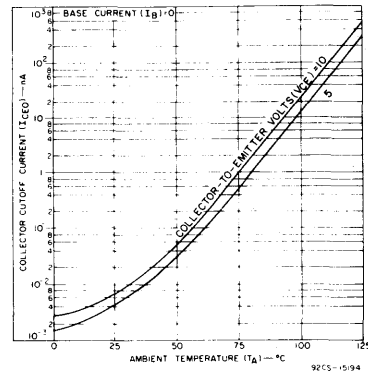


Fig.3 - Typical collector-to-emitter cutoff current vs ambient temperature for each transistor.

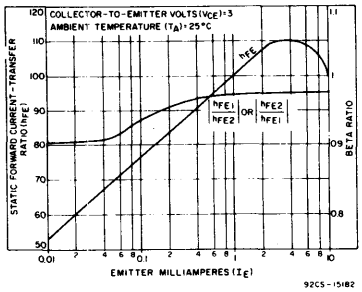


Fig.4 - Typical static forward current-transfer ratio and beta ratio for transistors Q_1 and Q_2 vs emitter current.

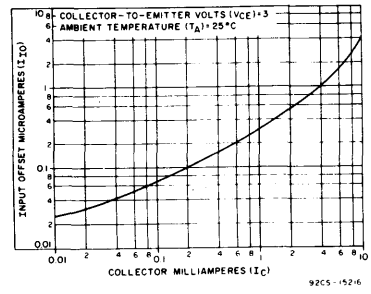


Fig.5 - Typical input offset current for matched transistor pair Q_1Q_2 vs collector current.

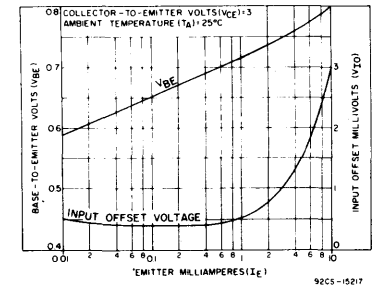


Fig.6 - Typical static base-to-emitter voltage characteristic and input offset voltage for differential pair and paired isolated transistors vs emitter current.

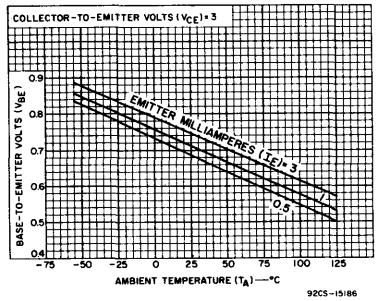


Fig. 7 — Typical base-to-emitter voltage characteristics vs ambient temperature for each transistor.

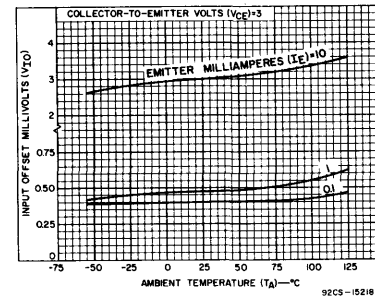


Fig. 8 — Typical input offset voltage characteristics for differential pair and paired isolated transistors vs ambient temperature.

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

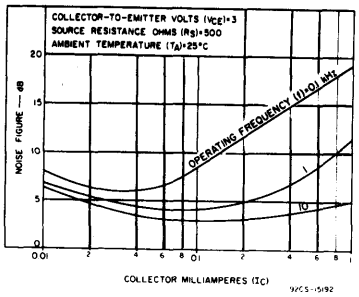


Fig.9(a) - Typical noise figure vs collector current.

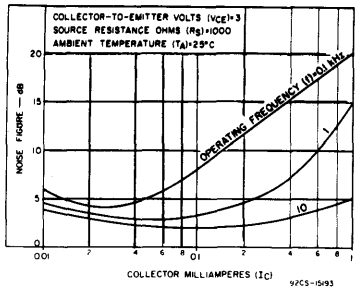


Fig.9(b) - Typical noise figure vs collector current.

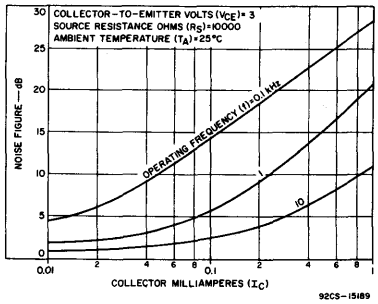


Fig.9(c) - Typical noise figure vs collector current.

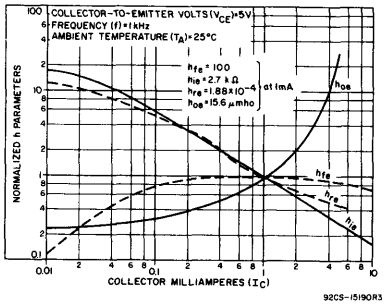


Fig.10 - Typical normalized forward current-transfer ratio, short-circuit input impedance, open-circuit output impedance, and open-circuit reverse voltage-transfer ratio vs collector current.

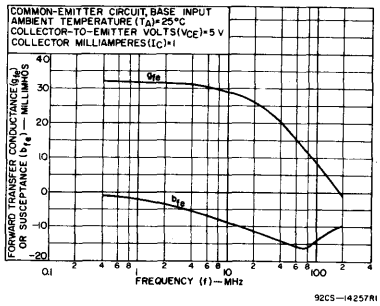


Fig.11 - Typical forward transfer admittance vs frequency.

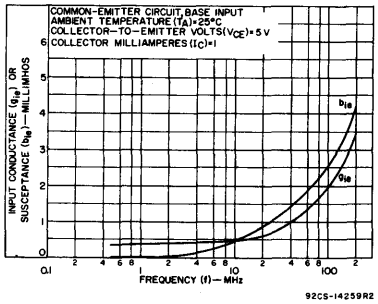


Fig.12 - Typical input admittance vs frequency.

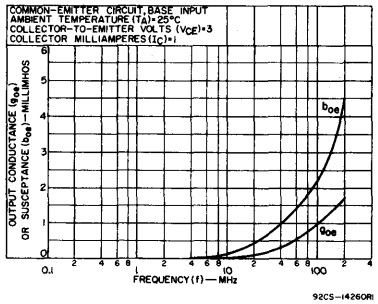


Fig.13 - Typical output admittance vs frequency.

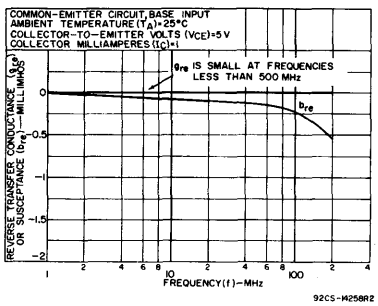


Fig.14 - Typical reverse transfer admittance vs frequency.

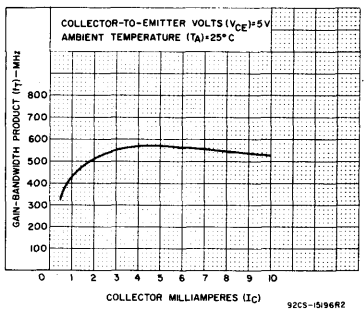


Fig.15 - Typical gain-bandwidth product vs collector current.

Amplifier Array
FOUR INDEPENDENT AC AMPLIFIERS

For Low-Noise and General AC Applications In Industrial Service

The RCA CA3048 is a silicon monolithic integrated circuit consisting of four independent identical AC amplifiers which can operate from a single-ended power supply.

The amplifiers include internal DC bias and feedback to provide temperature-stabilized operation. They may be used in a wide variety of AC applications in which operational amplifiers have previously been used.

Each high gain amplifier has a high impedance non-inverting input, and a lower impedance inverting input for the application of feedback. Two power-supply terminals and two ground terminals are provided to reduce internal and external coupling between amplifiers.

The CA3048 is supplied in a 16-lead dual-in-line plastic package.

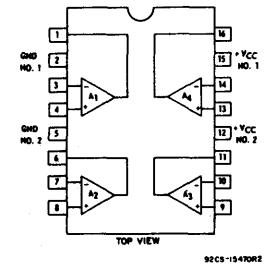


Fig.1 - Block diagram for CA3048.

ABSOLUTE-MAXIMUM RATINGS at T_A = 25°C:

DISSIPATION:
- At T_A = 55°C ..... 750 mW
- Above T_A = 55°C ..... Derate linearly at 7.7 mW/°C

TEMPERATURE RANGE:
- Operating ..... -40°C to +85°C
- Storage ..... -65°C to +150°C

LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ±0.79mm) from case for 10 seconds max. ..... +265°C

POWER SUPPLY VOLTAGE ..... +16 V
AC INPUT VOLTAGE ..... 0.5 V rms

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 4 is +2 to -3.6 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		+16 0	*	*	*	*	*	*	*	*	*	*	*	*	0 -16	*
2			*	+2 -3.6	0	*	*	+2 -3.6	-3.6	*	*	+16 0	+2 -3.6	*	+16 0	0 -16
3				+5 -5	*	*	*	*	*	*	*	*	*	*	*	*
4					+3.6 -2	*	*	*	*	*	*	*	*	*	*	*
5						0 -16	*	+2 -3.6	+2 -3.6	*	0 -16	+16 0	+2 -3.6	*	+16 0	*
6							*	*	*	*	*	*	0 -16	*	*	*
7								+5 -5	*	*	*	*	*	*	*	*
8									*	*	*	*	*	*	*	*
9										+5 -5	*	*	*	*	*	*
10											*	*	*	*	*	*
11												*	*	*	*	*
12													0 -16	*	*	*
13														+5 -5	*	*
14															*	*
15																+16 0
16																

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

FEATURES

- Four AC amplifiers on a common substrate
- Independently accessible inputs and outputs
- Operates from single-ended supply

EACH AMPLIFIER
- Noise figure at 1 kHz ..... 2 dB typ.
- High voltage gain ..... 53 dB min.
- High input resistance ..... 90 kΩ typ.
- Undistorted output voltage ..... 2 V rms min.
- Output impedance ..... 1 kΩ typ.
- Open-loop bandwidth ..... 300 kHz typ.

APPLICATIONS

- Multi-channel or cascade operation
- Low-level preamplifiers
- Equalizers
- Linear signal mixers
- Tone generators
- Multivibrators
- AC integrators

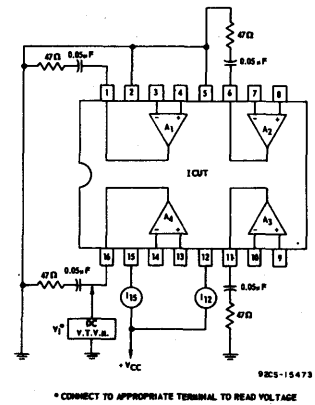


Fig.2 - Test circuit for measurement of collector supply voltage and currents.

CA3048

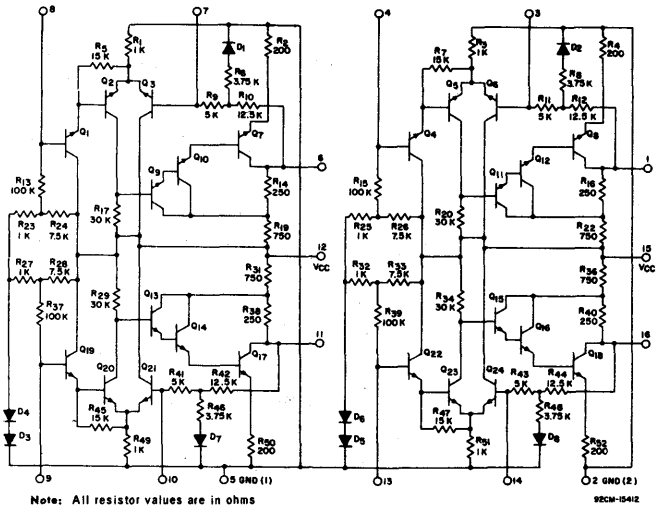


Fig.3 - Schematic diagram for CA3048.

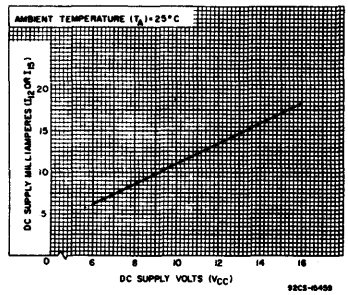


Fig.4 - Typical DC supply current vs supply voltage.

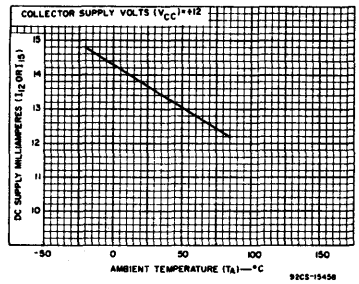
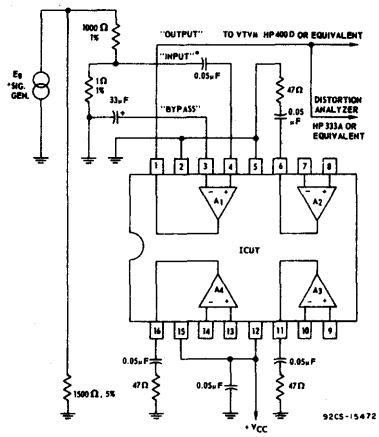


Fig.5 - Typical DC supply current vs ambient temperature.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIRCUIT FIG.	LIMITS CA3048 MIN.	LIMITS CA3048 TYP.	LIMITS CA3048 MAX.	UNITS	TYPICAL CHARACTERISTICS CURVES FIG.
STATIC								
Current drain per amplifier pair	I_{12} or I_{15}	V_{CC} = +12V	2	9.5	13.5	17.5	mA	4,5
DC Voltage at Output Terminals	V1, V6, V11, V16	V_{CC} = +12V	2	6.1	6.9	8.1	V	-
DC Voltage at Feedback Terminals	V3, V7, V10, V14	V_{CC} = +12V	2	1.7	2.0	2.3	V	-
DC Voltage at Input Terminals	V4, V8, V9, V13	V_{CC} = +12V	2	2.2	2.5	2.8	V	-
DYNAMIC (Characteristics given are for each amplifier with no AC feedback)								
Open-Loop Gain	A_{OL}	V_{CC} = +12V, E_{IN} = 2mV, f = 10kHz	6	53	58	-	dB	7,8
Output Voltage Swing	V_O(rms)	V_{CC} = +12V, f = 1kHz, THD = 5%	6	2.0	2.4	-	V	-
Open-Loop -3dB Bandwidth	BW	V_{CC} = +12V, E_{IN} = 2mV	6	250	300	-	kHz	9
Total Harmonic Distortion	THD	V_{CC} = +12V, f = 1kHz, E_{OUT} = 2V rms	6	-	0.65	-	%	10
Input Resistance	R_{IN}	OPEN LOOP Terminals 3, 7, 10, and 14 are bypassed to ground f = 1kHz	-	-	90	-	kΩ	-
Input Capacitance	C_{IN}	f = 1MHz	-	-	9	-	pF	-
Output Resistance	R_{OUT}	Terminals 3, 7, 10 and 14 are bypassed to ground	-	-	1	-	kΩ	-
Output Capacitance	C_{OUT}	f = 1MHz	-	-	18	-	pF	-
Feedback Capacitance (Output to non-inverting Input)	C_{FB}	V_{CC} = +12V, f = 1MHz	-	-	<0.1	-	pF	-
Broad-Band Output Noise Voltage	E_N	V_{CC} = +12V, R_S = 10kΩ, A = 40dB, Equivalent Noise BW = 50kHz	11	-	0.3	1	mV	-
Output Noise Voltage "Weighted"	$E_{N(WT)}$		12	-	0.5	2.2	mV	-
Noise Figure	NF (R_S = 5kΩ)	f = 10Hz	-	-	10	-	dB	-
		f = 100Hz	-	-	5.8	-	dB	
		f = 1kHz	-	-	2	-	dB	
		f = 10kHz	-	-	1.1	-	dB	
		f = 100kHz	-	-	0.6	-	dB	
Inter-Amplifier Audio Separation "Cross Talk"		V_{CC} = +12V, f = 1kHz, 0dB = 0.78V	13	-	<-45	-	dB	-
Inter-Amplifier Capacitance (Any amplifier output to any *other* amplifier input)	C	V_{CC} = +12V, f = 1MHz	-	-	<0.02	-	pF	-



* Sig Gen should be a low distortion type (0.2% THD or less) HP206A or equivalent.

● Adjustment of E_g to 2 volts will make E_s = 2mV.

Test Circuit shows Amplifier #1 under test, to test Amplifiers 2, 3, or 4; Connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig.6 - Test circuit for measurement of distortion, open-loop gain and bandwidth characteristics.

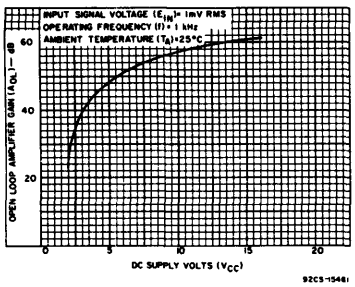


Fig.7 - Typical amplifier gain vs DC supply voltage.

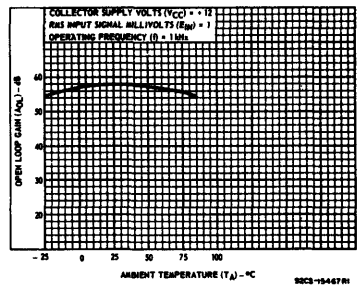


Fig.8 - Typical open-loop gain vs ambient temperature.

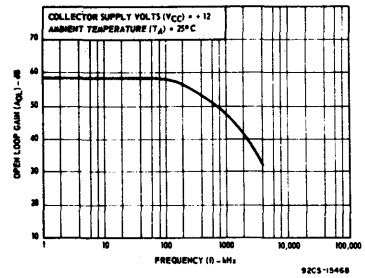


Fig.9 - Typical open-loop gain vs frequency.

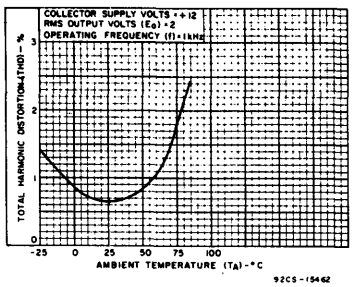
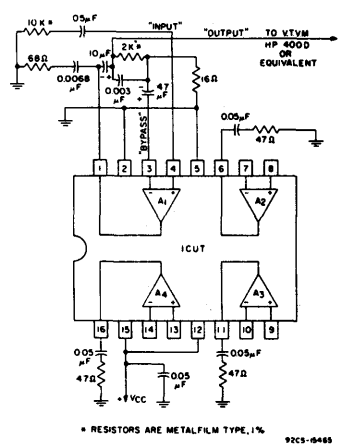


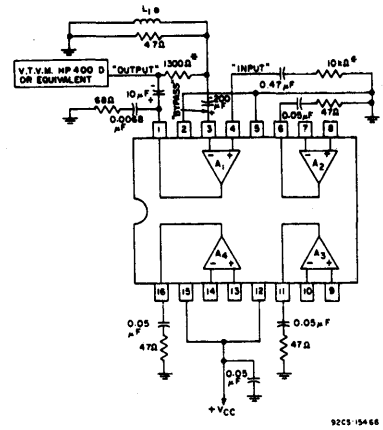
Fig.10 - Typical total harmonic distortion vs ambient temperature.



To test Amplifiers 1, 2, 3, or 4, connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig.11 - Test circuit for measurement of broadband noise characteristic.

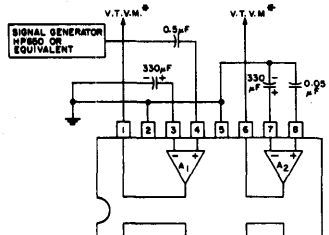


● L_1 - 2.5 millihenry inductor, dc resistance 0.3 ohms or less.

* Resistors metal film type, 1%. To test amplifiers, connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig.12 - Test circuit for measurement of "weighted" output noise voltage characteristic.



* V.T.V.M. - Hewlett-Packard Model 400D or equivalent.

Procedure:

1. Adjust Signal Generator for 0 dB output at reference terminal.
2. Read voltage at other output terminals (Figure shows terminal #1 used as reference).

Fig.13 - Test circuit for measurement of inter-amplifier audio separation "cross talk" characteristic.

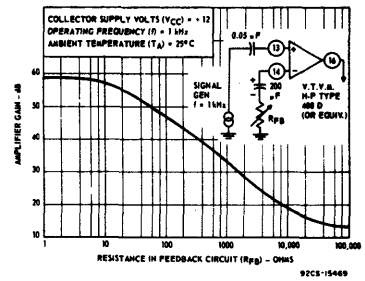


Fig.14 - Typical amplifier gain vs feedback resistance.

OPERATING CONSIDERATIONS

Economical Gain Control

The CA3048 is designed to permit flexibility in the methods by which amplifier gain can be controlled. Fig.14 shows a curve of the gain of an amplifier when the internal resistive feedback of the device is used in conjunction with an external resistor. Although measured gain of various amplifiers will not be uniform, because of tolerances of internal resistances, this method is very economical and easy to apply.

Stability

The CA3048, as in other devices having high gain-bandwidth product, requires some attention to circuit layout, design, and construction to achieve stability.

Should the CA3048 be left unterminated, socket capacitance alone will provide sufficient feedback to cause high frequency oscillations; therefore, all test circuits in this data bulletin include loading networks that provide stability under all conditions.

CA3049T, CA3102E

DUAL HIGH-FREQUENCY DIFFERENTIAL AMPLIFIERS

For Low-Power Applications at Frequencies up to 500 MHz

Features:

- **Power Gain 23 dB (typ.) at 200 MHz**
- **Noise Figure 4.6 dB (typ.) at 200 MHz**
- **Two differential amplifiers on a common substrate**
- **Independently accessible inputs and outputs**
- **Full military-temperature-range capability (−55°C to + 125°C) for the CA3102E and for the CA3049T**
- **The CA3049 is available in a sealed-junction Beam-Lead version (CA3049L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".**

RCA-CA3049T and CA3102E consist of two independent differential amplifiers with associated constant-current transistors on a common monolithic substrate. The six transistors which comprise the amplifiers are general-purpose devices which exhibit low I/f noise and a value of f_T in excess of 1 GHz. These features make the CA3049T and CA3102E useful from dc to 500 MHz. Bias and load resistors have been omitted to provide maximum application flexibility.

The monolithic construction of the CA3049T and CA3102E provides close electrical and thermal matching of the amplifiers. This feature makes these devices particularly useful in dual-channel applications where matched performance of the two channels is required.

The CA3102E is like the CA3049T except that it has a separate substrate connection for greater design flexibility. The CA3049T is supplied in the 12-lead TO-5 package; the CA3102E, in the 14-lead plastic dual-in-line package.

Applications

- **VHF amplifiers**
- **VHF mixers**
- **Multifunction combinations – RF/Mixer/Oscillator; Converter/IF**
- **IF amplifiers (differential and/or cascode)**
- **Product detectors**
- **Doubly balanced modulators and demodulators**
- **Balanced quadrature detectors**
- **Cascade limiters**
- **Synchronous detectors**
- **Balanced mixers**
- **Synthesizers**
- **Balanced (push-pull) cascode amplifiers**
- **Sense amplifiers**

MAXIMUM RATINGS, *ABSOLUTE-MAXIMUM VALUES, AT T_A = 25°C*

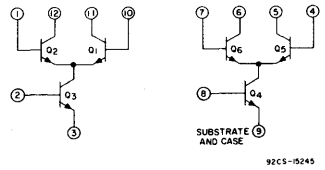
Power Dissipation, P:	CA3049T	CA3102E	
Any one transistor	300	300	mW
Total package	600	750	mW
For $T_A > 55°C$ Derate at:	5	6.67	mW/°C
Temperature Range:			
Operating.....	−55 to + 125	−55 to + 125	°C
Storage	−65 to + 150	−65 to + 150	°C

Lead Temperature (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

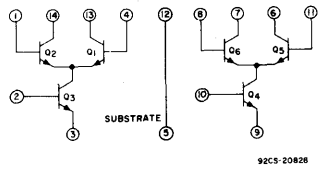
The following ratings apply for each transistor in the devices

Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

*The collector of each transistor of the CA3049T and CA3102E is isolated from the substrate by an integral diode. The substrate (terminal 9) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.



Schematic Diagram for CA3049T



Schematic Diagram for CA3102E

Typical Characteristics for CA3049T and CA3102E

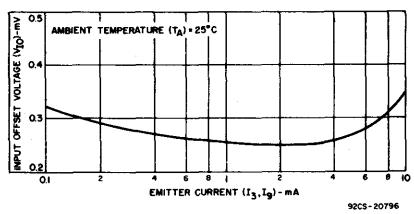


Fig. 4—Input offset voltage vs. emitter current.

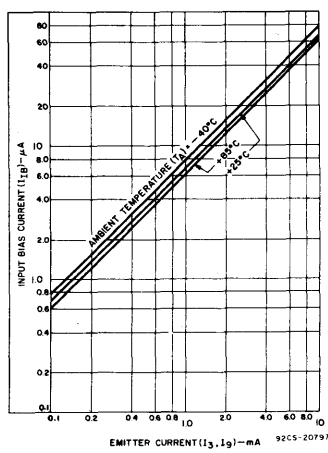


Fig. 5—Input bias current vs. emitter current.

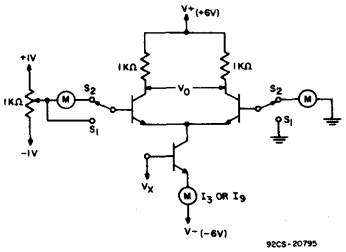


Fig. 1—Static characteristics test circuit for CA3102E.

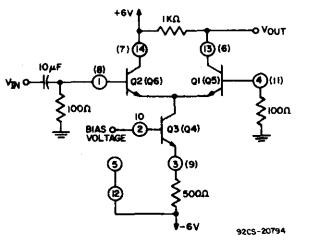
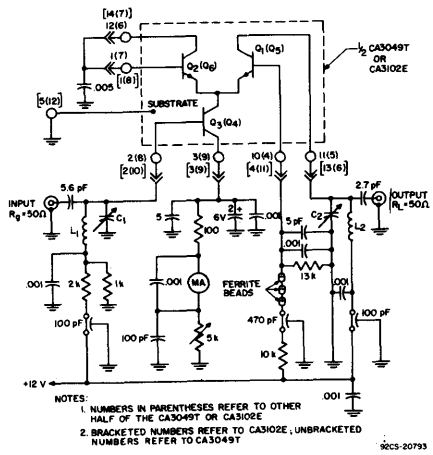


Fig. 2—AGC range and voltage gain test circuit for CA3102E.



L_1, L_2 – Approx. 1/2 Turn #18 Tinned Copper Wire, 5/8" Dia.
C_1, C_2 – 15 pF Variable Capacitors (Hammarlund, MAC-15; or Equivalent)
All Capacitors in μF Unless Otherwise Indicated
All Resistors in Ohms Unless Otherwise Indicated

Fig. 3—200 MHz cascode power gain and noise figure test circuit.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		TEST CIR-CUIT FIG.	CA3049T LIMITS MIN.	TYP.	MAX.	UNITS	TYPICAL CHARACTERISTICS CURVES FIG.
STATIC CHARACTERISTICS									
For Each Differential Amplifier									
Input Offset Voltage	V_{IO}			1	---	0.25	---	mV	-4
Input Offset Current	I_{IO}	$I_3 = I_9$ = 2 mA		1	---	0.3	---	μA	---
Input Bias Current	I_{IB}			1		13.5	33	μA	5
Temperature Coefficient Magnitude of Input Offset Voltage	$\lvert\frac{\Delta V_{IO}}{\Delta T}\rvert$			1	---	1.1	---	μV/°C	4
For Each Transistor									
DC Forward Base-to-Emitter Voltage	V_{BE}	V_{CE} = 6 V, I_C = 1 mA		---	---	774	---	mV	6
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 6 V, I_C = 1 mA		---	---	-0.9	---	mV/°C	6
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0		---	---	0.0013	100	nA	7
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0		---	15	24	---	V	---
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0		---	20	60	---	V	---
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_B = 0, I_E = 0		---	20	60	---	V	---
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0		---	5	7	---	V	---
DYNAMIC CHARACTERISTICS									
1/f Noise Figure (For Single Transistor)	NF	f = 100 KH$_3$, R_S = 500 Ω, I_C = 1 mA		---	---	1.5	---	dB	12
Gain-Bandwidth Product (For Single Transistor)	f_T	V_{CE} = 6 V, I_C = 5 mA		---	---	1.35	---	GH$_z$	11
Collector-Base Capacitance	C_{CB}	I_C = 0	V_{CB} = 5V	*	---	0.28	---	pF	8
				**	---	0.28	---	pF	
Collector-Substrate Capacitance	C_{CI}	I_C = 0	V_{CI} = 5V		---	1.65	---	pF	8
For Each Differential Amplifier									
Common-Mode Rejection Ratio	CMR	$I_3 = I_9$ = 2 mA		---	---	100	---	dB	---
AGC Range, One Stage	AGC	Bias Voltage = -6V		2	---	75	---	dB	---
Voltage Gain, Single-Ended Output	A	Bias Voltage = -4.2V, f = 10 MHz		2	---	22	---	dB	9, 10
Insertion Power Gain	G_p	f = 200 MHz	Cascode	3	---	23	---	dB	---
Noise Figure	NF	V_{CC} = 12V	Cascode	3	---	4.6	---	dB	---
Input Admittance	Y_{11}	For Cascode Configuration	Cascode	---	---	1.5 + j 2.45	---	mmho	14, 16, 18
		$I_3 = I_9$ = 2 mA	Diff.Amp.	---	---	0.878 + j 1.3	---		15, 17, 19
Reverse Transfer Admittance	Y_{12}	For Diff. Amplifier Configuration	Cascode	---	---	0 - j 0.008	---	mmho	---
		$I_3 = I_9$ = 4mA	Diff.Amp.	---	---	0 - j 0.013	---		---
Forward Transfer Admittance	Y_{21}	(each collector $I_C \simeq$ 2mA)	Cascode	---	---	17.9 - j 30.7	---	mmho	26, 28, 30
			Diff. Amp.	---	---	- 10.5 + j 13	---		27, 29, 31
Output Admittance	Y_{22}		Cascode	---	---	- 0.503 - j 15	---	mmho	20, 22, 24
			Diff.Amp.	---	---	0.071 + j 0.62	---		21, 23, 25

*Terminals 1 & 14, or 7 & 8. (CA3102E) 1 & 12 or 6 & 7 (CA3049T)
**Terminals 13 & 4, or 6 & 11. (CA3102E) 10 & 11 or 4 & 5 (CA3049T)

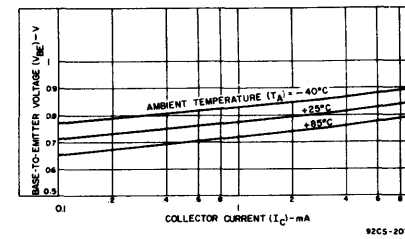


Fig. 6—Base-to-emitter voltage vs. collector current.

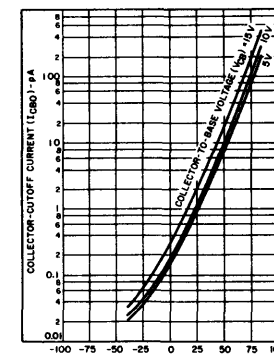


Fig. 7—Collector-cutoff current vs. temperature.

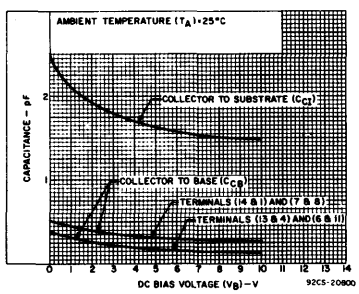


Fig. 8—Capacitance vs. dc bias voltage.

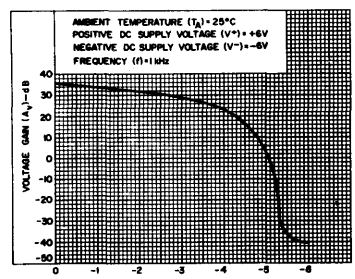


Fig. 9—Voltage gain vs. dc bias voltage.

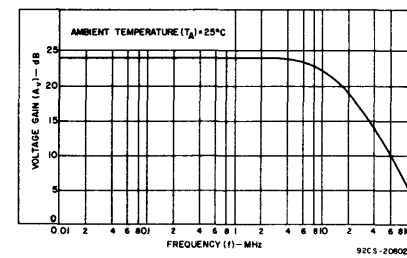


Fig. 10—Voltage gain vs. frequency.

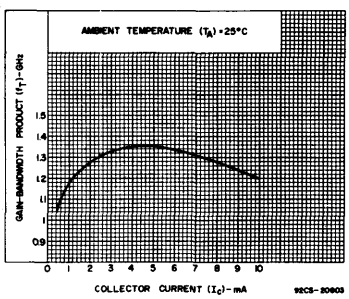


Fig. 11—Gain-bandwidth product vs. collector current.

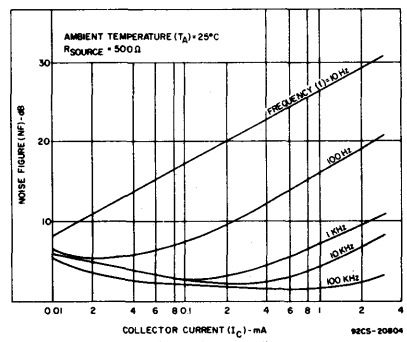


Fig. 12—1/f noise figure vs. collector current.

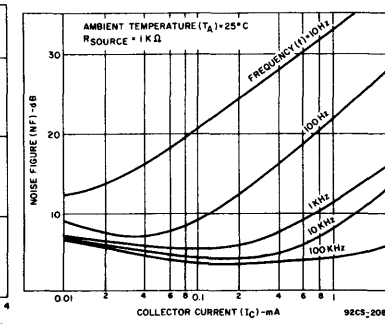


Fig. 13—1/f noise figure vs. collector current.

CA3049T, CA3102E

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		TEST CIR-CUIT FIG.	CA3102E LIMITS MIN.	TYP.	MAX.	UNITS	TYPICAL CHARAC-TERISTICS CURVES FIG.
STATIC CHARACTERISTICS									
For Each Differential Amplifier									
Input Offset Voltage	V_{IO}			1	...	0.25	5	mV	-4
Input Offset Current	I_{IO}	$I_3 = I_8$ = 2 mA		1	...	0.3	3	μA	...
Input Bias Current	I_{IB}			1	...	13.5	33	μA	5
Temperature Coefficient Magnitude of Input-Offset Voltage	$\lvert\Delta V_{IO}\rvert$ / ΔT			1	...	1.1	...	μV/°C	4
For Each Transistor									
DC Forward Base-to-Emitter Voltage	V_{BE}	V_{CE} = 6 V, I_C = 1 mA		...	674	774	874	mV	6
Temperature Coefficient of Base-to-Emitter Voltage	ΔV_{BE} / ΔT	V_{CE} = 6 V, I_C = 1 mA		...	...	-0.9	...	mV/°C	6
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0		...	...	0.0013	100	nA	7
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0		...	15	24	...	V	...
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0		...	20	60	...	V	...
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_B = 0, I_E = 0		...	20	60	...	V	...
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0		...	5	7	...	V	...
DYNAMIC CHARACTERISTICS									
1/f Noise Figure (For Single Transistor)	NF	f = 100 KH$_3$, R_S = 500 Ω, I_C = 1 mA		...	...	1.5	...	dB	12
Gain-Bandwidth Product (For Single Transistor)	f_T	V_{CE} = 6 V, I_C = 5 mA		...	...	1.35	...	GH$_z$	11
Collector-Base Capacitance	C_{CB}	I_C = 0	V_{CB} = 5V	*	...	0.28	...	pF	8
				**	..	0.15	...	pF	
Collector-Substrate Capacitance	C_{CI}	I_C = 0	V_{CI} = 5V		...	1.65	...	pF	8
For Each Differential Amplifier									
Common-Mode Rejection Ratio	CMR	$I_3 = I_8$ = 2 mA		.	...	100	...	dB	...
AGC Range, One Stage	AGC	Bias Voltage = -6V		2	...	75	...	dB	...
Voltage Gain, Single-Ended Output	A	Bias Voltage = -4.2V, f = 10 MHz		2	18	22	...	dB	9, 10
Insertion Power Gain	G_p	f = 200 MHz, V_{CC} = 12V	Cascode	3	...	23	...	dB	...
Noise Figure	NF		Cascode	3	...	4.6	...	dB	...
Input Admittance	Y_{11}	For Cascode Configuration, $I_3 = I_8$ = 2 mA	Cascode	...	...	1.5 + j 2.45	...	mmho	14, 16, 18
			Diff.Amp.	...	...	0.878 + j 1.3	...		15, 17, 19
Reverse Transfer Admittance	Y_{12}	For Diff. Amplifier Configuration, $I_3 = I_8$ = 4mA (each collector $I_C \simeq$ 2mA)	Cascode	...	...	0 - j 0.008	...	mmho	...
			Diff.Amp.	...	...	0 - j 0.013	...		...
Forward Transfer Admittance	Y_{21}		Cascode	...	...	17.9 - j 30.7	...	mmho	26, 28, 30
			Diff. Amp.	...	...	- 10.5 + j 13	...		27, 29, 31
Output Admittance	Y_{22}		Cascode	...	...	0.503 - j 15	...	mmho	20, 22, 24
			Diff.Amp.	...	...	0.071 + j 0.62	...		21, 23, 25

*Terminals 1 & 14, or 7 & 8. (CA3102E) 1 & 12 or 6 & 7 (CA3049T)
**Terminals 13 & 4, or 6 & 11. (CA3102E) 10 & 11 or 4 & 5 (CA3049T)

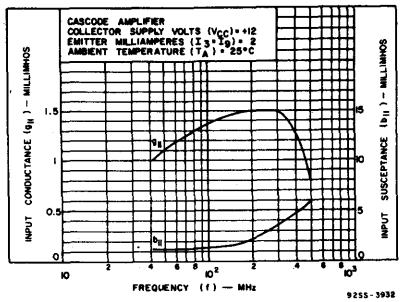


Fig. 14—Input admittance (Y_{11}) vs. frequency.

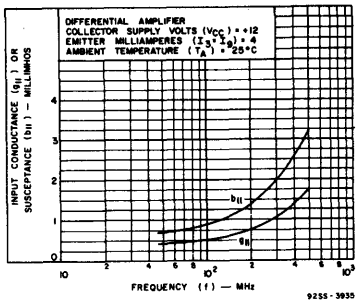


Fig. 15—Input admittance (Y_{11}) vs. frequency.

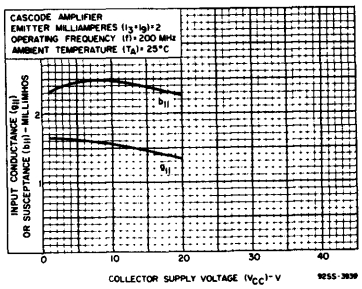


Fig. 16—Input admittance (Y_{11}) vs. collector supply voltage.

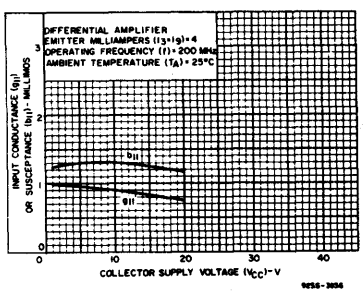


Fig. 17—Input admittance (Y_{11}) vs. collector supply voltage.

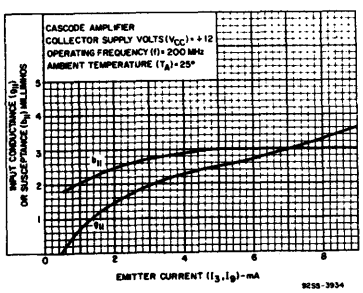


Fig. 18—Input admittance (Y_{11}) vs. emitter current.

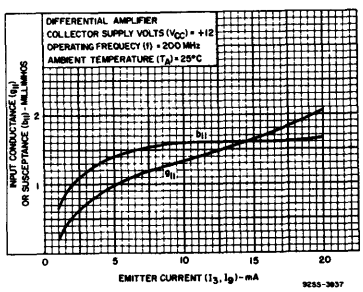


Fig. 19—Input admittance (Y_{11}) vs. emitter current.

Typical Output Admittance Characteristics for CA3049T and CA3102E

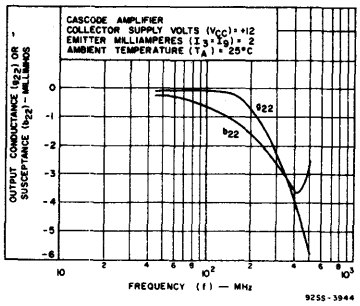


Fig. 20—Output admittance (Y_{22}) vs. frequency.

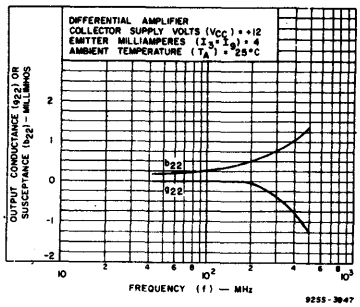


Fig. 21—Output admittance (Y_{22}) vs. frequency.

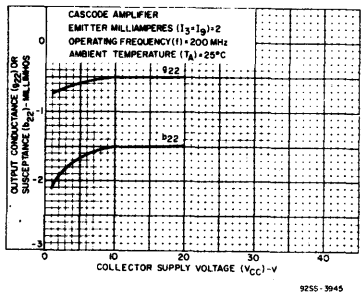


Fig. 22—Output admittance (Y_{22}) vs. collector supply voltage.

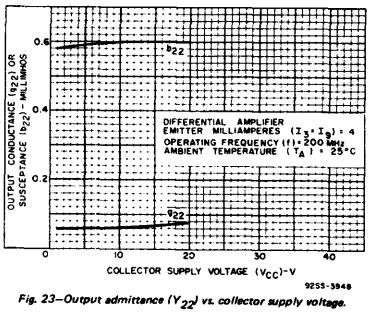


Fig. 23—Output admittance (Y_{22}) vs. collector supply voltage.

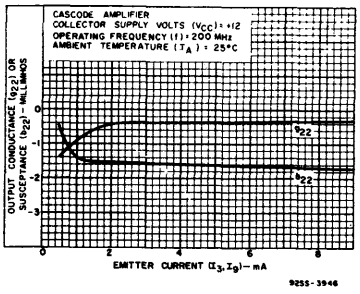


Fig. 24—Output admittance (Y_{22}) vs. emitter current.

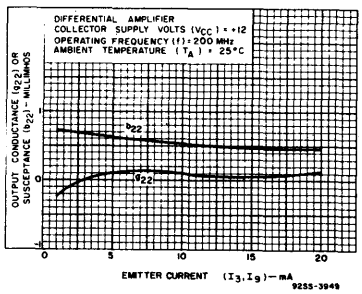


Fig. 25—Output admittance (Y_{22}) vs. emitter current.

Typical Forward Transfer Characteristics for CA3049T and CA3102E

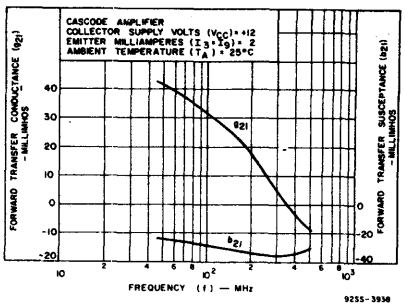


Fig. 26—Forward transfer admittance (Y_{21}) vs. frequency.

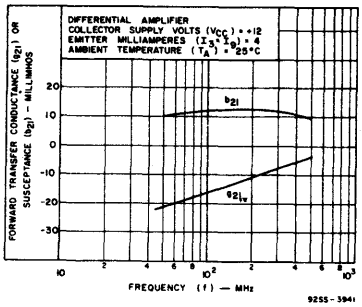


Fig. 27—Forward transfer admittance (Y_{21}) vs. frequency.

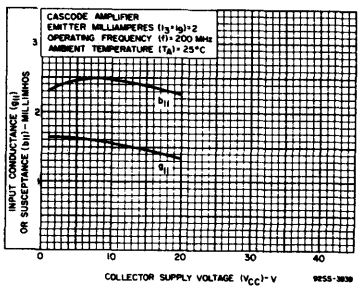


Fig. 28—Forward transfer admittance (Y_{21}) vs. collector supply voltage.

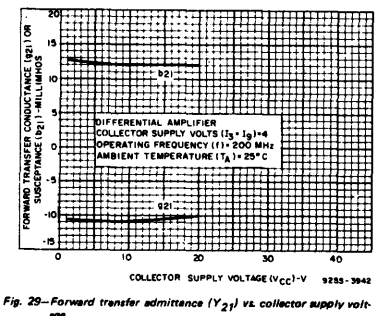


Fig. 29—Forward transfer admittance (Y_{21}) vs. collector supply voltage.

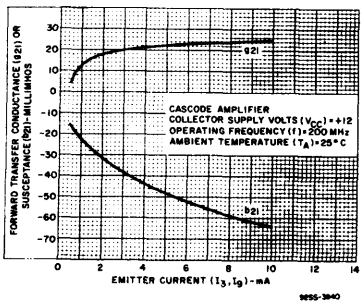


Fig. 30—Forward transfer admittance (Y_{21}) vs. emitter current.

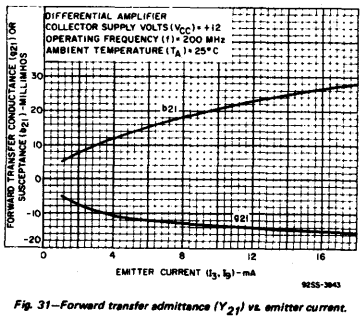


Fig. 31—Forward transfer admittance (Y_{21}) vs. emitter current.

CA3050, CA3051

Dual Differential Amplifiers

TWO DARLINGTON-CONNECTED DIFFERENTIAL AMPLIFIERS WITH DIODE BIAS STRING

For Low-Power Applications at Frequencies from DC to 20 MHz

The CA3050 and CA3051 each consists of two differential amplifiers with associated constant current transistors on a common substrate. Each amplifier is driven by Darlington-connected emitter follower inputs to provide high input impedance, low bias current, and low offset current. A string of diodes is included to provide temperature-compensated bias to the constant current transistors and a low impedance bias point for the inputs to the differential amplifiers when a single power supply is used.

APPLICATIONS

- Matched dual amplifiers
- Dual sense amplifiers
- Dual Schmitt triggers
- Dual multivibrators
- Doubly balanced detectors and modulators
- Balanced quadrature detectors
- Synthesizer mixers
- Product detectors

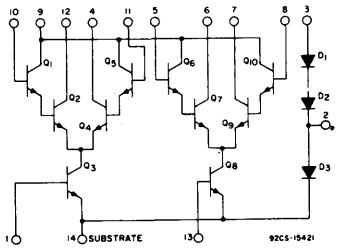


Fig.1 - Schematic diagram.

FEATURES

- Input offset current . . . . . 70 nA max.
- Input bias current . . . . . 500 nA max.
- Input offset voltage . . . . . 5 mV max.
- Input impedance . . . . . 460 kΩ typ.
- Independently accessible inputs and outputs

The CA3050 is supplied in the 14-lead dual-in-line ceramic package and the CA3051 is supplied in the 14-lead dual-in-line plastic package.

MAXIMUM RATINGS, ABSOLUTE-MAXIMUM VALUES, AT T_A = 25°C

	CA3050	CA3051	
Power Dissipation, P:			
Any one transistor	150	150	mW
Total package	900	750	mW
For T_A > 55°C, Derate at . .	8	6.67	mW/°C
Temperature Range:			
Operating	-55 to +125		°C
Storage	-65 to +150		°C
LEAD TEMPERATURE (During Soldering)			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C		

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

* The collector of each transistor of the CA3050 and CA3051 is isolated from the substrate by an integral diode. *The substrate (terminal 14) must be more negative than all collectors to maintain isolation between transistors and to provide for normal transistor action.*

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 3 is +5 to −2 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	–	•	•	•	•	•	•	•	•	•	•	•	•	+1 -5
2			+5 -2	•	•	•	•	•	•	•	•	•	•	+1 -1
3				•	•	•	•	•	•	•	•	•	•	+3 -1
4					•	•	•	•	•	+14 -2.5 Note 3	+14 -2.5 Note 4	•	•	+20 -1
5						+2.5 -14 Note 1	+2.5 -14 Note 1	+10 -10	+1 -20	•	•	•	•	+16 -1
6							•	+14 -2.5 Note 2	•	•	•	•	•	+20 -1
7								+14 -2.5 Note 2	•	•	•	•	•	+20 -1
8									+1 20	•	•	•	•	+16 -1
9										+20 -1	+20 -1	•	•	+20 -1
10											+10 -10	+2.5 -14 Note 3	•	+16 -1
11												+2.5 -14 Note 4	•	+16 -1
12														+20 -1
13														+1 -5
14														Ref. Substrate

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	5	0.1
2	50	50
3	50	1
4	50	1
5	5	0.1
6	50	1
7	50	1
8	5	0.1
9	50	1
10	5	0.1
11	5	0.1
12	50	1
13	5	0.1
14	100	5

NOTE 1: This rating is important only when terminal 5 is more positive than terminal 8.

NOTE 2: This rating is important only when terminal 8 is more positive than terminal 5.

NOTE 3: This rating is important only when terminal 10 is more positive than terminal 11.

NOTE 4: This rating is important only when terminal 11 is more positive than terminal 10.

*Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

TYPICAL STATIC CHARACTERISTICS

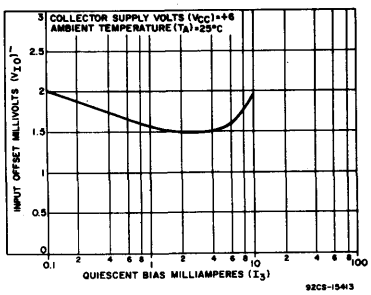


Fig.2(a) - Typical input offset voltage vs quiescent bias current.

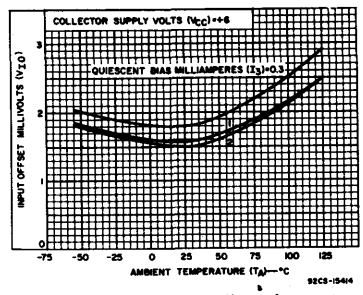


Fig.2(b) - Typical input offset voltage vs ambient temperature.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIR-CUIT FIG.	LIMITS CA3050/CA3051 MIN.	TYP.	MAX.	UNITS	TYPICAL CHARAC-TERISTICS CURVES FIG.
STATIC								
Amplifier Characteristics								
Input Offset Voltage	V_{IO}		–	–	1.5	5	mV	2a,b
Input Offset Current	I_{IO}		–	–	7	70	nA	3a,b
Input Bias Current	I_{IB}		–	–	200	500	nA	4a,b
Quiescent Operating Current Ratio	$\frac{(I_4+I_{12})}{I_3}$ or $\frac{(I_6+I_7)}{I_3}$	V_{CC} = + 6 V, I_3 = 2 mA	–	0.9	1.00	1.13	–	5a,b
DC Forward Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3 V, I_C = 50 μA	–	–	0.645	0.700	V	6
		V_{CE} = 3 V, I_C = 1 mA	–	–	0.725	0.800	V	6
		V_{CE} = 3 V, I_C = 3 mA	–	–	0.760	0.850	V	6
		V_{CE} = 3 V, I_C = 10 mA	–	–	0.805	0.900	V	6
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 3 V, I_C = 1 mA	–	–	-1.9	–	mV/°C	7
Transistor Characteristics								
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	–	–	0.002	100	nA	8
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	–	15	24	–	V	–
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	–	20	60	–	V	–
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	–	20	60	–	V	–
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	–	5	7	–	V	–
DYNAMIC								
Transistor Characteristics								
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 3 V, I_E = 0	–	–	0.78	–	pF	9
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 3 V, I_C = 0	–	–	0.47	–	pF	9
Collector-to-Substrate Capacitance	C_{CI}	V_{CS} = 3 V, I_C = 0	–	–	1.92	–	pF	9
Amplifier Characteristics								
Gain-Bandwidth Product (For Single Transistor)	f_T	V_{CE} = 5 V, I_C = 3 mA	–	–	600	–	MHz	10
Forward Transadmittance (With single-ended input and output)	$\lvert y_{21} \rvert$	V_{CC} = 10 V, I_3 = 2 mA f = 1 MHz	11	7	9	11	mmho	11
Bandwidth at -3 dB Point	BW	V_{CC} = 10 V, I_3 = 2 mA	11	–	4.3	–	MHz	11
Input Impedance	Z_I	V_{CC} = 10 V, I_3 = 2 mA f = 1 KHz	12	–	460	–	kΩ	12
Output Impedance	Z_O	I_3 = 2 mA, f = 1 KHz	13	–	170	–	kΩ	13
Common-Mode Rejection Ratio	CMR	I_3 = 2 mA, f = 1 KHz	–	–	65	–	dB	–
AGC Range	AGC	I_3 = 2 mA, f = 1 KHz Terminal No.3 Grounded	11	–	60	–	dB	–

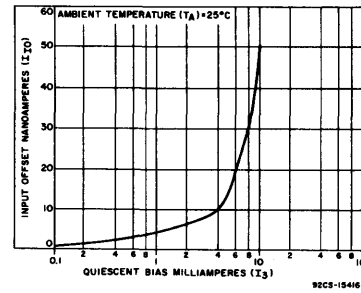


Fig.3(a) - Typical input offset current vs quiescent bias current.

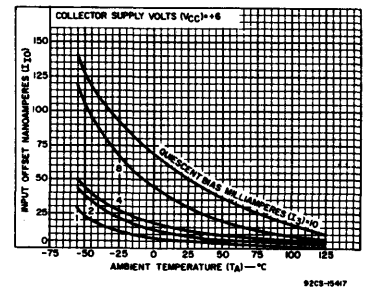


Fig.3(b) - Typical input offset current vs ambient temperature.

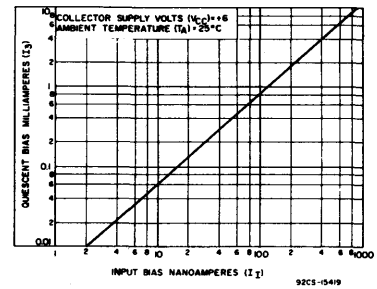


Fig.4(a) - Typical quiescent bias current vs input bias current.

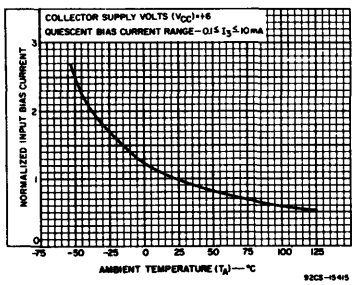


Fig.4(b) - Typical normalized input bias current vs ambient temperature.

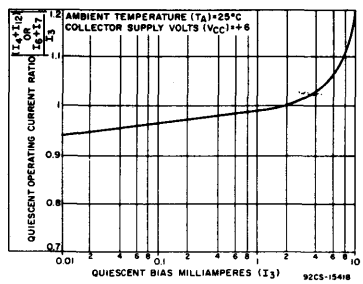


Fig.5(a) - Typical quiescent operating current ratio vs quiescent bias current.

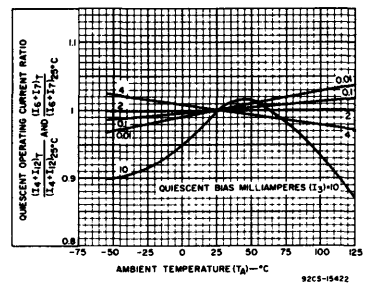


Fig.5(b) - Typical quiescent operating current ratio vs ambient temperature.

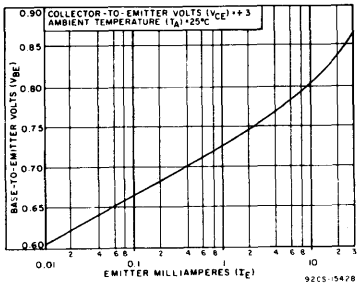


Fig.6 - Typical static base-to-emitter voltage characteristic vs emitter current for all transistors and forward diode voltage drops.

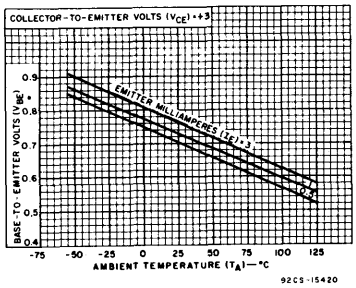


Fig.7 - Typical base-to-emitter voltage characteristic vs ambient temperature for each transistor.

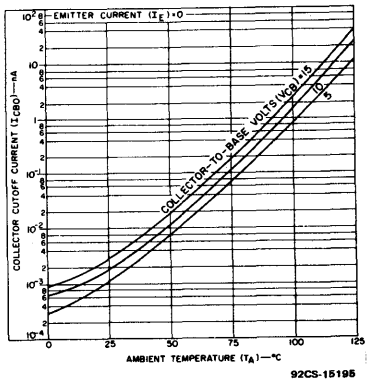


Fig.8 - Typical collector-to-base cutoff current vs ambient temperature for each transistor.

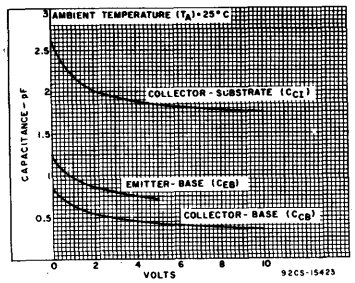


Fig.9 - Typical capacitance for each transistor.

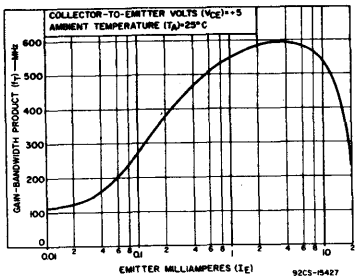


Fig.10 - Typical gain-bandwidth product (f_T) for each transistor vs emitter current.

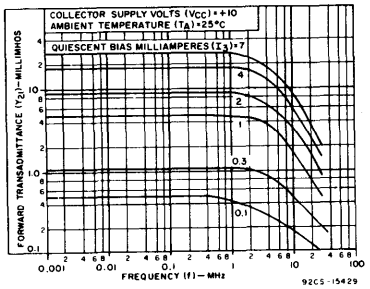


Fig.11(b) - Typical differential amplifier forward transadmittance with single-ended output vs frequency.

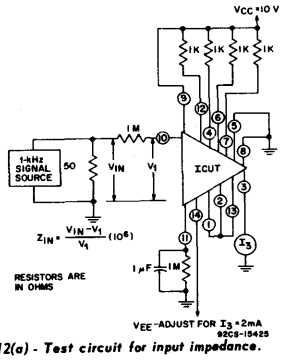
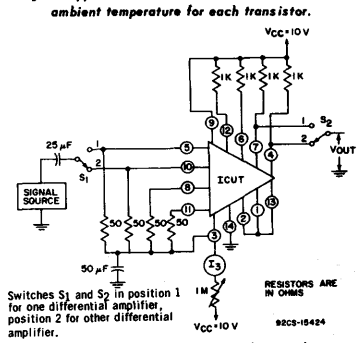


Fig.12(a) - Test circuit for input impedance.



Switches S_1 and S_2 in position 1 for one differential amplifier, position 2 for other differential amplifier.

Fig.11(a) - Test circuit for forward transadmittance, -3 dB bandwidth, and AGC range.

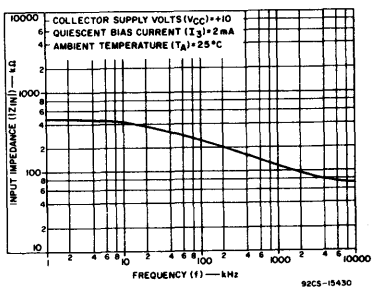
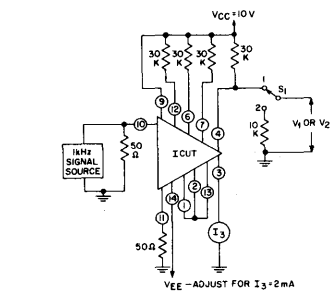


Fig.12(b) - Typical input impedance vs frequency with output short-circuited.



$$Z_{OUT} = \frac{(30K \times 10K)\frac{V_2}{V_1}}{\frac{V_2}{V_1}(30K + 10K) - 10K}$$

Fig.13(a) - Test circuit for output impedance.

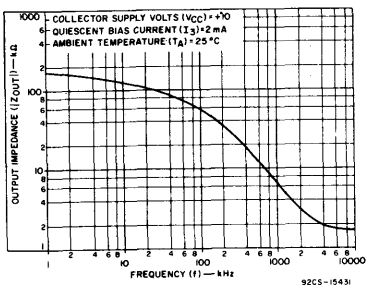


Fig.13(b) - Typical output impedance vs frequency with input short-circuited.

Zero-Voltage Switches

For 50/60 and 400 Hz Thyristor Control Applications

The RCA-CA3058, CA3059, and CA3079 zero-voltage switches are monolithic silicon integrated circuits designed to control a thyristor in a variety of AC power switching applications for AC input voltages of 24 V, 120 V, 208/230 V, and 277 V at 50/60 and 400 Hz. Each of the zero-voltage switches incorporates 4 functional blocks (see Fig. 1) as follows:

1. Limiter-Power Supply—Permits operation directly from an AC line.
2. Differential On/Off Sensing Amplifier—Tests the condition of external sensors or command signals. Hysteresis or proportional-control capability may easily be implemented in this section.
3. Zero-Crossing Detector—Synchronizes the output pulses of the circuit at the time when the AC cycle is at zero voltage point; thereby eliminating radio-frequency interference (RFI) when used with resistive loads.
4. Triac Gating Circuit—Provides high-current pulses to the gate of the power controlling thyristor.

In addition, the CA3058 and CA3059 provide the following important auxiliary functions (see Fig. 1):

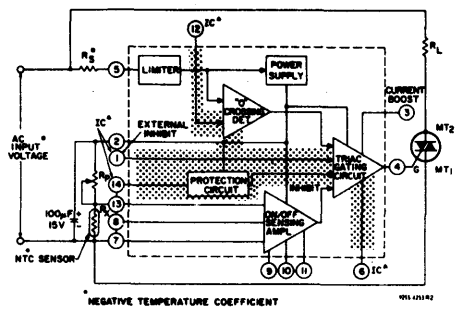
1. A built-in protection circuit that may be actuated to remove drive from the triac if the sensor opens or shorts.
2. Thyristor firing may be inhibited through the action of an internal diode gate connected to Terminal 1.
3. High-power dc comparator operation is provided by overriding the action of the zero-crossing detector. This is accomplished by connecting Terminal 12 to Terminal 7. Gate current to the thyristor is continuous when Terminal 13 is positive with respect to Terminal 9.

For an explanation of these functions see Operating Considerations. For detailed application information, see companion Application Note, ICAN-6182, "Features and Applications of RCA Integrated-Circuit Zero-Voltage Switches (CA3058, CA3059, and CA3079)".

The CA3058 is supplied in a hermetic 14-lead dual-in-line ceramic package. Types CA3059 and CA3079 are supplied in 14-lead dual-in-line plastic packages.

Applications:

- **Relay control**
- **Heater control**
- **Valve control**
- **Lamp control**
- **Synchronous switching of flashing lights**
- **On-off motor switching**
- **Differential comparator with self-contained power supply for industrial applications**
- **Photosensitive control**
- **Power one-shot control**



* NEGATIVE TEMPERATURE COEFFICIENT

AC Input Voltage (50/60 or 400 Hz) V AC	Input Series Resistor (R_S) k Ω	Dissipation Rating for R_S W
24	2	0.5
120	10	2
208/230	20	4
277	25	5

NOTE:
Circuitry within shaded areas, not included in CA3079

■ See chart

▲ IC = Internal Connection - - DO NOT USE (Terminal Restriction applies only to CA3079).

Fig. 1—Functional block diagram of CA3058, CA3059, and CA3079.

Features	CA3058	CA3059	CA3079
■ 24V, 120V, 208/230V, 277V at 50 60, or 400 Hz operation	√	√	√
■ Differential Input	√	√	√
■ Low Balance Input Current (max.) -μA	1	1	2
■ Built-in Protection Circuit for opened or shorted sensor (Term. 14)	√	√	
■ Sensor Range (R_X) - kΩ	2 to 100	2 to 100	2 to 50
■ DC Mode (Term 12)	√	√	
■ External Trigger (Term. 6)	√	√	
■ External Inhibit (Term. 1)	√	√	
■ DC Supply Volts (max.)	14	14	10
■ Operating Temperature Range - °C		−55 to +125	

MAXIMUM RATINGS,
Absolute-Maximum Values at $T_A = 25^oC$

DC SUPPLY VOLTAGE (BETWEEN TERMS. 2 AND 7):
- CA3058, CA3059 ..... 14 V
- CA3079 ..... 10 V

DC SUPPLY VOLTAGE (BETWEEN TERMS. 2 AND 8):
- CA3058, CA3059 ..... 14 V
- CA3079 ..... 10 V

PEAK SUPPLY CURRENT (TERMS. 5 AND 7) ..... ±50 mA

OUTPUT PULSE CURRENT (TERM. 4) ..... 150 mA

POWER DISSIPATION:
- Up to T_A=75°C – CA3058 ..... 700 mW
- Up to T_A=55°C – CA3059,CA3079 ... 700 mW
- Above T_A=75°C – CA3058 ..... Derate Linearly 8 mW/°C
- Above T_A=55°C – CA3059,CA3079 ..... Derate linearly 6.67 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating ..... −55 to +125°C
- Storage ..... −65 to +150°C

LEAD TEMPERATURE (DURING SOLDERING):
At a distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max. ..... +265°C

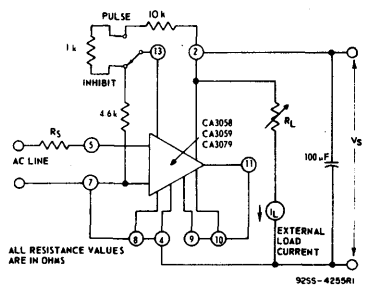


Fig. 2(a)—DC supply voltage test circuit for CA3058, CA3059, and CA3079.

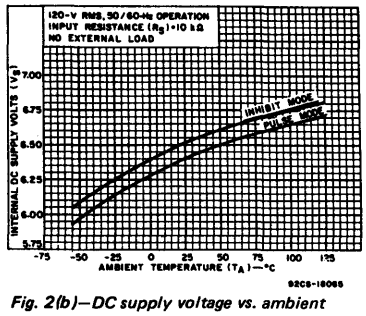


Fig. 2(b)—DC supply voltage vs. ambient temperature for CA3058, CA3059 and CA3079.

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

TERMINAL NO.	1 Note 3	2	3	4	5 Note 1	6 Note 3	7	8	9	10	11	12 Note 3	13	14 Note 2,3	MAXIMUM CURRENT RATINGS I_{IN} mA	I_{OUT} mA
1 Note 3		*	*	*	*	15 0	10 -2	*	*	*	*	*	*	*	10	0.1
2			0 -15	0 -15	2 -14	0 -14	0▲ -14	0▲ -14	0 -14	0 -14	0 -14	* .	0 -14	0 -14	150	10
3				0 -15	*	*	*	*	*	*	*	*	*	*	*	*
4					*	2 -10	*	*	*	*	*	*	*	*	0.1	150
5 Note 1						*	7 -7	*	*	*	*	*	*	*	50	10
6 Note 3							14 0	*	*	*	*	*	*	*	*	*
7								*	14 0	*	20 0	2.5 -2.5	14 0	6 -6	*	*
8									10 0	*	*	*	*	*	0.1	2
9										*	*	*	*	*	*	*
10											*	*	*	*	*	*
11												*	*	*	*	*
12 Note 3													*	*	50	50
13														*	*	*
14 Note 3															2	2

This chart gives the range of voltages which can be applied to the terminals listed horizontally with respect to the terminals listed vertically. For example, the voltage range of horizontal Terminal 6 to vertical Terminal 4 is 2 to −10 volts.

Note 1 – Resistance should be inserted between Terminal 5 and external supply or line voltage for limiting current into Terminal 5 to less than 50 mA.

Note 2 – Resistance should be inserted between Terminal 14 and external supply for limiting current into Terminal 14 to less than 2 mA.

Note 3 – For the CA3079 indicated terminal is internally connected and, therefore, should not be used.

▲For CA3079 (0 to −10 V).

*Voltages are not normally applied between these terminals; however, voltages appearing between these terminals are safe, if the specified voltage limits between all other terminals are not exceeded.

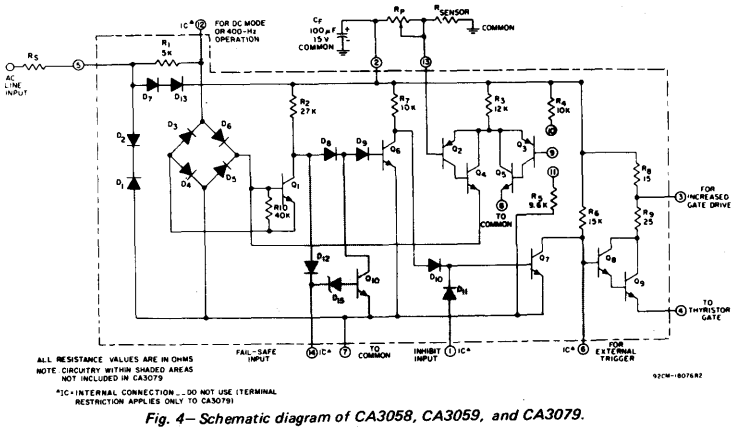


Fig. 4– Schematic diagram of CA3058, CA3059, and CA3079.

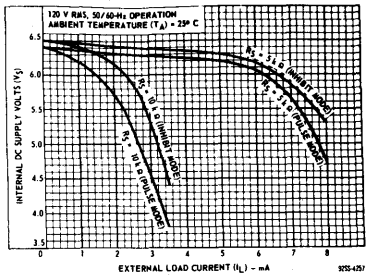


Fig. 2(c)–DC supply voltage vs. external load –current for CA3058, CA3059, and CA3079.

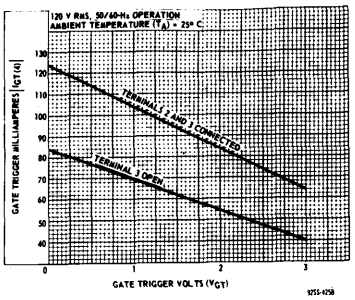


Fig. 3– Gate trigger current vs. gate trigger voltage for CA3058, CA3059, and CA3079.

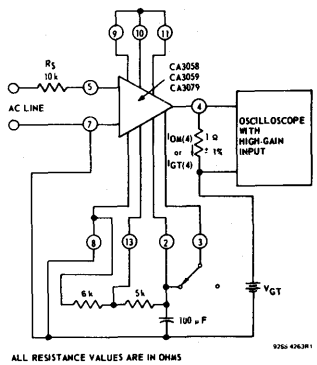


Fig. 5(a)–Peak output (pulsed) and gate trigger current with internal power supply test circuit for CA3058, CA3059, and CA3079.

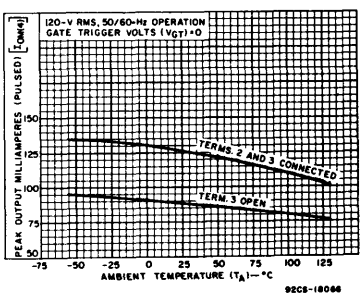


Fig. 5(b)–Peak output current (pulsed) vs. ambient temperature for CA3058, CA3059, and CA3079.

ELECTRICAL CHARACTERISTICS (For all types, unless indicated otherwise)
All voltages are measured with respect to Terminal 7.

CHARACTERISTIC	TEST CONDITIONS T_A = 25°C (Unless Indicated Otherwise)	LIMITS Min.	Typ.	Max.	UNITS
For Operating at 120 V rms, 50-60 Hz (AC Line Voltage)•					
DC Supply Voltage, V_S					
Inhibit Mode					
At 50/60 Hz	R_S = 8 kΩ, I_L = 0	6.1	6.5	7	V
At 400 Hz	R_S = 10 kΩ, I_L = 0	–	6.8	–	V
At 50/60 Hz	R_S = 5 kΩ, I_L = 2 mA	–	6.4	–	V
Pulse Mode					
At 50/60 Hz	R_S = 8 kΩ, I_L = 0	6	6.4	7	V
At 400 Hz	R_S = 10 kΩ, I_L = 0	–	6.7	–	V
At 50/60 Hz	R_S = 5 kΩ, I_L = 2 mA	–	6.3	–	V
At 50/60 Hz (CA3058) *See Fig. 2*	R_S = 8 kΩ, I_L = 0, T_A = −55 to +125°C	5.5	–	7.5	V
Gate Trigger Current, I_{GT}(4) *See Figs. 3, 5(a)*	Terms. 3 and 2 connected, V_{GT} = 1 V	–	105	–	mA
Peak Output Current (Pulsed), I_{OM}(4)	Term. 3 open, Gate Trigger Voltage (V_{GT}) = 0	50	84	–	mA
With Internal Power Supply	Terms. 3 and 2 connected, Gate Trigger Voltage (V_{GT})=0	90	124	–	mA
	Term. 3 open, V^+=12 V, V_{GT}=0	–	170	–	mA
With External Power Supply *See Figs. 5, 6*	Terms. 3 and 2 connected, V^+=12 V, V_{GT} = 0	–	240	–	mA
Inhibit Input Ratio, V_9/V_2					
All Types	Voltage Ratio of Term. 9 to 2	0.465	0.485	0.520	–
CA3058 *See Fig. 7*	T_A = −55 to +125°C	0.450	–	0.520	–
Total Gate Pulse Duration:*					
For positive dv/dt, t_P					
50-60 Hz	C_{EXT} = 0	70	100	140	μs
400 Hz	C_{EXT} = 0, R_{EXT} = ∞	–	12	–	μs
For negative dv/dt, t_N					
50-60 Hz	C_{EXT} = 0	70	100	140	μs
400 Hz *See Fig. 8*	C_{EXT} = 0, R_{EXT} = ∞	–	10	–	μs
Pulse Duration After Zero Crossing (50-60 Hz):					
For positive dv/dt, t_{P1}	C_{EXT} = 0	–	50	–	μs
For negative dv/dt, t_{N1} *See Fig. 8*	R_{EXT} = ∞	–	60	–	μs
Output Leakage Current, I_4					
Inhibit Mode:					
All Types		–	0.001	10	μA
CA3058 *See Fig. 9*	T_A = −55 to +125°C	–	–	20	μA
Input Bias Current, I_I					
CA3058, CA3059		–	220	1000	nA
CA3079 *See Fig. 10*		–	220	2000	nA

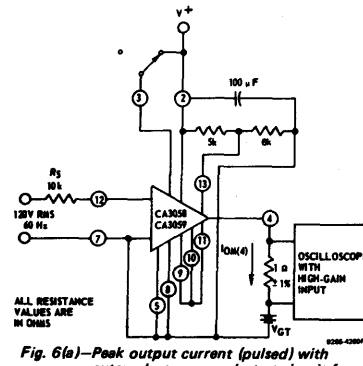


Fig. 6(a)—Peak output current (pulsed) with external power supply test circuit for CA3058 and CA3059.

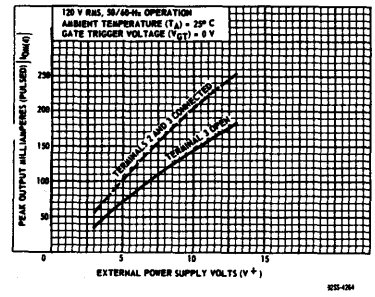


Fig. 6(b)—Peak output current (pulsed) vs. external power supply voltage for CA3058 and CA3059.

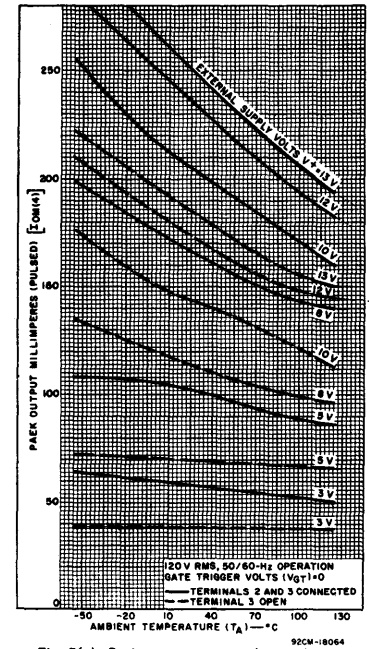


Fig. 6(c)—Peak output current (pulsed) vs. ambient temperature for CA3058 and CA3059.

ELECTRICAL CHARACTERISTICS (For all types, unless indicated otherwise) (Cont'd)
All voltages are measured with respect to Terminal 7.

CHARACTERISTIC	TEST CONDITIONS T_A = 25°C (Unless Indicated Otherwise)	LIMITS			UNITS
		Min.	Typ.	Max.	
For Operating at 120 V rms, 50-60 Hz (AC Line Voltage)●					
Common-Mode Input Voltage Range, V_{CMR}	Terms. 9 and 13 connected	–	1.5 to 5	–	V
Sensitivity, ΔV_{13}≠ (Pulse Mode) *See Figs. 5(a), 12*	Term. 12 open	–	6	–	mV

≠Required voltage change at Term. 13 to either turn OFF the triac when ON or turn ON the triac when OFF.

*Pulse duration in 50 Hz applications is approximately 15% longer than shown in Fig. 8(b).

● The values given in the Electrical Characteristics Chart at 120 V also apply for operation at input voltages of 24 V, 208/230 V, and 277 V, except for Pulse Duration. However, the series resistor (R_S) must have the indicated value, shown in the chart in Fig. 1, for the specified input voltage.

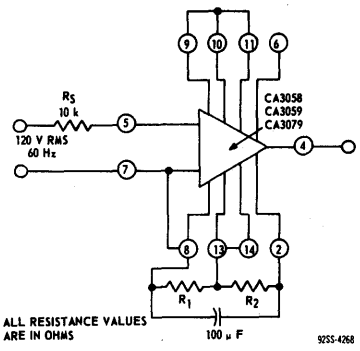


Fig. 7(a)—Input inhibit voltage ratio test circuit for CA3058, CA3059, and CA3079.

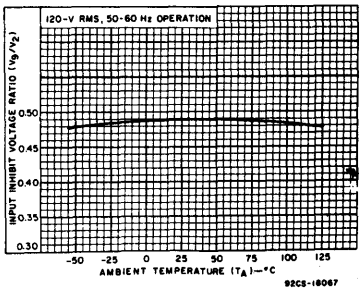


Fig. 7(b)—Input inhibit voltage ratio vs. ambient temperature for CA3058, CA3059, and CA3079.

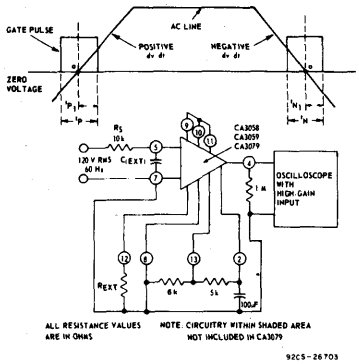


Fig. 8(a)—Gate pulse duration test circuit with associated waveform for CA3058, CA3059, and CA3079.

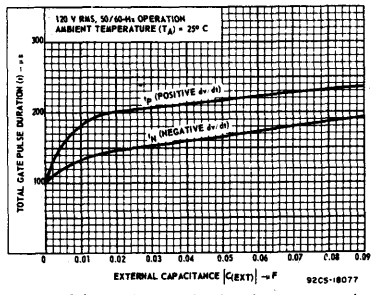


Fig. 8(b)—Total gate pulse duration vs. external capacitance for CA3058, CA3059, and CA3079.

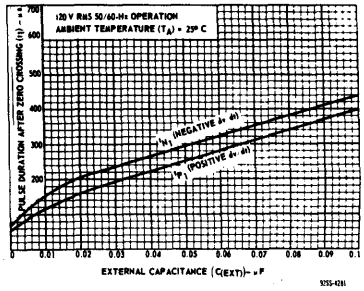


Fig. 8(c)—Pulse duration after zero crossing vs. external capacitance for CA3058, CA3059, and CA3079.

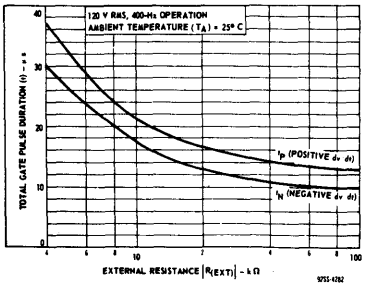


Fig. 8(d)—Total gate pulse duration vs. external resistance for CA3058 and CA3059.

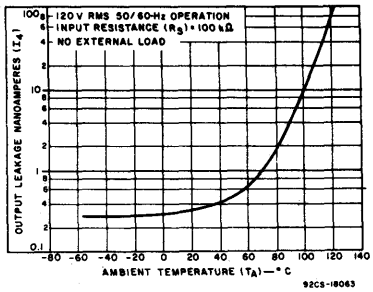


Fig. 9—Output leakage current (inhibit mode) vs. ambient temperature for CA3058, CA3059, and CA3079.

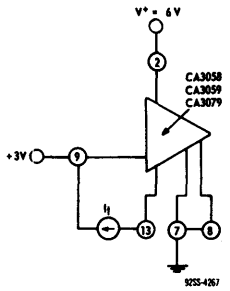


Fig. 10—Input bias current test circuit for CA3058, CA3059, and CA3079.

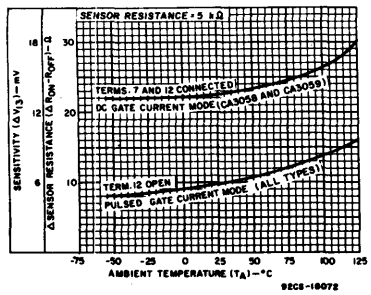


Fig. 12—Sensitivity vs. ambient temperature for CA3058, CA3059, and CA3079.

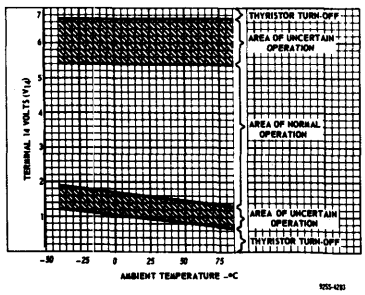


Fig. 13—Operating regions for built-in protection circuit for CA3058 and CA3059.

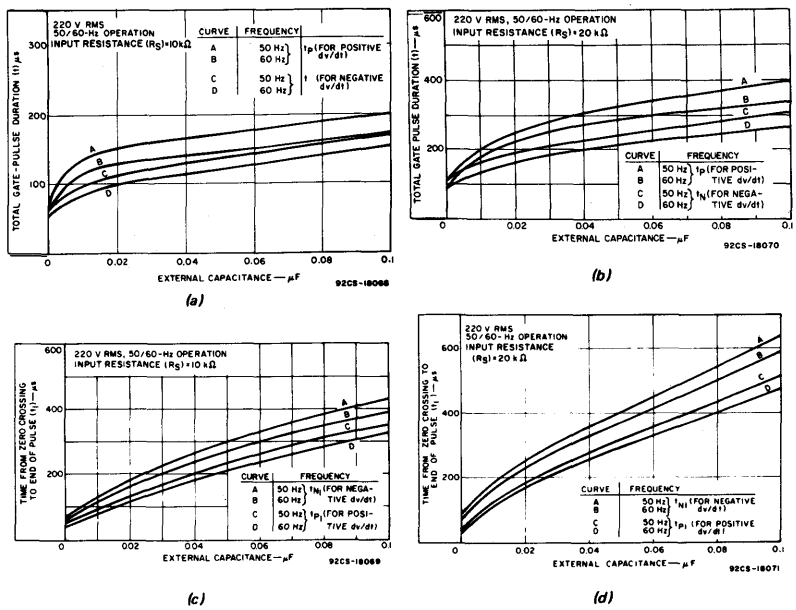


Fig. 11—Relative pulse width and location of zero crossing for 220-volt operation for CA3058, CA3059, and CA3079.

OPERATING CONSIDERATIONS

Power Supply Considerations for CA3058, CA3059, and CA3079

The CA3058, CA3059, and CA3079 are intended for operation as self-powered circuits with the power supplied from an AC line through a dropping resistor. The internal supply is designed to allow for some current to be drawn by the auxiliary power circuits. Typical power supply characteristics are given in Figs. 3(b) and 3(c).

Power Supply Considerations for CA3058 and CA3059

The output current available from the internal supply may not be adequate for higher power applications. In such applications an external power supply with a higher voltage should be used with a resulting increase in the output level. (See Fig. 5 for the peak output current characteristics). When an external power supply is used, Terminal 5 should be connected to Terminal 7 and the synchronizing voltage applied to Terminal 12 as illustrated in Fig. 5(a).

Operation of Built-in Protection for the CA3058, CA3059

A special feature of the CA3058 and CA3059 is the inclusion of a protection circuit which, when connected, removes power from the load if the sensor either shorts or opens. The protection circuit is activated by connecting Terminal 14 to Terminal 13 as shown in Fig. 1. To assure proper operation of the protection circuit the following conditions should be observed:

1. Use the internal supply and limit the external load current to 2 mA with a 5 kΩ dropping resistor.
2. Set the value of R_P and sensor resistance (R_X) between 2 kΩ and 100 kΩ.
3. The ratio of R_X to R_P, typically, should be greater than 0.33 and less than 3. If either of these ratios is not met with an unmodified sensor over the entire anticipated temperature range, then either a series or shunt resistor must be added to avoid undesired activation of the circuit.

If operation of the protection circuit is desired under conditions other than those specified above, then apply the data given in Fig. 13.

External Inhibit Function for the CA3058 and CA3059

A priority inhibit command may be applied to Terminal 1. The presence of at least +1.2 V at 10 μA will remove drive from the thyristor. This required level is compatible with DTL or T^2L logic. A logical 1 activates the inhibit function.

DC Gate Current Mode for the CA3058 and CA3059

Connecting Terminals 7 and 12 disables the zero-crossing detector and permits the flow of gate current on demand from the differential sensing amplifier. This mode of operation is useful when comparator operation is desired or when inductive loads are switched. Care must be exercised to avoid overloading the internal power supply when operating in this mode. A sensitive gate thyristor should be used with a resistor placed between Terminal 4 and the gate in order to limit the gate current.

CA3060, CA3060A Types

Operational Transconductance Amplifier Arrays

APPLICATIONS

- **For low power conventional operational amplifier applications**
- **Active filters**
- **Comparators**
- **Gyrators**
- **Mixers**
- **Modulators**
- **Multiplexers**
- **Multipliers**
- **Strobing and gating functions**
- **Sample and hold functions**

FEATURES

- **Low power consumption – as low as 100 μW per amplifier**
- **Independent biasing for each amplifier**
- **High forward transconductance**
- **Programmable range of input characteristics**
- **Low input bias and input offset current**
- **High input and output impedance**
- **No effect on device under output short-circuit conditions**
- **Zener diode bias regulator**

RCA-CA3060AD, CA3060BD, CA3060D, and CA3060E, monolithic integrated circuits, are arrays of three independent Operational Transconductance Amplifiers. This type of amplifier is a new circuit concept that has the generic characteristics of an operational voltage amplifier with the exception that the forward gain characteristic is best described by transconductance rather than voltage gain (open-loop voltage gain is the product of the transconductance and the load resistance, $g_m R_L$). When operated into a suitable load resistor and with provisions for feedback, these amplifiers are well suited for a wide variety of operational-amplifier and related applications. In addition, the extremely high output impedance makes these types particularly well suited for service in active filters.

The three amplifiers in the CA3060 family are identical push-pull Class A types which can be independently biased to achieve a wide range of characteristics for specific applications. The electrical characteristics of each amplifier are a function of the amplifier bias current (I_{ABC}). This feature offers the system designer maximum flexibility with regard to output current capability, power consumption, slew rate, input resistance, input bias current, and input offset current. The linear variation of the parameters with respect to bias and the ability to maintain a constant dc level between input and output of each amplifier also makes the CA3060 suitable for a variety of non-linear applications such as mixers, multipliers, and modulators.

In addition; the types in the CA3060 family incorporate a unique Zener diode regulator system that permits current regulation below supply voltages normally associated with such systems.

Generic applications of the OTA are described in ICAN-6668, Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers.

The CA3060AD, CA3060BD, and CA3060D are supplied in a hermetic 16-lead dual-in-line ceramic package which can be operated over the full military temperature range, -55°C to +125°C. The CA3060E is supplied in a 16-lead dual-in-line plastic package and is operational from -40°C to +85°C.

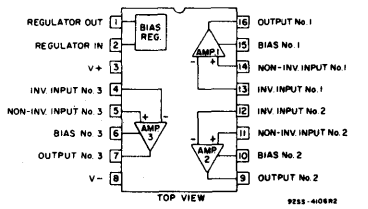


Fig. 1—Functional block diagram for each type in the CA3060 family.

MAXIMUM RATINGS, *Absolute Maximum Values at T_A = 25°C*

DC Supply Voltage (between V^+ and V^- terminals):
- CA3060AD, CA3060BD, CA3060E ..... 36V (±18V)
- CA3060D ..... 14 V (±7V)

Differential Input Voltage (each amplifier):
- CA3060AD, CA3060BD, CA3060E ..... ±5V
- CA3060D ..... ±5V

DC Input Voltage ..... V^+ to V^-

Input Signal Current (each amplifier of each type): ..... ±1 mA

Amplifier Bias Current (each amplifier of each type) ..... 2 mA

Bias Regulator Input Current ..... 5 mA

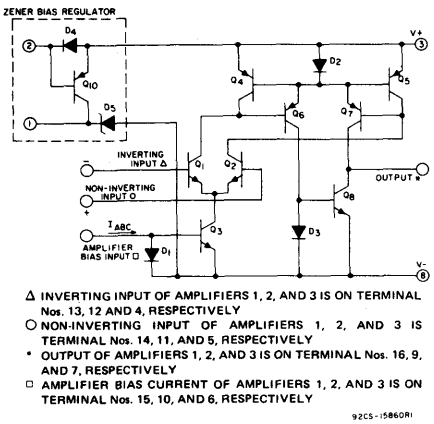
Output Short-Circuit Duration* ..... No limitation

*Short circuit may be applied to ground or to either supply.

Device Dissipation:
- Total Package of each type up to T_A = 75°C ..... 490 mW
- Above T_A = 75°C ..... Derate linearly 6.67 mW/°C

Temperature Range:
- Operating –
  - CA3060AD, CA3060BD, CA3060D ..... -55 to +125°C
  - CA3060E ..... -40 to +85°C
- Storage –
  - CA3060AD, CA3060BD, CA3060D, CA3060E ..... -65 to +150°C

Lead Temperature (During Soldering):
- At distance 1/16 ±1/32 in. (1.59 ±0.79 mm) from case for 10s max ..... +300°C



Δ INVERTING INPUT OF AMPLIFIERS 1, 2, AND 3 IS ON TERMINAL Nos. 13, 12 AND 4, RESPECTIVELY
O NON-INVERTING INPUT OF AMPLIFIERS 1, 2, AND 3 IS TERMINAL Nos. 14, 11, AND 5, RESPECTIVELY
* OUTPUT OF AMPLIFIERS 1, 2, AND 3 IS ON TERMINAL Nos. 16, 9, AND 7, RESPECTIVELY
□ AMPLIFIER BIAS CURRENT OF AMPLIFIERS 1, 2, AND 3 IS ON TERMINAL Nos. 15, 10, AND 6, RESPECTIVELY

92CS-15860R1

Fig. 2—Simplified schematic diagram showing bias regulator and one operational transconductance amplifier for each type of the CA3060 family.

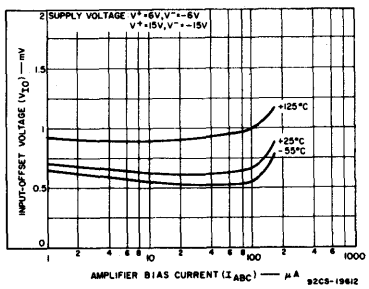


Fig. 3—Input offset voltage vs. amplifier bias current.

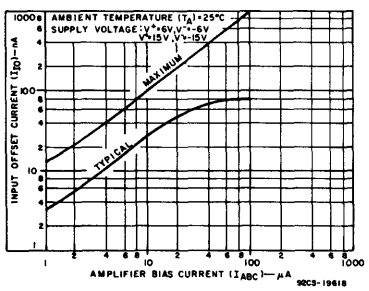


Fig. 4—Input offset current vs. amplifier bias current.

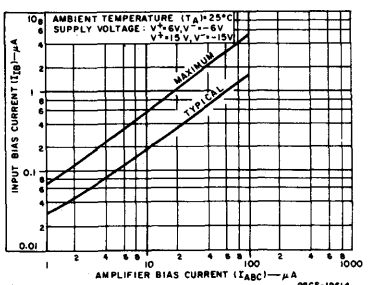


Fig. 5a—Input bias current vs. amplifier bias current

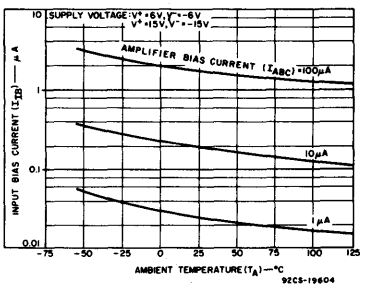


Fig. 5b—Input bias current vs. ambient temperature.

ELECTRICAL CHARACTERISTICS (CA3060D)
For each amplifier at T_A = 25°C, V^+ = 6 V, V^- = -6 V

CHARACTERISTIC	SYMBOL	TYPICAL CHARACTERISTICS CURVES Fig.	LIMITS Amplifier Bias Current I_{ABC} = 1 μA MIN.	TYP.	MAX.	I_{ABC} = 10 μA MIN.	TYP.	MAX.	I_{ABC} = 100 μA MIN.	TYP.	MAX.	UNITS
STATIC CHARACTERISTICS												
Input Offset Voltage	V_{IO}	3	–	1	5	–	1	5	–	1	5	mV
Input Offset Current	I_{IO}	4	–	3	14	–	30	100	–	250	1000	nA
Input Bias Current	I_{IB}	5a, b	–	33	70	–	300	550	–	2500	5000	nA
Peak Output Current	I_{OM}	6a, b	1.3	2.3	–	15	26	–	150	240	–	μA
Peak Output Voltage: Positive	V_{OM^+}	7	4.6	5	–	4.5	4.8	–	4.5	4.7	–	V
Negative	V_{OM^-}		5.8	5.95	–	5.8	5.95	–	5.7	5.9	–	
Amplifier Supply Current (each amplifier)	I_A	8a, b	–	8.5	14	–	85	120	–	850	1200	μA
Power Consumption (each amplifier)	P	–	–	0.10	0.17	–	1	1.45	–	10	14.5	mW
Input Offset-Voltage Sensitivity■: Positive	$\Delta V_{IO}/\Delta V^+$	–	–	1.5	120	–	2	120	–	2	120	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$		–	20	120	–	20	120	–	30	120	
Amplifier Bias Voltage*	V_{ABC}	9	–	0.54	–	–	0.60	–	–	0.66	–	V
DYNAMIC CHARACTERISTICS (at 1 kHz unless specified otherwise)												
Forward Transconductance (large signal)	921	10a, b	0.3	1.55	–	3	18	–	30	102	–	mmho
Common-Mode Rejection Ratio	CMRR	–	70	110	–	70	110	–	70	90	–	dB
Common-Mode Input-Voltage Range	V_{ICR}	–	4.4 to -5.1 min. 4.7 to -5.3 typ.			4.3 to -5 min. 4.6 to -5.2 typ.			4.3 to -5 min. 4.6 to -5.2 typ.			V
Slew Rate (Test ckt., Fig. 13	SR		–	0.1	–	–	1	–	–	8	–	V/μs
Open-Loop (g_{21}) Bandwidth	BW_{OL}	11	–	20	–	–	45	–	–	110	–	kHz
Input Impedance Components: Resistance	R_I	12	800	1600	–	90	170	–	10	20	–	kΩ
Capacitance at 1 MHz	C_I	–	–	2.7	–	–	2.7	–	–	2.7	–	pF
Output Impedance Components: Resistance	R_O	14	–	200	–	–	20	–	–	2	–	MΩ
Capacitance at 1 MHz	C_O	–	–	4.5	–	–	4.5	–	–	4.5	–	pF
ZENER BIAS REGULATOR CHARACTERISTICS (at T_A = 25°C, I_Z = 0.1 mA)				MIN.	TYP.	MAX.						
Voltage	V_Z	15	Temp. Coeff. = 3 mV/°C	6.2	6.7	7.9						V
Impedance	Z_Z	–			200	300						Ω

* Temperature-Coefficient: -2.2 mV/°C (at V_{ABC} = 0.54 V, I_{ABC} = 1 μA; -2.1 mV/°C (at V_{ABC} = 0.060 V, I_{ABC} = 10 μA); -1.9 mV/°C (at V_{ABC} = 0.66 V, I_{ABC} = 100 μA)

■ Conditions for Input Offset Voltage and Supply Sensitivity:
(a) Bias current derived from the regulator with an appropriate resistor connected from terminal No. 1 to the bias terminal on the amplifier under test —
V^+ is reduced to 5 volts for V^+ sensitivity
V^- is reduced to -5 volts for V^- sensitivity
(b) V^+ sensitivity in μV/V = $\frac{\text{Voffset - Voffset for +5 V and -6 V supplies}}{\text{1 volt}}$
V^- sensitivity in μV/V = $\frac{\text{Voffset - Voffset for -5 V and +6 V supplies}}{\text{1 volt}}$

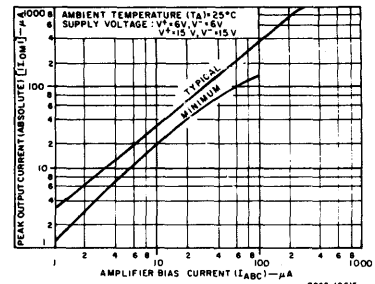


Fig.6a—Peak output current vs. amplifier bias current.

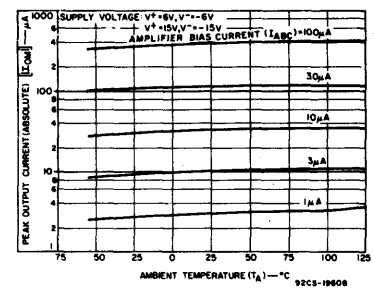


Fig.6b—Peak output current vs. ambient temperature.

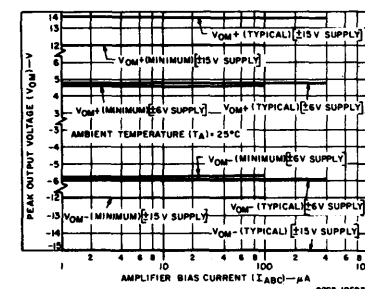


Fig.7—Peak output voltage vs. amplifier bias current.

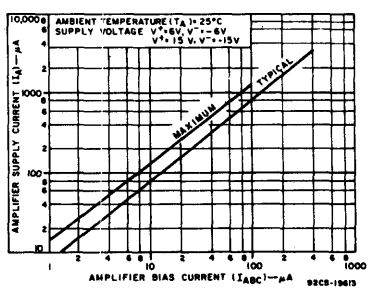


Fig.8a—Amplifier supply current (each amplifier) vs. amplifier bias current.

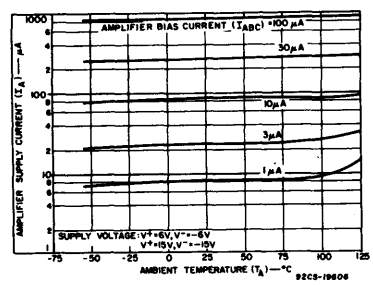


Fig.8b—Amplifier supply current (each amplifier) vs. ambient temperature.

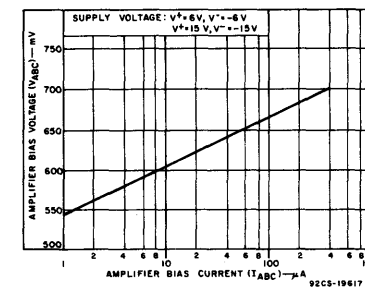


Fig.9—Amplifier bias voltage vs. amplifier bias current.

CA3060, CA3060A Types

ELECTRICAL CHARACTERISTICS (CA3060AD, CA3060BD, CA3060E)

For each amplifier at T_A = 25°C, V^+ = 15 V, V^- = -15 V

CHARACTERISTIC	SYMBOL	TYPICAL CHARACTERISTICS CURVE Fig.	LIMITS — Amplifier Bias Current I_{ABC} = 1 μA (CA3060BD) MIN.	TYP.	MAX.	I_{ABC} = 10 μA (CA3060BD) MIN.	TYP.	MAX.	I_{ABC} = 100 μA (CA3060AD, CA3060BD, CA3060E) MIN.	TYP.	MAX.	UNITS
STATIC CHARACTERISTICS												
Input Offset Voltage	V_{IO}	3	-	1	5	-	1	5	-	1	5	mV
Input Offset Current	I_{IO}	4	-	3	14	-	30	100	-	250	1000	nA
Input Bias Current	I_{IB}	5a,b	-	33	70	-	300	550	-	2500	5000	nA
Peak Output Current	I_{OM}	6a,b	1.3	2.3		15	26	-	150	240	-	μA
Peak Output Voltage: Positive	V_{OM^+}	7	12	13.6	-	12	13.6	-	12	13.6	-	V
Negative	V_{OM^-}		12	14.7	-	12	14.7	-	12	14.7	-	
Amplifier Supply Current (each amplifier)	I_A	8a,b	-	8.5	14	-	85	120	-	850	1200	μA
Power Consumption (each amplifier)	P		-	0.26	0.42	-	2.6	3.6	-	26	36	mW
Input Offset-Voltage Sensitivity■: Positive	$\Delta V_{IO}/\Delta V^+$	-	-	1.5	150	-	2	150	-	2	150	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$		-	20	150	-	20	150	-	30	150	
Amplifier Bias Voltage*	V_{ABC}	9	-	0.54	-	-	0.60	-	-	0.66	-	V
DYNAMIC CHARACTERISTICS (at 1 kHz unless specified otherwise)												
Forward Transconductance (large signal)	g_{21}	10a,b	0.3	1.55	-	3	18	-	30	102	-	mmho
Common-Mode Rejection Ratio	CMRR	-	70	110		70	110	-	70	90	-	dB
Common-Mode Input Voltage Range	V_{ICR}	-	+12 to -12 min. +13 to -14 typ.			+12 to -12 min. +13 to -14 typ.			+12 to -12 min. +13 to -14 typ.			V
Slew Rate (Test ckt., Fig. 13)	SR		-	0.1	-	-	1	-	-	8	-	V/μs
Open-Loop (g_{21}) Bandwidth	BW_{OL}	11	-	20	-	-	45	-	-	110	-	kHz
Input Impedance Components: Resistance	R_I	12	800	1600	-	90	170	-	10	20	-	kΩ
Capacitance at 1 MHz	C_I	-	-	2.7	-	-	2.7	-	-	2.7	-	pF
Output Impedance Components: Resistance	R_O	14	-	200	-	-	20	-	-	2	-	MΩ
Capacitance at 1 MHz	C_O	-	-	4.5	-	-	4.5	-	-	4.5	-	pF
ZENER BIAS REGULATOR CHARACTERISTICS (at T_A = 25°C, I_Z = 0.1 mA)						MIN.	TYP.	MAX.				
Voltage	V_Z	15	Temp. Coeff. = 3 mV/°C			6.2	6.7	7.9				V
Impedance	Z_Z	-					200	300				Ω

* Temperature-Coefficient; -2.2 mV/°C (at V_{ABC} = 0.54 V, I_{ABC} = 1 μA; -2.1 mV/°C (at V_{ABC} = 0.060 V, I_{ABC} = 10 μA); -1.9 mV/°C (at V_{ABC} = 0.66 V, I_{ABC} = 100 μA)

■ Conditions for Input Offset Voltage and Supply Sensitivity:

(a) Bias current derived from the regulator with an appropriate resistor connected from terminal No. 1 to the bias terminal on the amplifier under test ···

V^+ is reduced to 13 volts for V^+ sensitivity

V^- is reduced to −13 volts for V^- sensitivity

(b) V^+ sensitivity in μV/V = $\frac{\text{Voffset} - \text{Voffset for +13 V and −15 V supplies}}{1 \text{ volt}}$

V^- sensitivity in μV/V = $\frac{\text{Voffset} - \text{Voffset for −13 V and +15 V supplies}}{1 \text{ volt}}$

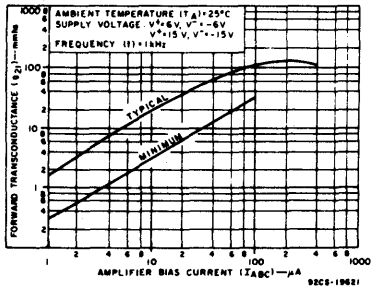


Fig. 10a—Forward transconductance vs. amplifier bias current.

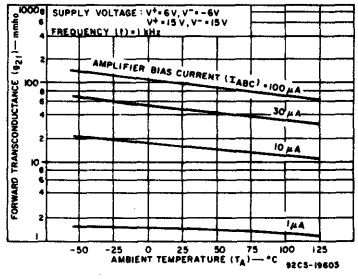


Fig. 10b—Forward transconductance vs. ambient temperature.

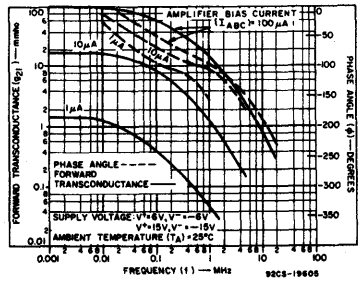


Fig. 11—Forward transconductance vs. frequency.

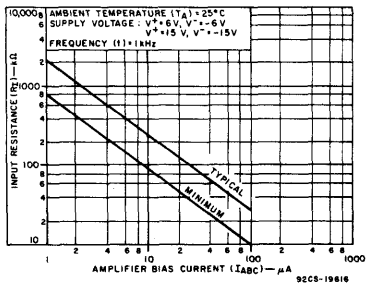
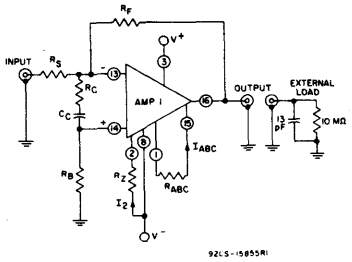


Fig. 12—Input resistance vs. amplifier bias current.



V_Z is measured between terminals 1 and 8.

V_{ABC} is measured between terminals 15 and 8.

$$R_Z = \frac{[(V^+)-(V^-)-0.7]}{I_Z}, \quad R_{ABC} = \frac{V_Z - V_{ABC}}{I_{ABC}}$$

Supply Voltage: for both ±6 V and ±15 V.

TYPICAL SLEW RATE TEST CIRCUIT PARAMETERS

I_{ABC}	SLEW RATE	I_Z	R_{ABC}	R_S	R_F	R_B	R_C	C_C
μA	V/μs	μA	ohms					μF
100	8	200	62 k	100k	100k	51k	100	0.02
10	1	200	620k	1M	1M	510k	1k	0.005
1	0.1	2	6.2M	10M	10M	5.1M	∞	0

Fig. 13—Slew rate test circuit for amplifier No. 1 of CA3060.

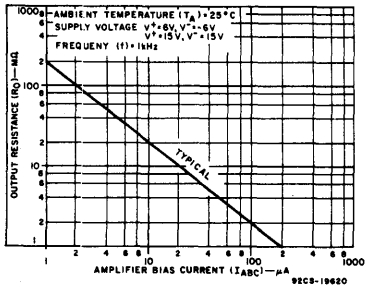


Fig.14—Output resistance vs. amplifier bias current.

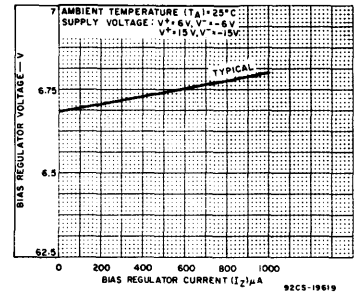


Fig.15—Bias regulator voltage vs. bias regulator current.

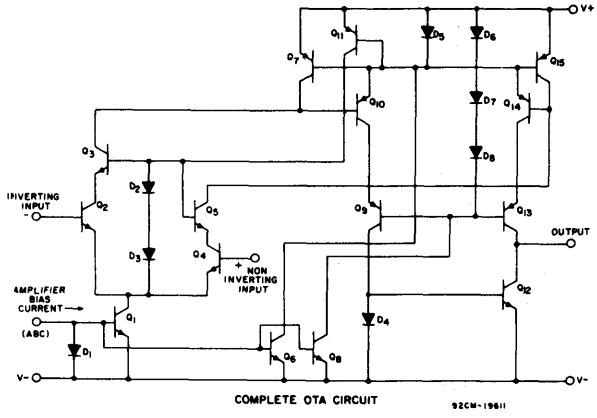


Fig.16 – Complete schematic diagram showing one of the three operational transconductance amplifiers.

OPERATING CONSIDERATIONS

The CA3060 consists of three operational amplifiers similar in form and application to conventional operational amplifiers but sufficiently different from the standard operational amplifier (op-amp) to justify some explanation of their characteristics. The amplifiers incorporated in the CA3060 are best described by the term Operational Transconductance Amplifier (OTA). The characteristics of an ideal OTA are similar to those of an ideal op-amp except that the OTA has an extremely high output impedance. Because of this inherent characteristic the output signal is best defined in terms of current which is proportional to the difference between the voltages of the two input terminals. Thus, the transfer characteristic is best described in terms of transconductance rather than voltage gain. Other than the difference given above, the characteristics tabulated on pages 3 and 4 of this data bulletin are similar to those of any typical op-amp.

The OTA circuitry incorporated in the CA3060 (See Fig. 16) provides the equipment designer with a wider variety of circuit arrangements than does the standard op-amp; because as the curves in the data bulletin indicate, the user may select the optimum circuit conditions for a specific application simply by varying the bias conditions of each amplifier. If low power consumption, low bias, and low offset current, or high input impedance are primary design requirements, then low current operating conditions may be selected. On the other hand, if operation into a moderate load impedance is the primary consideration, then higher levels of bias may be used.

Bias Considerations for Op-Amp Applications

The operational transconductance amplifiers allow the circuit designer to select and control the operating conditions of the circuit merely by the adjustment of the input bias current I_{ABC}. This enables the designer to have complete control over transconductance, peak output current and total power consumption independent of supply voltage.

In addition, the high output impedance makes these amplifiers ideal for applications where current summing is involved.

The design of a typical operational amplifier circuit (See Fig. 17) would proceed as follows:

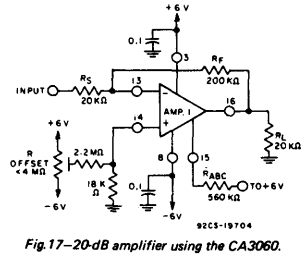


Fig.17—20-dB amplifier using the CA3060.

Circuit Requirements
Closed loop voltage gain = 10 (20 dB)
Offset voltage adjustable to zero
Current drain as low as possible
Supply voltage = ±6 V
Maximum input voltage = ±50 mV
Input resistance = 20 kΩ
Load resistance = 20 kΩ
Device: CA3060

Calculation

1. Required transconductance g_{21}.
   Assume that the open loop gain A_{OL} must be at least ten times the closed loop gain. Therefore, the forward transconductance required is given by

$$g_{21} = A_{OL}/R_L$$

$$\cong 100/18\ k\Omega$$

$$\cong 5.5\ mmho$$

(R_L = 20 kΩ in parallel with 200 kΩ

≅ 18 kΩ)

2. Selection of suitable amplifier bias current.
   The amplifier bias current is selected from the minimum value curve of transconductance (Fig. 10a) to assure that the amplifier will provide sufficient gain. For the required g_{21} of 5.5 mmho an amplifier bias current I_{ABC} of 20 μA is suitable.

3. Determination of Output Swing Capability.
   For a loop gain of 10 the output swing is ±0.5 V and the peak load current 25 μA. However, the amplifier must also supply the necessary current through the feedback resistor and for R_S = 20 kΩ than R_F = 200 kΩ if A_{OL} = 10. Therefore, the feedback loading = 0.5/200 kΩ = 2.5 μA.

   The total amplifier current output requirements are, therefore, ±27.5 μA. Referring to the data given in Fig. 6a we see that for an amplifier bias current of 20 μA the amplifier output current is ±40 μA. This is obviously adequate and it is not necessary to change the amplifier bias current I_{ABC}.

4. Calculation of bias resistance.
   For minimum supply current drain the amplifier bias current I_{ABC} should be fed directly from the supplies and not from the bias regulator. The value of the resistor R_{ABC} may be directly calculated using Ohm's law.

$$R_{ABC} = \frac{V_{SUP} \cdot V_{ABC}}{I_{ABC}}$$

$$R_{ABC} = \frac{12 - 0.63}{20 \times 10^{-6}}$$

$$= 568.5\ k\Omega \text{ or} \cong 560\ k\Omega$$

5. Calculation of offset adjustment circuit.
   In order to reduce the loading effect of the offset adjustment circuit on the power supply, the offset control should be arranged to provide the necessary offset current. The source resistance of the non-inverting input is made equal to the source resistance of the inverting input.

   i.e. $\dfrac{20 \times 200 \times 10^6 \text{ ohms}}{220 \times 10^3} \cong 18\ k\Omega$

Because the maximum offset voltage is 5 mV and an additional increment due to the offset current (Fig. 4) flowing through the source resistance

(i.e. $200 \times 10^{-9} \times 18 \times 10^3$ volts), therefore,

the Offset Voltage Range = 5 mV + 3.6 mV = ±8.6 mV

The current necessary to provide this offset is

$$\frac{8.6 \times 10^{-3}}{18 \times 10^3} \text{ or } 0.48\ \mu A$$

With a supply voltage of ±6 V, this current can be provided by a 10 MΩ resistor. However, the stability of such a resistor is often questionable and a more realistic value of 2.2 MΩ was used in the final circuit.

OTHER CONSIDERATIONS

Capacitance Effects

The CA3060 is designed to operate at such low power levels that high impedance circuits must be employed. In designing such circuits, particularly feedback amplifiers, stray circuit capacitance must always be considered because of its adverse effect on frequency response and stability. For example a 10-kΩ load with a stray capacitance of 15 pF has a time constant of 1 MHz. Fig. 18 illustrates how a 10-kΩ 15-pF load modifies the frequency characteristic.

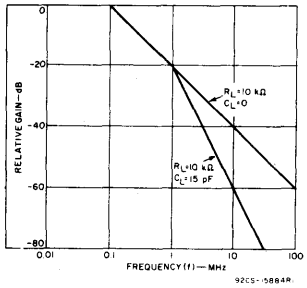


Fig.18—Effect of capacitive loading on frequency response.

Capacitive loading also has an effect on slew rate; because the peak output current is established by the amplifier bias current, I_{ABC} (see Fig. 6a), the maximum slew rate is limited to the maximum rate at which the capacitance can be charged by the I_{OM}. Therefore,

$$SR = dV/dt = I_{OM}/C_L$$

where C_L is the total load capacitance including strays. This relationship is shown graphically in Fig. 19. When measuring slew rate for this data bulletin, care was taken to keep the total capacitive loading to 13 pF.

Phase Compensation

In many applications phase compensation will not be required for the amplifiers of the CA3060. When needed, compensation may easily be accomplished by a simple RC network at the input of the amplifier as shown in Fig. 13. The values given in Fig. 13 provide stable operation for the critical unity gain condition, assuming that capacitive loading on the output is 13 pF or less. Input phase compensation is recommended in order to maintain the highest possible slew rate.

In applications such as integrators, two OTAs may be cascaded to improve current gain. Compensation is best accomplished in this case with a shunt capacitor at the output of the first amplifier. The high gain following compensation assures a high slew rate.

APPLICATIONS

Having determined the operating points of the CA3060 amplifiers, they can now function in the same manner as conventional op-amps, and thus, are well suited for most op-amp applications, including inverting and non-inverting amplifiers, integrators, differentiators, summing amplifiers etc.

TRI-LEVEL COMPARATOR

Tri-level comparator circuits are an ideal application for the CA3060 since it contains the requisite three amplifiers. A tri-level comparator has three adjustable limits. If either the upper or lower limit is exceeded, the appropriate output is activated until the input signal returns to a selected intermediate limit. Tri-level comparators are particularly suited to many industrial control applications.

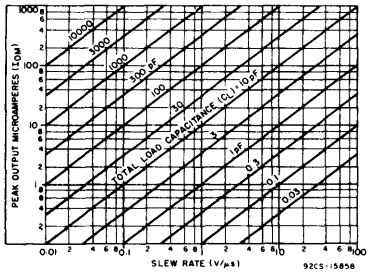


Fig.19—Effect of load capacitance on slew rate.

Circuit Description

Fig. 20 shows the block diagram of a tri-level comparator using the CA3060. Two of the three amplifiers are used to compare the input signal with the upper-limit and lower-limit reference voltages. The third amplifier is used to compare the input signal with a selected value of intermediate-limit reference voltage. By appropriate selection or resistance ratios this intermediate-limit may be set to any voltage between the upper-limit and lower-limit values. The output of the upper-limit and lower-limit comparator sets the corresponding upper or lower-limit flip-flop. The activated flip-flop retains its state until the third comparator (intermediate-limit) in the CA3060 initiates a reset function, thereby indicating that the signal voltage has returned to the intermediate-limit selected. The flip-flops employ two CA3086 transistor-array IC's, with circuitry to provide separate "SET" and "POSITIVE OUTPUT" terminals.

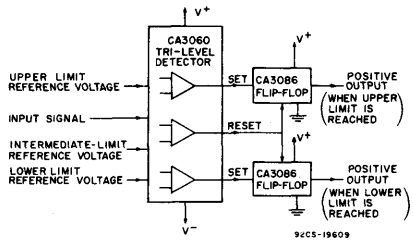


Fig.20—Functional block diagram of a tri-level comparator.

The circuit diagram of a tri-level comparator appears in Fig. 21. Power is provided for the CA3060 via terminals 3 and 8 by ±6-volt supplies and the built-in regulator provides amplifier-bias-current (I_{ABC}) to the three amplifiers via terminal 1. Lower-limit and upper-limit reference voltages are selected by appropriate adjustment of potentiometers R1 and R2, respectively. When resistors R3 and R4 are equal in value (as shown), the intermediate-limit reference voltage is automatically established at a value midway between the lower-limit and upper-limit values. Appropriate variation of resistors R3 and R4 permits selection of other values of intermediate-limit voltages. Input signal (E_S) is applied to the three comparators via terminals 5, 12, and 14. The "SET" output lines trigger the appropriate flip-flop whenever the input signal reaches a limit value. When the input signal returns to an intermediate-value, the common flip-flop "RESET" line is energized. The loads in the circuits, shown in Fig. 21 are 5-V, 25-mA lamps.

Active Filters — Using the CA3060 as a Gyrator

The high output impedance of the OTAs makes the CA3060 ideally suited for use as a gyrator in active filter applications. Fig. 22 shows two OTAs of the CA3060 connected as a gyrator in an active filter circuit. The OTAs in this circuit can make a 3-μF capacitor function as a floating 10-kilohenry inductor across Terminals A and B. The measured Q of 13 (at a frequency of 1 Hz) of this inductor compares favorably with a calculated Q of 16. The 20-kilohm to 2-megohm attenuators in this circuit extend the dynamic range of the OTA by a factor of 100. The 100-kilohm potentiometer, across V+ and V-, tunes the inductor by varying the g_{21} of the OTAs, thereby changing the gyration resistance.

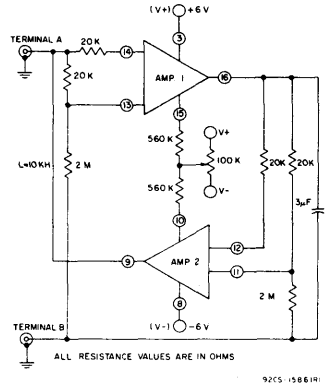


Fig.22—Two operational transconductance amplifiers of the CA3060 connected as a gyrator in an active filter circuit.

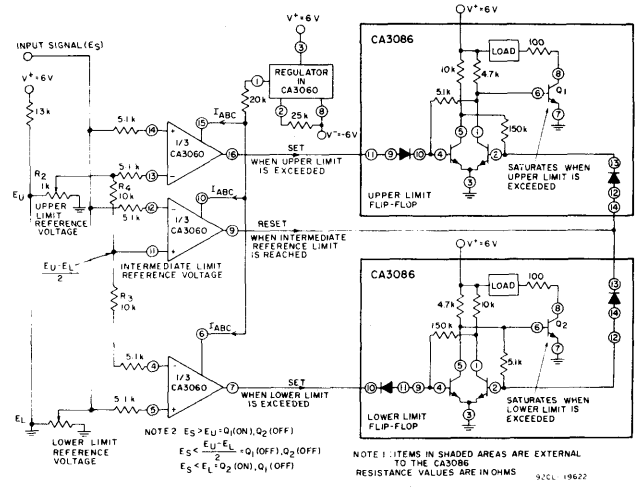


Fig.21—Tri-level comparator circuit.

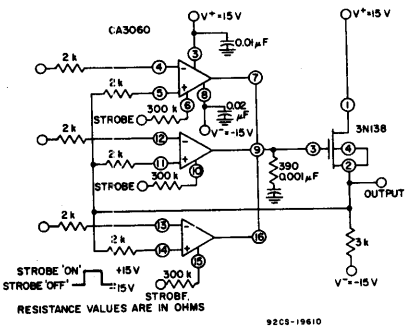


Fig.23—Three-channel multiplexer.

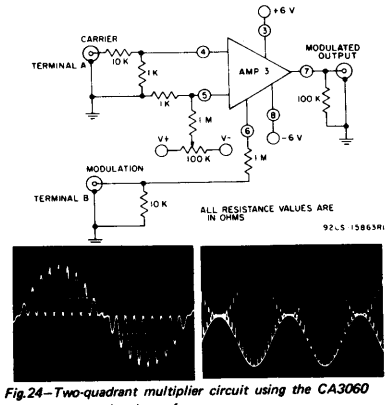


Fig.24—Two-quadrant multiplier circuit using the CA3060 with associated waveforms.

THREE CHANNEL MULTIPLEXER

Fig. 23 shows a schematic of a three channel multiplexer using a single CA3060 and a 3N138 MOS/FET as a buffer and power amplifier.

When the CA3060 is connected as a high-input impedance voltage follower, and strobe "ON," each amplifier is activated and the output swings to the level of the input of that amplifier. The cascade arrangement of each CA3060 amplifier with the MOS/FET provides an open loop voltage gain in excess of 100 dB, thus assuring excellent accuracy in the voltage follower mode with 100% feedback.

Operation at ±6 volts is also possible with several minor changes. First, the resistance in series with amplifier bias current (I_{ABC}) terminal of each amplifier should be decreased to maintain 100 μA of strobe—"ON" current at this lower supply voltage. Second, the drain resistance for the MOS/FET should be decreased to maintain the same value of source current. The low cost dual-gate protected MOS/FET, RCA-40841, may be used when operating at the low supply voltage.

The phase compensation network consists of a single 390Ω resistor and a 1000-pF capacitor, located at the interface of the CA3060 output and the MOS/FET gate. The bandwidth of the system is 1.5 MHz and the slew rate is 0.3 volts/μsec. The system slew rate is directly proportional to the value of the phase compensation capacitor. Thus, with higher gain settings where lower values of phase compensation capacitors are possible, the slew rate is proportionally increased.

NON LINEAR APPLICATIONS

AM Modulator (Two-Quadrant Multiplier)

Fig. 24 shows Amplifier No. 3 of the CA3060 used in an AM modulator or 2-quadrant multiplier circuit. When modulation is applied to the amplifier bias input, Terminal B, and the carrier frequency to the differential input, Terminal A, the waveform, shown in Fig. 24, is obtained. Fig. 24 is a result of adjusting the input offset control to balance the circuit so that no modulation can occur at the output without a carrier input. The linearity of the modulator is indicated by the solid trace of the superimposed modulating frequency. The maximum depth of modulation is determined by the ratio of the peak input modulating voltage to V^-.

The two-quadrant multiplier characteristic of this modulator is easily seen if modulation and carrier are reversed as shown in Fig. 24. The polarity of the output must follow that of the differential input; therefore, the output is positive only during the positive half cycle of the modulation and negative only in the second half cycle. Note, that both the input and output signals are referenced to ground. The output signal is zero when either the differential input or I_{ABC} are zero.

Four-Quadrant Multiplier

The CA3060 is also useful as a four-quadrant multiplier. A block diagram of such a multiplier, utilizing Amplifier Nos. 1, 2, and 3, is shown in Fig. 25 and a typical circuit is shown in Fig. 26. The multiplier consists of a single CA3060 and, as in the two-quadrant multiplier, exhibits no level shift between input and output. In Fig. 25, Amplifier No. 1 is connected as an inverting amplifier for the X-input signal. The output current of Amplifier No. 1 is calculated as follows:

$$I_O(1) = [-V_X]\ [g_{21}(1)] \qquad \text{(Eq.3)}$$

Ampl. No. 2 is a non-inverting amplifier so that

$$I_O(2) = [+V_X]\ [g_{21}(2)] \qquad \text{(Eq. 4)}$$

Because the amplifier output impedances are high, the load current is the sum of the two output currents, for an output voltage

$$V_O = V_X R_L\ [g_{21}(2) - g_{21}(1)] \qquad \text{(Eq. 5)}$$

The transconductance is approximately proportional to the amplifier bias current; therefore, by varying the bias current the g_{21} is also controlled. Amplifier No. 2 bias current is proportional to the Y-input signal and is expressed as

$$I_{ABC(2)} \approx \frac{(V\text{-}) + V_Y}{R_1} \qquad \text{(Eq. 6)}$$

Hence,

$$g_{21}(2) \approx k\ [\,(V\text{-}) + V_Y\,]. \qquad \text{(Eq. 7)}$$

Bias for Amplifier No. 1 is derived from the output of Amplifier No. 3 which is connected as a unity-gain inverting amplifier. $I_{ABC(1)}$, therefore, varies inversely with V_Y. And by the same reasoning as above

$$g_{21}(1) \approx k\ [(V\text{-}) - V_Y]. \qquad \text{(Eq. 8)}$$

Combining equation 5, 7, and 8 yields:

$$V_O \approx V_X \cdot k \cdot R_L \left\{[(V\text{-}) + V_Y] - [(V\text{-}) - V_Y]\right\} \text{ or}$$

$$V_O \approx 2\,k\,R_L\,V_X\,V_Y$$

Fig. 26 shows the actual circuit including all the adjustments associated with differential input and an adjustment for equalizing the gains of Amplifiers No. 1 and No. 2. Adjustment of the circuit is quite simple. With both the X and Y voltages at zero, connect Terminal 10 to Terminal 8. This procedure disables Amplifier No. 2 and permits adjusting the offset voltage of Amplifier No. 1 to zero by means of the 100-kΩ potentiometer. Next, remove the short between Terminals 10 and 8 and connect Terminal 15 to Terminal 8. This step disables Amplifier No. 1 and permits Amplifier No. 2 to be zeroed with the other potentiometer. With AC signals on both the X and Y input, R3 and R11 are adjusted for symmetrical output signals. Fig. 27 shows the output waveform with the multiplier adjusted. The voltage waveform in Fig. 27a shows suppressed carrier modulation of 1-kHz carrier with a triangular wave.

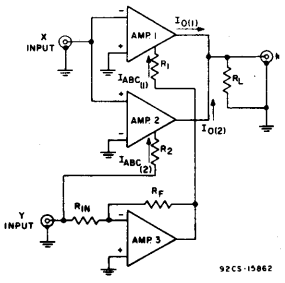


Fig.25—Four-quadrant multiplier using the CA3060.

Figures 27b and 27c, respectively, show the squaring of a triangular wave and a sine wave. Notice that in both cases the outputs are always positive and return to zero after each cycle.

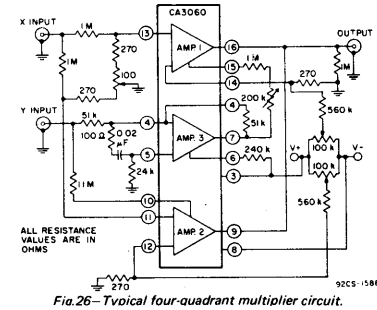


Fig.26—Typical four-quadrant multiplier circuit.

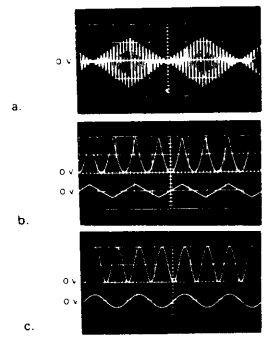


Fig.27—Voltage waveforms of four-quadrant multiplier circuit.

CA3075

FM IF Amplifier - Limiter, Detector, and Audio Preamplifier

For FM IF Amplifier Applications Up To 20 MHz In Communications Receivers And High-Fidelity Receivers

Features:

- Good sensitivity: Input limiting voltage (knee) = 250 μV typ. at 10.7 MHz
- Excellent AM rejection: 55 dB typ. at 10.7 MHz
- Internal Zener diode regulation for the IF amplifier section
- Low harmonic distortion
- Differential peak detection: Permits simplified single-coil tuning
- Audio preamplifier voltage gain: 21 dB typ.
- Minimum number of external parts required

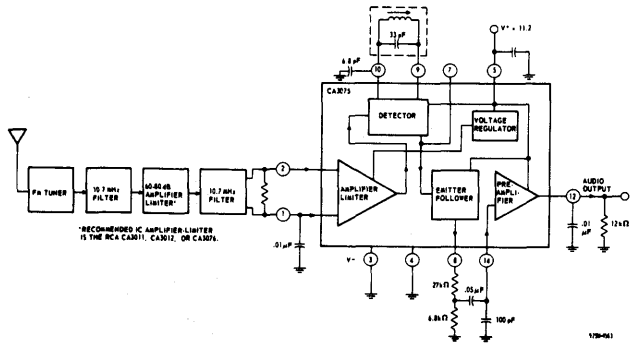


Fig. 1 - Block diagram of typical FM receiver utilizing the CA3075

RCA CA3075 is an integrated circuit which provides, in a single monolithic chip, an FM IF subsystem for Communications and High-Fidelity Receivers. This device, shown in the schematic diagram (Fig. 2), consists of a multistage IF amplifier-limiter section with a Zener regulated power supply, an FM detector stage, and an AF preamplifier section. A typical application of the CA3075, in FM receiver circuits, is shown in the block diagram (Fig. 1).

The three-stage, emitter-follower-coupled IF amplifier section provides a 60-dB typ. voltage gain at an operating frequency of 10.7 MHz and features, because of its transistor constant-current sink, an output stage with exceptionally good limiting characteristics.

The FM detector section, which utilizes a differential-peak-detection circuit, requires only a single coil in the associated outboard detector circuit; hence, tuning the detector circuit is a simple procedure.

The audio preamplifier circuit provides a 21-dB voltage gain with low impedance output for driving subsequent audio amplifier stages.

The CA3075 utilizes a 14-lead dual-in-line plastic package with leads in a special quad-formed arrangement.

MAXIMUM RATINGS, *Absolute-Maximum Values at* T_A = 25°C

DC Supply Voltage [between Terminals 5 (V^+) and 3 (V^-)]	12.5	V
DC Current (into Terminal 5)	30	mA
Device Dissipation:		
Up to T_A = 50°C	760	mW
Above T_A = 50°C	derate linearly 7.6	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During soldering for 10 s max.)	+265	°C

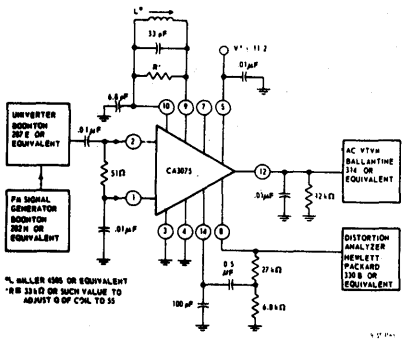


Fig. 2 - Test Circuit for input limiting voltage, recovered AF voltage, and total harmonic distortion

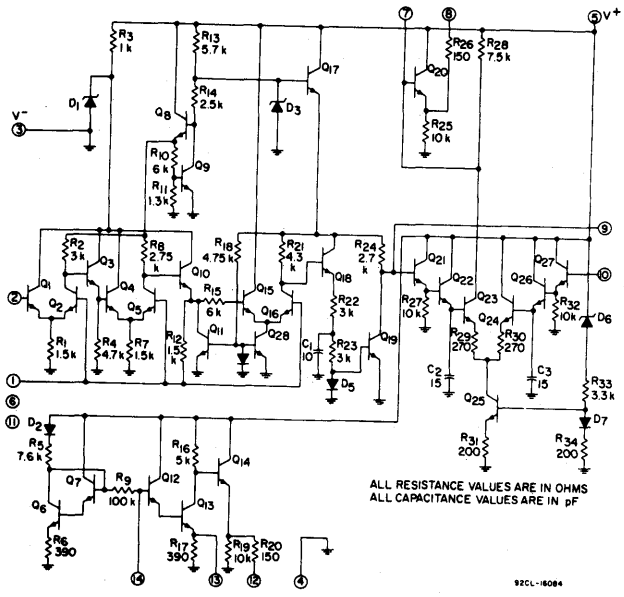


Fig.3 – Schematic diagram of CA3075

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST CIRCUIT FIG. NO.
Static Characteristics							
DC Voltage:							
At Terminal 7	V_7	V^+ = 11.2V	–	6.1	–	V	6
At Terminal 8	V_8		–	5.4	–	V	
At Terminal 12	V_{12}		–	5.2	–	V	
DC Current (into Terminal 5):							
At V^+ = 8.5 V	I_5	–	8.5	15	–	mA	6
At V^+ = 11.2 V			–	17.5	–	mA	
At V^+ = 12.5 V			–	19	29	mA	
Dynamic Characteristics at V^+ = 11.2							
IF AMPLIFIER							
Input Limiting Voltage (knee, – 3dB point)	V_I(lim)	f_O = 10.7 MHz, f(Modulation) = 400 Hz, Deviation = ±75 kHz	–	250	600	μV	3
AM Rejection	AMR	f_O = 10.7 MHz, f(Modulation) = 400 Hz, FM: Deviation = ±75 kHz, AM: Modulation = 30%	–	55	–	dB	5
Input Impedance Components:							
Parallel Resistance	R_I	f_O = 10.7 MHz	–	4.5	–	kΩ	–
Parallel Capacitance	C_I	V_{IN} = 10 mV RMS	–	4.5	–	pF	
DETECTOR							
Recovered AF Voltage (at Terminal 12)	V_O(AF)	f_O = 10.7 MHz, f(Modulation) = 400 Hz, Deviation = ± 75 kHz	–	1.5	–	V	3
Total Harmonic Distortion	THD		–	1	2	%	
AUDIO PREAMPLIFIER							
Voltage Gain	A(AF)	V_{IN} = 100 mV, f_O = 400 Hz	–	21	–	dB	4
Total Harmonic Distortion	THD	V_{OUT} = 2 V, f_O = 400 Hz	–	1.5	5	%	4

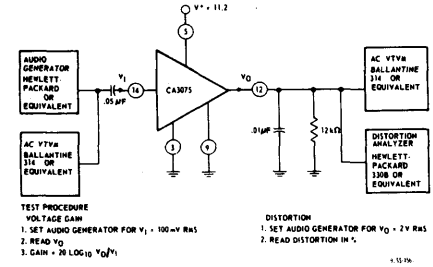


Fig.4 – Test circuit for audio preamplifier voltage gain and total harmonic distortion

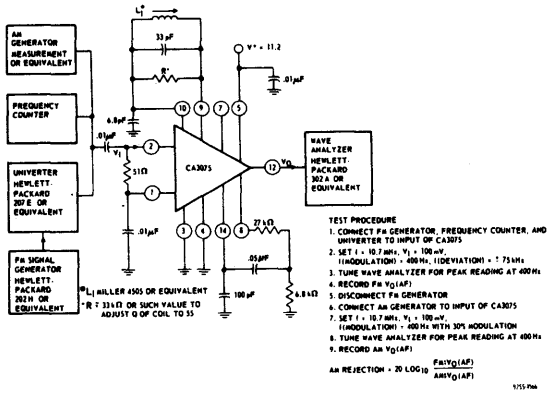


Fig.5 – Test circuit for AM rejection

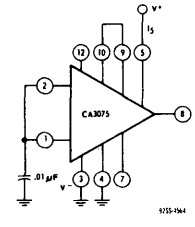


Fig. 6-Test circuit for static characteristics

CA3076

High-Gain Wide-Band IF Amplifier-Limiter

For FM IF Amplifier Applications in Communications Receivers

RCA CA3076, monolithic integrated circuit, is a high-gain wide-band amplifier-limiter for use in the IF sections of Communications and High-Fidelity FM Receivers. The CA3076, shown in the schematic diagram (Fig. 2), consists of a four stage IF amplifier-limiter section with a voltage regulator section. A typical application of the CA3076 in FM receiver circuits is shown in the block diagram (Fig. 1).

The four-stage emitter-follower-coupled IF amplifier section provides an 80-dB voltage gain with a 2-kilohm load at a frequency of 10.7 MHz. The output stage has exceptionally good limiting characteristics because of its transistor constant-current sink. The voltage regulator section provides zener-regulated, decoupled voltages for the IF amplifier.

The CA3076 utilizes an hermetically-sealed 8-lead TO-5 package.

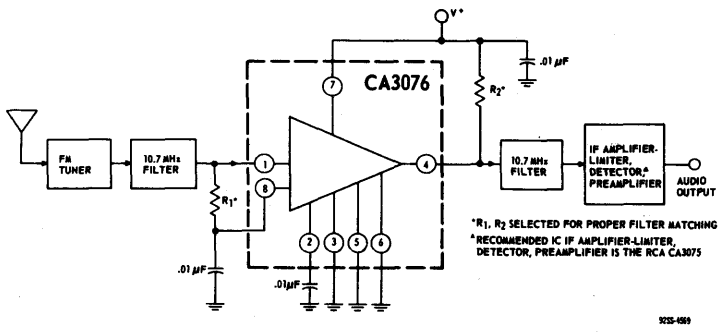


Fig. 1 - Block diagram of typical FM receiver utilizing the CA3076.

Features:

- exceptionally good sensitivity: input limiting voltage (knee) = 50 μV typ. at 10.7 MHz
- high gain: 80 dB with 2-kilohm load
- internal voltage supply regulator
- wide frequency capability: > 20 MHz

MAXIMUM RATINGS, *Absolute Maximum-Values at* $T_A = 25°C$

DC Supply Voltage [between Terminals 7 (V^+) and 3 (V^-)]	15	V
DC Current (into Terminal 7)	35	mA
Device Dissipation:		
Up to $T_A = 50°C$	500	mW
Above $T_A = 50°C$	derate linearly 5 mW/°C	
Ambient Temperature Range:		
Operating	- 55 to + 125	°C
Storage	- 65 to + 150	°C
Lead Temperature (During Soldering):		
At distance 1/32 in (3.17 mm) from seating plane for 10 s max	+ 265	°C

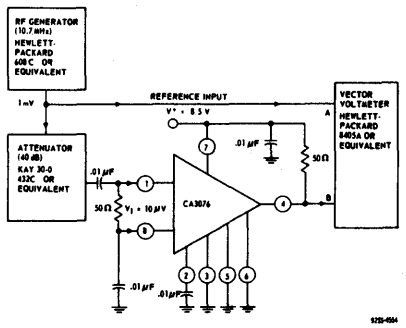


Fig. 2 - Forward transfer admittance (Y_{21}) test circuit

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Static Characteristics – $V^+ = 8.5$ V						
DC Current (into Term. 7)	I_7	–	10	15	24	mA
Quiescent Operating Current (into Term. 4)	I_4	–	–	0.65	–	mA
Dynamic Characteristics – $V^+ = 8.5$ V, $f_O = 10.7$ MHz						
Input Limiting Voltage (knee, - 3 dB point)	V_I (lim.)	–	–	50	200	μV
Output Voltage	V_O	$V_I = 20\mu V$	4	12	–	mV
Output Noise Voltage	V_N	$V_I = 0$	–	1	–	mV
Forward Transfer Admittance: Magnitude	$\lvert Y_{21}\rvert$	$V_I =$	–	6	–	mho
Phase	θ_{21}	$10\mu V$	–	80	–	degrees
Reverse Transfer Admittance: Magnitude	$\lvert Y_{12}\rvert$	–	–	0.1	–	μmho
Phase	θ_{12}		–	- 90	–	degrees
Input-Impedance Components: Parallel Resistance	R_I	–	–	7.5	–	kΩ
Parallel Capacitance	C_I		–	4	–	pF
Output-Impedance Components: Parallel Resistance	R_O	–	50	–	–	kΩ
Parallel Capacitance	C_O		–	1.7	–	pF

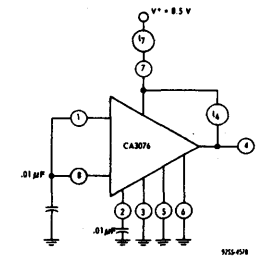
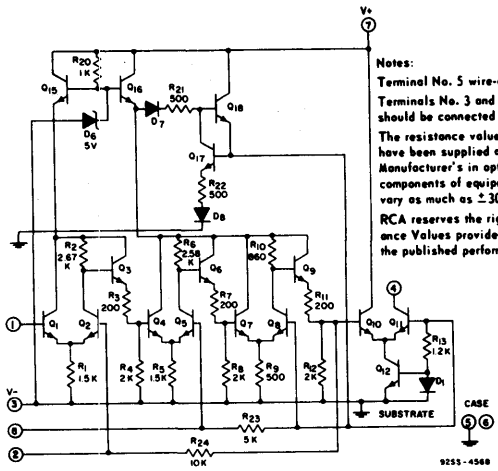


Fig. 3 - Test circuit for DC current (Terminal 7) and operating current (Terminal 4).



Notes:

Terminal No. 5 wire-connected to the case.

Terminals No. 3 and 6 which are connected to the substrate should be connected to the most negative point in the circuit.

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturer's in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ± 30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

Fig. 4 - Schematic diagram of CA3076.

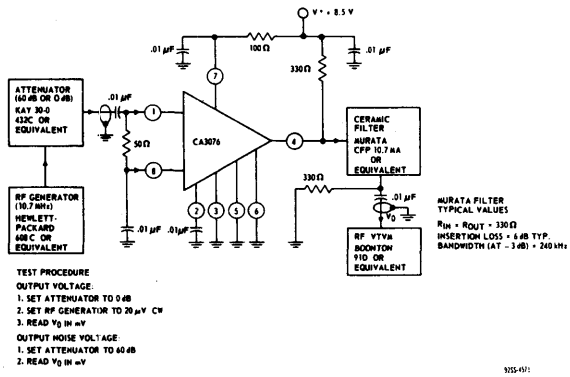


Fig. 5 - 10.7 MHz voltage gain and noise test circuit

CA3078, CA3078A Types
Micropower Operational Amplifier

The RCA CA3078T and CA3078AT are high-gain monolithic operational amplifiers which can deliver milliamperes of current yet only consume microwatts of standby power. Their operating points are externally adjustable and frequency compensation may be accomplished with one external capacitor. The CA3078T and CA3078AT provide the designer with the opportunity to tailor the frequency response and improve the slew rate without sacrificing power. Operation with a single 1.5-volt battery is a practical reality with these devices.

The CA3078AT is a premium device having a supply voltage range of $V^{\pm}$ = 0.75V to $V^{\pm}$ = 15V and an operating temperature range of -55°C to +125°C. The CA3078T has the same lower supply voltage limit but the upper limit is V^+ = +6V and V^- = -6V. The operating temperature range is from 0°C to +70°C.

The CA3078 and CA3078A are supplied in either the standard 8-lead TO-5 package ("T" suffix), or in the 8-lead dual-in-line formed-lead "DIL-CAN" package ("S" suffix).

Features:

- Low standby power: as low as 700 nW
- Wide supply voltage range: ±0.75 to ±15 V
- High peak output current: 6.5 mA min.
- Adjustable quiescent current
- Output short-circuit protection

Applications:

- Portable electronics
- Medical electronics
- Instrumentation
- Telemetry

MAXIMUM RATINGS, *Absolute Maximum Values at* T_A = 25°C

	CA3078AT	CA3078T
DC Supply Voltage (between V^+ and V^- terminal	36V	14V
Differential Input Voltage	±6V	±6V
DC Input Voltage	V^+ to V^-	V^+ to V^-
Input Signal Current	0.1 mA	0.1 mA
Output Short-Circuit Duration*	No Limitation	No Limitation
Device Dissipation	50 mW (up to 125°C)	500 mW (up to 70°C)
Temperature Range:		
Operating	-55 to +125°C	0 to +70°C ▲
Storage	-65 to +150°C	-65 to +150°C
Lead Temperature (During Soldering):		
At distance 1/16 ±1/32 in. (1.59 ±0.79 mm) from case for 10s max.	+300°C	+300°C

*Short circuit may be applied to ground or to either supply.

▲ Types CA3078S and T can be operated over the temperature range of -55 to +125°C, although the published limits for certain electrical specifications apply only over the temperature range of 0 to 70° C.

ELECTRICAL CHARACTERISTICS
For Equipment Design

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			CA3078T LIMITS R_{SET} = 1 MΩ, I_Q = 100 μA					CA3078AT LIMITS R_{SET} = 5.1 MΩ, I_Q = 20 μA					UNITS
		V^+ & V^-	R_S kΩ	R_L kΩ	T_A = 25°C			T_A = 0 to 70°C		T_A = 25°C			T_A = -55 to 125°C		
					MIN	TYP	MAX	MIN	MAX	MIN	TYP	MAX	MIN	MAX	
Input Offset Voltage	V_{IO}	6	≤10		-	1.3	4.5	-	5	-	0.70	3.5	-	4.5	mV
Input Offset Current	I_{IO}				-	6	32	-	40	-	0.50	2.5	-	5.0	nA
Input Bias Current	I_{IB}				-	60	170	-	200	-	7	12	-	50	nA
Open-Loop Diff. Voltage Gain	A_{OL}			≥10	88	92	-	86	-	92	100	-	90	-	dB
Total Quiescent Current	I_Q				-	100	130	-	150	-	20	25	-	45	μA
Device Dissipation	P_D				-	1200	1560	-	1800	-	240	300	-	540	μW
Maximum Output Voltage	V_{OM}			≥10	±5.1	±5.3	-	±5	-	±5.1	±5.3	-	±5	-	V
Common-Mode Input Voltage Range	V_{ICR}		≤10		-	-5.5 to +5.8	-	-5 to +5	-	-	-5.5 to +5.8	-	-5 to +5	-	V
Common-Mode Rejection Ratio	CMRR		≤10		80	110	-	-	-	80	115	-	-	-	dB
Maximum Output Current	I_{OM}^+ or I_{OM}^-					12	-	6.5	30		12	-	6.5	30	mA
Input Offset Voltage Sensitivity Positive	$\Delta V_{IO}/\Delta V^+$		≤10			22	150	-	-		6	150	-	-	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$					22	150		-		6	150	-	-	
										R_{SET} = 13 MΩ, I_Q = 20 μA					
Input Offset Voltage	V_{IO}	15	≤10					-	-		1.4	3.5	-	4.5	mV
Open-Loop Diff. Voltage Gain	A_{OL}		-	≥10	-	-	-	-	-	92	100	-	88	-	dB
Total Quiescent Current	I_Q				-	-	-	-	-		20	30	-	50	μA
Device Dissipation	P_D				-	-	-	-	-		600	750	-	1350	μW
Maximum Output Voltage	V_{OM}			≥10	-	-	-	-	-	±13.7	±14.1	-	±13.5	-	V
Common-Mode Rejection Ratio	CMRR		≤10		-	-	-	-	-	80	106	-	-	-	dB
Input Bias Current	I_{IB}				-	-	-	-	-		7	14	-	55	nA
Input Offset Current	I_{IO}				-	-	-	-	-		0.50	2.7	-	5.5	nA

OPERATING CONSIDERATIONS

Compensation Techniques

The CA3078AT and CA3078T can be phase-compensated with one or two external components depending upon the closed-loop gain, power consumption, and speed desired. The recommended compensation is a resistor in series with a capacitor connected from terminal 1 to terminal 8. Values of the resistor and capacitor required for compensation as a function of closed loop gain are shown in Figs. 24 and 25. These curves represent the compensation necessary at quiescent currents of 20 μA and 100 μA, respectively, for a transient response with 10% overshoot. Figs. 21 and 22 show the slew rates that can be obtained with the two different compensation techniques. Higher speeds can be achieved with input compensation, but this increases noise output.

Compensation can also be accomplished with a single capacitor connected from terminal 1 to terminal 8, with speed being sacrificed for simplicity. Table 1 gives an indication of slew rates that can be obtained with various compensation techniques at quiescent currents of 20 μA and 100 μA.

Single Supply Operation

The CA3078AT and CA3078T can operate from a single supply with a minimum total supply voltage of 1.5 volts. Figs. 27 and 28 show the CA3078AT or CA3078T in inverting and non-inverting 20-dB amplifier configurations utilizing a 1.5-volt type "AA" cell for a supply. The total power consumption for either circuit is approximately 675 nanowatts. The output voltage swing in this configuration is 300 mV p-p with a 20 kΩ load.

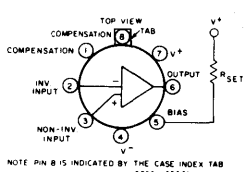


Fig. 1 – Functional diagram of the CA3078T and CA3078AT.

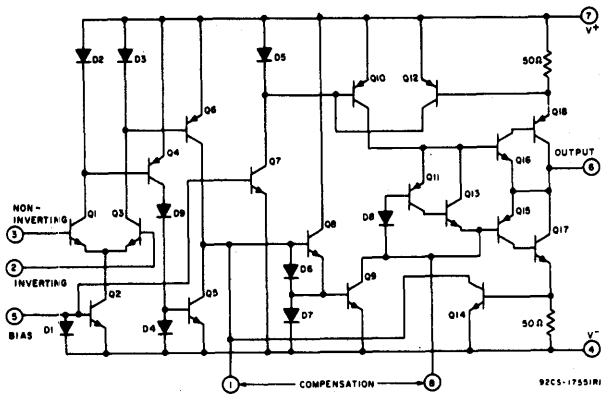


Fig.2-Schematic diagram of the CA3078T and CA3078AT.

Typical Values Intended Only for Design Guidance at T_A = 25°C and V^+ = +6 V, V^- = 6 V

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	CA3078AT		CA3078T	UNITS
			R_{SET} = 5.1 MΩ, I_Q = 20 μA	R_{SET} = 1 MΩ, I_Q = 100 μA	R_{SET} = 1 MΩ, I_Q = 100 μA	
Input Offset Voltage Drift	$\Delta V_{IO}/\Delta T_A$	R_S ≤10 KΩ	5	6	6	μV/°C
Input Offset Current Drift	$\Delta V_{IO}/\Delta T_A$	R_S ≤10 KΩ	0.3	70	70	pA/°C
Open-Loop Bandwidth	BW_{OL}	3dB pt	0.3	2	2	kHz
Slew Rate:						
Unity Gain	SR	See Figs.	0.027	0.04	0.04	V/μs
Comparator		20, 21	0.5	1.5	1.5	
Transient Response		10% to 90% Rise Time	3	2.5	2.5	μs
Input Resistance	R_I		7.4	1.7	0.87	MΩ
Output Resistance	R_O		1	0.8	0.8	KΩ
Equiv. Input Noise Voltage	e_N(10Hz)	R_S 0	40	–	25	nV/√Hz
Equiv. Input Noise Current	i_N(10Hz)	R_S 1 MΩ	0.25	–	1	pA/√Hz

ELECTRICAL CHARACTERISTICS, at T_A = 25°C
Typical Values Intended Only for Design Guidance

CHARACTERISTICS SYMBOLS	TYPICAL VALUES				UNITS
	CA3078AT		CA3078T		
	V^+ = +1.3V, V^-=-1.3V, R_{SET} = 2 MΩ, I_Q = 10 μA	V^+ = +0.75V, V^-=-0.75V, R_{SET} = 10 MΩ, I_Q = 1 μA	V^+ = +1.3V, V^-=-1.3V, R_{SET} = 2 MΩ, I_Q = 10 μA	V^+ = 0.75V, V^-=-0.75V, R_{SET} = 10 MΩ, I_Q = 1 μA	
V_{IO}	0.7	0.9	1.3	1.5	mV
I_{IO}	0.3	0.054	1.7	0.5	nA
I_{IB}	3.7	0.45	9	1.3	nA
A_{OL}	84	65	80	60	dB
I_Q	10	1	10	1	μA
P_D	26	1.5	26	1.5	μW
V_{OPP}	1.4	0.3	1.4	0.3	V
V_{ICR}	−0.8 to +1.1	−0.2 to +0.5	−0.8 to +1.1	−0.2 to +0.5	V
CMRR	100	90	100	90	dB
$I_{OM}^{\pm}$	12	0.5	12	0.5	mA
$\Delta V_{IO}/\Delta V^{\pm}$	20	50	20	50	μV/V

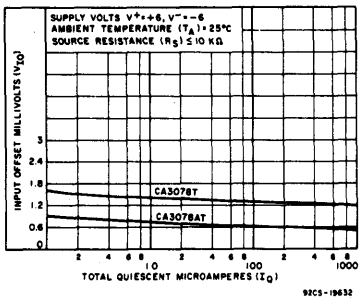


Fig.3 – Input offset voltage vs. total quiescent current.

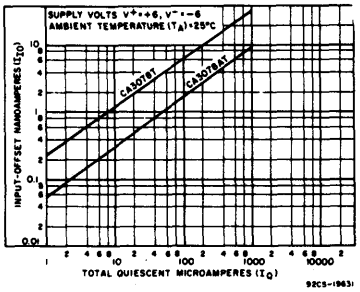


Fig.4 – Input offset current vs. total quiescent current.

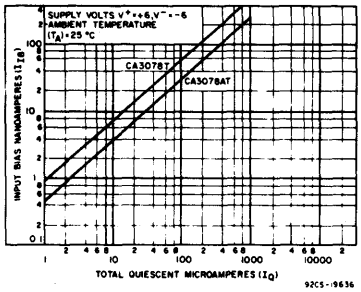


Fig.5 – Input bias current vs. total quiescent current.

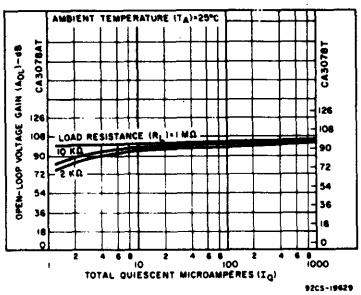


Fig.6 – Open-loop voltage gain vs. total quiescent current.

CA3078, CA3078A Types

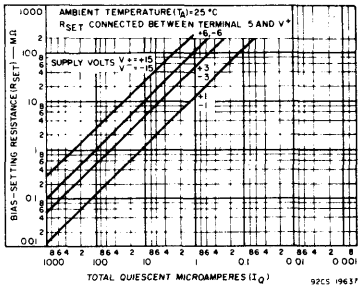


Fig. 7 – Bias-setting resistance vs. total quiescent current.

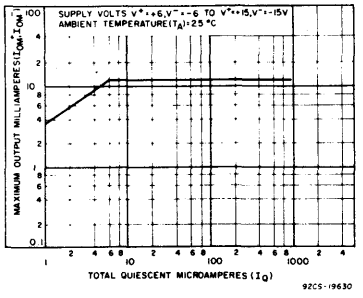


Fig. 8 – Maximum output current vs. total quiescent current.

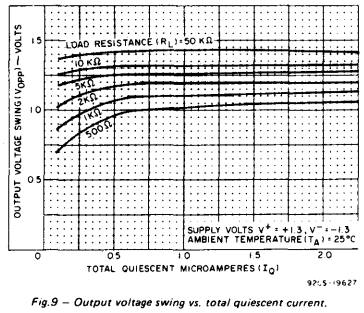


Fig. 9 – Output voltage swing vs. total quiescent current.

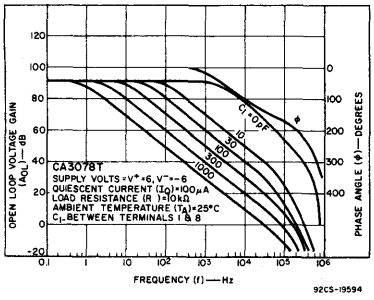


Fig. 10 – Open-loop voltage gain vs. frequency for I_Q = 100 μA – CA3078T.

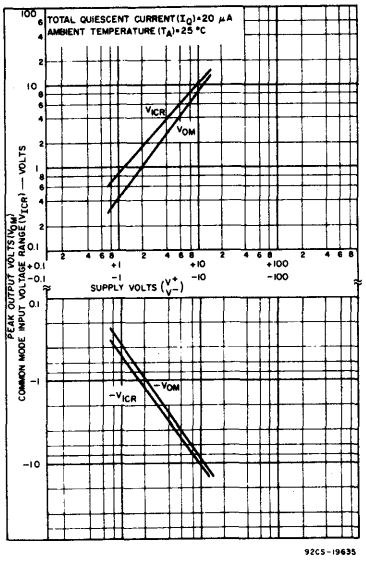


Fig. 11 – Output and common-mode voltage vs. supply voltage.

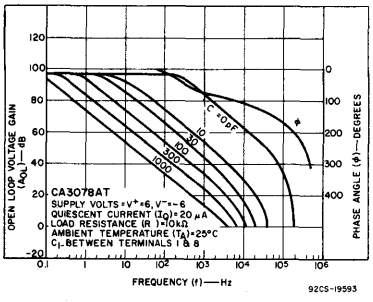


Fig. 12 – Open-loop voltage gain vs. frequency for I_Q = 20 μA – CA3078AT.

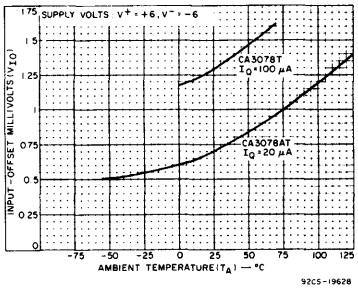


Fig. 13 – Input offset voltage vs. temperature.

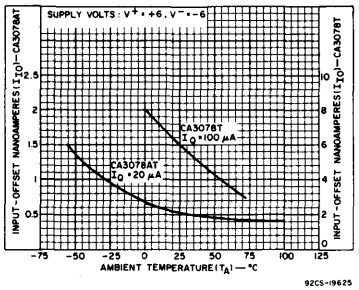


Fig. 14 – Input offset current vs. temperature.

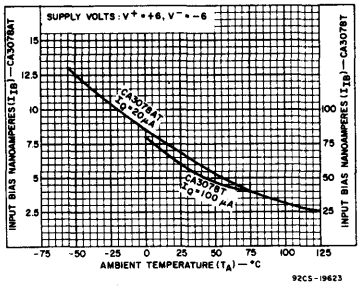


Fig. 15 – Input bias current vs. temperature.

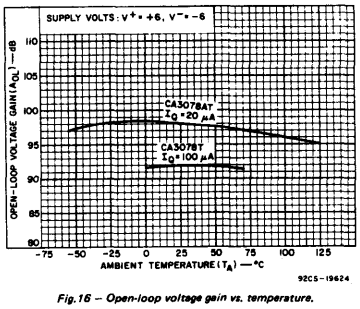


Fig. 16 – Open-loop voltage gain vs. temperature.

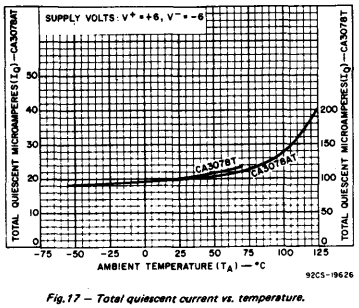


Fig. 17 – Total quiescent current vs. temperature.

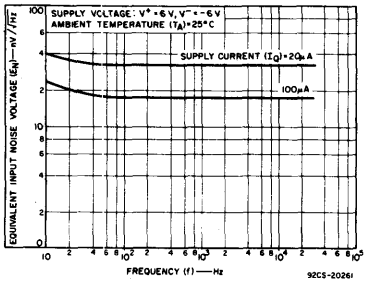


Fig.18 – Equivalent input noise voltage vs. frequency.

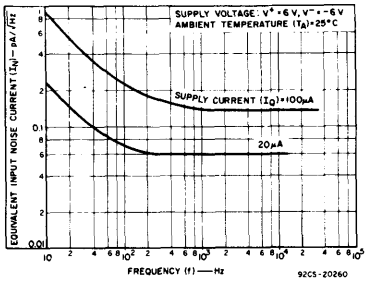


Fig.19 – Equivalent input noise current vs. frequency.

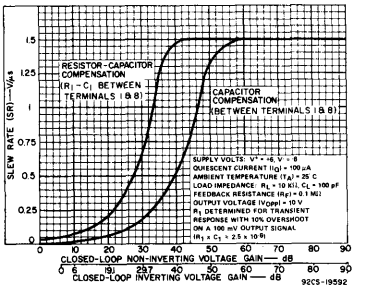


Fig.20 – Slew rate vs. closed-loop gain for I_Q = 100 μA – CA3078T.

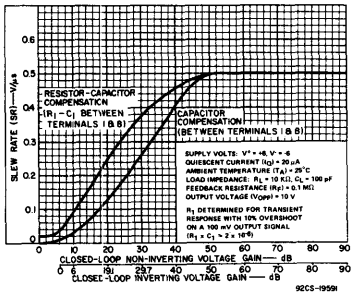


Fig.21 – Slew rate vs. closed-loop gain for I_Q = 20 μA – CA3078AT.

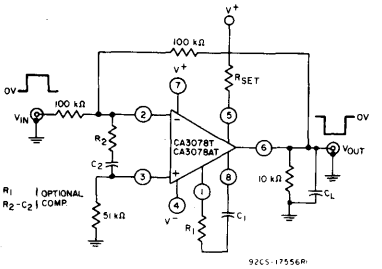


Fig.22 – Transient response and slew-rate, unity gain (inverting) test circuit.

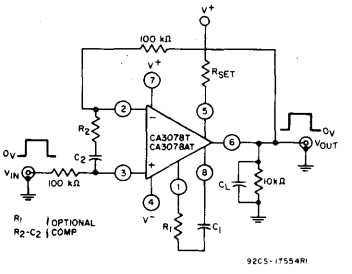


Fig.23 – Slew, rate, unity gain (non-inverting) test circuit.

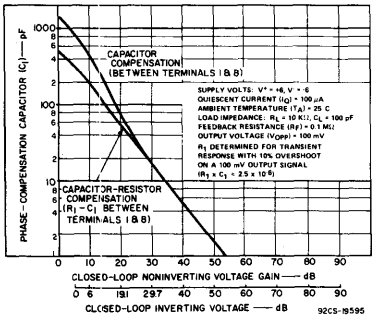


Fig.24 – Phase compensation capacitance vs. closed-loop gain – CA3078T.

Table I - Unity-gain slew rate vs. compensation – CA3078T and CA3078AT

SUPPLY VOLTS: V^+ = 6, V^- = -6
OUTPUT VOLTAGE (V_O) = ±5V
LOAD RESISTANCE (R_L) = 10 kΩ

TRANSIENT RESPONSE: 10% OVERSHOOT FOR AN OUTPUT VOLTAGE of 100 mV
AMBIENT TEMPERATURE (T_A) = 25°C

COMPENSATION TECHNIQUE	UNITY GAIN (INVERTING) Fig. 22					UNITY GAIN (NON-INVERTING) Fig. 23				
	R1	C1	R2	C2	SLEW RATE	R1	C1	R2	C2	SLEW RATE
CA3078T – I_Q = 100 μA	kΩ	pF	kΩ	μF	V/μs	kΩ	pF	kΩ	μF	V/μs
Single Capacitor	0	750	∞	0	0.0085	0	1500	∞	0	0.0095
Resistor & Capacitor	3.5	350	∞	0	0.04	5.3	500	∞	0	0.024
Input	∞	0	0.25	0.306	0.67	∞	0	0.311	0.45	0.67
CA3078AT – I_Q = 20 μA										
Single Capacitor	0	300	∞	0	0.0095	0	800	∞	0	0.003
Resistor & Capacitor	14	100	∞	0	0.027	34	125	∞	0	0.02
Input	∞	0	0.644	0.156	0.29	∞	0	0.77	0.4	0.4

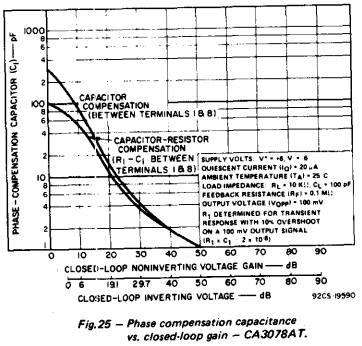


Fig.25 – Phase compensation capacitance vs. closed-loop gain – CA3078AT.

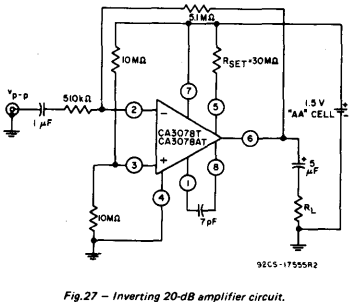


Fig.27 – Inverting 20-dB amplifier circuit.

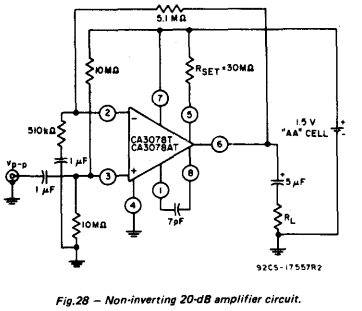


Fig.28 – Non-inverting 20-dB amplifier circuit.

Operational Transconductance Amplifiers (OTA's)

Gatable-Gain Blocks

The RCA-CA3080 and CA3080A types are Gatable-Gain Blocks which utilize the unique operational-transconductance-amplifier(OTA) concept described in Application Note ICAN-6668, "Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers".

The CA3080 and CA3080A types have differential input and a single-ended, push-pull, class A output. In addition, these types have an amplifier bias input which may be used either for gating or for linear gain control. These types also have a high output impedance and their transconductance (g_m) is directly proportional to the amplifier bias current (I_{ABC}).

The CA3080 and CA3080A types are notable for their excellent slew rate (50 V/μs), which makes them especially useful for multiplex and fast unity-gain voltage followers. These types are especially applicable for multiplex applications because power is consumed only when the devices are in the "ON" channel state.

The CA3080A is rated for operation over the full military-temperature range (−55 to +125°C) and its characteristics are specifically controlled for applications such as sample-hold, gain-control, multiplex, etc. Operational transconductance amplifiers are also useful in programmable power-switch applications, e.g., as described in Application Note ICAN-6048, "Some Applications of a Programmable Power Switch/Amplifier" (CA-3094, CA3094A, CA3094B).

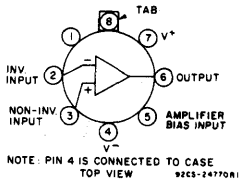
These types are supplied in the 8-lead TO-5 style package (CA3080, CA3080A), and in the 8-lead TO-5 style package with dual-in-line formed leads ("DIL-CAN", CA3080S, CA3080AS). The CA3080 is also supplied in the 8-lead dual-in-line plastic (MINI-DIP") package (CA3080E), and in chip form (CA3080H).

Features:

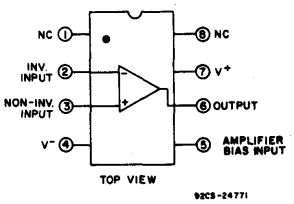
- **Slew rate (unity gain, compensated): 50 V/μs**
- **Adjustable power consumption: 10μW to 30 mW**
- **Flexible supply voltage range: ± 2 V to ± 15 V**
- **Fully adjustable gain: 0 to $g_m R_L$ limit**
- **Tight g_m spread: CA3080 (2:1), CA3080A (1.6:1)**
- **Extended g_m linearity: 3 decades**

Applications:

- **Sample and hold**
- **Multiplex**
- **Voltage follower**
- **Multiplier**
- **Comparator**



TO-5 Style Package



Plastic Package (CA3080E)

Fig.1 – Functional diagrams.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE (Between V+ and V− terminals)	36 V
DIFFERENTIAL INPUT VOLTAGE	±5 V
DC INPUT VOLTAGE	V+ to V−
INPUT SIGNAL CURRENT	1 mA
AMPLIFIER BIAS CURRENT	2 mA
OUTPUT SHORT-CIRCUIT DURATION*	Indefinite
DEVICE DISSIPATION	125 mW
TEMPERATURE RANGE:	
Operating	
CA3080, CA3080E, CA3080S	0 to + 70 °C
CA3080A, CA3080AS	−55 to + 125 °C
Storage	−65 to + 150 °C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 s max.	+ 265 °C

* Short circuit may be applied to ground or to either supply.

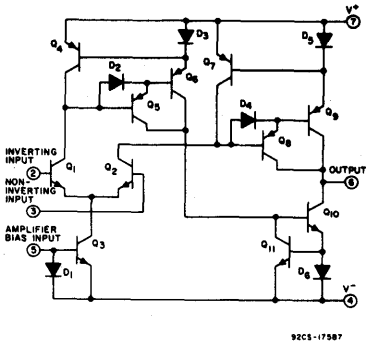


Fig.2 – Schematic diagram for CA3080 and CA3080A.

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS FOR THE CA3080 AND CA3080A

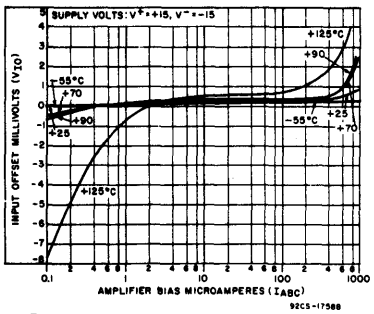


Fig.3 – Input offset voltage as a function of amplifier bias current.

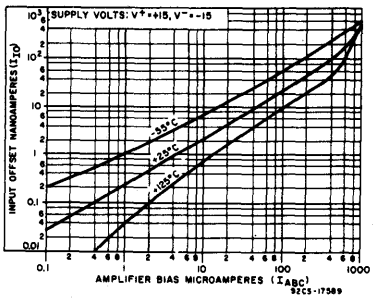


Fig.4 – Input offset current as a function of amplifier bias current.

ELECTRICAL CHARACTERISTICS
For Equipment Design

CHARACTERISTIC		TEST CONDITIONS: $V^+ = 15$ V, $V^- = -15$ V, $I_{ABC} = 500\ \mu A$, $T_A = 25^oC$ (unless indicated otherwise)	CA3080, CA3080E, CA3080S LIMITS Min.	Typ.	Max.	UNITS
Input Offset Voltage	V_{IO}		–	0.4	5	mV
		T_A = 0 to 70°C	–	–	6	
Input Offset Current	I_{IO}		–	0.12	0.6	μA
Input Bias Current	I_I		–	2	5	μA
		T_A = 0 to 70°C	–	–	7	
Forward Transconductance (large signal)	g_m		6700	9600	13000	μmho
		T_A = 0 to 70°C	5400	–	–	
Peak Output Current	$\|I_{OM}\|$	$R_L = 0$	350	500	650	μA
		$R_L = 0$, T_A = 0 to 70°C	300	–	–	
Peak Output Voltage: Positive	V^+_{OM}	$R_L = \infty$	12	13.5	–	V
Negative	V^-_{OM}		−12	−14.4	–	
Amplifier Supply Current	I_A		0.8	1	1.2	mA
Device Dissipation	P_D		24	30	36	mW
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V^+$		–	–	150	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$		–	–	150	
Common-Mode Rejection Ratio	CMRR		80	110	–	dB
Common-Mode Input-Voltage Range	V_{ICR}		12 to −12	13.6 to −14.6	–	V
Input Resistance	R_I		10	26	–	kΩ

ELECTRICAL CHARACTERISTICS
Typical Values Intended Only for Design Guidance

			CA3080, CA3080E, CA3080S	
Input Offset Voltage	V_{IO}	$I_{ABC} = 5\ \mu A$	0.3	mV
Input Offset Voltage Change	$\|\Delta V_{IO}\|$	$I_{ABC} = 500\ \mu A$ to $I_{ABC} = 5\ \mu A$	0.2	mV
Peak Output Current	I_{OM}	$I_{ABC} = 5\ \mu A$	5	μA
Peak Output Voltage: Positive	V^+_{OM}	$I_{ABC} = 5\ \mu A$	13.8	V
Negative	V^-_{OM}		−14.5	
Magnitude of Leakage Current		$I_{ABC} = 0$, $V_{TP} = 0$	0.08	nA
		$I_{ABC} = 0$, $V_{TP} = 36$ V	0.3	
Differential Input Current		$I_{ABC} = 0$, $V_{DIFF} = 4$ V	0.008	nA
Amplifier Bias Voltage	V_{ABC}		0.71	V
Slew Rate: Maximum (uncompensated)	SR		75	V/μs
Unity Gain (compensated)			50	
Open-Loop Bandwidth	BW_{OL}		2	MHz
Input Capacitance	C_I	f = 1 MHz	3.6	pF
Output Capacitance	C_O	f = 1 MHz	5.6	pF
Output Resistance	R_O		15	MΩ
Input-to-Output Capacitance	C_{I-O}	f = 1 MHz	0.024	pF

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS (Cont'd)

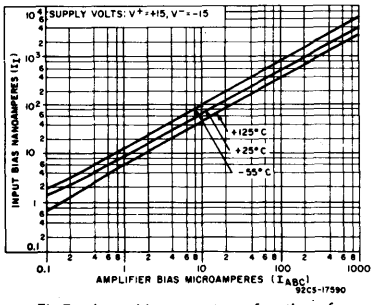


Fig.5 – Input bias current as a function of amplifier bias current.

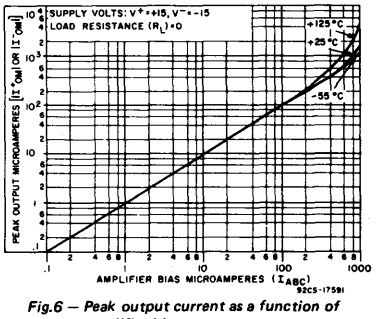


Fig.6 – Peak output current as a function of amplifier bias current.

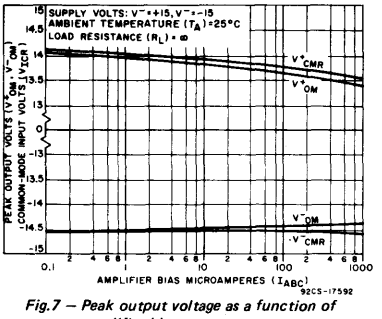


Fig.7 – Peak output voltage as a function of amplifier bias current.

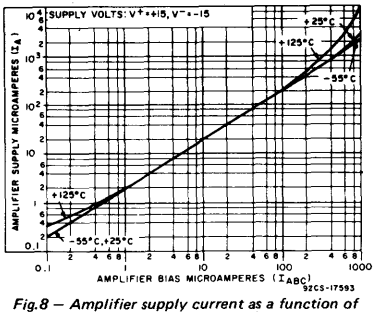


Fig.8 – Amplifier supply current as a function of amplifier bias current.

ELECTRICAL CHARACTERISTICS
For Equipment Design

CHARACTERISTIC		TEST CONDITIONS: $V^+ = 15$ V, $V^- = -15$ V, $I_{ABC} = 500\ \mu A$, $T_A = 25^oC$ (unless indicated otherwise)	CA3080A CA3080AS LIMITS Min.	Typ.	Max.	UNITS
Input Offset Voltage	V_{IO}	$I_{ABC} = 5\ \mu A$	–	0.3	2	mV
			–	0.4	2	
		$T_A = -55$ to $+125^oC$	–	–	5	
Input Offset Voltage Change	$\lvert\Delta V_{IO}\rvert$	$I_{ABC} = 500\ \mu A$ to $I_{ABC} = 5\ \mu A$	–	0.1	3	mV
Input Offset Current	I_{IO}		–	0.12	0.6	μA
Input Bias Current	I_I		–	2	5	μA
		$T_A = -55$ to $+125^oC$	–	–	8	
Forward Transconductance (large signal)	g_m		7700	9600	12000	μmho
		$T_A = -55$ to $+125^oC$	4000	–	–	
Peak Output Current	$\lvert I_{OM}\rvert$	$I_{ABC} = 5\ \mu A$, $R_L = 0$	3	5	7	μA
		$R_L = 0$	350	500	650	
		$R_L = 0, T_A = -55$ to $+125^oC$	300	–	–	
Peak Output Voltage:						V
Positive	V^+_{OM}	$I_{ABC} = 5\ \mu A$, $R_L = \infty$	12	13.8	–	
Negative	V^-_{OM}		−12	−14.5	–	
Positive	V^+_{OM}	$R_L = \infty$	12	13.5	–	
Negative	V^-_{OM}		−12	−14.4	–	
Amplifier Supply Current	I_A		0.8	1	1.2	mA
Device Dissipation	P_D		24	30	36	mW
Input Offset Voltage Sensitivity: |						μV/V
Positive	$\Delta V_{IO}/\Delta V^+$		–	–	150	
Negative	$\Delta V_{IO}/\Delta V^-$		–	–	150	
Magnitude of Leakage Current		$I_{ABC} = 0, V_{TP} = 0$	–	0.08	5	nA
		$I_{ABC} = 0, V_{TP} = 36$ V	–	0.3	5	
Differential Input Current		$I_{ABC} = 0, V_{DIFF} = 4$ V	–	0.008	5	nA
Common-Mode Rejection Ratio	CMRR		80	110	–	dB
Common-Mode Input-Voltage Range	V_{ICR}		12 to −12	13.6 to −14.6	–	V
Input Resistance	R_I		10	26	–	kΩ

ELECTRICAL CHARACTERISTICS
Typical Values Intended Only for Design Guidance

			CA3080A CA3080AS	
Amplifier Bias Voltage	V_{ABC}		0.71	V
Slew Rate:				
Maximum (uncompensated)	SR		75	V/μs
Unity Gain (compensated)			50	
Open-Loop Bandwidth	BW_{OL}	–	2	MHz
Input Capacitance	C_I	f = 1 MHz	3.6	pF
Output Capacitance	C_O	f = 1 MHz	5.6	pF
Output Resistance	R_O		15	MΩ
Input-to-Output Capacitance	C_{I-O}	f = 1 MHz	0.024	pF
Input Offset Voltage Temperature Drift	$\Delta V_{IO}/\Delta T$	$I_{ABC} = 100\ \mu A$, $T_A = -55$ to $+125^oC$	3	μV/°C

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS (Cont'd)

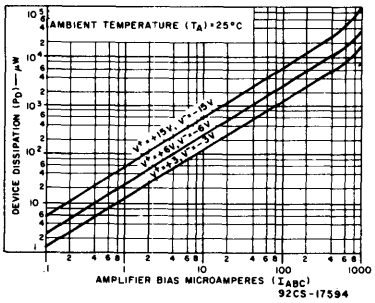


Fig.9 – Total power dissipation as a function of amplifier bias current.

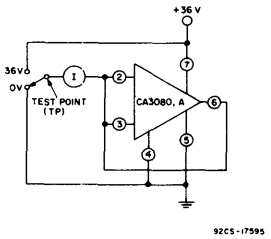


Fig.10 – Leakage current test circuit.

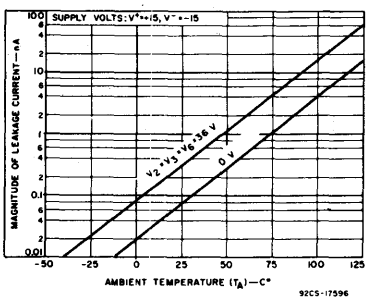


Fig.11 – Leakage current as a function of temperature.

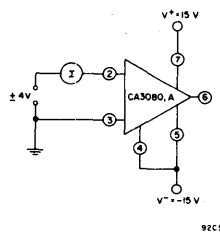


Fig.12 – Differential input current test circuit.

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS (Cont'd)

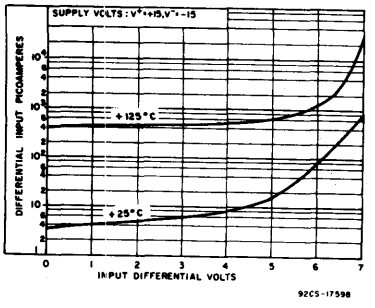


Fig. 13 – Input current as a function of input differential voltage.

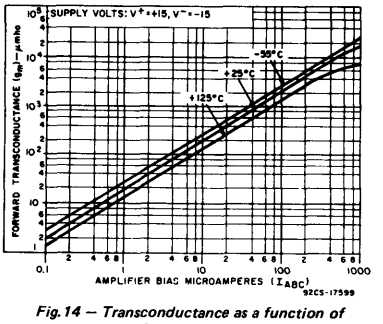


Fig. 14 – Transconductance as a function of amplifier bias current.

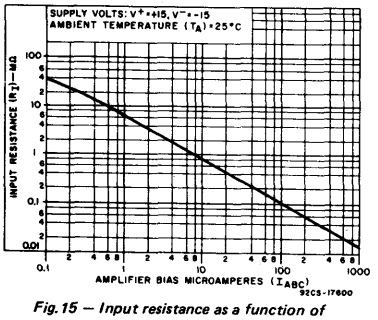


Fig. 15 – Input resistance as a function of amplifier bias current.

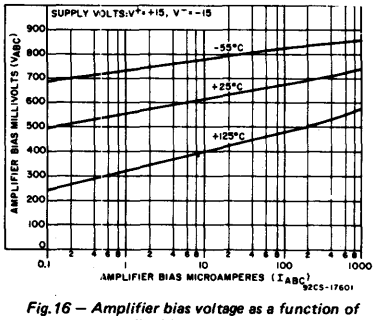


Fig. 16 – Amplifier bias voltage as a function of amplifier bias current.

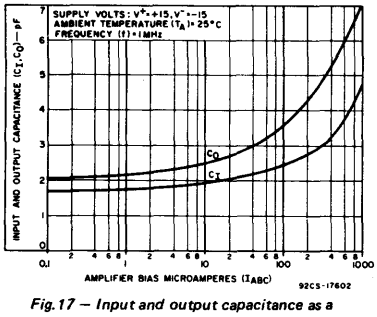


Fig. 17 – Input and output capacitance as a function of amplifier bias current.

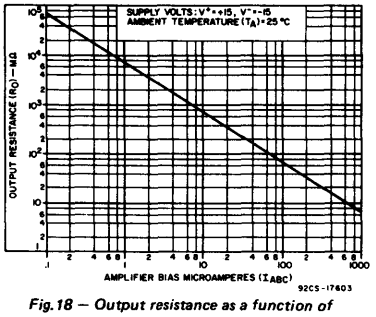


Fig. 18 – Output resistance as a function of amplifier bias current.

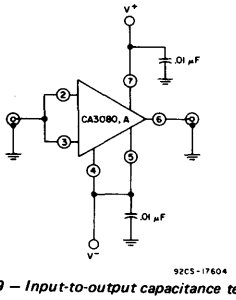


Fig. 19 – Input-to-output capacitance test circuit.

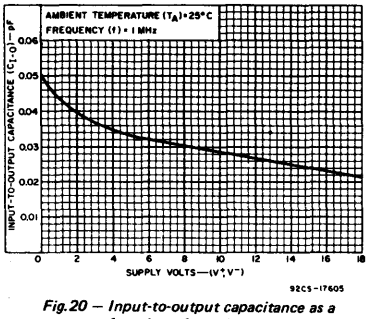


Fig. 20 – Input-to-output capacitance as a function of supply voltage.

APPLICATIONS

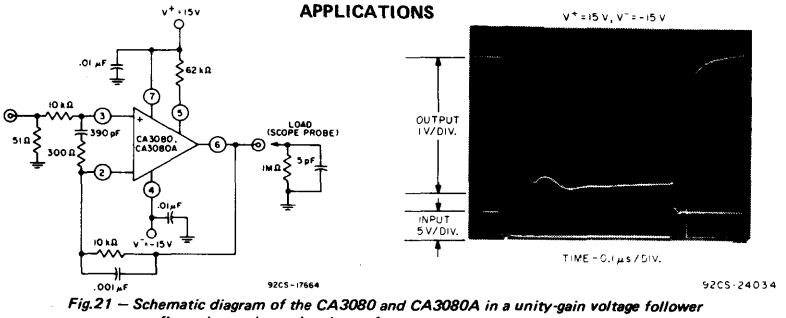
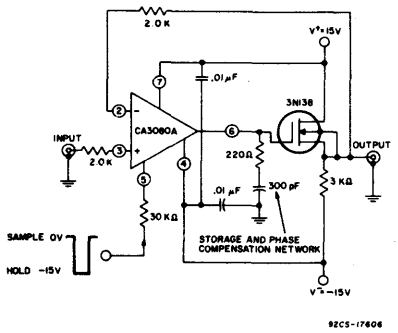


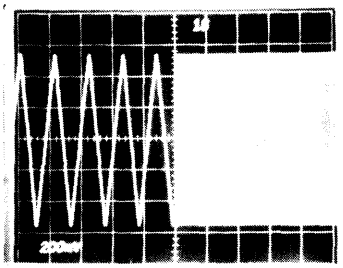
Fig. 21 – Schematic diagram of the CA3080 and CA3080A in a unity-gain voltage follower configuration and associated waveform.



SLEW RATE(IN SAMPLE MODE)=1.3 V/μs
ACQUISITION TIME*=3 μs

* TIME REQUIRED FOR OUTPUT TO SETTLE WITHIN ±3mV OF A 4-VOLT STEP

Fig.22 – Schematic diagram of the CA3080A in a sample-hold configuration.



(a) – Two-tone output signal from the function generator. A square-wave signal modulates the external sweeping input to produce 1 Hz and 1 MHz, showing the 1,000,000/1 frequency range of the function generator.

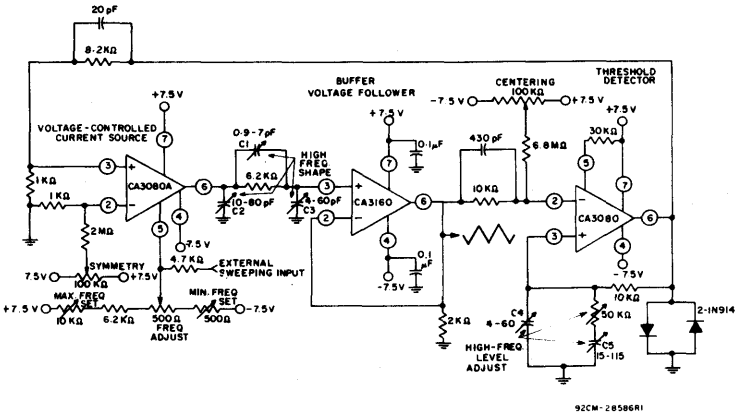
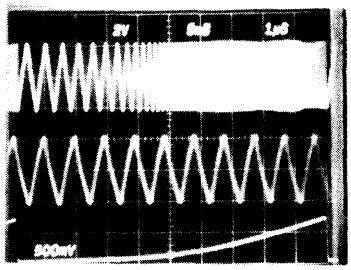


Fig.23 – 1,000,000/1 single-control function generator – 1 MHz to 1 Hz.



(b) – Triple-trace of the function generator sweeping to 1 MHz. The bottom trace is the sweeping signal and the top trace is the actual generator output. The center trace displays the 1 MHz signal via delayed oscilloscope triggering of the upper swept output signal.

Fig.24 – Function generator dynamic characteristics waveforms.

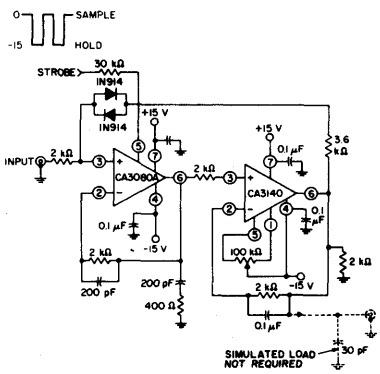
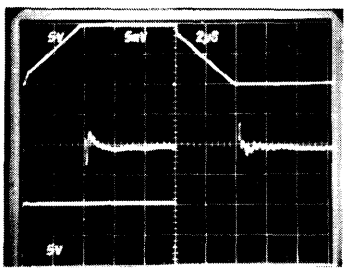
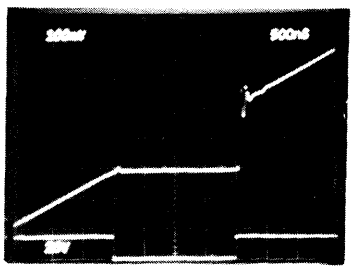


Fig.25 – Sample- and hold circuit.



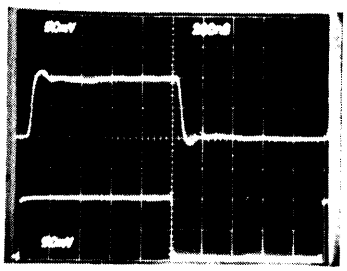
LARGE-SIGNAL RESPONSE AND SETTLING TIME
TOP TRACE: OUTPUT SIGNAL (5 V/DIV. AND 2 μs/DIV.)
BOTTOM TRACE: INPUT SIGNAL (5V/DIV. AND 2 μs/DIV.)
CENTER TRACE: DIFFERENCE OF INPUT AND OUTPUT SIGNALS THROUGH TEKTRONIX AMPLIFIER 7A13 (5mV/DIV. AND 2 μs/DIV.)
92CS-27884

Fig.26 – Large-signal response and settling time for circuit shown in Fig.25.



SAMPLING RESPONSE
TOP TRACE: SYSTEM OUTPUT (100 mV/DIV. AND 500 ns/DIV.)
BOTTOM TRACE: SAMPLING SIGNAL (20 V/DIV. AND 500 ns/DIV.)
92CS-27885

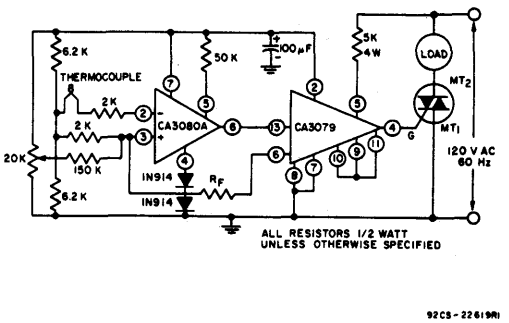
Fig.27 – Sampling response for circuit shown in Fig. 25.



TOP TRACE: OUTPUT
(50 mV/DIV. AND 200ns/DIV.)
BOTTOM TRACE: INPUT
(50 mV/DIV. AND 200 ns/DIV.)

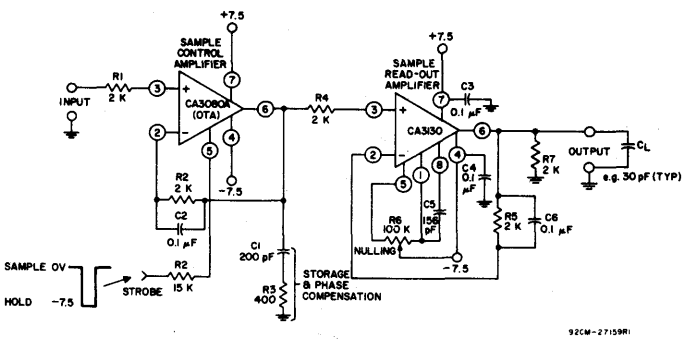
92CS-27883

Fig.28 – Input and output response for circuit shown in Fig. 25.



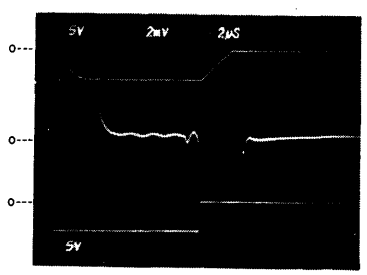
92CS-22619RI

Fig.29 – Thermocouple temperature control with CA3079 zero voltage switch as the output amplifier.



92CM-27159RI

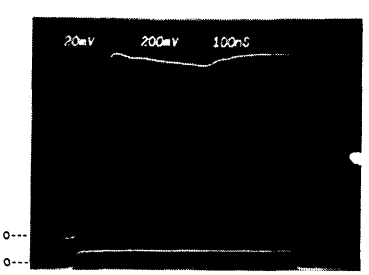
Fig.30 – Schematic diagram of the CA3080A in a sample-hold circuit with BiMos output amplifier.



TOP TRACE: OUTPUT—5 V/DIV. @ 2 μs/DIV.
CENTER TRACE: DIFFERENTIAL COMPARISON OF INPUT & OUTPUT—2 mV/DIV. @ 2 μs/DIV.
BOTTOM TRACE: INPUT—5 V/DIV. @ 2 μs/DIV.

92CS-27161

Fig.31 – Large-signal response for circuit shown in Fig. 30.



TOP TRACE: OUTPUT — 20 mV/DIV. @ 100 ns/DIV.
BOTTOM TRACE: INPUT—200 mV/DIV. @ 100 ns/DIV.

92CS-27160

Fig.32 – Small-signal response for circuit shown in Fig. 30.

CA3081, CA3082 Types

General-Purpose High-Current N-P-N Transistor Arrays

CA3081–Common-Emitter Array CA3082–Common-Collector Array

Directly Drive 7-Segment Incandescent Displays and Light-Emitting-Diode (LED) Displays

Features

- 7 transistors permit a wide range of applications in either a common-emitter (CA3081) or common-collector (CA3082) configuration
- High I_C: 100 mA max. ■ Low $V_{CE\ sat}$ (at 50 mA): 0.4 V typ.

Applications

- Drivers for:
  - Incandescent display devices (e.g. RCA NUMITRON DR2000 Series and lamps)
  - LED (e.g. RCA-SG1002 GaAs High-Efficiency Emitting Diode)
  - Relay control – Thyristor firing

RCA-CA3081* and CA3082* consist of seven high-current (to 100 mA) silicon n-p-n transistors on a common monolithic substrate. The CA3081 is connected in a common-emitter configuration and the CA3082 is connected in a common-collector configuration.

The CA3081 and CA3082 are capable of directly driving seven-segment displays, such as the RCA NUMITRON devices (DR2000 and DR2010), and light-emitting diode (LED) displays. These types are also well-suited for a variety of other driver applications, including relay control and thyristor firing.

The CA3081 and CA3082 are supplied in a 16-lead dual-in-line plastic package, and the CA3081F and CA3082F in a 16-lead dual-in-line frit-seal ceramic package, which includes a separate substrate connection for maximum flexibility in circuit design.

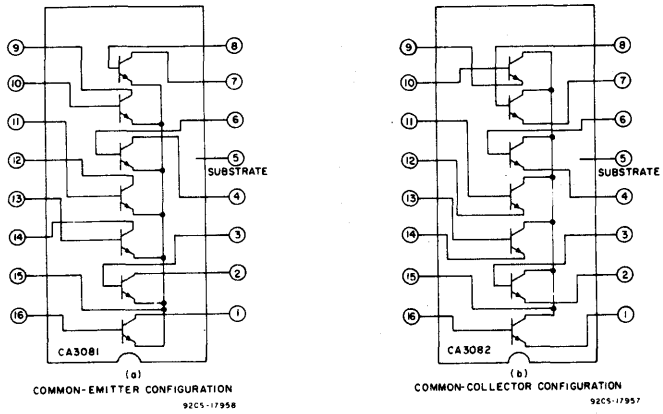


Fig.1–Functional diagrams of types CA3081 and CA3082.

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Power Dissipation:		
Any one transistor	500	mW
Total package	750	mW
Above 55°C	Derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating	–55 to +125	°C
Storage	–65 to + 150	°C
Lead Temperature (During Soldering):		
At distance 1/16" ±1/32" (1.59 mm ±0.79 mm) from case for 10 seconds max.	265	°C
The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage (V_{CEO})	16	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})■	20	V
Emitter-to-Base Voltage (V_{EBO})	5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	20	mA

■ The collector of each transistor of the CA3081 and CA3082 is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (5) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR OF TYPES CA3081 AND CA3082

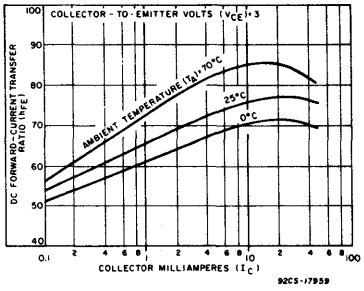


Fig.2–h_{FE} vs. I_C

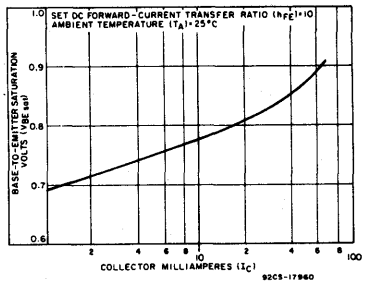


Fig.3–V_{BEsat} vs. I_C

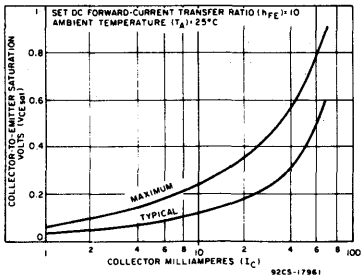


Fig.4–V_{CEsat} vs. I_C at T_A = 25°C.

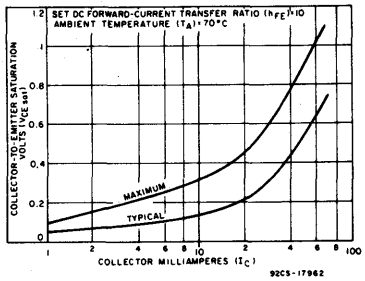


Fig.5–V_{CEsat} vs. I_C at T_A = 70°C.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	Typ. Char. Curve Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
Collector-to-Base Breakdown Voltage	$V_{(BR)CES}$	I_C = 500 μA, I_E = 0	–	20	60	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 500 μA, I_E = 0, I_B = 0	–	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	–	16	24	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_C = 500 μA	–	5	6.9	–	V
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 0.5 V, I_C = 30 mA	–	30	68	–	
		V_{CE} = 0.8 V, I_C = 50 mA	–	40	70	–	
Base-to-Emitter Saturation Voltage	$V_{BE\ sat}$	I_C = 30 mA, I_B = 1 mA	3	–	0.87	1.0	V
Collector-to-Emitter Saturation Voltage: CA3081, CA3082	$V_{CE\ sat}$	I_C = 30 mA, I_B = 1 mA	–	–	0.27	0.5	V
CA3081		I_C = 50 mA, I_B = 5 mA	4	–	0.4	0.7	
CA3082		I_C = 50 mA, I_B = 5 mA	4	–	0.4	0.8	
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10 V, I_B = 0	–	–	–	10	μA
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	–	–	–	1	μA

TYPICAL READ-OUT DRIVER APPLICATIONS

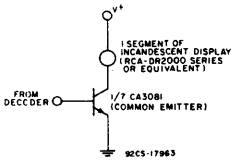


Fig.6—Schematic diagram showing one transistor of the CA3081 driving one segment of an incandescent display.

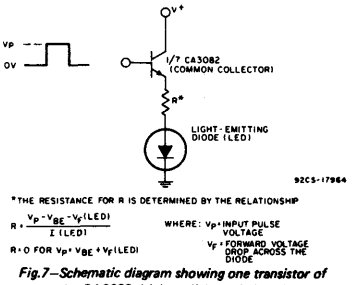


Fig.7—Schematic diagram showing one transistor of the CA3082 driving a light-emitting diode (LED).

CA3083

General-Purpose High-Current N-P-N Transistor Array

RCA-CA3083 is a versatile array of five high-current (to 100mA) n-p-n transistors on a common monolithic substrate. In addition, two of these transistors (Q1 and Q2) are matched at low currents (i.e. 1mA) for applications in which offset parameters are of special importance.

Independent connections for each transistor plus a separate terminal for the substrate permit maximum flexibility in circuit design. The CA3083 is supplied in a 16-lead dual-in-line plastic package, and the CA3083F in a 16-lead dual-in-line frit-seal ceramic package.

Features

- **High I_C: 100mA max.**
- **Low V_{CEsat} (at 50mA): 0.7V max.**
- **Matched pair (Q1 and Q2)–**
  V_{IO} (V_{BE} matched): ± 5 mV max.
  I_{IO} (at 1mA): 2.5 μA max.
- **5 independent transistors plus separate substrate connection**
- **The CA3083 is available in a sealed-junction Beam-Lead version (CA3083L). For further information see File No. 516, "Beam-Lead Devices for Hybrid Circuit Applications".**

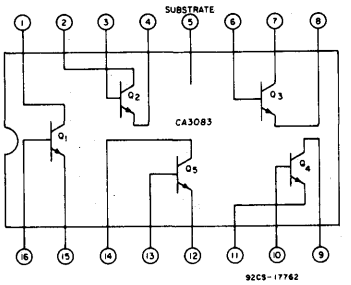


Fig.1–Functional diagram of the CA3083.

Applications

- **Signal processing and switching systems operating from DC to VHF**
- **Lamp and relay driver**
- **Differential amplifier**
- **Temperature-compensated amplifier**
- **Thyristor firing**
- **See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Circuit Transistor Array" for suggested applications**

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Power Dissipation:		
Any one transistor	500	mW
Total package	750	mW
Above 55°C	Derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating	-55 to +125	°C
Storage	65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16" ±1/32" (1.59 mm ±0.79 mm)		
from case for 10 seconds max.	265	°C
The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage (V_{CEO})	15	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})*	20	V
Emitter-to-Base Voltage (V_{EBO})	5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	20	mA

*The collector of each transistor of the CA3083 is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (5) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	Typ. Char. Curve Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
For Each Transistor:							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0	–	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0	–	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0	–	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0	–	5	6.9	–	V
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	–	10	μA
Collector-Cutoff-Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	–	–	1	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 10mA	2	40	76	–	
		V_{CE} = 3V, I_C = 50mA	2	40	75	–	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA	3	0.65	0.74	0.85	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_C = 50mA, I_B = 5mA	4	–	0.40	0.70	V
For Transistors Q1 and Q2 (As a Differential Amplifier):							
Absolute Input Offset Voltage	$\lvert V_{IO}\rvert$	V_{CE} = 3V, I_C = 1mA	7	–	1.2	5	mV
Absolute Input Offset Current	$\lvert I_{IO}\rvert$		8	–	0.7	2.5	μA

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

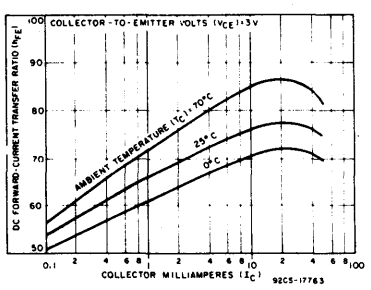


Fig.2 – h_{FE} vs I_C

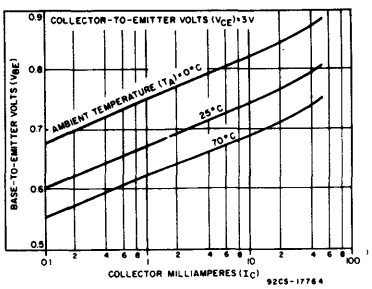


Fig.3 – V_{BE} vs I_C

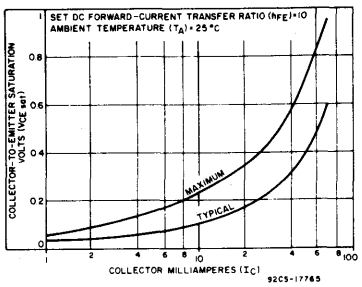


Fig.4 – V_{CEsat} vs I_C at 25°C

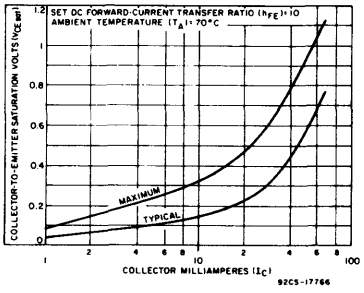


Fig.5 – V_{CEsat} vs I_C at 70°C

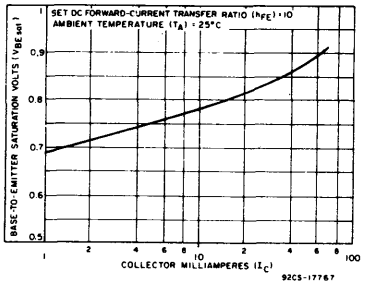


Fig.6 – V_{BEsat} vs I_C

TYPICAL STATIC CHARACTERISTICS FOR DIFFERENTIAL AMPLIFIER

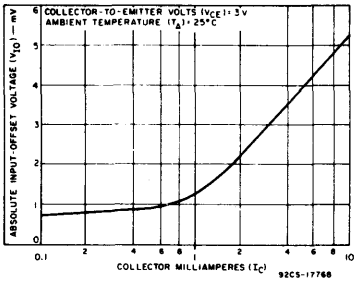


Fig.7 – V_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier).

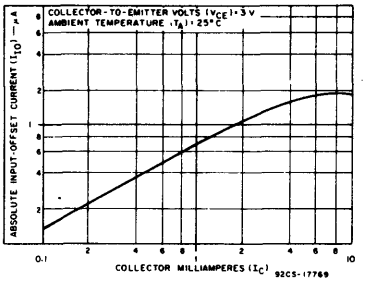


Fig.8 – I_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier).

CA3084

General-Purpose P-N-P Transistor Array

RCA-CA3084 is a general-purpose silicon p-n-p transistor array incorporating two independent transistors, a Darlington circuit, and a current-mirror pair with a shared diode.

The two independent transistors in the array may be used in a variety of circuit applications. The Darlington pair may be employed as the equivalent of a single high-beta transistor. The current-mirror pair is well suited for constant-current applications and can also be used as the active loads in a differential amplifier which uses n-p-n transistors.

The total array is especially useful for a wide range of applications in systems having low-power and low-frequency requirements. Although the transistors may be used as discrete units in conventional circuits, they offer the advantages inherent in integrated-circuit construction, that is, to provide close electrical and thermal matching.

The CA3084 utilizes the 14-lead dual-in-line plastic package.

FEATURES

- **Matched transistor pair (Q1 and Q2)**
  V_{IO} (V_{BE} matched): ± 6mV max.
  I_{IO} (at 100 μA): ± 0.6 μA
- **Wide operating current range**
- **Low noise figure - - 3.2 dB typ. at 1 kHz**
- **The CA3084 is available in a sealed-junction Beam-Lead version (CA3084L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".**

APPLICATIONS

- **General use in signal processing systems having low-power and low-frequency requirements**
- **Differential amplifiers**
- **Temperature compensated amplifiers**
- **Active loads for differential amplifiers using n-p-n transistors**
- **Complementary uses with RCA n-p-n transistor arrays**

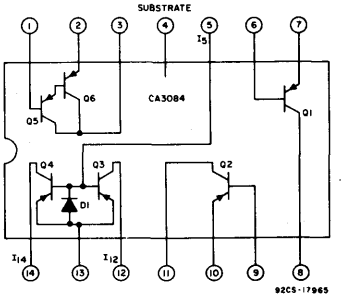


Fig.1 – Functional diagram of the CA3084.

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Dissipation:		
Any one transistor	200	mW*
Total package	750	mW
Above T_A = 55°C	derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering): At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C
The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage (V_{CEO})	−40	V
Collector-to-Base Voltage (V_{CBO})	−40	V
Base-to-Substrate Voltage (V_{BIO})*	40	V
Emitter-to-Base Voltage (V_{EBO})	−40	V
Collector Current (I_C)	−10	mA

*The base of each transistor of the CA3084 is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any base voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (4) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

STATIC CHARACTERISTICS FOR EACH TRANSISTOR

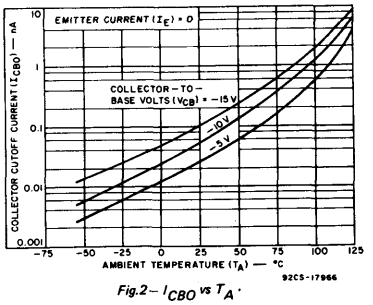


Fig.2 – I_{CBO} vs T_A.

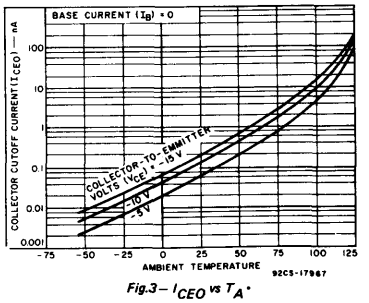


Fig.3 – I_{CEO} vs T_A.

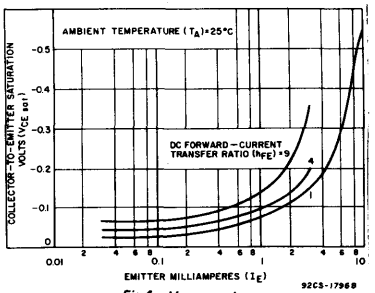


Fig.4 – V_{CEsat} vs I_E.

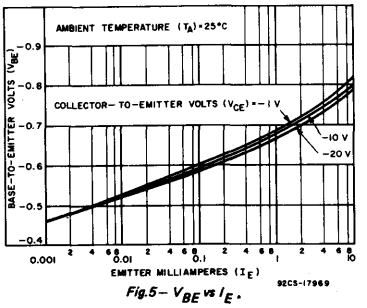


Fig.5 – V_{BE} vs I_E.

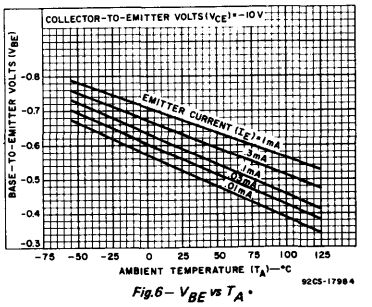


Fig.6 – V_{BE} vs T_A.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	Typ. Characteristics Curve Fig. No.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
For Each Transistor:							
Collector-Cutoff Current	I_{CBO}	V_{CB} = −10V, I_E = 0	2	–	−0.055	−100	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = −10V, I_B = 0	3	–	−0.12	−100	nA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_{CE} = −100μA, I_B = 0	–	−40	−70	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_{CB} = −100μA, I_E = 0	–	−40	−80	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_{EB} = −100μA, I_C = 0	–	−40	−100	–	V
Emitter-to-Substrate Breakdown Voltage	$V_{(BR)EIO}$	I_{EI} = 100μA	–	40	100	–	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_E = 1mA, I_B = 100μA	4	–	−0.125	−0.25	V
Base-to-Emitter Voltage	V_{BE}	I_E = 100μA, V_{CE} = −10V	5	−0.50	−0.59	−0.68	V
DC Forward-Current Transfer Ratio	h_{FE}	I_E = 100μA, V_{CE} = −10V	7	15	40	–	
For Transistors Q1 and Q2 (As a Differential Amplifier):							
Magnitude of Input Offset Voltage	$\lvert V_{IO} \rvert$	I_E = 100μA, V_{CE} = −10V	8	–	0.422	6	mV
Input Offset Current	I_{IO}	I_E = 100μA, V_{CE} = −10V	–	−0.6	0	0.6	μA
For Transistors Q3 and Q4 (Current-Mirror Configuration):							
Collector Current (Normalized)	I_C/I_5	V_{CE} = −5V, V_{CIO} = −5V, Term. 13 = Gnd. I_5 = −100μA,	10	0.85	1.00	1.15	
Magnitude of Collector Current Ratio	$\lvert I_{C(Q3)}/I_{C(Q4)} \rvert$	V_{CE} = −5V, V_{CIO} = −5V, Term. 13 = Gnd. I_5 = −100μA,	11	0.90	1.00	1.10	
For Transistors Q5 and Q6 (Darlington Configuration):							
Collector-Cutoff Current	I_{CEO}	V_{CE} = −10V, I_B = 0	–	–	–	−1.0	μA
Base-to-Emitter Voltage	V_{BE}	I_E = 100μA, V_{CE} = −10V	13	0.92	1.07	1.20	V
DC Forward-Current Transfer Ratio	h_{FE}	I_E = 100μA, V_{CE} = −10V	15	100	1230	–	

ELECTRICAL CHARACTERISTICS at T_A = 25°C
Typical Values Intended Only For Design Guidance

Magnitude of Temperature Coefficient:					
V_{BE} (for each transistor)	$\lvert \Delta V_{BE}/\Delta T \rvert$	I_E = 100μA,	6	−1.78	mV/°C
V_{IO} (as a differential amplifier)	$\lvert \Delta V_{IO}/\Delta T \rvert$	V_{CE} = −10V	9	0.54	μV/°C
V_{BE} (Darlington configuration)	$\lvert \Delta V_{BE}/\Delta T \rvert$		14	−3.7	mV/°C
For Each Transistor:					
Input Resistance	R_I	f = 1kHz, V_{CE} = −10V,	19	9	kΩ
Output Resistance	R_O	I_C = −100μA	20	– 600 –	kΩ
Forward Transconductance	g_m		22	– 3 –	mmho
Collector-to-Base Capacitance	C_{CBO}	I_{CB} = 0	23	– 3.3 –	pF
Collector-to-Emitter Capacitance	C_{CEO}	I_{CE} = 0	23	– 2.5 –	pF
Base-to-Substrate Capacitance	C_{BIO}	I_{CIO} = 0	23	– 4.5 –	pF

STATIC CHARACTERISTICS FOR EACH TRANSISTOR

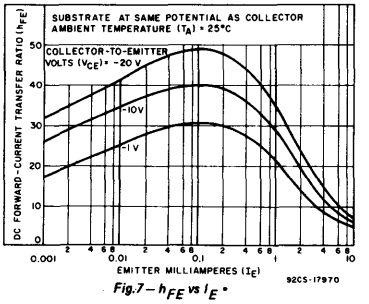


Fig.7 – h_{FE} vs I_E.

STATIC CHARACTERISTICS FOR DIFFERENTIAL AMPLIFIER

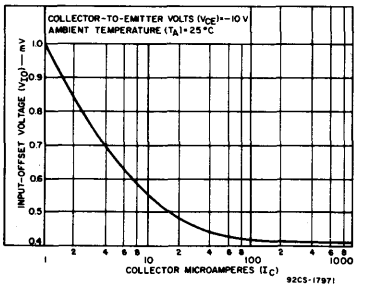


Fig.8–V_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier).

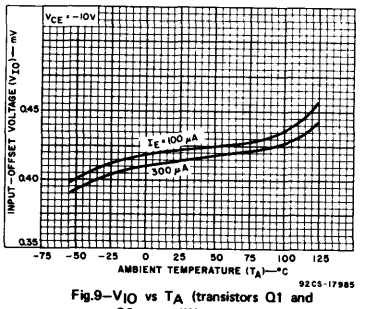


Fig.9–V_{IO} vs T_A (transistors Q1 and Q2 as a differential amplifier).

STATIC CHARACTERISTICS FOR CURRENT-MIRROR CONFIGURATION

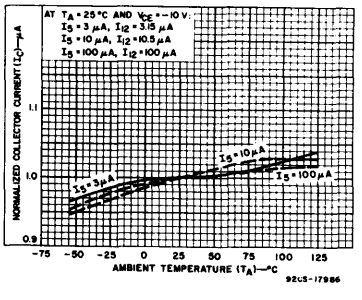


Fig.10–Normalized I_C vs T_A (transistors Q3 and Q4 in a current-mirror configuration.

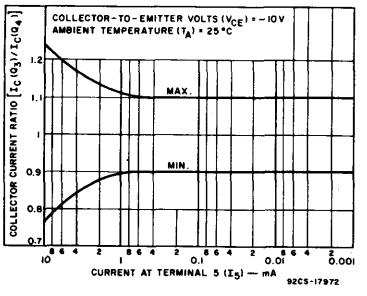


Fig.11–I_C ratio vs I_5 (transistors Q3 and Q4 in a current-mirror configuration.

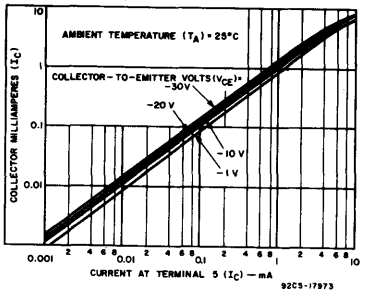


Fig.12 – I_C vs I_5 (transistors Q3 and Q4 in a current-mirror configuration).

STATIC CHARACTERISTICS FOR DARLINGTON CONFIGURATION

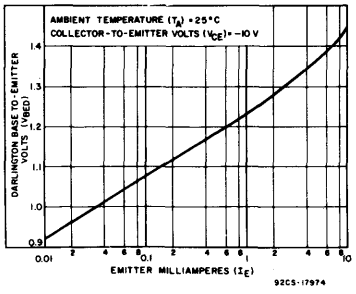


Fig.13 – V_{BE} vs I_E (transistors Q5 and Q6 in a darlingtron configuration).

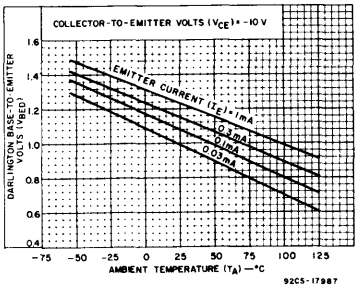


Fig.14 – V_{BE} vs T_A (transistors Q5 and Q6 in a darlington configuration).

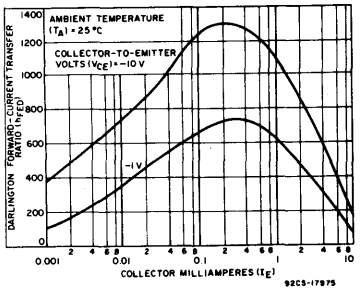


Fig.15 – h_{FE} vs. I_E (transistors Q5 and Q6 in a darlington configuration).

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

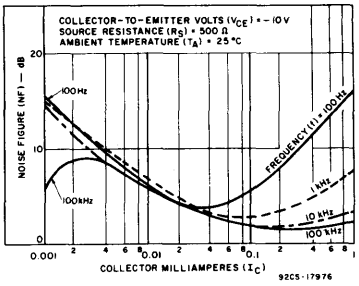


Fig.16 – NF vs I_C at R_S = 500 Ω

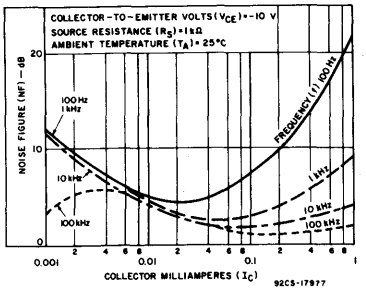


Fig.17 – NF bs I_C at R_S = 1 kΩ

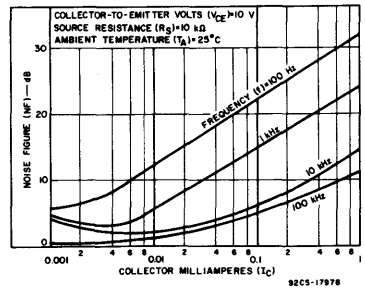


Fig.18 – NF vs I_C at R_S = 10kΩ

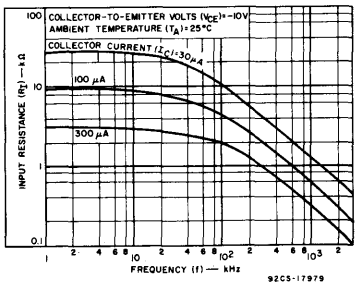


Fig.19 – R_I vs f

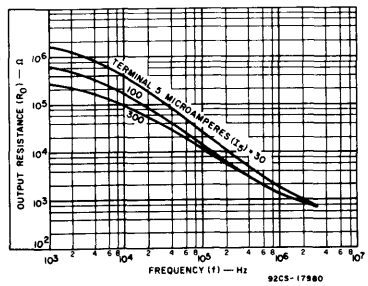


Fig.20 – R_O vs f

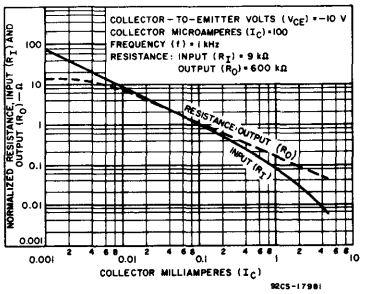


Fig.21 – Normalized R_I and R_O vs I_C

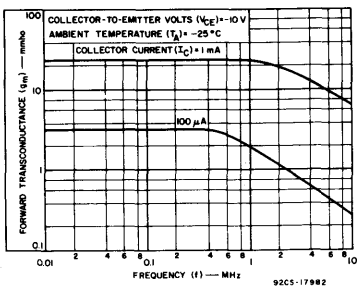


Fig.22 – g_m vs f

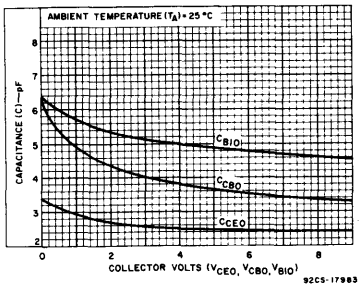


Fig.23 – Transistor capacitances vs collector voltages (V_{CEO}, V_{CBO}, V_{BIO})

Positive Voltage Regulators

For Regulated Voltages from 1.7 V to 46 V at Currents up to 100 mA

RCA-CA3085, CA3085A, and CA3085B are silicon monolithic integrated circuits designed specifically for service as voltage regulators at output voltages ranging from 1.7 to 46 volts at currents up to 100 milliamperes.

A block diagram of the CA3085 Series is shown in Fig. 1. The diagram shows the connecting terminals that provide access to the regulator circuit components. The voltage regulators provide important features such as: frequency compensation, short-circuit protection, temperature-compensated reference voltage, current limiting, and booster input. These devices are useful in a wide range of applications for regulating high-current, switching, shunt, and positive and negative voltages. They are also applicable for current and dual-tracking regulation.

The CA3085A and CA3085B have output current capabilities up to 100 mA and the CA3085 up to 12 mA without the use of external pass transistors. However, all the devices can provide voltage regulation at load currents greater than 100 mA with the use of suitable external pass transistors. The CA3085 Series has an unregulated input voltage ranging from 7.5 to 30 V (CA3085), 7.5 to 40 V (CA3085A), and 7.5 to 50 V (CA3085B) and a minimum regulated output voltage of 26 V (CA3085), 36 V (CA3085A), and 46 V (CA3085B).

The CA3085A is unilaterally interchangeable with the CA3055.

The CA3085 is available in a sealed-junction Beam-Lead version (CA3085L). For further information see File No. 515, "Beam-Lead Devices for Hybrid Circuit Applications".

Type	V_{IN} Range V	V_{OUT} Range V	Max. I_{OUT} mA	Max. Load Regulation % V_{OUT}
CA3085	7.5 to 30	1.8 to 26	12*	0.1
CA3085A	7.5 to 40	1.7 to 36	100	0.15
CA3085B	7.5 to 50	1.7 to 46	100	0.15

* This value may be extended to 100 mA; however, regulation is not specified beyond 12 mA.

These types are supplied in the 8-lead TO-5 style package (CA3085, CA3085A, CA3085B, and the 8-lead TO-5 with dual-in-line formed leads ("DIL-CAN", CA3085S, CA3085AS, CA3085BS). The CA3085 is also supplied in the 8-lead dual-in-line plastic package ("MINI-DIP", CA3085E), and in chip form (CA3085H).

Features

- **Up to 100 mA output current**
- **Input and output short-circuit protection**
- **Load and line regulation: 0.025%**
- **Pin compatible with LM100 Series**
- **Adjustable output voltage**
- Low noise

Applications

- **Shunt voltage regulator**
- **Current regulator**
- **Switching voltage regulator**
- **High-current voltage regulator**
- **Combination positive and negative voltage regulator**
- **Dual tracking regulator**
- See Application Note ICAN-6157 "Applications of the CA3085-Series Monolithic IC Voltage Regulators".

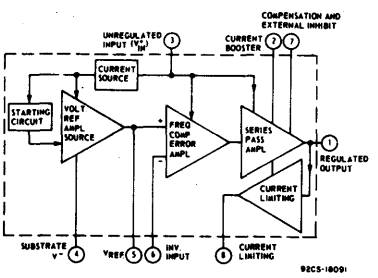


Fig. 1—Block diagram of CA3085 Series.

MAXIMUM RATINGS, ABSOLUTE-MAXIMUM VALUES at T_A = 25°C

POWER DISSIPATION: WITHOUT HEAT SINK	WITH HEAT SINK (TO-5 ONLY)
up to T_A = 55°C ..... 630 mW	up to T_C = 55°C .... 1.6 W
above T_A = 55°C derate linearly @ 6.67 mW/°C	above T_C = 55°C .... derate linearly at 16.7 mW/°C

TEMPERATURE RANGE:

Operating ..... −55 to +125°C
Storage ..... −65 to +150°C

UNREGULATED INPUT VOLTAGE:

CA3085 ..... 30 V
CA3085A ..... 40 V
CA3085B ..... 50 V

LEAD TEMPERATURE (DURING SOLDERING):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

Maximum Voltage Ratings

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical Terminal No. 7 and horizontal Terminal No. 1 is +3 to −10 volts.

MAXIMUM VOLTAGE RATINGS

TERMINAL No.	5	6	7	8	1	2	3	4
5	–	+5 −5	•	•	•	•	•	+10 0
6	–	–	•	•	•	•	•	•
7	–	–	–	+3 −10	+3 −10	•	•	+† 0
8	–	–	–	–	+5 −1	•	•	•
1	–	–	–	–	–	+10 −‡	0 −‡	+‡ 0
2	–	–	–	–	–	–	0 –	+‡ 0
3	–	–	–	–	–	–	–	+‡ 0
4	–	–	–	–	–	–	–	Substrate & Case

* Voltages are not normally applied between these terminals, however, voltages appearing between these terminals are safe, if the specified voltage limits between all other terminals are not exceeded.

‡ 30 V for CA3085
40 V for CA3085A
50 V for CA3085B

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
5	10	1.0
6	1.0	−0.1
7	1.0	−1.0
8	0.1	10
1	20	150
2	150	60
3	150	60
4	–	–

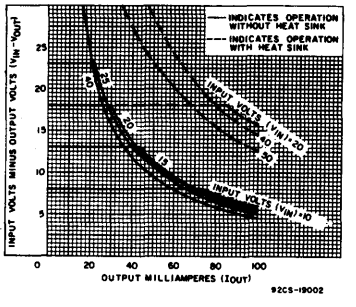


Fig. 3—Dissipation limitation ($V_{IN}-V_{OUT}$ vs. I_{OUT}).

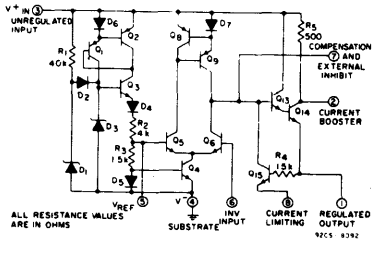


Fig. 2—Schematic diagram of CA3085 Series.

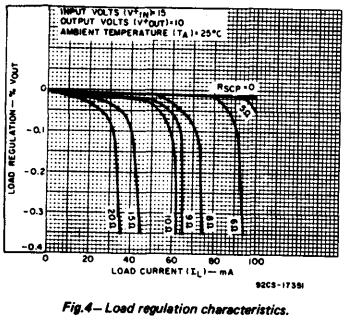


Fig. 4—Load regulation characteristics.

CA3085, CA3085A, CA3085B Types

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOL	TEST CONDITIONS: Test Circuit Fig. No.	TEST CONDITIONS: T_A = 25°C [Unless indicated otherwise]		LIMITS: CA3085 MIN.	CA3085 TYP.	CA3085 MAX.	CA3085A MIN.	CA3085A TYP.	CA3085A MAX.	CA3085B MIN.	CA3085B TYP.	CA3085B MAX.	UNITS
Reference Voltage	V_{REF}	15	V^+_{IN} = 15V		1.4	1.6	1.8	1.5	1.6	1.7	1.5	1.6	1.7	V
Quiescent Regulator Current	$I_{quiescent}$	15	V^+_{IN} = 30V		–	3.3	4.5	–	–	–	–	–	–	mA
			V^+_{IN} = 40V		–	–	–	–	3.65	5	–	–	–	
			V^+_{IN} = 50V		–	–	–	–	–	–	–	4.05	7	
Input Voltage Range	$V_{IN(range)}$	–	–		7.5	–	30	7.5	–	40	7.5	–	50	V
Maximum Output Voltage	$V_{O(max.)}$	15	V^+_{IN} = 30,40,50V#, R_L = 365Ω; Term. No. 6 to Gnd.		26	27	–	36	37	–	46	47	–	V
Minimum Output Voltage	$V_{O(min.)}$	15	V^+_{IN} = 30V		–	1.6	1.8	–	1.6	1.7	–	1.6	1.7	V
Input-Output Voltage Differential	V_{IN}-V_{OUT}	–	–		4	–	28	4	–	38	3.5	–	48	V
Limiting Current	I_{LIM}	16	V^+_{IN} = 16V, V^+_{OUT} = 10V, R_{SCP}* = 6Ω		–	96	120	–	96	120	–	96	120	mA
Load Regulation●	–	–	I_L = 1 to 100mA, R_{SCP} = 0		–	–	–	–	0.025	0.15	–	0.025	0.15	%V_{OUT}
		–	I_L = 1 to 100mA, R_{SCP} = 0, T_A = 0°C to +70°C		–	–	–	–	0.035	0.6	–	0.035	0.6	
		–	I_L = 1 to 12mA, R_{SCP} = 0		–	0.003	0.1	–	–	–	–	–	–	
Line Regulation▲	–	–	I_L = 1mA, R_{SCP} = 0		–	0.025	0.1	–	0.025	0.075	–	0.025	0.04	%/V
		–	I_L = 1mA, R_{SCP} = 0, T_A = 0°C to +70°C		–	0.04	0.15	–	0.04	0.1	–	0.04	0.08	
Equivalent Noise Output Voltage■	V_{NOISE}	12	V^+_{IN} = 25V	C_{REF} = 0	–	0.5	–	–	0.5	–	–	0.5	–	mV p-p
				C_{REF} = 0.22μF	–	0.3	–	–	0.3	–	–	0.3	–	
Ripple Rejection	–	13	V^+_{IN} = 25V, f = 1kHz	C_{REF} = 0	–	50	–	–	50	–	45	50	–	dB
				C_{REF} = 2μF	–	56	–	–	56	–	50	56	–	
Output Resistance	r_o	13	V^+_{IN} = 25V, f = 1kHz		–	0.075	1.1	–	0.075	0.3	–	0.075	0.3	Ω
Temperature Coefficient of Reference and Output Voltages	$\frac{\Delta V_{REF}}{\Delta V_O}$	–	I_L = 0, V_{REF} = 1.6V		–	0.0035	–	–	0.0035	–	–	0.0035	–	%/°C
Load Transient Recovery Time: Turn On	t_{ON}	11	V^+_{IN} = 25V, +50mA Step		–	1	–	–	1	–	–	1	–	μs
Turn Off	t_{OFF}		V^+_{IN} = 25V, –50mA Step		–	3	–	–	3	–	–	3	–	μs
Line Transient Recovery Time: Turn On	t_{ON}	–	V^+_{IN} = 25V, f = 1kHz, 2V Step		–	0.8	–	–	0.8	–	–	0.8	–	μs
Turn Off	t_{OFF}	–			–	0.4	–	–	0.4	–	–	0.4	–	μs

30V (CA3085), 40V(CA3085A), 50V(CA3085B)

* R_{SCP}: Short-circuit protection resistance

■ Bandwidth DC to 10 MHz.

● Load Regulation = $\frac{\Delta V_{OUT}}{V_{OUT(initial)}} \times 100\%$

▲ Line Regulation = $\frac{(\Delta V_{OUT})}{[V_{OUT(initial)}]\ (\Delta V_{IN})} \times 100\%$

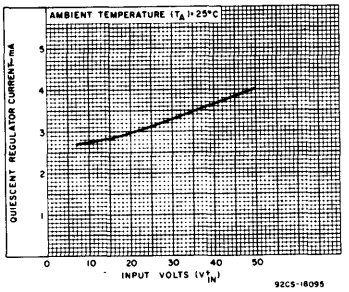


Fig.5–$I_{quiescent}$ vs. V^+_{IN}.

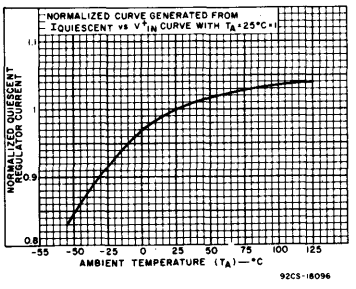


Fig.6–Normalized $I_{quiescent}$ vs. T_A.

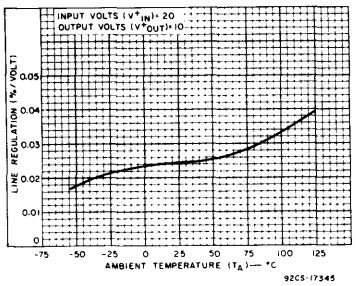


Fig.7–Line regulation temperature characteristics.

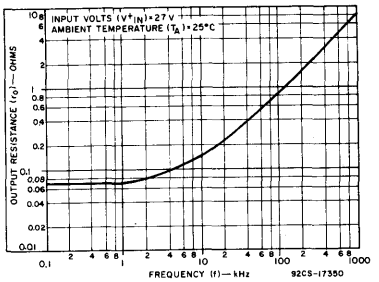


Fig.8–r_O vs. f.

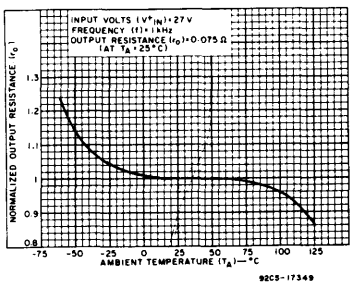


Fig.9–Normalized r_O vs. T_A.

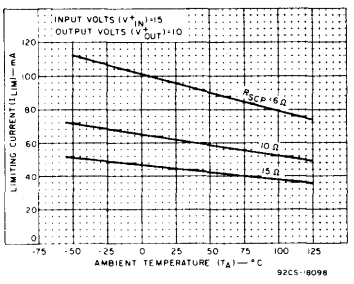


Fig.10–I_{LIM} vs. T_A.

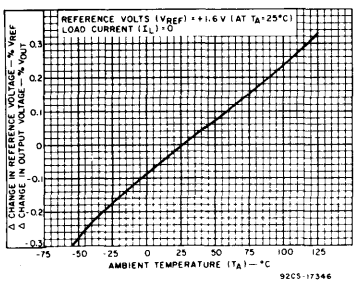


Fig.11–Temperature coefficient of V_{REF} and V_{OUT}.

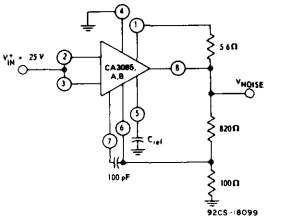


Fig.12–Test circuit for noise voltage.

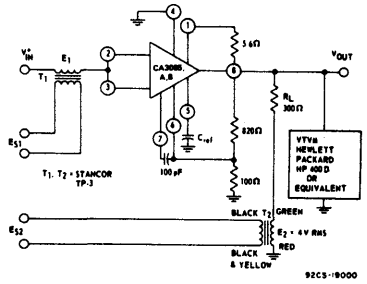


Fig.13—Test circuit for ripple rejection and output resistance.

TEST PROCEDURES FOR TEST CIRCUIT FOR RIPPLE REJECTION AND OUTPUT RESISTANCE

Output Resistance

Conditions:

1. V_{IN} = +25 V, C_{REF} = 0, Short E_1
2. Set E_{S2} at 1 kHz so that E_2 = 4 V rms
3. Read V_{OUT} on a VTVM, such as a Hewlett-Packard, HP400D or equivalent
4. Calculate R_{OUT} from $R_{OUT} = V_{OUT}(I_L R_L / E_2)$

Ripple Rejection — I

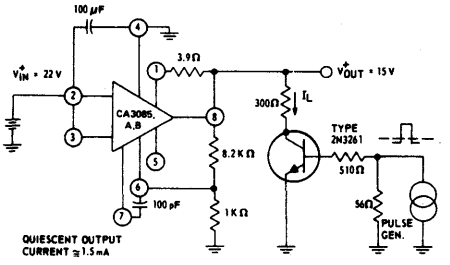
Conditions:

1. V_{IN} = +25 V, C_{REF} = 0, Short E_2
2. Set E_{S1} at 1 kHz so that E_1 = 3 V rms
3. Read V_{OUT} on a VTVM, such as a Hewlett-Packard, HP400D or equivalent
4. Calculate Ripple Rejection from 20 log (E_1/V_{OUT})

Ripple Rejection — II

Conditions:

1. Repeat Ripple Rejection I with C_{REF} = 2 μF



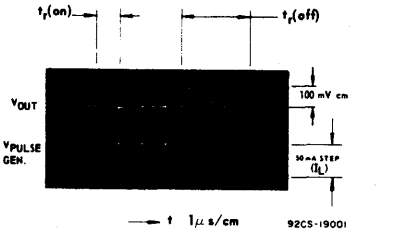
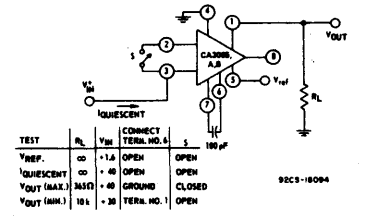
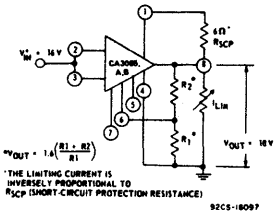


Fig.14—Turn-on and turn-off recovery time test circuit with associated waveforms.



TEST	R_L	V_{IN}	CONNECT TERM. NO. 6	S
V_{REF}	∞	+1.6	OPEN	OPEN
$I_{QUIESCENT}$	∞	+40	OPEN	OPEN
V_{OUT} (MAX.)	365Ω	+40	GROUND	CLOSED
V_{OUT} (MIN.)	10 k	+30	TERM. NO. 1	OPEN

Fig.15—Test circuit for V_{REF}, $I_{quiescent}$, $V_{OUT(max.)}$, $V_{OUT(min.)}$.



*THE LIMITING CURRENT IS INVERSELY PROPORTIONAL TO R_{SCP} (SHORT-CIRCUIT PROTECTION RESISTANCE)

Fig.16—Test circuit for limiting current

TYPICAL REGULATOR CIRCUITS USING THE CA3085 SERIES

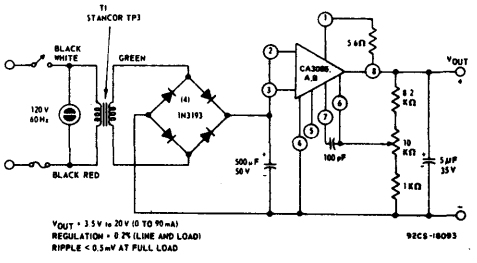
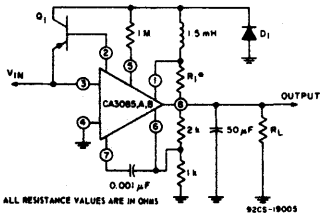
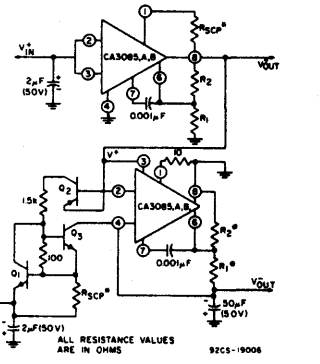


Fig.17—Application of the CA3085 Series in a typical power supply.



Q1 RCA-1N1763A OR EQUIVALENT
Q1 RCA-2N5322 OR EQUIVALENT
*R1 = 0.7 I_L (MAX.)

Fig.18—Typical switching regulator circuit.



ALL RESISTANCE VALUES ARE IN OHMS

Q1 RCA-2N2102 OR EQUIVALENT

Q2 ANY P-N-P SILICON TRANSISTOR (RCA-2N5322 OR EQUIVALENT)

Q3 ANY N-P-N SILICON TRANSISTOR THAT CAN HANDLE THE DESIRED LOAD CURRENT (RCA-2N3772 OR EQUIVALENT)

*$V_{OUT} = \left(\frac{R_1 + R_2}{R_1}\right)$

*R_{SCP} SHORT-CIRCUIT PROTECTION RESISTANCE

Fig.21— Combination positive and negative voltage regulator circuit.

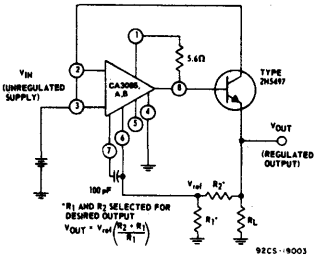


Fig.19— Typical high-current voltage regulator circuit.

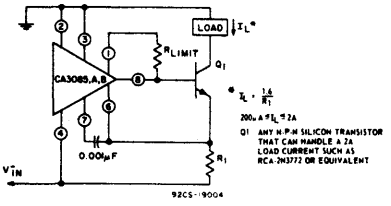


Fig.20— Typical current regulator circuit.

CA3086

General-Purpose N-P-N Transistor Array

Three Isolated Transistors and One Differentially– Connected Transistor Pair

For Low–Power Applications from DC to 120MHz

RCA-CA3086 consists of five general-purpose silicon n-p-n transistors on a common monolithic substrate. Two of the transistors are internally connected to form a differentially-connected pair.

The transistors of the CA3086 are well suited to a wide variety of applications in low-power systems at frequencies from DC to 120 MHz. They may be used as discrete transistors in conventional circuits. However, they also provide the very significant inherent advantages unique to integrated circuits, such as compactness, ease of physical handling and thermal matching.

The CA3086 is supplied in a 14-lead dual-in line plastic package. The CA3086F is supplied in a 14-lead dual-in-line hermetic (frit-seal) ceramic package.

Applications

- General-purpose use in signal processing systems operating in the DC to 120-MHz range
- Temperature compensated amplifiers
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Integrated-Circuit Transistor Array" for suggested applications.

MAXIMUM RATINGS, Absolute–Maximum Values at T_A = 25°C

DISSIPATION:		
Any one transistor	300	mW
Total package up to T_A = 55°C	750	mW
Above T_A = 55°C	derate linearly 6.67	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating	−55 to + 125	°C
Storage	−65 to + 150	°C
LEAD TEMPERATURE (During soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) From case for 10 seconds max.	+ 265	°C
The following ratings apply for each transistor in the device:		
COLLECTOR-TO-EMITTER VOLTAGE, V_{CEO}	15	V
COLLECTOR-TO-BASE VOLTAGE, V_{CBO}	20	V
COLLECTOR-TO-SUBSTRATE VOLTAGE, V_{CIO}*	20	V
EMITTER-TO-BASE VOLTAGE, V_{EBO}	5	V
COLLECTOR CURRENT, I_C	50	mA

* The collector of each transistor in the CA3086 is isolated from the substrate by an integral diode. The substrate (terminal 13) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action. To avoid undesirable coupling between transistors, the substrate (terminal 13) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	Typ. Characteristic Curves Fig. No.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	–	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0	–	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	–	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	–	5	7	–	V
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10V, I_E = 0	2	–	0.002	100	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10V, I_B = 0	3	—	See Curve	5	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 1mA	4	40	100	–	

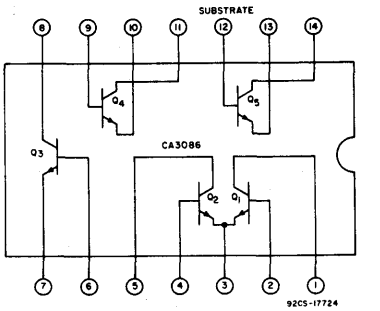


Fig.1 – Functional diagram of the CA3086.

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

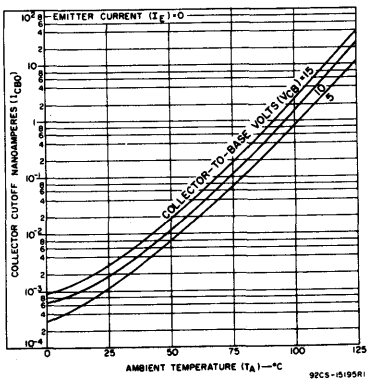


Fig.2– I_{CBO} vs T_A.

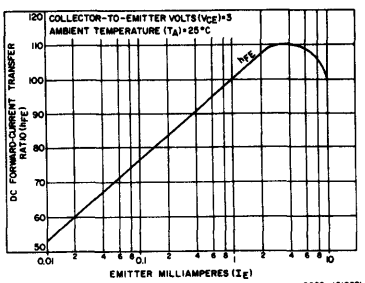


Fig.3– h_{FE} vs I_E.

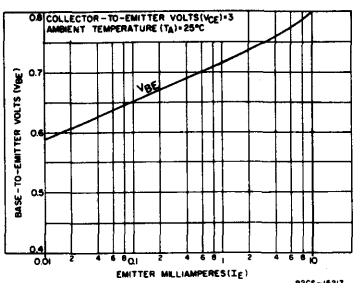


Fig.4 V_{BE} vs I_E.

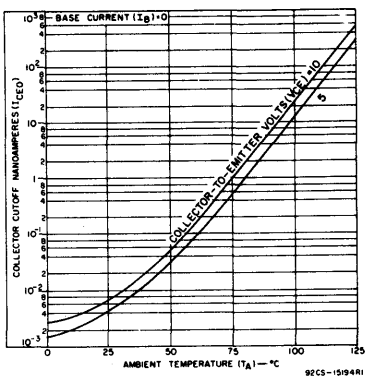


Fig.5– I_{CEO} vs T_A.

ELECTRICAL CHARACTERISTICS at T_A = 25°C Typical Values Intended Only for Design Guidance

CHARACTERISTICS	SYMBOL	TEST CONDITIONS		Typ. Characteristics Curves Fig. No.	TYPICAL VALUES	UNITS
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3 V	I_C = 10 mA	4	100	
			I_C = 10 μA	4	54	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3 V	I_E = 1 mA	5	0.715	V
			I_E = 10 mA	5	0.800	V
V_{BE} Temperature Coefficient	$\Delta V_{BE}/\Delta T$	V_{CE} = 3V, I_C = 1mA		6	−1.9	mV/°C
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_B = 1mA, I_C = 10mA		–	0.23	V
Noise Figure (low frequency)	NF	f = 1kHz, V_{CE} = 3V, I_C = 100μA, R_S = 1k Ω		–	3.25	dB
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:						
Forward Current-Transfer Ratio	h_{fe}	f = 1kHz, V_{CE} = 3V, I_C = 1mA		7	100	–
Short-Circuit Input Impedance	h_{ie}			7	3.5	kΩ
Open-Circuit Output Impedance	h_{oe}			7	15.6	μmho
Open-Circuit Reverse-Voltage Transfer Ratio	h_{re}			7	1.8×10^{-4}	–
Admittance Characteristics:						
Forward Transfer Admittance	Y_{fe}	f = 1MHz, V_{CE} = 3V, I_C = 1mA		8	31 − j1.5	mmho
Input Admittance	Y_{ie}			9	0.3 + j0.04	mmho
Output Admittance	Y_{oe}			10	0.001 + j0.03	mmho
Reverse Transfer Admittance	Y_{re}			11	See Curve	–
Gain-Bandwidth Product	f_T	V_{CE} = 3V, I_C = 3mA		12	550	MHz
Emitter-to-Base Capacitance	C_{EBO}	V_{EB} = 3V, I_E = 0		–	0.6	pF
Collector-to-Base Capacitance	C_{CBO}	V_{CB} = 3V, I_C = 0		–	0.58	pF
Collector-to-Substrate Capacitance	C_{CIO}	V_{CI} = 3V, I_C = 0		—	2.8	pF

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

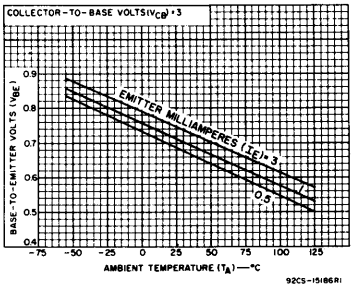


Fig.6 – V_{BE} vs T_A.

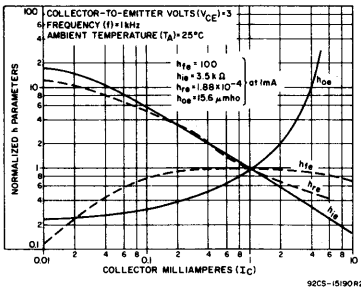


Fig.7 – Normalized h_{fe}, h_{ie}, h_{oe}, h_{re} vs I_C.

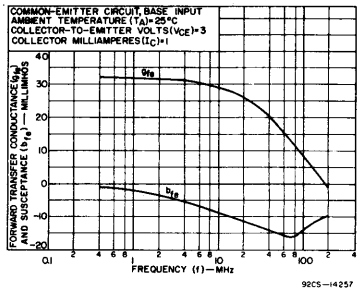


Fig.8 – y_{fe} vs f.

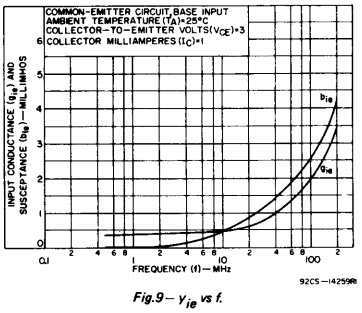


Fig.9 – y_{ie} vs f.

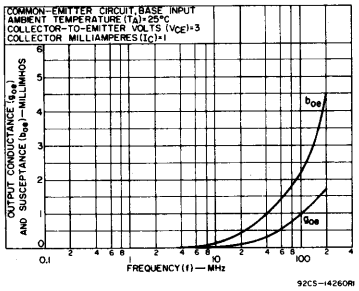


Fig.10 – y_{oe} vs f.

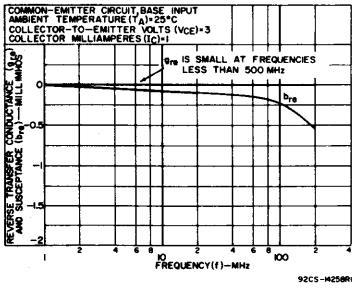


Fig.11 – y_{re} vs f.

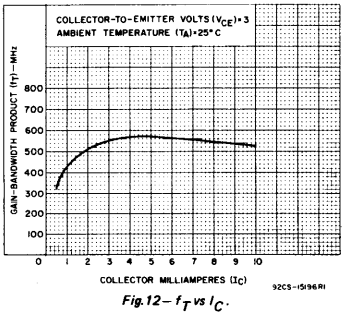


Fig.12 – f_T vs I_C.

CA3091D
Analog Multiplier

RCA-CA3091D, a monolithic silicon integrated circuit, is a four-quadrant multiplier that provides an output voltage that is the product of two input (x and y) voltages.

This device functions as a multiplier, divider, squarer, square rooter, and power-series approximator. In addition, this device is useful in applications such as ideal full-wave rectifiers, automatic level controllers, RMS converters, frequency discriminators, and voltage-controlled filters and oscillators.

The CA3091D comprises five basic circuits (See Fig. 1), including: a multiplier block, two linearity compensators, a current converter, a current source for biasing, and a regulator (reference voltage). A brief description of the operation, functions and typical applications is given in the section "Operating Considerations". In addition there is a separate section on "Symbols, Terms, and Definitions" that defines the terms and symbols used throughout the data bulletin.

The CA3091D is supplied in 14-lead dual-in-line ceramic package and operates over the full military temperature range of -55°C to +125°C.

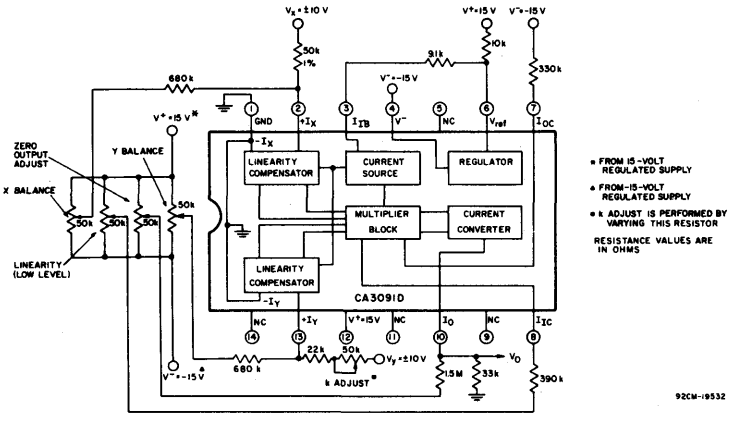


Fig.1–Functional block diagram of CA3091D with typical multiplier outboard(peripheral)circuitry.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

DC Supply Voltages:		
Between Terms. 12 and 1	+18	V
Between Terms. 4 and 1	−18	V
DC Supply Currents:		
At Term. 12 with DC Supply Voltage = +15 V	4	mA
At Term. 4 with DC Supply Voltage = −15 V	16	mA
Bias Current (At Term. 3)	1	mA
* Input Current	±1	mA
Output Short-Circuit Duration	No limitation	
Voltage Reference Current	10	mA
Linearity Correction Currents:		
At Terminals 7 and 8	10	mA
Device Dissipation (Up to 125°C)	200	mW
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (during soldering):		
At distance not less than 1/32 inch (0.79 mm) from case for 10 seconds max.	+265	°C

* External resistance is required to limit the current to the indicated ±1 mA value.

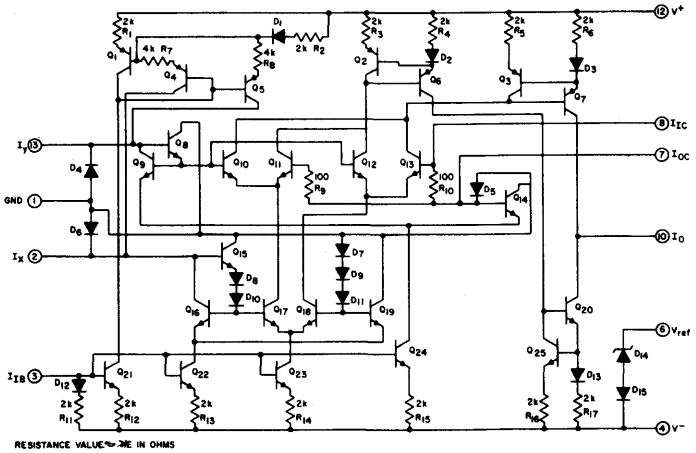


Fig.2–Schematic diagram of the CA3091D.

Features: ●

- **"Accuracy": ±4% (max.)**
- **"Linearity": 3.0% (max.)**
- **Feedthrough: 9 mV p-p (typ.)**
- **3-db bandwidth: 4.4 MHz**
- **Low power operation capability: ±6.0 V, 4 mW drain**
- **Low power-supply sensitivity: 36 mV/V typ.**
- **Smooth overload characteristics – no foldback if full-scale input signal is exceeded**
- **Negligible warm-up drift**
- **Broadband operation capability (flat to 1 MHz) – both inputs have similar characteristics for reduced high-frequency phase shift between the inputs**
- **Low-level linearity correction circuitry minimizes low-level feedthrough for improved small-signal accuracy**
- **All multiplication is performed with wideband circuitry – this permits two signals of frequencies much higher than the -3 db frequency of the multiplier to produce a difference frequency that is within the multiplier's bandwidth**
- **High immunity to parasitic oscillation**
- **Essentially free from excess peaking – provides improved frequency response**
- **Requires no level shifting at the output – current-source operation at the output permits output signal to be referenced to ground or other levels within the output voltage swing capabilities of the multiplier**
- **Internal bias regulator**

Applications:

- **Multiplier ■ Divider ■ Squarer ■ Square Rooter**
- **Power-series approximator**
- **Full-wave rectifier**
- **Automatic level controller**
- **RMS converter**
- **Frequency discriminator**
- **Voltage-controlled filters and oscillators**

ELECTRICAL CHARACTERISTICS, For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	Circuit and/or Char. Curve	LIMITS			UNITS
		T_A = 25°C, I_{IB} = 0.5 mA, V^+ = 15 V, V^- = -15 V		Min.	Typ.	Max.	
STATIC CHARACTERISTICS							
INPUT CIRCUIT							
Input Balance (Correction) Currents:							
At x Input	I_{IC}	x = 0	–	-20	-2.1	+20	μA
At y Input		y = 0	–	-20	-8.7	+20	μA
Feedthrough Linearity Balance (Correction) Current	I_{OC}		–	-34	-2.9	+34	μA
OUTPUT CIRCUIT							
Output Offset Current	I_{OO}	x & y = 0,	–	-10	-0.23	+10	μA
Output Offset Voltage	V_{OO}	I_{OO} thru R_L = 33kΩ	–	-0.330	-0.0076	+0.330	V
Output Peak Current Swing	$\|I_O\|$	Thru R_L = 24kΩ	3	0.41	0.45	–	mA
Output Peak Voltage Swing	$\|V_O\|$	Across R_L = 33kΩ	4	12	12.9	–	V
DC SUPPLIES & BIASING							
Current Drain (Idling):							
At Term. 4		V^- = -15 V	–	–	2.9	4.5	mA
At Term. 12		V^+ = +15 V	–	–	2.0	3.0	mA
Reference Voltage	V_{ref}	Measured across Terms. 6 & 4 at I = 1mA	–	5.5	6.1	6.7	V
DYNAMIC CHARACTERISTICS							
Output Current	I_O	With I = 0.2mA at each input	–	–	0.21	0.32	mA
Normalized k Factor $\left(k_N = \frac{k}{k_T}\right)$			11	0.69	1.0	1.7	
Accuracy		Worst case at 25°C	–	–	2.6	4.0	% of
Linearity				–	1.7	3.0	10 V
Feedthrough Voltage:							
At y = 20V p-p, x = 0			–	–	9	20	mV
At x = 20V p-p, y = 0			–	–	9	20	p-p

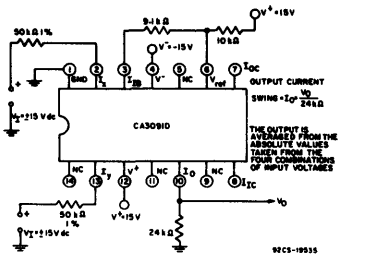


Fig.3–Test circuit for measurement of output current swing capability.

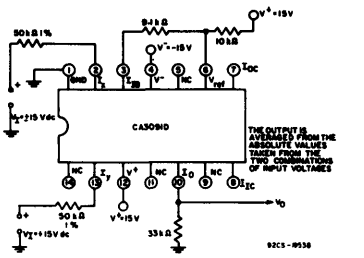


Fig.4–Test circuit for measurement of output voltage swing capability.

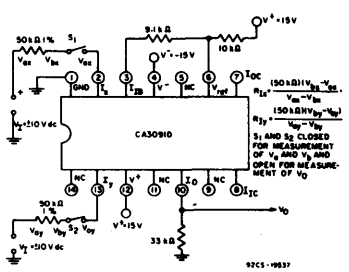


Fig.5–Test circuit for measurement of input resistance.

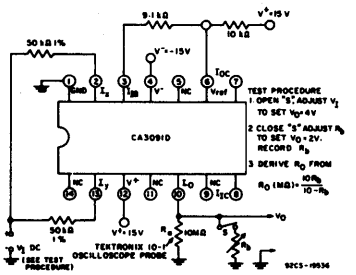


Fig.6–Test circuit for measurement of output resistance.

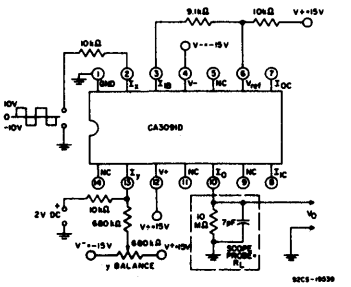


Fig.7–Test circuit for measurement of maximum slew rate.

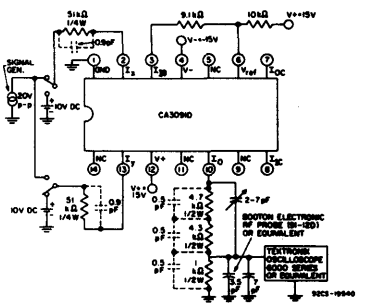


Fig.8–Test circuit for measurement of frequency response.

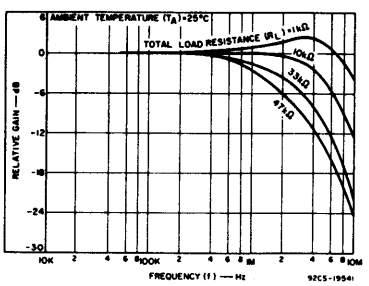
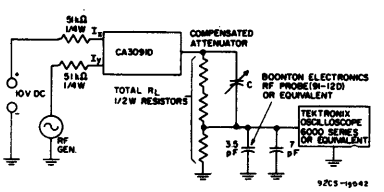


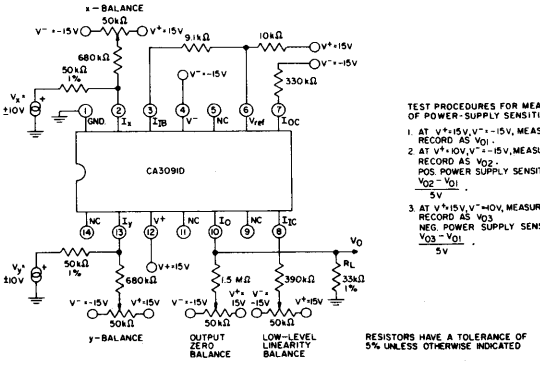
Fig.9- y–input frequency response characteristic curve with associated test circuit.



CA3091D

ELECTRICAL CHARACTERISTICS, Typical Values Intended Only for Design Guidance

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C, I_{IB} = 0.5 mA, V^+ = 15 V, V^- = −15 V	Circuit and/or Char. Curve	TYPICAL VALUES	UNITS
STATIC CHARACTERISTICS					
INPUT CIRCUIT					
Input Resistance:					
At x Input	R_I	$\lvert I_x \rvert \leq 0.2$ mA	5	1.3	kΩ
At y Input		$\lvert I_y \rvert \leq 0.2$ mA		0.5	kΩ
Input Capacitance:					
At x Input	C_I	at 1 MHz	–	5.8	pF
At y Input				5.8	pF
OUTPUT CIRCUIT					
Output Resistance	R_O		6	1.0	MΩ
Output Capacitance:	C_O	at 1 MHz		4.0	pF
DC Supply Voltage Sensitivity:					
At Term. 4	$\frac{\Delta V_O}{\Delta V^-}$		11	26	mV/V
At Term. 12	$\frac{\Delta V_O}{\Delta V^+}$			36	mV/V
DYNAMIC CHARACTERISTICS					
Bandwidth (At -3dB point):					
Through x Input	BW		8, 10	4.8	MHz
Through y Input			8, 9	4.4	MHz
3° Error Frequency:					
Through x Input			–	360	kHz
Through y Input				310	kHz
Maximum Slew Rate	SR	7pF in parallel with 10 MΩ load	7	27	V/μs
Temperature Coefficients:					
Output Offset Current	$\Delta I_{OO}/\Delta T$	x & y = 0	–	−0.021	μA/°C
x-Input Balance Current	$\Delta I_{IC}/\Delta T$	x = 0	–	−0.063	μA/°C
y-Input Balance Current		y = 0	–	−0.063	μA/°C
Normalized k Factor $\left(k_N = \frac{k}{k_r}\right)$	k_N		–	−0.76	%/%/°C
Accuracy			–	0.11	%/°C
Linearity			–	0.06	%/°C
Feedthrough:					
At x = 0			–	5.6	mV/°C
At y = 0				5.7	mV/°C



TEST PROCEDURES FOR MEASUREMENT OF POWER-SUPPLY SENSITIVITY

1. AT V^+=+15V, V^-=−15V, MEASURE V_O RECORD AS V_{O1}.
2. AT V^+=+10V, V^-=−15V, MEASURE V_O RECORD AS V_{O2}. POS. POWER SUPPLY SENSITIVITY = $\frac{V_{O2} - V_{O1}}{5V}$.
3. AT V^+=+15V, V^-=−10V, MEASURE V_O RECORD AS V_{O3}. NEG. POWER SUPPLY SENSITIVITY = $\frac{V_{O3} - V_{O1}}{5V}$.

k ≡ k FACTOR
$k_r \equiv 0.1$ REFERENCE OR ADJUSTED k FACTOR
$k_N = k/k_r = 0.1/V_O$ = NORMALIZED k FACTOR (i.e. k_N=1, IF V_x=V_y=V_O=10)
OUTPUT CURRENT (mA) [AT A CURRENT OF 0.2 mA AT BOTH INPUTS] = $V_O/33$ kΩ
OUTPUT VALUES ARE AVERAGED FOR 4 COMBINATIONS OF INPUTS (I_x = $\frac{V_O}{R_L}$ + $\frac{V_O/33\text{k}\Omega}{(0.2 \times 10^{-3})2}$ I_x I_y

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Fig.11—Test circuit for measurement of current gain and power-supply sensitivity.

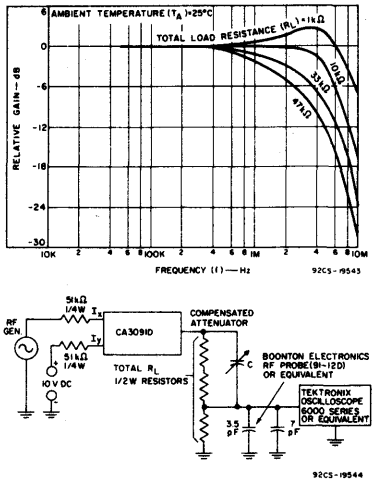


Fig.10- x-input frequency response characteristic curve with associated test circuit.

SYMBOLS, TERMS AND DEFINITIONS

Output Offset Current
The multiplier output current produced when both of the multiplier input signals are in the zero state.

Output Zero
Sets the output at the zero level when the x and y inputs are in the zero state. (It is implied that all other zeroing adjustments have been effected.)

R_I
Input Resistance — Converts the input voltage to an input current.

R_L
Output (Load) Resistance — Converts the output current to a voltage.

R_O
Output Resistance — See V_O and I_O for the equations associated with these properties.

Regulator Diode
A temperature compensated Zener diode, included in the multiplier circuit, to provide a stable I_{IB}.

Scale Factor or k factor (k)
Represents the basic gain of the multiplier as expressed in the equation $V_O = kV_xV_y$

The equation indicates the ideal transfer function for the multiplier. The normalized k factor is expressed by $k_N = k/k_{ref}$ where k_{ref} is the ideal or reference k factor. The ideal factor, k_{ref} is the value at which the k factor is set when the k-factor adjust control is trimmed. Optimum operation of the CA3091D is achieved when the k-factor is 0.1.

V_{IM}
The maximum ac sine-wave voltage to be applied to the multiplier; a 20-volt p-p sine wave is the nominal maximum swing voltage recommended for use with 50-kilohm input resistors.

V_{MID}
An ac or dc voltage that approximately satisfies the equation

$$V_{MID} = V_{IM}/\sqrt{2}.$$

V_O
The output product voltage derived from the expression

$$(kV_xV_y = V_O)$$

V_{ref}.
Temperature compensated zener connected to the −15 volt supply to provide a reference voltage as an aid in setting up a stable I_{IB}.

V_X, V_Y
The input voltages to be multiplied.

x-Balance Circuit
Sets the output to the zero level when the x-input is in the zero state.

y-Balance Circuit
Sets the output to the zero level when the y-input is in the zero state.

Accuracy
Accuracy defines the degree of error encountered in the operation of the multiplier. It is portrayed on a contour map by isomers (contour lines). Isomers with the highest values indicate "less-accurate" operation of the multiplier. (See illustrative Contour Map in Fig. 12.)

Contour Map
The contour map, shown in Fig. 12, is a graphical portrayal of the multiplier errors in the x, y input plane. Each contour line, termed "isomer", connects those points whose error values (in millivolts) are equal in magnitude. For example, a −20 mV contour line with points at V_X = 5V and V_Y = −3V indicates that the output voltage is 20 mV less than the theoretical output product (kV_XV_Y). This error voltage, presented in percent of full-scale input (±10 V), defines the "accuracy" of the device. Thus, a 20-mV error voltage represents an "accuracy" of 0.2% as derived from the equation:

$$\text{Accuracy} = 20\ \text{mV}/10 \times 100\% = 0.2\%.$$

A contour map provides a true indication of multiplier performance in each of the four quadrants. Each CA3091D is comprehensively tested and must provide the specified accuracy in the four quadrants.

Current Converter
This portion of the IC combines the multiplier's differential-amplifier output currents and converts them to a single-ended output current.

Current Sources
These circuits provide the biasing currents for the various circuits in the IC. The I_{IB} terminal provides the control current for the current-source circuit.

Feedthrough
Feedthrough occurs when an output signal is produced even though one of the input signals is zero. Consequently, feedthrough signal characteristics constitute a source of error in the operation of a multiplier. In the CA3091D, for example, the feedthrough signal output is specified to be less than 20 mV p-p when either terminal is set at 20 V p-p and the other terminal is set to zero.

I_{IB}
Circuit biasing control current.

I_{IC}
See I_{OC}.

I_O
Output product current ($k_I I_X I_Y = I_O$), where $k_I = kR_f^2 / R_L$

I_{OC}, I_{IC}
Compensatory input and output currents required to correct unlinearity along the x axis. (Optional for low-level signal use.)

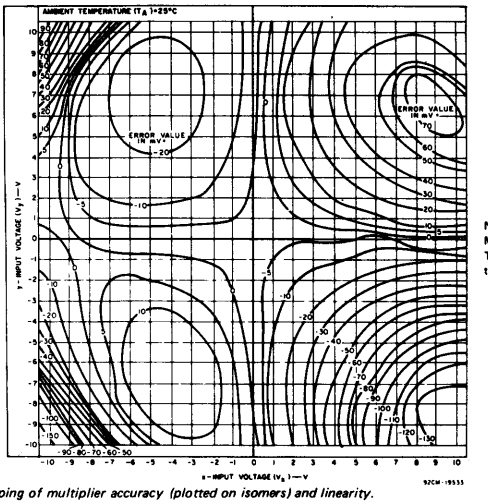
I_X, I_Y
Input currents to be multiplied.

k
Voltage Scale Factor (determines the gain of the multiplier).

k_I
Current Scale Factor (k_I) = $(R_f^2 / R_L)k$.

k adjust
Scale-Factor Adjustment.

Linearity
"Linearity" indicates the degree of multiplier error (i.e. deviation from "straight-line" characteristics) along each of the four boundaries of the input x, y field. These boundaries are formed when one input is held at one of the two maximum values (10 volts or −10 volts) and the other input is swept through the voltage range. (See Contour Map for additional information.)



Note: See "Contour Map" in "Symbols, Terms and Definitions" Section.

Fig.12—Contour mapping of multiplier accuracy (plotted on isomers) and linearity.

Linearity Adjust
An external circuit to provide vernier adjustment for optimum linearity. This control should be adjusted before adjusting the y-balance control.

Linearity Balance Circuit (Low-Level)
This circuit makes the multiplier's transfer function linear for low-level x-input signals.

Linearity Compensator
Internal circuitry that converts input current into a non-linear voltage, a requisite for producing a linear output in the differential amplifiers of the multiplier circuit.

Multiplier Circuitry
Provides the product of the two input voltages.

Multiplier Transfer Function
This function mathematically describes the interaction of the two inputs and the resulting output signal. The basic transfer function for a multiplier is

$$k(V_X + V_{Xe})(V_Y + V_{Ye}) = V_O + V_{Oe}$$

where: k = k factor and represents the basic gain of the multiplier

V_X, V_Y = the external inputs to be multiplied

V_O = the desired value of the product output signal

V_{Xe}, V_{Ye} = the "effective" errors that occur at the inputs of the multiplier and cause an output signal when either input is in a zero state.

V_{Oe} = the error voltage that develops at the output of the multiplier

DC correction factors are added to the multiplier inputs and output to compensate for the errors and offset variations. A complex linearity error term appears in the transfer function; however, this term is not included in the above equation for the purpose of clarity.

OPERATING CONSIDERATIONS

Operation of a Multiplier

A multiplier is, essentially, a gain-controlled amplifier (See Fig. 13) that multiplies the input signal (V_X) with the external gain controlling signal (V_Y) to produce the resultant output (V_O). The gain is externally adjustable by a coefficient (k). Stated simply, a multiplier produces an output voltage that is the linear product of two input voltages.

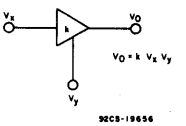
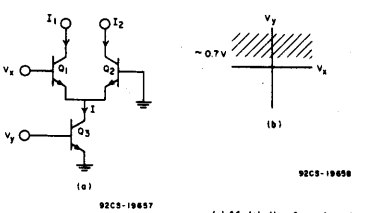


Fig.13—Gain-controlled amplifier.

The basic multiplier, shown in Fig. 14a, is a two-quadrant multiplier. The input signal (V_X) may have either a positive or negative polarity whereas, the external gain-controlling signal (V_Y) must be positive and greater than the base-to-emitter voltage (Fig. 14b). The output current ($I_1 - I_2$) of the differential amplifier, comprised of transistors Q1 and Q2, is related to both the input signal (V_X) and the current source (I). Since the current source (I) is related to the gain controlling signal (V_Y) the output current ($I_1 - I_2$), therefore, is related to both V_X and V_Y.



a) Basic circuit.

b) Multiplier functional only in shaded region.

Fig.14—Two-quadrant multiplier.

This relationship is essentially non-linear; thus an appropriate linearization circuit must be provided in the input stage to achieve the following linear relationship:

$$I_1 - I_2 = k' V_X V_Y \qquad \text{(Eq. 1)}$$

where k' is a constant

CA3091D

Figure 15 shows a typical arrangement of three differential amplifiers to form a four-quadrant multiplier. This arrangement incorporates the operating principles of the two-quadrant multiplier, but, in addition, it permits both of the input signals (V_X and V_Y) to have positive or negative polarities (or zero). When either input is zero, the output current ($I_1 - I_2$) must, theoretically, be zero as is shown by the following:

1. Assume $V_X = 0$,
   then $i_1 = i_2$ and $i_3 = i_4$
   therefore $i_1+i_4 = i_2+i_3$.
   Since $I_1 = i_1+i_4$ and $I_2 = i_2+i_3$,
   then $I_1 = I_2$.
   This equality is independent of V_Y
2. Now assume $V_Y = 0$,
   then $i_5 = i_6$.
   Sine $i_5 = i_1+i_2$ and $i_6 = i_3+i_4$,
   then $i_1+i_2 = i_3+i_4$.
   Since $i_1 = i_3$ and $i_2 = i_4$
   then $i_1+i_4 = i_3+i_2$.
   Therefore $I_1 = I_2$.
   This equality is independent of V_X.

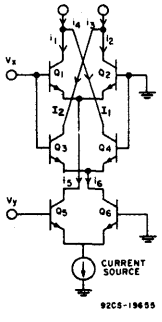


Fig.15—Basic four-quadrant multiplier.

The multiplying operation discussed in the previous section applies when neither V_X nor V_Y is zero. The output current ($I_1 - I_2$) then satisfies Equation 1,

$$I_1 - I_2 = k'V_XV_Y.$$

The multiplying action of the four-quadrant multiplier is dependent on current unbalance in the three differential amplifiers. Ideally, the multiplying operation should not occur if either V_X or V_Y is 0. However, in practical applications slight current unbalances do exist. It is necessary, therefore, to null out such unbalances with external potentiometers prior to operation.

TYPICAL OPERATING CONSIDERATIONS

The RCA-CA3091D, shown in Fig. 2, is a four-quadrant multiplier that incorporates the basic multiplier principle, previously discussed in "Operation of a Multiplier". Because the design of this multiplier is based on the multiplication of two input currents to produce an output current it is necessary to convert the input voltages to input currents and the output current to an output voltage by inserting resistors at both input and output terminals. Fig. 1 shows the four-quadrant multiplier with its peripheral circuitry for nulling current unbalances.

The Bias Current (I_{IB}) at Term. 3 sets the operating current level for the entire multiplier circuit by means of a current-source circuit. Therefore, it is essential that this bias current level remain constant under all operating conditions. To maintain this steady state, a temperature compensated zener diode is provided on the chip and connected to the Reference Voltage (Term.6).

Linearity of the differential amplifier transconductance function is accomplished by linearity compensators as shown in Fig. 1. To correct low-level signal unbalances that may occur between Differential Amplifiers A and B, an external potentiometer is connected to Terminals 7 and 8 (See Fig. 1). The Current Converter circuit, which consists of a set of current mirrors, supplies the output current ($I_1 - I_2$). It is important that circuit unbalances be corrected prior to operation. Table I describes the alignment procedures for correcting these unbalances.

A multifunctional circuit board (Figs. 16 and 17) is available for performing the four basic applications, such as, multiplying, dividing, squaring and taking the square root.

When the CA3091D is used as a multiplier (Fig. 18) or as a squarer (Fig. 18) only the basic pheripheral circuitry on the multifunctional circuit board is utilized and the general-purpose operational amplifier (CA3741T) is disabled from operation. Follow the ac alignment procedures for these two applications before operating the circuit.

When the CA3091D is used as a divider (Fig. 20), the operational amplifier is required in order to provide the proper negative feedback. The limitations for operation as a divider are that $0 < V_Y \leqslant 10V$ and $-10V \leqslant V_Z \leqslant 10V$. Note, the range of V_Y is limited to the positive polarity; if V_Y was permitted to go negative, the feedback loop would go positive and, thereby, create an unstable operating condition.

Alignment of the divider (Fig. 19) differs from multiplier and squarer alignment because of the additional variances introduced by the operational amplifier. A coupling capacitor is provided at the output of the divider alignment circuit in order to separate the ac signal from the dc signal and, thus, avoid interaction between the calibrating potentiometers.

The alignment procedure for the square-rooter function (Fig. 21) is identical to the alignment procedure for the divider function. The input voltage range is limited to $0 < V_I \leqslant 10V$. This limitation is necessary in order to prevent the output voltage (V_O) from latching to the negative output saturation voltage of the operational amplifier. Table II describes the divider alignment procedure.

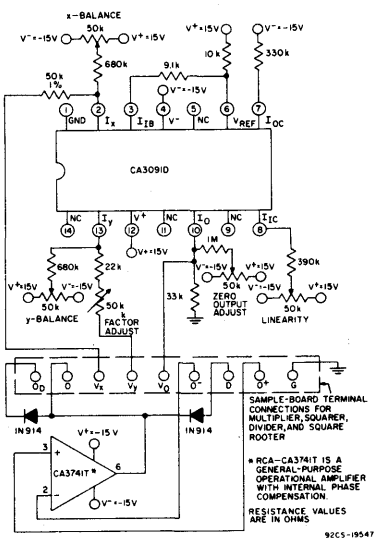
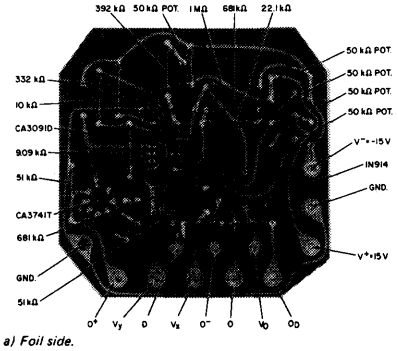
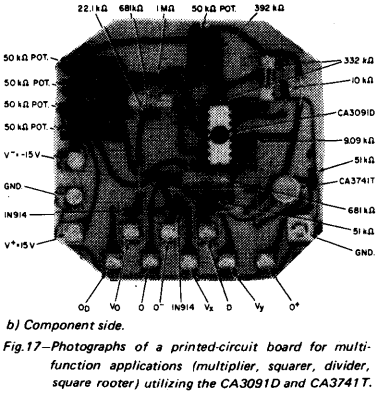


Fig.16—Typical multifunction circuit arrangement utilizing the CA3091D and CA3741T.



a) Foil side.



b) Component side.

Fig.17—Photographs of a printed-circuit board for multifunction applications (multiplier, squarer, divider, square rooter) utilizing the CA3091D and CA3741T.

Table I

AC Alignment Procedures For CA3091D, Four-Quadrant Multiplier

(Refer to Fig. 16, for circuit pertaining to following alignment procedures.)

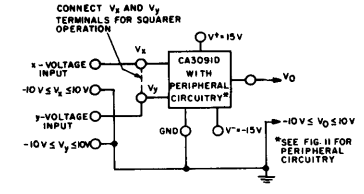
Step No.	Voltage Setting V_X	Voltage Setting V_Y	Control Adjust	Test Equipment Used	Measure	Notes
1	–	–	–	–	–	Set all potentiometers to center of range.
2	0	V_{IM}	x Balance	AC VM	V_O	Adjust for a minimum reading.
3	0	V_{IM}	Linearity	AC VM	V_O	Adjust for a minimum reading.
4	–	–	–	–	–	Repeat Steps 1 and 2 until no further improvement is noted.
5	V_{IM}	0	y Balance	AC VM	V_O	Adjust for a minimum reading.
6	0	0	Zero Output	DC VM	V_O	Adjust for zero output.
7	V_{MID}	V_{MID}	R_k	AC/DC VM	V_O	Adjust for $V^2_{MID}/10$ at the output.
8	–	–	–	–	–	Check multiplier for alignment in all four quadrants.

V_{IM} – Is the maximum AC swing of the sine wave that will be applied to the multiplier. A 20-volt p-p value is the nominal maximum swing of the AC sine wave with input resistors of 50 kilohms.

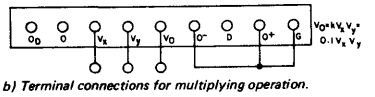
V_{MID} – An AC or DC voltage that approximately satisfies the equation $V_{MID} = V_{IM}/\sqrt{2}$. For example, if a 50-kilohm resistor is used with a 7-volt input, then R_k should be adjusted for a 4.9-volt output.

Table II — Divider Alignment Procedure

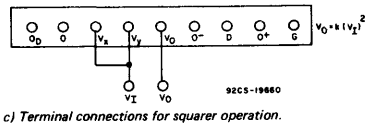
Step No.	Set V_z V	Set V_y V	Measure	Output Coupling	Test Equipment Used	Adjust	Notes
1	–	–	–	–	–	–	Set all potentiometers to center of range.
2	0	V_S	V_O	ac	ac – VM	O_{zero}	Adjust for minimum reading.
3	0	10V dc	V_O	dc	dc – VM	$x_{balance}$	Adjust for OV dc output.
4	V_S	V_S	V_O	ac	ac – VM	$y_{balance}$	Adjust for minimum reading.
5	5V dc	5V dc	V_O	dc	dc – VM	k_{adjust}	Adjust for 10V dc output.



a) Circuit arrangement for multiplier or squarer operation.



b) Terminal connections for multiplying operation.



c) Terminal connections for squarer operation.

Fig.18—Multifunction circuit-board arrangement with terminal connections for multiplier and squarer operation.

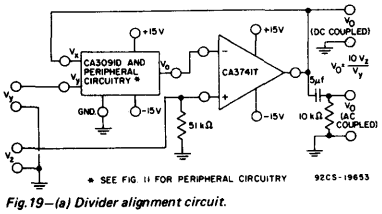


Fig.19—(a) Divider alignment circuit.

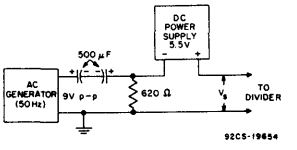
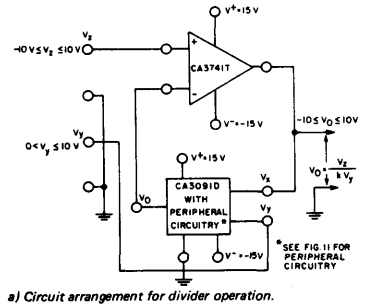
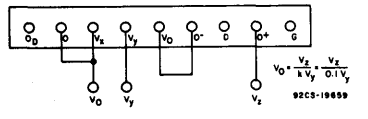


Fig.19—(b) Circuit to provide offset ac signal for use in divider alignment procedure.

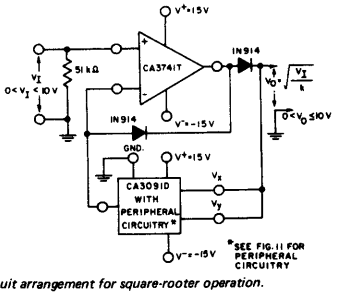


a) Circuit arrangement for divider operation.

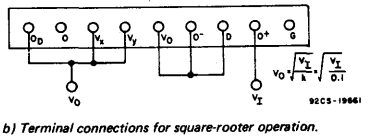


b) Terminal connections for divider operation.

Fig.20—Multifunction circuit-board arrangement with terminal connections for divider operation.



a) Circuit arrangement for square-rooter operation.



b) Terminal connections for square-rooter operation.

Fig.21—Multifunction circuit-board arrangement with terminal connections for square-rooter operation.

CA3093E

General-Purpose High-Current N-P-N Transistor-Zener Diode-Diode Array

RCA CA3093E* is a versatile array of three high-current (to 100mA) NPN transistors, two 10%-tolerance Zener diodes and one conventional diode, all on a common monolithic substrate. Two of the transistors (Q_1 and Q_2) are matched at 1 mA for applications in which offset parameters are of special importance. The combination of positive Zener voltage temperature coefficients and negative forward base-emitter voltage temperature coefficients provides a unique temperature compensation capability.

Independent connections for each transistor and diode plus a separate terminal for the substrate permit maximum flexibility in circuit design.

*Formerly developmental type TA6119

≠Z_1, Z_2 and D1 are transistors internally connected as shown below.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

Power Dissipation:		
Any one transistor	500	mW
Any one Zener Diode	250	mW
Total package	750	mW
Above 25°C ..... Derate linearly	6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C
The following maximum ratings apply for each transistor		
Collector-to-Emitter Voltage (V_{CEO})	15	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})*	20	V
Emitter-to-Base Voltage (V_{EBO})	5.5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	35	mA
The following maximum ratings apply for each Zener Diode or Diode		
Zener Diode dc Current (I_Z)	35	mA
Zener Diode-to-Substrate Voltage (V_{ZIO})*	20	V
Diode (D1) Forward Current (I_{DF})	50	mA
Diode (D1) Reverse Voltage (V_{DR})	5.5	V
Diode (D1)-to-Substrate Voltage (V_{DIO})*	20	V

*The collector of each transistor, the cathode of each Zener diode, and the anode of the diode are isolated from the substrate by an internal diode. The substrate must be connected to a voltage which is more negative than any of these isolated terminals in order to maintain isolation between devices and provide normal transistor action. To avoid undesired coupling between devices, the substrate terminal (5) should be maintained at either dc or signal (ac) ground. A suitable bypass capacitor can be used to establish a signal ground.

Features:

- 6 independent devices plus separate substrate connection
- Compensating temperature coefficients – V_{BE} and V_{D1} vs. V_Z

Transistors

- High I_C (100mA max)
- Matched pair (Q1 & Q_2)
  V_{IO} = ± 5mV max
  I_{IO} = 2.5 μA max } at I_C = 1mA
  $\Delta V_{IO}/\Delta T$ = 5 μV/°C typ
- h_{FE} = 40 min @ I_C = 10mA or 50mA
- Low V_{CEsat} . . . 0.7V max @ 50mA

Zener Diodes

- Two 1/4W Zeners
- V_Z = 7V ± 10%
- z_Z = 15Ω typ

Diode

- Close forward voltage match to V_{BE}'s of Q_1 and Q_2
- V_{PIV} = 5.5V min.

Applications

- Signal processing and switching systems operating from DC to VHF
- Lamp and relay driver
- Differential amplifier
- Temperature-compensated amplifier
- Thyristor firing
- Temperature-compensated shunt regulator
- Temperature-compensated series regulator
- Level shifting
- Voltage-level clamping
- Current regulator
- Voltage clamping
- Simple off-line regulated supply
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Circuit Transistor Array" for applications in addition to those given on pages 5 & 6 of this bulletin.

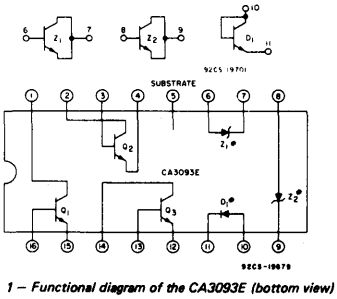


Fig. 1 – Functional diagram of the CA3093E (bottom view)

TYPICAL STATIC CHARACTERISTICS

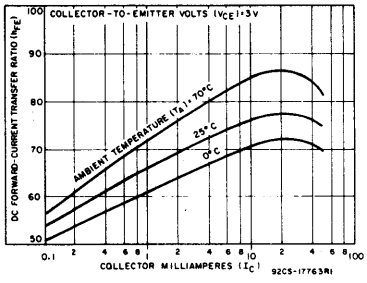


Fig. 2 – h_{FE} vs I_C

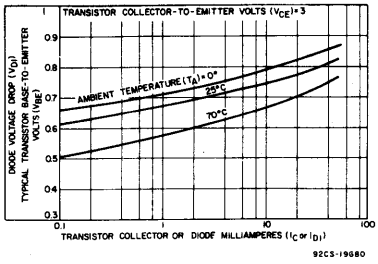


Fig. 3 – V_{BE} vs I_C and V_{D1} vs I_{D1}

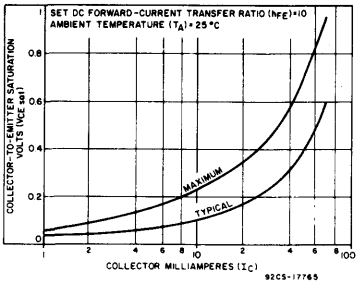


Fig. 4 – V_{CEsat} vs I_C at 25°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS Min.	Typ.	Max.	UNITS
For Each Transistor:						
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0	5.5	6.9	–	V
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	10	μA
Collector-Cutoff-Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	–	1	μA
DC Forward Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 10mA	40	76	–	
		V_{CE} = 3V, I_C = 50mA	40	75	–	
Forward Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA	0.65	0.74	0.85	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_C = 50mA, I_B = 5mA	–	0.40	0.70	V
Forward Base-to-Emitter Temp. Coefficient	$\Delta V_{BE}/\Delta T$	I_E = 10mA	–	-1.9	–	mV/°C
For Transistors Q1 and Q2 (As a Differential Amplifier):						
Absolute Input Offset Voltage	$\lvert V_{IO}\rvert$	V_{CE} = 3V, I_C = 1mA	–	1.2	5	mV
Absolute Input Offset Current	$\lvert I_{IO}\rvert$		–	0.7	2.5	μA
Temp. Coefficient of Offset Voltage	$\lvert\Delta V_{IO}/\Delta T\rvert$	–	–	5	–	μV/°C
For Each Zener Diode						
Zener Voltage	V_Z	I_Z = 10mA	6.3	7	7.7	V
Zener Impedance	^zZ	I_Z = 10mA, f = 1 kHz	–	15	25	Ω
Zener Reverse Current	I_{ZR}	V_Z = +5V	–	–	1	μA
Zener Voltage Temp. Coefficient	$\Delta V_Z/\Delta T$	I_Z = 10mA	– i.e.	+3.6 +.05	–	mV/°C %/°C
Zener-to-Substrate Breakdown Voltage	$V_{(BR)ZIO}$	I_Z = 100μA (Terminals 7 & 9)	20	60	–	V
Dissipation		Refer to Example in *Application "a"*	–	–	250	mW
For Diode (D1)						
Diode Forward Voltage	V_{DF}	I_C = 10mA, V_{CE} = 3V	0.65	0.74	0.85	V
Diode Forward Current	I_{DF}		–	–	50	mA
Diode Reverse-Breakdown Voltage	$V_{(BR)DR}$	I_{DR} = 500μA	5.5	6.9	–	V
Diode-to-Substrate Breakdown Voltage	$V_{(BR)DIO}$	I_{Diode} = 100μA (Terminal 10)	20	60	–	V
Diode Forward-Voltage Temp. Coefficient	$\Delta V_{DF}/\Delta T$	I_{DF} = 5mA	–	-1.9	–	mV/°C

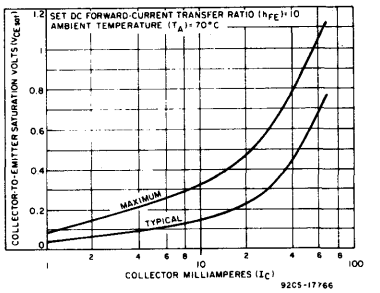


Fig. 5 – V_{CEsat} vs I_C at 70°C

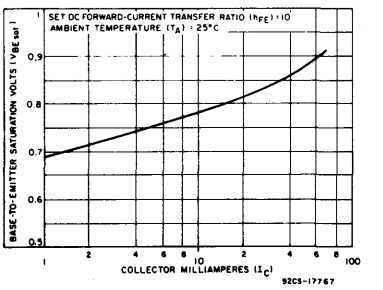


Fig. 6 – V_{BEsat} vs I_C

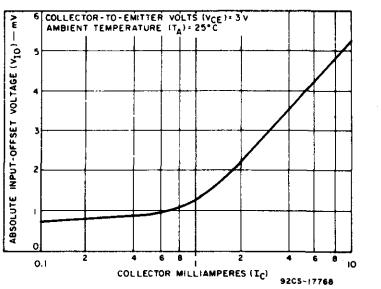


Fig. 7 – V_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier)

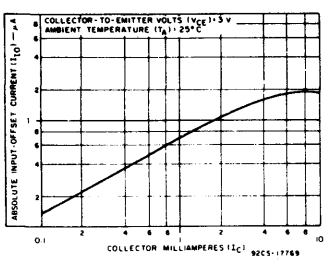


Fig. 8 – I_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier)

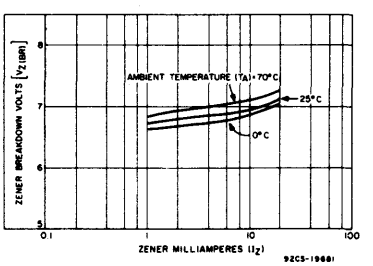


Fig. 9 – Typical Zener breakdown voltage vs current

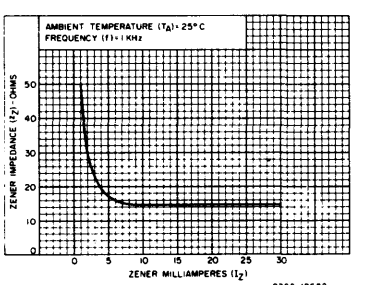
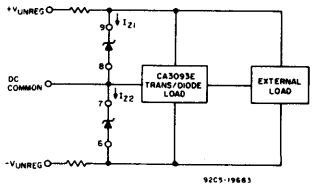


Fig. 10 – Typical Zener impedance vs current

CA3093E

TYPICAL APPLICATIONS

a) ±7V Regulator supplying CA3093E Transistors plus an external load.



Sample Computation for Determining Permissible Zener Dissipation at +25°C.

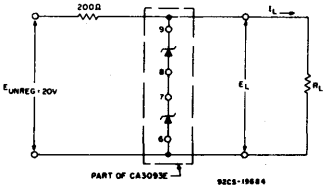
CA3093E Ratings at T_A = +25°C
Total Diss. Max = 750 mW (Derate @ 6.67 mW/°C above 25°C)
Each Zener Diss. Max = 250 mW
Max. Zener Current = 35 mA

Assume CA3093E Transistor/Diode Load Dissipation = 350 mW then max. total Zener Diss. ($P_{Z1} + P_{Z2}$) = 750 - 350 = 400 mW

$(I_{Z1} + I_{Z2})$ max = $\frac{400\ mW}{7V}$ = 57 mA

(Note: Max. current rating on each Zener is 35 mA)

b) 14V Regulator for Q1, Q2, Q3



Typical Load Regulation for I_L = 0 to 25 mA

$\Delta E_L/E_L \times 100 \approx -6\%$ (no load to full load)

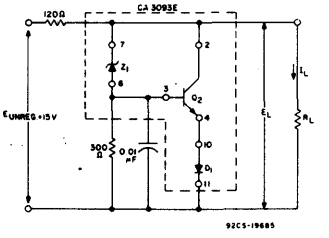
Typical Line Regulation

$\frac{(\Delta E_L/E_L) \times 100}{\Delta E\ unreg.} \approx \pm 0.9\%/V$

Typical Temperature Characteristic

$\frac{\Delta E_L/E_L}{\Delta T} \times 100 = +0.06\%/°C$

c) 8.6V Temp.-Compensated Shunt Regulator



Typical Temperature Characteristic @ R_L = 330Ω

$\frac{\Delta E_L/E_L}{\Delta T} \times 100 = \pm 0.007\%/°C$

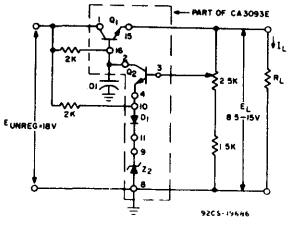
Typical Load Regulation I_L = 0 to 40 mA

$(\Delta E_L/E_L) \times 100 = -3\%$ (no load to full load)

Typical Line Regulation at R_L = 330Ω

$\frac{\Delta E_L/E_L}{\Delta E\ unreg.} \times 100 = \pm 0.55\%/V$

d) Temp.-Compensated Series Voltage Regulator



Typical Temperature Characteristic @ E_L = 12V

$\frac{\Delta E_L/E_L}{\Delta T} \times 100 = \pm\ 0.009\%/°C$

Typical Load Regulation @ E_L = 12V
I_L = 0 to 40 mA

$\frac{\Delta E_L}{E_L} \times 100 = \pm 0.4\%$ (no load to full load)

Typical Line Regulation @ E_L = 12V

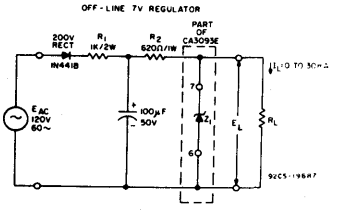
$\frac{(\Delta E_L/E_L) \times 100}{\Delta E\ unreg.} = \pm 0.45\%/V$

Typical E_L Ripple Voltage = 70 mV_{p-p}

Typical Load Regulation = $\frac{\Delta E_L}{E_L} \times 100 = -8.5\%$ (no load to full load)
I_L = 0 to 30 mA

Typical Line Regulation = $\frac{(\Delta E_L/E_L) \times 100}{\Delta E_{AC}} = \pm .075\%/V$

e) Off-Line 7V Regulator



Programmable Power Switch/Amplifier

For Control & General-Purpose Applications

CA3094T,S,E: For Operation Up to 24 Volts
CA3094AT,S,E: For Operation Up to 36 Volts
CA3094BT,S: For Operation Up to 44 Volts

The CA3094 is a differential-input power-control switch/amplifier with auxiliary circuit features for ease of programmability. For example, an error or unbalance signal can be amplified by the CA3094 to provide an on-off signal or proportional-control output signal up to 100 mA. This signal is sufficient to directly drive high-current thyristors, relays, dc loads, or power transistors. The CA3094 has the generic characteristics of the RCA-CA3080 operational amplifier directly coupled to an integral Darlington power transistor capable of sinking or driving currents up to 100 mA.

The gain of the differential input stage is proportional to the amplifier bias current (I_{ABC}), permitting programmable variation of the integrated circuit sensitivity with either digital and/or analog programming signals. For example, at an I_{ABC} of 100 μA, a one-millivolt change at the input will change the output from 0 to 100 mA (typical).

The CA3094 is intended for operation up to 24 volts and is especially useful for timing circuits, in automotive equipment, and in other applications where operation up to 24 volts is a primary design requirement (see Figs.28,29 and 30 in Applications Section). The CA3094A and CA3094B are like the CA3094 but are intended for operation up to 36 and 44 volts, respectively (single or dual supply).

These types are available in 8-lead TO-5 style packages with standard leads ("T" suffix) and with dual-in-line formed leads "DIL-CAN" ("S" suffix). Type CA3094 is also available in an 8-lead dual-in-line plastic package "MINI-DIP" ("E" suffix), and in chip form ("H" suffix).

Features:

- **Designed for *single* or dual power supply**
- **Programmable: strobing, gating, squelching, AGC capabilities**
- **Can deliver 3 watts (avg.) or 10 W (peak) to external load (in switching mode)**
- **High-power, single-ended class A amplifier will deliver power output of 0.6 watt (1.6 W device dissipation)**
- **Total harmonic distortion (THD) @ 0.6 W in class A operation — 1.4% typ.**
- **High current-handling capability — 100 mA (avg.), 300 mA (peak)**
- **Sensitivity controlled by varying bias current**
- **Output: "sink" or "drive" capability**

Applications:

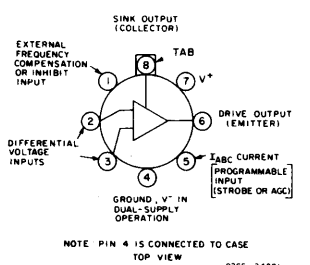
- **Error-signal detector: temperature control with thermistor sensor; speed control for shunt wound dc motor**
- **Over-current, over-voltage, over-temperature protectors**
- **Dual-tracking power supply with RCA-CA3085**
- **Wide-frequency-range oscillator ■ Analog timer**
- **Level detector ■ Alarm systems ■ Voltage follower**
- **Ramp-voltage generator ■ High-power comparator**
- **Ground-fault interrupter (GFI) circuits**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

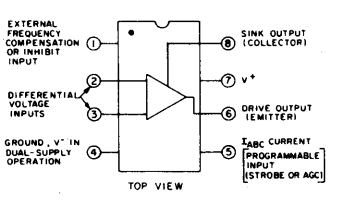
	CA3094	CA3094A	CA3094B	
DC SUPPLY VOLTAGE:				
Dual Supply	± 12 V	± 18 V	± 22 V	V
Single Supply	24 V	36 V	44 V	V
DC DIFFERENTIAL INPUT VOLTAGE (Terminals 2 and 3)	± 5*			V
DC COMMON-MODE INPUT VOLTAGE	Term. 4 ≤ Term. 2 & 3 ≤ Term. 7			
PEAK INPUT SIGNAL CURRENT (Terminals 2 and 3)	± 1			mA
PEAK AMPLIFIER BIAS CURRENT (Terminal 5)	2			mA
OUTPUT CURRENT:				
Peak	300			mA
Average	100			mA
DEVICE DISSIPATION:				
Up to T_A = 55°C:				
Without heat sink	630			mW
With heat sink	1.6			W
Above T_A = 55°C:				
Without heat sink derate linearly	6.67			mW/°C
With heat sink derate linearly	16.7			mW/°C
THERMAL RESISTANCE (Junction to Air)	140			°C/W
AMBIENT TEMPERATURE RANGE:				
Operating	−55 to +125			°C
Storage	−65 to +150			°C
LEAD TEMPERATURE (DURING SOLDERING): At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 s max.	+ 300			°C

*Exceeding this voltage rating will not damage the device unless the peak input signal current (1 mA) is also exceeded.

FUNCTIONAL DIAGRAMS



TO-5 Style Package



Plastic Package

ELECTRICAL CHARACTERISTICS at T_A = 25°C For Equipment Design

CHARACTERISTIC	TEST CONDITIONS: Single Supply V^+ = 30 V, Dual Supply V^+ = 15 V, V^- = 15 V, I_{ABC} = 100 μA Unless Otherwise Specified	LIMITS: Min.	Typ.	Max.	UNITS
INPUT PARAMETERS					
Input Offset Voltage V_{IO}	T_A = 25°C	–	0.4	5	mV
	T_A = 0 to 70°C	–	–	7	mV
Input-Offset-Voltage Change $\lvert\Delta V_{IO}\rvert$	Change in V_{IO} Between I_{ABC} = 100 μA and I_{ABC} = 5 μA	–	1	8	mV
Input Offset Current I_{IO}	T_A = 25°C	–	0.02	0.2	μA
	T_A = 0 to 70°C	–	–	0.3	μA
Input Bias Current I_I	T_A = 25°C	–	0.2	0.50	μA
	T_A = 0 to 70°C	–	–	0.70	μA
Device Dissipation P_D	I_{out} = 0	8	10	12	mW
Common-Mode Rejection Ratio CMRR		70	110	–	dB
Common-Mode Input-Voltage Range V_{ICR}	V^+ = 30 V, High	27	28.8	–	V
	V^+ = 30 V, Low	1.0	0.5	–	V
	V^+ = 15 V	+12	+13.8	–	V
	V^- = 15 V	−14	−14.5	–	V
Unity Gain-Bandwidth	I_C = 7.5 mA, V_{CE} = 15 V, I_{ABC} = 500 μA	–	30	–	MHz
Open-Loop Bandwidth At −3 dB Point BW_{OL}	I_C = 7.5 mA, V_{CE} = 15 V, I_{ABC} = 500 μA	–	4	–	kHz
Total Harmonic Distortion (Class A Operation) THD	P_D = 220 mW	–	0.4	–	%
	P_D = 600 mW	–	1.4	–	
Amplifier Bias Voltage V_{ABC} (Terminal (No.5 to Terminal No.4)		–	0.68	–	V
Input Offset Voltage Temperature Coefficient $\Delta V_{IO}/\Delta T$		–	4	–	μV/°C
Power-Supply Rejection $\Delta V_{IO}/\Delta V$		–	15	150	μV/V
1/F Noise Voltage E_N	f = 10 Hz, I_{ABC} = 50 μA	–	18	–	$\eta V/\sqrt{Hz}$
1/F Noise Current I_N	f = 10 Hz, I_{ABC} = 50 μA	–	1.8	–	$pA/\sqrt{Hz}$
Differential Input Resistance R_I	I_{ABC} = 20 μA	0.50	1	–	MΩ
Differential Input Capacitance C_I	f = 1 MHz, V^+ = 30 V	–	2.6	–	pF

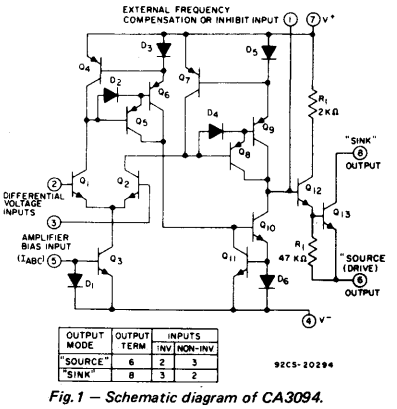


Fig.1 – Schematic diagram of CA3094.

TYPICAL CHARACTERISTICS CURVES

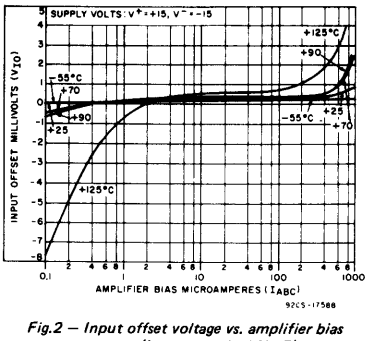


Fig.2 – Input offset voltage vs. amplifier bias current (I_{ABC}, terminal No.5).

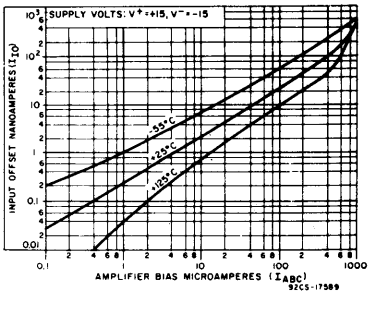


Fig.3 – Input offset current vs. amplifier bias current (I_{ABC}, terminal No.5).

ELECTRICAL CHARACTERISTICS at T_A = 25°C For Equipment Design

CHARACTERISTIC	TEST CONDITIONS: Single Supply V^+ = 30 V, Dual Supply V^+ = 15 V, V^- = 15 V, I_{ABC} = 100 μA Unless Otherwise Specified	LIMITS: Min.	Typ.	Max.	UNITS
OUTPUT PARAMETERS (Differential Input Voltage = 1V)					
Peak Output Voltage: (Terminal No. 6)	V^+ = 30 V				
With Q13 "ON" V^+_{OM}	R_L = 2 kΩ to ground	26	27	–	V
With Q13 "OFF" V_{-OM}		–	0.01	0.05	V
Peak Output Voltage: (Terminal No. 6)	V^+ = +15 V, V^- = -15 V			0.5	
Positive V^+_{OM}	R_L = 2 kΩ to −15 V	+11	+12	–	V
Negative V_{-OM}		–	−14.99	−14.95	V
Peak Output Voltage: (Terminal No. 8)	V^+ = 30 V				
With Q13 "ON" V^+_{OM}	R_L = 2 kΩ to 30 V	29.95	29.99	–	V
With Q13 "OFF" V_{-OM}		–	0.040	–	V
Peak Output Voltage: (Terminal No. 8)	V^+ = 15 V, V^- = − 15 V				
Positive V^+_{OM}	R_L = 2 kΩ to + 15 V	+14.95	+14.99	–	V
Negative V_{-OM}		–	14.96	–	V
Collector-to-Emitter Saturation Voltage (Terminal No. 8) $V_{CE(sat)}$	V^+ = 30 V, I_C = 50 mA, Terminal No.6 grounded	–	0.17	0.80	V
Output Leakage Current (Terminal No. 6 to Terminal No. 4)	V^+ = 30 V	–	2	10	μA
Composite Small-Signal Current Transfer Ratio (Beta) (Q_{12} and Q_{13}) h_{fe}	V^+ = 30 V, V_{CE} = 5 V, I_C = 50 mA	16,000	100,000	–	
Output Capacitance: Terminal No. 6 C_O	f = 1 MHz, All Remaining Terminals Tied to Terminal No. 4	–	5.5	–	pF
Terminal No. 8		–	17	–	pF
TRANSFER PARAMETERS					
Voltage Gain A	V^+ = 30 V, I_{ABC} = 100 μA, ΔV_{out} = 20 V, R_L = 2 kΩ	20,000	100,000	–	V/V
		86	100	–	dB
Forward Transconductance To Terminal No. 1 g_m		1650	2200	2750	μmhos
Slew Rate: Open Loop: Positive Slope	I_{ABC} = 500 μA, R_L = 2 kΩ	–	500	–	V/μs
Negative Slope		–	50	–	V/μs
Unity Gain (Non-Inverting, Compensated)	I_{ABC} = 500 μA, R_L = 2 kΩ	–	0.7	–	V/μs

TYPICAL CHARACTERISTICS CURVES (Cont'd)

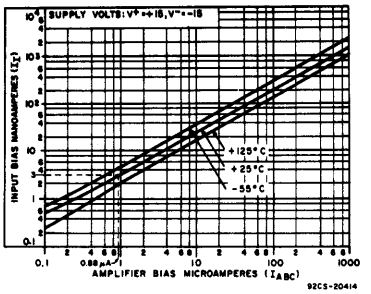


Fig.4 – Input bias current vs. amplifier bias current (I_{ABC}, terminal No.5).

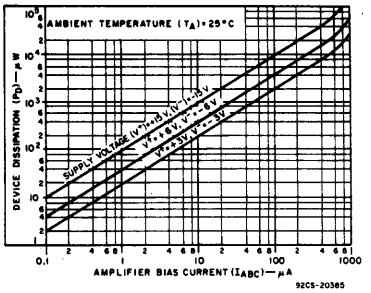


Fig.5 – Device dissipation vs. amplifier bias current (I_{ABC}, terminal No.5).

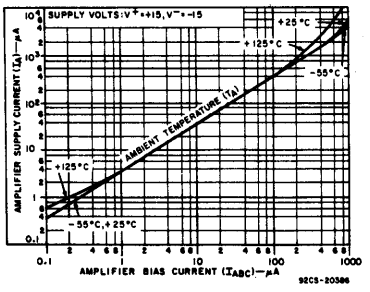


Fig.6 – Amplifier supply current vs. amplifier bias current (I_{ABC}, terminal No.5).

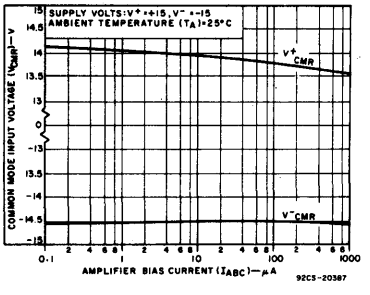


Fig.7 – Common mode input voltage vs. amplifier bias current (I_{ABC}, terminal No.5).

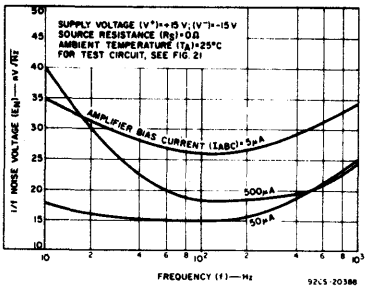


Fig.8 – I/F Noise voltage vs. frequency.

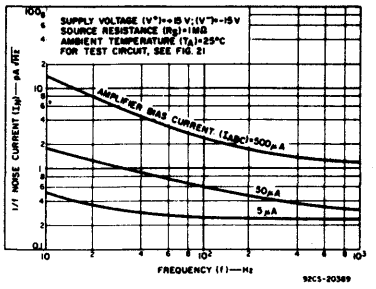


Fig.9 – I/F Noise current vs. frequency.

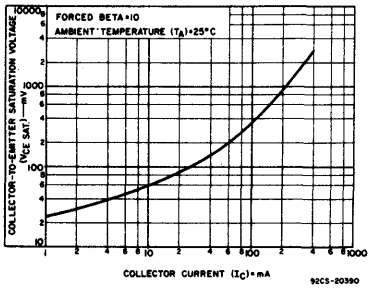


Fig.10 – Collector-emitter saturation voltage vs. collector current of output transistor Q_{13}.

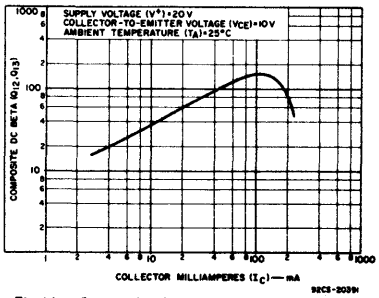


Fig.11 – Composite dc beta vs. collector current of Darlington-connected output transistors (Q_{12}, Q_{13}).

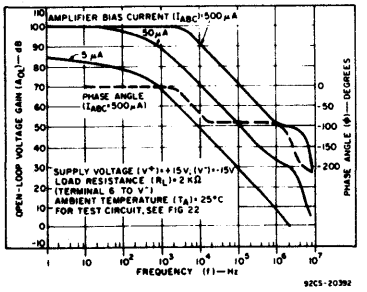


Fig.12 – Open-loop voltage gain vs. frequency.

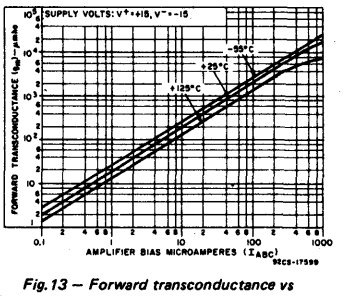


Fig.13 – Forward transconductance vs amplifier bias current.

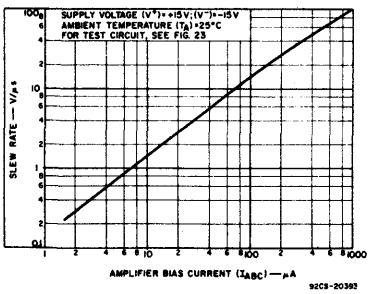


Fig.14 – Slew rate vs amplifier bias current.

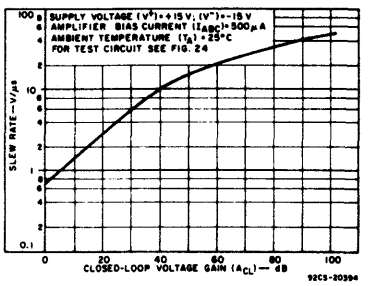


Fig.15 – Slew rate vs closed-loop voltage gain.

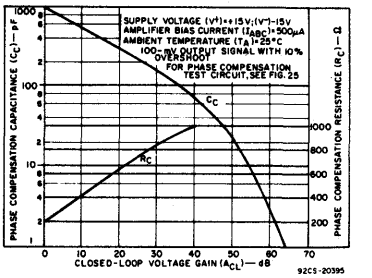


Fig.16 – Phase compensation capacitance and resistance vs closed-loop voltage gain.

OPERATING CONSIDERATIONS

The "Sink" Output (terminal No.8) and the "Drive" Output (terminal No.6) of the CA3094 are not inherently current (or power) limited. Therefore, if a load is connected between terminal No.6 and terminal No.4 (V^- or ground), it is important to connect a current-limiting resistor between terminal 8 and terminal No.7 (V^+) to protect transistor Q_{13} under shorted load conditions. Similarly, if a load is connected between terminal No.8 and terminal No.7, the current-limiting resistor should be connected between terminal No.6 and terminal No.4 or ground. In circuit applications where the emitter of the output transistor is not connected to the most negative potential in the system, it is recommended that a 100-ohm current-limiting resistor be inserted between terminal No.7 and the V^+ supply.

TEST CIRCUITS

I/F Noise Measurement Circuit

When using the CA3094, A, or B audio amplifier circuits, it is frequently necessary to consider the noise performance of the device. Noise measurements are made in the circuit shown in Fig.21. This circuit is a 30-dB, non-inverting amplifier with emitter-follower output and phase compensation from terminal No.2 to ground. Source resistors (Rs) are set to 0.Ω or 1 MΩ for E noise and I noise measurements, respectively. These measurements are made at frequencies of 10, Hz, 100 Hz, and 1 kHz with a 1-Hz measurement bandwidth. Typical values for 1/f noise at 10 Hz and 50 μA I_{ABC} are E_n = 18 nV/$\sqrt{Hz}$ and I_N = 1.8 pA/$\sqrt{Hz}$.

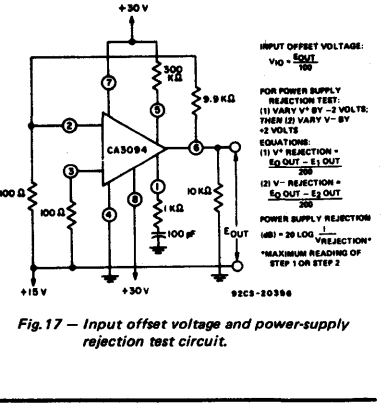


Fig.17 – Input offset voltage and power-supply rejection test circuit.

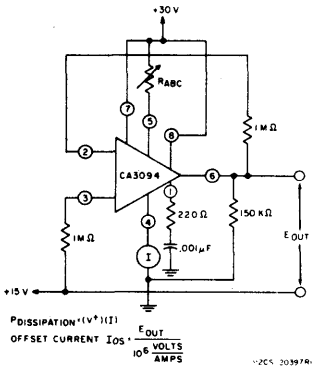


Fig. 18 – Input offset current test circuit.

TEST CIRCUITS (Cont'd)

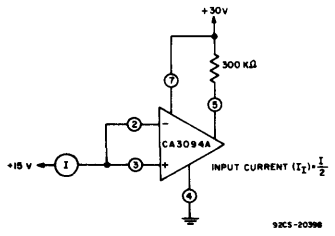


Fig. 19 – Input bias current test circuit.

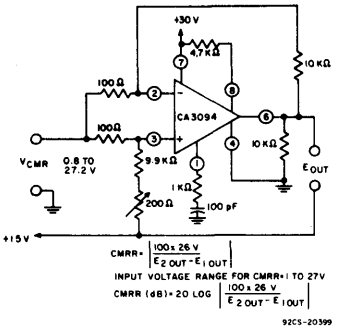


Fig. 20 – Common-mode range and rejection ratio test circuit.

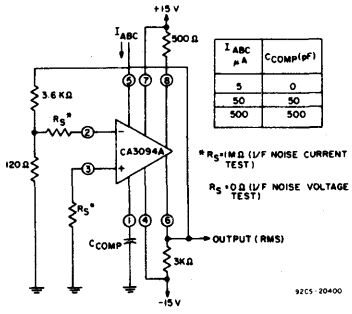


Fig. 21 – 1/F noise test circuit.

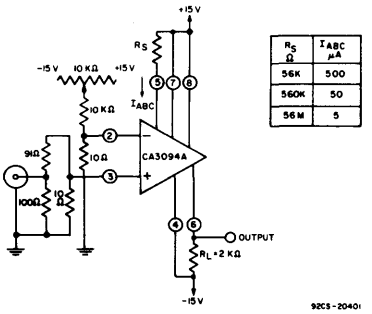


Fig. 22 – Open-loop gain vs frequency test circuit.

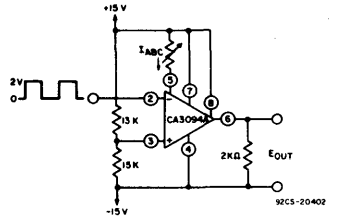


Fig. 23 – Open-loop slew rate vs I_{ABC} test circuit.

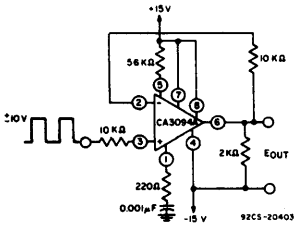


Fig. 24 – Slew rate vs. non-inverting unity gain test circuit.

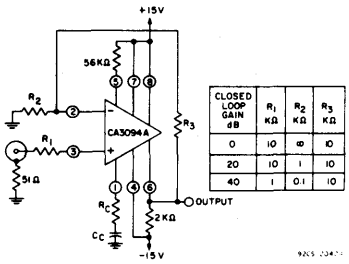


Fig. 25 – Phase compensation test circuit.

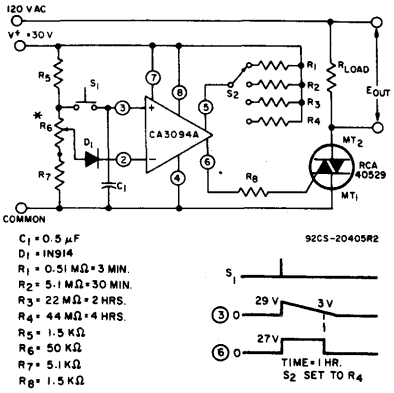


Fig. 26 – Presettable analog timer.

TYPICAL APPLICATIONS

For Additional Application Information, refer to Application Note ICAN-6048 "Some Applications of a Programmable Power/ Switch Amplifier IC".

Design Considerations

The selection of the optimum amplifier bias current (I_{ABC}) depends on –

1. The Desired Sensitivity – the higher the I_{ABC}, the higher the sensitivity – i.e., a greater-drive current capability at the output for a specific voltage change at the input.
2. Required Input Resistance – the lower the I_{ABC}, the higher the input resistance.

If the desired sensitivity and requred input resistance are not known and are to be experimentally determined, or the anticipated equipment design is sufficiently flexible to tolerate a wide range of these parameters, it is recommended that the equipment designer begin his calculations with an I_{ABC} of 100 μA, since the CA3094 is characterized at this value of amplifier bias current.

The CA3094 is extremely versatile and can be used in a wide variety of applications.

TYPICAL APPLICATIONS (Cont'd)

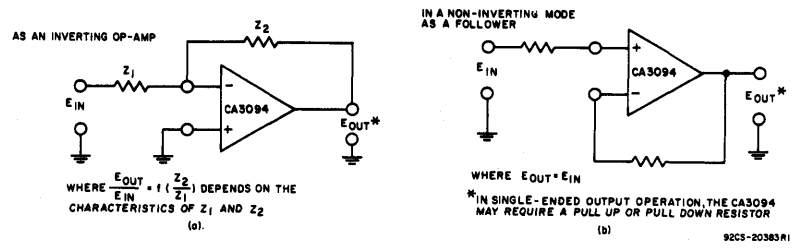
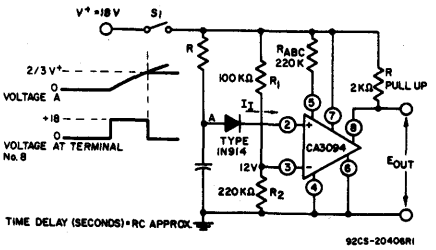


Fig.27 – Application of the CA3094: (a) as an inverting op-amp, and (b) in a non-inverting mode, as a follower.



Problem: To calculate the maximum value of R required to switch a 100-mA output current comparator

Given: $I_{ABC} = 5\ \mu A$, $R_{ABC} = 3.6\ M\Omega \approx \frac{18\ V}{5\ \mu A}$

$I_1 = 500$ nA @ $I_{ABC} = 100\ \mu A$ (from Fig.4)

$I_1 = 5\ \mu A$ can be determined by drawing a line on Fig.4 through $I_{ABC} = 100\ \mu A$ and $I_B = 500$ nA parallel to the typical $T_A = 25^oC$ curve.

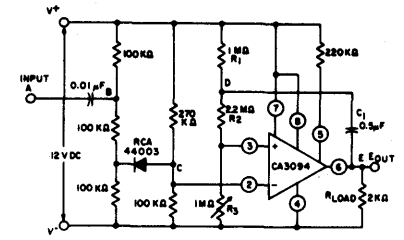
Then: $I_1 = 33$ nA @ $I_{ABC} = 5\ \mu A$

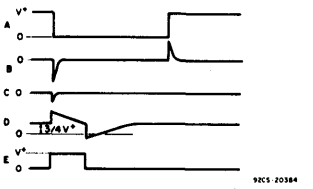
$$R_{max} = \frac{18-12\ volts}{33nA} = 180\ M\Omega\ @\ T_A = 25^oC$$

$$R_{max} = 180\ M\Omega \times 2/3^* = 120\ M\Omega\ @\ T_A = -55^oC$$

* Ratio of I_1 at $T_A = +25^oC$ to I_1 at $T_A = -55^oC$ for any given value of I_{ABC}.

Fig.28 – RC timer.





On a negative-going transient at input (A), a negative pulse at C will turn "on" the CA3094, and the output (E) will go from a low to a high level.

At the end of the time constant determined by C_1, R_1, R_2, R_3, the CA3094 will return to the "off" state and the output will be pulled low by R_{LOAD}. This condition will be independent of the interval when input A returns to a high level.

Fig.29 – RC timer triggered by external negative pulse.

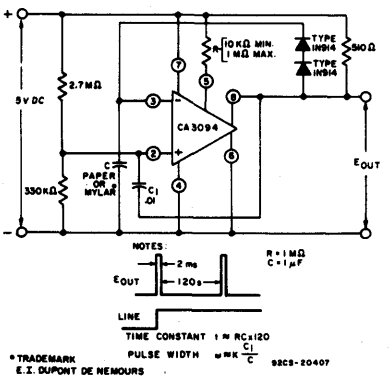


Fig.30 – Free-running pulse generator.

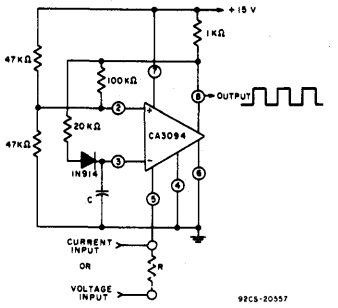


Fig.31 – Current or voltage-controlled oscillator.

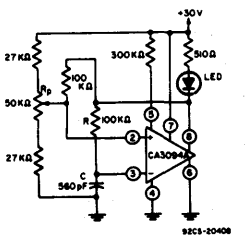


Fig.32 – Single-supply astable multivibrator.

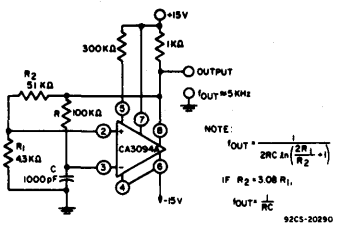
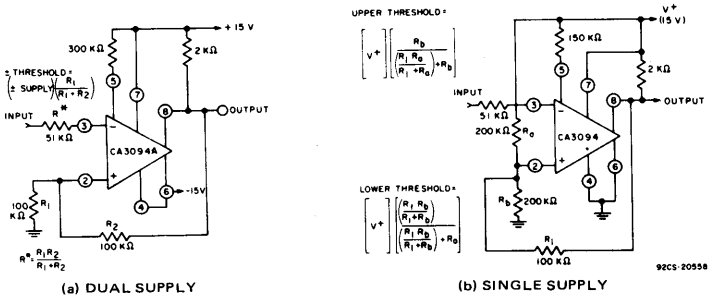


Fig.33 – Dual-supply astable multivibrator.

TYPICAL APPLICATIONS (Cont'd)



(a) DUAL SUPPLY (b) SINGLE SUPPLY

Fig.34 – Comparators (threshold detectors) –dual- and single-supply types.

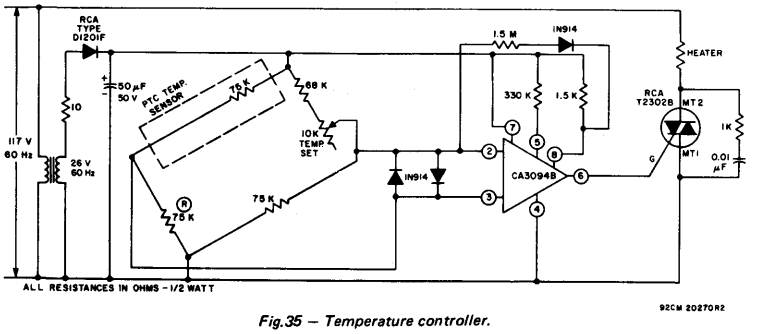


Fig.35 – Temperature controller.

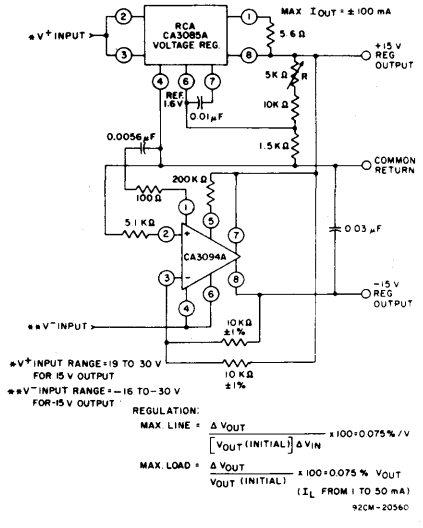


Fig.36 – Dual-voltage tracking regulator.

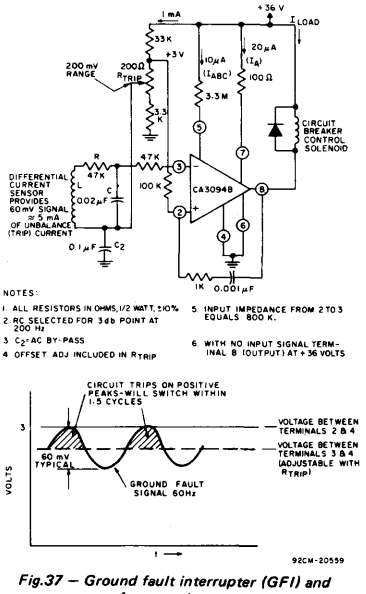
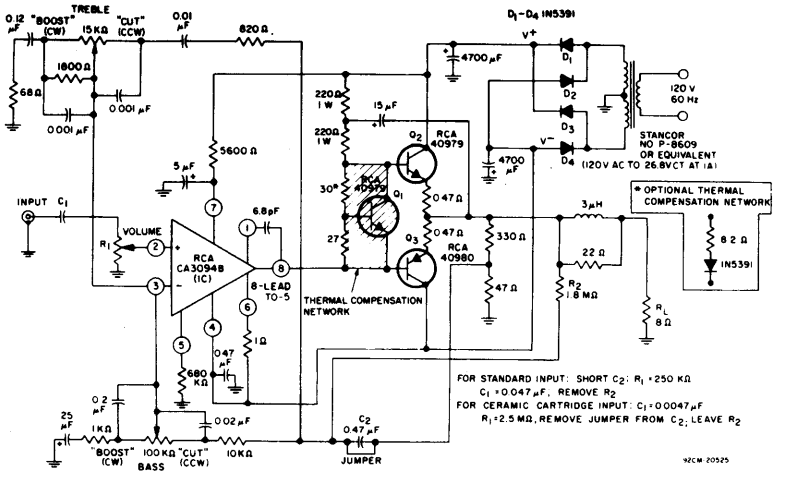


Fig.37 – Ground fault interrupter (GFI) and waveform pertinent to ground fault detector.



TYPICAL PERFORMANCE DATA
For 12-W Audio Amplifier Circuit

Power Output (8Ω load, Tone Control set at "Flat")		
Music (at 5% THD, regulated supply)	15	W
Continuous (at 0.2% IMD, 60 Hz & 2 kHz mixed in a 4:1 ratio, unregulated supply) See Fig. 8 in ICAN-6048	12	W
Total Harmonic Distortion		
At 1 W, unregulated supply	0.05	%
At 12 W, unregulated supply	0.57	%
Voltage Gain	40	dB
Hum and Noise (Below continuous Power Output)	83	dB
Input Resistance	250	kΩ
Tone Control Range	See Fig. 9 in ICAN-6048	

Fig. 38 – 12-watt amplifier circuit featuring true complementary-symmetry output stage with CA3094 in driver stage.

Super-Beta Transistor Array

Differential Cascode Amplifier Plus 3 Independent Transistors

RCA-CA3095E is a monolithic array of transistors connected as a super-beta differential cascode amplifier with three independent n-p-n transistors. (Refer to Fig. 1 for following description.)

The differential cascode amplifier incorporates two cascode amplifiers consisting of transistors Q1, Q3 and Q2, Q4, respectively, plus a voltage-limiting circuit, consisting of diodes D1, D2 and p-n-p transistor Q5. Two of these transistors, Q1 and Q2, are super-beta types that have an $h_{FE} > 1000$ and are capable of operating over a wide current range of 1 μA to 2 mA. Each of these types comprises the input section of its respective cascode amplifier. The output section of each cascode amplifier employs a conventional n-p-n transistor, Q3, Q4, respectively. The output signal is obtained at the collectors of these transistors. See Operating Considerations for bias considerations of the differential cascode amplifier.

The exceptionally high-beta characteristics of Q1 and Q2, plus the large signal-voltage swing capability of Q3 and Q4, make the composite differential cascode amplifier an excellent choice for a broad range of small-signal, high-input-impedance amplifier applications including low-noise video amplifiers. This amplifier is also recommended for use in long-interval timers, oscillators, and long-duration one-shot applications.

The independent transistors, Q6, Q7 and Q8, are high-voltage silicon n-p-n conventional types for general use in signal processing systems in the frequency range from dc through vhf. Separate terminals for each of these transistors permit maximum flexibility in circuit design.

The CA3095E is supplied in a 16-lead dual-in-line plastic package and operates over the ambient temperature range of −55°C to +125°C

Features

- Two super-beta n-p-n transistors – $h_{FE} > 1000$
- Voltage-limiting circuitry (D1, D2, Q5)
- Operation possible at I_{IB} down to < 1 nA
- Matched pair (Q1 and Q2) –
  V_{IO} = 5 mV max. at I_C = 100 μA dc
  I_{IO} = 20 nA max. at I_C = 100 μA dc
- Wide current range – < 1 μA to 2 mA

Independent Transistors:

- h_{FE} = 300 typ. for each transistor
- Wide current range – < 1 μA to 10 mA
- Matched general-purpose transistors
- High voltage – V_{CBO} = 45 V max.

Applications

Differential Cascode Amplifier:

- Super-beta pre-amplifier for op-amp
- High-impedance dc meter amplifier
- Low-noise video amplifier
- Piezoelectric transducer amplifier
- Long-interval timer
- Long-duration one-shot multivibrator
- Comparator with high-input impedance
- Long-time-constant integrator
- Photocell amplifier
- Low-noise amplifier—for operation from high-source impedances

Independent Transistors:

- General use in signal processing systems in dc through vhf range

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25 °C*

Power Dissipation:		
Any One Transistor	300	mW
Total Package–		
Up to 25 °C	750	mW
Above 25 °C ..... derate linearly	6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−55 to +150	°C
Lead Temperature (During Soldering):		
At distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C
Voltage and Current Ratings Apply for Each Specified Transistor:		
Super-Beta Transistors (Q1, Q2)–		
Collector-to-Base Voltage (V_{CBO})	6	V
Emitter-to-Base Voltage (V_{EBO})	6	V
Collector-to-Substrate Voltage (V_{CIO})*	45	V
Collector Current (I_C)	50	mA
Base Current (I_B)	20	mA
Conventional N-P-N Transistors (Q3, Q4, Q6, Q7, Q8)–		
Collector-to-Base Voltage (V_{CBO})	45	V
Collector-to-Emitter Voltage (V_{CEO})	35	V
Emitter-to-Base Voltage (V_{EBO})	6	V
Collector-to-Substrate Voltage (V_{CIO})*	45	V
Collector Current (I_C)	50	mA
Base Current (I_B)	20	mA
Conventional P-N-P Transistor (Q5)–		
Collector-to-Base Voltage (V_{CBO})	−45	V
Collector-to-Emitter Voltage (V_{CEO})	−35	V
Limiting Circuit Current ($I_{Pin\ 11}$)	20	mA

* The collector of each transistor is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal should be maintained at either dc or signal (ac) ground. A suitable bypass capacitor can be used to establish a signal ground.

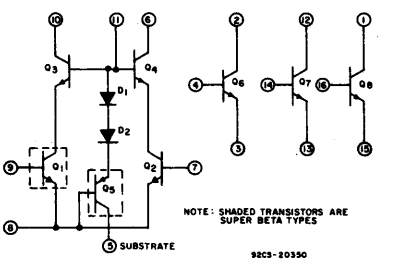


Fig. 1 – Functional diagram.

STATIC CHARACTERISTICS

Characteristics	Symbol	Test Conditions (T_A = 25 °C)		Limits Min.	Limits Typ.	Limits Max.	Units
Characteristics Apply for Each Super-Beta Cascode Amplifier Transistor Pair (Q1, Q3) and (Q2, Q4), Unless Indicated Otherwise							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0 See Note 1		6	–	–	V
Emitter-to-Base Breakdown Voltage (Applies only to Q1 & Q2)	$V_{(BR)EBO}$	I_E = 100 μA, I_C = 0	Term. 9 to 8 or Term. 7 to 8	6	8		V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 100 μA, I_B = I_E = 0		45	–	–	V
Collector Cutoff Current	I_{CER}	V_{6-8} or V_{10-8} = 10 V, I_{11} = 100 μA, R_{BE} = 100 MΩ			–	100	nA
DC Forward-Current Transfer Ratio	h_{FE}	V_{10-8} = 5 V or V_{6-8} = 5 V	I_C = 1 mA	–	1500	–	
			I_C = 100 μA	1000	2000	5000	
			I_C = 10 μA	–	1500	–	
Base-to-Emitter Voltage (Applies only to Q1 & Q2)	V_{BE}	I_C = 100 μA, V_{6-8} or V_{10-8} = 5 V		0.50	0.59	0.68	V
Saturation Voltage	V_{sat}	I_6 or I_{10} = 1 mA, I_{11} = 100 μA, I_7 or I_9 = 100 μA		–	0.22	0.7	V
For Cascode Amplifiers as a Differential Matched Pair							
Magnitude of Input-Offset Voltage	$\lvert I_{IO} \rvert$	I_C = 100 μA		–	1	5	mV
Magnitude of Input-Offset Current	$\lvert I_{IO} \rvert$	V_{6-8} = V_{10-8} = 5 V		–	4	20	nA
Magnitude of Input-Offset Voltage Drift (Temp. Coeff.)	$\lvert \frac{\Delta V_{IO}}{\Delta T} \rvert$			–	3.3	–	μV/°C
Magnitude of Input-Offset Current Drift (Temp. Coeff.)	$\lvert \frac{\Delta I_{IO}}{\Delta T} \rvert$			–	0.05	–	nA/°C

Note 1: Terminal No. 9 to terminals 10 and 11 connected or terminal No. 7 to terminals 6 and 11 connected.

Test Circuits for Measurement of Super-Beta Cascode Amplifier Characteristics

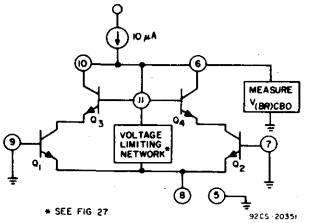


Fig. 2 – $V_{(BR)CBO}$ test circuit.

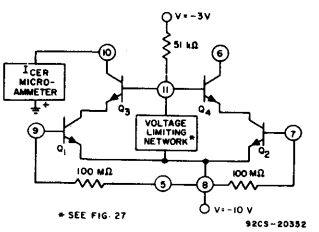


Fig. 3 – I_{CER} test circuit

CA3095E

STATIC CHARACTERISTICS (Cont'd)

Characteristics	Symbol	Test Conditions $T_A = 25\,^{\circ}C$		Limits Min.	Typ.	Max.	Units
For Each Conventional n-p-n Transistor (Q3, Q4, Q6, Q7, Q8)							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10\ \mu A$, $I_E = 0$		45	95	--	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$		35	50	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 100\ \mu A$, $I_C = 0$		6	8	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_{CI} = 100\ \mu A$, $I_B = I_E = 0$		45	95	–	V
Collector Cutoff Current	I_{CEO}	$V_{CE} = 10$ V, $I_B = 0$		–	–	100	nA
Collector Cutoff Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$		–	–	10	nA
DC Forward-Current Transfer Ratio	h_{FE}	$V_{CE} = 5$ V	$I_C = 10$ mA	–	210	–	
			$I_C = 1$ mA	150	300	500	
			$I_C = 10\ \mu A$	–	180	–	
Base-to-Emitter Voltage	V_{BE}	$I_C = 1$ mA, $V_{CE} = 5$ V		0.60	0.69	0.78	V
Collector-to-Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C = 10$ mA, $I_B = 1$ mA		–	0.22	0.7	V

Dynamic Characteristics

Characteristics	Symbol	Test Conditions $T_A = 25\,^{\circ}C$	Limits Min.	Typ.	Max.	Units
Characteristics Apply for Each Super-Beta Cascode Amplifier Transistor Pair (Q1, Q3), Unless Indicated Otherwise						
Gain-Bandwidth Product	f_T	$I_C = 100\ \mu A$, $V_{6-8} = V_{10-8} = 5$ V,	–	78	–	MHz
Noise Voltage (Referred to Input) For Differential Amplifier Operation	E_N	$I_C = 50\ \mu A$, f = 10 Hz	–	13	–	nV/$\sqrt{Hz}$
Noise Current (Referred to Input) For Differential Amplifier Operation	I_N	$I_C = 5\ \mu A$, f = 10 Hz	–	0.12	–	pA/$\sqrt{Hz}$
Collector-to-Base Capacitance	C_{CB}	$V_{6-7} = V_{10-9} = 5$ V, $I_E = 0$	–	0.3		pF
Collector-to-Substrate Capacitance	C_{CIO}	$V_{6-5} = V_{10-5} = 5$ V, $I_B = 0$	–	3.0	–	pF
For Each Conventional Transistor (Q3 through Q8)						
Gain-Bandwidth Product	f_T	$I_C = 100\ \mu A$, $V_{CE} = 5$ V	–	100	–	MHz
		$I_C = 3$ mA, $V_{CE} = 5$ V	–	320	–	
Noise Voltage (Referred to Input)	E_N	$I_C = 100\ \mu A$, $V_{CE} = 5$ V, f = 10 Hz	–	5	–	nV/$\sqrt{Hz}$
Noise Current (Referred to Input)	I_N	$I_C = 10\ \mu A$, $V_{CE} = 5$ V, f = 10 Hz	–	0.8	–	pA/$\sqrt{Hz}$
Collector-to-Base Capacitance	C_{CB}	$V_{CB} = 5$ V, $I_E = 0$	–	0.4	–	pF
Collector-to-Substrate Capacitance	C_{CIO}	$V_{CI} = 5$ V, $I_B = 0$	–	2	–	pF

* Curve plotted for I_{CEO} characteristic.

Test Circuits for Measurement of Super-Beta Cascode Amplifier Characteristics

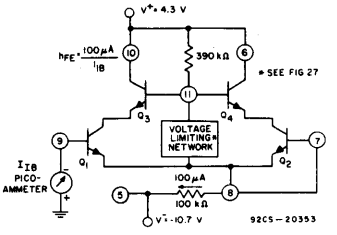


Fig.4–DC Beta (h_{FE}) test circuit.

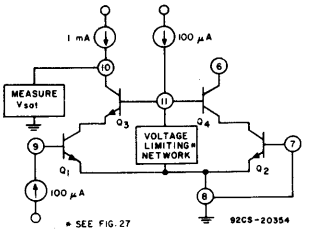


Fig.5–V_{sat} test circuit for super-beta cascode pairs.

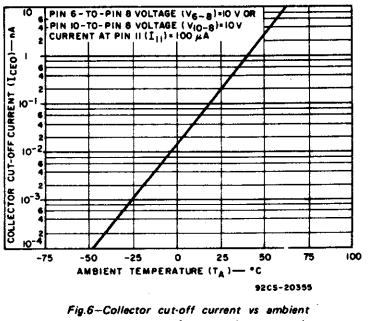


Fig.6–Collector cut-off current vs ambient temperature for super-beta cascode pairs.

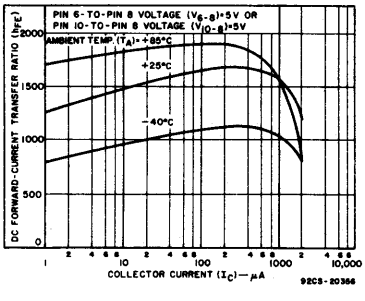


Fig.7–h_{FE} vs. I_C for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

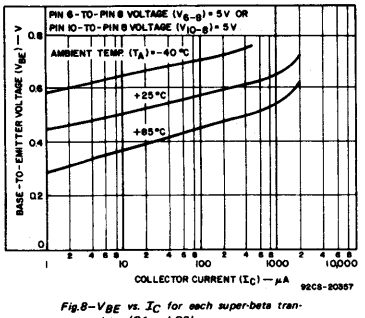


Fig.8–V_{BE} vs. I_C for each super-beta transistor (Q1 and Q2).

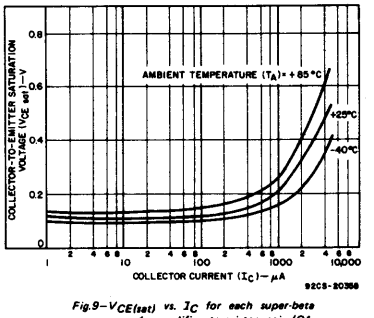


Fig.9–$V_{CE(sat)}$ vs. I_C for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

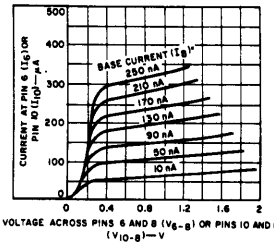


Fig.10–I-V characteristics for the super-beta cascode pairs.

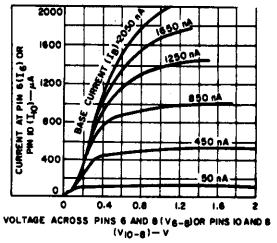


Fig.11–I-V characteristics for the super-beta cascode pairs.

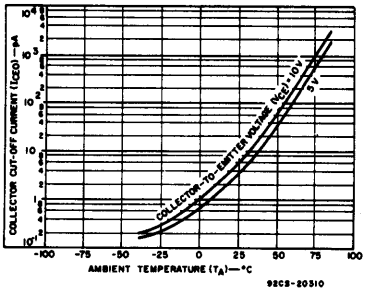


Fig.12–Collector cutoff current vs ambient temperature for the conventional transistors (V_{CE} = 5 V, 10 V).

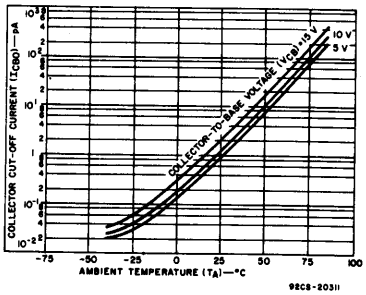


Fig.13–Collector cutoff current vs ambient temperature for the conventional transistors (V_{CB} = 5 V, 10 V, 15 V).

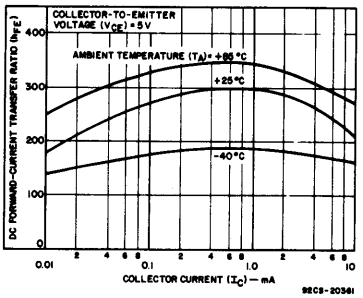


Fig.14–h_{FE} vs. I_C for each conventional transistor (Q6, Q7, Q8).

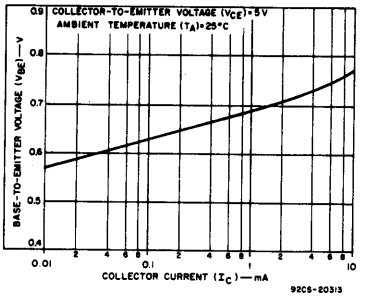


Fig.15–V_{BE} as a function of collector current for the conventional transistors.

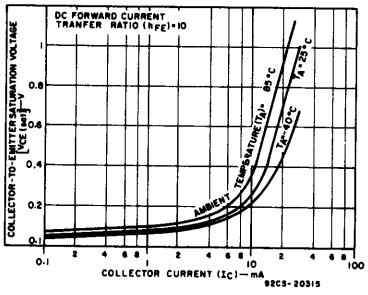


Fig.16–$V_{CE(sat)}$ as a function of collector current for the conventional transistors.

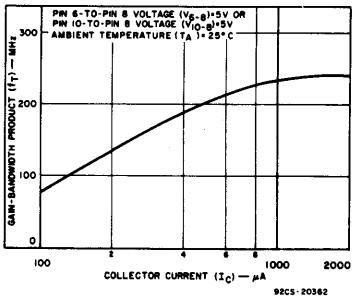


Fig.17–Gain bandwidth product vs collector current for the super-beta cascade pairs.

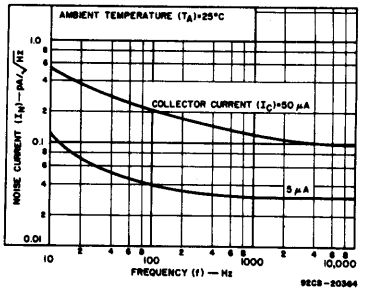


Fig.18–I_N vs. f for each super-beta cascade amplifier transistor pair (Q1, Q3) and (Q2, Q4).

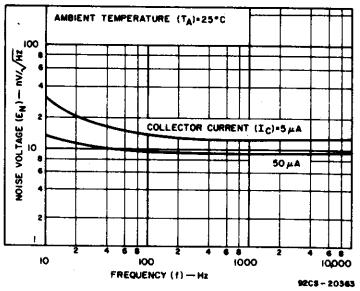


Fig.19–E_N vs. f for each super-beta cascade amplifier transistor pair (Q1, Q3) and (Q2, Q4).

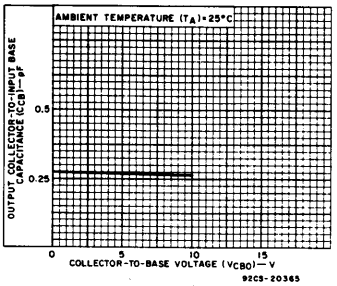


Fig.20–C_{CB} vs. V_{CBO} for each super-beta cascade amplifier transistor pair (Q1, Q3) and (Q2, Q4).

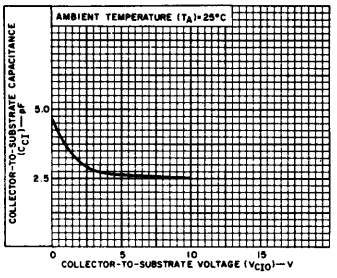


Fig.21–C_{CI} vs. V_{CIO} for each super-beta cascade amplifier transistor pair (Q1, Q3) and (Q2, Q4).

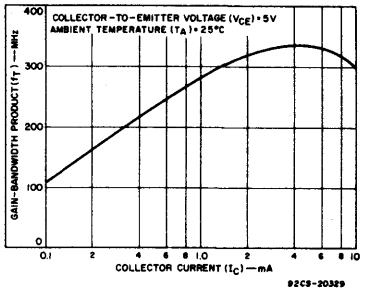


Fig.22–Gain bandwidth product vs collector current for the conventional transistors.

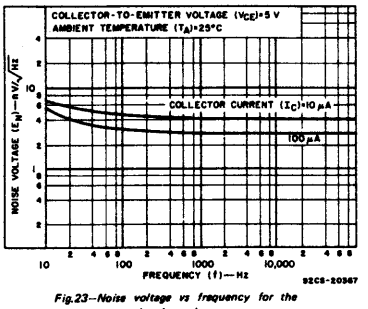


Fig.23–Noise voltage vs frequency for the conventional transistors.

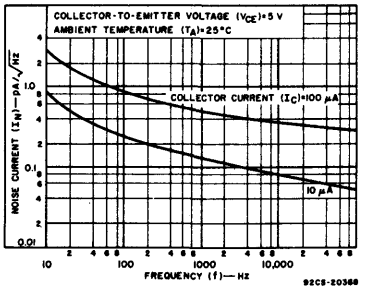


Fig.24–I_N vs. f for each conventional transistor (Q6, Q7, Q8).

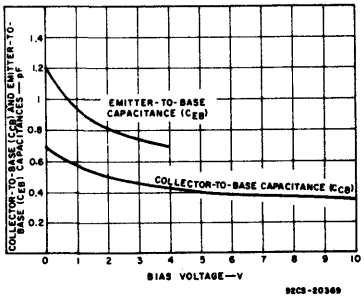


Fig.25–Collector-to-base and emitter-to-base capacitances vs bias voltage for the conventional transistors.

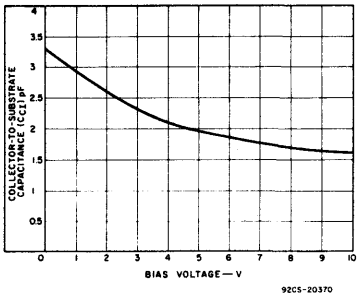


Fig.26–Collector-to-substrate capacitance vs bias voltage for the conventional transistors.

Operating Considerations

Operation Considerations for the Super-Beta Differential Cascode Amplifier

An internal voltage-limiting network (diodes D1, D2 and p-n-p transistor Q5) incorporated in the differential cascode amplifier, assures that the applied collector-to-emitter voltage of each super-beta unit is maintained below two volts. Fig. 27 shows a typical bias arrangement of the super-beta differential cascode amplifier.

Bias current for this network must be supplied by an external source. This bias current can be obtained by simply connecting a resistor from Pin 11 to the positive supply of the differential amplifier. The return path for most of the bias current is through the substrate, Pin 5, rather than through the common emitter, Pin 8. This arrangement provides superior common-mode and power-supply rejection. As a general rule-of-thumb, the current supplied into Pin 11 should be approximately 0.04 to 0.1 times the value of the quiescent current of Pin 8.

TYPICAL APPLICATIONS

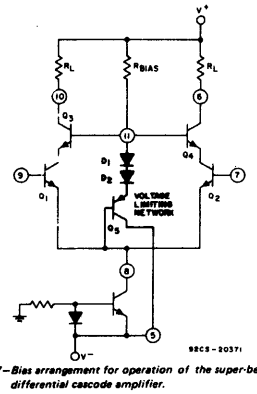


Fig.27–Bias arrangement for operation of the super-beta differential cascode amplifier.

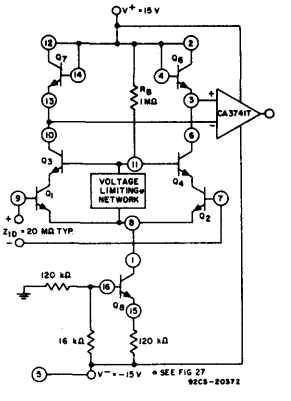


Fig.28–Super-beta Op-Amp with diode drive network.

TYPICAL APPLICATIONS (Cont'd)

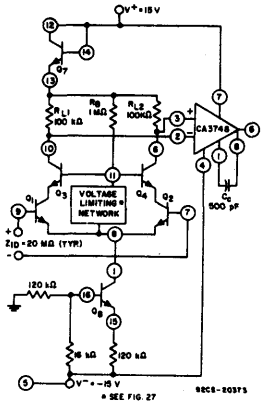


Fig.29–Super-beta Op-Amp with resistor drive network.

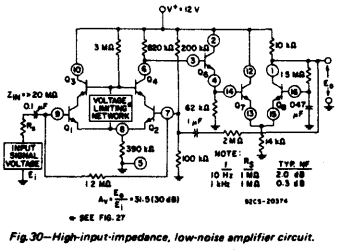


Fig.30–High-input-impedance, low-noise amplifier circuit.

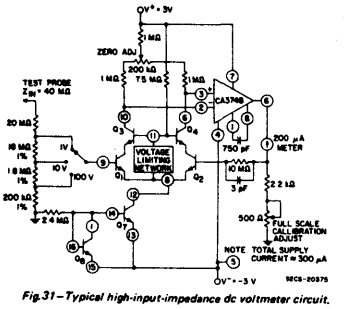


Fig.31–Typical high-input-impedance dc voltmeter circuit.

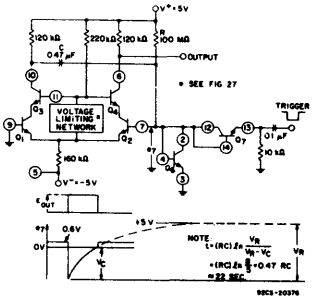


Fig.32–Long-delay monostable multivibrator circuit.

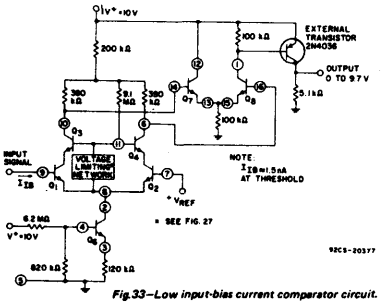


Fig.33–Low input-bias current comparator circuit.

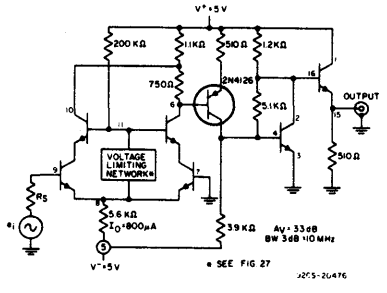


Fig.34–CA3095E wideband amplifier.

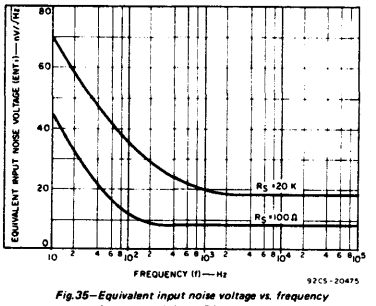


Fig.35–Equivalent input noise voltage vs. frequency for circuit of figure 34.

CA3096, CA3096A, CA3096C

N-P-N/P-N-P Transistor-Array

Five-Independent Transistors: Three n-p-n and Two p-n-p

RCA-CA3096CE, CA3096E, and CA3096AE are general-purpose high-voltage silicon transistor arrays. Each array consists of five independent transistors (two p-n-p and three n-p-n types) on a common substrate, which has a separate connection. Independent connections for each transistor permit maximum flexibility in circuit design.

Types CA3096AE, CA3096E, and CA3096CE are identical, except that the CA3096AE specifications include parameter matching and greater stringency in I_{CBO}, I_{CEO}, and V_{CE}(SAT). The CA3096CE is a relaxed version of the CA3096E.

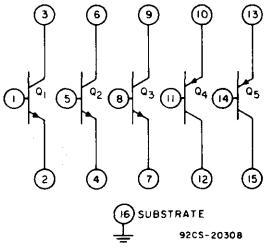
The CA3096CE, CA3096E, and CA3096AE are supplied in 16-lead dual-in-line plastic packages.

Applications:

- **Differential Amplifiers**
- **DC Amplifiers**
- **Sense Amplifiers**
- **Level Shifters**
- **Timers**
- **Lamp and Relay Drivers**
- **Thyristor Firing Circuits**
- **Temperature-Compensated Amplifiers**
- **Operational Amplifiers**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

	EACH N-P-N	EACH P-N-P	
COLLECTOR-TO-EMITTER VOLTAGE, V_{CEO}:			
CA3096AE, CA3096E	35	−40	V
CA3096CE	24	−24	V
COLLECTOR-TO-BASE VOLTAGE, V_{CBO}:			
CA3096AE, CA3096E	45	−40	V
CA3096CE	30	−24	V
COLLECTOR-TO-SUBSTRATE VOLTAGE, V_{CIO}:			
CA3096AE, CA3096E	45	–	V
CA3096CE	30	–	V
EMITTER-TO-SUBSTRATE VOLTAGE, V_{EIO}:			
CA3096AE, CA3096E	–	−40	V
CA3096CE	–	−24	V
EMITTER-TO-BASE VOLTAGE, V_{EBO}:			
CA3096E, CA3096E	6	−40	V
CA3096CE	6	−24	V
COLLECTOR CURRENT, I_C (All Types)	50	−10	mA
POWER DISSIPATION, P_D:			
Up to T_A = 55°C:			
Device (Total)	750		mW
Each Transistor	200		mW
Above T_A = 55°C derate linearly at	6.67		mW/°C
AMBIENT-TEMPERATURE RANGE, T_A:			
Operating		−55 to +125	°C
Storage		−65 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING):			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 s max.		265	°C



Schematic Diagram

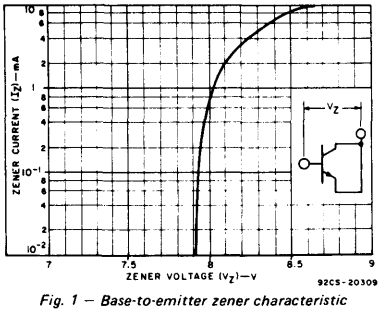


Fig. 1 – Base-to-emitter zener characteristic (n-p-n).

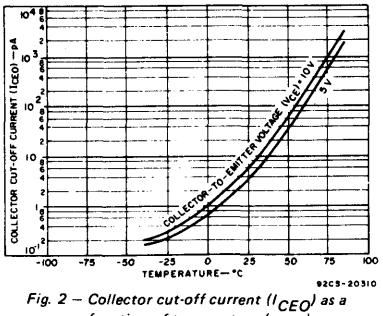


Fig. 2 – Collector cut-off current (I_{CEO}) as a function of temperature (n-p-n).

CA3096AE, CA3096E, CA3096CE ESSENTIAL DIFFERENCES

CHARACTERISTIC		CA3096AE	CA3096E	CA3096CE
$V_{(BR)CEO}$ (V) Min.	n-p-n	35	35	24
	p-n-p	−40	−40	−24
$V_{(BR)CBO}$ (V) Min.	n-p-n	45	45	30
	p-n-p	−40	−40	−24
h_{FE} @ 1 mA	n-p-n	150-500	150-500	100-670
	p-n-p	20-150	20-150	15-200
h_{FE} @ 100 μA	p-n-p	40-200	40-200	30-300
I_{CBO} (nA) Max.	n-p-n	40	100	100
	p-n-p	−40	−100	−100
I_{CEO} (nA) Max.	n-p-n	100	1000	1000
	p-n-p	−100	−1000	−1000
V_{CE}(SAT) (V) Max.	p-n-p	0.5	0.7	0.7
$\|V_{IO}\|$ (mV) Max.	n-p-n	5	–	–
	p-n-p	5	–	–
$\|I_{IO}\|$ (μA) Max.	n-p-n	0.6	–	–
	p-n-p	0.25	–	–

STATIC ELECTRICAL CHARACTERISTICS at $T_A = 25°C$
For Equipment Design

CHARACTERISTIC	TEST CONDITIONS	LIMITS									UNITS
		CA3096AE			CA3096E			CA3096CE			
		Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
For Each n-p-n Transistor											
I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$	–	0.001	40	–	0.001	100	–	0.001	100	nA
I_{CEO}	$V_{CE} = 10$ V, $I_B = 0$	–	0.006	100	–	0.006	1000	–	0.006	1000	nA
$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$	35	50	–	35	50	–	24	35	–	V
$V_{(BR)CBO}$	$I_C = 10\,\mu$A, $I_E = 0$	45	100	–	45	100	–	30	80	–	V
$V_{(BR)CIO}$	$I_{CI} = 10\,\mu$A, $I_B = I_E = 0$	45	100	–	45	100	–	30	80	–	V
$V_{(BR)EBO}$	$I_E = 10\,\mu$A, $I_C = 0$	6	8	–	6	8	–	6	8	–	V
V_Z	$I_Z = 10\,\mu$A	6	7.9	9.8	6	7.9	9.8	6	7.9	9.8	V
$V_{CE(SAT)}$	$I_C = 10$ mA, $I_B = 1$ mA	–	0.24	0.5	–	0.24	0.7	–	0.24	0.7	V
V_{BE}	$I_C = 1$ mA, $V_{CE} = 5$ V	0.6	0.69	0.78	0.6	0.69	0.78	0.6	0.69	0.78	V
h_{FE}		150	390	500	150	390	500	100	390	670	
$\lvert\Delta V_{BE}/\Delta T\rvert$	$I_C = 1$ mA, $V_{CE} = 5$ V	–	1.9	–	–	1.9	–	–	1.9	–	mV/°C

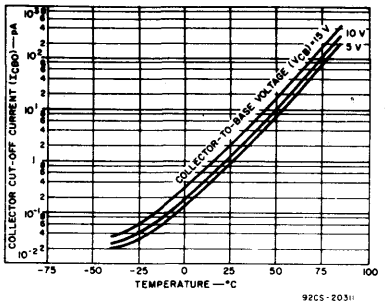


Fig. 3 – Collector cut-off current (I_{CBO}) as a function of temperature (n-p-n).

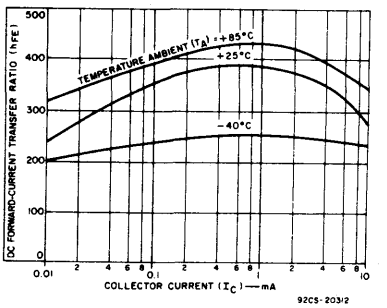


Fig. 4 – Transistor (n-p-n) h_{FE} as a function of collector current.

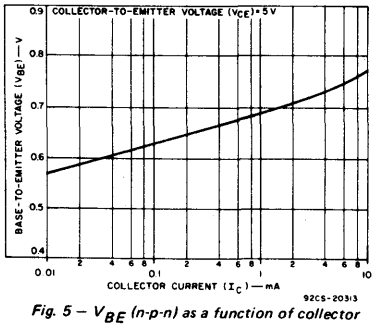


Fig. 5 – V_{BE} (n-p-n) as a function of collector current.

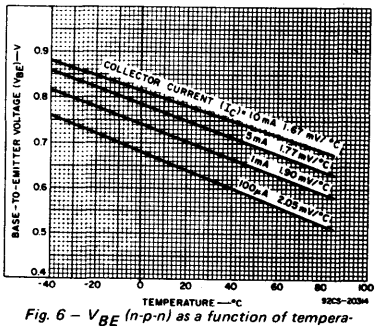


Fig. 6 – V_{BE} (n-p-n) as a function of temperature.

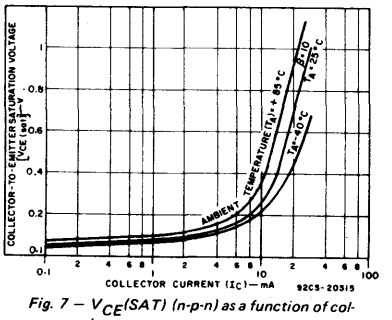


Fig. 7 – $V_{CE(SAT)}$ (n-p-n) as a function of collector current.

STATIC ELECTRICAL CHARACTERISTICS at T_A = 25°C (Cont'd)
For Equipment Design

CHARACTERISTIC	TEST CONDITIONS	LIMITS									UNITS
		CA3096AE			CA3096E			CA3096CE			
		Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
For Each p-n-p Transistor											
I_{CBO}	V_{CB} = −10 V, I_E = 0	–	−0.006	−40	–	−0.06	−100	–	−0.06	−100	nA
I_{CEO}	V_{CE} = −10V, I_B = 0	–	−0.12	−100	–	−0.12	−1000	–	−0.12	−1000	nA
$V_{(BR)CEO}$	I_C = −100 μA, I_B = 0	−40	−75	–	−40	−75	–	−24	−30	–	V
$V_{(BR)CBO}$	I_C = −10 μA, I_E = 0	−40	−80	–	−40	−80	–	−24	−60	–	V
$V_{(BR)EBO}$	I_E = −10 μA, I_C = 0	−40	−100	–	−40	−100	–	−24	−80	–	V
$V_{(BR)EIO}$	I_{EI} = 10 μA, I_B = I_C = 0	−40	−100	–	−40	−100	–	−24	−80	–	V
$V_{CE(SAT)}$	I_C = −1 mA, I_B = −100 μA	–	−0.16	−0.4	–	−0.16	−0.4	–	−0.16	−0.4	V
V_{BE}	I_C = −100 μA, V_{CE} = −5 V	−0.5	−0.6	−0.7	−0.5	−0.6	−0.7	−0.5	−0.6	−0.7	V
h_{FE}	I_C = −100 μA, V_{CE} = −5 V	40	85	200	40	85	200	30	85	300	
	I_C = −1 mA, V_{CE} = −5 V	20	47	150	20	47	150	15	47	200	
$\lvert\Delta V_{BE}/\Delta T\rvert$	I_C = −100 μA, V_{CE} = −5 V	–	2.2	–	–	2.2	–	–	2.2	–	mV/°C

STATIC ELECTRICAL CHARACTERISTICS at T_A = 25°C (CA3096AE Only)
For Equipment Design

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		CA3096AE			
		Min.	Typ.	Max.	
For Transistors Q1 and Q2 (as a Differential Amplifier)					
Absolute Input Offset Voltage, $\lvert V_{IO}\rvert$	V_{CE} = 5 V, I_C = 1 mA	–	0.3	5	mV
Absolute Input Offset Current, $\lvert I_{IO}\rvert$		–	0.07	0.6	μA
Absolute Input Offset Voltage Temperature Coefficient, $\frac{\lvert\Delta V_{IO}\rvert}{\Delta T}$		–	1.1	–	μV/°C
For Transistors Q4 and Q5 (As a Differential Amplifier)					
Absolute Input Offset Voltage, $\lvert V_{IO}\rvert$	V_{CE} = −5 V, I_C = −100 μA, R_S = 0	–	0.15	5	mV
Absolute Input Offset Current, $\lvert I_{IO}\rvert$		–	2	250	nA
Absolute Input Offset Voltage Temperature Coefficient, $\frac{\lvert\Delta V_{IO}\rvert}{\Delta T}$		–	0.54	–	μV/°C

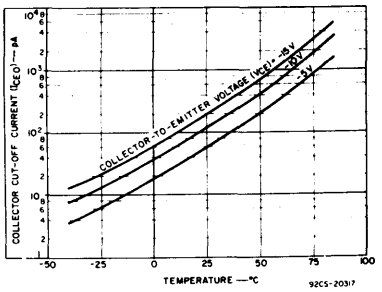


Fig. 8 – Collector cut-off current (I_{CEO}) as a function of temperature (p-n-p).

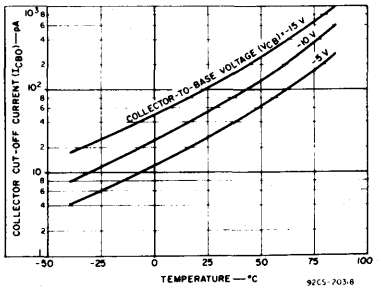


Fig. 9 – Collector cut-off current (I_{CBO}) as a function of temperature (p-n-p).

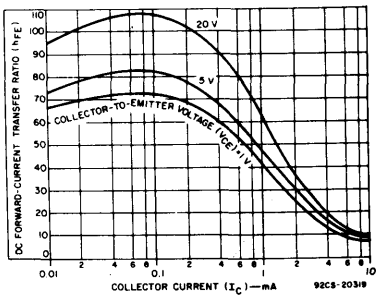


Fig. 10 – Transistor (p-n-p) h_{FE} as a function of collector current.

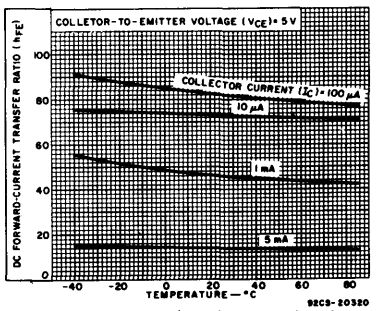


Fig. 11 – Transistor (p-n-p) h_{FE} as a function of temperature.

DYNAMIC

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$

Typical Values Intended Only for Design Guidance

CHARACTERISTICS	TEST CONDITIONS	TYPICAL VALUES	UNITS
For Each n-p-n Transistor			
Noise Figure (low frequency), NF	f = 1 kHz, V_{CE} = 5 V, I_C = 1 mA, R_S = 1 kΩ	2.2	dB
Low-Frequency, Input Resistance, R_i	f = 1.0 kHz, V_{CE} = 5 V, I_C = 1 mA	10	kΩ
Low-Frequency Output Resistance, R_o		80	kΩ
Admittance Characteristics:			
Forward Transfer Admittance, $\frac{g_{fe}}{y_{fe}\ b_{fe}}$	f = 1 MHz, V_{CE} = 5 V, I_C = 1 mA	7.5 −j13	mmho
Input Admittance, $\frac{g_{ie}}{y_{ie}\ b_{ie}}$		2.2 j3.1	mmho
Output Admittance, $\frac{g_{oe}}{y_{oe}\ b_{oe}}$		0.76 j2.4	mmho
Gain-Bandwidth Product, f_T	V_{CE} = 5 V, I_C = 1.0 mA	280	MHz
	V_{CE} = 5 V, I_C = 5 mA	335	
Emitter-to-Base Capacitance, C_{EB}	V_{EB} = 3 V	0.75	pF
Collector-to-Base Capacitance, C_{CB}	V_{CB} = 3 V	0.46	pF
Collector-to-Substrate Capacitance, C_{CI}	V_{CI} = 3 V	3.2	pF
For Each p-n-p Transistor			
Noise Figure (low frequency), NF	f = 1 kHz, I_C = 100 μA, R_S = 1 kΩ	3	dB
Low-Frequency Input Resistance, R_i	f = 1 kHz, V_{CE} = 5 V, I_C = 100 μA	27	kΩ
Low-Frequency Output Resistance, R_o		680	kΩ
Gain-Bandwidth Product, f_T	V_{CE} = 5 V, I_C = 100 μA	6.8	MHz
Emitter-to-Base Capacitance, C_{EB}	V_{EB} = −3 V	0.85	pF
Collector-to-Base Capacitance, C_{CB}	V_{CB} = −3 V	2.25	pF
Base-to-Substrate Capacitance, C_{BI}	V_{BI} = 3 V	3.05	pF

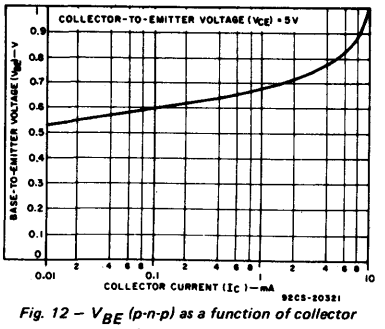


Fig. 12 – V_{BE} (p-n-p) as a function of collector current.

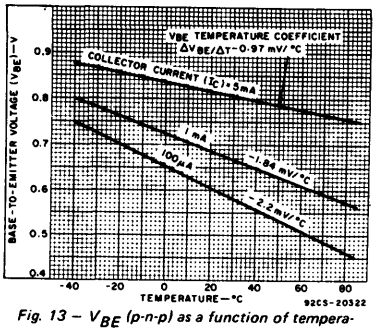


Fig. 13 – V_{BE} (p-n-p) as a function of temperature.

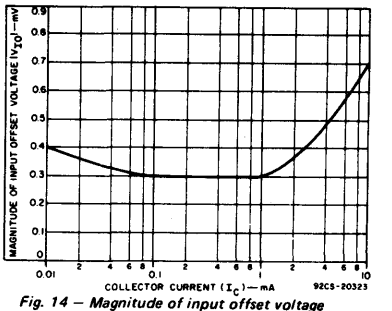


Fig. 14 – Magnitude of input offset voltage $|V_{IO}|$ as a function of collector current for n-p-n transistor Q_1–Q_2.

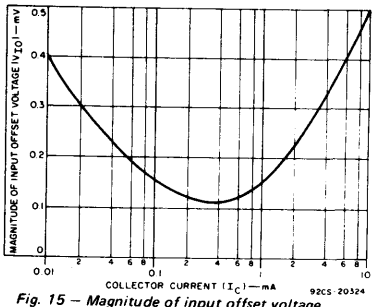


Fig. 15 – Magnitude of input offset voltage $|V_{IO}|$ as a function of collector current for p-n-p transistor Q_4–Q_5.

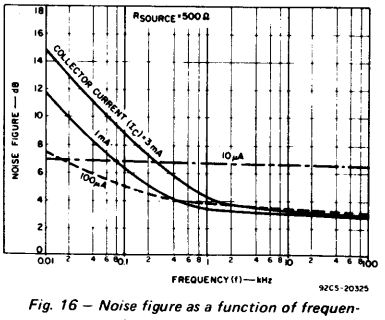


Fig. 16 – Noise figure as a function of frequency for n-p-n transistors.

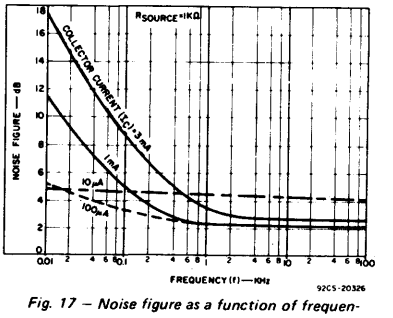


Fig. 17 – Noise figure as a function of frequency for n-p-n transistors.

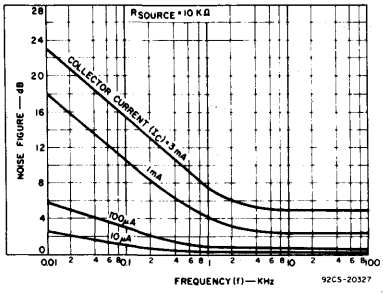


Fig. 18 – Noise as a function of frequency for n-p-n transistors.

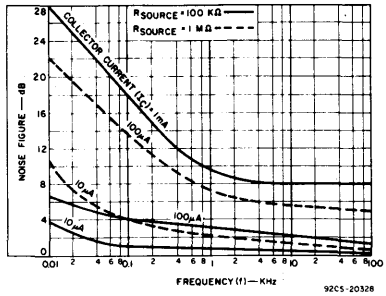


Fig. 19 – Noise figure as a function of frequency for n-p-n transistors.

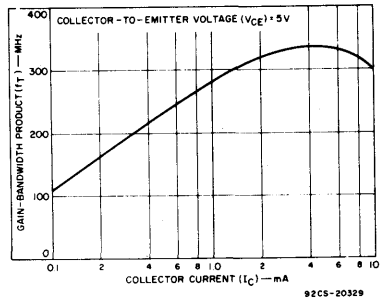


Fig. 20 – Gain-bandwidth product as a function of collector current (n-p-n).

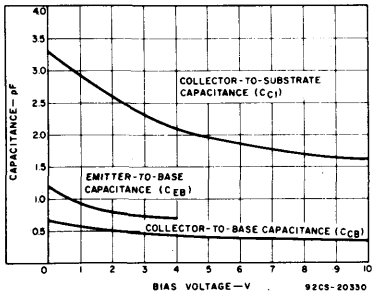


Fig. 21 – Capacitance as a function of bias voltage (n-p-n).

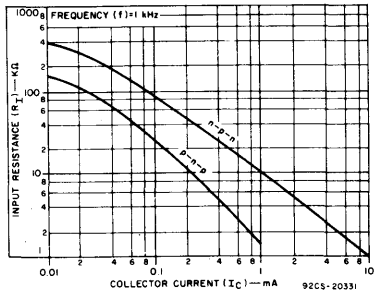


Fig. 22 – Input resistance as a function of collector current.

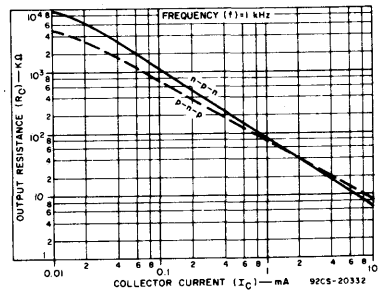


Fig. 23 – Output resistance as a function of collector current.

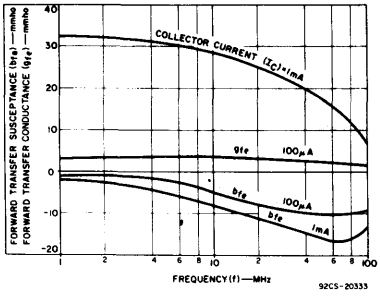


Fig. 24 – Forward transconductance as a function of frequency.

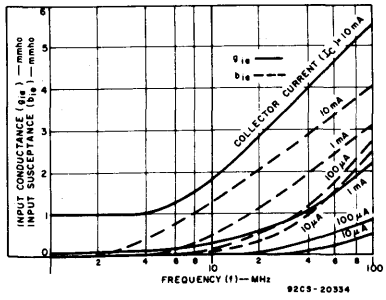


Fig. 25 – Input admittance as a function of frequency.

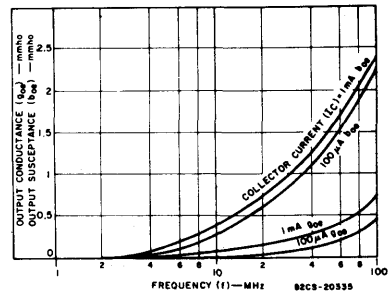


Fig. 26 – Output admittance as a function of frequency.

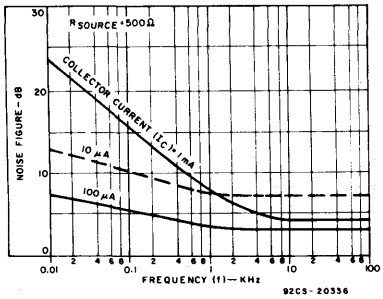


Fig. 27 – Noise figure as a function of frequency (p-n-p).

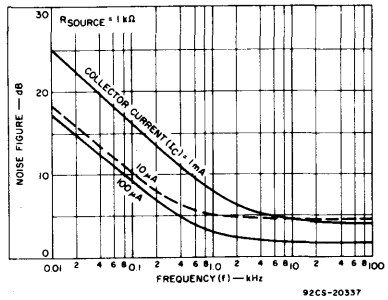


Fig. 28 – Noise figure as a function of frequency (p-n-p).

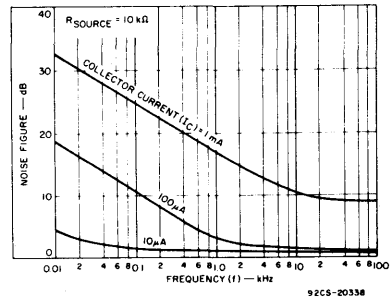


Fig. 29 – Noise figure as a function of frequency (p-n-p).

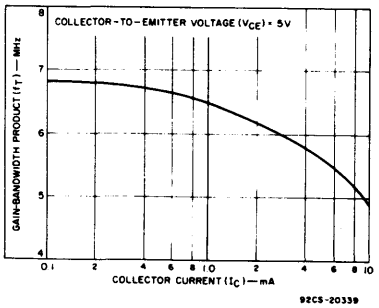


Fig. 30 – Gain-bandwidth product as a function of collector current (p-n-p).

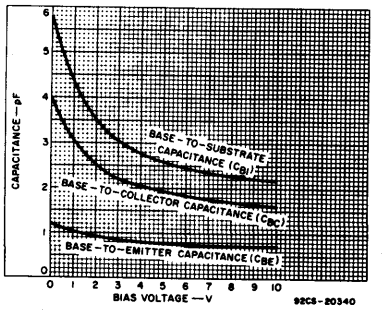


Fig. 31 – Capacitance as a function of bias voltage (p-n-p).

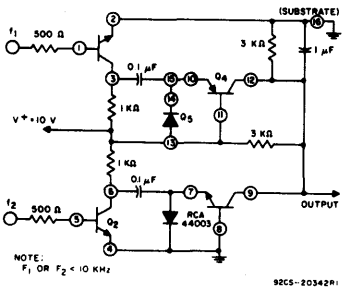


Fig. 32 – Frequency comparator using CA3096E.

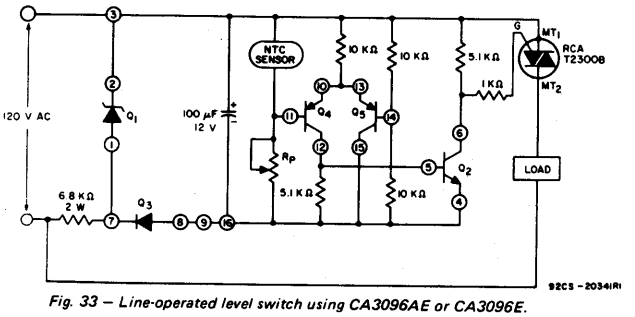


Fig. 33 – Line-operated level switch using CA3096AE or CA3096E.

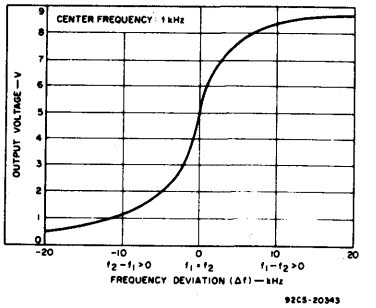


Fig. 34 – Frequency comparator characteristics.

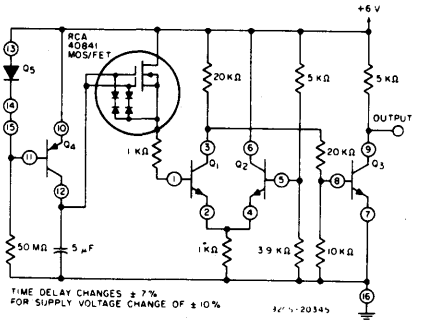
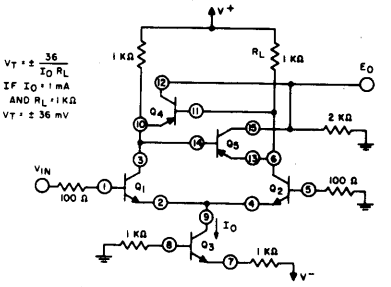


Fig. 35 – One-minute timer using CA3096AE and a MOS/FET.



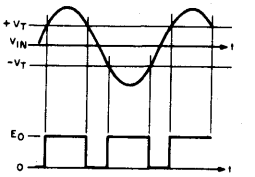


Fig. 36 – CA3096AE small-signal zero-voltage detector having noise immunity.

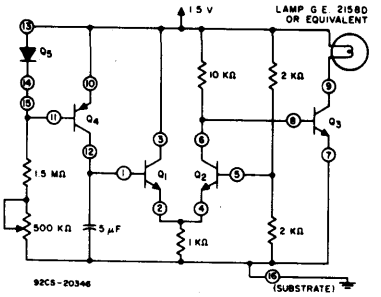


Fig. 37 – Ten-second timer operated form 1.5-volt supply using CA3096E.

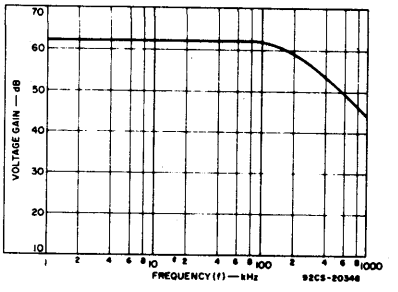
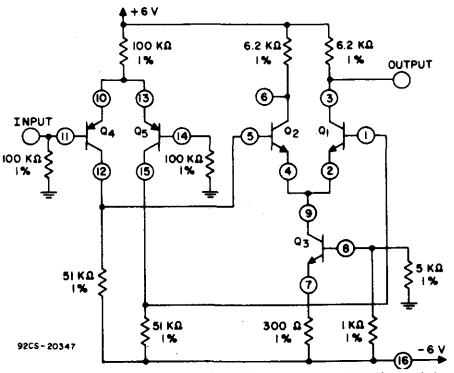


Fig. 38 – Gain-frequency characteristics.

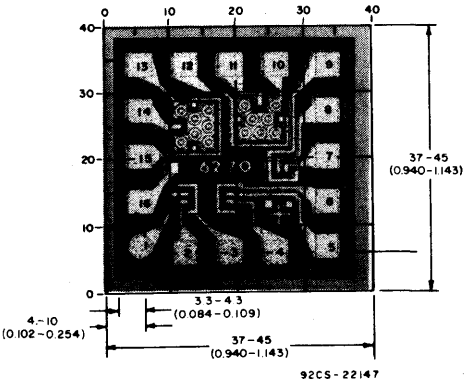
CA3096, CA3096A, CA3096C



Features:

1. Can be operated with either dual supply or single supply.
2. Wide-input common-mode range +5 V to −5 V.
3. Low bias current: $<1\ \mu A$.

Fig. 39 – Cascade of differential amplifiers using CA3096AE.



CA3096CH

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

The photographs and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Thyristor/Transistor Array

For Military, Commercial, and Industrial Applications

RCA-CA3097E Thyristor/Transistor Array is a monolithic integrated circuit that enables circuit designers to further integrate control systems. The CA3097E consists of five independent and completely isolated elements on one chip: an n-p-n transistor, a p-n-p/n-p-n transistor pair, a zener diode, a programmable unijunction transistor (PUT), and a sensitive-gate silicon controlled rectifier (SCR).

The CA3097 is supplied in either the 16-lead dual-in-line plastic package ("E" suffix) or the chip version ("H" suffix), and operates over the full military-temperature range of −55 to +125°C.

Includes:

- **Uncommitted n-p-n Transistor**
- **Sensitive-Gate Silicon Controlled Rectifier**
- **Programmable Unijunction Transistor (PUT)**
- **p-n-p/n-p-n Transistor Pair**
- **Zener Diode**
- **Separate Substrate Connection**

Features:

- **Complete isolation between elements**
- **n-p-n transistor – V_{CEO} = 30 V (min.) I_C = 100 mA (max.)**
- **p-n-p/n-p-n transistor pair – beta ≥ 8000 (typ.) @ I_C = 10 mA, individual p-n-p, n-p-n, or transistor pair operation**
- **Programmable unijunction transistor (PUT) – peak-point current = 15 nA (typ.) at R_G = 1 MΩ; V_{AK} = ±30 V**
- **(PUT) Extremely long RC time constants with low value of external capacitor**
- **Sensitive-gate silicon controlled rectifier (SCR) – 150 mA forward current (max.)**
- **Zener-diode impedance (Z_Z) = 15Ω (typ.) at 10 mA**

Applications:

- **Timers**
- **Light dimmers/motor controls**
- **Oscillators**
- **"One-shot" multivibrators**
- **Voltage regulators**
- **Comparators, Schmitt triggers**
- **Constant-current sources**
- **Amplifiers**
- **Logic circuits**
- **SCR triggering**
- **Pulse Circuits**

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

Isolation Voltage, any terminal to substrate*	+50 V
Dissipation, Total Package:	
Up to T_A = 55°C	750 mW
Above T_A = 55°C	derate linearly at 6.67 mW/°C
Ambient Temperature Range:	
Operating	−55 to +125°C
Storage	−65 to +150°C
Lead Temperature (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265 °C
Each n-p-n Transistor (Q3,Q5)	
The following ratings apply with terminals 6 & 9 connected together.	
Collector-to-Emitter Voltage (V_{CEO})	30 V
Collector-to-Base Voltage (V_{CBO})	50 V
Emitter-to-Base Voltage (V_{EBO})	5 V
Collector Current (I_C)	100 mA
Base Current (I_B)	20 mA
Dissipation (P_D)	500 mW
p-n-p Transistor (Q4)	
The following ratings apply with terminals 7 & 8 connected together.	
Collector-to-Emitter Voltage (V_{CEO})	−40 V
Collector-to-Base Voltage (V_{CBO})	−50 V
Emitter-to-Base Voltage (V_{EBO})	−40 V
Collector Current (I_C)	−10 mA
Base Current (I_B)	−3 mA
Dissipation (P_D)	200 mW
p-n-p/n-p-n Transistor Pair (Q3,Q4)	
Dissipation (P_D)	500 mW
Programmable Unijunction Transistor, PUT (Q1)	
Gate-to-Cathode Positive Voltage (V_{GK})	30 V
Gate-to-Cathode Negative Voltage (V_{GKR})	5 V
Gate-to-Anode Negative Voltage (V_{GA})	30 V
Anode-to-Cathode Voltage (V_{AK})	±30 V
DC Anode Current	150 mA
Peak Anode Non-Recurrent Forward (On-State) Current (10 μs pulse)	2 A
Total Average Dissipation	300 mW
Silicon Controlled Rectifier, SCR (Q2)	
Repetitive Peak Reverse Voltage (V_{RRXM}), R_{GK} = 1 KΩ	30 V
Repetitive Peak Off-State Voltage (V_{DRXM}), R_{GK} = 1 kΩ	30 V
DC On-State Current (I_{TDC})	150 mA
Peak Surge (Non-Repetitive) On-State Current (10 μs pulse)	2 A
Forward Peak Gate Current (I_{GFM})	20 mA
Peak Gate-to-Cathode Reverse Voltage (V_{GRM})	5 V
Total Average Dissipation	300 mW
Zener Diode, (Z1)	
DC Current (I_Z)	25 mA
Dissipation (P_D)	250 mW

* One or more of the terminals of each element of the CA3097E is isolated from the substrate by a junction diode. In order to maintain electrical isolation between elements, the substrate terminal must be connected to a voltage which is no more positive than that of any other terminal. To avoid undesirable coupling between elements, the substrate terminal (terminal 10) should be maintained at either dc or signal (sc) ground.

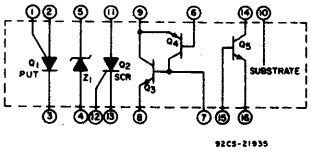


Fig. 1 – Schematic diagram of CA3097E.

TYPICAL CHARACTERISTICS

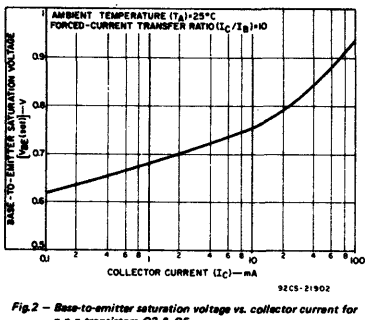


Fig.2 – Base-to-emitter saturation voltage vs. collector current for n-p-n transistors Q3 & Q5.

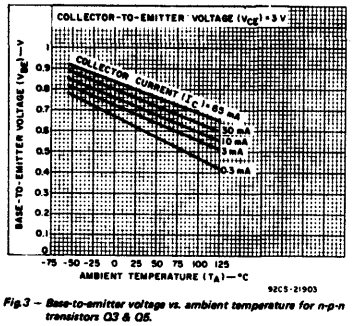


Fig.3 – Base-to-emitter voltage vs. ambient temperature for n-p-n transistors Q3 & Q5.

CA3097E

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS Ambient Temperature (T_A) = 25°C Unless Otherwise Specified	FIG. NO.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
n-p-n TRANSISTORS Q3, Q5 (TERMINALS 6 and 9 CONNECTED)							
COLLECTOR CUTOFF CURRENT	I_{CBO}	V_{CB} = 10 V, I_E = 0		–	–	1	μA
COLLECTOR CUTOFF CURRENT	I_{CEO}	V_{CE} = 10 V, I_B = 0		–	–	10	μA
COLLECTOR-TO-EMITTER BREAKDOWN VOLTAGE	$V_{(BR)CEO}$	I_C = 100μA, I_B = 0		30	–	–	V
COLLECTOR-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0		50	–	–	V
COLLECTOR-TO-SUBSTRATE BREAKDOWN VOLTAGE	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0		50	–	–	V
EMITTER-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)EBO}$	I_E = 100μA, I_C = 0		5	7.5	10	V
COLLECTOR-TO-EMITTER SATURATION VOLTAGE	$V_{CE(SAT)}$	I_C = 50mA, I_B = 5mA I_C = 10mA, I_B = 1 mA	5	– –	– 0.10	0.65 –	V
BASE-TO-EMITTER SATURATION VOLTAGE	$V_{BE(SAT)}$	I_C = 10mA, I_B = 1mA	2	–	0.76	–	V
BASE-TO-EMITTER VOLTAGE	V_{BE}	V_{CE} = 3V, I_C = 10mA	3	0.65	0.73	0.85	V
DC FORWARD-CURRENT TRANSFER RATIO	h_{FE}	V_{CE} = 3V, I_C = 10mA V_{CE} = 3V, I_C = 50mA	4	100 80	130 120	– –	
p-n-p TRANSISTOR Q4 (TERMINALS 7 and 8 CONNECTED)							
COLLECTOR CUTOFF CURRENT	I_{CBO}	V_{CB} = -10 V, I_E = 0		–	–	–1	μA
COLLECTOR CUTOFF CURRENT	I_{CEO}	V_{CE} = -10 V, I_B = 0		–	–	–10	μA
COLLECTOR-TO-EMITTER BREAKDOWN VOLTAGE	$V_{(BR)CEO}$	I_C = -100μA, I_B = 0		–40	–	–	V
COLLECTOR-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)CBO}$	I_C = -10μA, I_E = 0		–50	–	–	V
EMITTER-TO-SUBSTRATE BREAKDOWN VOLTAGE	$V_{(BR)EIO}$	I_{EI} = 10μA, I_B = 0, I_E = 0		–50	–	–	V
EMITTER-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)EBO}$	I_E = –10μA, I_C = 0		–40	–	–	V
COLLECTOR-TO-EMITTER SATURATION VOLTAGE	$V_{CE(SAT)}$	I_C = -1mA, I_B = -100μA	6	–	–	–0.33	V
BASE-TO-EMITTER SATURATION VOLTAGE	$V_{BE(SAT)}$	I_C = -1mA, I_B = –100μA·	7	–	–0.7	–	V
BASE-TO-EMITTER VOLTAGE	V_{BE}	V_{CE} = -3 V, I_C = –100μA	8	–0.5	–0.6	–0.7	V
DC FORWARD-CURRENT TRANSFER RATIO	h_{FE}	V_{CE} = – 3 V, I_C = –100μA V_{CE} = –3 V, I_C = –1 mA	9	30 40	60 –	– –	
n-p-n/p-n-p TRANSISTOR PAIR Q3, Q4							
DC FORWARD-CURRENT TRANSFER RATIO	h_{FE}	V_{CE} (n-p-n) = 3V, I_C = 10mA V_{CE} (n-p-n) = 3V, I_C = 50mA	10 10	– –	8000 6500	– –	

TYPICAL CHARACTERISTICS (CONT'D)

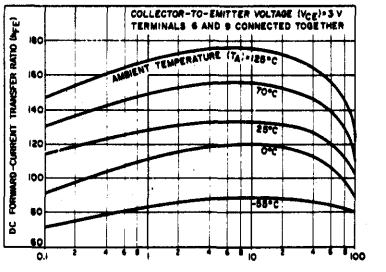


Fig.4 – DC forward-current transfer ratio vs. collector current for n-p-n transistors Q3 & Q5.

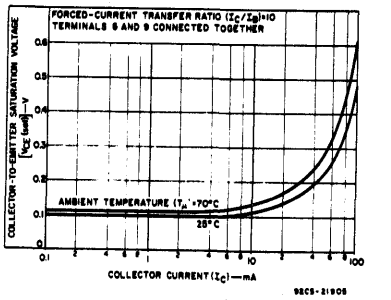


Fig.5 – Collector-to-emitter saturation voltage vs. collector for n-p-n transistors Q3 & Q5.

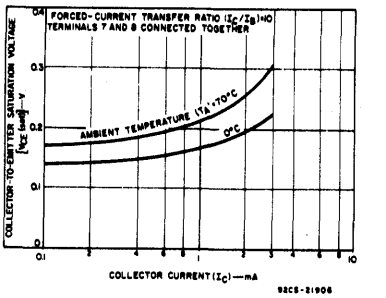


Fig.6 – Collector-to-emitter saturation voltage vs. collector current for p-n-p transistor Q4.

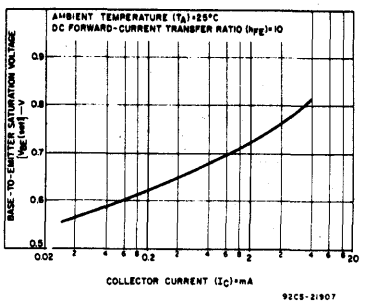


Fig.7 – Base-to-emitter saturation voltage vs. collector current for p-n-p transistor Q4.

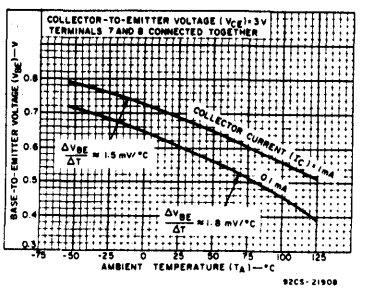


Fig.8 – Base-to-emitter voltage vs. ambient temperature for p-n-p transistor Q4.

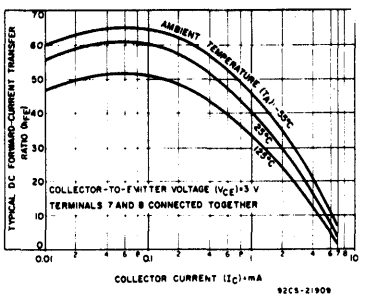


Fig.9 – DC forward-current transfer ratio vs. collector current for p-n-p transistor Q4.

ELECTRICAL CHARACTERISTICS (Cont'd.)

CHARACTERISTIC	SYMBOL	TEST CONDITIONS Ambient Temperature (T_A) = 25°C Unless Otherwise Specified	FIG. NO.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
PROGRAMMABLE UNIJUNCTION TRANSISTOR (PUT), Q1							
OFFSET VOLTAGE	V_T*	V_S = 10V, R_G = 10kΩ	11,22[a]	0.2	–	0.7	V
		V_S = 10V, R_G = 1MΩ		0.2	–	0.7	
ANODE-TO-CATHODE ON-STATE VOLTAGE	V_F	I_F = 50mA	12	–	0.90	1.5	V
		I_F = 100mA		–	1	–	
PEAK OUTPUT VOLTAGE	V_{OM}	C = 0.22μF Anode Supply Voltage = 20V	13,23	–	10	–	V
PEAK-POINT CURRENT	I_P	V_S = 10V, R_G = 10kΩ	14,22[a]	–	0.55	1	μA
		V_S = 10V, R_G = 1MΩ	–	–	0.015	0.15	
VALLEY-POINT CURRENT	I_V	V_S = 10V, R_G = 10kΩ	17,15	4	40	–	μA
		V_S = 10V, R_G = 1MΩ	16	–	–	25	
GATE REVERSE CURRENT	I_{GAO}	V_S = 30V	22[c]	–	0.02	–	nA
GATE REVERSE CURRENT	I_{GKS}	Anode-To-Cathode Short, V_S = 30V	22[d]	–	0.2	–	nA
OUTPUT PULSE RISE TIME	t_r	Anode Supply Voltage = 20V C = 0.22 μF	23	–	60	–	ns
SILICON CONTROLLED RECTIFIER (SCR), Q2							
PEAK OFF-STATE CURRENT: FORWARD	I_{DXM}	V_{DRXM} = 30V, R_{GK} = 1kΩ	24	–	–	2	μA
REVERSE	I_{RXM}	V_{RRXM} = 30V, R_{GK} = 1kΩ	24	–	–	2	
FORWARD DC VOLTAGE DROP	V_T	I_T = 50 mA	18	–	0.90	1.5	V
GATE-TO-SOURCE TRIGGER CURRENT	I_{GS}	T_A = 25°C	26	–	33	100	μA
		T_A = −55°C	26	–	50	–	
DC GATE-TRIGGER VOLTAGE	V_{GT}	V_L = 10V, R_L = 100Ω	19	–	0.55	0.75	V
HOLDING CURRENT	I_{HO}	R_{GK} = 1kΩ	20,24	–	1.2	–	mA
CRITICAL RATE-OF-RISE OF OFF-STATE VOLTAGE	dv/dt	EXPONENTIAL RISE, R_{GK} = 1kΩ, V_{DRXM} = 30V	25	–	150	–	V/μs
GATE-CONTROLLED TURN-ON TIME	t_{gt}	See Fig. 33	33	–	50	–	ns
CIRCUIT-COMMUTATED TURN-OFF TIME	t_q	See Fig. 33	33	–	10	–	μs
ZENER DIODE, Z1							
ZENER VOLTAGE	V_Z	I_Z = 10mA	21	7.2	8	8.8	V
ZENER IMPEDANCE	Z_Z	I_Z = 10mA, f = 1kHz		–	15	25	Ω
ZENER VOLTAGE TEMPERATURE COEFFICIENT	$(\Delta V_Z/V_Z)/\Delta T$	I_Z = 10mA		–	+0.05	–	%/°C
	$\Delta V_Z/\Delta T$			–	+4	–	mV/°C
ZENER-TO-SUBSTRATE BREAKDOWN VOLTAGE	$V_{(BR)ZIO}$	I_Z = 100μA TERM. 5 TO SUBSTRATE		50	80	–	V

* $V_T = V_P - V_S$ (Fig. 22)

TYPICAL CHARACTERISTICS (CONT'D)

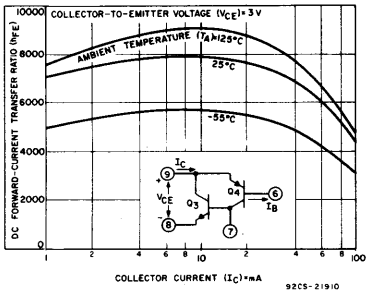


Fig.10 – DC forward-current transfer ratio vs. collector current for transistor pair Q3, Q4.

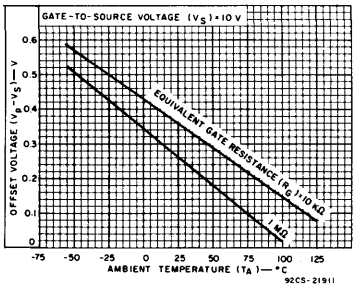


Fig.11 – Offset voltage vs. ambient temperature for Q1 (PUT).

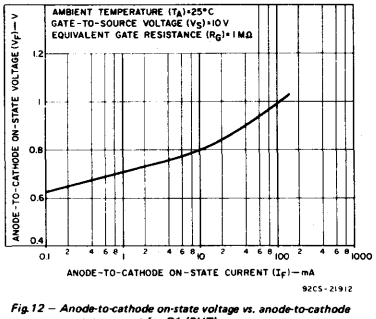


Fig.12 – Anode-to-cathode on-state voltage vs. anode-to-cathode on-state current for Q1 (PUT).

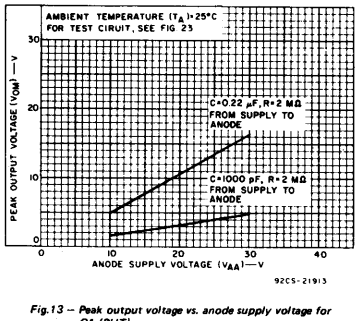


Fig.13 – Peak output voltage vs. anode supply voltage for Q1 (PUT).

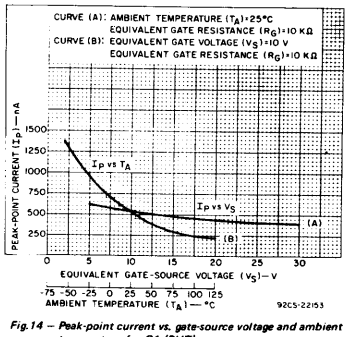


Fig.14 – Peak-point current vs. gate-source voltage and ambient temperature for Q1 (PUT).

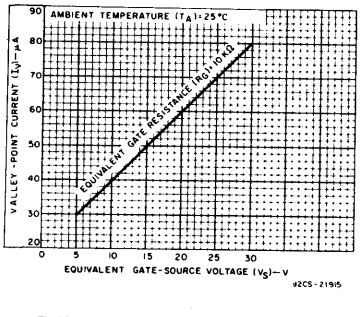


Fig.15 – Valley-point current vs. gate-source voltage for Q1 (PUT).

TYPICAL CHARACTERISTICS (CONT'D)

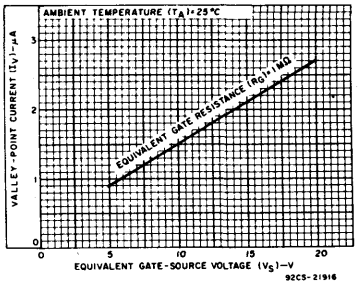


Fig.16 – Valley-point current vs. gate-source voltage for Q1 (PUT).

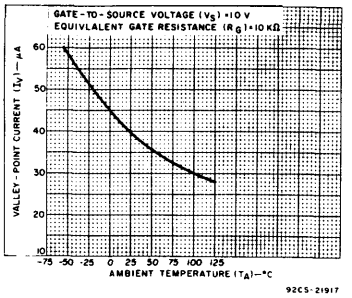


Fig.17 – Valley-point current vs. ambient temperature for Q1 (PUT).

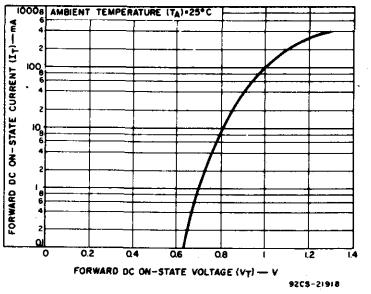


Fig.18 – Forward DC on-state current vs. on-state voltage for Q2 (SCR).

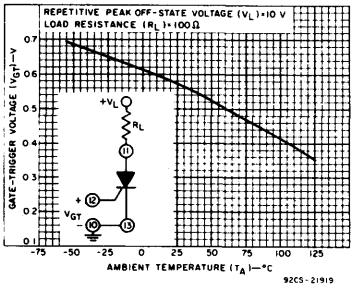


Fig.19 – Gate-trigger voltage vs. ambient temperature for Q2 (SCR).

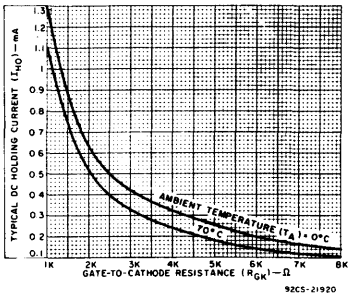


Fig.20 – Typical DC holding current vs. gate-to-cathode resistance for Q2 (SCR).

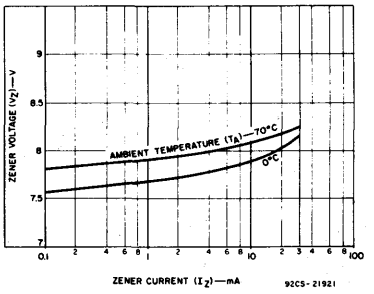


Fig.21 – Zener voltage vs. zener current for Z1.

OPERATING CONSIDERATIONS FOR CA3097E

1. Composite p-n-p/n-p-n Transistors Q3, Q4 (See Fig. 3)

To use Q3 as an individual n-p-n transistor, join terminals no. 6 and no. 9 to disable p-n-p transistor Q4.

The appropriate terminal connections are then:

Collector..... terminal 9
Base..... terminal 7
Emitter..... terminal 8

To use Q4 as an individual p-n-p transistor, join terminals no. 7 and no. 8 to disable n-p-n transistor Q3.

The appropriate terminal connections are then:

Collector..... terminal 7
Base..... terminal 6
Emitter..... terminal 9

To use Q3 and Q4 as a composite use terminals 6, 7, 8, and 9 as required.

2. Programmable Unijunction Transistor Q1 (PUT)

The programmable unijunction transistor is essentially an anode-gate SCR. The volt-ampere characteristic of the device is shown in Fig. 22. When an equivalent Thevenin source (V_S, R_G), as shown in Fig. 22, is applied to the gate terminal the device will be "off" if the anode-voltage is negative with respect to the gate voltage. Under this condition, any current flow is exclusively leakage current. When the anode voltage becomes more positive than the gate voltage by an increment equal to the threshold voltage (V_T = 0.4 V typ.), the device can turn "on" only if the current available at the anode terminal is **greater** than the specified peak-point current. The PUT will then switch through its negative-resistance region to the "on" state (low anode-to-gate voltage). It should be noted that I_P is not the maximum current allowed through the device, but is the current required at the peak of the V-I curve. I_P is typically a very low value of current.

After the PUT has switched to its low-impedance state, the device will remain "on" if the anode-current (I_A) exceeds the valley-point current (I_V). If $I_A < I_V$, the PUT will switch back to its high-impedance "off" state. Thus, the PUT can be made to "latch" or recover, depending on I_V. Since I_V is a function of the "on"-state gate current (which depends on R_G and V_S) a choice of R_G and/or V_S will determine the operating mode, i.e., "off" state→"on" state or "off" state → "on" state → "off" state. The value of I_V increases directly as a function of V_G and inversely with R_G. The PUT in the CA3097E has a low I_P.....I_P = 15 nA at V_S = 10 V, R_G = 1 MΩ. This low value of I_P indicates that an extremely large value of anode-supply resistor, e.g. 60 MΩ (typ.), can be used in timing circuits requiring long RC time constants. This becomes important when considering the size of the external timing capacitor to be used. Consequently, the use of the PUT in the CA3097E is advantageous since it has a lower I_P than most discrete PUT's.

Temperature Compensation of Switching Point

As described previously, the PUT will switch to its low-impedance state when its anode voltage is approximately a diode-drop above the gate voltage. Since the anode-to-gate threshold voltage vs. temperature characteristic is similar to that of a typical silicon-diode junction, a compensating series diode such as used in the circuit of Fig. 29 (Z1 connected as forward-biased diode) considerably reduces the effect of temperature on the switching point.

Bypassing Anode Current

If the PUT gate equivalent source is such that $I_A > I_V$, the PUT will remain "on". A method for turning the PUT off is by shunting current away from the anode until $I_A < I_V$. An example of this technique is the oscillator circuit of Fig. 29. Q3 transistor is turned "on" after the PUT fires and shunts current away from the anode, thereby forcing $I_A < I_V$. The PUT then turns "off" allowing C_T to recharge through R_T, to repeat the cycle.

Protecting The PUT Against Discharge Current Of The Capacitor

A current-limiting resistor in series with the PUT is normally required to dissipate capacitive discharge energy (see Figs. 23 and 29).

Silicon Controlled Rectifier, Q2 (SCR)

The SCR should be used with a 1 kΩ (or less) resistor connected between the cathode and gate terminals if the SCR is to be subjected to its maximum forward and reverse voltage ratings (V_{DXM} and V_{RXM}). Selecting a value for R_{GK} of 1 kΩ (or lower) increases the capability of the device to withstand greater dv/dt and increases the noise immunity of the SCR against false triggering at the gate. Practical considerations such as available current drive from the triggering devices (e.g., a PUT) will determine the lowest value of R_{GK} at which the SCR will fire with a $V_{GK} \approx 0.55$ V. With a value of 500Ω for R_{GK}, the trigger source must be capable of supplying 1.1 mA. R_{GK} should be non-inductive within the frequency band of the noise transients normally encountered in a particular application.

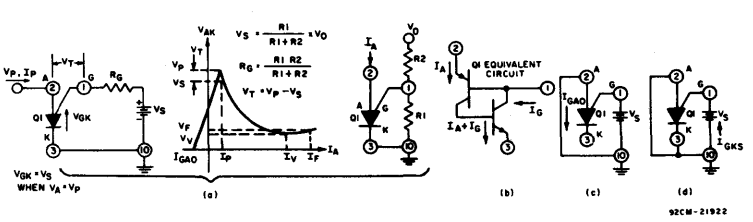


Fig. 22 – General anode characteristics for Q1 (PUT).

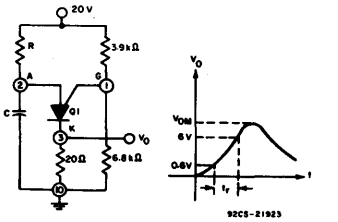


Fig. 23 – Output pulse characteristics for Q1 (PUT).

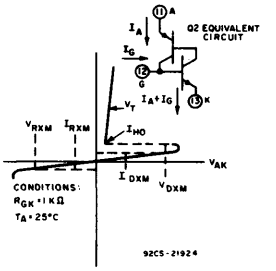


Fig. 24 – Principle voltage-current characteristics for Q2 (SCR).

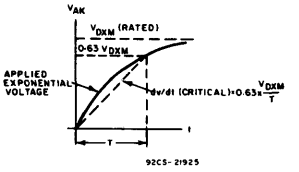


Fig. 25 – Definition of critical rate of rise of off-state voltage for Q2 (SCR).

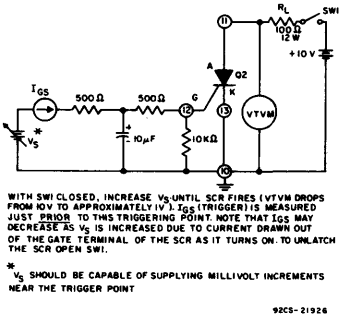


Fig. 26 – Test circuit for determining I_{GS} in Q2 (SCR).

APPLICATIONS CIRCUITS

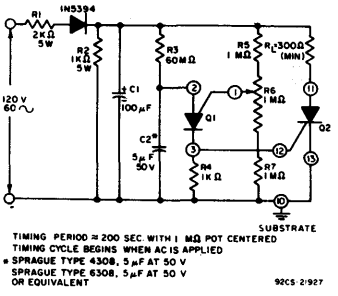


Fig. 27 – AC line-operated one-shot timer.

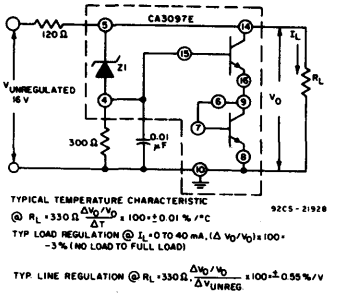


Fig. 28 – Temperature-compensated shunt regulator.

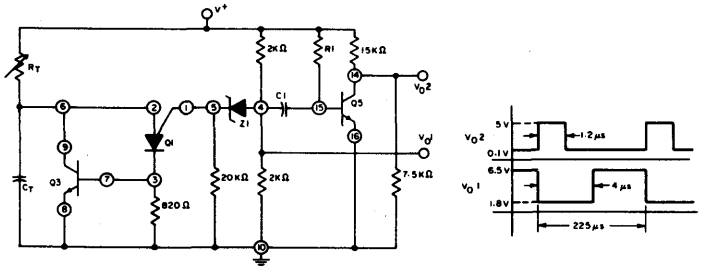
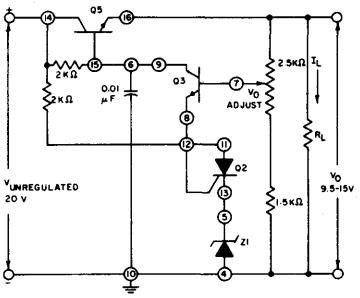


Fig. 29 – Pulse generator.

CA3097E

APPLICATIONS CIRCUITS (CONT'D)



TYPICAL LOAD REGULATION @ V_O=12 V, I_L=0 TO 40 mA

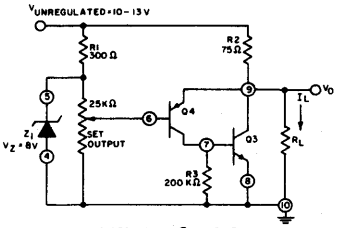
$\frac{\Delta V_O}{V_O} \times 100 = \pm 0.4\%$ (NO LOAD TO FULL LOAD)

TYPICAL LINE REGULATION @ V_O=12 V

$\frac{\Delta V_O / V_O}{\Delta V_{UNREG}} \times 100 = \pm 0.45\ \%/V$

92CS-21930

Fig. 30 – Series voltage regulator.



TYPICAL LOAD REGULATION @ V_O=7 V, I_L=0 TO 40 mA

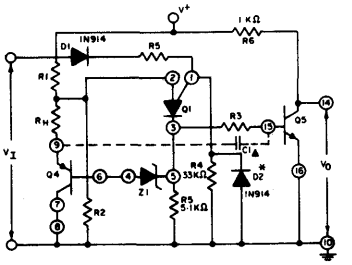
$\frac{\Delta V_O}{V_O} \times 100 = \pm 1.1\ \%$

TYPICAL LINE REGULATION @ V_O=7 V, I_L=20 mA

$\frac{\Delta V_O}{V_O / \Delta V_{UNREGULATED}} = \pm 0.85\ \%/VOLT$

92CS-21931

Fig. 31 – 5 to 7.5 V shunt regulator.



▲ OPTIONAL SPEED-UP CAPACITOR

* REQUIRED IF V_I SWINGS BELOW GROUND

TYPICAL OPERATING CONDITIONS:

FREQUENCY IN = 0-10 KHz

SUPPLY VOLTAGE (V^+) = 15 V

R1, R2, R_H = 5.1 KΩ

R3 = 6.2 KΩ, R5 = 300Ω

C_1 = 820 pF

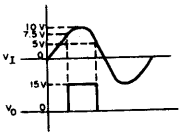
$V_{TH}U$ = 7.5 V, $V_{TH}L$ = 5 V

HYSTERESIS VOLTAGE = 2.5 V

UPPER THRESHOLD VOLTAGE $(V_{TH}U) \approx V^+ \frac{R2}{R1+R2}$

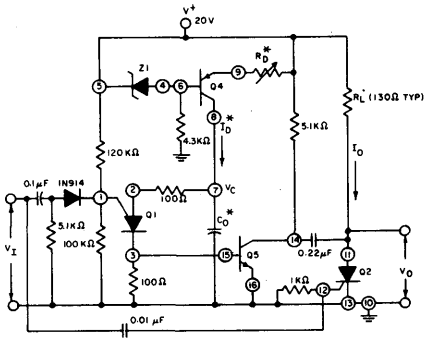
LOWER THRESHOLD VOLTAGE $(V_{TH}L) \approx \frac{(V^+)\left(\frac{R2\,R_H}{R2+R_H}\right)}{\frac{R2\,R_H}{R2+R_H}+R1}$

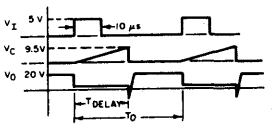
HYSTERESIS VOLTAGE = $V_{TH}U - V_{TH}L$



92CM-21932

Fig. 32 – Schmitt trigger.





*MONOSTABLE DELAY TIME SET BY ADJUSTMENT OF I_D (VARY R_D) OR BY C_O. I_O MUST BE GREATER THAN I_V OF Q1 (PUT) FOR MONOSTABLE OPERATION.

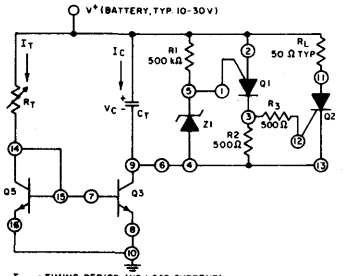
Q2 (SCR) SWITCHING TIMES:

GATE-CONTROLLED TURN-ON TIME (t_{gt}) = 50 ns (TYP)

CIRCUIT-COMMUTATED TURN-OFF TIME (t_q) = 10 μs (TYP)

92CM-21933

Fig. 33 – Monostable multivibrator with variable delay.



T_{OFF} = TIMING PERIOD (NO LOAD CURRENT)

PUT FIRES WHEN $V_C \approx 8$ V

$V_C = \frac{I_C (T_{OFF})}{C_T}$, $I_C \approx I_T$ (Q3, Q5 MATCHED)

I_T SET BY ADJUSTING R_T, $I_T \approx \frac{V^+ - 0.7}{R_T}$

T_{ON} = CAPACITOR DISCHARGE TIME THROUGH LOAD. LOAD TURNS OFF WHEN SCR ANODE CURRENT FALLS BELOW HOLDING CURRENT (I_{HO}). TYPICAL I_{HO} = 1.2 mA

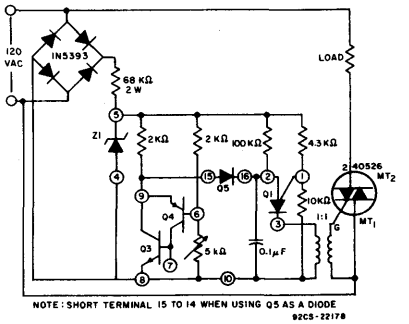
EXAMPLE: FOR TIMING PERIOD OF 8.3 MIN

C_T = 1000 μF, I_T = 16 μA

$R_T = \frac{V^+ - 0.7}{I_T}$ (FOR V^+ = 16 V, $R_T \approx 1$ MΩ)

92CS-21934

Fig.34 – Low-current-drain battery-operated long interval astable timer.



NOTE: SHORT TERMINAL 15 TO 14 WHEN USING Q5 AS A DIODE

92CS-22178

Fig.35 – Phase control circuit.

Programmable Schmitt Trigger — With Memory

—Dual-Input Precision Level Detectors

Applications:

- Control of relays, heaters, LED's lamps, photo-sensitive devices, thyristors, solenoids, etc.
- Signal reconditioning
- Phase and frequency modulators
- On/off motor switching
- Schmitt triggers, level detectors
- Time delays
- Overvoltage, overcurrent, overtemperature protection
- Battery-operated equipment
- Square and triangular-wave generators

Features:

- Programmable operating current
- Micropower standby dissipation
- Direct control of currents up to 150 mA
- Low input on/off current of less than 1 nA for programmable bias current of 1 μA
- Built-in hysteresis: 20 mV max.
- Programmable hysteresis: 20 mV to V^+
- Dual reference input
- High sensor range: 100 Ω to 100 MΩ
- Stable predictable switching levels
- Temperature-compensated reference voltage
- Power can be strobed off via term. 2

The RCA-CA3098 Programmable Schmitt Trigger is a monolithic silicon integrated circuit designed to control high-operating-current loads such as thyristors, lamps, relays, etc. The CA3098 can be operated with either a single power supply with maximum operating voltage of 16 volts, or a dual power supply with maximum operating voltage of ±8 volts. It can directly control currents up to 150 mA and operates with microwatt standby power dissipation when the current to be controlled is less than 30 mA. The CA3098 contains the following major circuit-function features (see Fig. 1):

1. Differential amplifiers and summer: the circuit uses two differential amplifiers, one to compare the input voltage with the "high" reference, and the other to compare the input with the "low" reference. The resultant output of the differential amplifiers actuates a summer circuit which delivers a trigger that initiates a change in state of a flip-flop.
2. Flip-flop: the flip-flop functions as a bistable "memory" element that changes state in response to each trigger command.
3. Driver and otuput stages: these stages permit the circuit to "sink" maximum peak load currents up to 150 mA at terminal 3.
4. Programmable operating current: the circuit incorporates access at terminal 2 to permit programming the desired quiescent operating current and performance parameters.

The CA3098 is supplied in the 8-lead dual-in-line plastic package ("Mini-Dip", E suffix), 8-lead TO-5 style package (T suffix), 8-lead TO-5-style package with formed leads "DIL-CAN" (S suffix), and in chip form (H suffix).

For information on another RCA Dual-Input Precision Level Detector, see the data bulletin for the RCA-CA3099E, File No. 620.

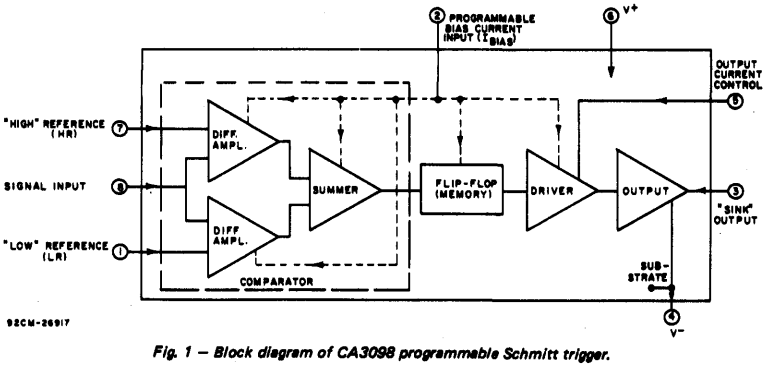


Fig. 1 – Block diagram of CA3098 programmable Schmitt trigger.

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

Supply Voltage Between Terminals 6 and 4,	16	V
Output Voltage Between Terminals 7 and 4, and 3 and 4	16	V
Differential Input Voltage Between Terminals 8 and 1, and Terminals 7 and 8	10	V
Operating Voltage Range:		
Term. 8	V^- to V^+	
Term. 7	(V^- plus 2.0 V) to V^+	
Term. 1	(V^-) to (V^+ minus 2.0 V)	
Load Current (Term. 3)	150	mA
Input Current to Voltage Regulator (Term. 5)	25	mA
Programmable Bias Current (Term. 2)	1	mA
Output Current Control (Term. 5)	15	mA
Power Dissipation:		
Without Heat Sink:		
Up to T_A = 55°C		
CA3098S, CA3098T	630	mW
CA3098E	630	mW
Above T_A = 55°C Derate linearly at	6.67	mW/°C
With Heat Sink:		
Up to T_A = 55°C		
CA3098S, CA3098T	1.6	W
Above T_A = 55°C		
CA3098S, CA3098T Derate linearly at	16.67	mW/°C
Ambient Temperature Range (All Packages):		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	265	°C

CA3098 Types

General Description of Circuit Operation (Refer to Figs. 2, 3, 4)

When the signal-input voltage of the CA3098 is equal to or less than the "low" reference voltage (LR), current flows from an external power supply through a load connected to terminal 3 ("sink" output). This condition is maintained until the signal-input voltage rises to or exceeds the "high" reference voltage (HR), thereby effecting a change in the state of the flip-flop (memory) such that the output stage interrupts current flow in the external load. This condition, in turn, is maintained until such time as the signal again becomes equal to or less than the "low" reference voltage (VR).

The CA3098 comparator is unique in that it contains circuit provisions to permit programmability. This feature provides flexibility to the designer to optimize quiescent power consumption, input-circuit characteristics, hysteresis, and additionally permits independent control of the comparator, namely, pulsing, strobing, keying, squelching, etc. Programmability is accomplished by means of the bias current (I_{bias}) supplied to terminal 2.

An auxiliary means of controlling the magnitude of load-current flow at terminal 3 is provided by "sinking" current into terminal 5. Figs. 3 and 4 highlight the operation of the CA3098 when connected as a simple hysteresis switch (Schmitt trigger).

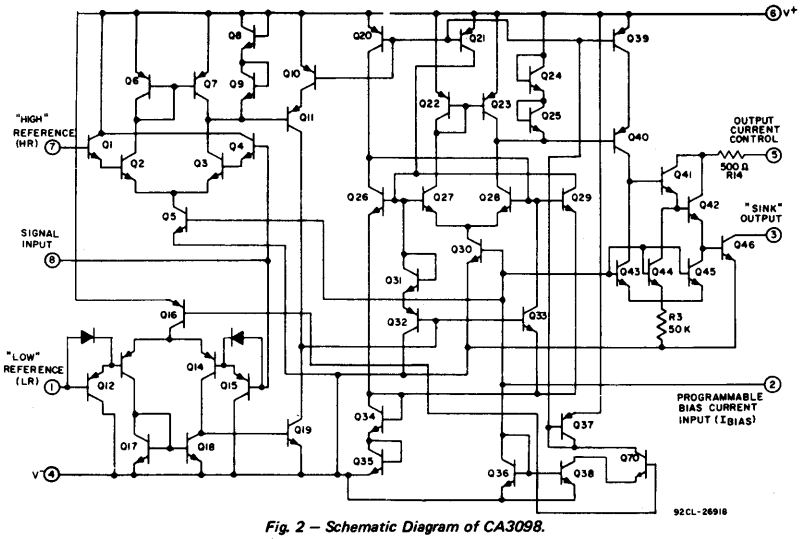


Fig. 2 – Schematic Diagram of CA3098.

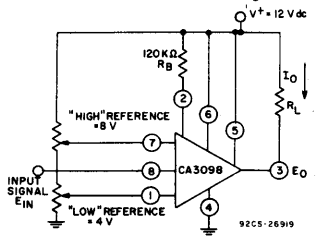


Fig. 3 – Basic hysteresis switch (Schmitt trigger).

Sequence	Input Signal Level	Output Voltage (V) (Term. 3)
1	$4 \geqslant E_{IN} > 0$	0
2	$8 \geqslant E_{IN} > 4$	0
3	$E_{IN} > 8$	12
2	$8 \geqslant E_{IN} > 4$	12
1	$4 \geqslant E_{IN} > 0$	0

Fig. 4 – Resultant output states of the CA3098, shown in Fig. 3 as a function of various input signal levels.

TYPICAL CHARACTERISTIC CURVES

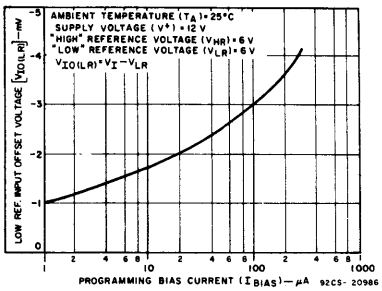


Fig. 5 – Input-offset voltage ("low" reference) vs. programming bias current.

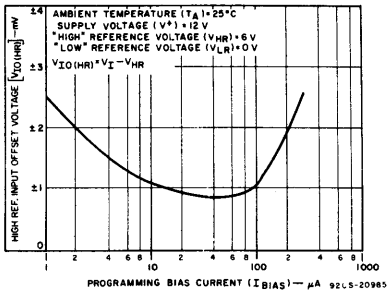


Fig. 6 – Input-offset voltage ("high" reference) vs. programming bias current.

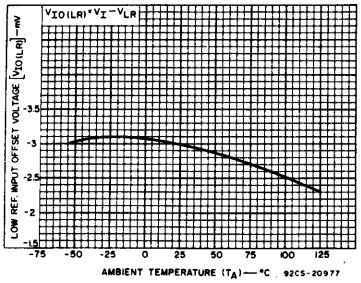


Fig. 7 – Input-offset voltage ("low reference) vs. ambient temperature.

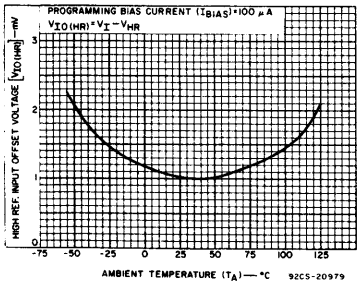


Fig. 8 – Input-offset voltage ("high reference) vs. ambient temperature.

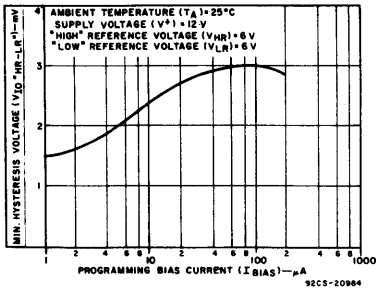


Fig. 9 – Min. hysteresis voltage vs. programming bias current.

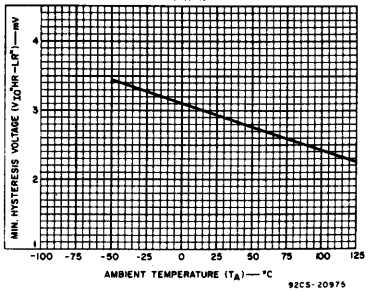


Fig. 10 – Min. hysteresis voltage vs. ambient temperature.

ELECTRICAL CHARACTERISTICS at T_A = 25°C Unless Otherwise Specified

CHARACTERISTIC	TEST CONDITIONS	Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
Input Offset Voltage: "Low" Ref., $V_{IO(LR)}$	V_{LR} = Gnd, V_{HR} = 3 V, I_{BIAS} = 100 μA	5	−15	−3	6	mV
"High" Ref., $V_{IO(HR)}$	V_{HR} = Gnd, V_{LR} = −3 V, I_{BIAS} = 100 μA	6	−10	±10	10	
Temp. Coeff: "Low" Ref.	−55 °C to + 125 °C	7	–	4.5	–	μV/°C
"High" Ref.	−55 °C to + 125 °C	8	–	±8.2	–	
Min. Hysteresis Voltage $V_{IO(HR-LR)}$:	V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	9	–	3	20	mV
Temp. Coeff.	−55°C to + 125 °C	10	–	6.7	–	μV/°C
Output Saturation Voltage, V_{CE}(SAT)	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	11,12	–	0.72	1.2	V
Total Supply Current, I_{TOTAL}: "ON"	V_I = 4 V, V_{REG} = 6 V; V^+ = 12 V, I_{BIAS} = 100 μA	13,14	500	710	800	μA
"OFF"	V_I = 8 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA		400	560	750	μA
Input Bias Current, I_{IB}: $I_{B(p\text{-}n\text{-}p)}$	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	15	–	42	100	nA
$I_{B(n\text{-}p\text{-}n)}$	V_I = 8 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA		–	28	100	nA
Output Leakage Current, $I_{CE(OFF)}$	Current from Term. 3 when Q46 is "OFF"	–	–	–	10	μA
Switching Times: Delay, t_d	I_C = 100 μA, I_{BIAS} = 100 μA, V^+ = 5 V, V_{REG} = 2.5 V	18	–	600	–	ns
Fall, t_f			–	50	–	ns
Rise, t_r			–	500	–	ns
Storage, t_s			–	4.5	–	μs
Output Current, I_O	V^+ = 12 V, I_{BIAS} = 50 μA	–	100	–	–	mA

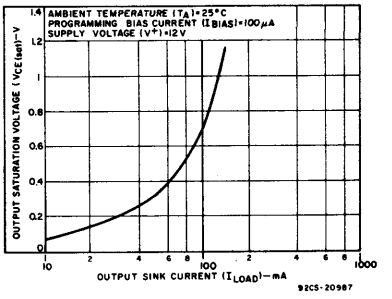


Fig. 11 – Output saturation voltage vs. output sink current.

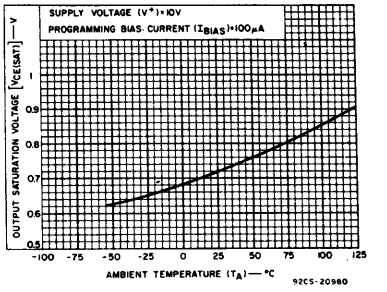


Fig. 12 – Output saturation voltage vs. ambient temperature.

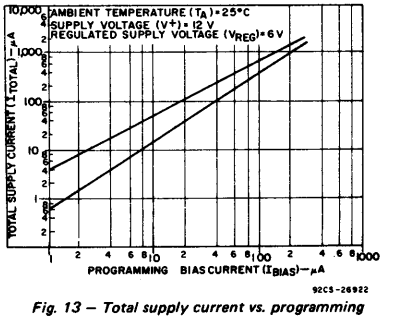


Fig. 13 – Total supply current vs. programming bias current.

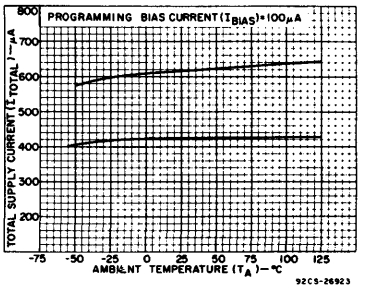


Fig. 14 – Total supply current vs. ambient temperature.

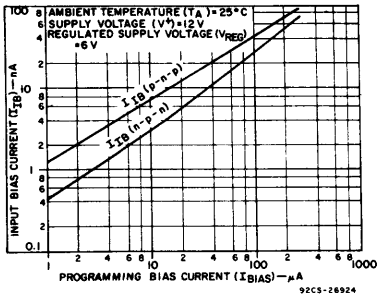


Fig. 15 – Input bias current vs. programming bias current.

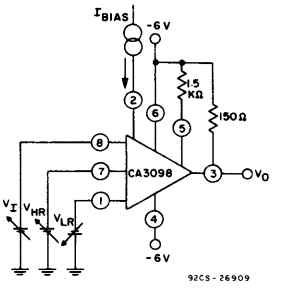


Fig. 16 – Input-offset voltage test circuit.

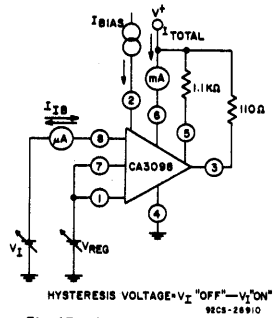


Fig. 17 – Min. hysteresis voltage, total supply current, and input-bias-current test circuit.

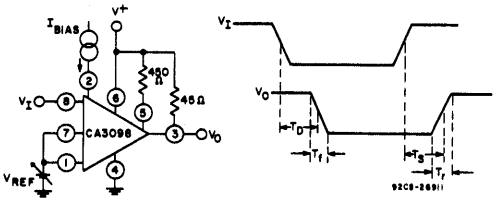


Fig. 18 – Switching time test circuit.

TYPICAL APPLICATIONS

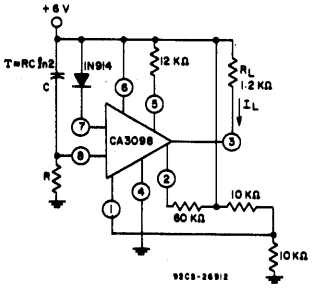


Fig. 19 – Time delay circuit: Terminal 3 "sinks" after τ seconds.

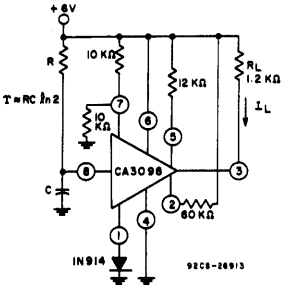


Fig. 20 – Time delay circuit: "sink" current interrupted after T seconds.

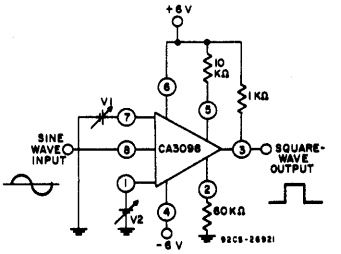
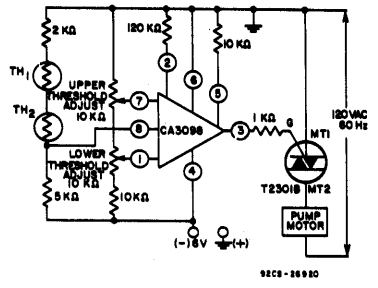


Fig. 21 – Sine-wave to square-wave converter with duty-cycle adjustment (V_1 and V_2).



Notes (a) Motor pump is "ON" when water level rises above thermistor TH_2.
(b) Motor pump remains "ON" until water level falls below thermistor TH_1.
(c) Thermistors, operate in self-heating mode.

Fig. 22(a) – Water-level control.

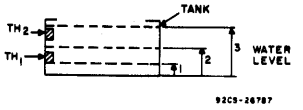


Fig. 22(b) – Water level diagram for circuit of Fig. 22(a).

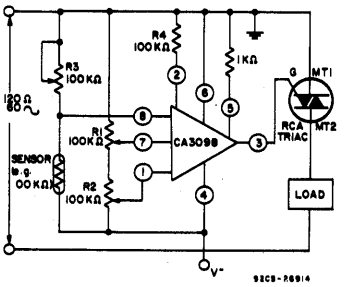


Fig. 23 – OFF/ON control of triac with programmable hysteresis.

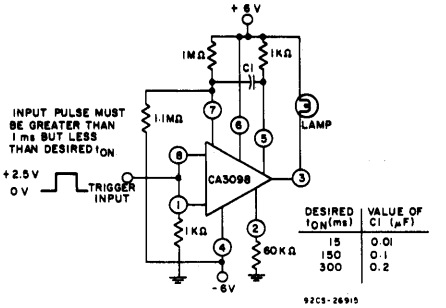


Fig. 24 – One-shot multivibrator.

Programmable Comparator - - With Memory

RCA-CA3099E Programmable Comparator is a monolithic silicon integrated circuit designed to control high-operating-current loads such as thyristors, lamps, relays, etc. The CA3099E can be operated with either a single power supply with maximum operating voltage of 16 volts, or a dual power supply with a maximum operating voltage of ± 8 volts. It can directly control currents up to 150 mA. It operates with microwatt standby power dissipation when the current to be controlled is less than 30 mA. The CA3099E contains the following six (6) major circuit-function features (Figure 1):

1. **Differential amplifiers and summer;** the circuit uses two differential amplifiers, one to compare the input voltage with the "high" reference, and the other to compare the input with the "low" reference. The resultant output of the *differential* amplifiers actuates a summer circuit which delivers a trigger that initiates a change in state of a flip-flop.
2. **Flip-flop;** the flip-flop functions as a bistable "memory" element that changes state in response to each trigger command.
3. **Driver and output stages;** these stages permit the circuit to "sink" maximum peak load currents up to 150 mA at terminal 3.
4. **Programmable operating current;** the circuit incorporates a separate terminal to permit programming the desired quiescent operating current and performance parameters.
5. **Internal sources of reference voltage and programmable bias current;** an integral circuit supplies a temperature-compensated reference voltage (V_b/2) which is about 1/2 of the externally applied bias voltage (V_b). Additionally, integral circuitry can optionally be used to supply an uncompensated constant-current source of bias (I_{bias}).
6. **Voltage regulator;** provides optional on-chip voltage regulation when power for the CA3099E is provided by an unregulated supply.

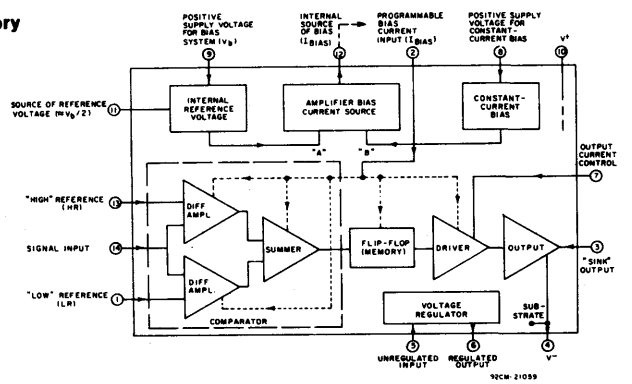


Fig.1–Block diagram of CA3099E programmable comparator.
(See page 3 for general description of circuit operation.)

Features:

- Programmable operating current
- Micro-power standby dissipation
- Directly controls current up to 150 mA
- Low input on/off current of less than 1 nA for programmable bias current of 1 μA
- Built-in hysteresis: 10 mV max.
- Programmable hysteresis: 10 mV to V^+
- Dual reference input
- High sensor range: 100 Ω to 100 MΩ
- Stable predictable switching levels
- Temperature-compensated reference voltage

Applications:

- Control of relays, heaters, LED's, lamps, photo-sensitive devices, thyristors, solenoids, etc.
- Signal reconditioning
- Phase and frequency modulators
- On/off motor switching
- Schmitt triggers, level detectors
- Time delays
- Overvoltage, overcurrent, overtemperature protection
- Battery-operated equipment
- Square and triangular-wave generators

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

Supply Voltage Between Terminals 10 and 4, 9 and 4, 8 and 4	16	V
Output Voltage Between Terminals 7 and 4, and 3 and 4	16	V
Differential Input Voltage Between Terminals 14 and 1, and Terminals 13 and 14	10	V
Operating Voltage Range:		
Term. 14	0 V to V^+	
Term. 13	2.0 V to V^+	
Term. 1	0 V to V^+ minus 2.0 V	
Load Current (Term. 3)	150	mA
Input Current to Voltage Regulator (Term. 5)	25	mA
Programming *Bias Current* (Term. 2)	1	mA
Output Current Control (Term. 7)	15	mA
Power Dissipation:		
Up to T_A = 55°C	750	mW
Above T_A = 55°C Derate Linearly at	6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Termperature (During Soldering): At distance not less than 1/32 inch (0.79 mm) from seating plane for 10 s maximum	+265	°C

ELECTRICAL CHARACTERISTICS AT T_A = 25°C *(Unless otherwise indicated)*

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C Unless Otherwise Indicated	FIG. No.	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNIT
Reference Voltage	V_{REF}	Term. 9 = 12 V, Term.4 = Grd, Term.11 = Test	–	5.7	6	6.3	V
Reference Voltage Temperature Coefficient			–	–	100	–	μV/°C
Regulated Supply Voltage	V_{REG}	Term.5 1K to 12V, Term.4 = Grd, Term.6 10K to Grd	5	6	7.2	8	V
Regulated Supply Voltage Temperature Coefficient			5	–	2.9	–	mV/°C
Input Offset Voltage: "Low" Reference	V_{IO} (LR)	V_{LR} = Grd, V_{HR} = 3 V, I_{BIAS} = 100 μA	20, 6	–8	–3	2	mV
"High" Reference	V_{IO} (HR)	V_{HR} = Grd, V_{LR} = −3 V, I_{BIAS} = 100 μA	20, 7	–5	±1	5	
"Low" Reference Temp. Coefficient		−55°C to +125°C	20, 8	–	4.5	20	μV/°C
"High" Reference Temp. Coefficient		−55°C to +125°C	20, 9	–	±8.2	±20	
Min. Hysteresis Voltage	$V_{IO(HR-LR)}$	V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21, 10	–	3	10	mV
Min. Hysteresis Voltage Temperature Coefficient		−55°C to +125°C	11	–	6.7	20	μV/°C
Output Saturation Voltage	V_{CE}(SAT)	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,12,13	–	0.72	1.2	V
Total Supply Current: I_{TOTAL} "ON"	I_{TOTAL}	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,14,15	600	710	800	μA
I_{TOTAL} "OFF"		V_I = 8 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,14,15	420	560	750	
Input Bias Current: $I_{B(p-n-p)}$	I_{IB}	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,16,17	–	33	200	nA
$I_{B(n-p-n)}$		V_I = 8 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,16,17	–	20	60	
Output Leakage Current	$I_{CE(OFF)}$	Current from Term.3 when Q46 is "OFF"	–	–	–	10	
Internal Bias Current	I_{IBC}		18,19	120	200	280	μA
Switching Times:							
Delay	t_d	I_C = 100 μA, I_{BIAS} = 100 μA, V^+ = 5 V, V_{REG} = 2.5 V	22	–	600	–	ns
Fall	t_f		22	–	50	–	
Rise	t_r		22	–	500	–	
Storage	t_s		22	–	4.5	–	μs

CA3099E

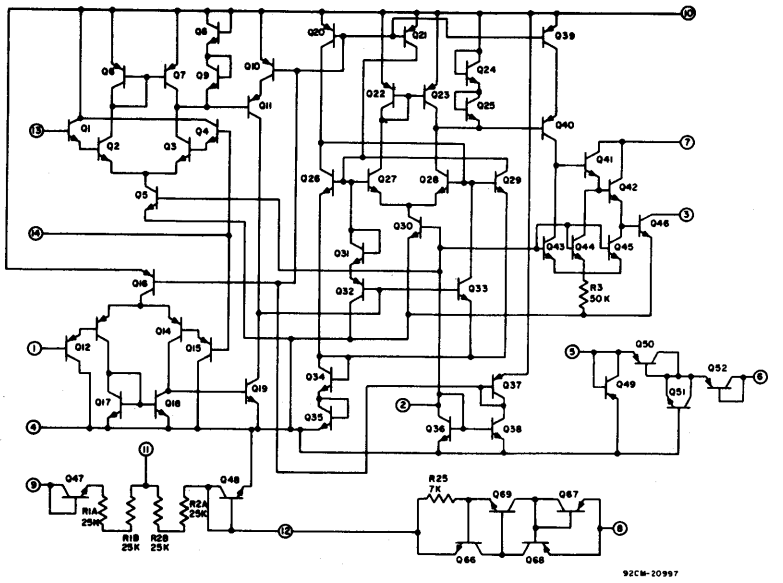


Fig.2–Schematic diagram of CA3099E.

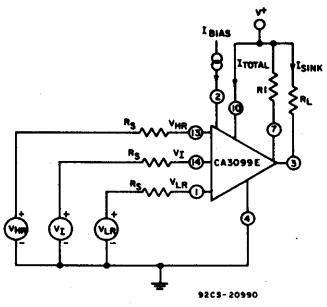


Fig. 3 – Functional diagram.

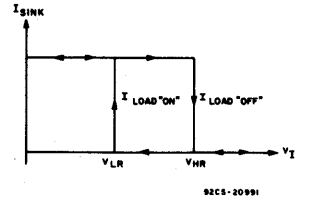


Fig. 4 – Logic diagram.

General Description of Circuit Operation (Refer to Fig.1)

When the signal-input voltage of the CA3099E is equal to or less than the "low" reference voltage (LR), current flows from an external power supply through a load connected to terminal 3 ("sink" output). This condition is maintained until the signal-input voltage rises to or exceeds the "high" reference voltage (HR), thereby effecting a change in the state of the flip-flop (memory) such that the output stage interrupts current flow in the external load. This condition, in turn, is maintained until such time as the signal again becomes equal to or less than the "low" reference voltage (VR).

The CA3099E comparator is unique in that it contains circuit provisions to permit programmability. This feature provides flexibility to the designer to optimize quiescent power consumption, input-circuit characteristics, hysteresis, and additionally permits independent control of the comparator, namely, pulsing, strobing, keying, squelching, etc. Programmability is accomplished by means of the bias current (I_{bias}) supplied to terminal 2. As an alternative to externally supplied bias current, the CA3099E contains an internal source of regulated bias current accessible at terminal 12. This internal source of bias current is developed by two alternative methods; in the first method, bias voltage (V_b) applied at terminal 9 develops a source of temperature-compensated reference voltage ($\approx V_b/2$) at terminal 11 and additionally supplies a source of bias current at terminal 12 via line "A". Alternately, when a positive supply voltage is applied at terminal 8, a source of constant-current biasing is provided at terminal 12 via line "B".

An auxiliary means of controlling the magnitude of load-current flow at terminal 3 is provided by "sinking" current into terminal 7. The CA3099E contains an on-chip voltage regulator which may optionally be used to regulate the voltages and bias currents (exclusive of the load current at terminal 3) needed for the operation of the IC.

Fig. 2 is the schematic diagram of the CA3099E. Figs. 3 and 4 are, respectively, functional and logic diagrams of CA3099E operation.

TYPICAL CHARACTERISTIC CURVES

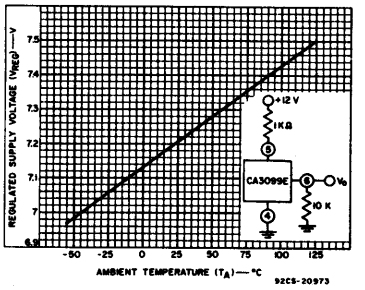


Fig. 5 – Regulated supply voltage vs. ambient temperature.

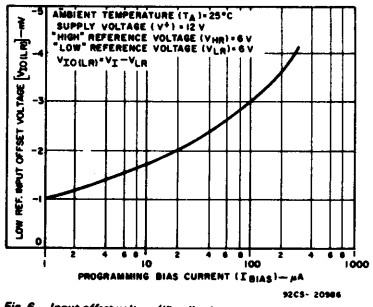


Fig. 6 – Input-offset voltage ("low" reference) vs. programming bias current.

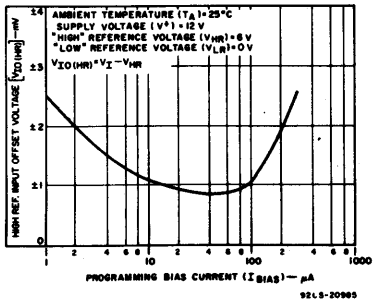


Fig. 7 – Input-offset voltage ("high" reference) vs. programming bias current.

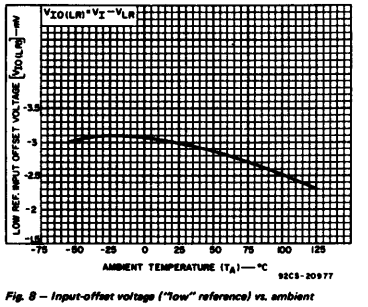


Fig. 8 – Input-offset voltage ("low" reference) vs. ambient temperature.

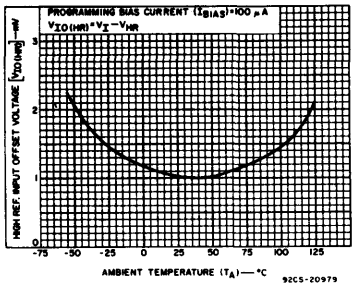


Fig. 9 – Input-offset voltage ("high" reference) vs. ambient temperature.

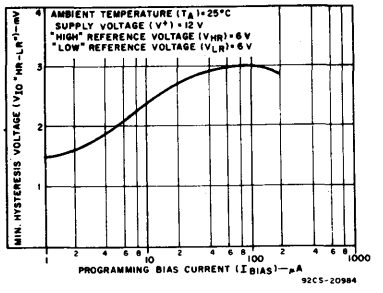


Fig. 10 – Min. hysteresis voltage vs. programming bias current.

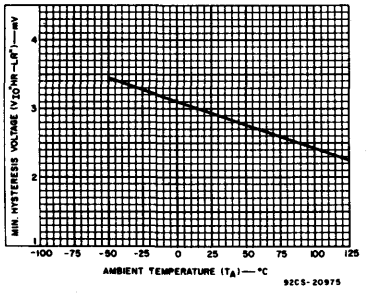


Fig. 11 – Min. hysteresis voltage vs. ambient temperature.

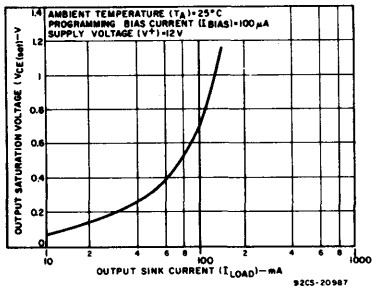


Fig. 12 – Output saturation voltage vs. output sink current.

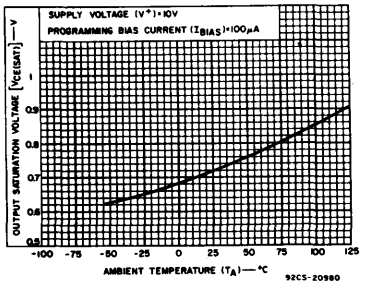


Fig. 13 – Output saturation voltage vs. ambient temperature.

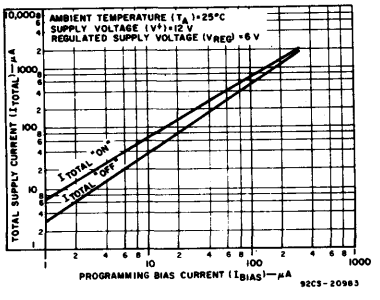


Fig. 14 – Total supply current vs. programming bias current.

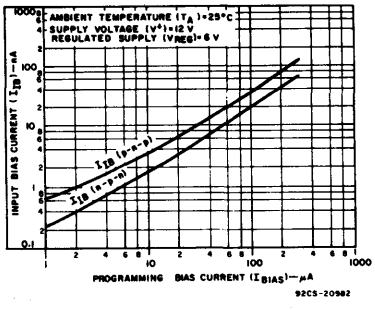


Fig. 15 - Input bias current vs. programming bias current.

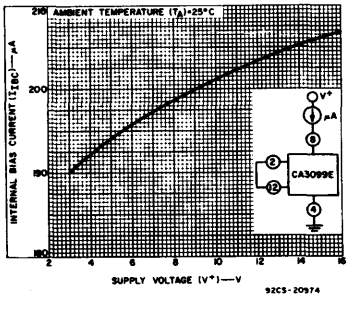


Fig. 16 – Internal bias current vs. supply voltage.

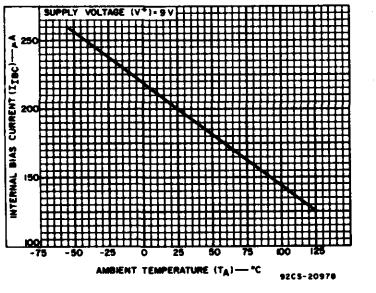


Fig. 17 – Internal bias current vs. ambient temperature.

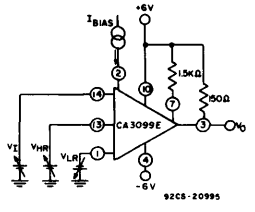


Fig. 18 – Input-offset voltage test circuit.

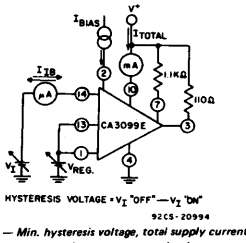


Fig. 19 – Min. hysteresis voltage, total supply current, and input bias current test circuit.

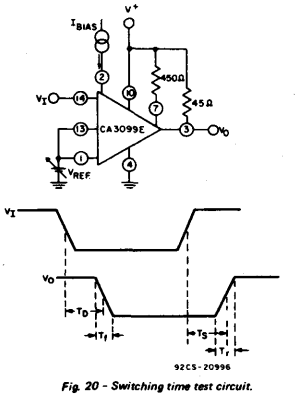


Fig. 20 - Switching time test circuit.

For application information, see Data Bulletin File No. 620.

Wideband Operational Amplifier

The RCA-CA3100 is a large-signal wideband, high-speed operational amplifier which has a unity gain crossover frequency (f_T) of approximately 38 MHz and an open-loop, 3 dB corner frequency of approximately 110 kHz. It can operate at a total supply voltage of from 14 to 36 volts (±7 to ±18 volts when using split supplies) and can provide at least 18 V p-p and 30 mA p-p at the output when operating from ±15 volt supplies. The CA3100 can be compensated with a single external capacitor and has dc offset adjust terminals for those applications requiring offset null. (See Fig. 15).

The CA3100 circuit contains both bipolar and P-MOS transistors on a single monolithic chip. This circuit is supplied in the standard 8-lead TO-5 package (T suffix), the 8-lead TO-5 dual-in-line formed-lead "DIL-CAN" package (S suffix), the 8-lead Mini-DIP (E Suffix), or in chip form (H suffix).

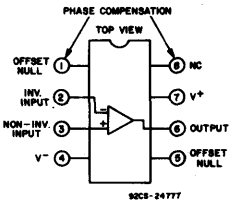
Features:

- High open-loop gain at video frequencies – 42 dB typ. at 1 MHz
- High unity-gain crossover frequency (f_T) – 38 MHz typ.
- Wide power bandwidth – V_O = 18 V p-p typ. at 1.2 MHz
- High slew rate – 70 V/μs (typ.) in 20 dB amplifier
  25 V/μs (typ.) in unity-gain amplifier
- Fast settling time – 0.6 μs typ.
- High output current – ±15 mA min.
- LM118, 748/LM101 pin compatibility
- Single capacitor compensation
- Offset null terminals

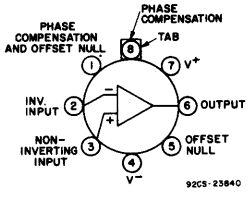
Applications:

- Video amplifiers
- Fast peak detectors
- Meter-driver amplifiers
- High-frequency feedback amplifiers
- Video pre-drivers
- Oscillators
- Multivibrators
- Voltage-controlled oscillator
- Fast comparators

TERMINAL ASSIGNMENTS



E Suffix



S & T Suffixes

MAXIMUM RATINGS, *Absolute-Maximum Values:*

Supply Voltage (between V+ and V⁻ terminals)	36	V
Differential Input Voltage	±12	V
Input Voltage to Ground*	±15	V
Offset Terminal to V⁻ Terminal Voltage	±0.5	V
Output Current	50	mA•
Device Dissipation:		
Up to T_A = 55°C	630	mW
Above T_A = 55°C	6.67	mW/°C
Ambient Temperature Range:		
Operating:		
E Type	−40 to +85	°C
S and T Types	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 s max.	+265	°C

* If the supply voltage is less than ±15 volts, the maximum input voltage to ground is equal to the supply voltage.

• CA3100S, CA3100T does not contain circuitry to protect against short circuits in the output.

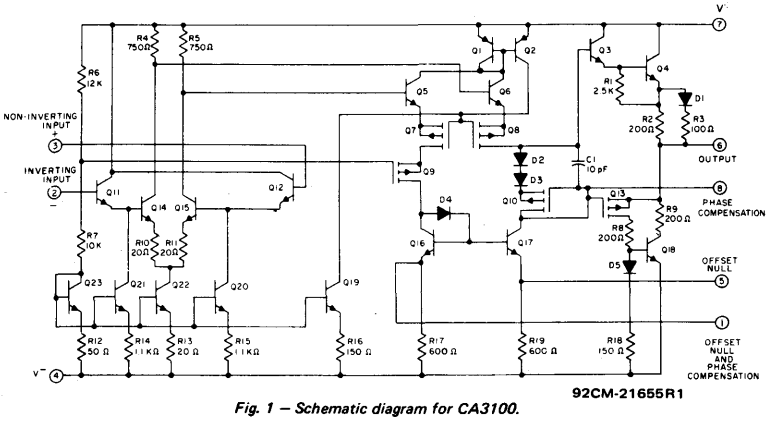


Fig. 1 – Schematic diagram for CA3100.

TYPICAL CHARACTERISTIC CURVES

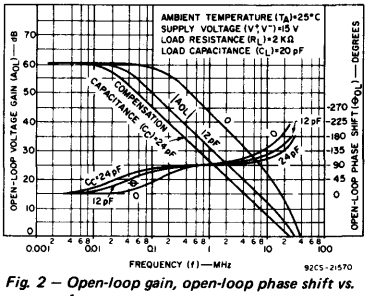


Fig. 2 – Open-loop gain, open-loop phase shift vs. frequency.

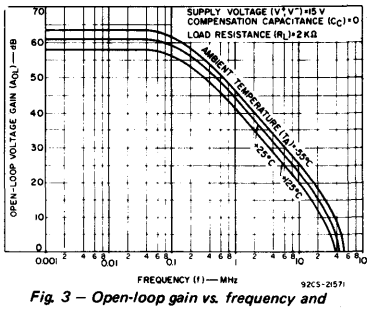


Fig. 3 – Open-loop gain vs. frequency and temperature.

ELECTRICAL CHARACTERISTICS, At T_A = 25°C:

CHARACTERISTICS	TEST CONDITIONS SUPPLY VOLTAGE (V^+,V^-)=15 V UNLESS OTHERWISE SPECIFIED	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
STATIC					
Input Offset Voltage, V_{IO}	V_O = 0 ± 0.1 V	–	±1	±5	mV
Input Bias Current, I_{IB}	V_O = 0 ± 1 V	–	0.7	2	μA
Input Offset Current, I_{IO}		–	±0.05	±0.4	μA
Low-Frequency Open-Loop Voltage Gain, A_{OL}●	V_O = ± 1 V Peak, F = 1 kHz	56	61	–	dB
Common-Mode Input, Voltage Range, V_{ICR}	CMRR ≥ 76 dB	±12	+14 −13	–	V
Common-Mode Rejection Ratio, CMRR	V_I Common Mode = ± 12 V	76	90	–	dB
Maximum Output Voltage: Positive, V_{OM}^+	Differential Input Voltage = 0 ± 0.1 V, R_L = 2 KΩ	+9	+11	–	V
Negative, V_{OM}^-		−9	−11	–	
Maximum Output Current: Positive, I_{OM}^+	Differential Input Voltage = 0 ± 0.1 V, R_L = 250 Ω	+15	+30	–	mA
Negative, I_{OM}^-		−15	−30	–	
Supply Current, I^+	V_O = 0 ± 0.1 V, R_L ≥ 10 KΩ	–	8.5	10.5	mA
Power-Supply Rejection Ratio, PSRR	ΔV^+ = ± 1 V, ΔV^- = ± 1 V	60	70	–	dB
DYNAMIC					
Unity-Gain Crossover Frequency, f_T	C_C = 0, V_O = 0.3 V (P-P)	–	38	–	MHz
1-MHz Open-Loop Voltage Gain, A_{OL}	f = 1 MHz, C_C = 0, V_O = 10 V (P-P)	36	42	–	dB
Slew Rate, SR: 20-dB Amplifier	A_V = 10, C_C = 0, V_I = 1 V (Pulse)	50	70	–	V/μs
Follower Mode	A_V = 1, C_C = 10 pF, V_I = 10 V (Pulse)	–	25	–	
Power Bandwidth, PBW▲: 20-dB Amplifier	A_V = 10, C_C = 0, V_O = 18 V (P-P)	0.8	1.2	–	MHz
Follower Mode	A_V = 1, C_C = 10 pF, V_O = 18 V (P-P)	–	0.4	–	
Open-Loop Differential Input Impedance, Z_I	F = 1 MHz	–	30	–	KΩ
Open-Loop Output Impedance, Z_O	F = 1 MHz	–	110	–	Ω
Wideband Noise Voltage Referred to Input, e_N(Total)	BW = 1 MHz, R_S = 1 KΩ	–	8	–	μVRMS
Settling Time, t_s To Within ± 50 mV of 9 V Output Swing	R_L = 2 KΩ, C_L = 20 pF	–	0.6	–	μs

▲ Power Bandwidth = $\frac{\text{Slew Rate}}{\pi V_O \text{ (P-P)}}$ ● Low-frequency dynamic characteristic

TYPICAL CHARACTERISTIC CURVES (Cont'd)

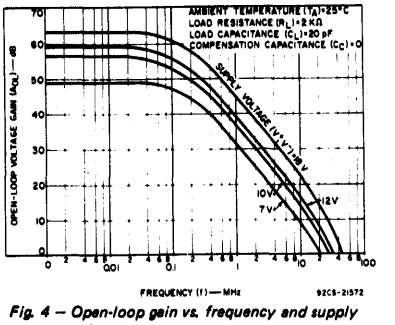


Fig. 4 – Open-loop gain vs. frequency and supply voltage.

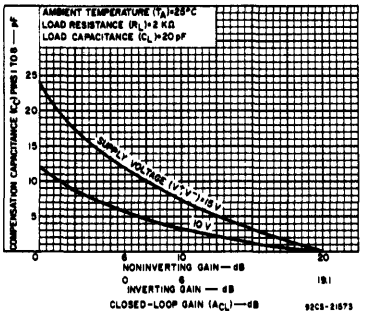


Fig. 5 – Required compensation capacitance vs. closed-loop gain.

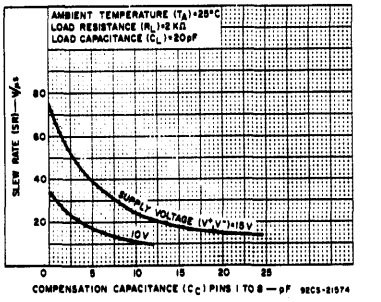


Fig. 6 – Slew rate vs. compensation capacitance.

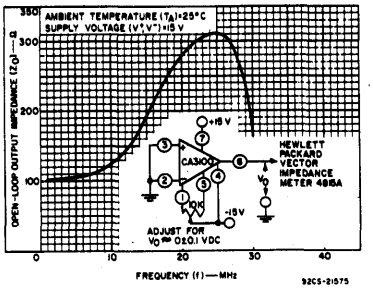


Fig. 7 – Typical open-loop output impedance vs. frequency.

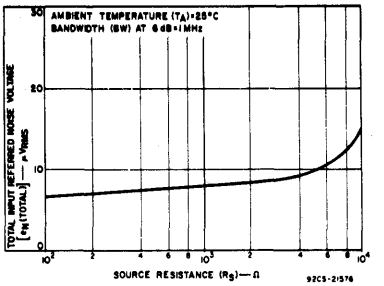


Fig. 8 – Wideband input noise voltage vs. source resistance.

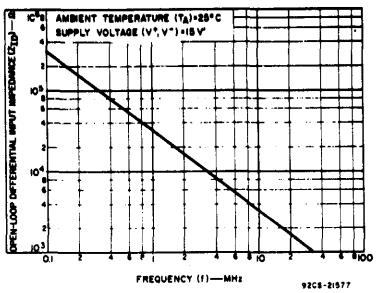


Fig. 9 – Typical open-loop differential input impedance vs. frequency.

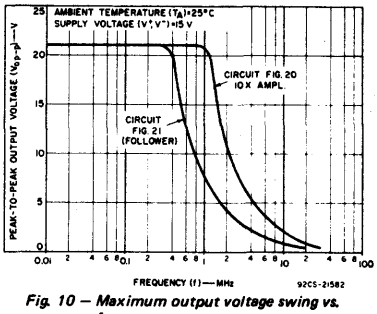


Fig. 10 – Maximum output voltage swing vs. frequency.

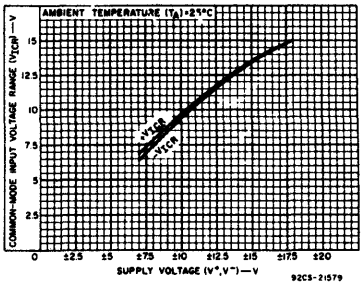


Fig. 11 – Common-mode input voltage range vs. supply voltage.

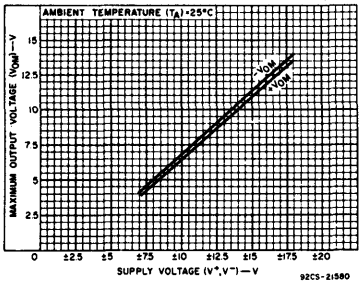


Fig. 12 – Maximum output voltage vs. supply voltage.

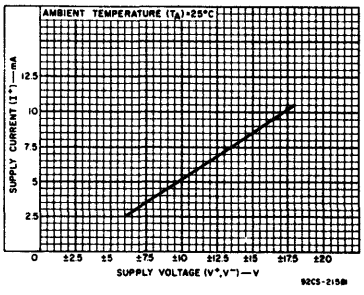


Fig. 13 – Supply current vs. supply voltage.

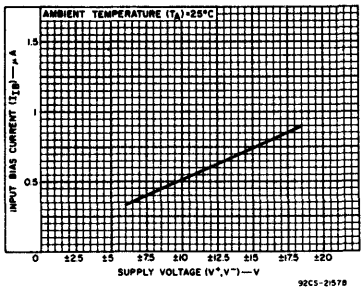


Fig. 14 – Input bias current vs. supply voltage.

TEST CIRCUITS

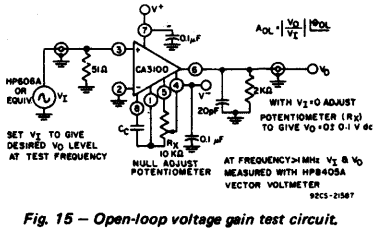


Fig. 15 – Open-loop voltage gain test circuit.

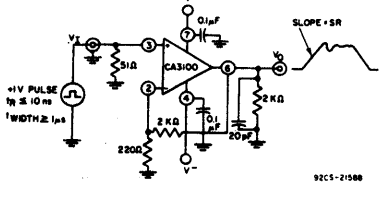


Fig. 16 – Slew rate in 10X amplifier test circuit.

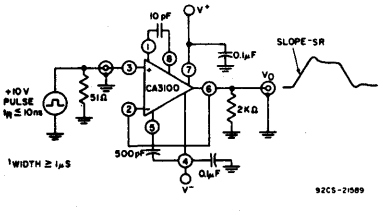


Fig. 17 – Follower slew rate test circuit.

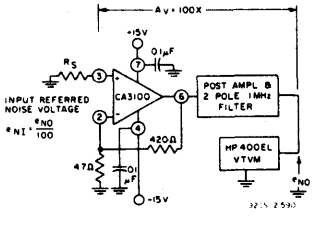


Fig. 18 – Wideband input noise voltage test circuit.

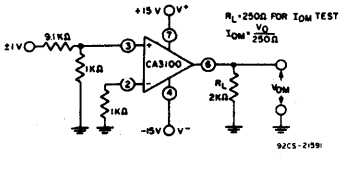


Fig. 19 – Output voltage swing (V_{OM}), output current swing (I_{OM}) test circuit.

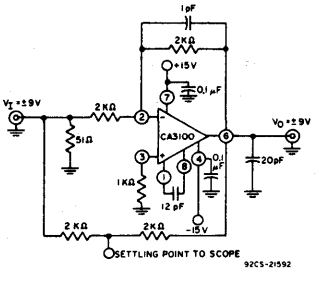


Fig. 20 – Settling time test circuit.

TYPICAL APPLICATIONS

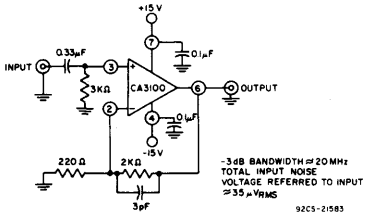


Fig. 21 – 20 dB video amplifier.

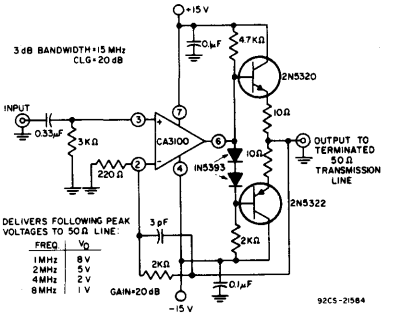


Fig. 22 – 20 dB video line driver.

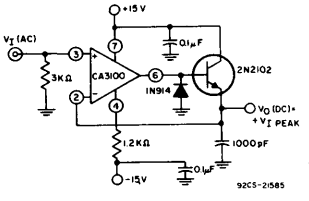


Fig. 23 – Fast positive peak detector.

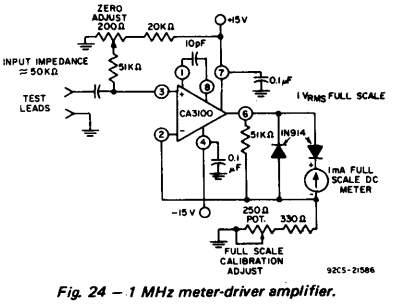
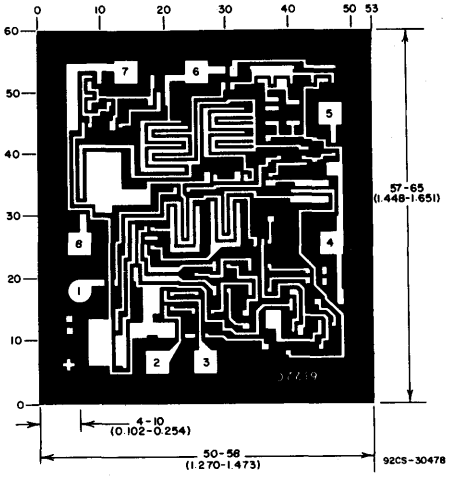


Fig. 24 – 1 MHz meter-driver amplifier.

Chip Dimensions and Pad Layout



CA3100H Chip

The photographs and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

CA3118, CA3146, CA3183 Types

High-Voltage Transistor Arrays

RCA-CA3118AT, CA3118T, CA3146AE, CA3146E, CA-3183AE, and CA3183E are general-purpose high-voltage silicon n-p-n transistor arrays on a common monolithic substrate.

Types CA3118AT and CA3118T consist of four transistors with two of the transistors connected in a Darlington configuration. These types are well suited for a wide variety of applications in low-power systems in the DC through VHF range. Both types are supplied in a hermetically sealed 12-lead TO-5 type package and operate over the full military temperature range. (CA3118AT and CA3118T are high-voltage versions of the popular predecessor type CA3018.

Types CA3146AE and CA3146E consist of five transistors with two of the transistors connected to form a differentially-connected pair. These types are recommended for low-power applications in the DC through VHF range.

Types CA3183AE and CA3183E consist of five high-current transistors with independent connections for each transistor. In addition two of these transistors (Q1 and Q2) are matched at low-current (i.e. 1mA) for applications where offset parameters are of special importance. A special substrate terminal is also included for greater flexibility in circuit design.

The types with an "A" suffix are premium versions of their non-"A" counterparts and feature tighter control of breakdown voltages making them more suitable for higher voltage applications.

For detailed application information, see companion Application Note, ICAN-5296 "Application of the RCA CA3018 Integrated Circuit Transistor Array."

Features

- Matched general-purpose transistors
- V_{BE} matched ±.5mV max.
- Operation from DC to 120 MHz (CA3118AT, T; CA3146AE, E)
- Low-noise figure: 3.2dB typ. at 1kHz (CA3118AT, T; CA3146AE, E)
- High I_C: 75mA max. (CA3183AE, E)

Applications

- General use in signal processing systems in DC through VHF range
- Custom designed differential amplifiers
- Temperature compensated amplifiers
- Lamp and relay drivers (CA3183AE, E)
- Thyristor firing (CA3183AE, E)

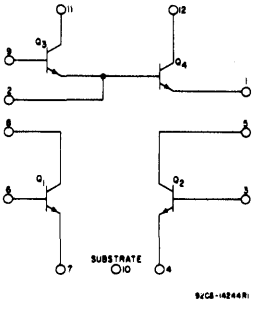
TYPE	P_T● max. mW	I_C max. mA	V_{CEO} max. V	V_{CBO} max. V	V_{CE} sat. at 10 mA typ. V	h_{FE} at 1mA, & V_{CE}=5V typ.	V_{IO} Diff. Pair at 1mA max. mV	I_{IO} Diff. Pair at 1mA max. µA	T_A Range (Operating) °C
VALUES APPLY FOR EACH TRANSISTOR									
CA3118AT	300	50	40	50	0.33	95	±5	2	−55 – +125
CA3118T	300	50	30	40	0.33	95	±5	2	
CA3146AE	300	50	40	50	0.33	95	±5	2	
CA3146E	300	50	30	40	0.33	95	±5	2	
CA3183AE	500	75	40	50	0.16	75	±5	2.5	
CA3183E	500	75	30	40	0.16	75	±5	2.5	

● Caution on Total Package Power Dissipation: The maximum total package dissipation rating for the CA3118 Series circuits is 450 mW at temperatures up to +85°C, then derate linearly at 5 mW/°C. The maximum total package dissipation rating for the CA3146 and CA3183 Series circuits is 750 mW at temperatures up to +55°C, then derate linearly at 6.67 mW/°C.

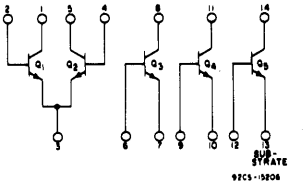
MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Power Dissipation:		
Any one transistor –		
CA3118AT, CA3118T, CA3146AE, CA3146E	300	mW
CA3183AE, CA3183E	500	mW
Total package –		
Up to 85°C (CA3118AT, CA3118T)	450	mW
Up to 55°C (CA3146AE, CA3146E, CA3183AE, CA3183E)	750	mW
Above 85°C (CA3118AT, CA3118T)	derate linearly 5	mW/°C
Above 55°C (CA3146AE, CA3146E, CA3183AE, CA3183E)	derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating –	−55 to +125	°C
Storage (all types)	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C
The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage (V_{CEO}):		
CA3118AT, CA3146AE, CA3183AE	40	V
CA3118T, CA3146E, CA3183E	30	V
Collector-to-Base Voltage (V_{CBO}):		
CA3118AT, CA3146AE, CA3183AE	50	V
CA3118T, CA3146E, CA3183E	40	V
Collector-to-Substrate Voltage (V_{CIO}): ■		
CA3118AT, CA3146AE, CA3183AE	50	V
CA3118T, CA3146E, CA3183E	40	V
Emitter-to-Base Voltage (V_{EBO}) all types	5	V
Collector Current –		
CA3118AT, CA3118T, CA3146AE, CA3146E	50	mA
CA3183AE, CA3183E	75	mA
Base Current (I_B) – CA3183AE, CA3183E	20	mA

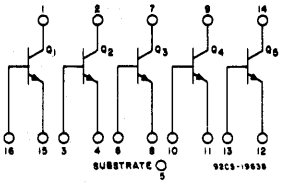
■ The collector of each transistor is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.



CA3118AT, CA3118T



CA3146AE, CA3146E



CA3183AE, CA3183E

Fig. 1 – Schematic diagrams of high-voltage arrays.

The CA3146AE and CA3146E are supplied in the 14-lead dual-in-line plastic package; the CA3183AE and CA3183E are supplied in the 16-lead dual-in-line plastic package.

COMPARISON OF RELATED PREDECESSOR TYPE WITH TYPES IN THIS DATA BULLETIN

	DATA FILE NO.	V_{CEO} min.	V_{CBO} min.	V_{CE} sat. typ. V I_C=10 mA	V_{BE} typ. V I_C=1 mA	I_C max. mA	C_{CB} typ. pF	C_{CI} typ. pF	C_{EB} typ. pF
CA3018	338	15	20	0.23	0.715	50	0.58	2.8	0.6
CA3018A	338	15	20	0.23	0.715	50	0.58	2.8	0.6
CA3118AT		40	50	0.33	0.730	50	0.37	2.2	0.7
CA3118T		30	40	0.33	0.730	50	0.37	2.2	0.7
				I_C=10 mA	I_C=1 mA				
CA3046	341	15	20	0.23	0.715	50	0.58	2.8	0.6
CA3146AE		40	50	0.33	0.730	50	0.37	2.2	0.7
CA3146E		30	40	0.33	0.730	50	0.37	2.2	0.7
				I_C=50 mA	I_C=10 mA				
CA3083	481	15	20	0.4	0.74	100	–	–	–
CA3183AE		40	50	1.7	0.75	75	–	–	–
CA3183E		30	40	1.7	0.75	75	–	–	–

STATIC ELECTRICAL CHARACTERISTICS – CA3118 and CA3146 Series

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C		LIMITS CA3118AT, CA3146AE Min.	CA3118AT, CA3146AE Typ.	CA3118AT, CA3146AE Max.	CA3118T, CA3146E Min.	CA3118T, CA3146E Typ.	CA3118T, CA3146E Max.	UNITS
For Each Transistor:										
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0		50	72	–	40	72	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0		40	56	–	30	56	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 10μA, I_B = 0, I_E = 0		50	72	–	40	72	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10μA, I_C = 0		5	7	–	5	7	–	V
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10V, I_B = 0		–	see curve	5	–	see curve	5	μA
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10V, I_E = 0		–	0.002	100	–	0.002	100	nA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE}=5V	I_C=10 mA	–	85	–	–	85	–	–
			I_C=1 mA	30	100	–	30	100	–	
			I_C=10μA	–	90	–	–	90	–	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 1mA		0.63	0.73	0.83	0.63	0.73	0.83	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_C = 10mA, I_B = 1mA		–	0.33	–	–	0.33	–	V
For transistors Q3 and Q4 (Darlington Configeration):										
Collector-Cutoff Current (CA3118AT and CA3118T only)	I_{CEO}	V_{CE} = 10V, I_B = 0		–	–	5	–	–	–	μA
DC Forward-Current Transfer Ratio (CA3118AT and CA3118T only)	h_{FE}	V_{CE} = 5V, I_C = 1mA		1500	9000	–	1500	9000	–	–
Base-to-Emitter (Q3 to Q4)	V_{BE}	V_{CE}=5V	I_E = 10mA	–	1.46	–	–	1.46	–	V
			I_E = 1mA	–	1.32	–	–	1.32	–	
Magnitude of Base-to-Emitter Temperature Coefficient	$\left\lvert\frac{\Delta V_{BE}}{\Delta T}\right\rvert$	V_{CE} = 5V, I_E = 1mA		–	4.4	–	–	4.4	–	mV/°C
For transistors Q1 and Q2 (AS a Differential Amplifier):										
Magnitude of Input Offset Voltage $\lvert V_{BE1} - V_{BE2}\rvert$	$\lvert V_{IO}\rvert$	V_{CE} = 5V, I_E = 1mA		–	0.48	5	–	0.48	5	mV
Magnitude of h_{FE} Ratio (CA3118AT and CA3118T only)		V_{CE} = 5V, $I_{C1} = I_{C2}$ = 1mA		0.9	1.0	1.1	0.9	1.0	1.1	–
Magnitude of Base-to-Emitter Temperature Coefficient	$\left\lvert\frac{\Delta V_{BE}}{\Delta T}\right\rvert$	V_{CE} = 5V, I_E = 1mA		–	1.9	–	–	1.9	–	mV/°C
Magnitude of V_{IO} ($V_{BE1} - V_{BE2}$) Temperature Coefficient	$\left\lvert\frac{\Delta V_{IO}}{\Delta T}\right\rvert$	V_{CE} = 5V, $I_{C1} = I_{C2}$ = 1mA		–	1.1	–	–	1.1	–	μV/°C
Magnitude of Input Offset Current $\lvert I_{IO_1} - I_{IO_2}\rvert$ (CA3146AE and CA3146E only)	I_{IO}	V_{CE} = 5V, $I_{C_1} = I_{C_2}$ = 1 mA		–	0.3	2	–	0.3	2	μA

TYPICAL STATIC CHARACTERISTICS CURVES– CA3118 and CA3146 SERIES (cont'd Fig.2 to 12)

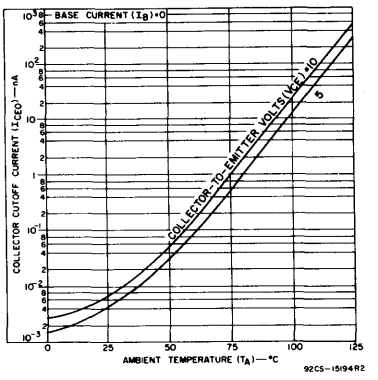


Fig. 2 – I_{CEO} vs. T_A for any transistor.

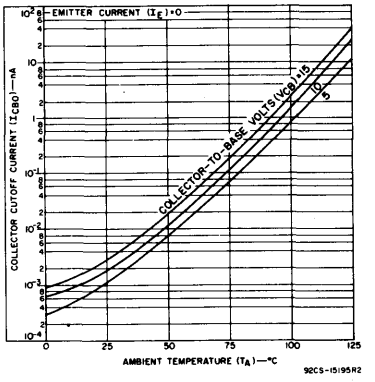


Fig. 3 – I_{CBO} vs. T_A for any transistor.

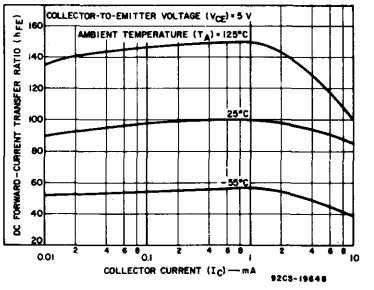


Fig. 4 – h_{FE} vs. I_C for any transistor.

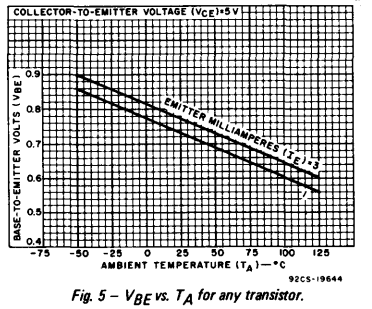


Fig. 5 – V_{BE} vs. T_A for any transistor.

CA3118, CA3146, CA3183 Types

DYNAMIC ELECTRICAL CHARACTERISTICS – CA3118 and CA3146 Series

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C	CA3118AT CA3146AE			CA3118T CA3146E			UNITS
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Low Frequency Noise Figure	NF	f = 1kHz, V_{CE} = 5V, I_C = 100μA, Source resistance = 1 kΩ	–	3.25	–	–	3.25	–	dB
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:									
Foward-Current Transfer Ratio	h_{fe}	f = 1kHz, V_{CE} = 5V, I_C = 1mA	–	100	–	–	100	–	–
Short-Circuit Input Impedance	h_{ie}		–	2.7	–	–	3.5	–	kΩ
Open-Circuit Output Impedance	h_{oe}		–	15.6	–	–	15.6	–	μmho
Open-Circuit Reverse Voltage Transfer Ratio	h_{re}		–	$1.8x10^{-4}$	–	–	$1.8x10^{-4}$	–	–
Admittance Characteristics:									
Foward Transfer Admittance	Y_{fe}	f = 1MHz, V_{CE} = 5V, I_C = 1mA	–	31-j1.5	–	–	31-j1.5	–	mmho
Input Admittance	Y_{ie}		–	0.35+j0.04	–	–	0.3+j0.04	–	mmho
Output Admittance	Y_{oe}		–	0.001+j0.03	–	–	0.001+j0.03	–	mmho
Reverse Transfer Admittance	Y_{re}			See curve			See curve		mmho
Gain-Bandwidth Product	f_T	V_{CE} = 5V, I_C = 3mA	300	500	–	300	500	–	MHz
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 5V, I_E = 0	–	0.70	–	–	0.70	–	pF
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 5V, I_C = 0	–	0.37	–	–	0.37	–	pF
Collector-to-Substrate Capacitance	C_{CI}	V_{CI} = 5V, I_C = 0	–	2.2	–	–	2.2	–	pF

STATIC ELECTRICAL CHARACTERISTICS – CA3183 Series

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C	LIMITS CA3183AE			LIMITS CA3183E			UNITS
			Min.	Typ.	Max.	Min.	Typ.	Max.	
For Each Transistor:									
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C=100μA, I_E=0	50	–	–	40	–	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C=1mA, I_B=0	40	–	–	30	–	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI}=100μA, I_B=0, I_E = 0	50	–	–	40	–	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0	5	–	–	5	–	–	V
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	10	–	–	10	μA
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	–	1	–	–	1	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3 V, I_C = 10mA	40	–	–	40	–	–	–
		V_{CE} = 5V, I_C = 50mA	40	–	–	40	–	–	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA	0.65	0.75	0.85	0.65	0.75	0.85	V
Collector-to-Emitter Saturation Voltage	*V_{CEsat}	I_C = 50mA, I_B = 5mA	–	1.7	3.0	–	1.7	3.0	V
For Transistors Q1 and Q2 (As a Differential Amplifier):									
Absolute Input Offset Voltage	$\lvert V_{IO}\rvert$	V_{CE} = 3V, I_C = 1mA	–	0.47	5	–	0.47	5	mV
Absolute Input Offset Current	$\lvert I_{IO}\rvert$		–	0.78	2.5	–	0.78	2.5	μA

* A maximum dissipation of 5 transistors x 150mW = 750mW is possible for a particular application.

TYPICAL STATIC CHARACTERISTICS CURVES– CA3118 and CA3146 SERIES (cont'd Fig.2 to 12)

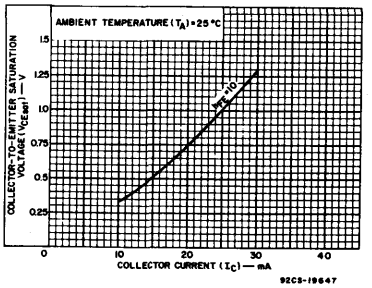


Fig. 6 – V_{CE} sat vs. I_C for any transistor.

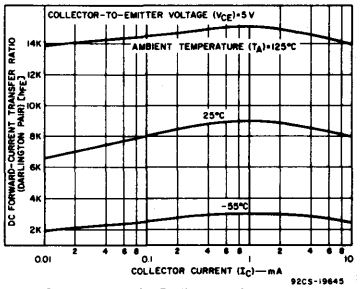


Fig. 7 – h_{FE} vs. I_C for Darlington pair (Q3 and Q4) for types CA3118AT and CA3118T.

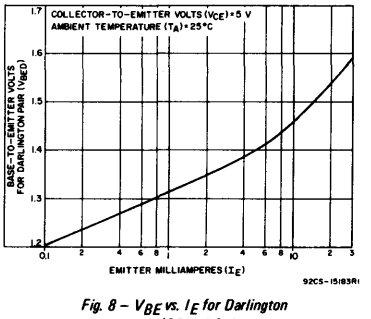


Fig. 8 – V_{BE} vs. I_E for Darlington pair (Q3 and Q4).

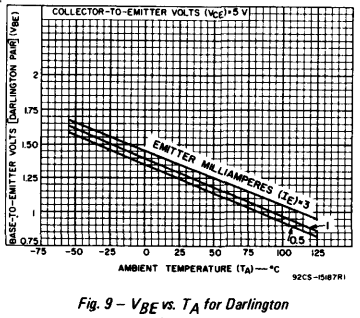


Fig. 9 – V_{BE} vs. T_A for Darlington pair (Q3 and Q4).

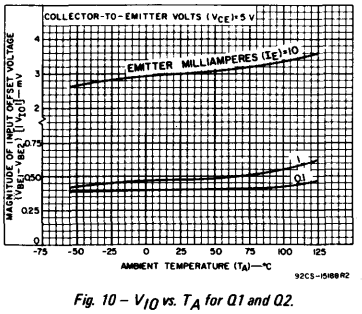


Fig. 10 – V_{IO} vs. T_A for Q1 and Q2.

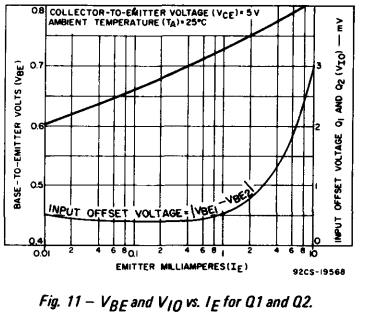


Fig. 11 – V_{BE} and V_{IO} vs. I_E for Q1 and Q2.

TYPICAL STATIC CHARACTERISTICS CURVES– CA3118 and CA3146 SERIES (Fig.2 to 12)

TYPICAL DYNAMIC CHARACTERISTICS CURVES (FOR ANY TRANSISTOR)–CA3118,CA3146 SERIES (Fig. 13 to 22)

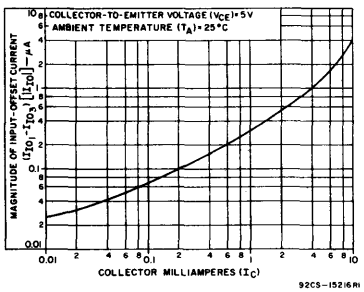


Fig. 12 – I_{IO} vs. I_C (Q1 and Q2) for types CA3146AE and CA3146E.

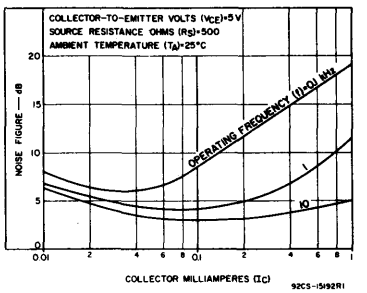


Fig. 13 – NF vs. I_C @ R_S = 500Ω.

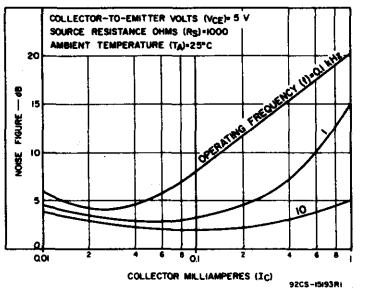


Fig. 14 – NF vs. I_C @ R_S = 1kΩ.

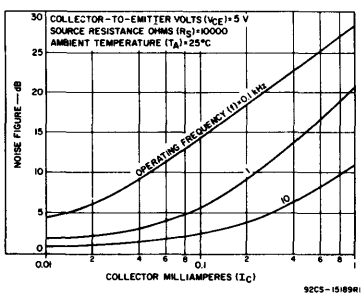


Fig. 15 – NF vs. I_C @ R_S = 10kΩ.

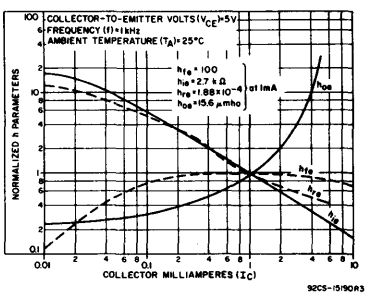


Fig. 16 – h_{fe}, h_{ie}, h_{oe}, h_{re} vs. I_C.

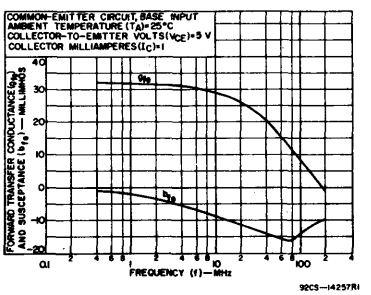


Fig. 17 – y_{fe} vs. f.

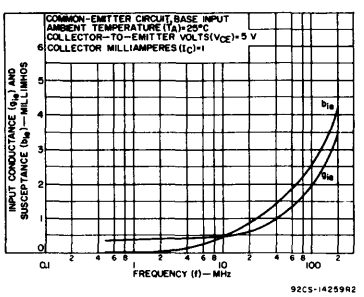


Fig. 18 – y_{ie} vs. f.

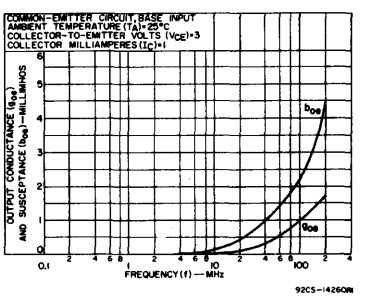


Fig. 19 – y_{oe} vs. f.

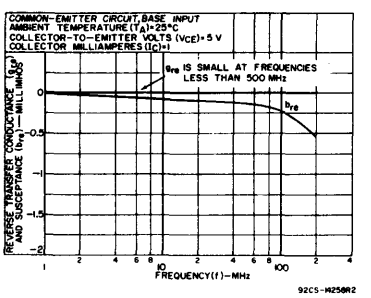


Fig. 20 – y_{re} vs. f.

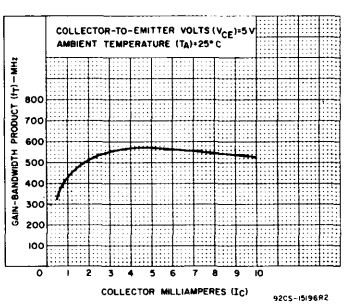


Fig. 21 – f_T vs. I_C.

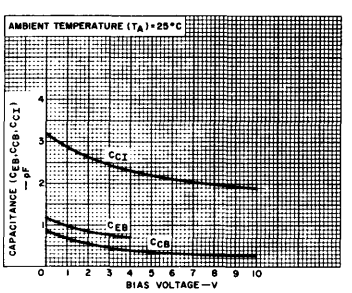


Fig. 22 – C_{EB}, C_{CB}, C_{CI} vs. bias voltage

TYPICAL STATIC CHARACTERISTICS CURVES—CA3183 SERIES (Fig. 23 to 30)

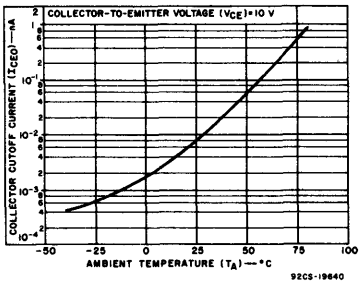


Fig. 23 – I_{CEO} vs. T_A for any transistor.

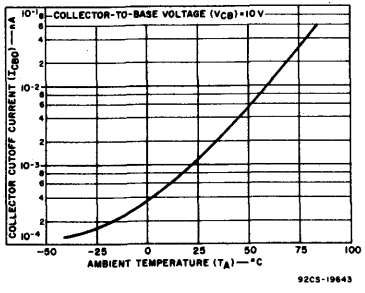


Fig. 24 – I_{CBO} vs. T_A for any transistor.

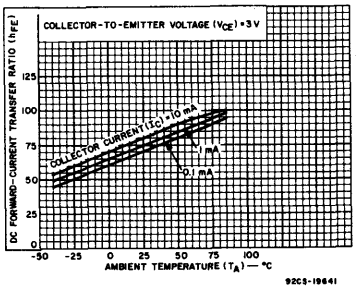


Fig. 25 – h_{FE} vs. T_A for any transistor.

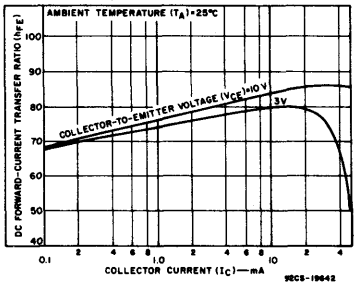


Fig. 26 – h_{FE} vs. I_C for any transistor.

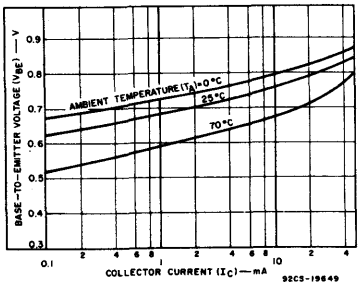


Fig. 27 – V_{BE} vs. I_C for any transistor.

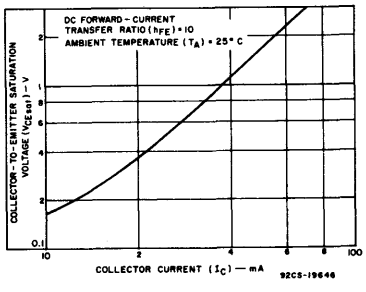


Fig. 28 – $V_{CE\ sat}$ vs. I_C for any transistor.

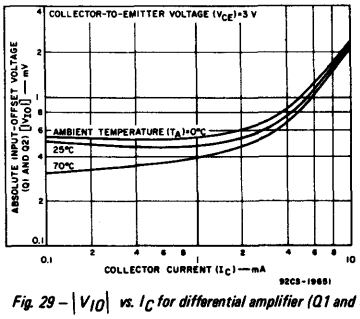


Fig. 29 – $|V_{IO}|$ vs. I_C for differential amplifier (Q1 and Q2).

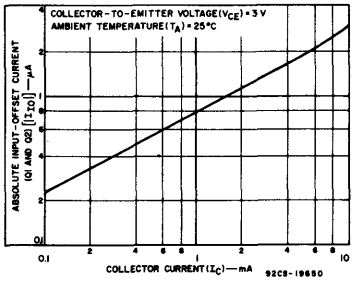


Fig. 30 – $|I_{IO}|$ vs. I_C for differential amplifier (Q1 and Q2).

High-Frequency N-P-N Transistor Array

For Low-Power Applications at Frequencies up to 500 MHz

RCA-CA3127E* consists of five general-purpose silicon n-p-n transistors on a common monolithic substrate. Each of the completely isolated transistors exhibits low 1/f noise and a value of f_T in excess of 1 GHz, making the CA3127E useful from dc to 500 MHz. Access is provided to each of the terminals for the individual transistors and a separate substrate connection has been provided for maximum application flexibility. The monolithic construction of the CA3127E provides close electrical and thermal matching of the five transistors.

The CA3127E is supplied in a 16-lead dual-in-line plastic package and operates over the full military temperature range of −55 to +125°C.

* Formerly RCA Dev. No. TA6206.

Features:

- Gain-Bandwidth Product (f_T) > 1 GHz
- Power Gain = 30 dB (typ.) at 100 MHz
- Noise Figure = 3.5 dB (typ.) at 100 MHz
- Five independent transistors on a common substrate

Applications:

- VHF amplifiers
- Multifunction combinations—RF/mixer/oscillator
- Sense amplifiers
- Synchronous detectors
- VHF mixers
- IF Converter
- IF amplifiers
- Synthesizers
- Cascade amplifiers

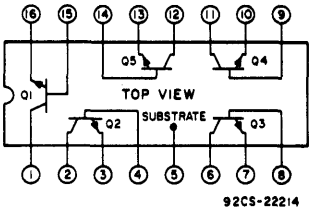


Fig. 1 – Schematic diagram of CA3127E.

MAXIMUM RATINGS,
Absolute-Maximum Values:

POWER DISSIPATION, P_D:
- Any one transistor . . . . . 85 mW
- Total Package:
- For T_A up to 75°C . . . . . 425 mW
- For T_A > 75°C Derate Linearly at . . . . . 6.67 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating . . . . . −55 to +125°C
- Storage . . . . . −65 to +125°C

LEAD TEMPERATURE (DURING SOLDERING):
- At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

The following ratings apply for each transistor in the device:
- Collector-to-Emitter Voltage, V_{CEO} . . . 15 V
- Collector-to-Base Voltage, V_{CBO} . . . . 20 V
- Collector-to-Substrate Voltage, V_{CIO}* . . 20 V
- Collector Current, I_C . . . . . 20 mA

*The collector of each transistor of the CA3127E is isolated from the substrate by an integral diode. The substrate (terminal 5) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.

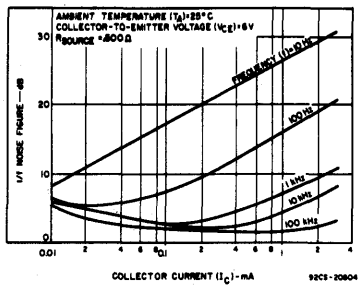


Fig. 2 – 1/f noise figure as a function of collector current at R_{SOURCE} = 500 Ω.

STATIC ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	TEST CONDITIONS		LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
For Each Transistor:						
Collector-to-Base Breakdown Voltage	I_C = 10 μA, I_E = 0		20	32	–	V
Collector-to-Emitter Breakdown Voltage	I_C = 1 mA, I_B = 0		15	24	–	V
Collector-to-Substrate Breakdown Voltage	I_{C1} = 10 μA, I_B = 0, I_E = 0		20	60	–	V
Emitter-to-Base Breakdown Voltage*	I_E = 10 μA, I_C = 0		4	5.7	–	V
Collector-Cutoff-Current	V_{CE} = 10 V, I_B = 0		–	–	0.5	μA
Collector-Cutoff-Current	V_{CB} = 10 V, I_E = 0		–	–	40	nA
DC Forward-Current Transfer Ratio	V_{CE} = 6 V	I_C = 5 mA	35	88	–	
		I_C = 1 mA	40	90	–	
		I_C = 0.1 mA	35	85	–	
Base-to-Emitter Voltage	V_{CE} = 6 V	I_C = 5 mA	0.71	0.81	0.91	V
		I_C = 1 mA	0.66	0.76	0.86	
		I_C = 0.1 mA	0.60	0.70	0.80	
Collector-to-Emitter Saturation Voltage	I_C = 10 mA, I_B = 1 mA		–	0.26	0.50	V
Magnitude of Difference in V_{BE}	Q_1 & Q_2 Matched		–	0.5	5	mV
Magnitude of Difference in I_B	V_{CE} = 6 V, I_C = 1 mA		–	0.2	3	μA

*When used as a zener for reference voltage, the device must not be subjected to more than 0.1 millijoule of energy from any possible capacitance or electrostatic discharge in order to prevent degradation of the junction. Maximum operating zener current should be less than 10 mA.

CA3127E

DYNAMIC CHARACTERISTICS at $T_A = 25°C$

CHARACTERISTICS	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
I/F Noise Figure	f = 100 kHz, R_S = 500 Ω, I_C = 1 mA	–	1.8	–	dB
Gain-Bandwidth Product	V_{CE} = 6 V, I_C = 5 mA	–	1.15	–	GHz
Collector-to-Base Capacitance	V_{CB} = 6 V, f = 1 MHz	–	See	–	pF
Collector-to-Substrate Capacitance	V_{CI} = 6 V, f = 1 MHz	–	Fig.	–	pF
Emitter-to-Base Capacitance	V_{BE} = 4 V, f = 1 MHz	–	5	–	pF
Voltage Gain	V_{CE} = 6 V, f = 10 MHz R_L = 1 kΩ, I_C = 1 mA	–	28	–	dB
Power Gain	Cascode Configuration f = 100 MHz, V^+ = 12 V	27	30	–	dB
Noise Figure	I_C = 1 mA	–	3.5	–	dB
Input Resistance	Common-Emitter	–	400	–	Ω
Output Resistance	Configuration	–	4.6	–	kΩ
Input Capacitance	V_{CE} = 6 V	–	3.7	–	pF
Output Capacitance	I_C = 1 mA	–	2	–	pF
Magnitude of Forward Transadmittance	f = 200 MHz	–	24	–	mmho

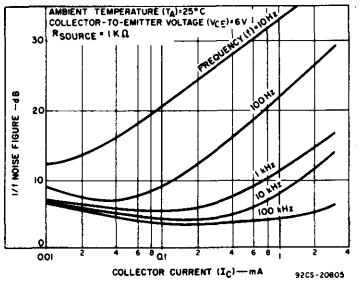


Fig. 3 – 1/f noise figure as a function of collector current at R_{SOURCE} = 1 kΩ.

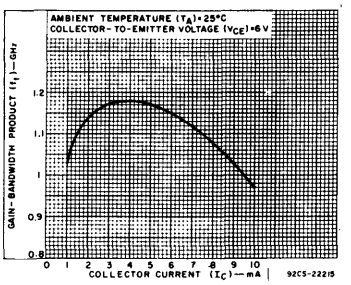


Fig. 4 – Gain-bandwidth product as a function of collector current.

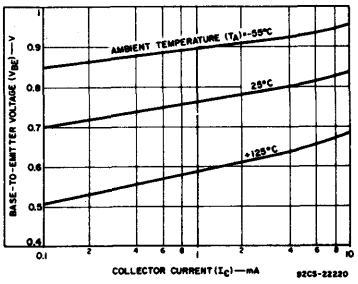


Fig. 5 – Base-to-emitter voltage as a function of collector current.

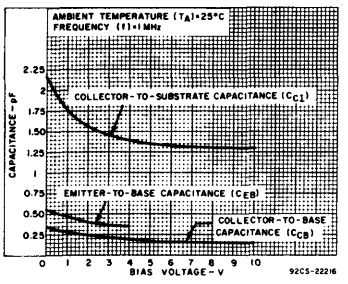


Fig. 6(a) – Capacitance as a function of bias voltage for Q_2.

Transistor	Capacitance (pF)							
	C_{CB}		C_{CE}		C_{EB}		C_{CI}	
	Pkg.	Total	Pkg.	Total	Pkg.	Total	Pkg.	Total
Bias Voltage	–	6 V	–	6 V	–	4 V	–	6 V
Q1	0.025	0.190	0.090	0.125	0.366	0.610	0.475	1.65
Q2	0.015	0.170	0.225	0.265	0.130	0.360	0.085	1.35
Q3	0.040	0.200	0.215	0.240	0.360	0.625	0.210	1.40
Q4	0.040	0.190	0.225	0.270	0.365	0.610	0.085	1.25
Q5	0.010	0.165	0.095	0.115	0.140	0.365	0.090	1.35

Fig. 6(b) – Typical capacitance values at f = 1 MHz. Three terminal measurement. Guard all terminals except those under test.

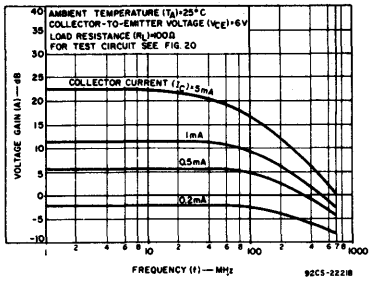


Fig. 7 – Voltage gain as a function of frequency at R_L = 100 Ω.

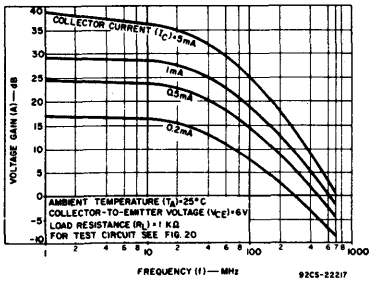


Fig. 8 – Voltage gain as a function of frequency at R_L = 1 kΩ.

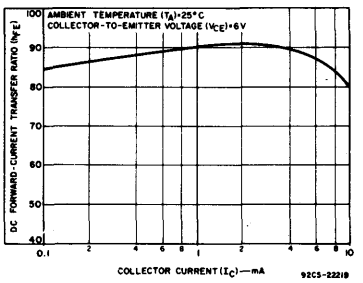


Fig. 9 – DC forward-current transfer ratio as a function of collector current.

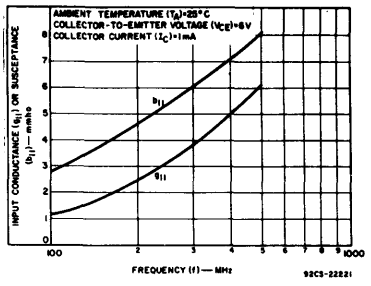


Fig. 10 – Input admittance (Y_{11}) as a function of frequency.

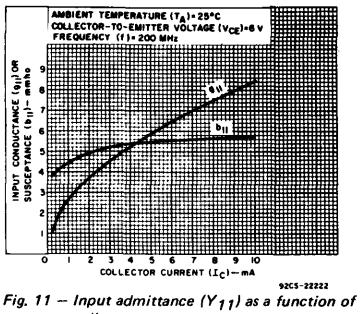


Fig. 11 – Input admittance (Y_{11}) as a function of collector current.

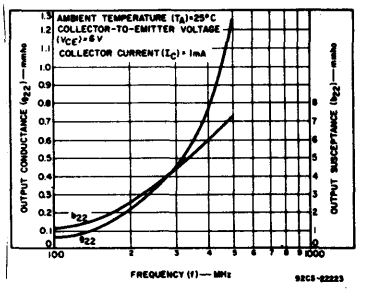


Fig. 12 – Output admittance (Y_{22}) as a function of frequency.

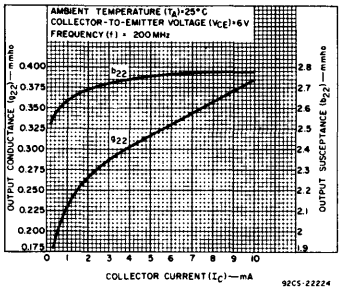


Fig. 13 – Output admittance (Y_{22}) as a function of collector current.

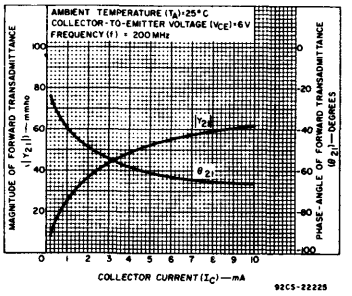


Fig. 14 – Forward transadmittance (Y_{21}) as a function of collector current.

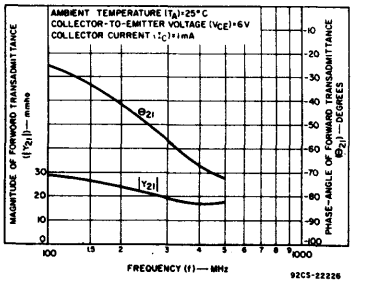


Fig. 15 – Forward transadmittance (Y_{21}) as a function of frequency.

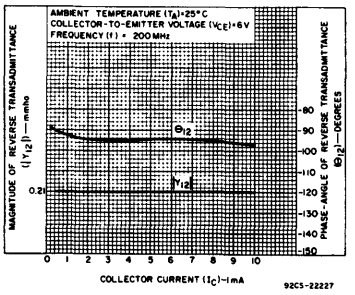


Fig. 16 – Reverse transadmittance (Y_{12}) as a function of collector current.

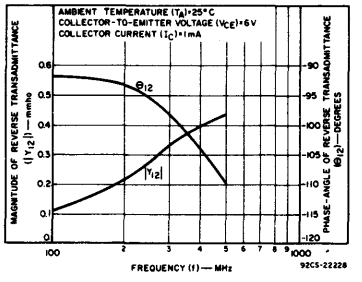


Fig. 17 – Reverse transadmittance (Y_{12}) as a function of frequency.

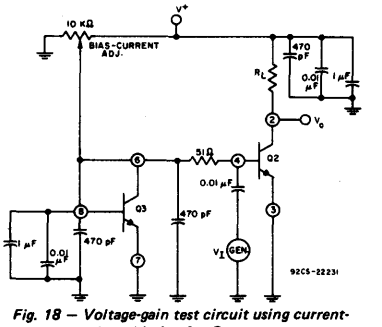


Fig. 18 – Voltage-gain test circuit using current-mirror biasing for Q_2.

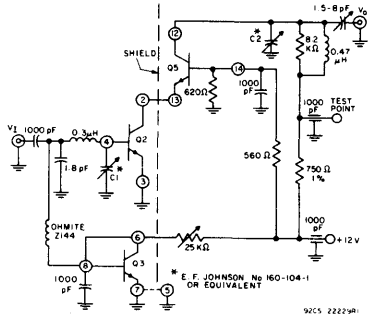


Fig. 19 – 100-MHz power-gain and noise-figure test circuit.

This circuit was chosen because it conveniently represents a close approximation in performance to a properly unilateralized single transistor of this type. The use of Q3 in a current-mirror configuration facilitates simplified biasing. The use of the cascode circuit in no way implies that the transistors cannot be used individually.

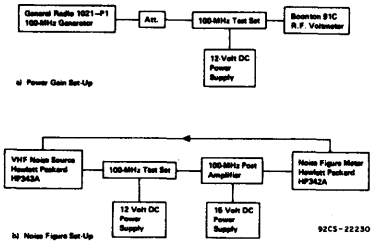


Fig. 20 – Block diagrams of power-gain and noise-figure test set-ups.

CA3130, CA3130A, CA3130B Types

BiMOS Operational Amplifiers

With MOS/FET Input, COS/MOS Output

RCA-CA3130T, CA3130E, CA3130S, CA-3130AT,CA3130AS,CA3130AE,CA3130BT, and CA3130BS are integrated-circuit operational amplifiers that combine the advantages of both COS/MOS and bipolar transistors on a monolithic chip.

Gate-protected p-channel MOS/FET (PMOS) transistors are used in the input circuit to provide very-high-input impedance, very-low-input current, and exceptional speed performance. The use of PMOS field-effect transistors in the input stage results in common-mode input-voltage capability down to 0.5 volt below the negative-supply terminal, an important attribute in single-supply applications.

A complementary-symmetry MOS (COS/MOS) transistor-pair, capable of swinging the output voltage to within 10 millivolts of either supply-voltage terminal (at very high values of load impedance), is employed as the output circuit.

The CA3130 Series circuits operate at supply voltages ranging from 5 to 16 volts, or ±2.5 to ±8 volts when using split supplies. They can be phase compensated with a single external capacitor, and have terminals for adjustment of offset voltage for applications requiring offset-null capability. Terminal provisions are also made to permit strobing of the output stage.

The CA3130 Series is supplied in standard 8-lead TO-5 style packages (T suffix), 8-lead dual-in-line formed lead TO-5 style "DIL-CAN" packages (S suffix). The CA3130 is available in chip form (H suffix). The CA3130 and CA3130A are also available in the Mini-DIP 8-lead dual-in-line plastic package (E suffix). All types operate over the full military-temperature range of −55°C to +125°C. The CA3130B is intended for applications requiring premium-grade specifications and with limits established for: input current, temperature coefficient of input-offset voltage, and gain over the range of −55°C to +125°C. The CA3130A offers superior input characteristics over those of the CA3130.

Features:

- MOS/FET input stage provides:
  very high Z_I = 1.5 TΩ (1.5 x 10^{12} Ω) typ.
  very low I_I = 5 pA typ. at 15-V operation
  2 pA typ. at 5-V operation
- Common-mode input-voltage range includes negative supply rail; input terminals can be swung 0.5 V below negative supply rail
- COS/MOS output stage permits signal swing to either (or both) supply rails

} Ideal for single-supply applications

- Low V_{IO}: 2 mV max. (CA3130B)
- Wide BW: 15 MHz typ. (unity-gain crossover)
- High SR: 10 V/μs typ. (unity-gain follower)
- High output current (I_O): 20 mA typ.
- High A_{OL}: 320,000 (110 dB) typ.
- Compensation with single external capacitor

Applications:

- Ground-referenced single-supply amplifiers
- Fast sample-hold amplifiers
- Long-duration timers/monostables
- High-input-impedance comparators (ideal interface with digital COS/MOS)
- High-input-impedance wideband amplifiers
- Voltage followers (e.g., follower for single-supply D/A converter)
- Voltage regulators (permits control of output voltage down to zero volts)
- Peak detectors
- Single-supply full-wave precision rectifiers
- Photo-diode sensor amplifiers

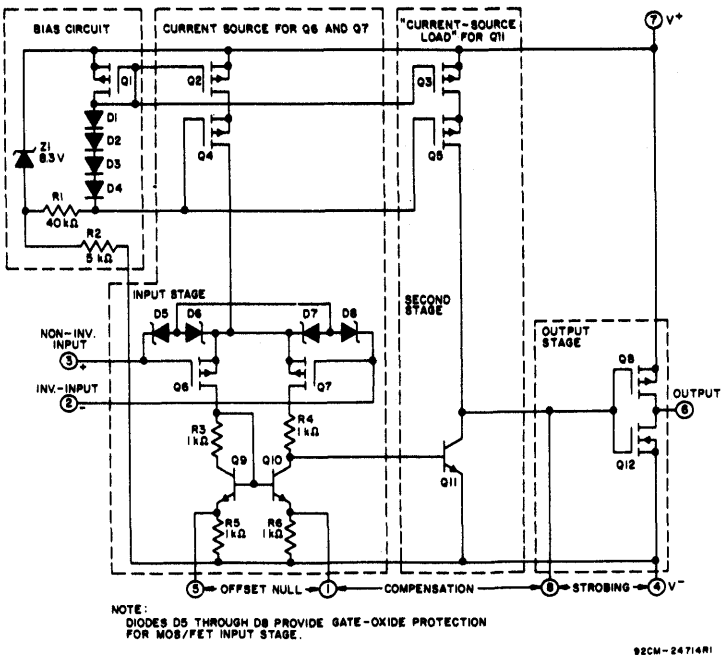
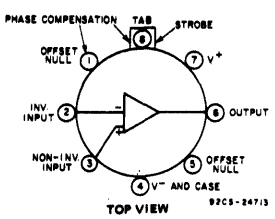
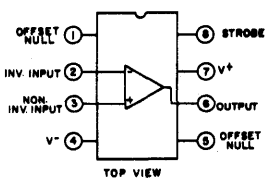


Fig. 1 – Schematic diagram of the CA3130 Series.



S and T Suffixes



E Suffix

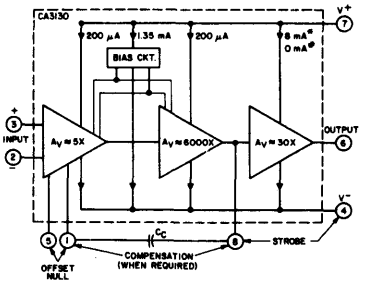
Fig. 2 – Functional diagrams for the CA3130 series.

MAXIMUM RATINGS, *Absolute-Maximum Values*

DC SUPPLY VOLTAGE (Between V^+ and V^- Terminals) ..... 16 V
DIFFERENTIAL-MODE INPUT VOLTAGE ..... ±8 V
COMMON-MODE DC INPUT VOLTAGE ... (V^+ +8 V) to (V^- −0.5 V)
INPUT-TERMINAL CURRENT ..... 1 mA
DEVICE DISSIPATION:
WITHOUT HEAT SINK –
UP TO 55°C ..... 630 mW
ABOVE 55°C .... Derate linearly 6.67 mW/°C
WITH HEAT SINK –
AT 125°C ..... 418 mW
BELOW 125°C ... Derate linearly 16.7 mW/°C

TEMPERATURE RANGE:
OPERATING (all types) ..... −55 to + 125°C
STORAGE (all types) ..... −65 to + 150°C
OUTPUT SHORT-CIRCUIT DURATION * ..... INDEFINITE
LEAD TEMPERATURE (DURING SOLDERING):
AT DISTANCE 1/16 ± 1/32 INCH (1.59 ± 0.79 mm) FROM CASE FOR 10 SECONDS MAX. ..... +265°C

*Short circuit may be applied to ground or to either supply.



TOTAL SUPPLY VOLTAGE (FOR INDICATED VOLTAGE GAINS) = 15 V
*WITH INPUT TERMINALS BIASED SO THAT TERM. 6 POTENTIAL IS +7.5 V ABOVE TERM. 4.
*WITH OUTPUT TERMINAL DRIVEN TO EITHER SUPPLY RAIL.
92CS-24715

Fig. 3 – Block diagram of the CA3130 Series.

ELECTRICAL CHARACTERISTICS at T_A=25°C, V^+=15 V, V^- = 0 V (Unless otherwise specified)

CHARACTERISTIC	LIMITS									Units
	CA3130B (T,S)			CA3130A (T,S,E)			CA3130 (T,S,E)			
	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage, $\|V_{IO}\|$, $V^{\pm}$=±7.5 V	–	0.8	2	–	2	5	–	8	15	mV
Input Offset Current, $\|I_{IO}\|$, $V^{\pm}$=±7.5 V	–	0.5	10	–	0.5	20	–	0.5	30	pA
Input Current, I_I $V^{\pm}$=±7.5 V	–	5	20	–	5	30	–	5	50	pA
Large-Signal Voltage Gain, A_{OL}	100 k	320 k	–	50 k	320 k	–	50 k	320 k	–	V/V
V_O=10 V_{p-p}, R_L=2 kΩ	100	110	–	94	110	–	94	110	–	dB
Common-Mode Rejection Ratio, CMRR	86	100	–	80	90	–	70	90	–	dB
Common-Mode Input-Voltage Range, V_{ICR}	0	−0.5 to 12	10	0	−0.5 to 12	10	0	−0.5 to 12	10	V
Power-Supply Rejection Ratio, $\Delta V_{IO}/\Delta V^{\pm}$ $V^{\pm}$=±7.5 V	–	32	100	–	32	150	–	32	320	μV/V
Maximum Output Voltage:										
At R_L=2 kΩ V_{OM}^+	12	13.3	–	12	13.3	–	12	13.3	–	V
At R_L=2 kΩ V_{OM}^-	–	0.002	0.01	–	0.002	0.01	–	0.002	0.01	V
At R_L=∞ V_{OM}^+	14.99	15	–	14.99	15	–	14.99	15	–	V
At R_L=∞ V_{OM}^-	–	0	0.01	–	0	0.01	–	0	0.01	V
Maximum Output Current:										
I_{OM}^+ (Source) @ V_O = 0 V	12	22	45	12	22	45	12	22	45	mA
I_{OM}^- (Sink) @ V_O = 15 V	12	20	45	12	20	45	12	20	45	mA
Supply Current, I^+:										
V_O=7.5 V, R_L=∞	–	10	15	–	10	15	–	10	15	mA
V_O = 0 V, R_L = ∞	–	2	3	–	2	3	–	2	3	mA
Input Current, I_I*	–	Fig.12	15	–	Fig.12	–	–	Fig.12	–	nA
Input Offset Voltage Temp. Drift, $\Delta V_{IO}/\Delta T$*	–	5	15	–	10	–	–	10	–	μV/°C
Large-Signal Voltage Gain, A_{OL}*	50 k	320 k	–	–	320 k	–	–	320 k	–	V/V
	94	110	–	–	110	–	–	110	–	dB

* T_A = −55 to +125°C, $V^{\pm}$ = ±7.5 V (I_I and $\Delta V_{IO}/\Delta T$), V_O = 10 V_{p-p} and R_L = 2 kΩ (A_{OL}).

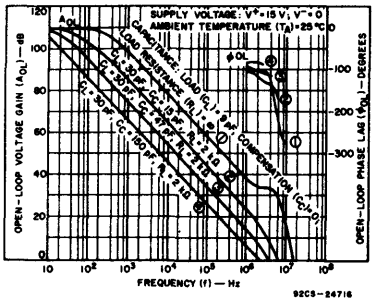


Fig. 4 – Open-loop voltage gain and phase shift vs. frequency for various values of C_L, C_C, and R_L.

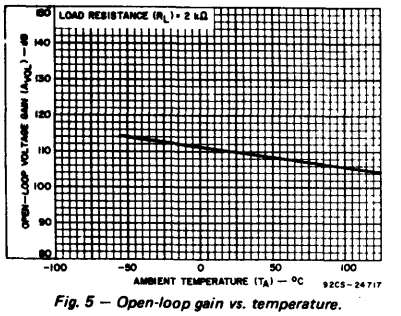


Fig. 5 – Open-loop gain vs. temperature.

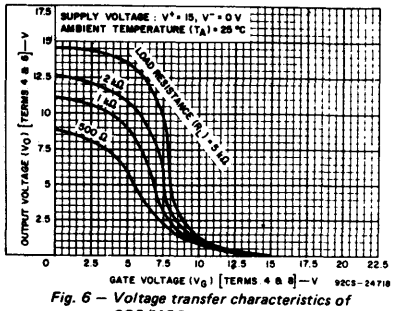


Fig. 6 – Voltage transfer characteristics of COS/MOS output stage.

TYPICAL VALUES INTENDED ONLY FOR DESIGN GUIDANCE

CHARACTERISTIC	TEST CONDITIONS $V^+ = +7.5$ V, $V^- = -7.5$ V, $T_A = 25^oC$ (Unless Otherwise Specified)	CA3130B (T,S)	CA3130A (T,S,E)	CA3130 (T,S,E)	UNITS
Input Offset Voltage Adjustment Range	10 kΩ across Terms. 4 and 5 or 4 and 1	±22	±22	±22	mV
Input Resistance, R_I		1.5	1.5	1.5	TΩ
Input Capacitance, C_I	f = 1 MHz	4.3	4.3	4.3	pF
Equivalent Input Noise Voltage, e_n	BW = 0.2 MHz, R_S = 1 MΩ*	23	23	23	μV
Unity Gain Crossover Frequency, f_T	C_C = 0	15	15	15	MHz
	C_C = 47 pF	4	4	4	
Slew Rate, SR: Open Loop	C_C = 0	30	30	30	V/μs
Closed Loop	C_C = 56 pF	10	10	10	
Transient Response: Rise Time, t_r	C_C = 56 pF, C_L = 25 pF, R_L = 2 kΩ (Voltage Follower)	0.09	0.09	0.09	μs
Overshoot		10	10	10	%
Settling Time (4 V_{p-p} Input to <0.1%)		1.2	1.2	1.2	μs

* Although a 1-MΩ source is used for this test, the equivalent input noise remains constant for values of R_S up to 10 MΩ.

CHARACTERISTIC	TEST CONDITIONS $V^+ = 5$ V, $V^- = 0$ V, $T_A = 25^oC$ (Unless Otherwise Specified)	CA3130B (T,S)	CA3130A (T,S,E)	CA3130 (T,S,E)	UNITS
Input Offset Voltage, V_{IO}		1	2	8	mV
Input Offset Current, I_{IO}		0.1	0.1	0.1	pA
Input Current, I_I		2	2	2	pA
Common-Mode Rejection Ratio, CMRR		100	90	80	dB
Large-Signal Voltage Gain, A_{OL}	V_O = 4 V_{p-p}, R_L = 5 kΩ	100 k	100 k	100 k	V/V
		100	100	100	dB
Common-Mode Input Voltage Range, V_{ICR}		0 to 2.8	0 to 2.8	0 to 2.8	V
Supply Current, I^+	V_O = 5 V, R_L = ∞	300	300	300	μA
	V_O = 2.5 V, R_L = ∞	500	500	500	
Power Supply Rejection Ratio, $\Delta V_{IO}/\Delta V^+$		200	200	200	μV/V

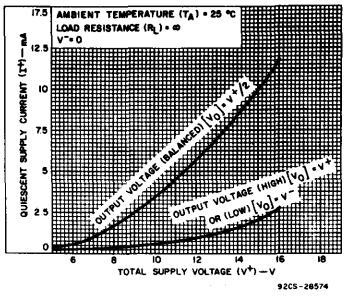


Fig. 7 – Quiescent supply current vs. supply voltage.

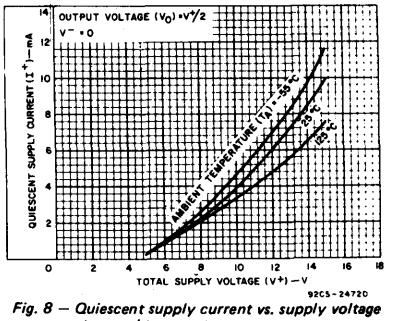


Fig. 8 – Quiescent supply current vs. supply voltage at several temperatures.

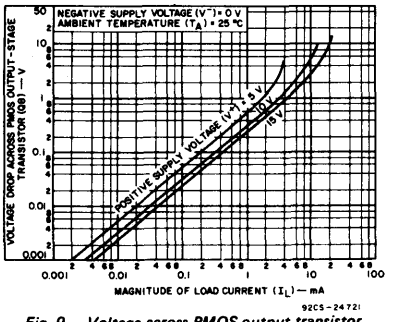


Fig. 9 – Voltage across PMOS output transistor (Q8) vs. load current.

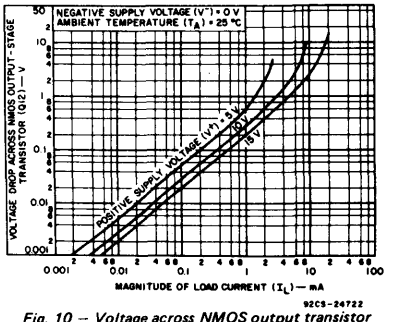


Fig. 10 – Voltage across NMOS output transistor (Q12) vs. load current.

CIRCUIT DESCRIPTION

Fig. 3 is a block diagram of the CA3130 Series COS/MOS Operational Amplifiers. The input terminals may be operated down to 0.5 V below the negative supply rail, and the output can be swung very close to either supply rail in many applications. Consequently, the CA3130 Series circuits are ideal for single-supply operation. Three Class A amplifier stages, having the individual gain capability and current consumption shown in Fig. 3, provide the total gain of the CA3130. A biasing circuit provides two potentials for common use in the first and second stages. Term. 8 can be used both for phase compensation and to strobe the output stage into quiescence. When Term. 8 is tied to the negative supply rail (Term. 4) by mechanical or electrical means, the output potential at Term. 6 essentially rises to the positive supply-rail potential at Term. 7. This condition of essentially zero current drain in the output stage under the strobed "OFF" condition can only be achieved when the ohmic load resistance presented to the amplifier is very high (e.g., when the amplifier output is used to drive COS/MOS digital circuits in comparator applications).

Input Stages—The circuit of the CA3130 is shown in Fig. 1. It consists of a differential-input stage using PMOS field-effect transistors (Q6, Q7) working into a mirror-pair of bipolar transistors (Q9, Q10) functioning as load resistors together with resistors R3 through R6. The mirror-pair transistors also function as a differential-to-single-ended converter to provide base drive to the second-stage bipolar transistor (Q11). Offset nulling, when desired, can be effected by connecting a 100,000-ohm potentiometer across Terms. 1 and 5 and the potentiometer slider arm to Term. 4. Cascode-connected PMOS transistors Q2, Q4 are the constant-current source for the input stage. The biasing circuit for the constant-current source is subsequently described. The small diodes D5 through D8 provide gate-oxide protection against high-voltage transients, e.g., including static electricity during handling for Q6 and Q7.

Second-Stage—Most of the voltage gain in the CA3130 is provided by the second amplifier stage, consisting of bipolar transistor Q11 and its cascode-connected load resistance provided by PMOS transistors Q3 and Q5. The source of bias potentials for these PMOS transistors is subsequently described. Miller-Effect compensation (roll-off) is accomplished by simply connecting a small capacitor between Terms. 1 and 8. A 47-picofarad capacitor provides sufficient compensation for stable unity-gain operation in most applications.

Bias-Source Circuit—At total supply voltages, somewhat above 8.3 volts, resistor R2 and zener diode Z1 serve to establish a voltage of 8.3 volts across the series-connected circuit, consisting of resistor R1, diodes D1 through D4, and PMOS transistor Q1. A tap at the junction of resistor R1 and diode D4 provides a gate-bias potential of about 4.5 volts for PMOS transistors Q4 and Q5 with respect to Term. 7. A potential of about 2.2 volts is developed across diode-connected PMOS transistor Q1 with respect to Term. 7 to provide gate bias for PMOS transistors Q2 and Q3. It should be noted that Q1 is "mirror-connected"† to both Q2 and Q3. Since transistors Q1, Q2, Q3 are designed to be identical, the approximately 200-microampere current in Q1 establishes a similar current in Q2 and Q3 as constant-current sources for both the first and second amplifier stages, respectively.

At total supply voltages somewhat less than 8.3 volts, zener diode Z1 becomes nonconductive and the potential, developed across series-connected R1, D1-D4, and Q1, varies directly with variations in supply voltage, Consequently, the gate bias for Q4, Q5 and Q2, Q3 varies in accordance with supply-voltage variations. This variation results in deterioration of the power-supply-rejection ratio (PSRR) at total supply voltages below 8.3 volts. Operation at total supply voltages below about 4.5 volts results in seriously degraded performance.

Output Stage—The output stage consists of a drain-loaded inverting amplifier using COS/MOS transistors operating in the Class A mode. When operating into very high resistance loads, the output can be swung within millivolts of either supply rail. Because the output stage is a drain-loaded amplifier, its gain is dependent upon the load impedance. The transfer characteristics of the output stage for a load returned to the negative supply rail are shown in Fig. 6. Typical op-amp loads are readily driven by the output stage. Because large-signal excursions are nonlinear, requiring feedback for good waveform reproduction, transient delays may be encountered. As a voltage follower, the amplifier can achieve 0.01 per cent accuracy levels, including the negative supply rail.

Input Current Variation with Common-Mode Input Voltage

As shown in the Table of Electrical Characteristics, the input current for the CA3130 Series Op-Amps is typically 5 pA at T_A=25°C when terminals 2 and 3 are at a common-mode potential of +7.5 volts with respect to negative supply Terminal 4. Fig.11 contains data showing the variation of input current as a function of common-mode input voltage at T_A = 25°C. These data show that circuit designers can advantageously exploit these characteristics to design circuits which typically require an input current of less than 1 pA, provided the common-mode input voltage does not exceed 2 volts. As previously noted, the input current is essentially the result of the leakage current through the gate-protection diodes in the input circuit and, therefore, a function of the applied

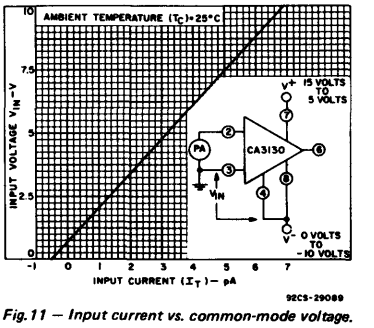


Fig.11 – Input current vs. common-mode voltage.

voltage. Although the finite resistance of the glass terminal-to-case insulator of the TO-5 package also contributes an increment of leakage current, there are useful compensating factors. Because the gate-protection network functions as if it is connected to Terminal 4 potential, and the TO-5 case of the CA3130 is also internally tied to Terminal 4, input terminal 3 is essentially "guarded" from spurious leakage currents.

Offset Nulling

Offset-voltage nulling is usually accomplished with a 100,000-ohm potentiometer connected across Terms. 1 and 5 and with the potentiometer slider arm connected to Term. 4. A fine offset-null adjustment usually can be effected with the slider arm positioned in the mid-point of the potentiometer's total range.

Input-Current Variation with Temperature

The input current of the CA3130 Series circuits is typically 5 pA at 25°C. The major portion of this input current is due to leakage current through the gate-protective diodes in the input circuit. As with any semiconductor-junction device, including op amps with a junction-FET input stage, the leakage current approximately doubles for every 10°C increase in temperature. Fig.12 provides data on the typical variation of input bias current as a function of temperature in the CA3130.

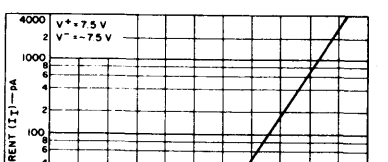


Fig.12 – Input current vs. ambient temperature.

In applications requiring the lowest practical input current and incremental increases in

†For general information on the characteristics of COS/MOS transistor-pairs in linear-circuit applications, see File No. 619, data bulletin on CA3600E "COS/MOS Transistor Array".

current because of "warm-up" effects, it is suggested that an appropriate heat sink be used with the CA3130. In addition, when "sinking" or "sourcing" significant output current the chip temperature increases, causing an increase in the input current. In such cases, heat-sinking can also very markedly reduce and stabilize input current variations.

Input-Offset-Voltage (V_{IO}) Variation with DC Bias vs. Device Operating Life

It is well known that the characteristics of a MOS/FET device can change slightly when a dc gate-source bias potential is applied to the device for extended time periods. The magnitude of the change is increased at high temperatures. Users of the CA3130 should be alert to the possible impacts of this effect if the application of the device involves extended operation at high temperatures with a significant differential dc bias voltage applied across Terms. 2 and 3. Fig.13 shows typical data pertinent to shifts in offset voltage encountered with CA3130 devices (TO-5 package) during life testing. At lower temperatures (TO-5 and plastic), for example at 85°C, this change in voltage is considerably less. In typical linear applications where the differential voltage is small and symmetrical, these incremental changes are of about the same magnitude as those encountered in an operational amplifier employing a bipolar transistor input stage. The two-volt dc differential voltage example represents conditions when the amplifier output stage is "toggled", e.g., as in comparator applications.

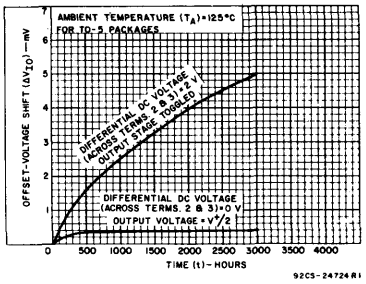


Fig.13 – Typical incremental offset-voltage shift vs. operating life.

Power-Supply Considerations

Because the CA3130 is very useful in single-supply applications, it is pertinent to review some considerations relating to power-supply current consumption under both single- and dual-supply service. Figs. 14a and 14b show the CA3130 connected for both dual- and single-supply operation.

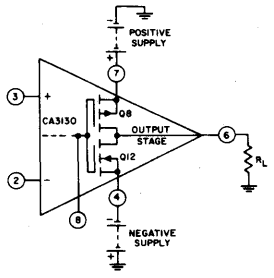
Dual-supply operation: When the output voltage at Term. 6 is zero-volts, the currents supplied by the two power supplies are equal. When the gate terminals of Q8 and Q12 are driven increasingly positive with respect to ground, current flow through Q12 (from the negative supply) to the load is increased and current flow through Q8 (from the positive supply) decreases correspondingly. When the gate terminals of Q8 and Q12 are driven increasingly negative with respect to ground, current flow through Q8 is increased and current flow through Q12 is decreased accordingly.

Single-supply operation: Initially, let it be assumed that the value of R_L is very high (or disconnected), and that the input-terminal bias (Terms. 2 and 3) is such that the output terminal (No. 6) voltage is at $V^+/2$, i.e., the voltage-drops across Q8 and Q12 are of equal magnitude. Fig. 7 shows typical quiescent supply-current vs. supply-voltage for the CA3130 operated under these conditions. Since the output stage is operating as a Class A amplifier, the supply-current will remain constant under dynamic operating conditions as long as the transistors are operated in the linear portion of their voltage-transfer characteristics (see Fig. 6). If either Q8 or Q12 are swung out of their linear regions toward cut-off (a non-linear region), there will be a corresponding reduction in supply-current. In the extreme case, e.g., with Term. 8 swung down to ground potential (or tied to ground), NMOS transistor Q12 is completely cut off and the supply-current to series-connected transistors Q8, Q12 goes essentially to zero. The two preceding stages in the CA3130, however, continue to draw modest supply-current (see the lower curve in Fig. 7) even though the output stage is strobed off. Fig. 14a shows a dual-supply arrangement for the output stage that can also be strobed off, assuming $R_L=\infty$, by pulling the potential of Term. 8 down to that of Term. 4.

Let it now be assumed that a load-resistance of nominal value (e.g., 2 kilohms) is connected between Term. 6 and ground in the circuit of Fig.14b. Let it further be assumed again that the input-terminal bias (Terms. 2 and 3) is such that the output terminal (No. 6) voltage is a $V^+/2$. Since PMOS transistor Q8 must now supply quiescent current to both R_L and transistor Q12, it should be apparent that under these conditions the supply-current must increase as an inverse function of the R_L magnitude. Fig. 9 shows the voltage-drop across PMOS transistor Q8 as a function of load current at several supply-voltages. Fig. 6 shows the voltage-transfer characteristics of the output stage for several values of load resistance.

Wideband Noise

From the standpoint of low-noise performance considerations, the use of the CA3130 is most advantageous in applications where in the source resistance of the input signal is in the order of 1 megohm or more. In this case, the total input-referred noise voltage is typically only 23 μV when the test-circuit amplifier of Fig. 15 is operated at a total supply voltage of 15 volts. This value of total input-referred noise remains essentially constant, even though the value of source



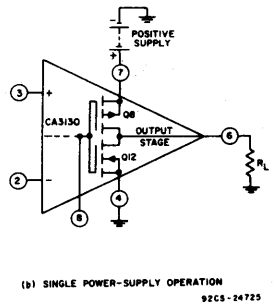


Fig.14 – CA3130 output stage in dual and single power-supply operation.

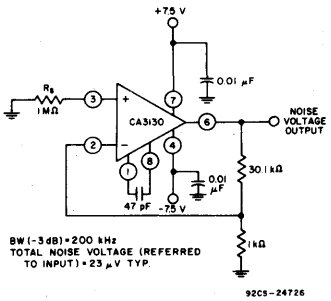


Fig.15 – Test-circuit amplifier (30-dB gain) used for wideband noise measurements.

resistance is raised by an order of magnitude. This characteristic is due to the fact that reactance of the input capacitance becomes a significant factor in shunting the source resistance. It should be noted, however, that for values of source resistance very much greater than 1 megohm, the total noise voltage generated can be dominated by the thermal noise contributions of both the feedback and source resistors.

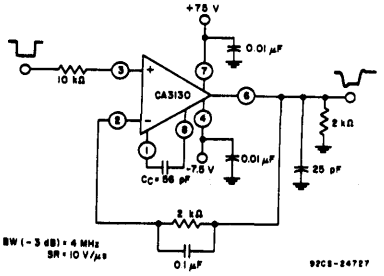
TYPICAL APPLICATIONS

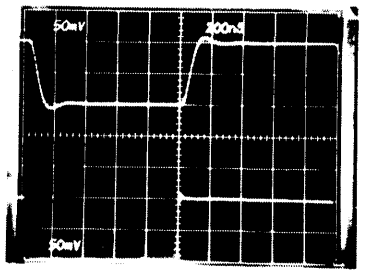
Voltage Followers

Operational amplifiers with very high input resistances, like the CA3130, are particularly

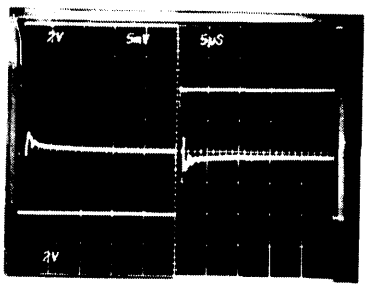
suited to service as voltage followers. Fig. 16 shows the circuit of a classical voltage follower, together with pertinent waveforms using the CA3130 in a split-supply configuration.

A voltage follower, operated from a single supply, is shown in Fig. 17, together with related waveforms. This follower circuit is linear over a wide dynamic range, as illustrated by the reproduction of the output waveform in Fig. 17a with input-signal ramping. The waveforms in Fig. 17b show that the follower does not lose its input-to-output phase-sense, even though the input is being swung 7.5 volts below ground potential. This unique characteristic is an important attribute in both operational amplifier and comparator applications. Fig. 17b also shows the manner in which the COS/MOS output stage permits the output signal to swing down to the negative supply-rail potential (i.e., ground in the case shown). The digital-to-analog converter (DAC) circuit, described in the following section, illustrates the practical use of the CA3130 in a single-supply voltage-follower application.





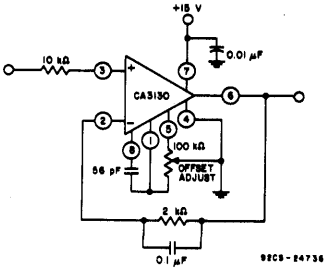
Top Trace: Output
Bottom Trace: Input
(a) Small-signal response (50 mV/div. and 200 ns/div.)

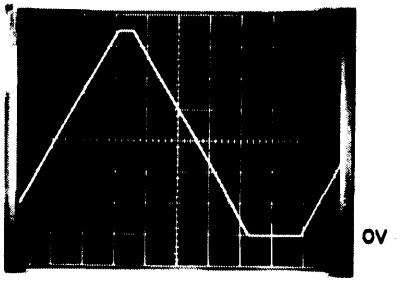


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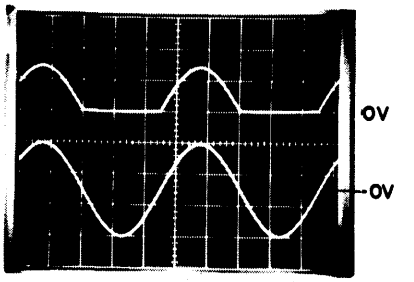
Top Trace: Output signal (2 V/div. and 5 µs/div.)
Center Trace: Difference signal (5 mV/div. and 5 µs/div.)
Bottom Trace: Input signal (2 V/div. and 5 µs/div.)
(b) Input-output difference signal showing settling time (Measurement made with Tektronix 7A13 differential amplifier)

Fig. 16 – Split-supply voltage follower with associated waveforms.





(a) Output-waveform with input-signal ramping (2 V/div. and 500 µs/div.)



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Top Trace: Output (5 V/div. and 200 µs/div.)
Bottom Trace: Input (5 V/div. and 200 µs/div.)
(b) Output-waveform with ground-reference sine-wave input

Fig. 17 – Single-supply voltage-follower with associated waveforms. (e.g., for use in single-supply D/A converter; see Fig.9 in ICAN-6080).

9-Bit COS/MOS DAC

A typical circuit of a 9-bit Digital-to-Analog Converter (DAC)* is shown in Fig.18 This system combines the concepts of multiple-switch COS/MOS IC's, a low-cost ladder network of discrete metal-oxide-film resistors, a CA3130 op amp connected as a follower, and an inexpensive monolithic regulator in a simple single power-supply arrangement. An additional feature of the DAC is that it is readily interfaced with COS/MOS input logic, e.g., 10-volt logic levels are used in the circuit of Fig.18.

The circuit uses an R/2R voltage-ladder network, with the output potential obtained directly by terminating the ladder arms at either the positive or the negative power-supply terminal. Each CD4007A contains three "inverters", each "inverter" functioning as a single-pole double-throw switch to terminate an arm of the R/2R network at either the positive or negative power-supply terminal. The resistor ladder is an assembly of one per cent tolerance metal-oxide film resistors. The five arms requiring the highest accuracy are assembled with series and parallel combinations of 806,000-ohm resistors from the same manufacturing lot.

A single 15-volt supply provides a positive bus for the CA3130 follower amplifier and feeds the CA3085 voltage regulator. A "scale-adjust" function is provided by the regulator output control, set to a nominal 10-volt level in this system. The line-voltage regulation (approximately 0.2%) permits a 9-bit accuracy to be maintained with variations of several volts in the supply. The flexibility afforded by the COS/MOS building blocks simplifies the design of DAC systems tailored to particular needs.

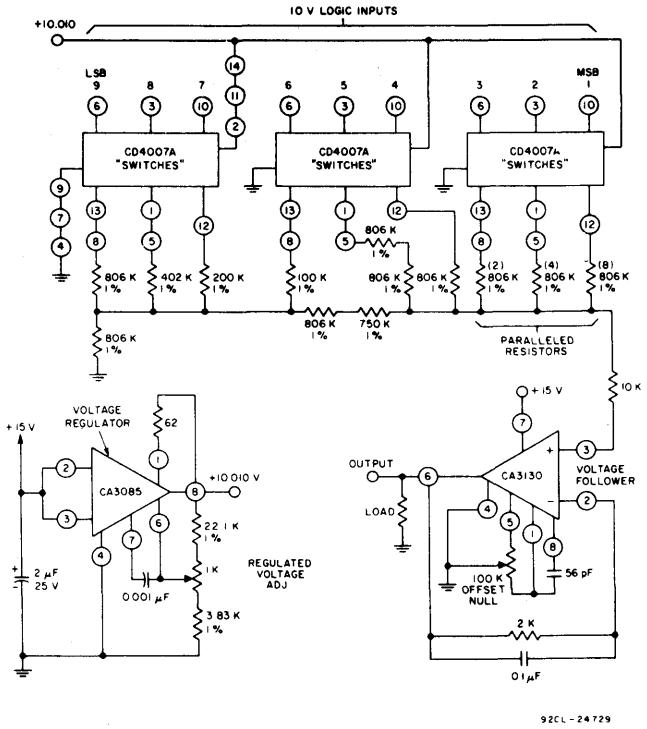
Single-Supply, Absolute-Value, Ideal Full-Wave Rectifier

The absolute-value circuit using the CA3130 is shown in Fig. 19. During positive excursions, the input signal is fed through the feedback network directly to the output. Simultaneously, the positive excursion of the input signal also drives the output terminal (No. 6) of the inverting amplifier in a negative-going excursion such that the 1N914 diode effectively disconnects the amplifier from the signal path. During a negative-going excursion of the input signal, the CA3130 functions as a normal inverting amplifier with a gain equal to −R2/R1. When the equality of the two equations shown in Fig. 19 is satisfied, the full-wave output is symmetrical.

Peak Detectors

Peak-detector circuits are easily implemented with the CA3130, as illustrated in Fig. 20 for both the peak-positive and the peak-negative circuit. It should be noted that with large-signal inputs, the bandwidth of the

*"Digital-to-Analog Conversion Using the RCA-CD4007A COS/MOS IC", Application Note ICAN-6080.



BIT	REQUIRED RATIO-MATCH
1	STANDARD
2	±0.1%
3	±0.2%
4	±0.4%
5	±0.8%
6-9	±1% ABS

ALL RESISTANCES IN OHMS

Fig. 18 — 9-bit DAC using COS/MOS digital switches and CA3130.

peak-negative circuit is much less than that of the peak-positive circuit. The second stage of the CA3130 limits the bandwidth in this case. Negative-going output-signal excursion requires a positive-going signal excursion at the collector of transistor Q11, which is loaded by the intrinsic capacitance of the associated circuitry in this mode. On the other hand, during a negative-going signal excursion at the collector of Q11, the transistor functions in an active "pull-down" mode so that the intrinsic capacitance can be discharged more expeditiously.

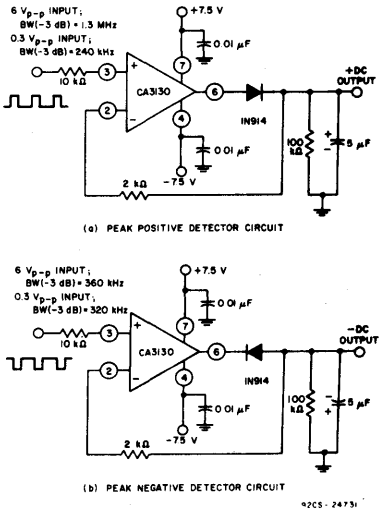
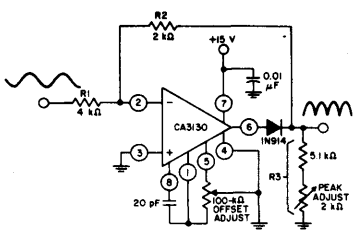
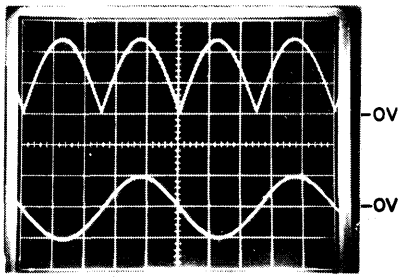


Fig. 20 — Peak-detector circuits.





Top Trace: Output signal (2 V/div.)
Bottom Trace: Input signal (10 V/div.)
Time base on both traces: 0.2 ms/div.

Fig. 19 — Single-supply, absolute-value, ideal full-wave rectifier with associated waveforms.

Error-Amplifier in Regulated-Power Supplies

The CA3130 is an ideal choice for error-amplifier service in regulated power supplies since it can function as an error-amplifier when the regulated output voltage is required to approach zero. Fig. 21 shows the schematic diagram of a 40-mA power supply capable of providing regulated output voltage by continuous adjustment over the range from 0 to 13 volts. Q3 and Q4 in IC2 (a CA3086 transistor-array IC) function as zeners to provide supply-voltage for the CA3130 comparator (IC1). Q1, Q2, and Q5 in IC2 are configured as a low impedance, temperature-compensated source of adjustable reference voltage for the error amplifier. Transistors Q1, Q2, Q3, and Q4 in IC3 (another CA3086 transistor-array IC) are connected in parallel as the series-pass element. Transistor Q5 in IC3 functions as a current-limiting device by diverting base drive from the series-pass transistors, in accordance with the adjustment of resistor R2.

Fig. 22 contains the schematic diagram of a regulated power-supply capable of providing regulated output voltage by continuous ad-

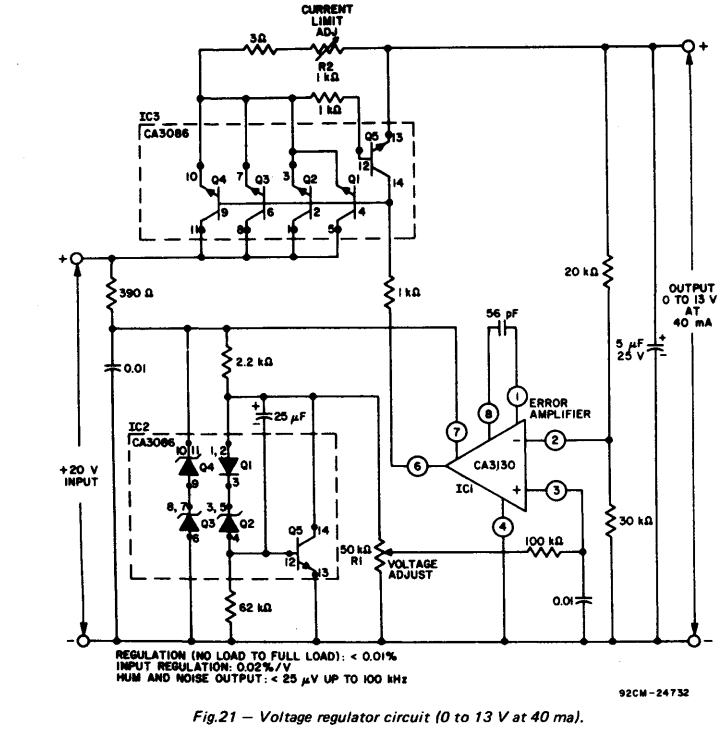


Fig.21 – Voltage regulator circuit (0 to 13 V at 40 ma).

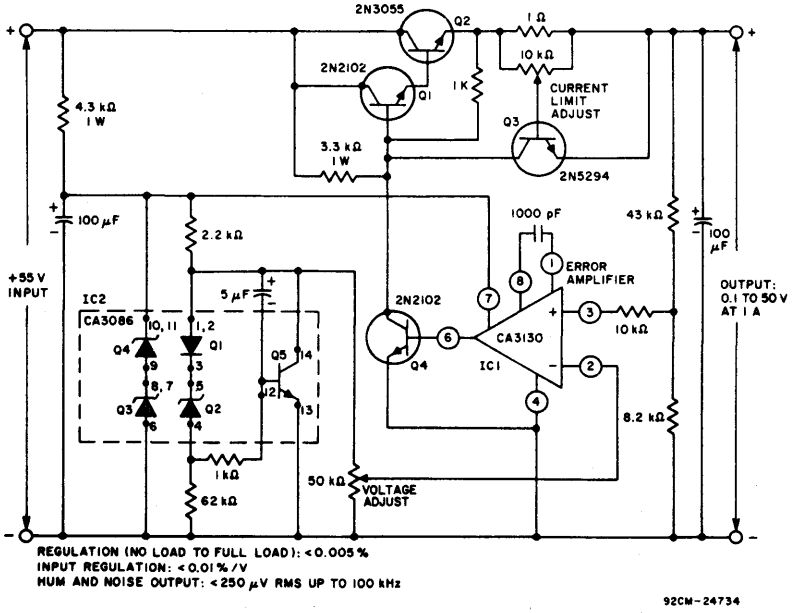


Fig.22 – Voltage regulator circuit (0.1 to 50 V at 1 A).

justment over the range from 0.1 to 50 volts and currents up to 1 ampere. The error amplifier (IC1) and circuitry associated with IC2 function as previously described, although the output of IC1 is boosted by a discrete transistor (Q4) to provide adequate base drive for the Darlington-connected series-pass transistors Q1, Q2. Transistor Q3 functions in the previously described current-limiting circuit.

Multivibrators

The exceptionally high input resistance presented by the CA3130 is an attractive feature for multivibrator circuit design because it permits the use of timing circuits with high R/C ratios. The circuit diagram of a pulse generator (astable multivibrator), with provisions for independent control of the "on" and "off" periods, is shown in Fig. 23. Resistors R1 and R2 are used to bias the CA3130 to the mid-point of the supply-voltage and R3 is the feedback resistor. The pulse repetition rate is selected by positioning S1 to the desired position and the rate remains essentially constant when the resistors which determine "on-period" and "off-period" are adjusted.

Function Generator

Fig. 24 contains a schematic diagram of a function generator using the CA3130 in the integrator and threshold detector functions. This circuit generates a triangular or square-wave output that can be swept over a 1,000,000:1 range (0.1 Hz to 100 kHz) by means of a single control, R1. A voltage-control input is also available for remote sweep-control.

The heart of the frequency-determining system is an operational-transconductance-amplifier (OTA)*, IC1, operated as a voltage-controlled current-source. The output, I_O, is a current applied directly to the integrating capacitor, C1, in the feedback loop of the integrator IC2, using a CA3130, to provide the triangular-wave output. Potentiometer R2 is used to adjust the circuit for slope symmetry of positive-going and negative-going signal excursions.

Another CA3130, IC3, is used as a controlled switch to set the excursion limits of the triangular output from the integrator circuit. Capacitor C2 is a "peaking adjustment" to optimize the high-frequency square-wave performance of the circuit.

Potentiometer R3 is adjustable to perfect the "amplitude symmetry" of the square-wave output signals. Output from the threshold detector is fed back via resistor R4 to the input of IC1 so as to toggle the current source from plus to minus in generating the linear triangular wave.

*See File No. 475 and ICAN-6668.

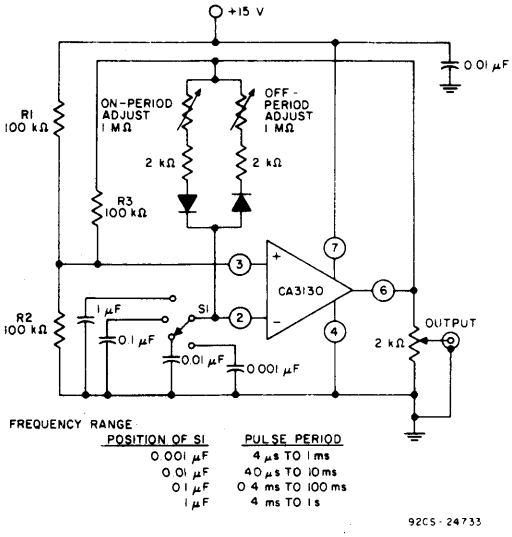


Fig.23 – Pulse generator (astable multivibrator) with provisions for independent control of "ON" and "OFF" periods.

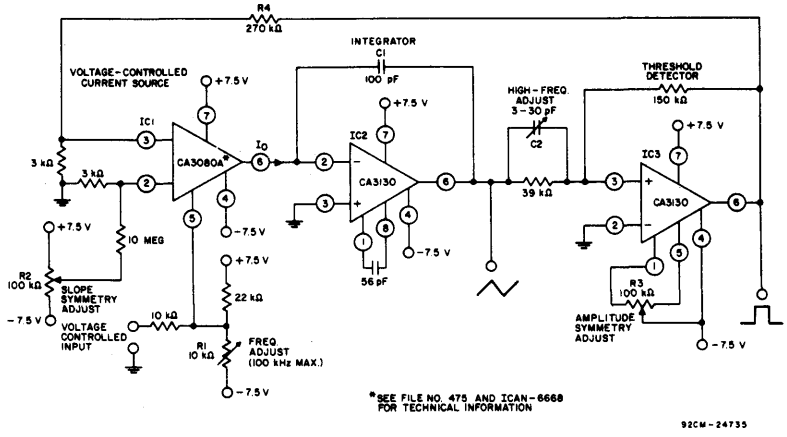


Fig. 24 – Function generator (frequency can be varied 1,000,000/1 with a single control).

Operation with Output-Stage Power-Booster

The current-sourcing and -sinking capability of the CA3130 output stage is easily supplemented to provide power-boost capability. In the circuit of Fig. 25, three COS/MOS transistor-pairs in a single CA3600E* IC array are shown parallel connected with the output stage in the CA3130. In the Class A mode of CA3600E shown, a typical device consumes 20 mA of supply current at 15-V operation. This arrangement boosts the current-handling capability of the CA3130 output stage by about 2.5X.

The amplifier circuit in Fig. 25 employs feedback to establish a closed-loop gain of 48 dB. The typical large-signal bandwidth (−3 dB) is 50 kHz.

*See File No. 619 for technical information.

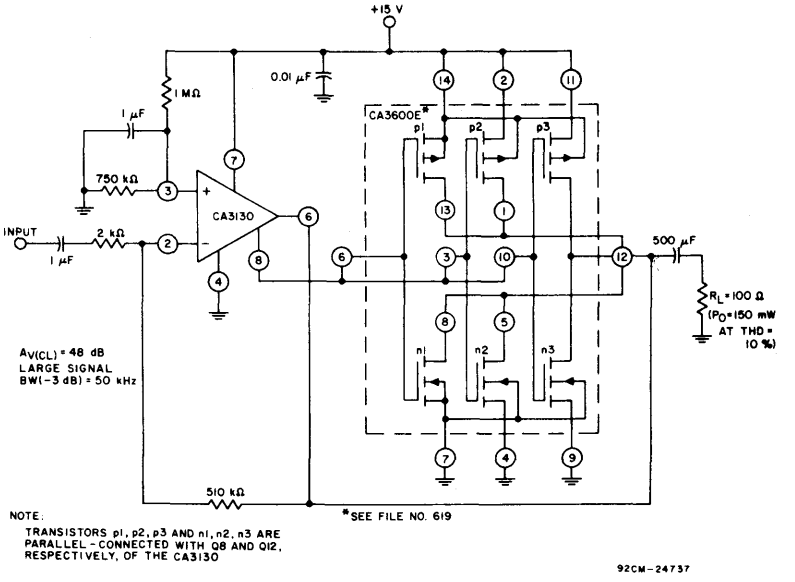
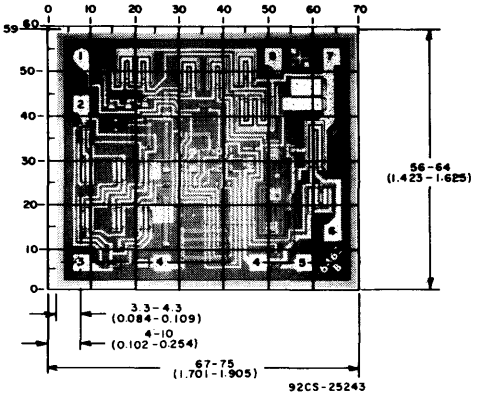


Fig. 25 – COS/MOS transistor array (CA3600E) connected as power-booster in the output stage of the CA3130.



Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

The photographs and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Dimensions and Pad Layout for CA3130H.

CA3138G, CA3138AG Types

Preliminary Data

High-Current, High-Beta N-P-N Transistor Arrays

Hermetic Gold-CHIP Type For Industrial, Commercial, and Military Applications

Four Isolated Discrete Sealed-Junction High-Current N-P-N Transistors

Features:

- **High Current –1 A**
- **High Beta – 95 min. at I_C = 500 mA, V_{CE} = 5 V**
- **Low V_{CE}(SAT) –0.4 V max. at I_C = 500 mA, I_B = 12.5 mA**
- **"Hermetic Gold-CHIP" Construction**
- **Silicon Nitride Passivated**
- **Platinum Silicide Ohmic Contacts**
- **Gold-CHIP Interconnect Metallization and Gold Bonding Pads**

Applications:

- **High-Current LED Driver**
- **Relay and Solenoid Driver**
- **Lamp Driver**

The RCA-CA3138G and CA3138AG are high-current n-p-n transistor arrays containing four isolated (discrete) sealed-junction high-current n-p-n transistors. They are intended for high-current, high-speed switching and driver applications.

The CA3138AG has all the features and characteristics of the CA3138G but is intended for applications requiring premium grade specifications –– higher rating for V_{CBO} of 25 volts and limits established for I_{CEO}, I_{EBO}, and h_{FE} at 10 mA.

The CA3138G and CA3138AG are supplied in a 14-lead dual-in-line plastic package and operate over the full military temperature range of –55°C to +125°C. The transistor chip used for these types is of the sealed-junction type to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

The semiconductor junctions are sealed by a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal connection system of unique design is employed.

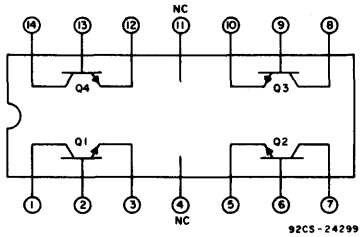


Fig. 1 – Terminal diagram (top view).

MAXIMUM RATINGS, *Absolute-Maximum Values*

COLLECTOR-TO-EMITTER VOLTAGE ..... With Base Open (V_{CEO})	15	V
COLLECTOR-TO-BASE VOLTAGE With Emitter Open (V_{CBO})		
CA3138G	20	V
CA3138AG	25	V
EMITTER-TO-BASE VOLTAGE ..... With Collector Open (V_{EBO})	5	V
COLLECTOR CURRENT (I_C)	1	A
POWER DISSIPATION (P_D):		
At T_A up to 25°C:		
For Each Transistor	1	W
Total Package	2	W
At T_A above 25°C derate linearly	20	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating	–55 to +125	°C
Storage	–65 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	265	°C

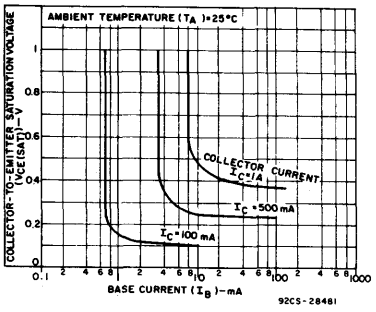


Fig. 2 – V_{CE}(sat) vs I_B

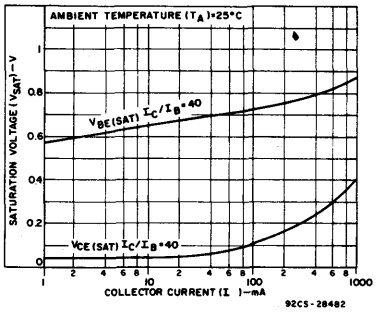


Fig. 3 – V_{sat} vs I_C

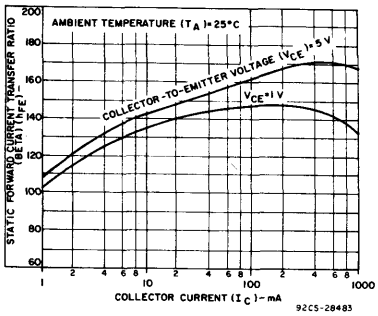


Fig. 4 – h_{FE} vs I_C

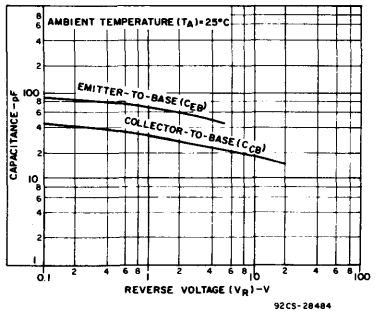


Fig. 5 – C_{CB}, C_{CE} vs V_R

ELECTRICAL CHARACTERISTICS at T_A = 25°C

Characteristic		Test Conditions	LIMITS						Units
			CA3138G			CA3138AG			
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Collector-to-Emitter Sustaining Voltage, $V_{CEO}(sus)$*		I_C = 1 mA, I_B = 0	15	20	–	15	20	–	V
Collector-to-Emitter Breakdown Voltage, $V_{(BR)CES}$		I_C = 10 μA	20	55	–	25	60	–	V
Collector-to-Base Breakdown Voltage, $V_{(BR)CBO}$		I_C = 10 μA, I_E = 0	20	55	–	25	60	–	V
Emitter-to-Base Breakdown Voltage, $V_{(BR)EBO}$		I_E = 10 μA, I_C = 0	5	7.2	–	5	7.2	–	V
Base-to-Emitter Saturation Voltage, $V_{BE}(sat)$*		I_C = 500 mA, I_B = 12.5 mA	0.7	0.81	1.1	0.7	0.81	1.1	V
Collector-to-Emitter Saturation Voltage, $V_{CE}(sat)$*		I_C=500 mA, I_B = 12.5 mA	–	0.26	0.4	–	0.26	0.4	V
Collector-Cutoff Current	I_{CBO}	V_{CB} = 15 V	–	0.03	1	–	0.02	0.1	μA
	I_{CEO}	V_{CE} = 10 V	–	0.5	–	–	0.3	1.0	
	I_{EBO}	V_{EB} = 4 V	–	0.01	–	–	0.01	0.1	
Static Forward-Current Transfer Ratio (Beta), h_{FE}*		I_C=10 mA, V_{CE} = 5 V	–	–	–	35	140	–	
		I_C = 100 mA, V_{CE} = 5 V	80	160	450	80	160	450	
		I_C = 500 mA, V_{CE} = 5 V	95	170	500	95	170	500	
		I_C = 1 A, V_{CE} = 5 V	40	170	–	40	170	–	
Small-Signal Forward Current Transfer Ratio, h_{fe}		I_C = 50 mA, V_{CE} = 10 V, f = 100 MHz	2	–	–	2	–	–	
Collector-to-Base Capacitance, C_{CB}		V_{CB} = 10 V, I_E = 0	–	18	–	–	18	–	pF
Emitter-to-Base Capacitance, C_{EB}		V_{EB} = 0.5 V, I_C = 0	–	77	–	–	77	–	pF
Rise Time (See Test Ckt. Fig. 6), t_r		I_C = 570 mA	–	6	–	–	6	–	ns
Fall Time (See Test Ckt. Fig. 6), t_f		I_{B1} = 30 mA	–	100	–	–	100	–	ns
Delay Time (See Test Ckt. Fig. 6), t_d		I_{B2} = 0	–	7.5	–	–	7.5	–	ns
Storage Time (See Test Ckt. Fig. 6), t_s			–	850	–	–	850	–	ns

*Pulse Conditions: width = 300 μs; duty cycle = 1%.

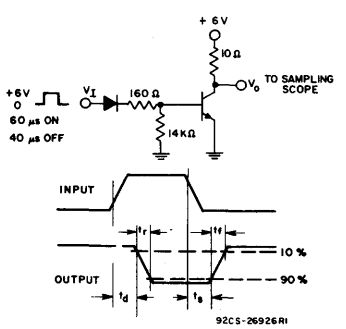


Fig. 6 – Switching time test circuit and waveforms.

CA3140, CA3140A, CA3140B Types

BiMOS Operational Amplifiers

With MOS/FET Input, Bipolar Output

The CA3140B, CA3140A, and CA3140 are integrated-circuit operational amplifiers that combine the advantages of high-voltage PMOS transistors with high-voltage bipolar transistors on a single monolithic chip. Because of this unique combination of technologies, this device can now provide designers, for the first time, with the special performance features of the CA3130 COS/MOS operational amplifiers and the versatility of the 741 series of industry-standard operational amplifiers.

The CA3140, CA3140A, and CA3140 BiMOS operational amplifiers feature gate-protected MOS/FET (PMOS) transistors in the input circuit to provide very-high-input impedance, very-low-input current, and high-speed performance. The CA3140B operates at supply voltages from 4 to 44 volts; the CA3140A and CA3140 from 4 to 36 volts (either single or dual supply). These operational amplifiers are internally phase-compensated to achieve stable operation in unity-gain follower operation, and, additionally, have access terminals for a supplementary external capacitor if additional frequency roll-off is desired. Terminals are also provided for use in applications requiring input offset-voltage nulling.

The use of PMOS field-effect transistors in the input stage results in common-mode input-voltage capability down to 0.5 volt below the negative-supply terminal, an important attribute for single-supply applications. The output stage uses bipolar transistors and includes built-in protection against damage from load-terminal short-circuiting to either supply-rail or to ground.

The CA3140 Series has the same 8-lead terminal pin-out used for the "741" and other industry-standard operational amplifiers. They are supplied in either the standard 8-lead TO-5 style package (T suffix), or in the 8-lead dual-in-line formed-lead TO-5 style package "DIL-CAN" (S suffix). The CA3140 is available in chip form (H suffix). The CA3140A and CA3140 are also available in an 8-lead dual-in-line plastic package (Mini-DIP-E suffix). The CA3140B is intended for operation at supply voltages ranging from 4 to 44 volts, for applications requiring premium-grade specifications and with electrical limits established for operations over the range from −55°C to +125°C. The CA3140A and CA3140 are for operation at supply voltages up to 36 volts (±18 volts). The CA3140 ages up to 36 volts (±18 volts). All types can be operated safely over the temperature range from −55°C to +125°C.

Features:

- **MOS/FET Input Stage**
  - **(a) Very high input impedance (Z_{IN}) − 1.5 TΩ typ.**
  - **(b) Very low input current (I_I) − 10 pA typ. at ± 15 V**
  - **(c) Low input-offset voltage (V_{IO}) − to 2 mV max.**
  - **(d) Wide common-mode input-voltage range (V_{ICR}) − can be swung 0.5 volt below negative supply-voltage rail**
  - **(e) Output swing complements input common-mode range**
  - **(f) Rugged input stage − bipolar diode protected**
- **Directly replaces industry type 741 in most applications**
- **Includes numerous industry operational amplifier categories such as general-purpose, FET input, wideband (high slew rate)**
- **Operation from 4-to-44 volts Single or Dual supplies**
- **Internally compensated**
- **Characterized for ± 15-volt operation and for TTL supply systems with operation down to 4 volts**
- **Wide bandwidth − 4.5 MHz unity gain at ± 15 V or 30 V; 3.7 MHz at 5 V**
- **High voltage-follower slew rate − 9 V/μs**
- **Fast settling time − 1.4 μs typ. to 10 mV with a 10-V_{p-p} signal**
- **Output swings to within 0.2 volt of negative supply**
- **Strobable output stage**

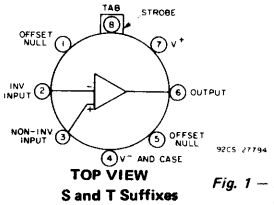
MAXIMUM RATINGS, *Absolute-Maximum Values:*

	CA3140, CA3140A	CA3140B
DC SUPPLY VOLTAGE (BETWEEN V^+ AND V^- TERMINALS)	36 V	44 V
DIFFERENTIAL-MODE INPUT VOLTAGE	±8 V	±8 V
COMMON-MODE DC INPUT VOLTAGE	(V^+ +8 V) to (V^- −0.5 V)	
INPUT-TERMINAL CURRENT	1 mA	
DEVICE DISSIPATION:		
WITHOUT HEAT SINK −		
UP TO 55°C	630 mW	
ABOVE 55°C	Derate linearly 6.67 mW/°C	
WITH HEAT SINK −		
Up to 55°C	1 W	
Above 55°C	Derate linearly 16.7 mW/°C	
TEMPERATURE RANGE:		
OPERATING (ALL TYPES)	−55 to + 125°C	
STORAGE (ALL TYPES)	−65 to +150°C	
OUTPUT SHORT-CIRCUIT DURATION*	INDEFINITE	
LEAD TEMPERATURE (DURING SOLDERING): AT DISTANCE 1/16 ± 1/32 INCH (1.59 ± 0.79 MM) FROM CASE FOR 10 SECONDS MAX.	+265°C	

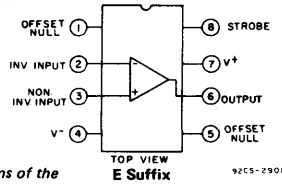
* Short circuit may be applied to ground or to either supply.

Applications:

- **Ground-referenced single-supply amplifiers in automobile and portable instrumentation**
- **Sample and hold amplifiers**
- **Long-duration timers/multivibrators (microseconds−minutes−hours)**
- **Photocurrent instrumentation**
- **Peak detectors** ■ **Active filters**
- **Comparators**
- **Interface in 5 V TTL systems & other low-supply voltage systems**
- **All standard operational amplifier applications**
- **Function generators** ■ **Tone controls**
- **Power supplies** ■ **Portable instruments**
- **Intrusion alarm systems**



TOP VIEW
S and T Suffixes



E Suffix

Fig. 1 − Functional diagrams of the CA3140 series.

TYPICAL ELECTRICAL CHARACTERISTICS

CHARACTERISTIC		TEST CONDITIONS $V^+ = +15$ V, $V^- = -15$ V, $T_A = 25°C$		CA3140B (T,S)	CA3140A (T,S,E)	CA3140 (T,S,E)	UNITS
Input Offset Voltage Adjustment Resistor		Typ. Value of Resistor Between Term. 4 and 5 or 4 and 1 to Adjust Max. V_{IO}		43	18	4.7	kΩ
Input Resistance	R_I			1.5	1.5	1.5	TΩ
Input Capacitance	C_I			4	4	4	pF
Output Resistance	R_O			60	60	60	Ω
Equivalent Wideband Input Noise Voltage (See Fig. 39)	e_n	BW = 140 kHz, R_S = 1 MΩ		48	48	48	μV
Equivalent Input Noise Voltage (See Fig.10)	e_n	f = 1 kHz	R_S = 100 Ω	40	40	40	$nV/\sqrt{Hz}$
		f = 10 kHz		12	12	12	
Short-Circuit Current to Opposite Supply	Source I_{OM}^+			40	40	40	mA
	Sink I_{OM}^-			18	18	18	mA
Gain-Bandwidth Product, (See Figs. 5 &18)	f_T			4.5	4.5	4.5	MHz
Slew Rate, (See Fig.6)	SR			9	9	9	V/μs
Sink Current From Terminal 8 To Terminal 4 to Swing Output Low				220	220	220	μA
Transient Response: Rise Time	t_r	R_L = 2 kΩ, C_L = 100 pF		0.08	0.08	0.08	μs
Overshoot (See Fig. 37)				10	10	10	%
Settling Time at 10 V_{p-p}, (See Fig.17)	1 mV / 10 mV t_s	R_L = 2 kΩ, C_L = 100 pF, Voltage Follower		4.5 / 1.4	4.5 / 1.4	4.5 / 1.4	μs

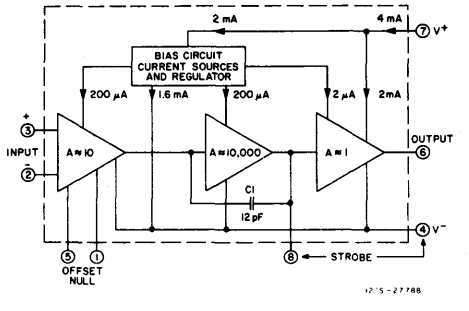


Fig.2 – Block diagram of CA3140 series.

CIRCUIT DESCRIPTION

Fig.2 is a block diagram of the CA3140 Series PMOS Operational Amplifiers. The input terminals may be operated down to 0.5 V below the negative supply rail. Two class A amplifier stages provide the voltage gain, and a unique class AB amplifier stage provides the current gain necessary to drive low-impedance loads.

A biasing circuit provides control of cascoded constant-current flow circuits in the first and second stages. The CA3140 includes an on-chip phase-compensating capacitor that is sufficient for the unity gain voltage-follower configuration.

Input Stages – The schematic circuit diagram of the CA3140 is shown in Fig.3. It consists of a differential-input stage using PMOS field-effect transistors (Q9, Q10) working into a mirror pair of bipolar transistors (Q11, Q12) functioning as load resistors together with resistors R2 through R5. The mirror-pair transistors also function as a differential-to-single-ended converter to provide base-current drive to the second-stage bipolar transistor (Q13). Offset nulling, when desired, can be effected with a 10-kΩ potentiometer connected across terminals 1 and 5 and with its slider arm connected to terminal 4. Cascode-connected bipolar transistors Q2, Q5 are the constant-current source for the input stage. The base-biasing circuit for the constant-current source is described subsequently. The small diodes D3, D4, D5 provide gate-oxide protection against high-voltage transients, e.g., static electricity.

Second Stage – Most of the voltage gain in the CA3140 is provided by the second amplifier stage, consisting of bipolar transistor Q13 and its cascode-connected load resistance provided by bipolar transistors Q3, Q4. On-chip phase compensation, sufficient for a majority of the applications is provided by C1. Additional Miller-Effect compensation (roll-off) can be accomplished, when desired, by simply connecting a small capacitor between terminals 1 and 8. Terminal 8 is also used to strobe the output stage into quiescence. When terminal 8 is tied to the negative supply rail (terminal 4) by mechanical or electrical means, the output terminal 6 swings low, i.e., approximately to terminal 4 potential.

Output Stage – The CA3140 Series circuits employ a broadband output stage that can sink loads to the negative supply to complement the capability of the PMOS input stage when operating near the negative rail. Quiescent current in the emitter-follower cascade circuit (Q17, Q18) is established by transistors (Q14, Q15) whose base-currents are "mirrored" to current flowing through diode D2 in the bias circuit section. When the CA3140 is operating such that output terminal 6 is sourcing current, transistor Q18 functions as an emitter-follower to source current from the V+ bus (terminal 7), via D7, R9, and R11. Under these conditions, the collector potential of Q13 is sufficiently high to permit the necessary flow of base current to emitter follower Q17 which, in turn, drives Q18.

ELECTRICAL CHARACTERISTICS FOR EQUIPMENT DESIGN

At $V^+ = 15$ V, $V^- = 15$ V, $T_A = 25^oC$ Unless Otherwise Specified

CHARACTERISTIC	LIMITS									UNITS
	CA3140B			CA3140A			CA3140			
	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage, $\|V_{IO}\|$	–	0.8	2	–	2	5	–	5	15	mV
Input Offset Current, $\|I_{IO}\|$	–	0.5	10	–	0.5	20	–	0.5	30	pA
Input Current, I_I	–	10	30	–	10	40	–	10	50	pA
Large-Signal Voltage Gain, A_{OL}● (See Figs. 4,18)	50 k	100 k	–	20 k	100 k	–	20 k	100 k	–	V/V
	94	100	–	86	100	–	86	100	–	dB
Common-Mode Rejection Ratio, CMRR (See Fig.9)	–	20	50	–	32	320	–	32	320	μV/V
	86	94	–	70	90	–	70	90	–	dB
Common-Mode Input-Voltage Range, V_{ICR} (See Fig.20)	−15	−15.5 to +12.5	12	−15	−15.5 to +12.5	12	−15	−15.5 to +12.5	11	V
Power-Supply Rejection Ratio, PSRR $\Delta V_{IO}/\Delta V$ (See Fig.11)	–	32	100	–	100	150	–	100	150	μV/V
	80	90	–	76	80	–	76	80	–	dB
Max. Output Voltage■ V_{OM}^+ (See Figs.13,20) V_{OM}^-	+12	13	–	+12	13	–	+12	13	–	V
	−14	−14.4	–	−14	−14.4	–	−14	−14.4	–	
Supply Current, I^+ (See Fig.7)	–	4	6	–	4	6	–	4	6	mA
Device Dissipation, P_D	–	120	180	–	120	180	–	120	180	mW
Input Current, I_I▲ (See Fig.19)	–	10	30	–	10	–	–	10	–	nA
Input Offset Voltage $\|V_{IO}\|$▲	–	1.3	3	–	3	–	–	10	–	mV
Input Offset Voltage Temp. Drift, $\Delta V_{IO}/\Delta T$	–	5	–	–	6	–	–	8	–	μV/°C
Large-Signal Voltage Gain, A_{OL}▲ (See Figs.4,18)	20 k	100 k	–	–	100 k	–	–	100 k	–	V/V
	86	100	–	–	100	–	–	100	–	dB
Max. Output Voltage,★ V_{OM}^+	+19	+19.5	–	–	–	–	–	–	–	V
V_{OM}^-	−21	−21.4	–	–	–	–	–	–	–	
Large-Signal Voltage Gain, A_{OL}♦★	20 k	50 k	–	–	–	–	–	–	–	V/V
	86	94	–	–	–	–	–	–	–	dB

● At $V_O = 26V_{p\text{-}p}$, +12V, −14V and $R_L = 2$ kΩ.

■ At $R_L = 2$ kΩ.

▲ At $T_A = -55^oC$ to $+125^oC$, $V^+ = 15$ V, $V^- = 15$ V, $V_O = 26V_{p\text{-}p}$, $R_L = 2$ kΩ.

♦ At $V_O = +19$ V, −21 V, and $R_L = 2$ kΩ.

★ At $V^+ = 22$ V, $V^- = 22$ V.

When the CA3140 is operating such that output terminal 6 is sinking current to the V− bus, transistor Q16 is the current-sinking element. Transistor Q16 is mirror-connected to D6, R7, with current fed by way of Q21, R12, and Q20. Transistor Q20, in turn, is biased by current-flow through R13, zener D8, and R14. The dynamic current-sink is controlled by voltage-level sensing. For purposes of explanation, it is assumed that output terminal 6 is quiescently established at the potential mid-point between the V+ and V− supply rails. When output-current sinking-mode operation is required, the collector potential of transistor Q13 is driven below its quiescent level, thereby causing Q17, Q18 to decrease the output voltage at terminal 6. Thus, the gate terminal of PMOS transistor Q21 is displaced toward the V− bus, thereby reducing the channel resistance of Q21. As a consequence, there is an incremental increase in current flow through Q20, R12, Q21, D6, R7, and the base of Q16. As a result, Q16 sinks current from terminal 6 in direct response to the incremental change in output voltage caused by Q18. This sink current flows regardless of load; any excess current is internally supplied by the emitter-follower Q18. Short-circuit protection of the output circuit is provided by Q19, which is driven into conduction by the high voltage drop developed across R11 under output short-circuit conditions. Under these conditions, the collector of Q19 diverts current from Q4 so as to reduce the base-current drive from Q17, thereby limiting current flow in Q18 to the short-circuited load terminal.

Bias Circuit – Quiescent current in all stages (except the dynamic current sink) of the CA3140 is dependent upon bias current flow in R1. The function of the bias circuit is to establish and maintain constant-current flow through D1, Q6, Q8 and D2. D1 is a diode-connected transistor mirror-connected in parallel with the base-emitter junctions of Q1, Q2, and Q3. D1 may be considered as a current-sampling diode that senses the emitter current of Q6 and automatically adjusts the base current of Q6 (via Q1) to maintain a constant current through Q6, Q8, D2. The base-currents in Q2, Q3 are also determined by constant-current flow D1. Furthermore, current in diode-connected transistor D2 establishes the currents in transistors Q14 and Q15.

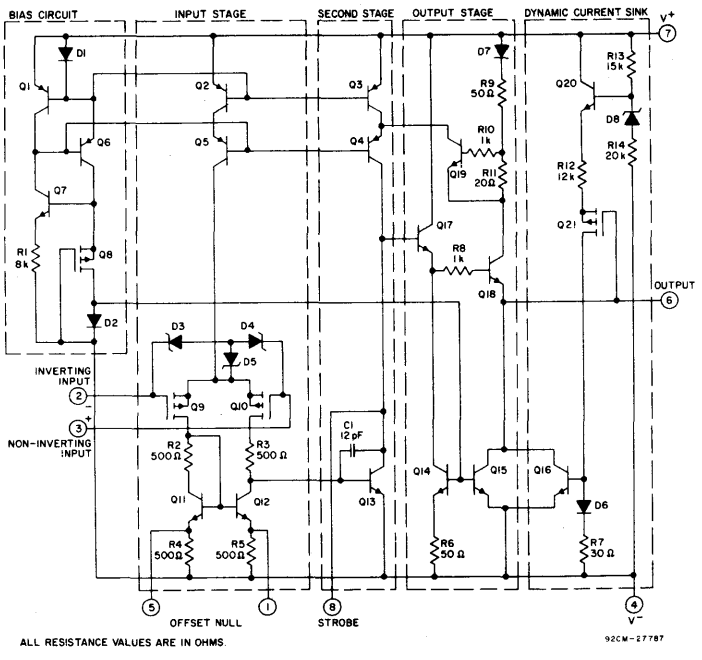


Fig.3 – Schematic diagram of CA3140 series.

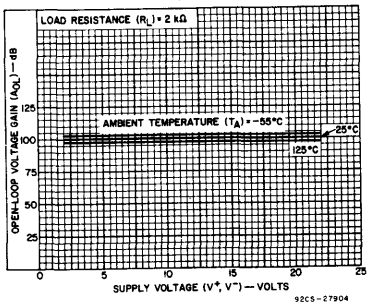


Fig.4 – Open-loop voltage gain vs supply voltage and temperature.

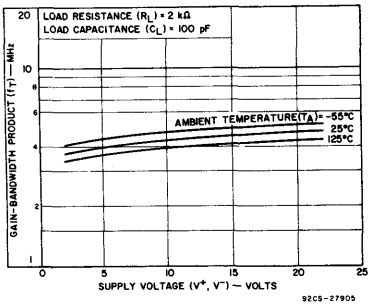


Fig.5 – Gain-bandwidth product vs supply voltage and temperature.

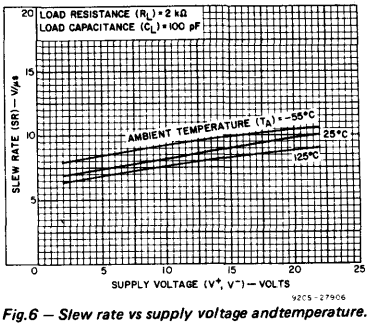


Fig.6 – Slew rate vs supply voltage andtemperature.

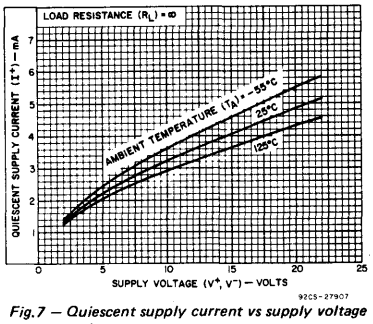


Fig.7 – Quiescent supply current vs supply voltage and temperature.

TYPICAL ELECTRICAL CHARACTERISTICS FOR DESIGN GUIDANCE

At $V^+ = 5$ V, $V^- = 0$ V, $T_A = 25^oC$

CHARACTERISTIC		CA3140B (T,S)	CA3140A (T,S,E)	CA3140 (T,S,E)	UNITS
Input Offset Voltage	$\|V_{IO}\|$	0.8	2	5	mV
Input Offset Current	$\|I_{IO}\|$	0.1	0.1	0.1	pA
Input Current	I_I	2	2	2	pA
Input Resistance		1	1	1	TΩ
Large-Signal Voltage Gain	A_{OL}	100 k	100 k	100 k	V/V
(See Figs.4,18)		100	100	100	dB
Common-Mode Rejection Ratio,	CMRR	20	32	32	μV/V
		94	90	90	dB
Common-Mode Input-Voltage Range	V_{ICR}	−0.5	−0.5	−0.5	V
(See Fig.20)		2.6	2.6	2.6	
Power-Supply Rejection Ratio	$\Delta V_{IO}/\Delta V^+$	32	100	100	μV/V
		90	80	80	dB
Maximum Output Voltage	V_{OM}^+	3	3	3	V
(See Figs.13,20)	V_{OM}^-	0.13	0.13	0.13	
Maximum Output Current:					
Source	I_{OM}^+	10	10	10	mA
Sink	I_{OM}^-	1	1	1	
Slew Rate (See Fig.6)		7	7	7	V/μs
Gain-Bandwidth Product (See Fig.5)	f_T	3.7	3.7	3.7	MHz
Supply Current (See Fig.7)	I^+	1.6	1.6	1.6	mA
Device Dissipation	P_D	8	8	8	mW
Sink Current from Term. 8 to Term. 4 to Swing Output Low		200	200	200	μA

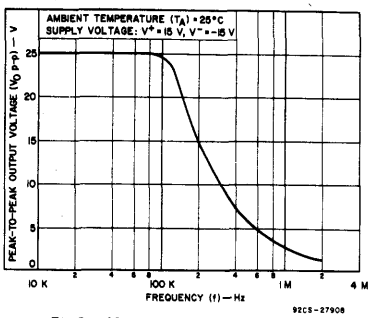


Fig.8 – Maximum output voltage swing vs frequency.

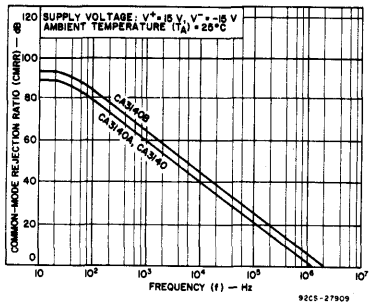


Fig.9 – Common-mode rejection ratio vs frequency.

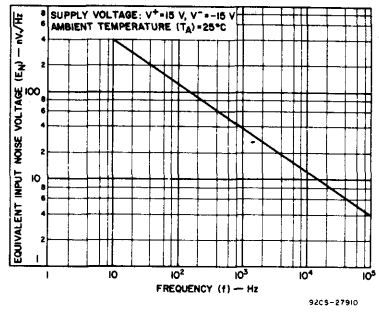


Fig.10 – Equivalent input noise voltage vs frequency.

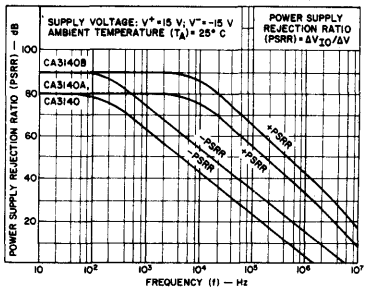


Fig.11 – Power supply rejection ratio vs frequency.

APPLICATIONS CONSIDERATIONS

Wide dynamic range of input and output characteristics with the most desirable high input-impedance characteristic is achieved in the CA3140 by the use of an unique design based upon the PMOS-Bipolar process. Input-common-mode voltage range and output-swing capabilities are complementary, allowing operation with the single supply down to four volts.

The wide dynamic range of these parameters also means that this device is suitable for many single-supply applications, such as, for example, where one input is driven below the potential of terminal 4 and the phase sense of the output signal must be maintained – a most important consideration in comparator applications.

OUTPUT CIRCUIT CONSIDERATIONS

Excellent interfacing with TTL circuitry is easily achieved with a single 6.2-volt zener diode connected to terminal 8 as shown in Fig.12. This connection assures that the maximum output signal swing will not go more positive than the zener voltage minus two base-to-emitter voltage drops within the CA3140. These voltages are independent of the operating supply voltage.

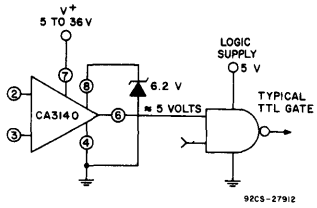


Fig.12 – Zener clamping diode connected to terminals 8 and 4 to limit CA3140 output swing to TTL levels.

Fig.13 shows output current-sinking capabilities of the CA3140 at various supply voltages. Output voltage swing to the negative supply rail permits this device to operate both power transistors and thyristors directly without the need for level-shifting circuitry usually associated with the 741 series of operational amplifiers.

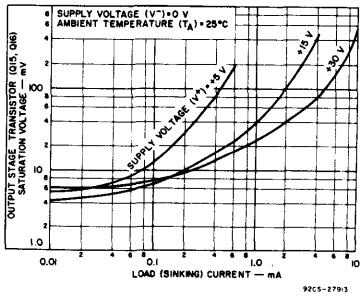


Fig.13 – Voltage across output transistors Q15 and Q16 vs load current.

Fig.16 show some typical configurations. Note that a series resistor, R_L, is used in both cases to limit the drive available to the driven device. Moreover, it is recommended that a series diode and shunt diode be used at the thyristor input to prevent large negative transient surges that can appear at the gate of thyristors, from damaging the integrated circuit.

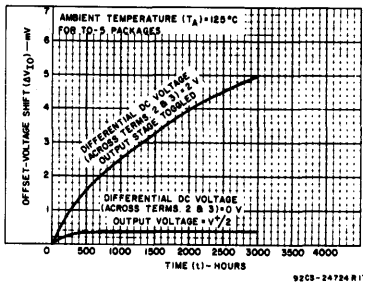


Fig.14 – Typical incremental offset-voltage shift vs operating life.

OFFSET-VOLTAGE NULLING

The input-offset voltage can be nulled by connecting a 10-kΩ potentiometer between terminals 1 and 5 and returning its wiper arm to terminal 4, see Fig.15a. This technique, however, gives more adjustment range than required and therefore, a considerable portion of the potentiometer rotation is not fully utilized. Typical values of series resistors that may be placed at either end of the potentiometer, see Fig.15b, to optimize its utilization range are given in the table "Typical Electrical Characteristics" shown in this bulletin.

An alternate system is shown in Fig.15c. This circuit uses only one additional resistor of approximately the value shown in the table. For potentiometers, in which the resistance does not drop to zero ohms at either end of rotation, a value of resistance 10% lower than the values shown in the table should be used.

LOW-VOLTAGE OPERATION

Operation at total supply voltages as low as 4 volts is possible with the CA3140. A current regulator based upon the PMOS threshold voltage maintains reasonable constant operating current and hence consistent performance down to these lower voltages.

The low-voltage limitation occurs when the upper extreme of the input common-mode voltage range extends down to the voltage at terminal 4. This limit is reached at a total supply voltage just below 4 volts. The output voltage range also begins to extend down to the negative supply rail, but is slightly higher than that of the input. Fig.20 shows these characteristics and shows that with 2-volt dual supplies, the lower extreme of the input common-mode voltage range is below ground potential.

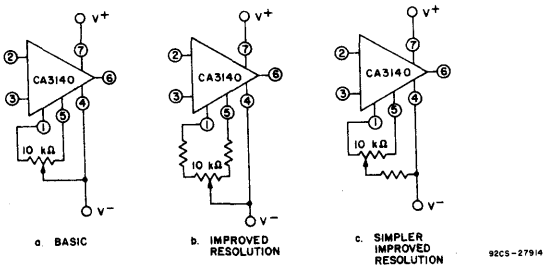


Fig. 15 – Three offset-voltage nulling methods.

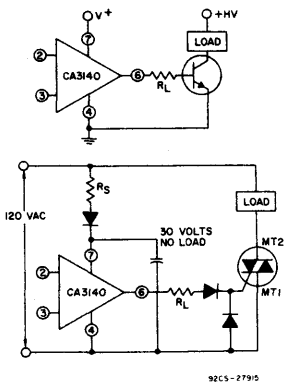


Fig. 16 – Methods of utilizing the $V_{CE(sat)}$ sinking-current capability of the CA3140 series.

BANDWIDTH AND SLEW RATE

For those cases where bandwidth reduction is desired, for example, broadband noise reduction, an external capacitor connected between terminals 1 and 8 can reduce the open-loop −3 dB bandwidth. The slew rate will, however, also be proportionally reduced by using this additional capacitor. Thus, a 20% reduction in bandwidth by this technique will also reduce the slew rate by about 20%.

Fig. 17 shows the typical settling time required to reach 1 mV or 10 mV of the final value for various levels of large signal inputs for the voltage-follower and inverting unity-gain amplifiers. The exceptionally fast settling time characteristics shown in Fig. 18 are largely due to the high combination of high gain and wide bandwidth of the CA3140.

INPUT CIRCUIT CONSIDERATIONS

As mentioned previously, the amplifier inputs can be driven below the terminal 4 potential, but a series current-limiting resistor is recommended to limit the maximum input terminal current to less than 1 mA to prevent damage to the input protection circuitry.

Moreover, some current-limiting resistance should be provided between the inverting input and the output when the CA3140 is used as a unity-gain voltage follower. This resistance prevents the possibility of extremely large input-signal transients from forcing a signal through the input-protection network and directly driving the internal constant-current source which could result in positive feedback via the output terminal. A 3.9-kΩ resistor is sufficient.

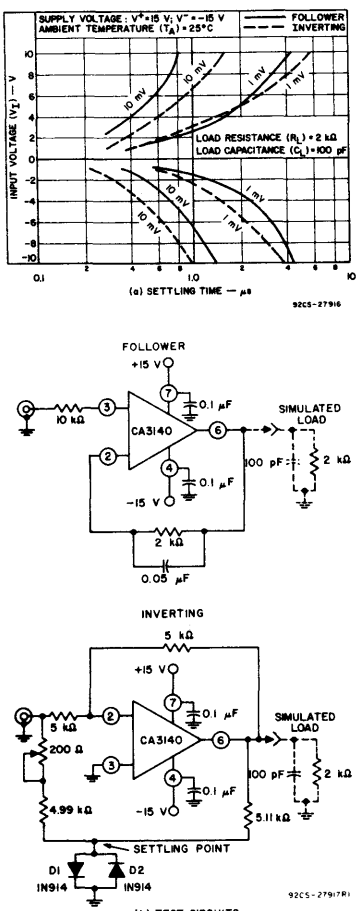


Fig. 17 – Input voltage vs settling time.

The typical input current is in the order of 10 pA when the inputs are centered at nominal device dissipation. As the output supplies load current, device dissipation will increase, raising the chip temperature and resulting in increased input current. Fig. 19 shows typical input-terminal current versus ambient temperature for the CA3140.

It is well known that MOS/FET devices can exhibit slight changes in characteristics (for example, small changes in input offset voltage) due to the application of large differential input voltages that are sustained over long periods at elevated temperatures.

Both applied voltage and temperature accelerate these changes. The process is reversible and offset voltage shifts of the opposite polarity reverse the offset. Fig. 14 shows the typical offset voltage change as a function of various stress voltages at the maximum rating of 125°C (for TO-5); at lower temperatures (TO-5 and plastic), for example, at 85°C, this change in voltage is considerably less. In typical linear applications, where the differential voltage is small and symmetircal, these incremental changes are of about the same magnitude as those encountered in an operational amplifier employing a bipolar a transistor input stage.

SUPER SWEEP FUNCTION GENERATOR

A function generator having a wide tuning range is shown in Fig. 21. The 1,000,000/1 adjustment range is accomplished by a single variable potentiometer or by an auxiliary sweeping signal. The CA3140 functions as a non-inverting read-out amplifier of the tri-

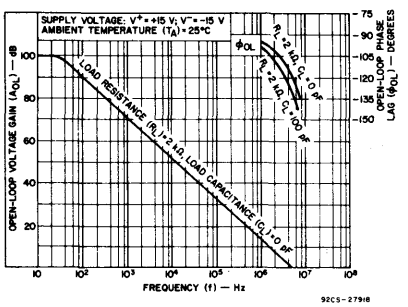


Fig. 18 – Open-loop voltage gain and phase lag vs frequency.

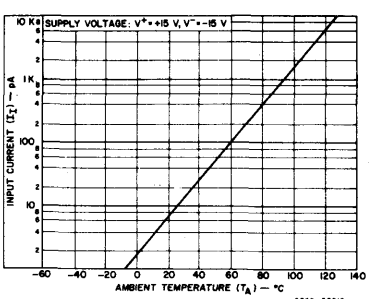


Fig. 19 – Input current vs ambient temperature.

angular signal developed across the integrating capacitor network connected to the output of the CA3080A current source.

Buffered triangular output signals are then applied to a second CA3080 functioning as a high-speed hysteresis switch. Output from the switch is returned directly back to the

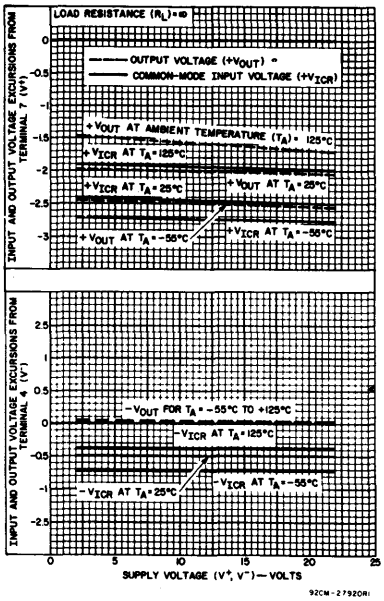


Fig.20 – Output-voltage-swing capability and common-mode input-voltage range vs supply voltage and temperature.

input of the CA3080A current source, thereby, completing the positive feedback loop.

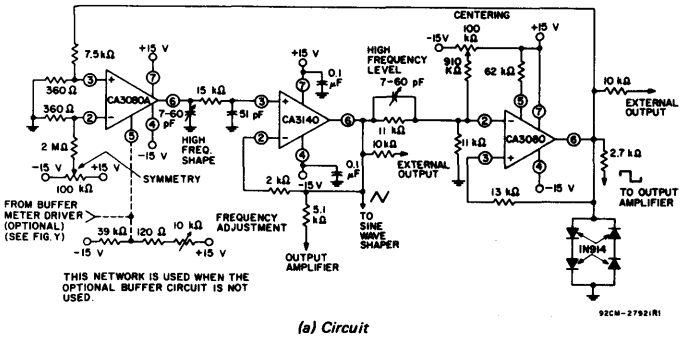
The triangular output level is determined by the four 1N914 level-limiting diodes of the second CA3080 and the resistor-divider network connected to terminal No.2 (input) of the CA3080. These diodes establish the input trip level to this switching stage and, therefore, indirectly determine the amplitude of the output triangle.

Compensation for propagation delays around the entire loop is provided by one adjustment on the input of the CA3080. This adjustment, which provides for a constant generator amplitude output, is most easily made while the generator is sweeping. High-frequency ramp linearity is adjusted by the single 7-to-60 pF capacitor in the output of the CA3080A.

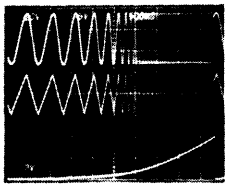
It must be emphasized that only the CA-3080A is characterized for maximum output linearity in the current-generator function.

METER DRIVER AND BUFFER AMPLIFIER

Fig. 22 shows the CA3140 connected as a meter driver and buffer amplifier. Low driving impedance is required of the CA-3080A current source to assure smooth operation of the Frequency Adjustment



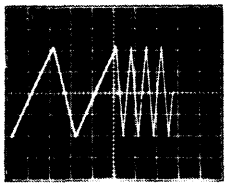
(a) Circuit



TOP TRACE: OUTPUT AT JUNCTION OF 2.7 Ω AND 51 Ω RESISTORS 5 V/DIV AND 500 ms/DIV
CENTER TRACE: EXTERNAL OUTPUT OF TRIANGULAR FUNCTION GENERATOR 2 V/DIV AND 500 ms/DIV
BOTTOM TRACE: OUTPUT OF "LOG" GENERATOR 10 V/DIV AND 500 ms/DIV

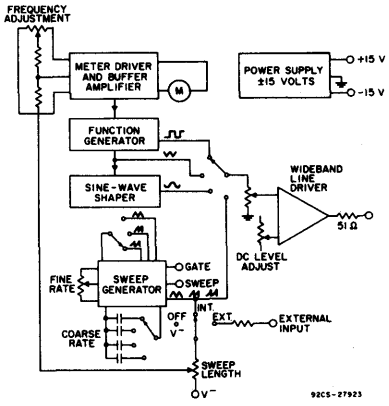
92CS-27922

(b1) Function generator sweeping



92CS-27937

(b2) Function generator with fixed frequencies



(c) Interconnections

Fig.21 – Function generator.

Control. This low-driving impedance requirement is easily met by using a CA3140 connected as a voltage follower. Moreover, a meter may be placed across the input to the CA3080A to give a logarithmic analog indication of the function generators frequency.

Analog frequency readout is readily accomplished by the means described above because the output current of the CA3080A varies approximately one decade for each 60-mV change in the applied voltage, V_{ABC} (voltage between terminals 5 and 4 of the CA3080A of the function generator). Therefore, six decades represent 360-mV change in V_{ABC}.

Now, only the reference voltage must be established to set the lower limit on the meter. The three remaining transistors from the CA3086 Array used in the sweep generator are used for this reference voltage. In addition, this reference generator arrangement tends to track ambient temperature variations, and thus compensates for the effects of the normal negative temperature coefficient of the CA3080A V_{ABC} terminal voltage.

Another output voltage from the reference generator is used to insure temperature tracking of the lower end of the Frequency Adjustment Potentiometer. A large series resistance simulates a current source, assuring similar temperature coefficients at both ends of the Frequency Adjustment Control.

To calibrate this circuit, set the Frequency Adjustment Potentiometer at its low end. Then adjust the Minimum Frequency Calibration Control for the lowest frequency. To

1V/DIV and 1 sec/DIV

Three tone test signals, highest frequency ⩾ 0.5 MHz. Note the slight assymmetry at the three-second/cycle signal. This assymmetry is due to slightly different positive and negative integration from the CA3080A and from the pc board and component leakages at the 100-pA level.

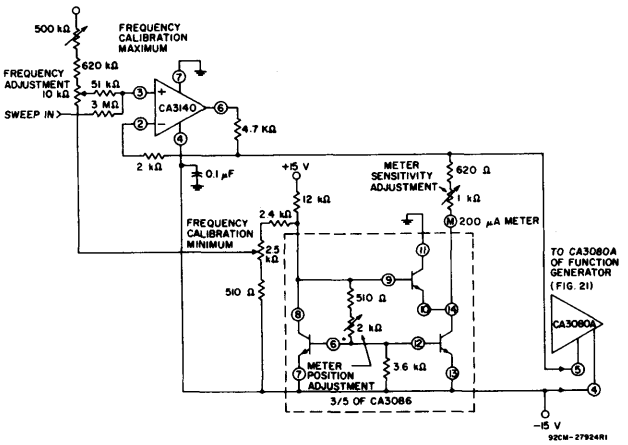


Fig. 22 – Meter driver and buffer amplifier.

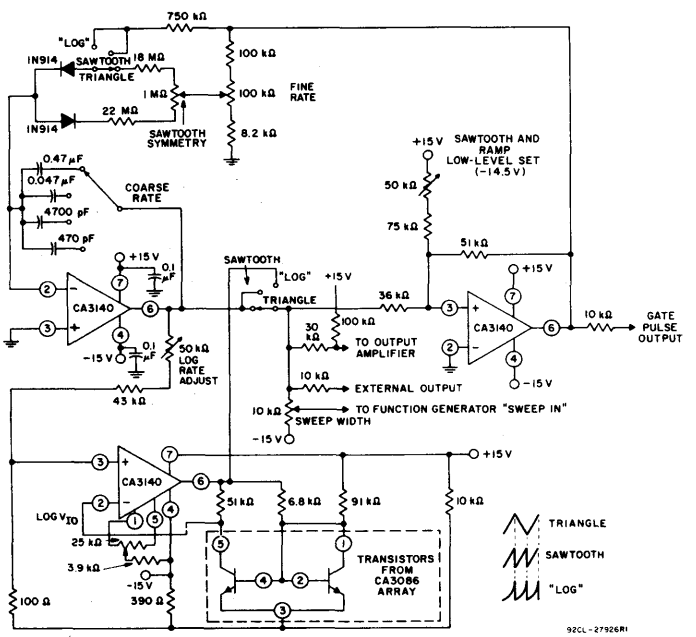


Fig. 24 – Sweeping generator.

establish the upper frequency limit, set the Frequency Adjustment Potentiometer to its upper end and then adjust the Maximum Frequency Calibration Control for the maximum frequency. Because there is interaction among these controls, repetition of the adjustment procedure may be necesary.

Two adjustments are used for the meter. The meter sensitivity control sets the meter-scale width of each decade, while the meter position control adjusts the pointer on the scale with negligible effect on the sensitivity adjustment. Thus, the meter sensitivity adjustment control calibrates the meter so that it deflects 1,6 of full scale for each decade change in frequency.

SINE-WAVE SHAPER

The circuit shown in Fig. 23 uses a CA3140 as a voltage follower in combination with diodes from the CA3019 Array to convert the triangular signal from the function generator to a sine-wave output signal having typically less than 2% THD. The basic zero-crossing slope is established by the 10-kΩ

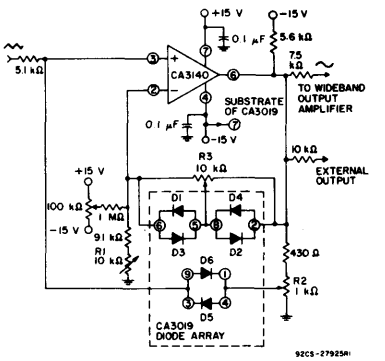


Fig. 23 – Sine-wave shaper.

potentiometer connected between terminals 2 and 6 of the CA3140 and the 9.1-kΩ resistor and 10-kΩ potentiometer from terminal 2 to ground. Two break points are established by diodes D_1 through D_4. Positive feedback via D_5 and D_6 establishes the zero slope at the maximum and minimum levels of the sine wave. This technique is necessary because the voltage-follower configuration approaches unity gain rather than the zero gain required to shape the sine wave at the two extremes.

This circuit can be adjusted most easily with a distortion analyzer, but a good first approximation can be made by comparing the output signal with that of a sine-wave generator. The initial slope is adjusted with the potentiometer R_1, followed by an adjustment of R_2. The final slope is established by adjusting R_3, thereby adding additional segments that are contributed by these diodes. Because there is some interaction among these controls, repetition of the adjustment procedure may be necessarv

SWEEPING GENERATOR

Fig. 24 shows a sweeping generator. Three CA3140's are used in this circuit. One CA3140 is used as an integrator, a second device is used as a hysteresis switch that determines the starting and stopping points of the sweep. A third CA3140 is used as a logarithmic shaping network for the log function. Rates and slopes, as well as sawtooth, triangle, and logarithmic sweeps are generated by this circuit.

WIDEBAND OUTPUT AMPLIFIER

Fig. 25 shows a high-slew-rate, wideband amplifier suitable for use as a 50-ohm transmission-line driver. This circuit, when used in conjunction with the function generator and sine-wave shaper circuits shown in Figs. 21 and 23 provides 18 volts peak-to-peak output open-circuited, or 9 volts peak-to-peak output when terminated in 50 ohms. The slew rate required of this amplifier is 28 volts/μs (18 volts peak-to-peak x π x 0.5 MHz).

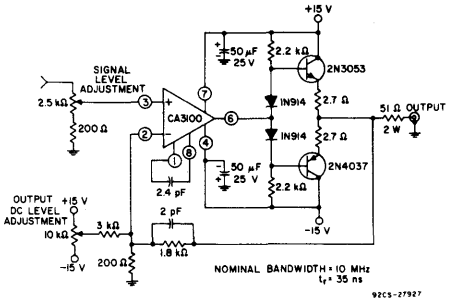


Fig. 25 – Wideband output amplifier.

POWER SUPPLIES

High input-impedance, common-mode capability down to the negative supply and high output-drive current capability are key factors in the design of wide-range output-voltage supplies that use a single input voltage to provide a regulated output voltage that can be adjusted from essentially 0 to 24 volts.

Unlike many regulator systems using comparators having a bipolar transistor-input stage, a high-impedance reference-voltage divider from a single supply can be used in connection with the CA3140 (see Fig. 26).

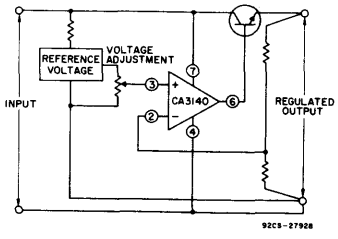


Fig. 26 – Basic single-supply voltage regulator showing voltage-follower configuration.

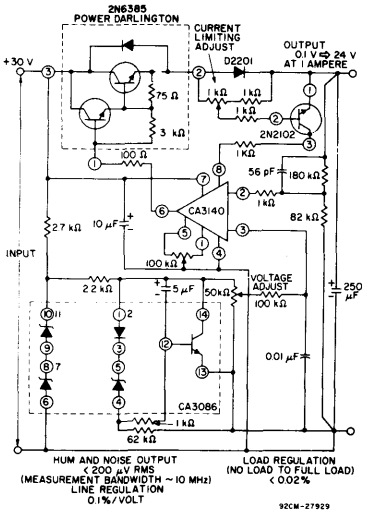


Fig. 27 – Regulated power supply.

Essentially, the regulators, shown in Figs. 27 and 28, are connected as non-inverting power operational amplifiers with a gain of 3.2. An 8-volt reference input yields a maximum output voltage slightly greater than 25 volts. As a voltage follower, when the reference input goes to 0 volts the output will be 0 volts. Because the offset voltage is also multiplied by the 3.2 gain factor, a potentiometer is needed to null the offset voltage.

Series pass transistors with high I_{CBO} levels will also prevent the output voltage from reaching zero because there is a finite voltage drop (V_{CE}sat) across the output of the CA3140 (see Fig.13). This saturation voltage level may indeed set the lowest voltage obtainable.

The high impedance presented by terminal 8 is advantageous in effecting current limiting. Thus, only a small signal transistor is required for the current-limit sensing amplifier. Resistive decoupling is provided for this transistor to minimize damage to it or the CA3140 in the event of unusual input or output transients on the supply-rail.

Figs. 27 and 28, show circuits in which a D2201 high-speed diode is used for the current sensor. This diode was chosen for its slightly higher forward-voltage drop characteristic thus giving greater sensitivity. It must be emphasized that heat sinking of this diode is essential to minimize variation of the current trip point due to internal heating of the diode. That is, 1 ampere at 1 volt forward drop represents one watt which can result in significant regenerative changes in the current trip point as the diode temperature rises. Placing the small-signal reference amplifier in the proximity of the current-sensing diode also helps minimize the variability in the trip level due to the negative temperature coefficient of the diode.

In spite of those limitations, the current limiting point can easily be adjusted over the range from 10 mA to 1 ampere with a single adjustment potentiometer. If the temperature stability of the current-limiting

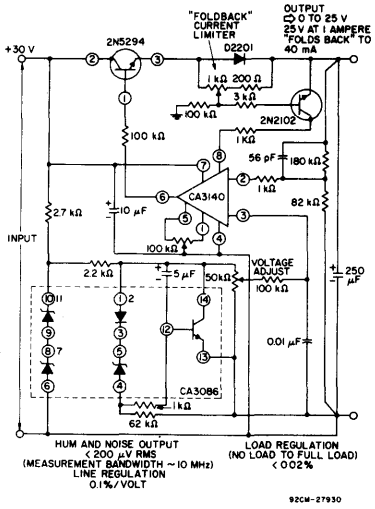
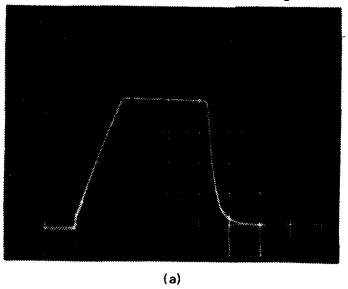
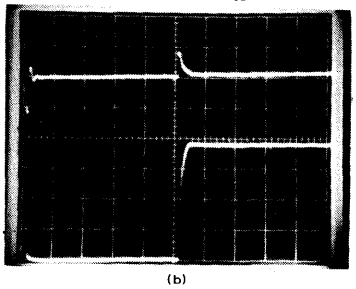


Fig. 28 – Regulated power supply with "foldback" current limiting.



(a)

SUPPLY TURN-ON AND TURN-OFF CHARACTERISTICS
(5 VOLTS / DIV AND -1 s/DIV.)
92CS-27882



(b)

TRANSIENT RESPONSE
TOP TRACE: OUTPUT VOLTAGE (200 mV/DIV AND 5 µs/DIV)
BOTTOM TRACE: COLLECTOR OF LOAD SWITCHING TRANSISTOR, LOAD = 1 AMPERE (5 VOLTS/DIV AND 5 µs/DIV)
92CS-27881

Fig. 29 – Waveforms of dynamic characteristics of power supply currents shown in Figs. 29 and 30.

system is a serious consideration, the more usual current-sampling resistor-type of circuitry should be employed.

A power Darlington transistor (in a heat sink TO-3 case), is used as the series-pass element for the conventional current-limiting system, Fig. 27, because high-power Darlington dissipation will be encountered at low output voltage and high currents.

A small heat-sink VERSAWATT transistor is used as the series-pass element in the foldback current system, Fig.28, since dissipation levels will only approach 10 watts. In this system, the D2201 diode is used for current sampling. Foldback is provided by the 3 kΩ and 100 kΩ divider network connected to the base of the current-sensing transistor.

Both regulators, Figs. 27 and 28, provide better than 0.02% load regulation. Because there is constant loop gain at all voltage settings, the regulation also remains constant. Line regulation is 0.1% per volt. Hum and noise voltage is less than 200 μV as read with a meter having a 10-MHz bandwidth.

Fig.31 (a) shows the turn ON and turn OFF characteristics of both regulators. The slow turn-on rise is due to the slow rate of rise of the reference voltage. Fig. 29 (b) shows the transient response of the regulator with the switching of a 20-Ω load at 20 volts output.

TONE CONTROL CIRCUITS

High-slew-rate, wide-bandwidth, high-output voltage capability and high input impedance are all characteristics required of tone-control amplifiers. Two tone control circuits that exploit these characteristics of the CA3140 are shown in Figs. 30 and 31.

The first circuit, shown in Fig. 31, is the Baxandall tone-control circuit which provides unity gain at midband and uses standard linear potentiometers. The high input impedance of the CA3140 makes possible the use of low-cost, low-value, small-size capacitors, as well as reduced load of the driving stage.

Bass treble boost and cut are ± 15 dB at 100 Hz and 10 kHz, respectively. Full peak-to-peak output is available up to at least 20 kHz due to the high slew rate of the CA3140. The amplifier gain is −3 dB down from its "flat" position at 70 kHz.

Fig. 30 shows another tone-control circuit with similar boost and cut specifications. The wideband gain of this circuit is equal to the ultimate boost or cut plus one, which in this case is a gain of eleven. For 20-dB boost and cut, the input loading of this circuit is essentially equal to the value of the resistance from terminal No.3 to ground. A detailed analysis of this circuit is given in "An IC Operational Transconductance Amplifier (OTA) With Power Capability" by L, Kaplan and H. Wittlinger, IEEE Transactions on Broadcast and Television Receivers, Vol. BTR-18, No.3, August, 1972.

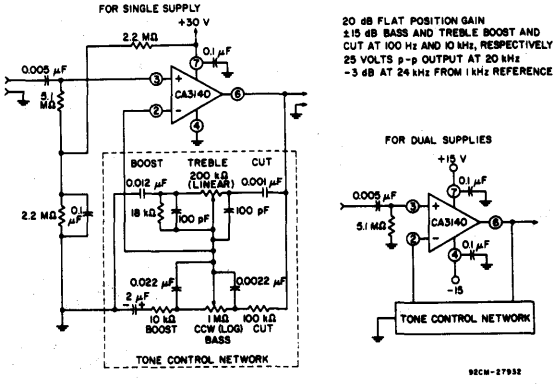


Fig. 30 – Tone control circuit using CA3130 series (20-dB midband gain).

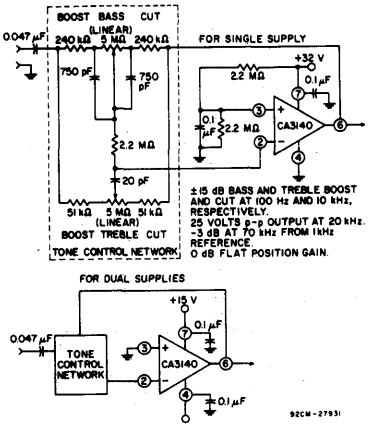


Fig. 31 – Baxandall tone control circuit using CA3140 series.

WIEN BRIDGE OSCILLATOR

Another application of the CA3140 that makes excellent use of its high input-impedance, high-slew-rate, and high-voltage qualities is the Wien Bridge sine-wave oscillator. A basic Wien Bridge oscillator is shown in Fig. 32. When $R_1 = R_2 = R$ and $C_1 = C_2 = C$, the frequency equation reduces to the familiar $f = 1/2\pi RC$ and the gain required for oscillation, A_{OSC} is equal to 3. Note that if C_2 is increased by a factor of four and R_2 is reduced by a factor of four, the gain required for oscillation becomes 1.5, thus permitting a potentially higher operating frequency closer to the gain-bandwidth product of the CA3140.

Oscillator stabilization takes on many forms. It must be precisely set, otherwise the amplitude will either diminish or reach some form of limiting with high levels of distortion. The element, R_S, is commonly replaced with some variable resistance element. Thus, through some control means, the value of R_S is adjusted to maintain constant oscillator output. A FET channel resistance, a thermistor, a lamp bulb, or other device whose resistance is made to increase as the output amplitude is increased are a few of the elements often utilized.

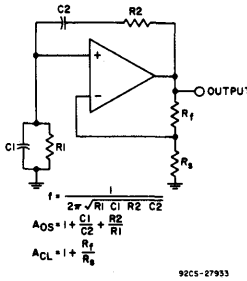


Fig. 32 – Basic Wien bridge oscillator circuit using an operational amplifier.

Fig. 33 shows another means of stabilizing the oscillator with a zener diode shunting the feedback resistor (R_f of Fig. 32). As the output signal amplitude increases, the zener diode impedance decreases resulting in more feedback with consequent reduction in gain; thus stabilizing the amplitude of the output signal. Furthermore, this combination of a monolithic zener diode and bridge rectifier circuit tends to provide a zero temperature coefficient for this regulating system. Because this bridge rectifier system has no time constant, i.e., thermal time constant for the lamp bulb, and RC time constant for filters often used in detector networks, there is no lower frequency limit. For example, with 1-μF polycarbonate capacitors and 22 MΩ for the frequency determining network, the operating frequency is 0.007 Hz.

As the frequency is increased, the output amplitude must be reduced to prevent the output signal from becoming slew-rate limited. An output frequency of 180 kHz will reach a slew rate of approximately 9 volts/μs when its amplitude is 16 volts peak-to-peak.

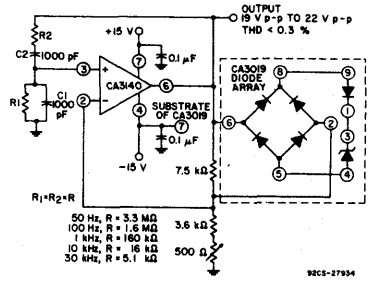


Fig. 33 – Wien bridge oscillator circuit using CA3140 series.

SIMPLE SAMPLE-AND-HOLD SYSTEM

Fig. 34 shows a very simple sample-and-hold system using the CA3140 as the readout amplifier for the storage capacitor. The CA3080A serves as both input buffer amplifier and low feed-through transmission switch.* System offset nulling is accomplished with the CA3140 via its offset nulling terminals. A typical simulated load of 2 kΩ and 30 pF is shown in the schematic.

In this circuit, the storage compensation capacitance (C_1) is only 200 pF. Larger value capacitors provide longer "hold" periods but with slower slew rates. The slew rate

$$\frac{dv}{dt} = \frac{i}{c} = 0.5 \text{ mA}/200 \text{ pF} = 2.5 \text{ V}/\mu\text{s}.$$

* ICAN-6668 "Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers".

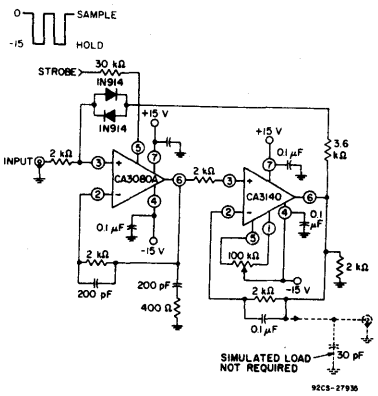
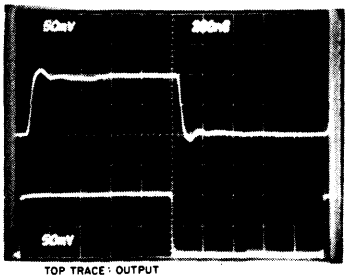
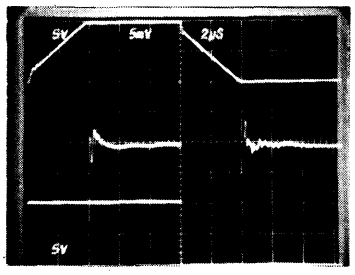


Fig. 34 – Sample- and hold circuit.

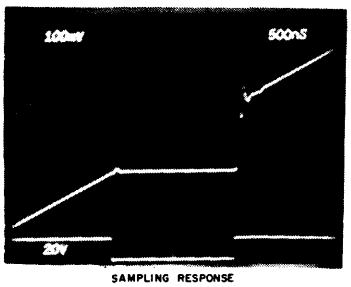
Pulse "droop" during the hold interval is 170 pA/200 pF which is = 0.85 μV/μs; (i.e., 170 pA/200 pF). In this case, 170 pA represents the typical leakage current of the CA3080A when strobed off. If C_1 were increased to 2000 pF, the "hold-droop" rate will decrease to 0.085 μV/μs, but the slew rate would decrease to 0.25 V/μs. The parallel diode network connected between terminal 3 of the CA3080A and terminal 6 of the CA3140 prevents large input-signal feed-through across the input terminals of the CA3080A to the 200 pF storage capacitor when the CA3080A is strobed off. Fig. 35 shows dynamic characteristic waveforms of this sample-and-hold system.



TOP TRACE: OUTPUT
(50 mV/DIV. AND 200 ns/DIV.)
BOTTOM TRACE: INPUT
(50 mV/DIV. AND 200 ns/DIV.)
92CS-27883



LARGE-SIGNAL RESPONSE AND SETTLING TIME
TOP TRACE: OUTPUT SIGNAL
(5 V/DIV. AND 2 μs/DIV.)
BOTTOM TRACE: INPUT SIGNAL
(5 V/DIV. AND 2 μs/DIV.)
CENTER TRACE: DIFFERENCE OF INPUT AND OUTPUT SIGNALS THROUGH TEKTRONIX AMPLIFIER 7A13
(5 mV/DIV. AND 2 μs/DIV.) 92CS-27884



SAMPLING RESPONSE
TOP TRACE: SYSTEM OUTPUT
(100 mV/DIV. AND 500 ns/DIV.)
BOTTOM TRACE: SAMPLING SIGNAL
(20 V/DIV. AND 500 ns/DIV.)
92CS-27885

Fig. 35 – Sample- and hold system dynamic characteristics waveforms.

CURRENT AMPLIFIER

The low input-terminal current needed to drive the CA3140 makes it ideal for use in current-amplifier applications such as the one shown in Fig. 36.• In this circuit, low current is supplied at the input potential as the power supply to load resistor R_L. This load current is increased by the multiplication factor R2/R1, when the load current is monitored by the power supply meter M. Thus, if the load current is 100 nA, with values shown, the load current presented to the supply will be 100 μA; a much easier current to measure in many systems.

Note that the input and output voltages are transferred at the same potential and only the output current is multiplied by the scale factor.

The dotted components show a method of decoupling the circuit from the effects of high output-load capacitance and the potential oscillation in this situation. Essentially, the necessary high-frequency feedback is provided by the capacitor with the dotted series resistor providing load decoupling.

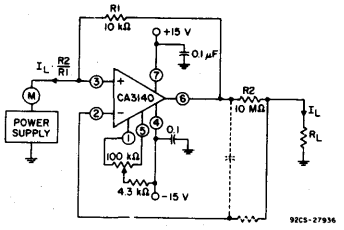


Fig. 36 – Basic current amplifier for low-current measurement systems.

Fig. 37 shows a single-supply, absolute-value, ideal full-wave rectifier with associated waveforms. During positive excursions, the input signal is fed through the feedback network directly to the output. Simultaneously, the positive excursion of the input signal also drives the output terminal (No.6) of the inverting amplifier in a negative-going excursion such that the 1N914 diode effectively disconnects the amplifier from the signal path. During a negative-going excursion of the input signal, the CA3140 functions as a normal inverting amplifier with a gain equal to −R2/R1. When the equality of the two equations shown in Fig. 37 is satisfied, the full-wave output is symmetrical.

• "Operational Amplifiers Design and Applications", J. G. Graeme, McGraw-Hill Book Company, page 308 – "Negative Immittance Converter Circuits".

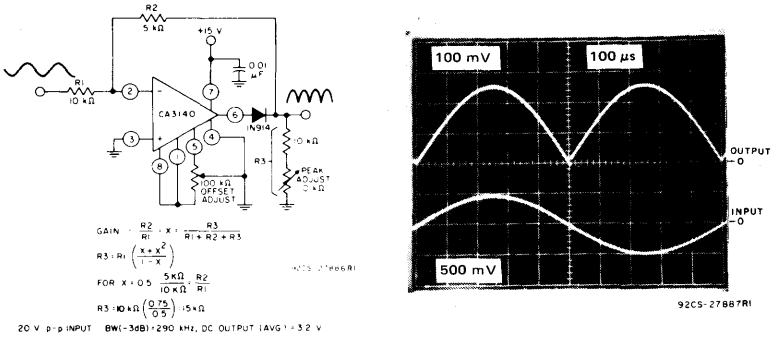


Fig. 37 – Single-supply, absolute-value, ideal full-wave rectifier with associated waveforms.

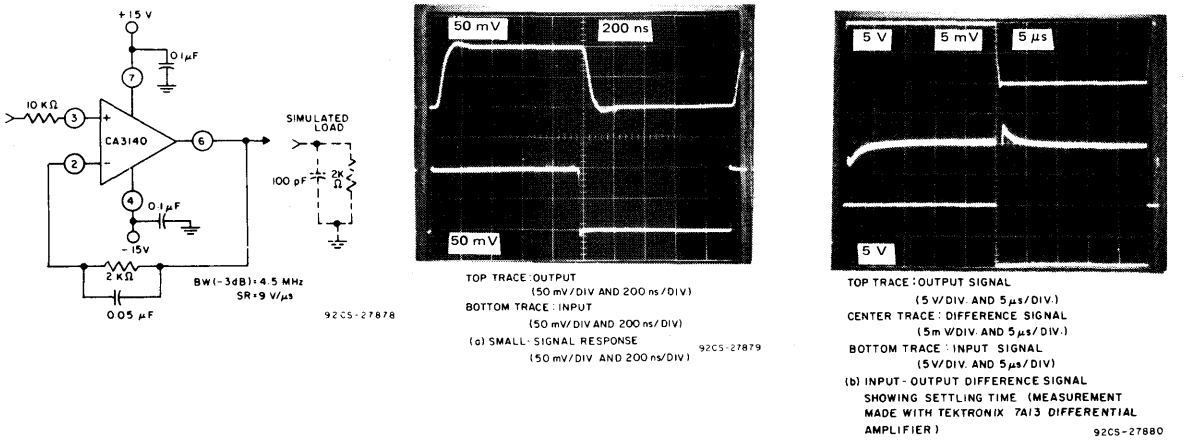


Fig. 38 – Split-supply voltage-follower test circuit and associated waveforms.

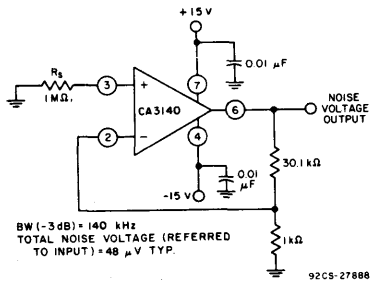
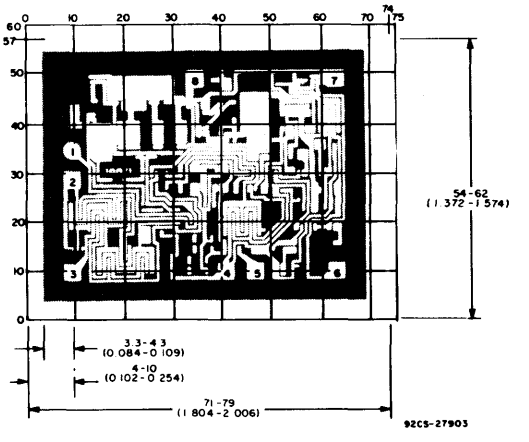


Fig. 39 – Test circuit amplifier (30-dB gain) used for wideband noise measurement.



CA3140H Chip

The photographs and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

CA3141E

High-Voltage Diode Array

For Commercial, Industrial, and Military Applications

Features:

- **Matched monolithic construction – V_F for each diode pair matched to within 0.55 mV (typ.) at I_F = 1 mA**
- **Low diode capacitance – 0.3 pF (typ.) at V_R = 2 V**
- **High diode-to-substrate breakdown voltage – 30 V (min.)**
- **Low reverse (leakage) current – 100 nA (max.)**

Applications:

- **Balanced modulators or demodulators**
- **Analog switches**
- **High-voltage diode gates**
- **Current ratio detectors**

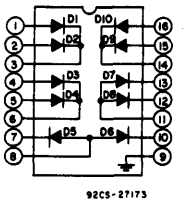


Fig. 1 – Terminal assignment.

The RCA-CA3141E High-Voltage Diode Array consists of ten general-purpose high-reverse-breakdown diodes. Six diodes are internally connected to form three common-cathode diode pairs, and the remaining four diodes are internally connected to form two common-anode diode pairs. Integrated circuit construction assures excellent static and dynamic matching of the diodes, making the CA3141E extremely useful for a wide variety of applications in communications and switching systems.

The CA3141E is supplied in the 16-lead dual-in-line plastic package (E suffix), and in chip form (H suffix).

MAXIMUM RATINGS, *Absolute Maximum Values*

PEAK INVERSE VOLTAGE (PIV)	30 V
PEAK DIODE-TO-SUBSTRATE VOLTAGE	30 V
PEAK FORWARD SURGE CURRENT [I_F (SURGE)]	100 mA
DC FORWARD CURRENT (I_F)	25 mA
DISSIPATION:	
Any one diode unit	50 mW
Total Package:	
Up to 55°C	650 mW
For $T_A > 55°C$	Derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10s max.	+265°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	TEST CONDITIONS		LIMITS Min.	Typ.	Max.	UNIT
DC Forward Voltage Drop, V_F	I_F (Anode)	100 μA	–	0.7	0.9	V
		1 mA	–	0.78	1	
		10 mA	–	0.93	1.2	
DC Reverse Breakdown Voltage, $V_{(BR)R}$	I_F = −10 μA		30	50	–	V
DC Breakdown Voltage Between Any Diode and Substrate, $V_{(BR)DI}$	I_{DI} = 10 μA		30	50	–	V
DC Reverse (Leakage) Current, I_R	V_F = −20 V		–	–	100	nA
DC Reverse (Leakage) Current Between Any Diode and Substrate, I_{DI}	V_{DI} = 20 V		–	–	100	nA
Magnitude of Diode Offset Voltage Between Diode Pairs	V_{DI} = 20 V, I_{FA} = 1 mA		–	0.55	–	mV
Temperature Coefficient of Forward Voltage Drop, $\Delta V_F/\Delta T$	I_F = 1 mA		–	−1.5	–	mV/°C
Reverse Recovery Time, t_{rr}	I_F = 2 mA, I_R = 2 mA		–	50	–	ns
Diode Capacitance, C_D			See Fig. 5			pF
Diode Anode-to-Substrate Capacitance, C_{DAI}			See Fig. 6			pF
Diode Cathode-to-Substrate Capacitance, C_{DCI}			See Fig. 7			pF
Magnitude of Cathode-to-Anode Current Ratio, $\lvert I_{FC}/I_{FA} \rvert$	I_{FA} = 1 mA, V_{DS} = 10 V		0.9	0.96	–	

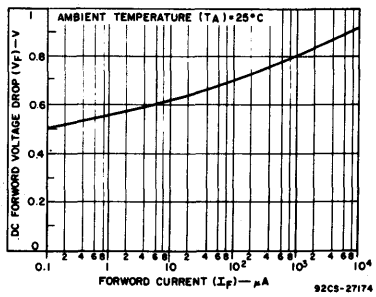


Fig. 2 – DC forward voltage drop vs. forward current.

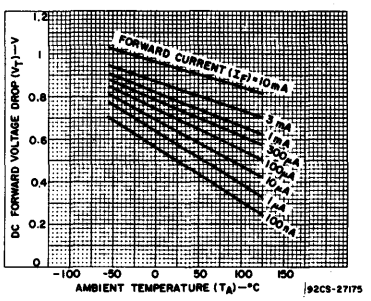


Fig. 3 – DC forward voltage drop vs. ambient temperature.

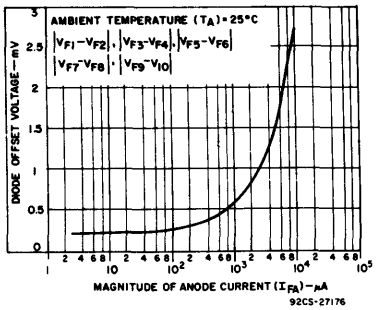


Fig. 4 – Diode offset voltage vs. magnitude of anode current.

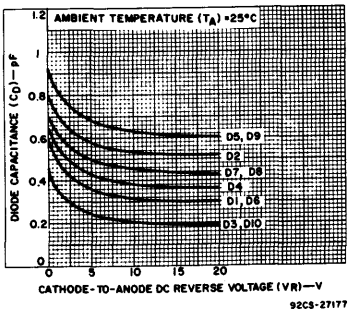


Fig. 5 – Diode capacitance vs. cathode-to-anode reverse voltage.

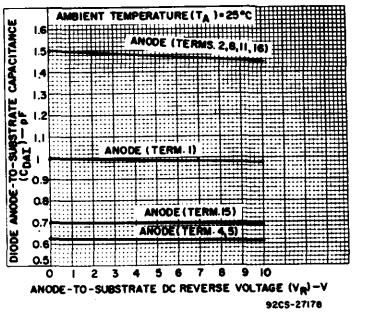


Fig. 6 – Diode anode-to-substrate capacitance vs. reverse voltage.

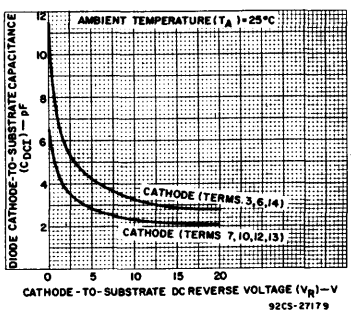


Fig. 7 – Diode cathode-to-substrate capacitance vs. cathode-to-substrate DC reverse voltage.

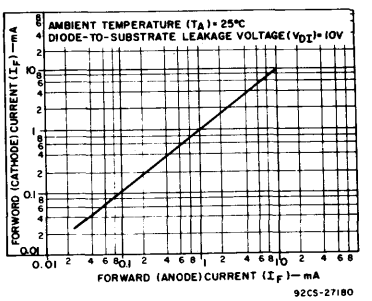


Fig. 8 – Forward (cathode) current vs. forward (anode) current.

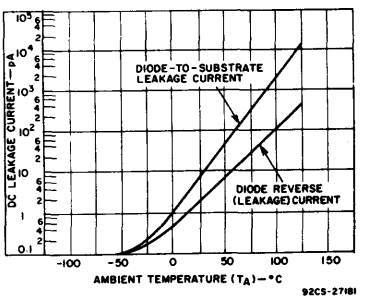


Fig. 9 – DC leakage current vs. ambient temperature.

CA3160, CA3160A, CA3160B Types
BiMOS Operational Amplifiers

With MOS/FET Input, COS/MOS Output

The RCA-CA3160T, CA3160S, CA3160E; CA3160AT, CA3160AS, CA3160AE; and CA3160BT, CA3160BS are integrated-circuit operational amplifiers that combine the advantages of both COS/MOS and bipolar transistors on a monolithic chip. The CA3160 series circuits are frequency-compensated versions of the popular CA3130 series.

Gate-protected p-channel MOS/FET (PMOS) transistors are used in the input circuit to provide very-high-input impedance, very-low-input current, and exceptional speed performance. The use of PMOS field-effect transistors in the input stage results in common-mode input-voltage capability down to 0.5 volt below the negative-supply terminal, an important attribute in single-supply applications.

A complementary-symmetry MOS (COS/MOS) transistor-pair, capable of swinging the output voltage to within 10 millivolts of either supply-voltage terminal (at very high values of load impedance), is employed as the output circuit.

The CA3160 Series circuits operate at supply voltages ranging from 5 to 16 volts, or +2.5 to +8 volts when using split supplies, and have terminals for adjustment of offset voltage for applications requiring offset-null capability. Terminal provisions are also made to permit strobing of the output stage.

The CA3160 series is supplied in standard 8-lead TO-5-style packages (T suffix) and 8-lead dual-in-line formed-lead TO-5 style "DIL-CAN" packages (S suffix). The CA3160 is available in chip form (H suffix).

The CA3160A and CA3160 are also available in the 8-lead dual-in-line plastic package (Mini-DIP-E suffix). All types operate over the full military-temperature range of −55°C to +125°C. The CA3160B is intended for applications requiring preminium-grade specifications and with limits established for: input current, temperature coefficient of input-offset voltage, and gain over the range of −55°C to +125°C. The CA3160A offers superior input characteristics over those of the CA3160.

Features:

- **Similar to CA3130 but has internal compensation**
- **MOS/FET input stage provides:**
  very high Z_I = 1.5 TΩ (1.5 x 10^{12} Ω) typ.
  very low I_I = 5 pA typ. at 15-V operation
  2 pA typ. at 5-V operation
- **Common-mode input-voltage range includes negative supply rail; input terminals can be swung 0.5 V below negative supply rail**
- **COS/MOS output stage permits signal swing to either (or both) supply rails**

} **Ideal for single-supply applications**

- **Low V_{IO}: 2 mV max. (CA3160B)**
- **Wide BW: 4 MHz typ. (unity-gain crossover)**
- **High SR: 10 V/μs typ. (unity-gain follower)**
- **High output current (I_O): 20 mA typ.**
- **High A_{OL}: 320,000 (110 dB) typ.**
- **Internal phase compensation for unity gain (With terminal access for supplementary external phase compensation network if desired)**

Applications:

- **Ground-referenced single-supply amplifiers**
- **Fast sample-hold amplifiers**
- **Long-duration timers/monostables**
- **Ideal interface with digital COS/MOS**
- **High-input-impedance wideband amplifiers**
- **Voltage followers (e.g., follower for single-supply D/A converter)**
- **Voltage regulators (permits control of output voltage down to zero volts)**
- **Wien-Bridge oscillators**
- **Voltage-controlled oscillators**
- **Photo-diode sensor amplifiers**

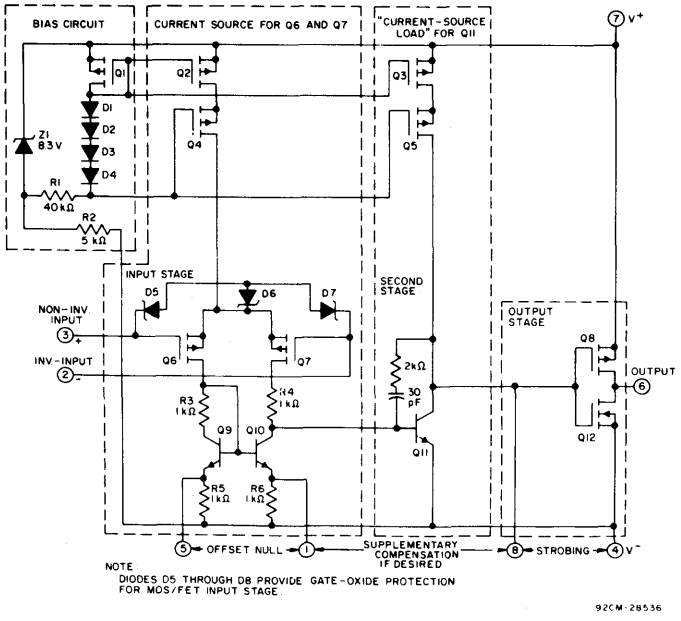
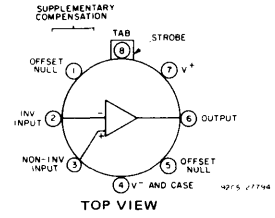
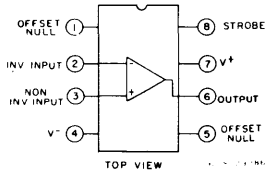


Fig. 1 – Schematic diagram of the CA3160 Series.



S and T Suffixes



E Suffix

CA3160 Series devices have an on-chip frequency-compensation network. Supplementary phase-compensation or frequency roll-off (if desired) can be connected externally between terminals 1 and 8.

Fig.2 – Functional diagrams of the CA3160 Series.

MAXIMUM RATINGS, *Absolute-Maximum Values*

DC SUPPLY VOLTAGE (Between V^+ and V^- Terminals) ..... 16 V
DIFFERENTIAL-MODE INPUT VOLTAGE ..... ±8 V
COMMON-MODE DC INPUT VOLTAGE... (V^+ +8 V) to (V^- −0.5 V)
INPUT-TERMINAL CURRENT ..... 1 mA
DEVICE DISSIPATION:
WITHOUT HEAT SINK −
UP TO 55°C ..... 630 mW
ABOVE 55°C .... Derate linearly 6.67 mW/°C
WITH HEAT SINK −
AT 125°C ..... 418 mW
BELOW 125°C ... Derate linearly 16.7 mW/°C

TEMPERATURE RANGE:
OPERATING (All Types).... −55 to +125°C
STORAGE (All Types) ..... −65 to +150°C
OUTPUT SHORT-CIRCUIT DURATION* ..... INDEFINITE
LEAD TEMPERATURE (DURING SOLDERING):
AT DISTANCE 1/16 ± 1/32 INCH (1.59 ± 0.79 MM) FROM CASE FOR 10 SECONDS MAX. ..... +265°C

*Short circuit may be applied to ground or to either supply.

ELECTRICAL CHARACTERISTICS at T_A=25°C, V^+=15 V, V^- = 0 V (Unless otherwise specified)

CHARACTERISTIC	LIMITS									Units
	CA3160B (T, S)			CA3160A (T, S, E)			CA3160 (T, S, E)			
	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage, $\|V_{IO}\|$, $V^{\pm}$=±7.5 V	–	0.8	2	–	2	5	–	6	15	mV
Input Offset Current, $\|I_{IO}\|$, $V^{\pm}$=±7.5 V	–	0.5	10	–	0.5	20	–	0.5	30	pA
Input Current, I_I $V^{\pm}$=±7.5 V	–	5	20	–	5	30	–	5	50	pA
Large-Signal Voltage Gain, A_{OL} V_O=10 V_{p-p}, R_L=2 kΩ	100 k	320 k	–	50 k	320 k	–	50 k	320 k	–	V/V
	100	110	–	94	110	–	94	110	–	dB
Common-Mode Rejection Ratio, CMRR	86	100	–	80	95	–	70	90	–	dB
Common-Mode Input-Voltage Range, V_{ICR}	0	−0.5 to 12	10	0	−0.5 to 12	10	0	−0.5 to 12	10	V
Power-Supply Rejection Ratio, $\Delta V_{IO}/\Delta V^{\pm}$ $V^{\pm}$=±7.5 V	–	32	100	–	32	150	–	32	320	μV/V
Maximum Output Voltage: At R_L=2 kΩ V_{OM}^+	12	13.3	–	12	13.3	–	12	13.3	–	V
At R_L=2 kΩ V_{OM}^-	–	0.002	0.01	–	0.002	0.01	–	0.002	0.01	V
At R_L=∞ V_{OM}^+	14.99	15	–	14.99	15	–	14.99	15	–	V
At R_L=∞ V_{OM}^-	–	0	0.01	–	0	0.01	–	0	0.01	V
Maximum Output Current: I_{OM}^+ (Source) @ V_O = 0 V	12	22	45	12	22	45	12	22	45	mA
I_{OM}^- (Sink) @ V_O = 15 V	12	20	45	12	20	45	12	20	45	mA
Supply Current, I^+: V_O=7.5 V, R_L=∞	–	10	15	–	10	15	–	10	15	mA
V_O = 0 V, R_L = ∞	–	2	3	–	2	3	–	2	3	mA
Input Current, I_I*	–	Fig.11	15	–	Fig.11	–	–	Fig.11	–	nA
Input Offset Voltage Temp. Drift, $\Delta V_{IO}/\Delta T$*	–	5	15	–	6	–	–	8	–	μV/°C
Large-Signal Voltage Gain, A_{OL}*	50 k	320 k	–	–	320 k	–	–	320 k	–	V/V
	94	110	–	–	110	–	–	110	–	dB

* T_A = −55 to +125°C, $V^{\pm}$ = ±7.5 V (I_I and $\Delta V_{IO}/\Delta T$), V_O = 10 V_{p-p} and R_L = 2 kΩ (A_{OL}).

CIRCUIT DESCRIPTION

Fig.3 is a block diagram of the CA3160 series COS/MOS Operational Amplifiers. The input terminals may be operated down to 0.5 V below the negative supply rail, and the output can be swung very close to either supply rail in many applications. Consequently, the CA3160 series circuits are ideal for single-supply operation. Three class A amplifier stages, having the individual gain capability and current consumption shown in Fig.3, provide the total gain of the CA3160. A biasing circuit provides two potentials for common use in the first and second stages. Terminals 8 and 1 can be used to supplement the internal phase compensation network if additional phase compensation or frequency roll-off is desired. Terminals 8 and 4 can also be used to strobe the output stage into a low quiescent current state. When Terminal 8 is tied to the negative supply rail (Terminal 4) by mechanical or electrical means, the output potential at Terminal 6 essentially rises to the positive supply-rail potential at Terminal 7. This condition of essentially zero current drain in the output stage under the strobed "OFF" condition can only be achieved when the ohmic load resistance presented to the amplifier is very high (e.g., when the amplifier output is used to drive COS/MOS digital circuits in comparator applications).

Input Stages – The circuit of the CA3160 is shown in Fig.1. It consists of a differential-input stage using PMOS field-effect transistors (Q6, Q7) working into a mirror-pair of bipolar transistors (Q9, Q10) functioning as load resistors together with resistors R3 through R6. The mirror-pair transistors also function as a differential-to-single-ended converter to provide base drive to the second-stage bipolar transistor (Q11). Offset nulling, when desired, can be effected by connecting a 100,000-ohm potentiometer across Terms. 1 and 5 and the potentiometer slider arm to Term. 4. Cascode-connected PMOS transistors Q2, Q4, are the constant-current source for the input stage. The biasing circuit for the constant-current source is subsequently described. The small diodes D5 through D7 provide gate-oxide protection against high-voltage transients, e.g., including static electricity during handling for Q6 and Q7.

Second-Stage – Most of the voltage gain in the CA3160 is provided by the second amplifier stage, consisting of bipolar transistor Q11 and its cascode-connected load resistance provided by PMOS transistors Q3 and Q5. The source of bias potentials for these PMOS transistors is described later. Miller Effect compensation (roll off) is accomplished by means of the 30-pF capacitor and 2-kΩ resistor connected between the base and collector of transistor Q11. These internal components provide sufficient compensation for unity gain operation in most applications. However, additional compensation, if desired, may be used between Terminals 1 and 8.

Bias-Source Circuit – At total supply voltages, somewhat above 8.3 volts, resistor R2 and zener diode Z1 serve to establish a voltage of 8.3 volts across the series-connected

TYPICAL VALUES INTENDED ONLY FOR DESIGN GUIDANCE

CHARACTERISTIC	TEST CONDITIONS		CA3160B (T, S)	CA3160A (T, S, E)	CA3160 (T, S, E)	UNITS
	$V^+ = +7.5$ V, $V^- = -7.5$ V, $T_A = 25°C$ (Unless Otherwise Specified)					
Input Offset Voltage Adjustment Range	10 kΩ across Terms. 4 and 5 or 4 and 1		±22	±22	±22	mV
Input Resistance, R_I			1.5	1.5	1.5	TΩ
Input Capacitance, C_I	f = 1 MHz		4.3	4.3	4.3	pF
Equivalent Input Noise Voltage, e_n	BW= 0.2 MHz	R_S=1 MΩ	40	40	40	μV
		R_S=10MΩ	50	50	50	
Equivalent Input Noise Voltage, e_n	R_S= 100 Ω	1 kHz	72	72	72	$nV\sqrt{Hz}$
		10 kHz	30	30	30	
Unity Gain Crossover Frequency, f_T			4	4	4	MHz
Slew Rate, SR:			10	10	10	V/μs
Transient Response: Rise Time, t_r	C_L = 25 pF, R_L = 2 kΩ (Voltage Follower)		0.09	0.09	0.09	μs
Overshoot			10	10	10	%
Settling Time (4 V_{p-p} Input to <0.1%)			1.8	1.8	1.8	μs

CHARACTERISTIC	TEST CONDITIONS	CA3160B (T, S)	CA3160A (T, S, E)	CA3160 (T, S, E)	UNITS
	$V^+ = 5$ V, $V^- = 0$ V, $T_A = 25°C$ (Unless Otherwise Specified)				
Input Offset Voltage, V_{IO}		1	2	6	mV
Input Offset Current, I_{IO}		0.1	0.1	0.1	pA
Input Current, I_I		2	2	2	pA
Common-Mode Rejection Ratio, CMRR		100	90	80	dB
Large-Signal Voltage Gain, A_{OL}	$V_O = 4\ V_{p-p}$, R_L = 5 kΩ	100 k	100 k	100 k	V/V
		100	100	100	dB
Common-Mode Input Voltage Range, V_{ICR}		0 to 2.8	0 to 2.8	0 to 2.8	V
Supply Current, I^+	V_O = 5 V, $R_L = \infty$	300	300	300	μA
	V_O = 2.5 V, $R_L = \infty$	500	500	500	
Power Supply Rejection Ratio, $\Delta V_{IO}/\Delta V^+$		200	200	200	μV/V

CIRCUIT DESCRIPTION (cont'd)

circuit, consisting of resistor R1, diodes D1 through D4, and PMOS transistor Q1. A tap at the junction of resistor R1 and diode D4 provides a gate-bias potential of about 4.5 volts for PMOS transistors Q4 and Q5 with respect to Terminal 7. A potential of about 2.2 volts is developed across diode-connected PMOS transistor Q1 with respect to Terminal 7 to provide gate bias for PMOS transistors Q2 and Q3. It should be noted that Q1 is "mirror-connected"† to both Q2 and Q3. Since transistors Q1, Q2, Q3 are designed to be identical, the approximately 200-microampere current in Q1 establishes a similar current in Q2 and Q3 as constant-current sources for both the first and second amplifier stages, respectively.

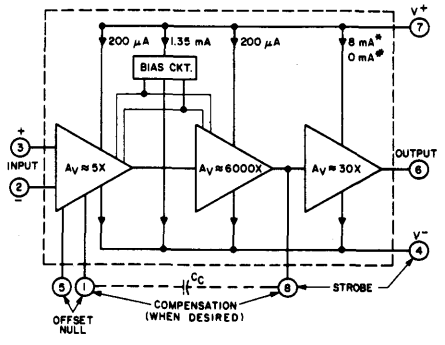
At total supply voltages somewhat less than 8.3 volts, zener diode Z1 becomes non-conductive and the potential, developed across series-connected R1, D1-D4, and Q1, varies directly with variations in supply voltage. Consequently, the gate bias for Q4, Q5 and Q2, Q3 varies in accordance with supply-voltage variations. This variation results in deterioration of the power-supply-rejection ratio (PSRR) at total supply voltages below 8.3 volts. Operation at total supply voltages below about 4.5 volts results in seriously degraded performance.

Output Stage – The output stage consists of a drain-loaded inverting amplifier using COS/MOS transistors operating in the Class A mode. When operating into very high resistance loads, the output can be swung within millivolts of either supply rail. Because the output stage is a drain-loaded amplifier, its gain is dependent upon the load impedance. The transfer characteristics of the output stage for a load returned to the negative supply rail are shown in Fig.6. Typical op-amp loads are readily driven by the output stage. Because large-signal excursions are non-linear, requiring feedback for good waveform reproduction, transient delays may be encountered. As a voltage follower, the amplifier can achieve 0.01 per cent accuracy levels, including the negative supply rail.

Offset Nulling

Offset-voltage nulling is usually accomplished with a 100,000-ohm potentiometer connected across Terminals 1 and 5 and with the potentiometer slider arm connected to Terminal 4. A fine offset-null adjustment usually can be effected with the slider arm positioned in the mid-point of the potentiometer's total range.

† For general information on the characteristics of COS/MOS transistor-pairs in linear-circuit applications, see File No. 619, data bulletin on CA3600E "COS/MOS Transistor Array".



TOTAL SUPPLY VOLTAGE (FOR INDICATED VOLTAGE GAINS) = 15 V
*WITH INPUT TERMINALS BIASED SO THAT TERM. 6 POTENTIAL IS +7.5 V ABOVE TERM. 4.
#WITH OUTPUT TERMINAL DRIVEN TO EITHER SUPPLY RAIL.
92CS-28573

Fig. 3 – Block diagram of the CA3160 Series.

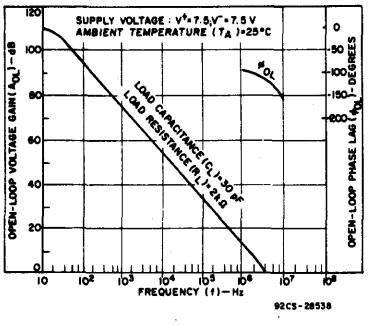


Fig.4 – Open-loop voltage gain and phase shift vs. frequency for various values of C_L and R_L.

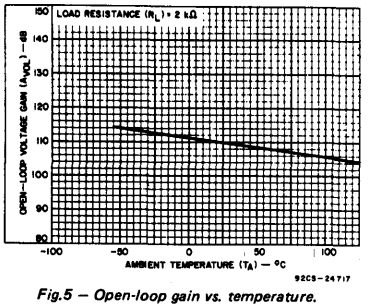


Fig.5 – Open-loop gain vs. temperature.

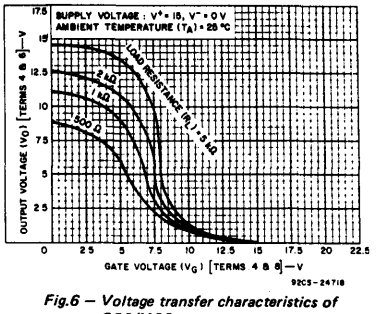


Fig.6 – Voltage transfer characteristics of COS/MOS output stage.

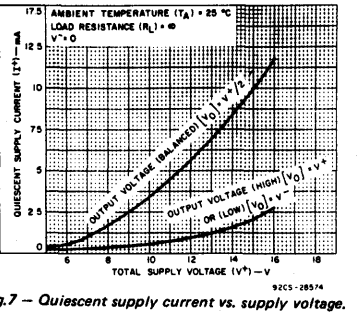


Fig.7 – Quiescent supply current vs. supply voltage.

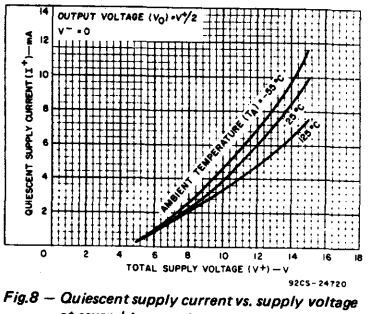


Fig.8 – Quiescent supply current vs. supply voltage at several temperatures.

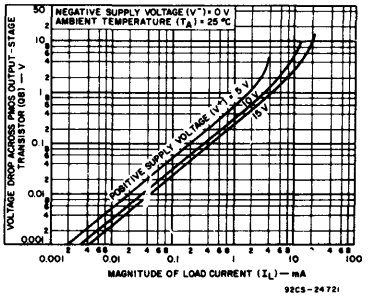


Fig.9 – Voltage across PMOS output transistor (Q8) vs. load current.

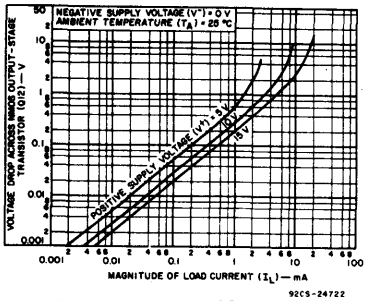


Fig.10 – Voltage across NMOS output transistor (Q12) vs. load current.

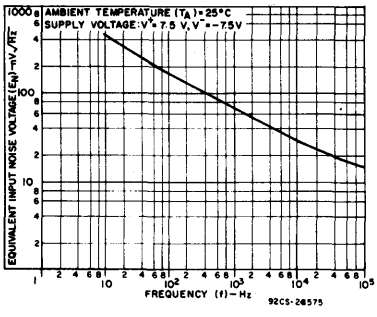


Fig.11 – Equivalent noise voltage vs. frequency.

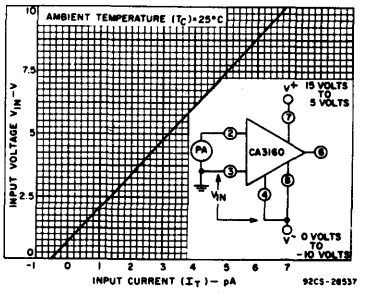


Fig.12 – Input current vs. common-mode voltage.

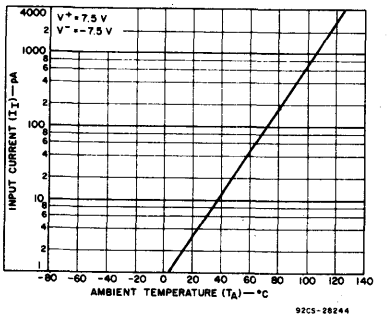


Fig. 13 – Input current vs. ambient temperature.

Input Current Variation with Common-Mode Input Voltage

As shown in the Table of Electrical Characteristics, the input current for the CA3160 Series Op-Amps is typically 5 pA at T_A=25°C when Terminals 2 and 3 are at a common-mode potential of +7.5 volts with respect to negative supply Terminal 4. Fig. 12 contains data showing the variation of input current as a function of common-mode input voltage at T_A=25°C. These data show that circuit designers can advantageously exploit these characteristics to design circuits which typically require an input current of less than 1 pA, provided the common-mode input voltage does not exceed 2 volts. As previously noted, the input current is essentially the result of the leakage current through the gate-protection diodes in the input circuit and, therefore, a function of the applied voltage. Although the finite resistance of the glass terminal-to-case insulator of the TO-5 package also contributes an increment of leakage current, there are useful compensating factors. Because the gate-protection network functions as if it is connected to Terminal 4 potential, and the TO-5 case of the CA3160 is also internally tied to Terminal 4, input terminal 3 is essentially "guarded" from spurious leakage currents.

Input-Current Variation with Temperature

The input current of the CA3160 Series circuits is typically 5 pA at 25°C. The major portion of this input current is due to leakage current through the gate-protective diodes in the input circuit. As with any semiconductor-junction device, including op amps with a junction-FET input stage, the leakage current approximately doubles for every 10°C increase in temperature. Fig. 13 provides data on the typical variation of input bias current as a function of temperature in the CA3160.

In applications requiring the lowest practical input current and incremental increases in current because of "warm-up" effects, it is suggested that an appropriate heat sink be used with the CA3160. In addition, when "sinking" or "sourcing" significant output current the chip temperature increases, causing an increase in the input current. In such cases, heat-sinking can also very markedly reduce and stabilize input current variations.

Input-Offset-Voltage (V_{IO}) Variation with DC Bias vs. Device Operating Life

It is well known that the characteristics of a MOS/FET device can change slightly when a dc gate-source bias potential is applied to the device for extended time periods. The magnitude of the change is increased at high temperatures. Users of the CA3160 should be alert to the possible impacts of this effect if the application of the device involves extended operation at high temperatures with a significant differential dc bias voltage applied across Terminals 2 and 3. Fig. 14 shows typical data pertinent to shifts in offset voltage encountered with CA3160 devices in TO-5 packages during life testing. At lower temperatures (TO-5 and plastic) for example at 85°C, this change in voltage is considerably less. In typical linear applications where the differential voltage is small and symmetrical, these incremental changes are of about the same magnitude as those encountered in an operational amplifier employing a bipolar transistor input stage. The two-volt dc differential voltage example represents conditions when the amplifier output state is "toggled", e.g., as in comparator applications.

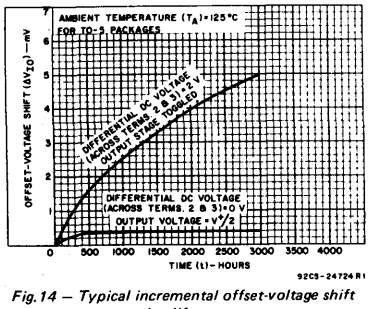
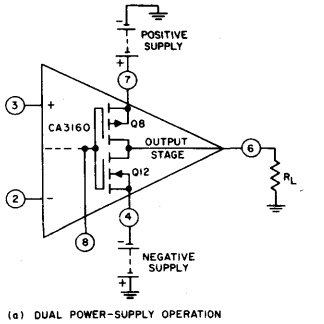


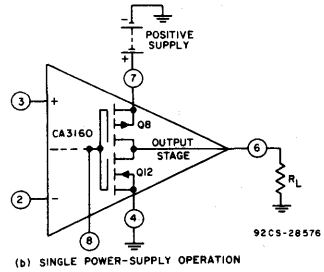
Fig. 14 — Typical incremental offset-voltage shift vs. operating life.

Power-Supply Considerations

Because the CA3160 is very useful in single-supply applications, it is pertinent to review some considerations relating to power-supply current consumption under both single- and dual-supply service. Figs. 15(a) and 15(b) show the CA3160 connected for both dual- and single-supply operation.



(a) DUAL POWER-SUPPLY OPERATION



(b) SINGLE POWER-SUPPLY OPERATION

Fig. 15 — CA3160 output stage in dual and single power-supply operation.

Dual-supply operation: When the output voltage at Terminal 6 is zero-volts, the currents supplied by the two power supplies are equal. When the gate terminals of Q8 and Q12 are driven increasingly positive with respect to ground, current flow through Q12 (from the negative supply) to the load is increased and current flow through Q8 (from the positive supply) decreases correspondingly. When the gate terminals of Q8 and Q12 are driven increasingly negative with respect to ground, current flow through Q8 is increased and current flow through Q12 is decreased accordingly.

Single-supply operation: Initially, let it be assumed that the value of R_L is very high (or disconnected), and that the input-terminal bias (Terminals 2 and 3) is such that the output terminal (No. 6) voltage is at $V^+/2$, i.e., the voltage-drops across Q8 and Q12 are of equal magnitude. Fig. 7 shows typical quiescent supply-current vs. supply-voltage for the CA3160 operated under these conditions.

Since the output stage is operating as a Class A amplifier, the supply-current will remain constant under dynamic operating conditions as long as the transistors are operated in the linear portion of their voltage-transfer characteristics (see Fig. 6). If either Q8 or Q12 are swung out of their linear regions toward cut-off (a non-linear region), there will be a corresponding reduction in supply-current. In the extreme case, e.g., with Terminal 8 swung down to ground potential (or tied to ground), NMOS transistor Q12 is completely cut off and the supply-current to series-connected transistors Q8, Q12 goes essentially to zero. The two preceding stages in the CA3160, however, continue to draw modest supply-current (see the lower curve in Fig. 7) even though the output stage is strobed off. Fig. 15(a) shows a dual-supply arrangement for the output stage that can also be strobed off, assuming R_L=∞, by pulling the potential of Terminal 8 down to that of Terminal 4.

Let it now be assumed that a load-resistance of nominal value (e.g., 2 kilohms) is connected between Terminal 6 and ground in the circuit of Fig. 15(b). Let it further be assumed again that the input-terminal bias (Terminals 2 and 3) is such that the output terminal (No. 6) voltage is a $V^+/2$. Since PMOS transistor Q8 must now supply quiescent current to both R_L and transistor Q12, it should be apparent that under these conditions the supply-current must increase as an inverse function of the R_L magnitude. Fig. 9 shows the voltage-drop across PMOS transistor Q8 as a function of load current at several supply voltages. Fig. 6 shows the voltage-transfer characteristics of the output stage for several values of load resistance.

Wideband Noise

From the standpoint of low-noise performance considerations, the use of the CA3160 is most advantageous in applications where in the source resistance of the input signal is in the order of 1 megohm or more. In this case, the total input-referred noise voltage is typically only 40 μV when the test-circuit amplifier of Fig.16 is operated at a total supply voltage of 15 volts. This value of total input-referred noise remains essentially constant, even though the value of source resistance is raised by an order of magnitude.

This characteristic is due to the fact that reactance of the input capacitance becomes a significant factor in shunting the source resistance. It should be noted, however, that for values of source resistance very much greater than 1 megohm, the total noise voltage generated can be dominated by the thermal noise contributions of both the feedback and source resistors.

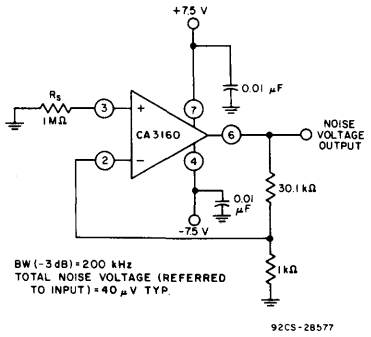


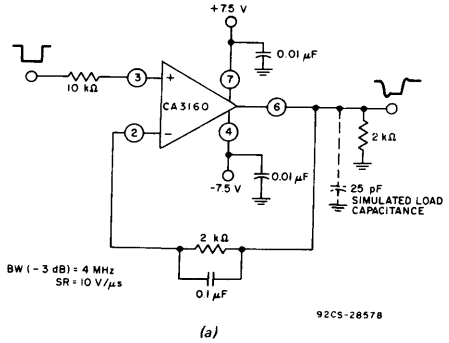
Fig. 16 – Test-circuit amplifier (30-dB gain) used for wideband noise measurements.

TYPICAL APPLICATIONS

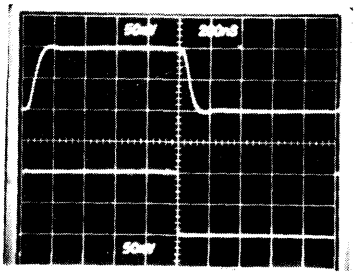
Voltage Followers

Operational amplifiers with very high input resistances, like the CA3160, are particularly suited to service as voltage followers. Fig.17 shows the circuit of a classical voltage follower, together with pertinent waveforms using the CA3160 in a split-supply configuration.

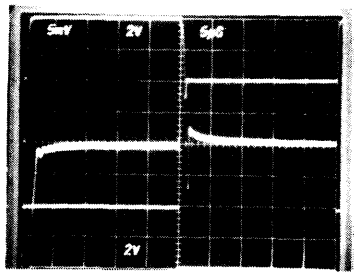
A voltage follower, operated from a single-supply, is shown in Fig.18 together with related waveforms. This follower circuit is linear over a wide dynamic range, as illustrated by the reproduction of the output waveform in Fig.18b with input-signal ramping. The waveforms in Fig.18c show that the follower does not lose its input-to-output phase-sense, even though the input is being swung 7.5 volts below ground potential. This unique characteristic is an important attribute in both operational amplifier and comparator applications. Fig.18c also shows the manner in which the COS/MOS output stage permits the output signal to swing down to the negative supply-rail potential (i.e., ground in the case shown). The digital-to-analog converter (DAC) circuit, described in the following section, illustrates the practical use of the CA3160 in a single-supply voltage-follower application.



(a)



(b) Small Signal Response
Top Trace: Output
Bottom Trace: Input



(c) Input-Output Difference Signal Showing Settling Time
Top Trace: Output Signal
Center Trace: Difference Signal 5 mV/div
Bottom Trace: Input Signal

Fig. 17 – Split-supply voltage follower with associated waveforms.

9-Bit COS/MOS DAC

A typical circuit of a 9-bit Digital-to-Analog Converter (DAC)* is shown in Fig.19. This system combines the concepts of multiple-switch COS/MOS IC's, a low-cost ladder network of discrete metal-oxide-film resistors, a CA3160 op amp connected as a follower, and an inexpensive monolithic regulator in a simple single power-supply arrangement. An additional feature of the DAC is that it is readily interfaced with COS/MOS input logic, e.g., 10-volt logic levels are used in the circuit of Fig.19.

The circuit uses an R/2R voltage-ladder network, with the output-potential obtained directly by terminating the ladder arms at either the positive or the negative power-supply terminal. Each CD4007A contains three "inverters", each "inverter" functioning as a single-pole double-throw switch to terminate an arm of the R/2R network at either the positive or negative power-supply terminal. The resistor ladder is an assembly of one per cent tolerance metal-oxide film resistors. The five arms requiring the highest accuracy are assembled with series and parallel combinations of 806,000-ohm resistors from the same manufacturing lot.

A single 15-volt supply provides a positive bus for the CA3160 follower amplifier and feeds the CA3085 voltage regulator. A "scale-adjust" function is provided by the regulator output control, set to a nominal 10-volt level in this system. The line-voltage regulation (approximately 0.2%) permits a 9-bit accuracy to be maintained with variations of several volts in the supply. The flexibility afforded by the COS/MOS building blocks simplifies the design of DAC systems tailored to particular needs.

Error-Amplifier in Regulated Power Supplies

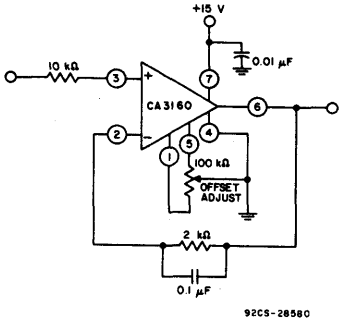
The CA3160 is an ideal choice for error-amplifier service in regulated power supplies since it can function as an error-amplifier when the regulated output voltage is required to approach zero.

The circuit shown in Fig.20 uses a CA3160 as an error amplifier in a continuously adjustable 1-ampere power supply. One of the key features of this circuit is its ability to regulate down to the vicinity of zero volts with only one dc power supply input.

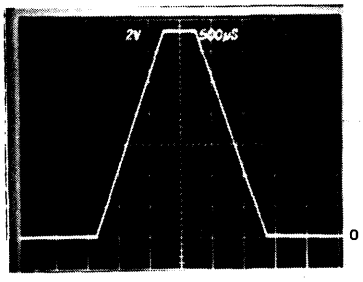
An RC network, connected between the base of the output drive transistor and the input voltage, prevents "turn-on overshoot", a condition typical of many operational-amplifier regulator circuits. As the amplifier becomes operational, this RC network ceases to have any influence on the regulator performance.

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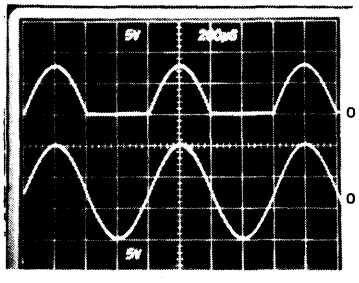
* "Digital-to-Analog Conversion Using the RCA-CD4007A COS/MOS IC", Application Note ICAN-6080.



(a)



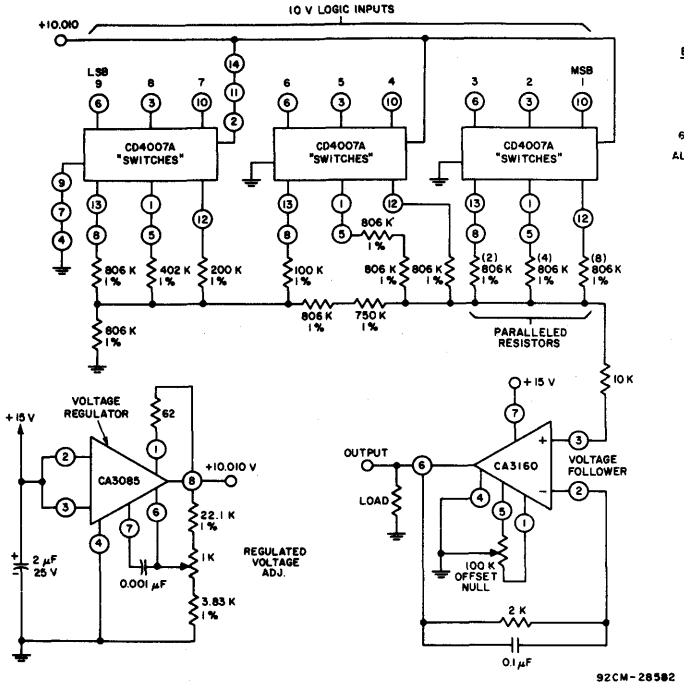
(b) Output signal with input-signal ramping.



92CS-28581R1

(c) Output-Waveform with Ground-Reference
Sine-Wave Input
Top Trace: Output
Bottom Trace: Input

Fig. 18 — Single-supply voltage-follower with associated waveforms. (e.g., for use in single-supply D/A converter; see Fig.9 in ICAN-6080.)



BIT	REQUIRED RATIO-MATCH
1	STANDARD
2	±0.1%
3	±0.2%
4	±0.4%
5	±0.8%
6-9	±1% ABS.

ALL RESISTANCES IN OHMS

Fig. 19 — 9-bit DAC using COS/MOS digital switches and CA3160.

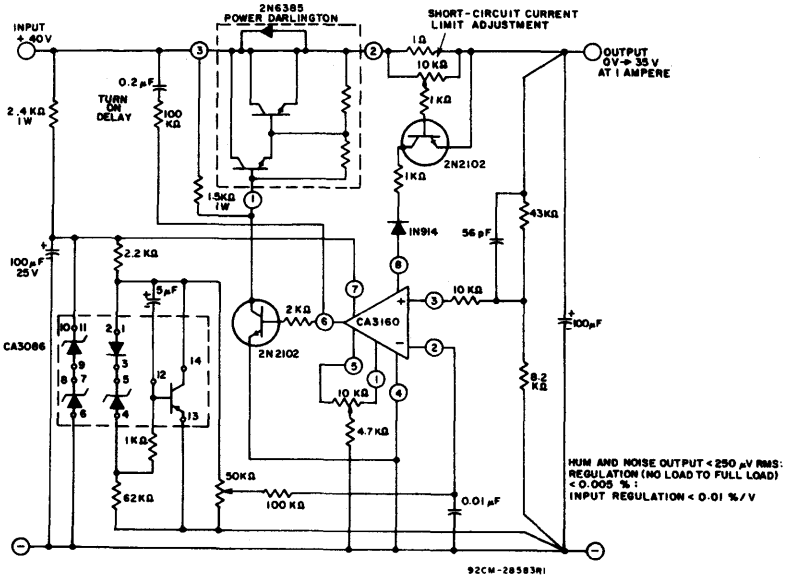


Fig.20 – Voltage regulator circuit (0.1 to 35 V at 1 A).

Precision Voltage-Controlled Oscillator

The circuit diagram of a precision voltage-controlled oscillator is shown in Fig.21. The oscillator operates with a tracking error in the order of 0.02 percent and a temperature coefficient of 0.01%/°C. A multivibrator (A_1) generates pulses of constant amplitude (V) and width (T_2). Since the output (terminal 6) of A_1 (a CA3130) can swing within about 10 millivolts of either supply-rail, the output pulse amplitude (V) is essentially equal to V+. The average output voltage ($E_{avg} = V\ T_2/T_1$) is applied to the non-inverting input terminal of comparator A_2 via an integrating network R_3, C_2. Comparator A_2 operates to establish circuit conditions such that $E_{avg} = V1$. This circuit condition is accomplished by feeding an output signal from terminal 6 of A_2 through R_4, D_4 to the inverting terminal (terminal 2) of A_1, thereby adjusting the multivibrator interval, T_3.

Voltmeter With High Input Resistance

The voltmeter circuit shown in Fig.22 illustrates an application in which a number of the CA3160 characteristics are exploited. Range-switch SW1 is ganged between input and output circuitry to permit selection of the proper output voltage for feedback to Terminal 2 via 10 KΩ current-limiting resistor. The circuit is powered by a single 8.4-volt mercury battery. With zero input signal, the circuit consumes somewhat less than 500 microamperes plus the meter current required to indicate a given voltage. Thus, at full-scale input, the total supply current rises to slightly more than 1500 microamperes.

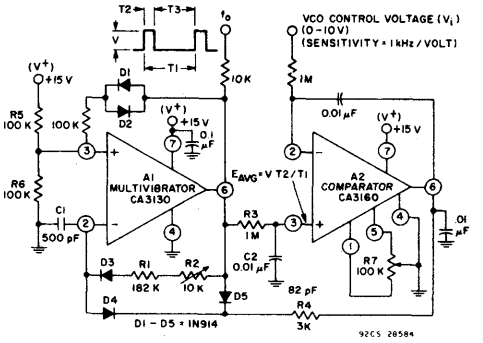


Fig.21 – Voltage-controlled oscillator.

Function Generator

A function generator having a wide tuning range is shown in Fig.23. The adjustment range, in excess of 1,000,000/1, is accomplished by a single potentiometer. Three operational amplifiers are utilized: a CA3160 as a voltage follower, a CA3080 as a high-speed comparator, and a second CA3080A as a programmable current source. Three variable capacitors C1, C2, and C3 shape the triangular signal between 500 kHz and 1 MHz. Capacitors C4, C5, and the trimmer potentiometer in series with C5 maintain essentially constant (±10%) amplitude up to 1 MHz.

Staircase Generator

Fig.24 shows a staircase generator circuit utilizing three COS/MOS operational amplifiers. Two CA3130's are used; one as a multivibrator, the other as a hysteresis switch. The third amplifier, a CA3160, is used as a linear staircase generator.

Picoammeter Circuit

Fig. 25 is a current-to-voltage converter configuration utilizing a CA3160 and CA3140 to provide a picoampere meter for ±3 pA full-scale meter deflection. By placing Terminals 2 and 4 of the CA3160 at ground potential,

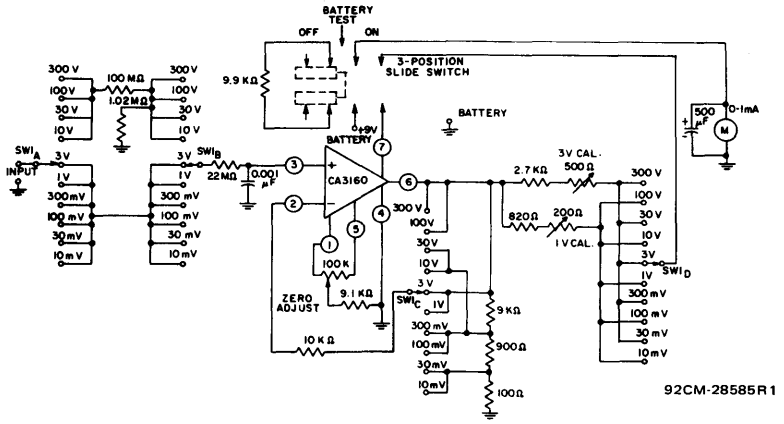
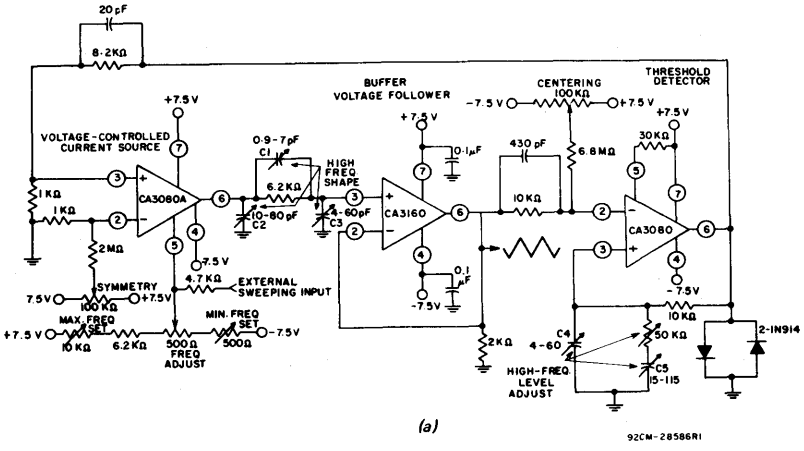
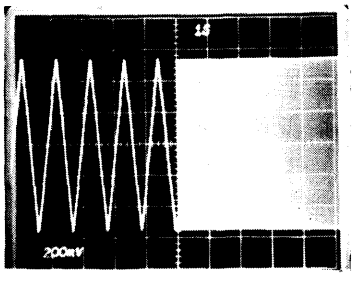


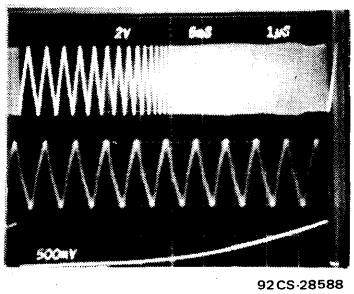
Fig.22 – High-input-resistance DC voltmeter.



(a)



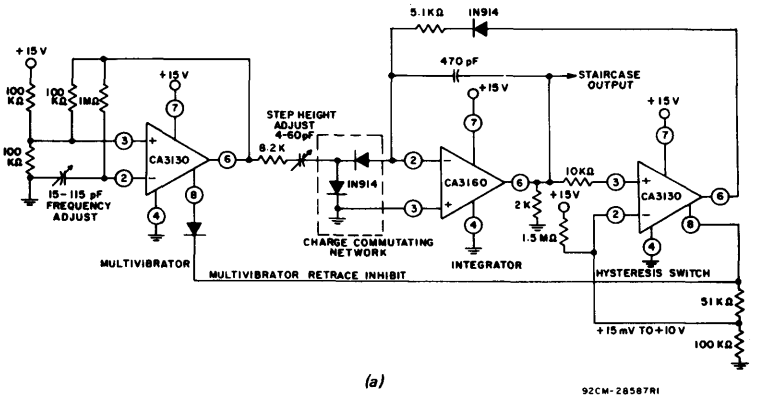
(b) – Two-tone output signal from the function generator. A square-wave signal modulates the external sweeping input to produce 1 Hz and 1 MHz, showing the 1,000,000/1 frequency range of the function generator.



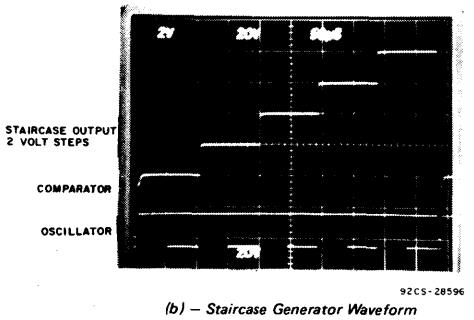
92CS-28588

(c) – Triple-trace of the function generator sweeping to 1 MHz. The bottom trace is the sweeping signal and the top trace is the actual generator output. The center trace displays the 1 MHz signal via delayed oscilloscope triggering of the upper swept output signal.

Fig. 23 – 1,000,000/1 single-control function generator – 1 MHz to 1 Hz.



(a)



(b) – Staircase Generator Waveform
Top Trace: Staircase Output
2 Volt Steps
Center Trace: Comparator
Bottom Trace: Oscillator

Fig. 24 – Staircase generator.

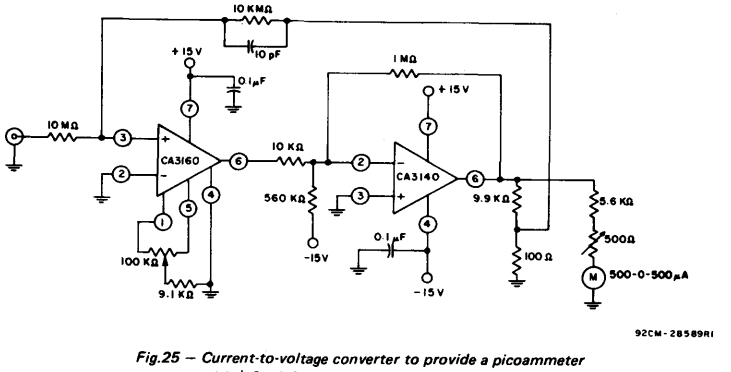


Fig.25 – Current-to-voltage converter to provide a picoammeter with ± 3 pA full-scale deflection.

output (Terminal 6) near ground, thus markedly reducing the dissipation by reducing the supply current to the device.

The CA3140 stage serves as a X100 gain stage to provide the required plus and minus output swing for the meter and feedback network. A 100-to-1 voltage divider network consisting of a 9.9-KΩ resistor in series with a 100-ohm resistor sets the voltage at the 10-KMΩ resistor (in series with Terminal 3) to ±30 mV full-scale deflection. This 30-mV signal results from ±3 volts appearing at the top of the voltage divider network which also drives the meter circuitry.

By utilizing a switching technique in the meter circuit and in the 9.9 KΩ and 100-ohm network similar to that used in voltmeter circuit shown in Fig. 22, a current range of 3 pA to 1 nA full scale can be handled with the single 10-KMΩ resistor.

Single-Supply Sample-and-Hold System

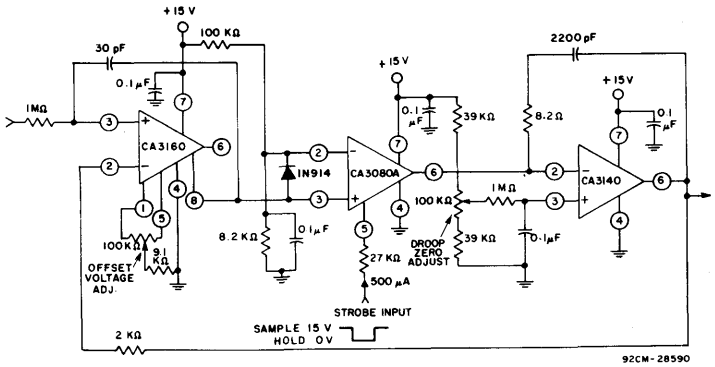
Fig. 26 shows a single-supply sample-and-hold system using a CA3160 to provide a high input impedance and an input-voltage range of 0 to 10 volts. The output from the input buffer integrator network is coupled to a CA3080A. The CA3080A functions as a strobeable current source for the CA3140 output integrator and storage capacitor. The CA3140 was chosen because of its low output impedance and constant gain-bandwidth product. Pulse "droop" during the hold interval can be reduced to zero by adjusting the 100-KΩ bias-voltage potentiometer on the positive input of the CA3080A. This zero adjustment sets the CA3080A output voltage at its zero current position. In this sample-and-hold circuit it is essential that the amplifier bias current be reduced to zero to minimize output signal current during the hold mode. Even with 320 mV at the amplifier bias circuit terminal (5) at least ± 100 pA of output current will be available.

Wien Bridge Oscillator

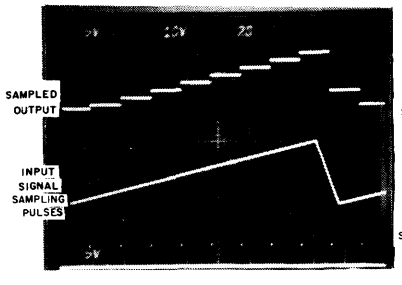
A simple, single-supply Wien Bridge oscillator using a CA3160 is shown in Fig. 27. A pair of parallel-connected 1N914 diodes comprise the gain-setting network which standardizes the output voltage at approximately 1.1 volts. The 500-ohm potentiometer is adjusted so that the oscillator will always start and the oscillation will be maintained. Increasing the amplitude of the voltage may lower the threshold level for starting and for sustaining the oscillation, but will introduce more distortion.

Operation with Output-Stage Power-Booster

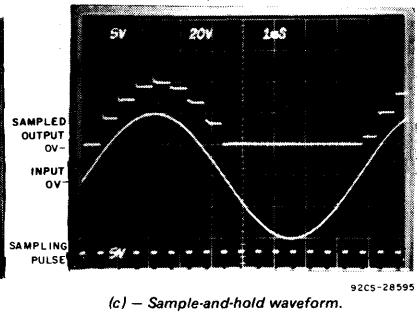
The current sourcing and sinking capability of the CA3160 output stage is easily supplemented to provide power-boost capability. In the circuit of Fig.28, three COS/MOS transistor-pairs in a single CA3600 IC array are shown parallel-connected with the output stage in the CA3160. In the Class A mode of CA3600E shown, a typical device consumes



(a)



(b) – Sample-and-hold waveform.
Top Trace: Sampled Output
Center Trace: Input Signal
Bottom Trace: Sampling Pulses



(c) – Sample-and-hold waveform.
Top Trace: Sampled Output
Center Trace: Input
Bottom Trace: Sampling Pulse

Fig. 26 – Single-supply sample-and-hold system–input 0-to-10 volts.

the CA3160 input is operated in the "guarded mode". Under this operating condition, even slight leakage resistance present between Terminals 3 and 2 or between Terminals 3 and 4 would result in zero voltage across this leakage resistance, thus substantially reducing the leakage current.

If the CA3160 is operated with the same voltage on input Terminals 3 and 2 as on Terminal 4, a further reduction in the input current to the less than one picoampere level can be achieved as shown in Fig. 12.

To further enhance the stability of this circuit, the CA3160 can be operated with its 20 mA of supply current at 15-V operation. This arrangement boosts the current-handling capability of the CA3160 output stage by about 2.5X.

The amplifier circuit in Fig. 28 employs feedback to establish a closed-loop gain of 20 dB. The typical large-signal-bandwidth (–3 dB) is 190 kHz.

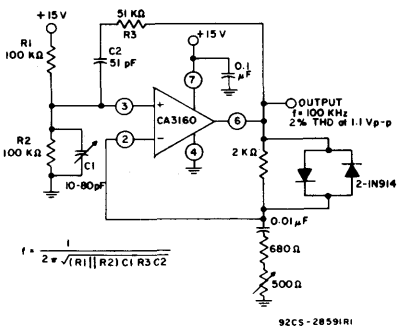


Fig.27 – Single-supply Wien Bridge oscillator.

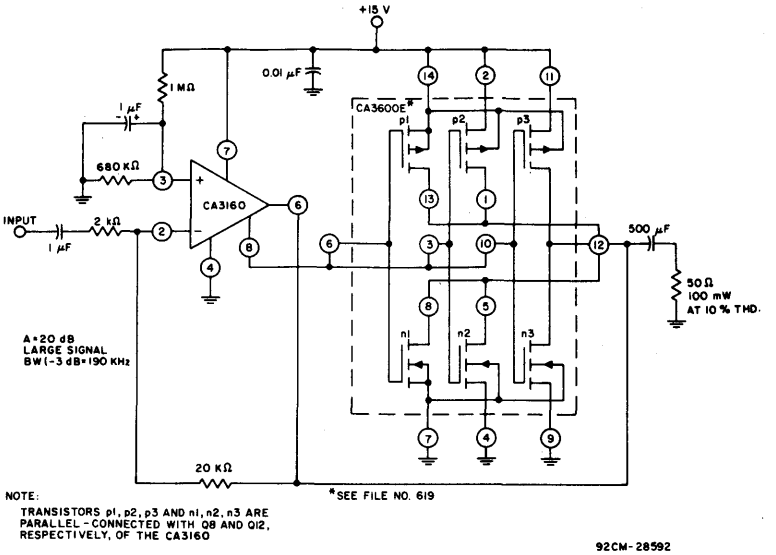
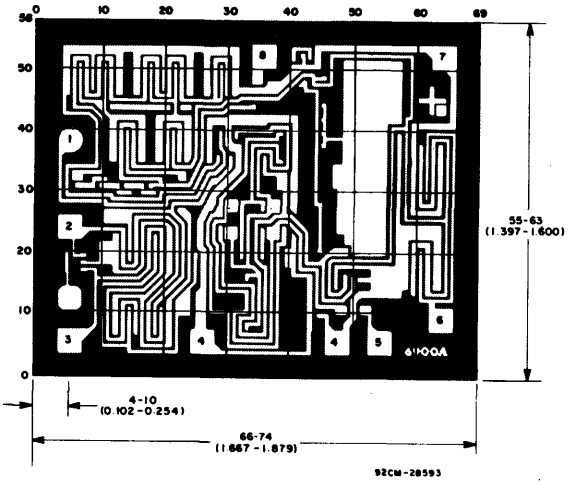


Fig.28 – COS/MOS transistor array (CA3600E) connected as power booster in the output stage of the CA3160.



Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

The photograph and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

CA3161E

BCD-to-Seven-Segment Decoder/Driver

The RCA-CA3161E is a monolithic integrated circuit that performs the BCD-to-seven-segment decoding function and features constant-current segment drivers. When used with the CA3162E A/D Converter* the CA3161E provides a complete digital readout system with a minimum number of external parts.

* The CA3162E is described in RCA data bulletin File No. 1080.

Features:

- **TTL-compatible input logic levels**
- **25-mA (typ.) constant-current segment outputs**
- **Eliminates need for output current-limiting resistors**
- **Pin compatible with other industry standard decoders**
- **Low standby power dissipation — 18 mW (typ.)**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE (between terminals 1 and 10) . . . . . +7 V
INPUT VOLTAGE (terminals 1, 2, 6, 7) . . . . . +5.5 V
OUTPUT VOLTAGE:
Output "Off" . . . . . +7 V
Output "On" (See note 1) . . . . . +10 V
DEVICE DISSIPATION:
Up to T_A = +55°C 1W
Above T_A = +55°C . . . . . derate linearly at 10.5 mW/°C
AMBIENT TEMPERATURE RANGE:
Operating . . . . . 0 to +75°C
Storage . . . . . −65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

NOTE 1: This is the maximum output voltage for any single output. The output voltage must be consistent with the maximum dissipation and derating curve for worst-case conditions. Example: All segments "on", 100% duty cycle.

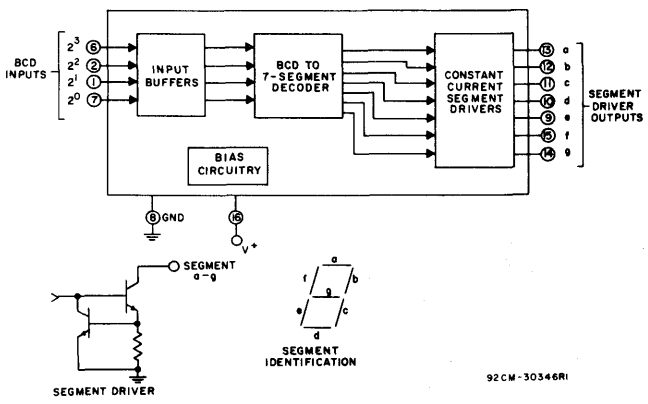
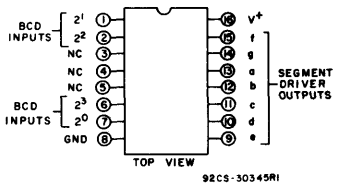
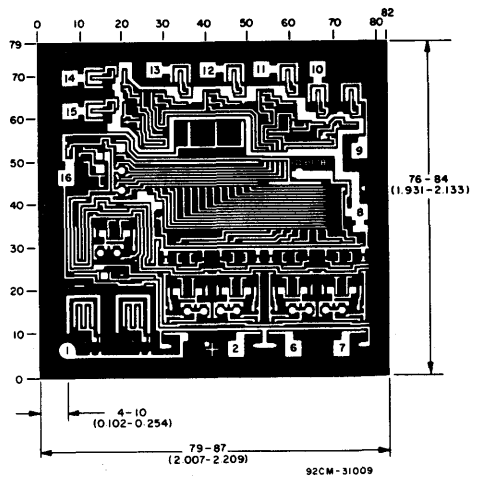


Fig. 1 — Functional block diagram of the CA3161E.



TERMINAL ASSIGNMENT
CA3161E



ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC		LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Voltage Operating Range, V^+		4.5	5	5.25	V
Supply Current, I^+ (all inputs high)		–	3.5	8	mA
Output Current Low (V_O = 2 V)		18	25	32	mA
Output Current High (V_O = 5.5 V)		–	–	250	μA
Input Voltage High (logic "1" level)		2	–	–	V
Input Voltage Low (logic "0" level)		–	–	0.8	V
Input Current High (logic "1")	2 V	−30	–	–	μA
Input Current Low (logic "0")	0 V	−40	–	–	μA
Propagation Delay Time	t_{PHL}	–	2.6	–	μs
	t_{PLH}	–	1.4	–	

The photographs and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

TRUTH TABLE

BINARY STATE	INPUTS				OUTPUTS							DISPLAY
	2^3	2^2	2^1	2^0	a	b	c	d	e	f	g	
0	L	L	L	L	L	L	L	L	L	L	H	0
1	L	L	L	H	H	L	L	H	H	H	H	1
2	L	L	H	L	L	L	H	L	L	H	L	2
3	L	L	H	H	L	L	L	L	H	H	L	3
4	L	H	L	L	H	L	L	H	H	L	L	4
5	L	H	L	H	L	H	L	L	H	L	L	5
6	L	H	H	L	L	H	L	L	L	L	L	6
7	L	H	H	H	L	L	L	H	H	H	H	7
8	H	L	L	L	L	L	L	L	L	L	L	8
9	H	L	L	H	L	L	L	L	H	L	L	9
10	H	L	H	L	H	H	H	H	H	H	L	—
11	H	L	H	H	L	H	H	L	L	L	L	E
12	H	H	L	L	H	L	L	H	L	L	L	H
13	H	H	L	H	H	H	H	L	L	L	H	L
14	H	H	H	L	L	L	H	H	L	L	L	P
15	H	H	H	H	H	H	H	H	H	H	H	BLANK

CA3162E

A/D Converter for 3-Digit Digital Readout System

The CA3162E is a monolithic integrated circuit that comprises the A/D converter section of a 3-digit digital readout system. It is used with the CA3161E BCD-to-Seven-Segment Decoder/Driver* and a minimum of external parts to implement a complete system.

* The CA3161E is described in RCA data bulletin File No. 1079.

Features:

- **Dual-slope A/D conversion**
- **Ultra-stable internal band-gap voltage reference**
- **Capable of reading 99 mV below ground with single supply**
- **Differential input**
- **Internal timing — no external clock required**
- **Choice of low-speed (4-Hz) or high-speed (96-Hz) conversion rate**
- **"Hold" inhibits conversion but maintains display**
- **Multiplexed operation for high efficiency**
- **Overrange indication — "EEE" for reading greater than +999 mV, "___" for reading more negative than −99 mV when used with CA3161E BCD-to-Seven Segment Decoder/Driver**

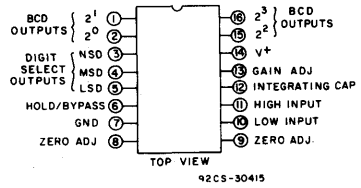
ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 5 V, Zero pot centered, gain pot = 2.4 kΩ unless otherwise stated

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Operating Supply Voltage Range, V^+		4.5	5	5.5	V
Supply Current, I^+	100 kΩ to V^+ on terms. 3,4,5	–	–	17	mA
Input Impedance, Z_I		–	100	–	MΩ
Input Bias Current, I_{IB}	Terms. 10 and 11	–	−80	–	nA
Unadjusted Zero Offset	$V_{11}-V_{10}$ = 0V, read decoded output	−12	–	+12	mV
Unadjusted Gain	$V_{11}-V_{10}$ = 900 mV, read decoded output	846	–	954	mV
Linearity	See Notes 1 and 2	−1	–	+1	Count
Conversion Rate: Slow Mode	Term. 6 = open or gnd	–	4	–	Hz
Fast Mode	Term. 6 = 5 V	–	96	–	Hz
Conversion Control Voltage (Hold Mode) at Terminal 6		0.8	1.2	1.6	V
Common-Mode Input Voltage Range, V_{ICR}	See Note 3	−0.2	–	+0.2	V
BCD Sink Current at terms. 1,2,15,16	$V_{BCD} \leq$ 0.5 V, at logic zero state	0.4	1.6	–	mA
Digit Select Sink Current at terms. 3,4,5	$V_{Digit\ Select}$ = 4V at logic zero state	1.6	2.5	–	mA
Zero Temperature Coefficient	V_I = 0V, zero pot centered	–	10	–	μV/°C
Gain Temperature Coefficient	V_I = 900 mV, gain pot = 2.4 kΩ	–	0.005	–	%/°C

NOTES:

1. Apply zero volts across V_{11} to V_{10}. Adjust zero potentiometer to give 000 mV reading. Apply 900 mV to input and adjust gain potentiometer to give 900 mV reading.
2. Linearity is measured as a difference from a straight line drawn through zero and positive full scale. Limits do not include ±0.5 count bit digitizing error.
3. For applications where negative terminal 10 is not operated at terminal 7 potential, a return path of not more than 100 kΩ resistance must be provided for input bias currents.

TERMINAL ASSIGNMENT
CA3162E



Circuit Description

The functional block diagram of the CA3162E is shown in Fig. 1. The heart of the system is the V/I converter and reference-current generator. The V/I converter converts the input voltage applied between terminals 10 and 11 to a current that charges the integrating capacitor on terminal 12 for a predetermined time interval. At the end of the charging interval, the V/I converter is disconnected from the integrating capacitor, and a band-gap reference constant-current source of opposite polarity is connected. The number of clock counts that elapse before the charge is restored to its original value is a direct measure of the signal induced current.

The restoration is sensed by the comparator, which in turn latches the counter. The count is then multiplexed to the BCD outputs.

The timing for the circuit is derived from a 786-kHz ring oscillator. The oscillator frequency is divided by 2048 to provide the multiplex rate of 384 Hz. This rate is further divided by 96 to obtain the slow-speed conversion rate of 4 Hz (terminal 6 open or grounded). When the "hold" terminal (terminal 6) is biased to ±1.2 V, conversion ceases, but multiplex continues and the reading is held and displayed continuously.

When terminal 6 is biased at +5 V, a portion of the divide-by-96 circuitry is disabled so that the conversion rate increases to 24 times the slow-speed rate (4 x 24 = 96 Hz). Note

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE (between terminals 7 and 14) . . . . . +7 V
INPUT VOLTAGE (terminal 10 or 11 to ground) . . . . . ±15 V
DEVICE DISSIPATION:
Up to T_A = +55°C . . . . . 750 mW
Above T_A = +55°C . . . . . derate linearly at 7.9 mW/°C
AMBIENT TEMPERATURE RANGE:
Operating . . . . . 0 to +75°C
Storage . . . . . −65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):
At distance 1/16 ± 1/32 inch (1.59 ± 0/79 mm) from case for 10 seconds max. . . . . +265°C

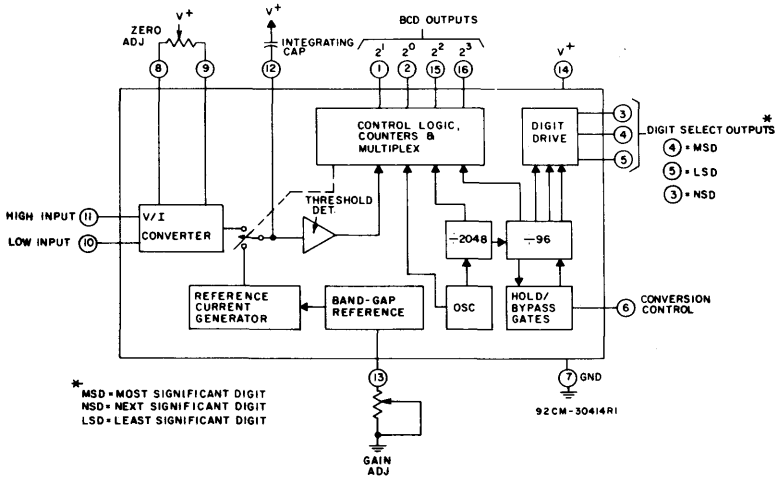


Fig. 1 – Functional block diagram of the CA3162E.

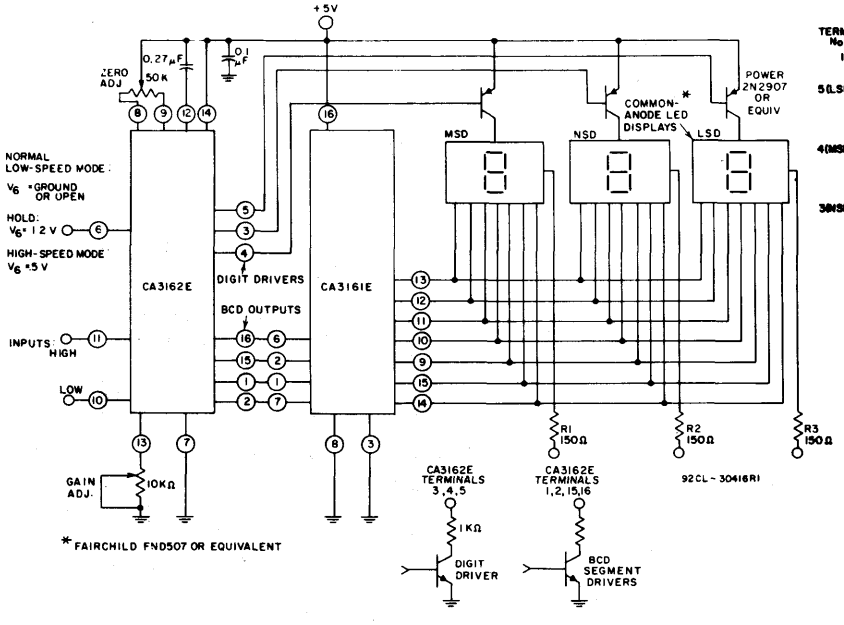


Fig. 2 – Basic digital readout system using the CA3162E and the CA3161E.

that the multiplex rate is unchanged. Fig. 3 shows the timing of conversion and digit select pulses for the high-speed mode. Note that the basic A/D conversion process requires approximately 5 ms in both modes.

The "EEE" or "___" displays indicate that the range of the system has been exceeded in the positive or negative direction, respectively. Negative voltages to −99 mV are displayed with the minus sign in the MSD. The BCD code is 1010 for a negative overrange (___) and 1011 for a positive overrange (EEE).

System Application

Fig. 2 is the block diagram of a basic system using the CA3162E and the CA3161E. An actual-size PC board layout for this circuit is shown in Fig. 4. The BCD outputs of the CA3162E drive the BCD inputs of the CA3161E BCD-to-7-segment decoder directly. The seven-segment outputs are multiplexed to the three LED displays. The digits are selected by terminals 3, 4, and 5 (CA3162E), which provide base current to the external p-n-p transistors. The p-n-p's, in turn, provide current to the anodes of the display. Adjustment procedures for the gain and zero potentiometers are given in Note 1 of the Electrical Characteristics chart.

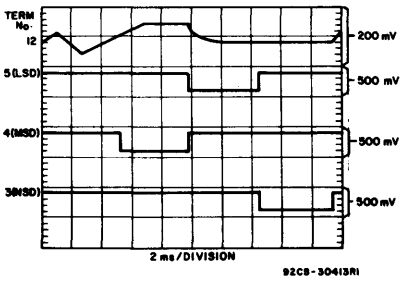


Fig. 3 – High speed mode timing diagram.

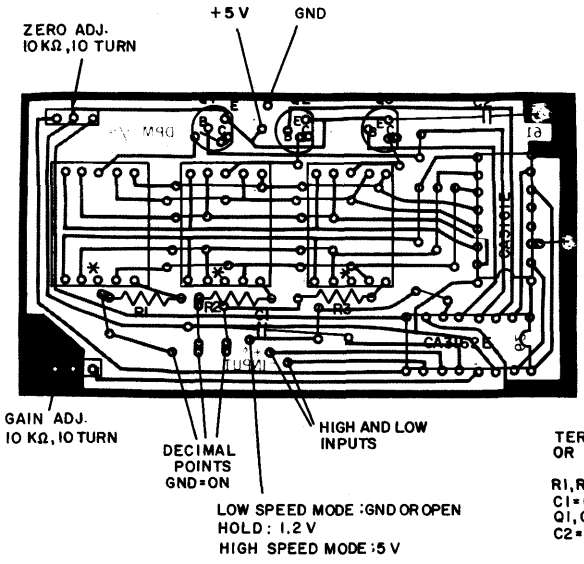
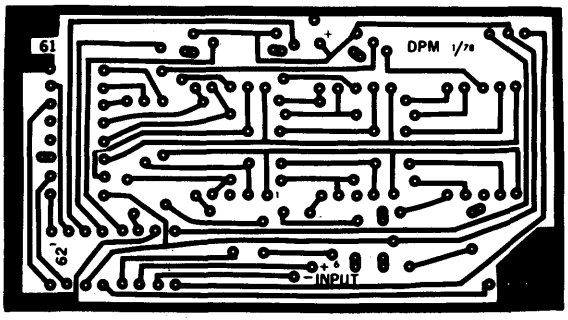


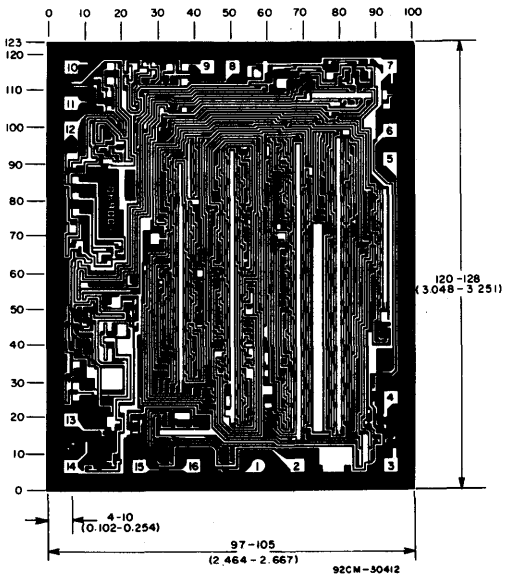
Fig. 4 — Component side of p.c. board 92CS-31017



92CS-31015

Fig. 5 — P.C. board layout for a basic digital readout system using the CA3162E and CA3161E.

Dimensions and pad layout for the CA3162H Chip.



92CM-30412

The photographs and dimensions of each Linear chip represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

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